bury radioactive and chemical wastes from the accident and to entomb the remains of the disabled reactor in a steel and concrete sarcophagus. Many of these workers are now ill, suffering from a variety of problems including immune suppression, increased rates of cancer, and reproductive disorders.

Radiation is a known mutagen, causing damage to DNA. More than 13,000 children in the area surrounding Chernobyl were exposed to the radioactive isotope iodine-131; many had exposures 400 times the maximum annual radiation exposure recommended for workers in the nuclear industry. The rate of thyroid cancer among children in the Ukraine is now 10 times the pre-Chernobyl levels. Chromosome mutations have been detected in the cells of many people who resided near Chernobyl at the time of the accident, and birth defects in the population have increased significantly.

To examine germ-line mutations (those passed on to future generations) resulting from the Chernobyl accident, geneticists collected blood samples from 79 families who resided in heavily contaminated districts. These families included children born in 1994 who had not been exposed to radiation but who might possess mutations acquired from their parents. DNA sequences from these parents and children were analyzed, allowing the researchers to identify possible germ-line mutations. The germ-line mutation rate in these families was found to be twice as high as that in a control group of families in Britain. Furthermore, the mutation rate was correlated with the level of surface radiation: families in which the parents had resided in more-contaminated districts had higher mutation rates than those from less-contaminated districts.

This chapter is about the infidelity of DNA—about how errors arise in genetic instructions and how those errors are sometimes repaired. The Chernobyl catastrophe illustrates one cause of mutations (radiation) and the detrimental effects that DNA damage can have.

We begin with a brief examination of the different types of mutations, including their phenotypic effects, how they may be suppressed, and mutation rates. The next section explores how mutations spontaneously arise in the course of replication and afterward, as well as how chemicals and radiation induce mutations. We then consider the analysis of mutations. Finally, we take a look at DNA repair and some of the diseases that arise when DNA repair is defective. Throughout the chapter, it will be useful to keep in mind that mutations, by definition, are *inherited* changes in the DNA sequence—they must be passed on. Mutation requires both that the structure of a DNA molecule be changed and that this change is replicated.

www.whfreeman.com/pierce More information about the health effects of radiation released in the Chernobyl accident

The Nature of Mutation

DNA is a highly stable molecule that replicates with amazing accuracy (see Chapters 10 and 12), but changes in DNA structure and errors of replication do occur. A **mutation** is defined as an inherited change in genetic information; the descendants may be cells produced by cell division or individual organisms produced by reproduction.

The Importance of Mutations

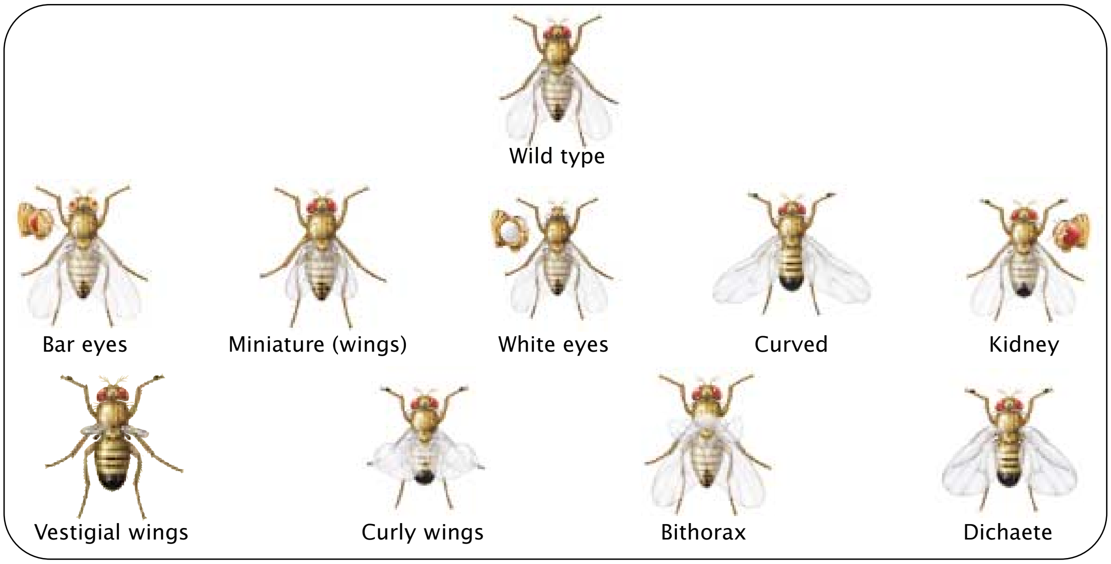
Mutations are both the sustainer of life and the cause of great suffering. On the one hand, mutation is the source of all genetic variation, the raw material of evolution. Without mutations and the variation that they generate, organisms could not adapt to changing environments and would risk extinction. On the other hand, most mutations have detrimental effects, and mutation is the source of many human diseases and disorders.

Much of genetics focuses on how variants produced by mutation are inherited; genetic crosses are meaningless if all individuals are identically homozygous for the same alleles. Mutations serve as important tools of genetic analysis; the solution to almost any genetic problem begins with a good set of mutants. Much of Gregor Mendel's success in unraveling the principles of inheritance can be traced to his use of carefully selected variants of the garden pea; similarly, Thomas Hunt Morgan and his students discovered many basic principles of genetics by analyzing mutant fruit flies (FIGURE 17.1).

Mutations are also useful for probing fundamental biological processes. Finding mutations that affect different components of a biological system and studying their effects can often lead to an understanding of the system. This method, referred to as genetic dissection, is analogous to figuring out how an automobile works by breaking different parts of a car and observing the effects—for example, smash the radiator and the engine overheats, revealing that the radiator cools the engine. The disruption of function in individual organisms bearing particular mutations likewise can be a source of insight into biological processes. For example, geneticists have begun to unravel the molecular details of development by studying mutations that interrupt various embryonic stages in *Drosophila* (see Chapter 21). Although this method of breaking "parts" to determine their function might seem like a crude approach to understanding a system, it is actually very powerful and has been used extensively in biochemistry, developmental biology, physiology, and behavioral science (but this method is *not* recommended for learning how your car works).

Concepts

Mutations are heritable changes in the genetic coding instructions of DNA. They are essential to the study of genetics and are useful in many other biological fields.

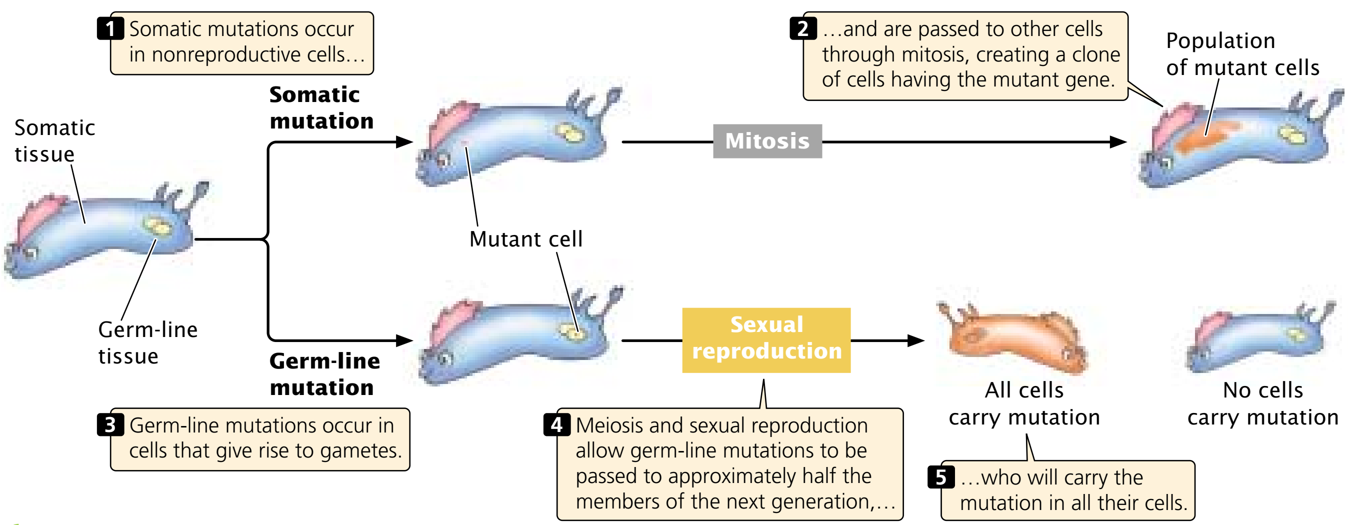


17.1 Morgan and his students discovered many principles of heredity by studying mutation in *Drosophila melanogaster.* Shown here are several common mutations.

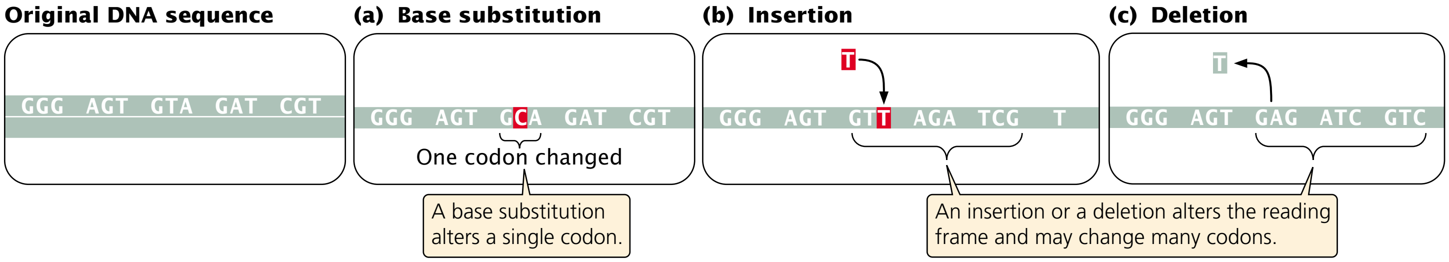
Categories of Mutations

In multicellular organisms, we can distinguish between two broad categories of mutations: somatic mutations and germ-line mutations. **Somatic mutations** arise in somatic tissues, which do not produce gametes (Figure 17.2). These mutations are passed on to other cells through the process of mitosis, which leads to a population of genetically identical cells (a clone). The earlier in development that a somatic mutation occurs, the larger the clone of cells within that individual organism that will contain the mutation.

Because of the huge number of cells present in a typical eukaryotic organism, somatic mutations must be numerous. For example, there are about 10^{14} cells in the human body. If a mutation arises only once in every million cell divisions (a fairly typical rate of mutation), hundreds of millions of somatic mutations must arise in each person. The effect of these mutations depends on many factors, including the type of cell in which they occur and the developmental stage at which they arise. Many somatic mutations have no obvious effect on the phenotype of the organism, because the function of the mutant cell (even the cell itself) is replaced by that of normal cells. However, cells with a somatic mutation that stimulates cell division can increase in number and spread; this type of mutation can give rise to cells with a selective advantage and is the basis for all cancers (see Chapter 21).



17.2 There are two basic classes of mutations: somatic mutations and germ-line mutations.



17.3 Three basic types of gene mutations are base substitutions, insertions, and deletions.

Germ-line mutations arise in cells that ultimately produce gametes. These mutations can be passed to future generations, producing individual organisms that carry the mutation in all their somatic and germ-line cells (see Figure 17.2). When we speak of mutations in multicellular organisms, we're usually talking about germ-line mutations. In single-cell organisms, however, there is no distinction between germ-line and somatic mutations, because cell division results in new individuals.

Historically, mutations have been partitioned into those that affect a single gene, called *gene mutations,* and those that affect the number or structure of chromosomes, called *chromosome mutations.* This distinction arose because chromosome mutations could be observed directly, by looking at chromosomes with a microscope, whereas gene mutations could be detected only by observing their phenotypic effects. Now, with the development of DNA sequencing, gene mutations and chromosome mutations are distinguished somewhat arbitrarily on the basis of the size of the DNA lesion. Nevertheless, it is useful to use the term *chromosome mutation* for a large-scale genetic alteration that affects chromosome structure or the number of chromosomes and the term **gene mutation** for a relatively small DNA lesion that affects a single gene. This chapter focuses on gene mutations; chromosome mutations were discussed in Chapter 9.

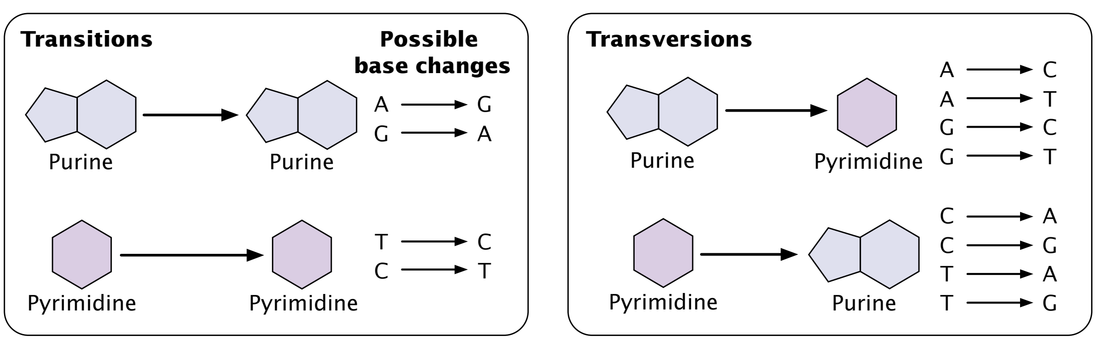
Types of Gene Mutations

There are a number of ways to classify gene mutations. Some classification schemes are based on the nature of the phenotypic effect—whether the mutation alters the amino acid sequence of the protein and, if so, how. Other schemes are based on the causative agent of the mutation, and still others focus on the molecular nature of the defect. The most appropriate scheme depends on the reason for studying the mutation. Here, we will categorize mutations primarily on the basis of their molecular nature, but we will also encounter some terms that relate the causes and the phenotypic effects of mutations.

Base substitutions The simplest type of gene mutation is a **base substitution,** the alternation of a single nucleotide in the DNA (FIGURE 17.3a). Because of the complementary nature of the two DNA strands (see Figure 10.14), when the base of one nucleotide is altered, the base of the corresponding nucleotide on the opposite strand also will be altered in the next round of replication. A base substitution therefore usually leads to a base-pair substitution.

Base substitutions are of two types. In a **transition,** a purine is replaced by a different purine or, alternatively, a pyrimidine is replaced by a different pyrimidine (FIGURE 17.4). In a **transversion,** a purine is replaced by a pyrimidine or a pyrimidine is replaced by a purine. The number of possible transversions (see Figure 17.4) is twice the number of possible transitions, but transitions usually arise more frequently.

Insertions and deletions The second major class of gene mutations contains **insertions** and **deletions**—the addition or the removal, respectively, of one or more nucleotide pairs (FIGURE 17.3b and c). Although base substitutions are often assumed to be the most common type of mutation, molecular analysis has revealed that insertions and deletions are more frequent. Insertions and deletions within

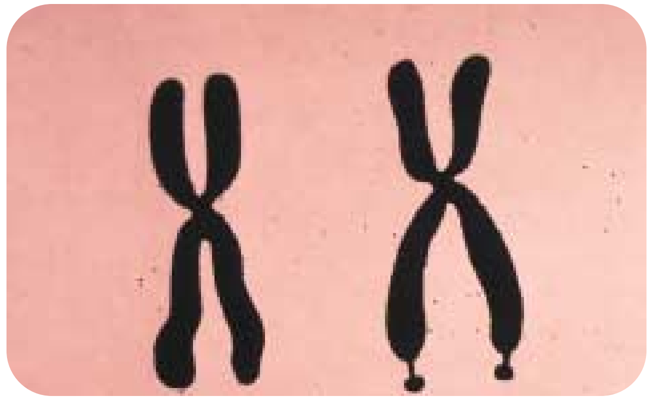


17.4 A transition is the substitution of a purine for a purine or a pyrimidine for a pyrimidine; a transversion is the substitution of a pyrimidine for a purine or a purine for a pyrimidine.

sequences that encode proteins may lead to **frameshift mutations,** changes in the reading frame (see p. 000 in Chapter 15) of the gene. The initiation codon in mRNA sets the reading frame: after the initiation codon, other codons are read as successive nonoverlapping groups of three nucleotides. The addition or deletion of a nucleotide usually changes the reading frame, altering all amino acids encoded by codons following the mutation (see Figure 17.3b and c). Many amino acids can be affected; so frameshift mutations generally have drastic effects on the phenotype. Not all insertions and deletions lead to frameshifts, however; because codons consist of three nucleotides, insertions and deletions consisting of any multiple of three nucleotides will leave the reading frame intact, although the addition or removal of one or more amino acids may still affect the phenotype. These mutations are called **in-frame insertions** and **deletions,** respectively.

Concepts

Gene mutations consist of changes in a single gene and may be base substitutions (a single pair of nucleotides is altered) or insertions or deletions (nucleotides are added or removed). A base substitution may be a transition (substitution of like bases) or a transversion (substitution of unlike bases). Insertions and deletions often lead to a change in the reading frame of a gene.



17.5 The fragile-X chromosome is associated with a characteristic constriction (fragile site) on the long arm. (Visuals Unlimited.)

Expanding trinucleotide repeats In 1991, an entirely novel type of mutation was discovered. This mutation occurs in a gene called *FMR-1* and causes fragile-X syndrome, the most common hereditary cause of mental retardation. The disorder is so named because, in specially treated cells of persons having the condition, the tip of the X chromosome is attached only by a slender thread (Figure 17.5). The *FMR-1* gene contains a number of adjacent copies of the trinucleotide CGG. The normal *FMR-1* allele (not containing the mutation) has 60 or fewer copies of this trinucleotide but, in persons with fragile-X syndrome, the allele may har-

Table 17.1 Examples of genetic diseases caused by expanding trinucleotide repeats

		Number of Copies of Repeat	
Disease	Repeated Sequence	Normal Range	Disease Range
Spinal and bulbar muscular atrophy	CAG	11–33	40–62
Fragile-X syndrome	CGG	6–54	50–1500
Jacobsen syndrome	CGG	11	100–1000
Spinocerebellar ataxia (several types)	CAG	4–44	21–130
Autosomal dominant cerebellar ataxia	CAG	7–19	37–~220
Myotonic dystrophy	CTG	5–37	44–3000
Huntington disease	CAG	9–37	37–121
Friedreich ataxia	GAA	6–29	200–900
Dentatorubral-pallidoluysian atrophy	CAG	7–25	49–75
Myoclonus epilepsy of the Unverricht-Lundborg type*	CCCGCCCCGCG	2–3	12–13

*Technically not a trinucleotide repeat but does entail a multiple of three nucleotides that expands and contracts in similar fashion to trinucleotide repeats.

bor hundreds or even thousands of copies. Mutations in which copies of a trinucleotide may increase greatly in number are called **expanding trinucleotide repeats.**

Expanding trinucleotide repeats have been found in several other human diseases (Table 17.1). The number of copies of the trinucleotide repeat often correlates with the severity or age of onset of the disease. The number of copies of the repeat also correlates with the instability of trinucleotide repeats—when more repeats are present, the probability of expansion to even more repeats increases. This instability leads to a phenomenon known as anticipation (see p. 000 in Chapter 5), in which diseases caused by trinucleotide-repeat expansions become more severe in each generation. Less commonly, the number of trinucleotide repeats may decrease within a family.

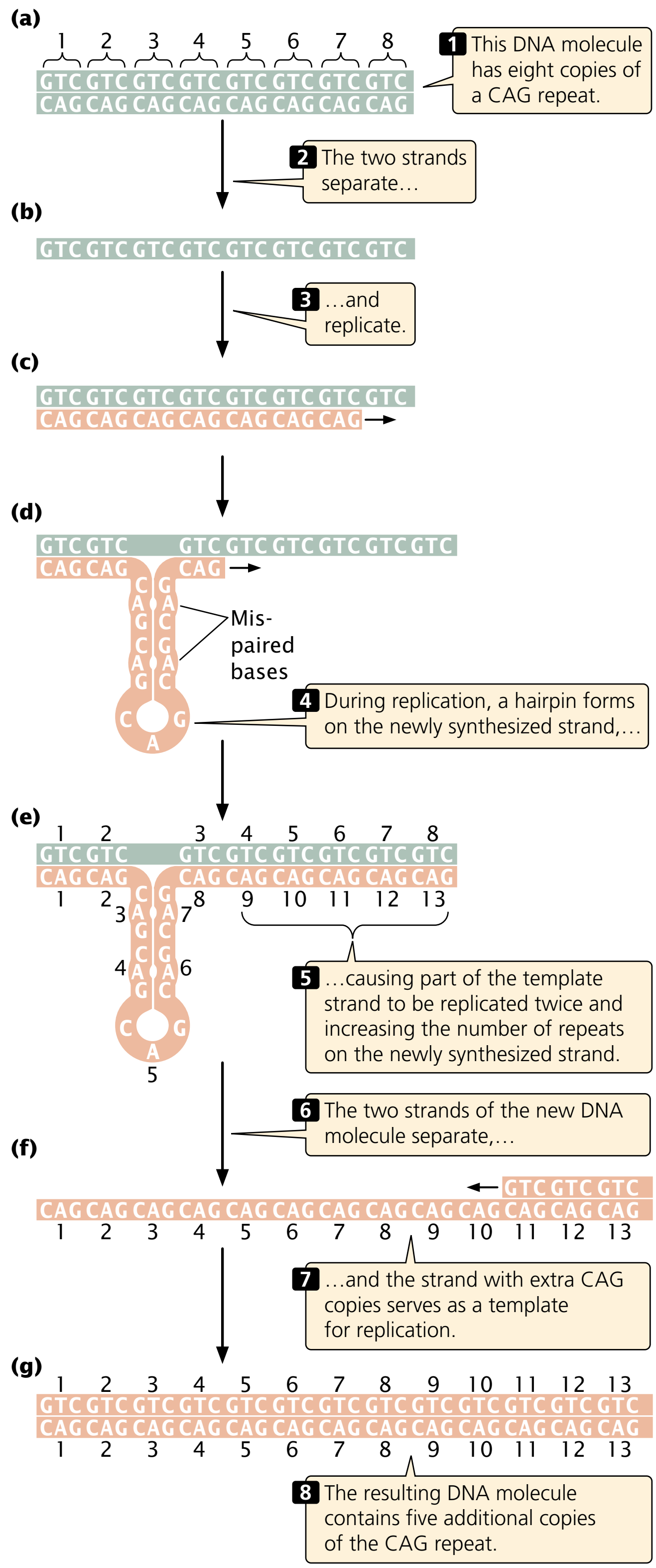
How an increase in the number of trinucleotides produces disease symptoms is not yet clear. In several of the diseases (e.g., Huntington disease), the trinucleotide CAG expands within the coding part of a gene, producing a toxic protein that has extra glutamine residues (the amino acid encoded by CAG). In other diseases (e.g., fragile-X syndrome and myotonic dystrophy), the repeat is outside the coding region of the gene and therefore must have some other mode of action. At least one disease (a rare type of epilepsy) has now been associated with an expanding repeat of a 12-bp sequence. Although this repeat is not a trinucleotide, it is included as a type of expanding trinucleotide because its repeat is a multiple of three.

The mechanism that leads to the expansion of trinucleotide repeats is still unclear. Strand slippage in DNA replication (see Figure 17.14) and crossing over between misaligned repeats (see Figure 17.15) are two possible sources of expansion. Single-stranded regions of some trinucleotide repeats are known to fold into hairpins (FIGURE 17.6) and other special DNA structures. Such structures may promote strand slippage in replication and may prevent these errors from being recognized and corrected, as described later in this chapter in the section on mismatch repair.

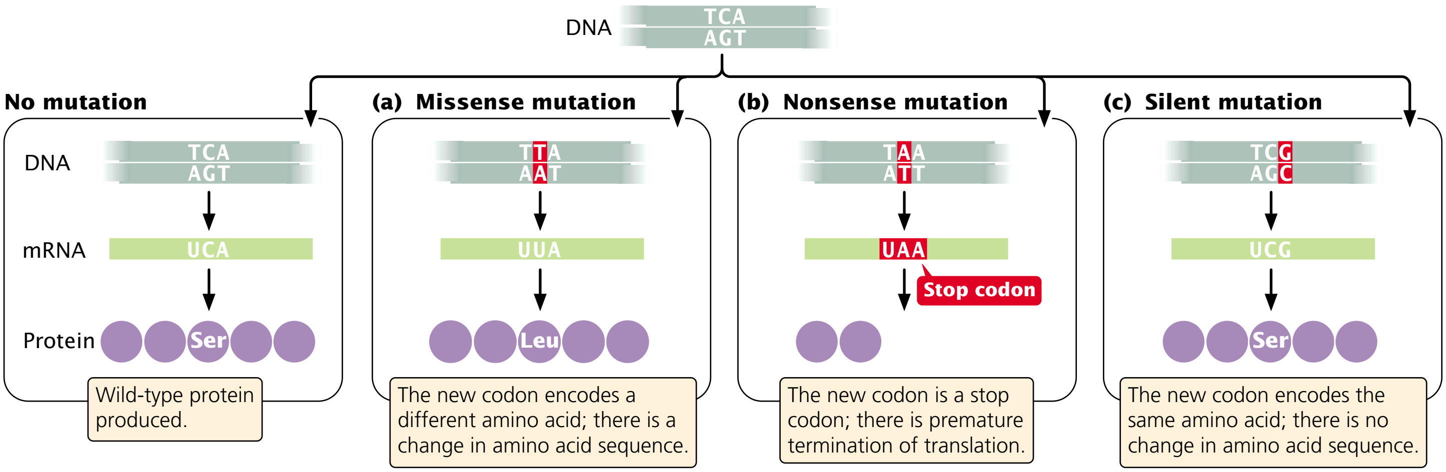
Concepts

Expanding trinucleotide repeats are regions of DNA that consist of repeated copies of three nucleotides. Increased numbers of trinucleotide repeats are associated with several genetic diseases.

Phenotypic effects of mutations Mutations have a variety of phenotypic effects. The effect of a mutation must be considered with reference to a phenotype against which the mutant can be compared, which is usually the wild-type phenotype—that is, the most common phenotype in natural populations of the organism. For example, most *Drosophila melanogaster* in nature have red eyes; so



17.6 **The number of copies of a trinucleotide may increase by strand slippage in replication.**



17.7 Base substitutions can cause (a) missense, (b) nonsense, and (c) silent mutations.

red eyes are considered the wild-type eye color; any other genetically determined eye color in fruit flies is considered to be a mutant. A mutation that alters the wild-type phenotype is called a **forward mutation,** where as a **reverse mutation** (a *reversion*) changes a mutant phenotype back into the wild type.

Geneticists use special terms to describe the phenotypic effects of mutations. A base substitution that alters a codon in the mRNA, resulting in a different amino acid in the protein, is referred to as a **missense mutation** (FIGURE 17.7a). A **nonsense mutation** changes a sense codon (one that specifies an amino acid) into a nonsense codon (one that terminates translation; FIGURE 17.7b). If a nonsense mutation occurs early in the mRNA sequence, the protein will be greatly shortened and will usually be nonfunctional. A **silent mutation** alters a codon but, thanks to the redundancy of the genetic code, the codon still specifies the same amino acid (FIGURE 17.7c). A **neutral mutation** is a missense mutation that alters the amino acid sequence of the protein but does not change its function. Neutral mutations occur when one amino acid is replaced by another that is chemically similar or when the affected amino acid has little influence on protein function.

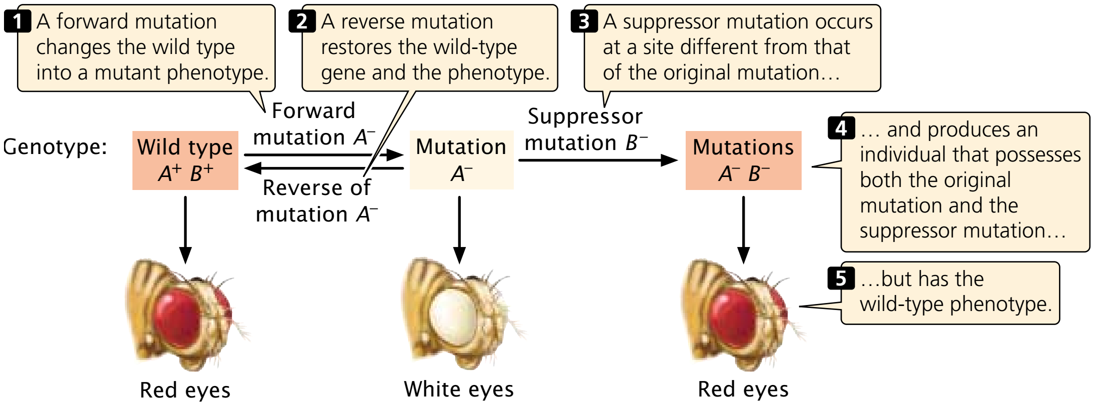
Loss-of-function mutations cause the complete or partial absence of normal function. A loss-of-function mutation so alters the structure of the protein that the protein no longer works correctly or the mutation can occur in regulatory regions that affect the transcription, translation, or splicing of the protein. Loss-of-function mutations are frequently recessive, and diploid individuals must be homozygous for the mutation before they can exhibit the effects of the loss of the functional protein. In contrast, a **gain-of-function mutation** produces an entirely new trait or it causes a trait to appear in inappropriate tissues or at inappropriate times in development. These mutations are frequently dominant in their expression. Still other types of mutations are **conditional mutations,** which are expressed only under certain conditions, and **lethal mutations,** which cause premature death.

Suppressor mutations A **suppressor mutation** is a genetic change that hides or suppresses the effect of another mutation. This type of mutation is distinct from a reverse mutation, in which the mutated site changes back into the original wild-type sequence (FIGURE 17.8). A suppressor mutation occurs at a site that is distinct from the site of the original mutation; thus, an individual organism with a suppressor mutation is a double mutant, possessing both the original mutation and the suppressor mutation but exhibiting the phenotype of an unmutated wild type.

Geneticists distinguish between two classes of suppressor mutations: intragenic and intergenic. An **intragenic suppressor** is in the same gene as that containing the mutation being suppressed and may work in several ways. The suppressor may change a second nucleotide in the same codon that was altered by the original mutation, producing a codon that specifies the same amino acid as the original, unmutated codon (FIGURE 17.9). Intragenic suppressors may also work by suppressing a frameshift mutation. If the original mutation is a one-base deletion, then the addition of a single base elsewhere in the gene will restore the former reading frame (see Figure 17.9). Consider the following nucleotide sequence in DNA and the amino acids that it encodes:

DNA	AAA	TCA	CTT	GGC	GTA	CAA
Amino acids	Phe	Ser	Glu	Pro	His	Val

Suppose a one-base deletion occurs in the first nucleotide of the second codon. This deletion shifts the reading frame by one nucleotide and alters all the amino acids that follow the mutation.



17.8 Relation of forward, reverse, and suppressor mutations.

One-nucleotide deletion

AAA ~~T~~CAC TTG GCG TAC AA
Phe Val Asn Arg Stop

If a single nucleotide is added to the third codon (the suppressor mutation), the reading frame is restored, although two of the amino acids differ from those specified by the original sequence.

One-nucleotide duplication

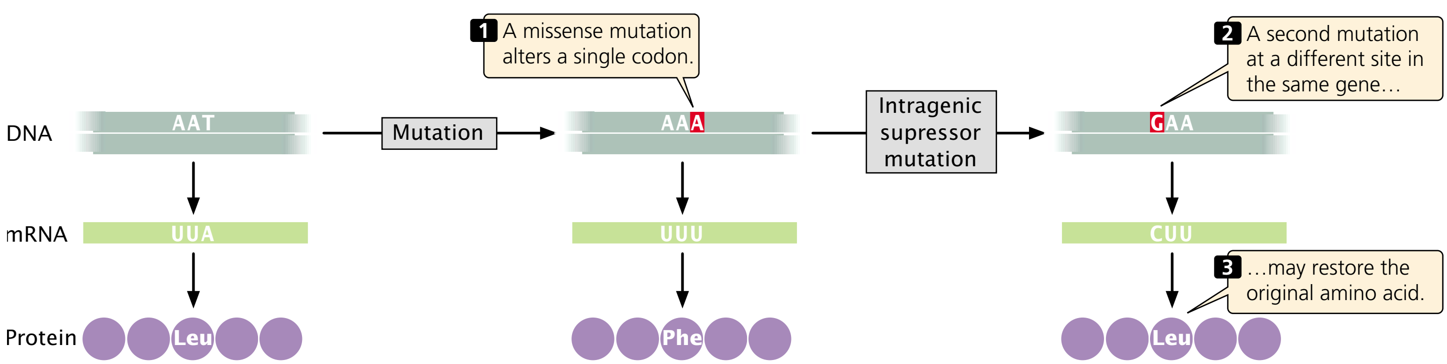
AAA CAC TT~~T~~ GGC GTA CAA
Phe Val Lys Pro His Val

Similarly, a mutation due to an insertion may be suppressed by a subsequent deletion in the same gene.

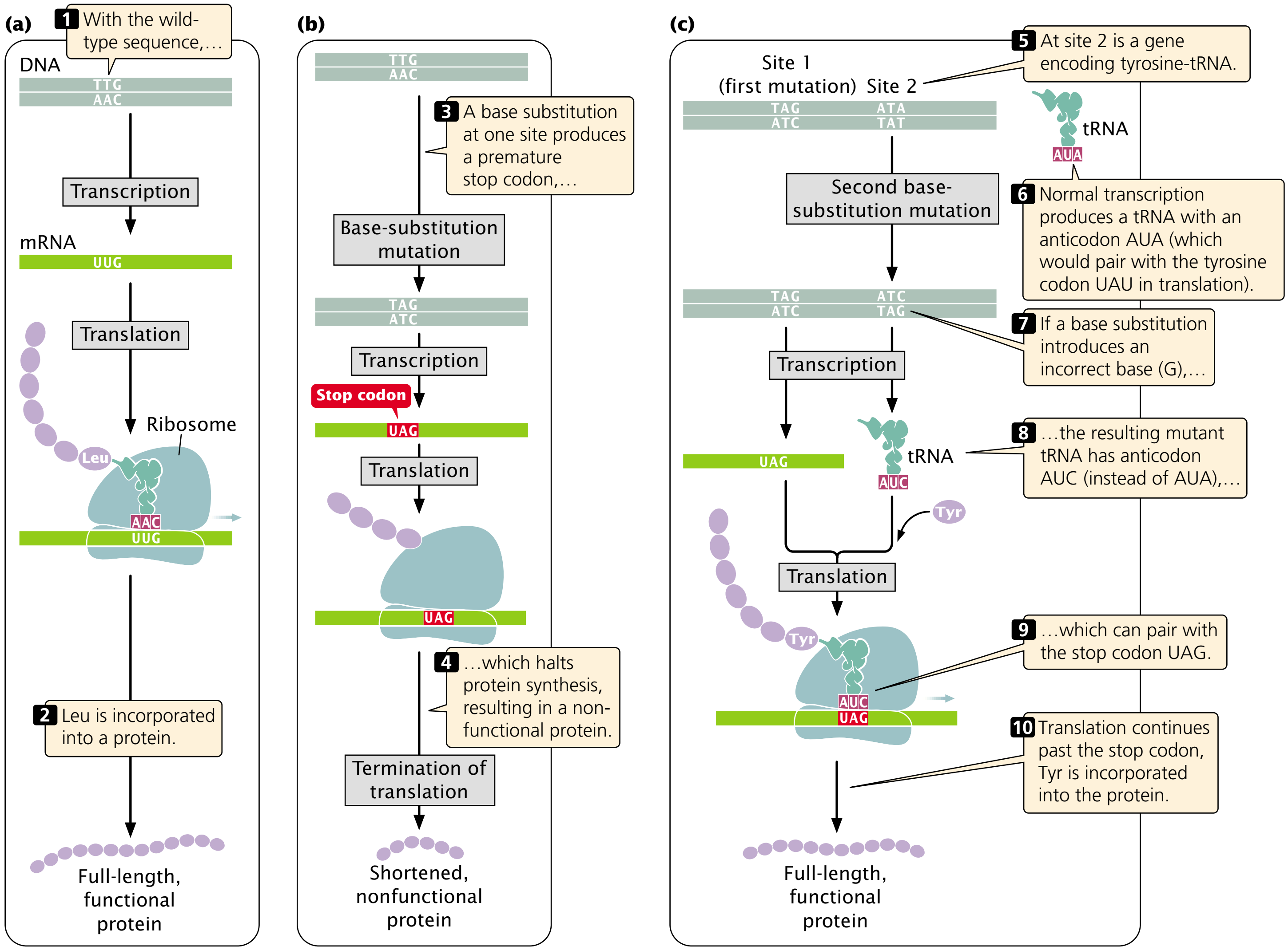
A third way in which an intragenic suppressor may work is by making compensatory changes in the protein. A first missense mutation may alter the folding of a polypeptide chain by changing the way in which amino acids in the protein interact with one another. A second missense mutation at a different site (the suppressor) may recreate the original folding pattern by restoring interactions between the amino acids.

Intergenic suppressors, in contrast, occur in a gene that is different from the one bearing the original mutation. These suppressors sometimes work by changing the way that the mRNA is translated. In the example illustrated in (FIGURE 17.10), the original DNA sequence is AAC (UUG in the mRNA) and specifies leucine. This sequence mutates to A<u>T</u>C (U<u>A</u>G in mRNA), a termination codon. The ATC nonsense mutation could be suppressed by a mutation in a gene that encodes a tRNA molecule by changing the anticodon on the tRNA so that it is capable of pairing with the UAG termination codon. For example, the gene that encodes the tRNA for tyrosine ($tRNA^{Tyr}$), which has the anticodon AUA, might be mutated to have the anticodon AU<u>C</u>, which will then pair with the UAG stop codon. Instead of translation terminating at the UAG codon, tyrosine would be inserted into the protein and a full-length protein would be produced, although tyrosine would now substitute for leucine. The effect of this change would depend on the role of this amino acid in the overall structure of the protein, but the effect is likely to be less detrimental than the effect of the nonsense mutation, which would halt translation prematurely.

Because cells in many organisms have multiple copies of tRNA genes, other unmutated copies of $tRNA^{Tyr}$ would



17.9 An intragenic suppressor mutation occurs in the same gene that contains the mutation being suppressed.



17.10 An intergenic suppressor mutation occurs in a different gene from the one bearing the original mutation. (a) The wild-type sequence produces a full-length, functional protein. (b) A base substitution at a site in one gene produces a premature stop codon, resulting in a shortened, nonfunctional protein. (c) A base substitution at a site in another gene, which in this case encodes tRNA, alters the anticodon of $tRNA^{Tyr}$ so that $tRNA^{Tyr}$ can pair with the stop codon produced by the original mutation, allowing tyrosine to be incorporated into the protein and translation to continue. Tyrosine replaces the leucine residue present in the original protein.

remain available to recognize the tyrosine codons. However, we might expect that the tRNAs that have undergone a suppressor mutation would also suppress the normal termination codons at the ends of coding sequences, resulting in the production of longer-than-normal proteins, but this event does not usually take place. Mutations in tRNA genes can also suppress missense and frameshift mutations.

Intergenic suppressors can also work through genic interactions (see p. 000 in Chapter 5). Polypeptide chains which are produced by two genes may interact to produce a functional protein. A mutation in one gene may alter the encoded polypeptide so that the interaction is destroyed and then a functional protein is no longer produced. A suppressor mutation in the second gene may produce a compensatory change in its polypeptide therefore restoring the original interaction. Characteristics of some of the different types of mutations are summarized in Table 17.2.

Table 17.2 Characteristics of different types of mutations

Type of Mutation	Definition
Base substitution	Changes the base of a single DNA nucleotide
Transition	Base substitution in which a purine replaces a purine or a pyrimidine replaces a pyrimidine
Transversion	Base substitution in which a purine replaces a pyrimidine or a pyrimidine replaces a purine
Insertion	Addition of one or more nucleotides
Deletion	Deletion of one or more nucleotides
Frameshift mutation	Insertion or deletion that alters the reading frame of a gene
In-frame deletion or insertion	Insertion or deletion of a multiple of three nucleotides that does not alter the reading frame
Expanding trinucleotide repeats	Repeated sequence of three nucleotides (trinucleotide) in which the number of copies of the trinucleotide increases
Forward mutation	Changes the wild-type phenotype to a mutant phenotype
Reverse mutation	Changes a mutant phenotype back to the wild-type phenotype
Missense mutation	Changes a sense codon into a different sense codon, resulting in the incorporation of a different amino acid in the protein
Nonsense mutation	Changes a sense codon into a nonsense codon, causing premature termination of translation
Silent mutation	Changes a sense codon into a synonymous codon, leaving unchanged the amino acid sequence of the protein
Neutral mutation	Changes the amino acid sequence of a protein without altering its ability to function
Loss-of-function mutation	Causes a complete or partial loss of function
Gain-of-function mutation	Causes the appearance of a new trait or function or causes the appearance of a trait in inappropriate tissues or at inappropriate times
Lethal mutation	Causes premature death
Suppressor mutation	Suppresses the effect of an earlier mutation at a different site
Intragenic suppressor mutation	Suppresses the effect of an earlier mutation within the same gene
Intergenic suppressor mutation	Suppresses the effect of an earlier mutation in another gene

Concepts

A suppressor mutation overrides the effect of an earlier mutation at a different site. An intragenic suppressor mutation occurs within the *same* gene, as that containing the original mutation, whereas an intergenic suppressor mutation occurs in a *different* gene.

www.whfreeman.com/pierce Descriptions and illustrations of different types of mutations

Mutation Rates

The frequency with which a gene changes from the wild type to a mutant is referred to as the **mutation rate** and is generally expressed as the number of mutations per biological unit, which may be mutations per cell division, per

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Achondroplasia

by Ron Green

Achondroplasia is an inherited autosomal dominant condition that causes diminished growth in the long bones of the legs, leading to dwarfism. Several years ago, the gene for achondroplasia was identified and cloned. If two people with achondroplasia marry, and each of them is heterozygous for achondroplasia (has one of two possible copies of the gene), chances are that two of every four children that they have will also be heterozygous and dwarfs. On average, one child in every four born to the couple will not inherit the achondroplasia gene and will be of average height, and one child in four will be homozygous for the gene. Homozygosity for this gene is lethal, and these children usually die in infancy.

A researcher who helped identify the gene understandably felt that he had made a significant contribution by allowing short-statured parents the option of aborting fetuses with the lethal double dose of the gene. To his surprise, shortly after news of the discovery was published, he received a call from one member of an achondroplasia couple, asking whether it was possible to test for both the *presence* and the *absence* of the gene. The couple wanted this information, they said, because they planned to abort not just all fetuses homozygous for the achondroplasia gene, but any completely unaffected ones as well. They were intent on having only short-statured children like themselves.

This case poses at least two major conflicts for genetic professionals. First, there is the conflict between respect for parental autonomy, which would ordinarily encourage acceding to the parents' request for assistance and information, and the medical professional's desire not to visit harm on a child. Children born with achondroplasia frequently must undergo a series of surgical procedures to correct serious bone problems. Throughout life, they also face many social and physical obstacles because of their short stature. Is it right for parents to deliberately bring a child into existence with this condition? Is it appropriate for health professionals to assist such efforts? How do we balance respect for parental autonomy against nonmaleficence?

Matters become more complex when we realize that some people with achondroplasia reject the idea that any harm is being done by the parents in this case. They maintain that most of the problems that they face are socially constructed and are due to society's marginalization and neglect of those who are different. Some also reject medical or genetic "solutions" to their problems. The proper response, they believe, is not to prevent the birth of a child with a genetic condition but to eliminate the social handicaps and discriminatory attitudes. Thus, the parents in this case may be driven not merely by their personal wishes but by a commitment to social justice.

Traditional, nondirective genetic counseling has assumed that people seek prenatal testing to prevent the birth of a child with a genetic disease, to prepare for the birth and treatment of a child with a recognized genetic disorder, or to reconsider their reproductive plans. What this case reveals is that genetics is opening up the possibility of shaping our children's lives in ways that go far beyond what is normally associated with the healing role. Somewhat less dramatic, but perhaps more worrisome is the fact that the identification of the genetic basis of many traits that are not considered diseases (e.g., height, intelligence, temperament) will offer parents a new range of choices in the "genetic design" of their children. At this moment, research is underway to identify and replace disease-causing genes in human embryos. In the future, such embryonic gene therapy will open up the possibility of *enhancing* children's capabilities. Beginning with genes that improve a child's resistance to cancer or AIDS, genetic interventions may make it possible to increase a child's height, stamina, or IQ. Science could offer parents who yearn for a champion basketball player or world-class swimmer the means to realize their dreams.

A family of three who have achondroplasia. (Gail Burton/AP.)

As complex as it may seem, the science here is the easy part. Far more difficult are the ethical questions. To begin with, there is the question of whether we will ever have enough knowledge to "play God" in this way. Do we dare alter the course of human evolution? The history of twentieth-century science is littered with well-intended technologies—from DDT to nuclear power—that eventually brought unforeseen harms. Will our genetic interventions follow this path? Will our clumsy attempts to "improve" the human genome unleash an epidemic of new genetic diseases? And what of the child's rights in all this? Is it fair to "engineer" a child into a parent's dream of perfection?

gamete, or per round of replication. For example, the mutation rate for achondroplasia (a type of hereditary dwarfism) is about four mutations per 100,000 gametes, usually expressed more simply as 4×10^{-5}. In contrast, **mutation frequency** is defined as the incidence of a specific type of mutation within a group of individual organisms. For achondroplasia, the mutation frequency in the United States is about 2×10^{-4}, which means that about 1 of every 20,000 persons in the U.S. population carries this mutation.

Mutation rates are affected by three factors. First, they depend on the frequency with which primary changes take place in DNA. Primary change may arise from spontaneous molecular changes in DNA or it may be induced by chemical or physical agents in the environment.

A second factor influencing the mutation rate is the probability that, when a change takes place, it will be repaired. Most cells possess a number of mechanisms to repair altered DNA; so most alterations are corrected before they are replicated. If these repair systems are effective, mutation rates will be low; if they are faulty, mutation rates will be elevated. Some mutations increase the overall rate of mutation at other genes; these mutations usually occur in genes that encode components of the replication machinery or DNA repair enzymes.

A third factor, one that influences our ability to calculate mutation rates, is the probability that a mutation will be recognized and recorded. When DNA is sequenced, all mutations are potentially detectable. In practice, however, sequencing is expensive; so mutations are usually detected by their phenotypic effects. Some mutations may appear to arise at a higher rate simply because they are easier to detect.

Mutation rates vary among organisms and among genes within organisms (Table 17.3), but we can draw several general conclusions about mutation rates. First, spontaneous mutation rates are low for all organisms studied. Typical mutation rates for viral and bacterial genes range from about 1 to 100 mutations per 10 billion cells (1×10^{-8} to 1×10^{-10}). The mutation rates for most eukaryotic genes are a bit higher, from about 1 to 10 mutations per million gametes (1×10^{-5} to 1×10^{-6}). These higher values in eukaryotes may be due to the fact that the rates are calculated *per gamete,* and several cell divisions are required to produce a gamete, whereas mutation rates in prokaryotic cells and viruses are calculated *per cell division.*

Within each major class of organisms, mutation rates vary considerably. These differences may be due to differing abilities to repair mutations, unequal exposures to mutagens, or biological differences in rates of spontaneously arising mutations. Even within a single species, spontaneous rates of mutation vary among genes. The reason for this variation is not entirely understood, but some regions of DNA are known to be more susceptible to mutation than others.

Table 17.3 Mutation rates of different genes in different organisms

Organism	Mutation	Rate	Unit
Bacteriophage T2	Lysis inhibition	1×10^{-8}	Per replication
	Host range	3×10^{-9}	
Escherichia coli	Lactose fermentation	2×10^{-7}	Per cell division
	Histidine requirement	2×10^{-8}	
Neurospora crassa	Inositol requirement	8×10^{-8}	Per asexual spore
	Adenine requirement	4×10^{-8}	
Corn	Kernel color	2.2×10^{-6}	Per gamete
Drosophila	Eye color	4×10^{-5}	Per gamete
	Allozymes	5.14×10^{-6}	
Mouse	Albino coat color	4.5×10^{-5}	Per gamete
	Dilution coat color	3×10^{-5}	
Human	Huntington disease	1×10^{-6}	Per gamete
	Achondroplasia	1×10^{-5}	
	Neurofibromatosis (Michigan)	1×10^{-4}	
	Hemophilia A (Finland)	3.2×10^{-5}	
	Duchenne muscular dystrophy (Wisconsin)	9.2×10^{-5}	

Concepts

Mutation rate is the frequency with which a specific mutation arises, whereas mutation frequency is the incidence of a mutation within a defined group of individual organisms. Rates of mutations are generally low and are affected by environmental and genetic factors.

Causes of Mutations

Mutations result from both internal and external factors. Those that are a result of natural changes in DNA structure are termed **spontaneous mutations,** whereas those that result from changes caused by environmental chemicals or radiation are **induced mutations.**

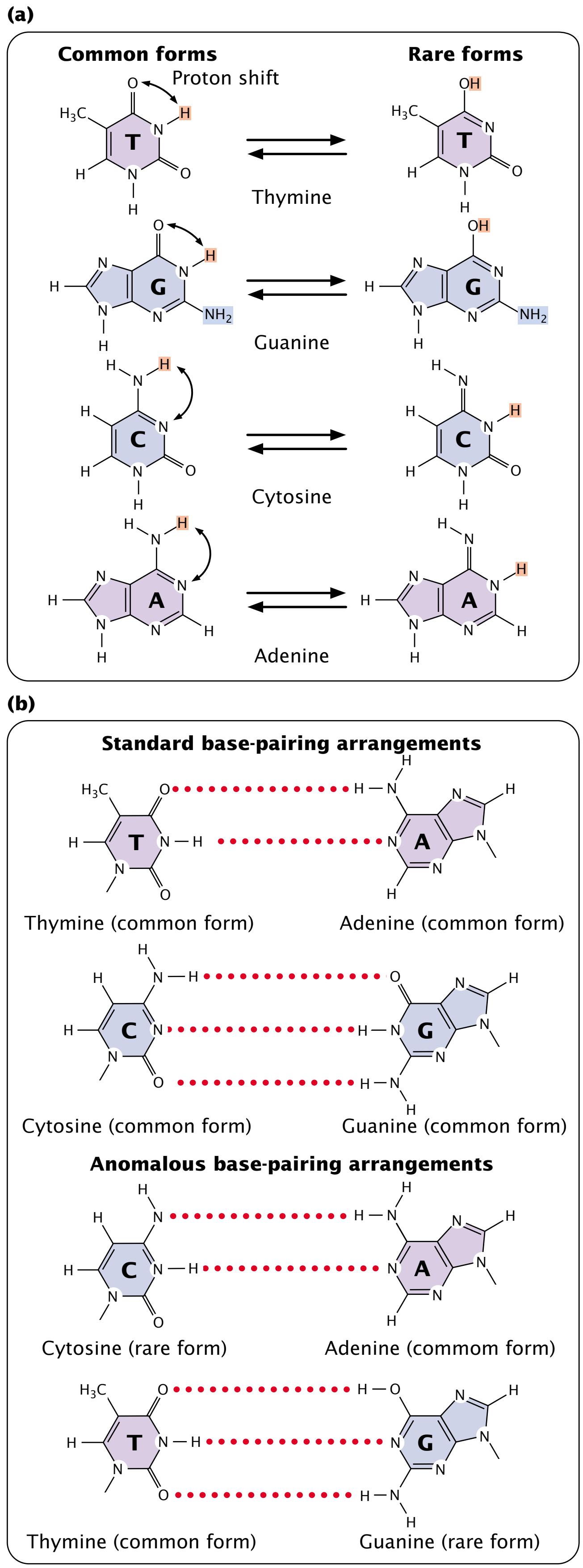
Spontaneous Replication Errors

Replication is amazingly accurate: fewer than one in a billion errors are made in the course of DNA synthesis (Chapter 12). However, spontaneous replication errors do occasionally occur.

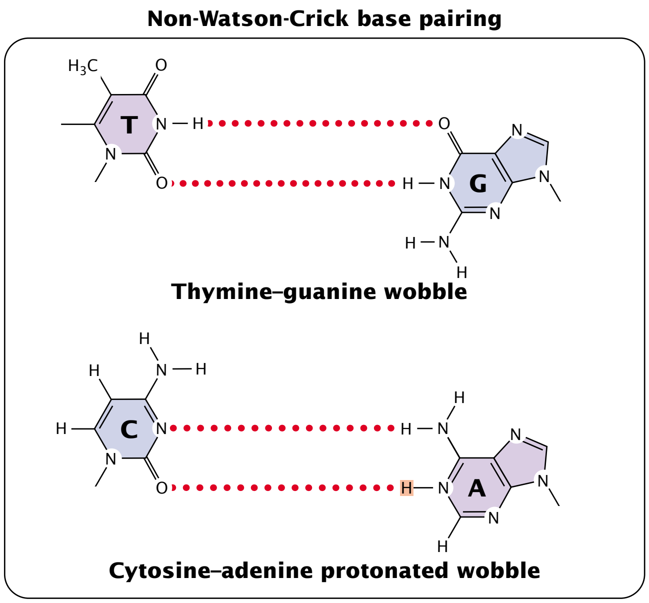
The primary cause of spontaneous replication errors was formerly thought to be tautomeric shifts, in which the positions of protons in the DNA bases change. Purine and pyrimidine bases exist in different chemical forms called tautomers (FIGURE 17.11a). The two tautomeric forms of each base are in dynamic equilibrium, although one form is more common than the other. The standard Watson and Crick base pairings—adenine with thymine, and cytosine with guanine—are between the common forms of the bases, but, if the bases are in their rare tautomeric forms, other base pairings are possible (FIGURE 17.11b).

Watson and Crick proposed that tautomeric shifts might produce mutations, and for many years their proposal was the accepted model for spontaneous replication errors, but there has never been convincing evidence that the rare tautomers are the cause of spontaneous mutations. Furthermore, research now shows little evidence of these structures in DNA.

Mispairing can also occur through wobble, in which normal, protonated, and other forms of the bases are



17.11 Purine and pyrimidine bases exist in different forms called tautomers. (a) A tautomeric shift occurs when a proton changes its position, resulting in a rare tautomeric form. (b) Standard and anomalous base-pairing arrangements occur if bases are in the rare tautomeric forms. Base mispairings due to tautomeric shifts were originally thought to be a major source of errors in replication, but such structures have not been detected in DNA, and most evidence now suggests that other types of anomalous pairings (see Figure 17.14) are responsible for replication errors.



17.12 Nonstandard base pairings can occur as a result of the flexibility in DNA structure. Thymine and guanine can pair through wobble between normal bases. Cytosine and adenine can pair through wobble when adenine is protonated (has an extra hydrogen).

able to pair because of flexibility in the DNA helical structure (FIGURE 17.12). These structures have been detected in DNA molecules and are now thought to be responsible for many of the mispairings in replication.

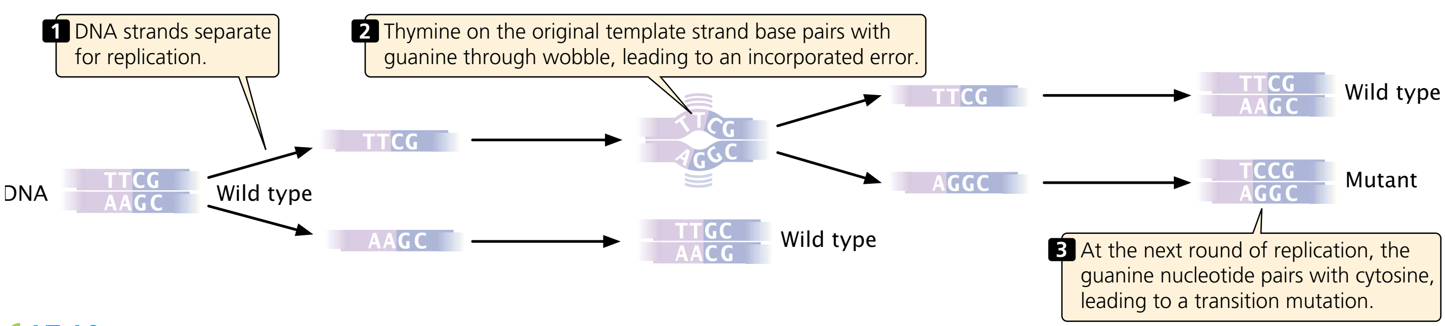
When a mismatched base has been incorporated into a newly synthesized nucleotide chain, an **incorporated error** is said to have occurred. Suppose that, in replication, thymine (which normally pairs with adenine) mispairs with guanine through wobble (FIGURE 17.13). In the next round of replication, the two mismatched bases separate, and each serves as template for the synthesis of a new nucleotide strand. This time, thymine pairs with adenine, producing another copy of the original DNA sequence. On the other strand, however, the incorrectly incorporated guanine serves as the template and pairs with cytosine, producing a new DNA molecule that has an incorporated error—a C·G pair in place of the original T·A pair (a T·A→C·G base substitution). The original incorporated error leads to a **replication error,** which creates a permanent mutation, because all the base pairings are correct and there is no mechanism for repair systems to detect the error.

Mutations due to small insertions and deletions also may arise spontaneously in replication and crossing over. **Strand slippage** may occur when one nucleotide strand forms a small loop (FIGURE 17.14). If the looped-out nucleotides are on the newly synthesized strand, an insertion results. At the next round of replication, the insertion will be incorporated into both strands of the DNA molecule. If the looped-out nucleotides are on the template strand, then there is a deletion on the newly replicated strand, and this deletion will be perpetuated in subsequent rounds of replication.

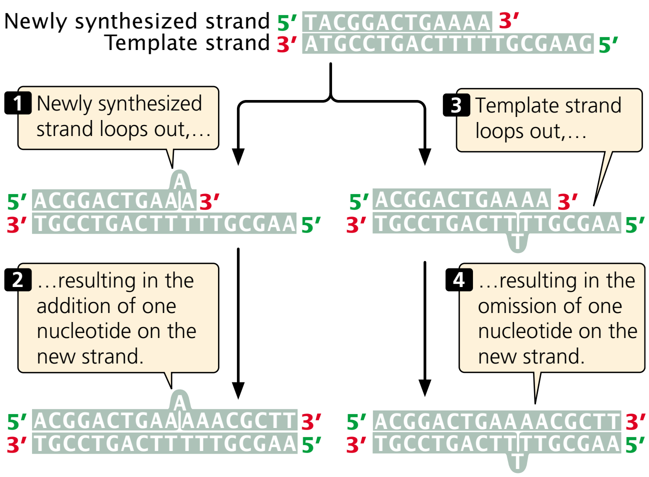
During normal crossing over, the homologous sequences of the two DNA molecules align, and crossing over produces no net change in the number of nucleotides in either molecule. Misaligned pairing may cause **unequal crossing over,** which results in one DNA molecule with an insertion and the other with a deletion (FIGURE 17.15). Some DNA sequences are more likely than others to undergo strand slippage or unequal crossing over. Stretches of repeated sequences, such as trinucleotide repeats or homopolymeric repeats (more than five repeats of the same base in a row), are prone to strand slippage. Stretches with more repeats are more likely to undergo strand slippage. Duplicated or repetitive sequences may misalign during pairing, leading to unequal crossing over. Both strand slippage and unequal crossing over produce duplicated copies of sequences, which in turn promote further strand slippage and unequal crossing over. This chain of events may explain the phenomenon of anticipation often observed for expanding trinucleotide repeats.

Concepts

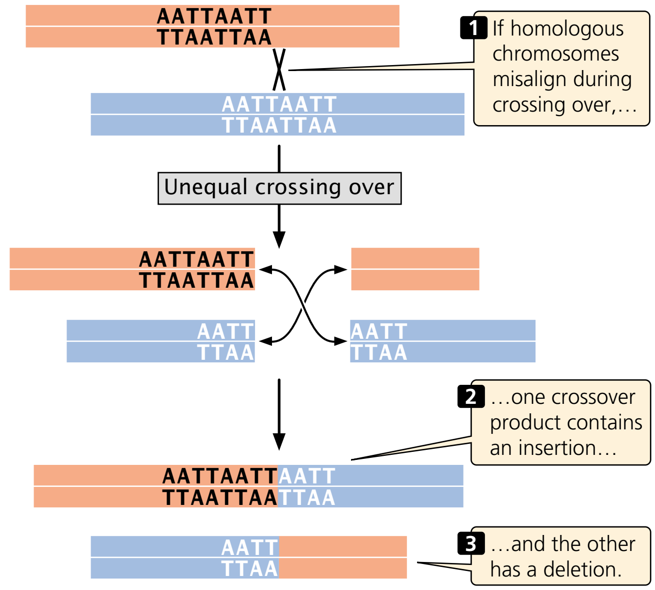
Spontaneous replication errors arise from altered base structures and from wobble base pairing. Small insertions and deletions may occur through strand slippage in replication and through unequal crossing over.



17.13 Wobble base pairing leads to a replicated error.



17.14 **Insertions and deletions may result from strand slippage.**

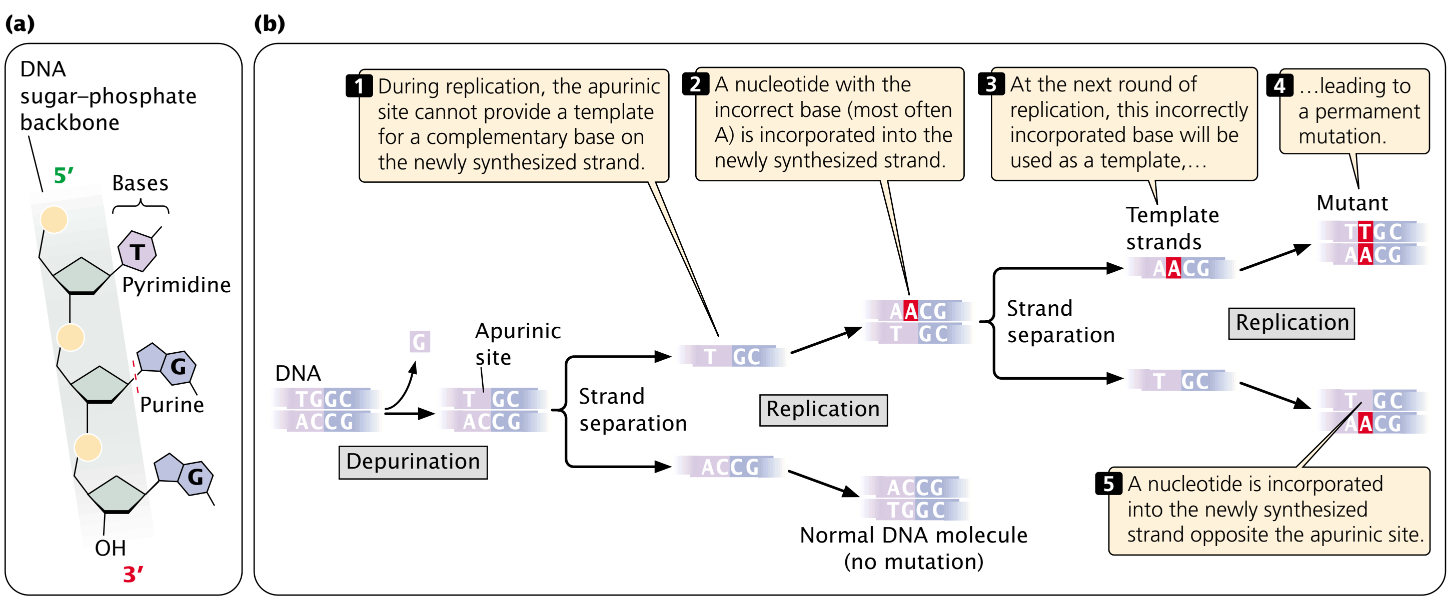


17.15 **Unequal crossing over produces insertions and deletions.**

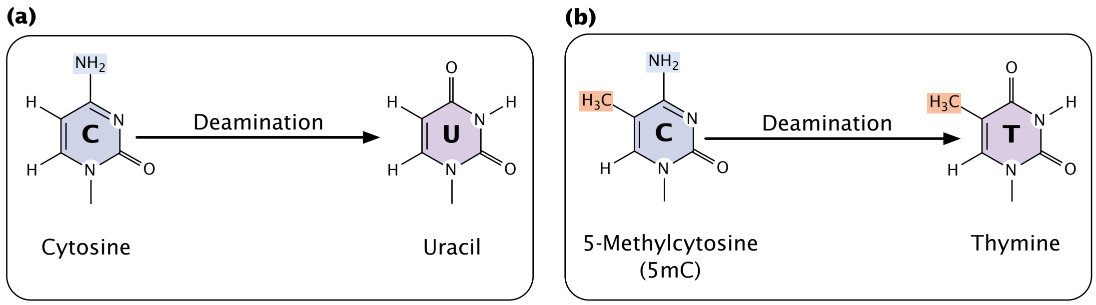
Spontaneous Chemical Changes

In addition to spontaneous mutations that arise in replication, mutations also result from spontaneous chemical changes in DNA. One such change is **depurination,** the loss of a purine base from a nucleotide. Depurination results when the covalent bond connecting the purine to the 1′-carbon atom of the deoxyribose sugar breaks (FIGURE 17.16a), producing an apurinic site—a nucleotide that lacks its purine base. An apurinic site cannot act as a template for a complementary base in replication. In the absence of base-pairing constraints, an incorrect nucleotide (most often adenine) is incorporated into the newly synthesized DNA strand opposite the apurinic site (FIGURE 17.16b), frequently leading to an incorporated error. The incorporated error is then transformed into a replication error at the next round of replication. Depurination is a common cause of spontaneous mutation; a mammalian cell in culture loses approximately 10,000 purines every day.

Another spontaneously occurring chemical change that takes place in DNA is **deamination,** the loss of an amino group (NH_2) from a base. Deamination may occur spontaneously or be induced by mutagenic chemicals.



17.16 **Depurination, loss of a purine base from the nucleotide, produces an apurinic site.**



17.17 Deamination alters DNA bases.

Deamination may alter the pairing properties of a base: the deamination of cytosine, for example, produces uracil (FIGURE 17.17a), which pairs with adenine during replication. After another round of replication, the adenine will pair with thymine, creating a T·A pair in place of the original C·G pair (C·G→U·A→T·A); this chemical change is a transition mutation. This type of mutation is usually repaired by enzymes that remove uracil whenever it is found in DNA. The ability to recognize the product of cytosine deamination may explain why thymine, not uracil, is found in DNA. Some cytosine bases in DNA are naturally methylated and exist in the form of 5-methylcytosine (5mC; see p. 000 in Chapter 10 and Figure 10.19), which when deaminated becomes thymine (FIGURE 17.17b). Because thymine pairs with adenine in replication, the deamination of 5-methylcytosine changes an original C·G pair to T·A (C·G→5mC·A→T·A). This change cannot be detected by DNA repair systems, because it produces a normal base. Consequently, C·G→T·A transitions occur frequently in eukaryotic cells.

Concepts

Some mutations arise from spontaneous alterations to DNA structure, such as depurination and deamination, which may alter the pairing properties of the bases and cause errors in subsequent rounds of replication.

Chemically Induced Mutations

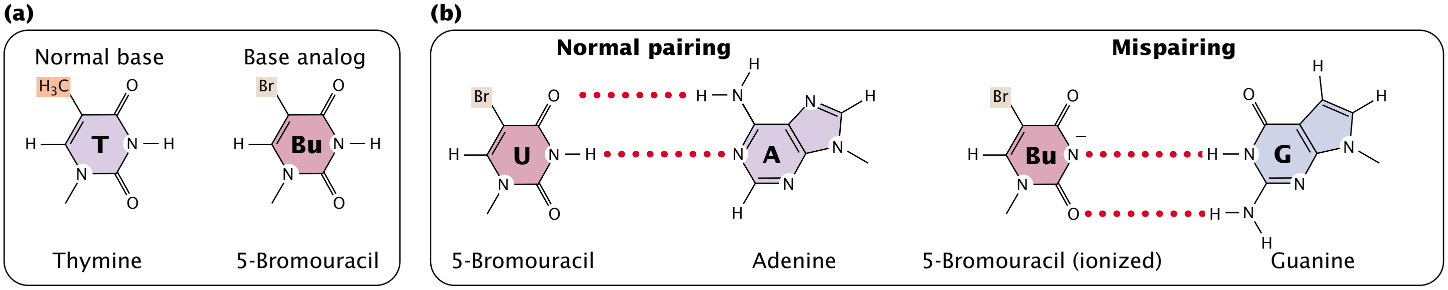
Although many mutations arise spontaneously, a number of environmental agents are capable of damaging DNA, including certain chemicals and radiation. Any environmental agent that significantly increases the rate of mutation above the spontaneous rate is called a **mutagen.**

The first discovery of a chemical mutagen was made by Charlotte Auerbach, who was born in Germany to a Jewish family in 1899. After attending university in Berlin and doing research, she spent several years teaching at various schools in Berlin. Faced with increasing anti-Semitism in Nazi Germany, Auerbach immigrated to Britain, where she conducted research on the development of mutants in *Drosophila*. There she met Herman Muller, who had shown that radiation induces mutations; he suggested that Auerbach try to obtain mutants by treating *Drosophila* with chemicals. Her initial attempts met with little success. Other scientists were conducting top-secret research on mustard gas (used as a chemical weapon in World War I) and noticed that it produced many of the same effects as radiation. Auerbach was asked to determine whether mustard gas was mutagenic.

Collaborating with pharmacologist J. M. Robson, Auerbach studied the effects of mustard gas on *Drosophila melanogaster*. The experimental conditions were crude. They heated liquid mustard gas over a Bunsen burner on the roof of the pharmacology building, and the flies were exposed to the gas in a large chamber. After developing serious burns on her hands from the gas, Auerbach let others carry out the exposures, and she analyzed the flies. Auerbach and Robson showed that mustard gas is indeed a powerful mutagen, reducing the viability of gametes and increasing the numbers of mutations seen in the offspring of exposed flies. Because the research was part of the secret war effort, publication of their findings was delayed until 1947.

www.whfreeman.com/pierce A brief history of Herman Muller

Base analogs One class of chemical mutagens consists of **base analogs**, chemicals with structures similar to that of any of the four standard bases of DNA. DNA polymerases cannot distinguish these analogs from the standard bases; so, if base analogs are present during replication, they may be incorporated into newly synthesized DNA molecules. For example, 5-bromouracil (5BU) is an analog of thymine; it has the same structure as that of thymine except that it has a bromine (Br) atom on the 5-carbon atom instead of a methyl group (FIGURE 17.18a). Normally, 5-bromouracil pairs with adenine just as thymine does, but it occasionally mispairs with guanine (FIGURE 17.18b), leading to a transition (T·A→5BU·A→5BU·G→C·G), as shown in



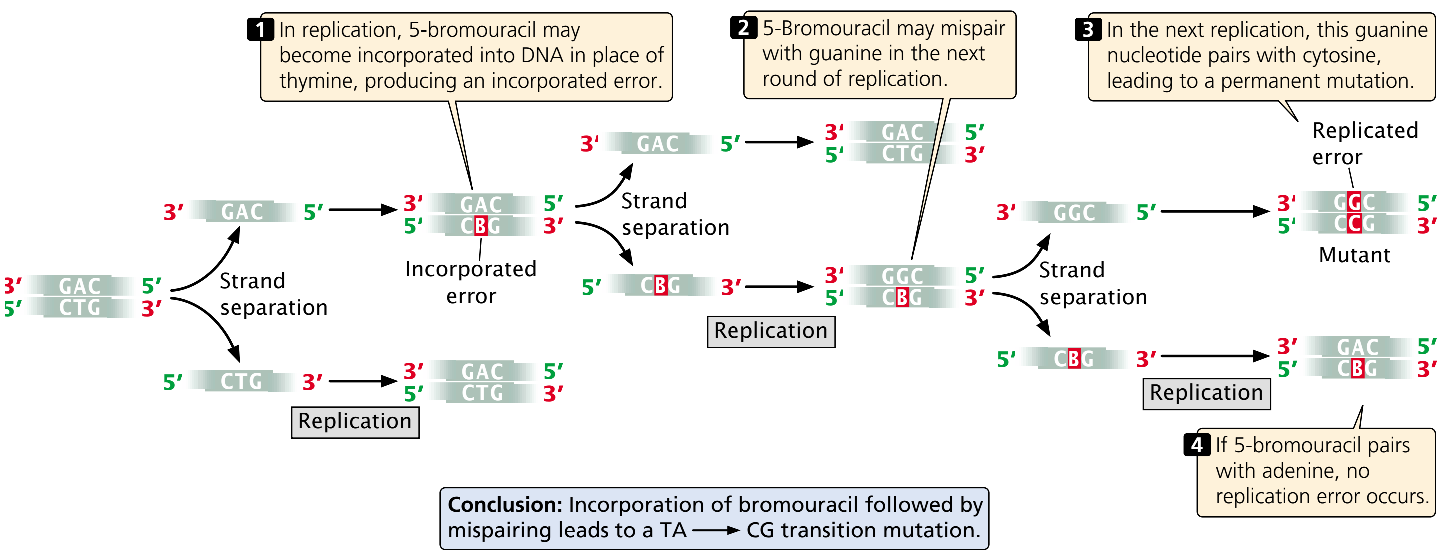
17.18 5-Bromouracil (a base analog) resembles thymine, except that it has a bromine atom in place of a methyl group on the 5-carbon atom. Because of the similarity in their structures, 5-bromouracil may be incorporated into DNA in place of thymine. Like thymine, 5-bromouracil normally pairs with adenine but, when ionized, it may pair with guanine through wobble.

FIGURE 17.19. Through mispairing, 5-bromouracil may also be incorporated into a newly synthesized DNA strand opposite guanine. In the next round of replication, 5-bromouracil may pair with adenine, leading to another transition (G·C→G·5BU→A·5BU→A·T).

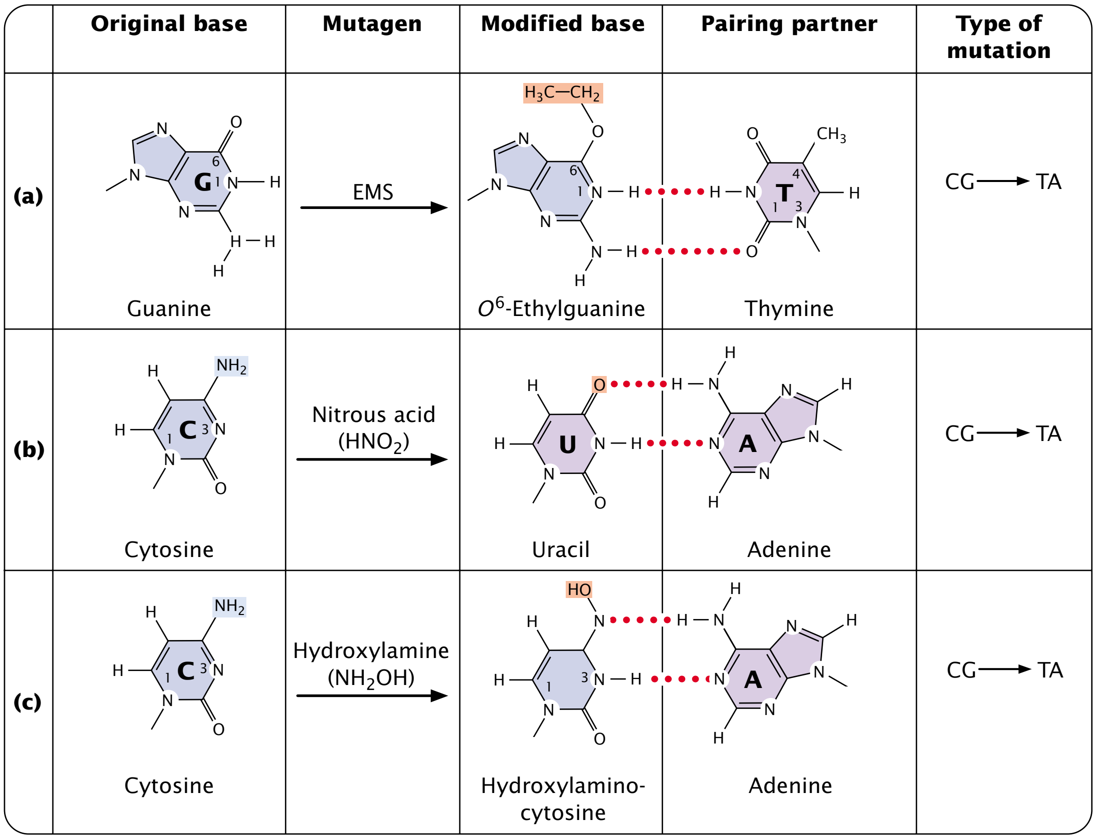
Another mutagenic chemical is 2-aminopurine (2AP), which is a base analog of adenine (FIGURE 17.20). Normally, 2-aminopurine base pairs with thymine, but it may mispair with cytosine, causing a transition mutation (T·A→T·2AP→C·2AP→C·G). Alternatively, 2-aminopurine may be incorporated through mispairing into the newly synthesized DNA opposite cytosine and later pair with thymine, leading to a C·G→C·2AP→T·2AP→T·A transition.

Thus, both 5-bromouracil and 2-aminopurine can produce transition mutations. In the laboratory, mutations by base analogs can be reversed by treatment with the same analog or by treatment with a different analog.

Alkylating agents Alkylating agents are chemicals that donate alkyl groups. These agents include methyl (CH_3) and ethyl (CH_3-CH_2) groups, which are added to nucleotide bases by some chemicals. For example, ethylmethanesulfonate (EMS) adds an ethyl group to guanine, producing 6-ethylguanine, which pairs with thymine (see Figure 17.20a). Thus, EMS produces C·G→T·A transitions. EMS is also capable of adding an ethyl group to thymine, producing 4-ethylthymine, which then pairs with guanine, leading to a T·A→C·G transition. Because EMS produces both C·G→T·A and T·A→C·G transitions, mutations produced by EMS can be reversed by additional treatment with EMS. Mustard gas is another alkylating agent.



17.19 5-Bromouracil can lead to a replicated error.



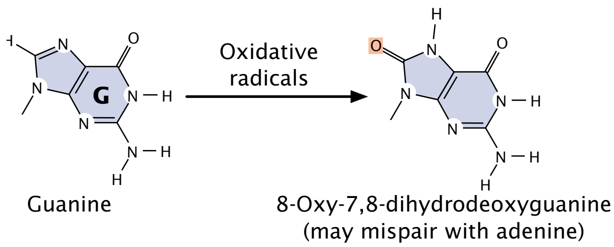
17.20 Chemicals may alter DNA bases. (a) The alkylating agent ethylmethanesulfonate (EMS) adds an ethyl group to guanine, producing 6-ethylguanine, which pairs with thymine, producing a C·G→T·A transition mutation. (b) Nitrous acid deaminates cytosine to produce uracil, which pairs with adenine, producing a C·G→T·A transition mutation. (c) Hydroxylamine converts cytosine into hydroxylaminocytosine, which frequently pairs with adenine, leading to a C·G→T·A transition mutation.

Deamination In addition to its spontaneous occurrence (see Figure 17.17), deamination can be induced by some chemicals. For instance, nitrous acid deaminates cytosine, creating uracil, which in the next round of replication pairs with adenine (see Figure 17.20b), producing a C·G→T·A transition mutation. Nitrous acid changes adenine into hypoxanthine, which pairs with cytosine, leading to a T·A→C·G transition. Nitrous acid also deaminates guanine, producing xanthine, which pairs with cytosine just as guanine does; however xanthine may also pair with thymine, leading to a C·G→T·A transition. Nitrous acid produces exclusively transition mutations and, because both C·G→T·A and T·A→C·G transitions are produced, these mutations can be reversed with nitrous acid.

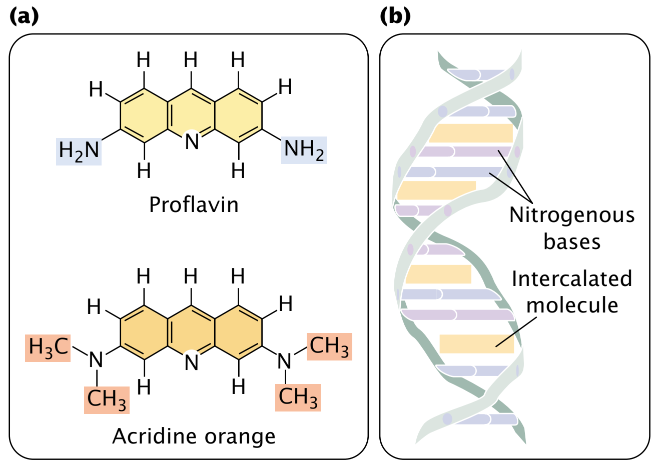
Hydroxylamine Hydroxylamine is a very specific base-modifying mutagen that adds a hydroxyl group to cytosine, converting it into hydroxylaminocytosine (see Figure 17.20c). This conversion increases the frequency of a rare tautomer that pairs with adenine instead of guanine and leads to C·G→T·A transitions. Because hydroxylamine acts only on cytosine, it will *not* generate T·A→C·G transitions; thus, hydroxylamine will not reverse the mutations that it produces.

Oxidative reactions Reactive forms of oxygen (including superoxide radicals, hydrogen peroxide, and hydroxyl radicals) are produced in the course of normal aerobic metabolism, as well as by radiation, ozone, peroxides, and certain drugs. These reactive forms of oxygen damage DNA and induce mutations by bringing about chemical changes to DNA. For example, oxidation converts guanine into 8-oxy-7,8-dihydrodeoxyguanine (FIGURE 17.21), which frequently mispairs with adenine instead of cytosine, causing a G·C→T·A transversion mutation.

Intercalating agents **Intercalating agents,** such as proflavin, acridine orange, ethidium bromide, and dioxin are



17.21 Oxidative radicals convert guanine into 8-oxy-7,8-dihydrodeoxyguanine, which frequently mispairs with adenine instead of cytosine, producing a C·G→T·A transversion.



17.22 Intercalating agents such as proflavin and acridine orange insert themselves between adjacent bases in DNA, distorting the three-dimensional structure of the helix and causing single-nucleotide insertions and deletions in replication.

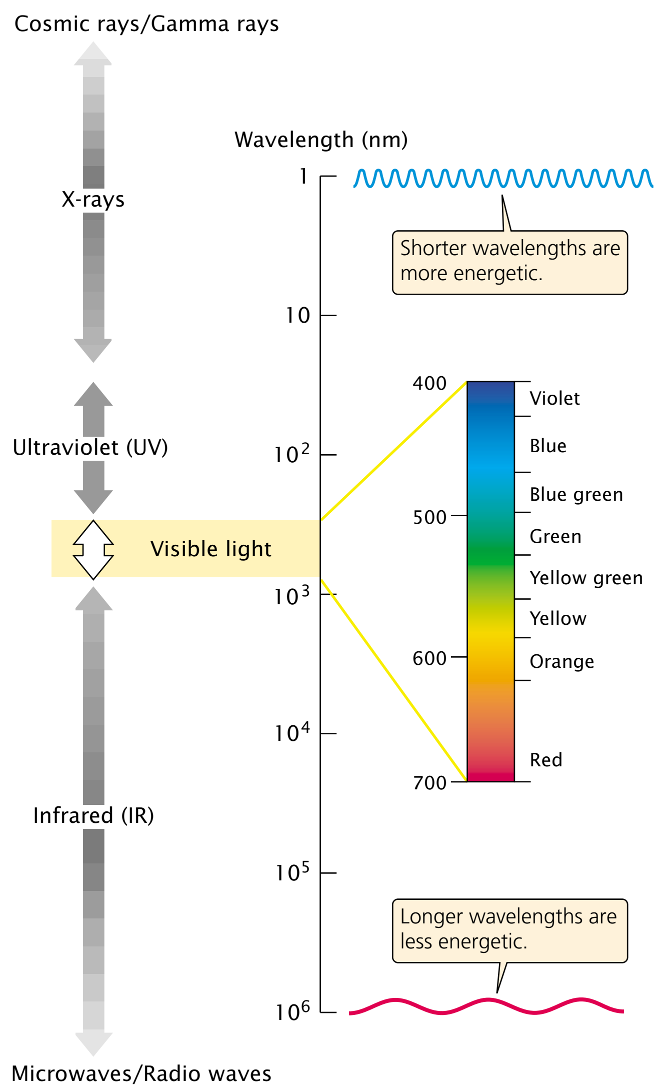
about the same size as a nucleotide (FIGURE 17.22a). They produce mutations by sandwiching themselves (intercalating) between adjacent bases in DNA, distorting the three-dimensional structure of the helix and causing single-nucleotide insertions and deletions in replication (FIGURE 17.22b). These insertions and deletions frequently produce frameshift mutations (which change all amino acids downstream of the mutation), and so the mutagenic effects of intercalating agents are often severe. Because intercalating agents generate both additions and deletions, they can reverse the effects of their own mutations.

Concepts

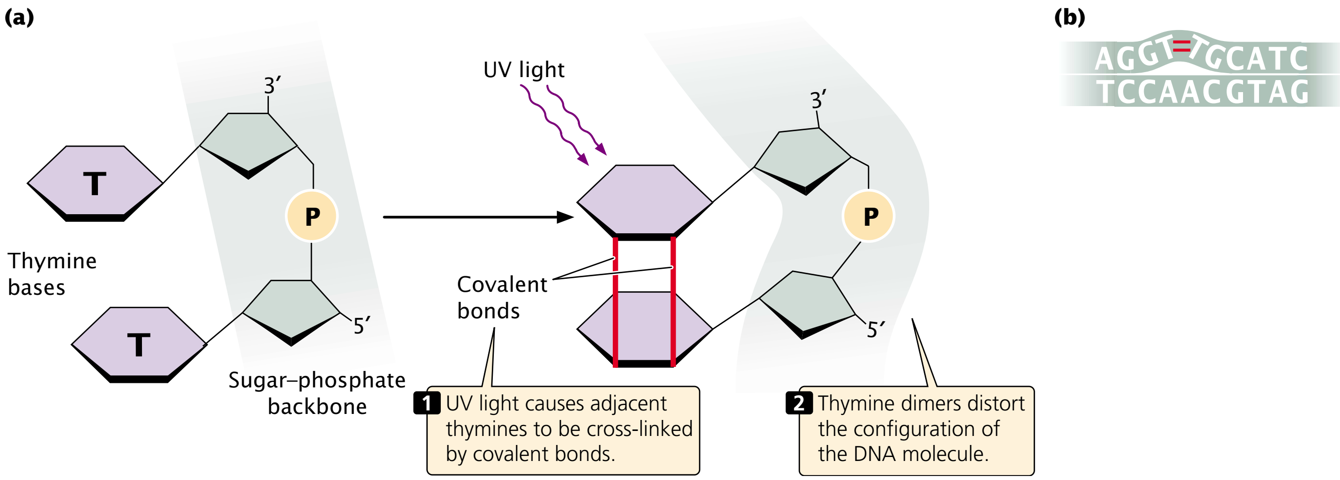
Chemicals can produce mutations by a number of mechanisms. Base analogs are inserted into DNA and frequently pair with the wrong base. Alkylating agents, deaminating chemicals, hydroxylamine, and oxidative radicals change the structure of DNA bases, thereby altering their pairing properties. Intercalating agents wedge between the bases and cause single-base insertions and deletions in replication.

Radiation

In 1927, Herman Muller demonstrated that mutations in fruit flies could be induced by X-rays. The results of subsequent studies showed that X-rays greatly increase mutation rates in all organisms. The high energies of X-rays, gamma rays, and cosmic rays (FIGURE 17.23) are all capable of penetrating tissues and damaging DNA. These forms of radiation, called ionizing radiation, dislodge electrons from the atoms that they encounter, changing stable molecules into free radicals and reactive ions, which then alter the structures of bases and break phosphodiester



17.23 In the electromagnetic spectrum, as wavelength decreases, energy increases. (Adapted from *Life 6e*, figure 8.5).



17.24 Pyrimidine dimers result from Ultraviolet light. (a) Formation of thymine dimer. (b) Distorted DNA.

bonds in DNA. Ionizing radiation also frequently results in double-strand breaks in DNA. Attempts to repair these breaks can produce chromosome mutations (discussed in Chapter 9).

Ultraviolet light has less energy than that of ionizing radiation and does not eject electrons and cause ionization but is nevertheless highly mutagenic. Purine and pyrimidine bases readily absorb UV light, resulting in the formation of chemical bonds between adjacent pyrimidine molecules on the same strand of DNA and in the creation of structures called **pyrimidine dimers** (FIGURE 17.24a). Pyrimidine dimers consisting of two thymine bases (called thymine dimers) are most frequent, but cytosine dimers and thymine–cytosine dimers also can form. These dimers distort the configuration of DNA (FIGURE 17.24b) and often block replication. Most pyrimidine dimers are immediately repaired by mechanisms discussed later in this chapter, but some escape repair and inhibit replication and transcription.

When pyrimidine dimers block replication, cell division is inhibited and the cell usually dies; for this reason, UV light kills bacteria and is an effective sterilizing agent. For a mutation—a hereditary error in the genetic instructions—to occur, the replication block must be overcome. How do bacteria and other organisms replicate despite the presence of thymine dimers?

Bacteria can circumvent replication blocks produced by pyrimidine dimers and other types of DNA damage by means of the **SOS system.** This system allows replication blocks to be overcome, but in the process makes numerous mistakes and greatly increases the rate of mutation. Indeed, the very reason that replication can proceed in the presence of a block is that the enzymes in the SOS system do not strictly adhere to the base-pairing rules. The trade-off is that replication may continue and the cell survives, but only by sacrificing the normal accuracy of DNA synthesis.

The SOS system is complex, including the products of at least 25 genes. A protein called RecA binds to the damaged DNA at the blocked replication fork and becomes activated. This activation promotes the binding of a protein called LexA, which is a repressor of the SOS system. The activated RecA complex induces LexA to undergo self-cleavage, destroying its repressive activity. This inactivation enables other SOS genes to be expressed, and the products of these genes allow replication of the damaged DNA to proceed. The SOS system allows bases to be inserted into a new DNA strand in the absence of bases on the template strand, but these insertions result in numerous errors in the base sequence.

Eukaryotic cells have a specialized DNA polymerase called polymerase η (eta) that bypasses pyrimidine dimers. Polymerase η preferentially inserts AA opposite a pyrimidine dimer. This strategy seems to be reasonable because about two-thirds of pyrimidine dimers are thymine dimers. However, the insertion of AA opposite a CT dimer results in a C·G→A·T transversion. Polymerase η is therefore said to be an error-prone polymerase.

Concepts

Ionizing radiation such as X-rays and gamma rays damage DNA by dislodging electrons from atoms; these electrons then break phosphodiester bonds and alter the structure of bases. Ultraviolet light causes mutations primarily by producing pyrimidine dimers that disrupt replication and transcription. The SOS system enables bacteria to overcome replication blocks but introduces mistakes in replication.

The Study of Mutations

Because mutations often have detrimental effects, they have been the subject of intense study by geneticists. These studies have included the analysis of reverse mutations, which are often sources of important insight into how mutations cause DNA damage; the development of tests to determine the mutagenic properties of chemical compounds; and the investigation of human populations tragically exposed to high levels of radiation.

The Analysis of Reverse Mutations

The study of reverse mutations (reversions) can provide useful information about how mutagens alter DNA structure. For example, any mutagen that produces both A·T→G·C and G·C→A·T transitions should be able to reverse its own mutations. However, if the mutagen produces only G·C→A·T transitions, then reversion by the same mutagen is not possible. Hydroxylamine (see Figure 17.20c) exhibits this type of one-way mutagenic activity; it causes G·C→A·T transitions but is incapable of reversing the mutations that it produces; so we know that it does not produce A·T→G·C transitions. Ethylmethanesulfonate (see Figure 17.20a), on the other hand, produces G·C→A·T transitions and reverses its own mutations; so we know that it also produces T·A→C·G transitions.

Analyses of the ability of different mutagens to cause reverse mutations can be sources of insight into the molecular nature of the mutations. We can use reverse mutations to determine whether a mutation results from a base substitution or a frameshift. Base analogs such as 2-aminopurine cause transitions, and intercalating agents such as acridine orange (see Figure 17.22) produce frameshifts. If a chemical reverses mutations produced by 2-aminopurine but not those produced by acridine orange, we can conclude that the chemical causes transitions and not frameshifts. If nitrous acid (which produces both G·C→A·T and A·T→G·C transitions) reverses mutations produced by the chemical but hydroxylamine (which causes *only* G·C→A·T transitions) does not, we know that, like hydroxylamine, the chemical produces only G·C→A·T transitions. Table 17.4 illustrates the reverse mutations that are theoretically possible among several mutagenic agents. The actual ability of mutagens to produce reversals is more complex than suggested by Table 17.4 and depends on environmental conditions and the organism tested.

Concepts

The study of the ability of mutagenic agents to produce reverse mutations provides important information about how mutagens alter DNA.

Detecting Mutations with the Ames Test

Humans in industrial societies are surrounded by a multitude of artificially produced chemicals: more than 50,000 different chemicals are in commercial and industrial use today, and from 500 to 1000 new chemicals are introduced each year. Some of these chemicals are potential carcinogens and may cause potential harm to humans. How can we determine which chemicals are hazardous? In a few instances, previous human exposure to a specific chemical is correlated with an increase in cancer incidence, providing good evidence that the chemical is a carcinogen. But, ideally, we would like to know which chemicals are hazardous before we are exposed to them. One method for testing the cancer-causing potential of chemicals is to administer them to laboratory animals (rats or mice) and compare the inci-

Table 17.4 Theoretical reverse mutations possible by various mutagenic agents

		Reversal of Mutations by					
Mutagen	**Type of Mutation**	**5-Bromo-uracil**	**2-Amino-purine**	**Ethyl methane sulfonate**	**Nitrous acid**	**Hydroxyl-amine**	**Acridine orange**
5-Bromouracil	C·G↔T·A	+	+	+	+	+/−	−
2-Aminopurine	C·G↔T·A	+	+	+	+	+/−	−
Nitrous acid	C·G↔T·A	+	+	+	+	+/−	−
Ethylmethane sulfonate	C·G↔T·A	+	+	+	+	+/−	−
Hydroxylamine	C·G↔T·A	+	+	+	+	−	−
Acridine orange	Frameshift	−	−	−	−	−	+

Note: + indicates that reverse mutations occur, − indicates that reverse mutations do not occur, and +/− indicates that only some mutations are reversed. Not all reverse mutations are equally likely.

dence of cancer in the treated animals with that of control animals. These tests are unfortunately time consuming and expensive. Furthermore, the ability of a substance to cause cancer in rodents is not always indicative of its effect on humans. After all, we aren't rats!

In 1974, Bruce Ames developed a simple test for evaluating the potential of chemicals to cause cancer. The **Ames test** is based on the principle that both cancer and mutations result from damage to DNA, and the results of experiments have demonstrated that 90% of known carcinogens are also mutagens. Ames proposed that mutagenesis in bacteria could serve as an indicator of carcinogenesis in humans.

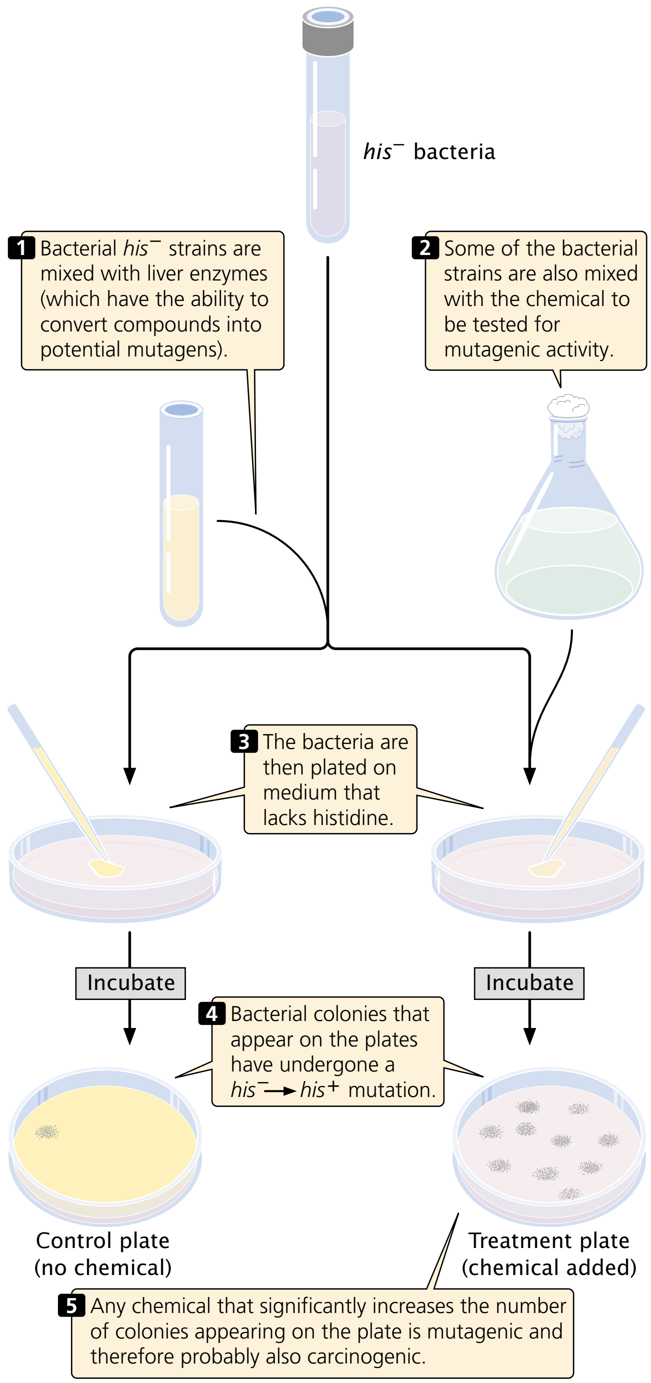
The Ames test uses four strains of the bacterium *Salmonella typhimurium* that have defects in the lipopolysaccharide coat, which normally protects the bacteria from chemicals in the environment. Furthermore, their DNA repair system has been inactivated, enhancing their susceptibility to mutagens.

One of the four strains used in the Ames test detects base-pair substitutions; the other three detect different types of frameshift mutations. Each strain carries a mutation that renders it unable to synthesize the amino acid histidine (*his*$^-$), and the bacteria are plated onto medium that lacks histidine (FIGURE 17.25). Only bacteria that have undergone a reverse mutation of the histidine gene (*his*$^- \rightarrow$ *his*$^+$) are able to synthesize histidine and grow on the medium. Different dilutions of a chemical to be tested are added to plates inoculated with the bacteria, and the number of mutant bacterial colonies that appear on each plate is compared with the number that appear on control plates with no chemical (arose through spontaneous mutation). Any chemical that significantly increases the number of colonies appearing on a treated plate is mutagenic and is probably also carcinogenic.

Some compounds are not active carcinogens but may be converted into cancer-causing compounds in the body. To make the Ames test sensitive for such *potential* carcinogens, a compound to be tested is first incubated in mammalian liver extract that contains metabolic enzymes. The Ames test has been applied to thousands of chemicals and commercial products. An early demonstration of its usefulness was the discovery, in 1975, that most hair dyes sold in the United States contained compounds that were mutagenic to bacteria. These compounds were then removed from most hair dyes.

Concepts

The Ames test uses *his*$^-$ strains of bacteria to test chemicals for their ability to produce *his*$^- \rightarrow$ *his*$^+$ mutations. Because mutagenic activity and carcinogenic potential are closely correlated, the Ames test is widely used to screen chemicals for their cancer-causing potential.



17.25 The Ames test is used to identify chemical mutagens.

www.whfreeman.com/pierce More information on the Ames test

Radiation Exposure in Humans

People are routinely exposed to low levels of radiation from cosmic, medical, and environmental sources, but there have also been tragic events that produced exposures of much higher degree.

17.26 Hiroshima was destroyed by an atomic bomb on August 6, 1945. The atomic explosion produced many somatic mutations among the survivors. (Stanley Troutman/AP.)

On August 6, 1945, a high-flying American plane dropped a single atomic bomb on the city of Hiroshima, Japan. The explosion devastated 4.5 square miles of the city, killed from 90,000 to 140,000 people, and injured almost as many (FIGURE 17.26). Three days later, the United States dropped an atomic bomb on the city of Nagasaki, this time destroying 1.5 square miles of city and killing between 60,000 and 80,000 people. Huge amounts of radiation were released during these explosions and many people were exposed.

After the war, a joint Japanese–U.S. effort was made to study the biological effects of radiation exposure on the survivors of the atomic blasts and their children. Somatic mutations were examined by studying radiation sickness and cancer among the survivors; germ-line mutations were assessed by looking at birth defects, chromosome abnormalities, and gene mutations in children born to people that had been exposed to radiation.

Geneticist James Neel and his colleagues examined almost 19,000 children of parents who were within 2000 meters of the center of the atomic blast at Hiroshima or Nagasaki, along with a similar number of children whose parents did not receive radiation exposure. Radiation doses were estimated for the child's parents on the basis of careful assessment of the parents' location, posture, and position at the time of the blast. A blood sample was collected from each child, and gel electrophoresis was used to investigate amino acid substitutions in 28 proteins. When rare variants were detected, blood samples from the child's parents also were analyzed to establish whether the variant was inherited or a new mutation.

Of a total of 289,868 genes examined by Neel and his colleagues, only one mutation was found in the children of exposed parents; no mutations were found in the control group. From these findings, a mutation rate of 3.4×10^{-6} was estimated for the children whose parents were exposed to the blast, which is within the range of spontaneous mutation rates observed for other eukaryotes. Neel and his colleagues also examined the frequency of chromosome mutations, sex ratios of children born to exposed parents, and frequencies of chromosome aneuploidy. There was no evidence in any of these assays for increased mutations among the children of the people who were exposed to radiation from the atomic explosions, suggesting that germ-line mutations were not elevated.

Animal studies clearly show that radiation causes germ-line mutations; so why was there no apparent increase in germ-line mutations among the inhabitants of Hiroshima and Nagasaki? The exposed parents did exhibit an increased incidence of leukemia and other types of cancers; so somatic mutations were clearly induced. The answer to the question is not known, but the lack of germ-line mutations may be due to the fact that those persons who received the largest radiation doses died soon after the blasts.

www.whfreeman.com/pierce Information on studies of the health effects of the nuclear blasts at Hiroshima and Nagasaki

The Techa River in southern Russia is another place where people have been tragically exposed to high levels of radiation. The Mayak nuclear facility, located 60 miles from the city of Chelyabinsk, produced plutonium for nuclear warheads in the early days of the Cold War. Between 1949 and 1956, this plant dumped some 76 million cubic meters of radioactive sludge into the Techa River. People downstream used the river for drinking water and crop irrigation; some received radiation doses 1700 times the annual amount considered safe by today's standards. Radiation in the area was further elevated by a series of nuclear accidents at the Mayak plant; the worst was an explosion of a radioactive liquid storage tank in 1957, which showered radiation over a 27,000-square-kilometer area.

Although Soviet authorities suppressed information about the radiation problems along the Techa until the 1990s, Russian physicians lead by Mira Kossenko quietly began studying cancer and other radiation-related illnesses among the inhabitants in the 1960s. They found that the overall incidence of cancer was elevated among people who lived on the banks of the Techa River.

Most data on radiation exposure in humans are from the intensive study of the survivors of the atomic bombing of Hiroshima and Nagasaki. However, the inhabitants of Hiroshima and Nagasaki were exposed in one intense burst of radiation, and these data may not be appropriate for understanding the effects of long-term low-dose radiation. Today, U.S. and Russian scientists are studying the people of the Techa River region, as well as those exposed to radiation in the Chernobyl accident (see the story at the beginning of this chapter), in an attempt to better understand the effects of chronic radiation exposure on human populations.

DNA Repair

The integrity of DNA is under constant assault from radiation, chemical mutagens, and spontaneously arising changes. In spite of this onslaught of damaging agents, the rate of mutation remains remarkably low, thanks to the efficiency with which DNA is repaired. It has been estimated that fewer than one in a thousand DNA lesions becomes a mutation; all the others are corrected.

There are a number of complex pathways for repairing DNA, but several general statements can be made about DNA repair. First, most DNA repair mechanisms require two nucleotide strands of DNA because most replace whole nucleotides, and a template strand is needed to specify the base sequence. The complementary, double-stranded nature of DNA not only provides stability and efficiency of replication, but also enables either strand to provide the information necessary for correcting the other.

A second general feature of DNA repair is redundancy, meaning that many types of DNA damage can be corrected by more than one pathway of repair. This redundancy testifies to the extreme importance of DNA repair to the survival of the cell: it ensures that almost all mistakes are corrected. If a mistake escapes one repair system, it's likely to be repaired by another system.

We will consider four general mechanisms of DNA repair: mismatch repair, direct repair, base-excision repair, and nucleotide-excision repair (Table 17.5).

Table 17.5 Summary of common DNA repair mechanisms

Repair System	Type of Damage Repaired
Mismatch	Replication errors, including mispaired bases and strand slippage
Direct	Pyrimidine dimers; other specific types of alterations
Base-excision	Abnormal bases, modified bases, and pyrimidine dimers
Nucleotide-excision	DNA damage that distorts the double helix, including abnormal bases, modified bases, and pyrimidine dimers

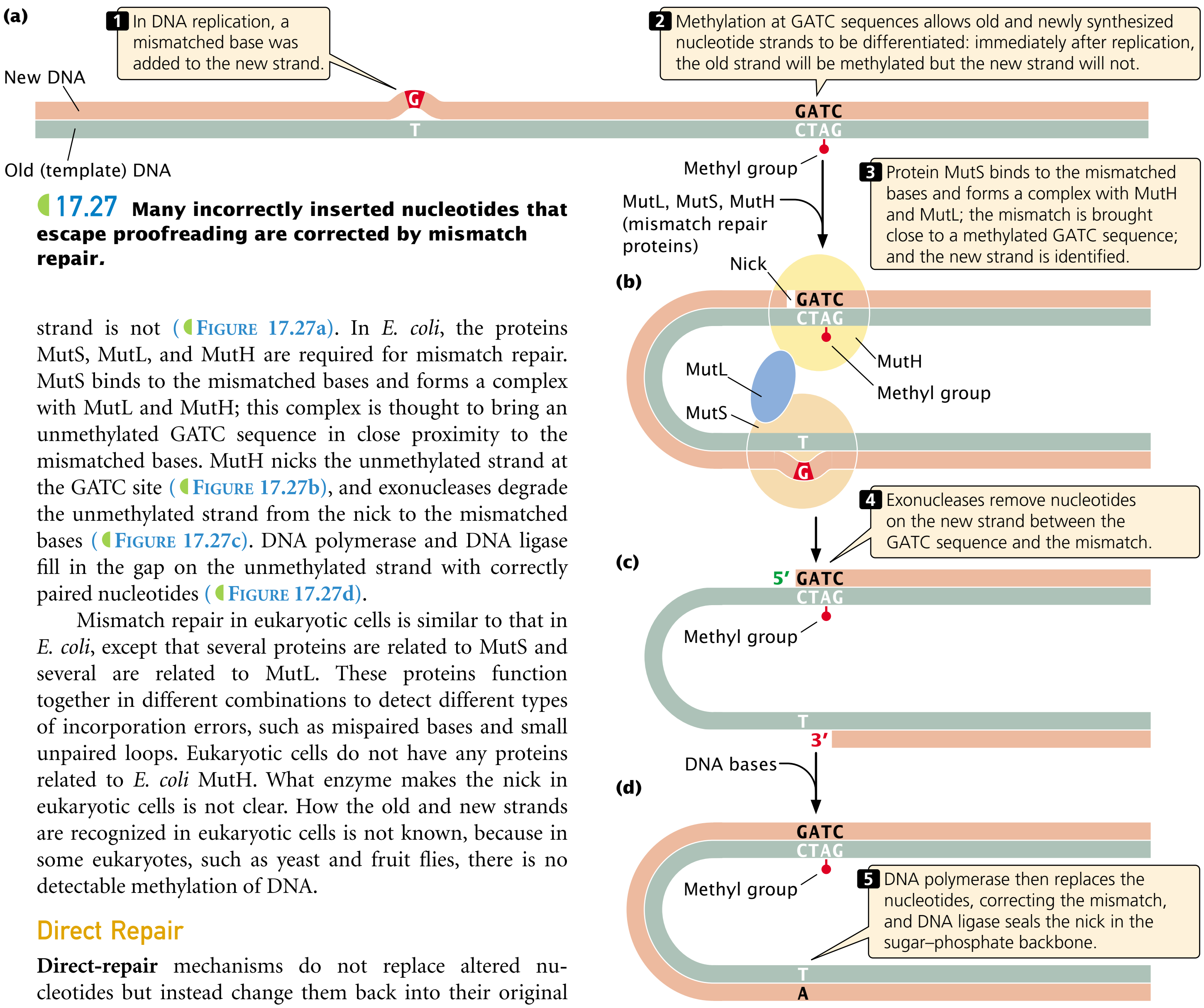
Mismatch Repair

Replication is extremely accurate: each new copy of DNA has only one error per billion nucleotides. However, in the process of replication, mismatched bases are incorporated into the new DNA with a frequency of about 10^{-4} to 10^{-5}; so most of the errors that initially arise are corrected and never become permanent mutations. Some of these corrections are made in proofreading (see p. 000 in Chapter 12). DNA polymerases have the capacity to recognize and correct mismatched nucleotides. When a mismatched nucleotide is added to a newly synthesized DNA strand, the polymerase stalls. It then uses its $3' \rightarrow 5'$ exonuclease activity to back up and remove the incorrectly inserted nucleotide before continuing with $5' \rightarrow 3'$ polymerization.

Many incorrectly inserted nucleotides that escape detection by proofreading are corrected by *mismatch repair* (see p. 000 in Chapter 12). Incorrectly paired bases distort the three-dimensional structure of DNA, and mismatch-repair enzymes detect these distortions. In addition to detecting incorrectly paired bases, the mismatch-repair system corrects small unpaired loops in the DNA, such as those caused by strand slippage in replication (see Figure 17.14). Some trinucleotide repeats may form secondary structures on the unpaired strand (see Figure 17.6d), allowing them to escape detection by the mismatch-repair system.

After the incorporation error has been recognized, mismatch-repair enzymes cut out the distorted section of the newly synthesized strand and fill the gap with new nucleotides, by using the original DNA strand as a template. For this strategy to work, mismatch repair must have some way of distinguishing between the old and the new strands of the DNA so that the incorporation error, and not part of the original strand, is removed.

The proteins that carry out mismatch repair in *E. coli* differentiate between old and new strands by the presence of methyl groups on special sequences of the old strand. After replication, adenine nucleotides in the sequence GATC are methylated by an enzyme called Dam methylase. The process of methylation is delayed and so, immediately after replication, the old strand is methylated and the new



17.27 Many incorrectly inserted nucleotides that escape proofreading are corrected by mismatch repair.

strand is not (FIGURE 17.27a). In *E. coli*, the proteins MutS, MutL, and MutH are required for mismatch repair. MutS binds to the mismatched bases and forms a complex with MutL and MutH; this complex is thought to bring an unmethylated GATC sequence in close proximity to the mismatched bases. MutH nicks the unmethylated strand at the GATC site (FIGURE 17.27b), and exonucleases degrade the unmethylated strand from the nick to the mismatched bases (FIGURE 17.27c). DNA polymerase and DNA ligase fill in the gap on the unmethylated strand with correctly paired nucleotides (FIGURE 17.27d).

Mismatch repair in eukaryotic cells is similar to that in *E. coli*, except that several proteins are related to MutS and several are related to MutL. These proteins function together in different combinations to detect different types of incorporation errors, such as mispaired bases and small unpaired loops. Eukaryotic cells do not have any proteins related to *E. coli* MutH. What enzyme makes the nick in eukaryotic cells is not clear. How the old and new strands are recognized in eukaryotic cells is not known, because in some eukaryotes, such as yeast and fruit flies, there is no detectable methylation of DNA.

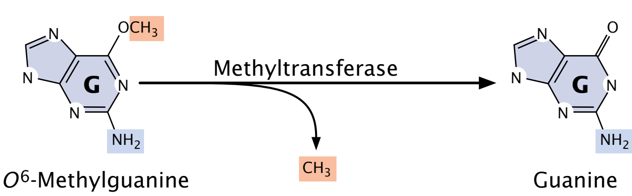
Direct Repair

Direct-repair mechanisms do not replace altered nucleotides but instead change them back into their original (correct) structures. One of the best-characterized direct-repair mechanisms is photoreactivation of UV-induced pyrimidine dimers. *E. coli* and some eukaryotic cells possess an enzyme called photolyase, which uses energy captured from light to break the covalent bonds that link the pyrimidines in a dimer.

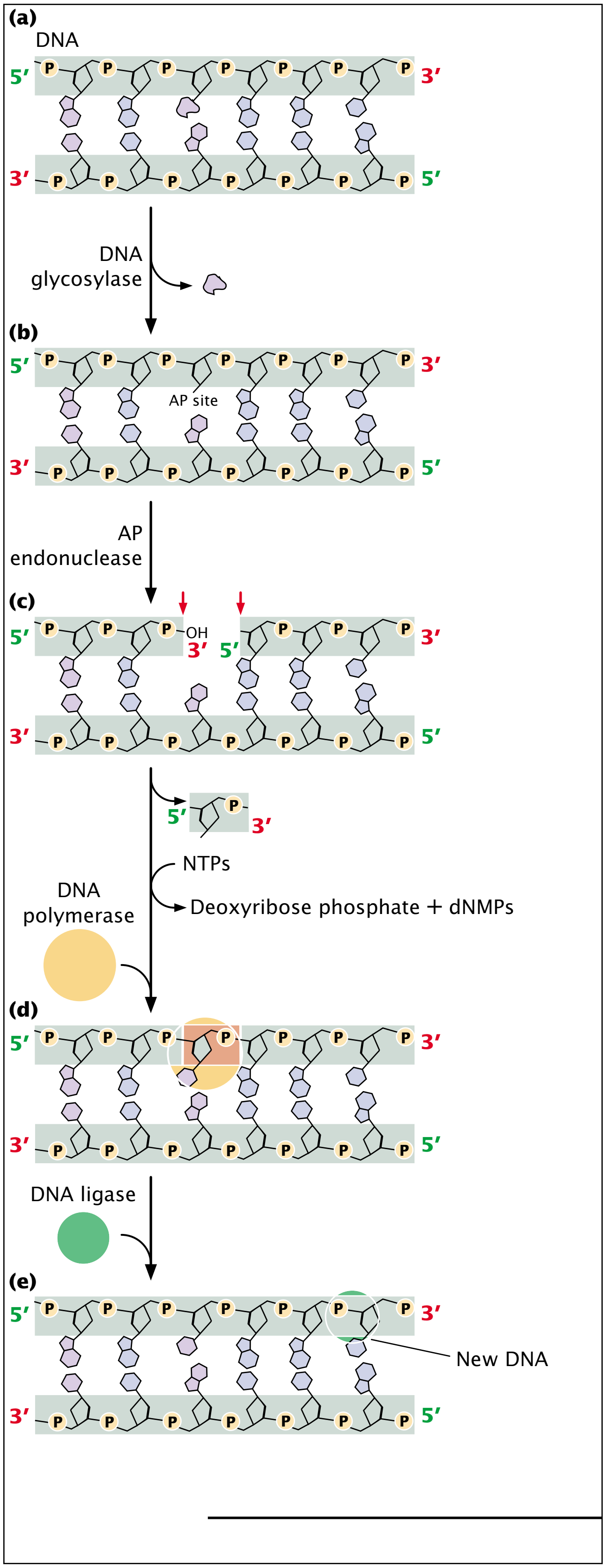
Direct repair also corrects O^6-methylguanine, an alkylation product of guanine that pairs with adenine, producing G·C→T·A transversions. An enzyme called O^6-methylguanine-DNA methyltransferase removes the methyl group from O^6-methylguanine, restoring the base to guanine (FIGURE 17.28).

Base-Excision Repair

In **base-excision repair,** modified bases are first excised and then the entire nucleotide is replaced. The excision of modified bases is catalyzed by a set of enzymes called DNA glycosylases, each of which recognizes and removes a specific type of modified base by cleaving the bond that links that base to the 1′-carbon atom of deoxyribose (FIGURE 17.29a). Uracil glycosylase, for example, recognizes and removes uracil produced by the deamination of cytosine. Other glycosylases



17.28 Direct repair changes nucleotides back into their original structures.



17.29 **Base-excision repair excises modified bases and then replaces the entire nucleotide.**

recognize hypoxanthine, 3-methyladenine, 7-methylguanine, and other modified bases.

After the base has been removed, an enzyme called AP (apurinic or apyrimidinic) endonuclease cuts the phosphodiester bond, and other enzymes remove the deoxyribose sugar (FIGURE 17.29b). DNA polymerase then adds new nucleotides to the exposed 3′-OH group (FIGURE 17.29c), replacing a section of nucleotides on the damaged strand. The nick in the phosphodiester backbone is sealed by DNA ligase (FIGURE 17.29d), and the original intact sequence is restored (FIGURE 17.29e).

Nucleotide-Excision Repair

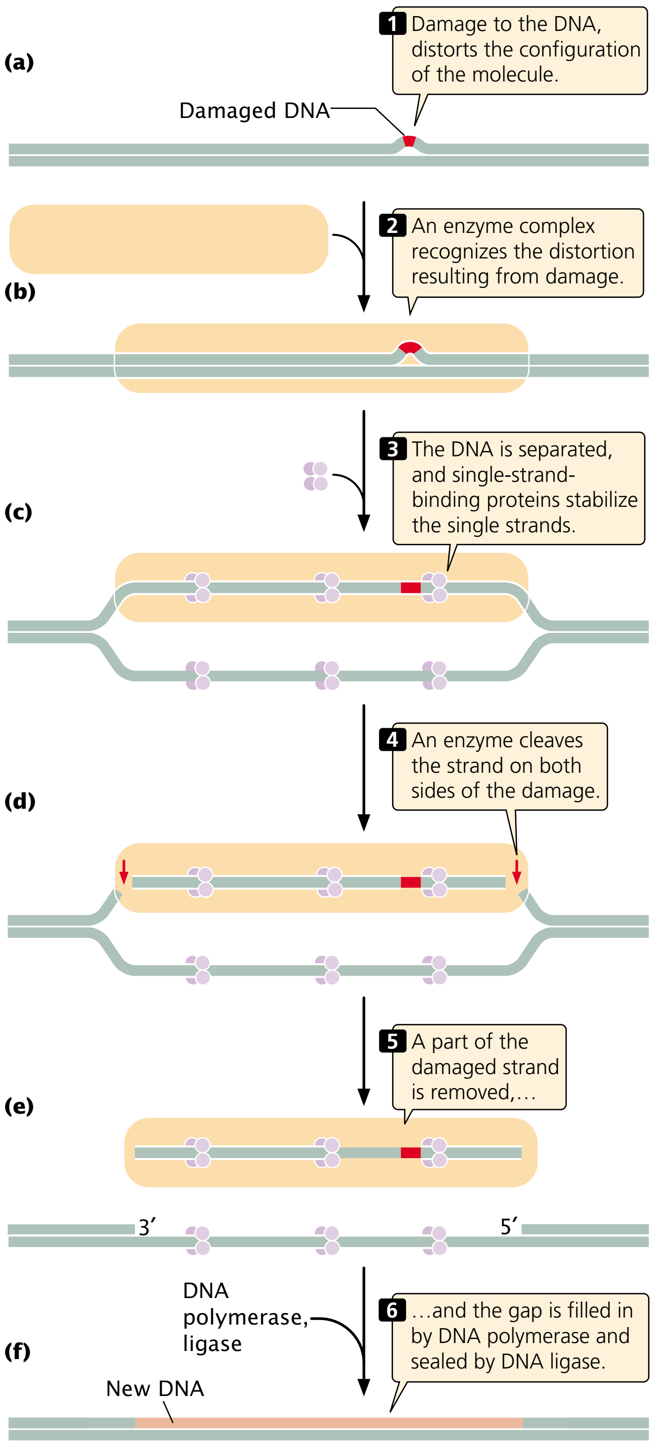
The final repair pathway that we'll consider is **nucleotide-excision repair,** which removes bulky DNA lesions that distort the double helix, such as pyrimidine dimers or large hydrocarbons attached to the DNA. Nucleotide-excision repair is quite versatile and can repair many different types of DNA damage. It is found in cells of all organisms from bacteria to humans and is one of the most important of all repair mechanisms.

The process of nucleotide excision is complex; in humans, a large number of genes take part. First, a complex of enzymes scans DNA, looking for distortions of its three-dimensional configuration (FIGURE 17.30a and b). When a distortion is detected, additional enzymes separate the two nucleotide strands at the damaged region, and single-strand-binding proteins stabilize the separated strands (FIGURE 17.30c). Next, the sugar–phosphate backbone of the damaged strand is cleaved on both sides of the damage. One cut is made 5 nucleotides upstream (on the 3′ side) of the damage, and the other cut is made 8 nucleotides (in prokaryotes) or from 21 to 23 nucleotides (in eukaryotes) downstream (on the 5′ side) of the damage (FIGURE 17.30d). Part of the damaged strand is peeled away (FIGURE 17.30e), and the gap is filled in by DNA polymerase and sealed by DNA ligase (FIGURE 17.30f).

Other Types of DNA Repair

The DNA repair pathways described so far respond to damage that is limited to one strand of a DNA molecule, leaving the other strand to be used as a template for the synthesis of new DNA during the repair process. Some types of DNA damage, however, affect both strands of the molecule and therefore pose a more severe challenge to the DNA repair machinery. Ionizing radiation frequently results in double-strand breaks in DNA. The repair of double-strand breaks is frequently by homologous recombination. Models for homologous recombination were described in Chapter 12.

Another type of damage that affects both strands is an interstrand cross-link, which arises when the two strands of a duplex are connected through covalent bonds. Interstrand cross-links are extremely toxic to cells because they halt replication. Several drugs commonly used in chemotherapy, including cisplatin, mitomycin C, psoralen, and nitrogen mustard, cause interstrand cross-links. Nitrogen mustard, which is structurally related to the mustard gas used by Charlotte Auerbach to induce mutations in *Drosophila*, was the first chemical agent to be used in chemotherapy treatment. Little is known about how interstrand cross-links are repaired. One model proposes that double-strand breaks are made on each side of the cross-link and are subsequently repaired by the pathways that repair double-strand breaks.



17.30 Nucleotide-excision repair consists of four steps: detection of damage, excision of damage, polymerization, and ligation.

Connecting Concepts

The Basic Pathway of DNA Repair

We have now examined several different mechanisms of DNA repair. What do these methods have in common? How are they different? Most methods of DNA repair depend on the presence of two strands, because nucleotides in the damaged area are removed and replaced. Nucleotides are replaced in mismatch repair, base excision repair, and nucleotide-excision repair, but are not replaced by direct-repair mechanisms.

Repair mechanisms that include nucleotide removal utilize a common four-step pathway:

1. **Detection:** The damaged section of the DNA is recognized.
2. **Excision:** DNA repair endonucleases nick the phosphodiester backbone on one or both sides of the DNA damage.
3. **Polymerization:** DNA polymerase adds nucleotides to the newly exposed 3′-OH group by using the other strand as a template and replacing damaged (and frequently some undamaged) nucleotides.
4. **Ligation:** DNA ligase seals the nicks in the sugar–phosphate backbone.

The primary differences in the mechanisms of mismatch, base-excision, and nucleotide-excision repair are in the details of detection and excision. In base-excision and mismatch repair, a single nick is made in the sugar–phosphate backbone on one side of the damaged strand; in nucleotide-excision repair, nicks are made on both sides of the DNA lesion. In base-excision repair, DNA polymerase displaces the old nucleotides as it adds new nucleotides to the 3′ end of the nick; in mismatch repair, the old nucleotides are degraded; and, in nucleotide-excision repair, nucleotides are displaced by helicase enzymes. All three mechanisms use DNA polymerase and ligase to fill in the gap produced by the excision and removal of damaged nucleotides.

www.whfreeman.com/pierce Additional information on DNA repair

Genetic Diseases and Faulty DNA Repair

Several human diseases are connected to defects in DNA repair. These diseases are often associated with high incidences of specific cancers, because defects in DNA repair

17.31 **Xeroderma pigmentosum is a human disease that results from defects in DNA repair.** (Ken Greer/Visuals Unlimited.)

lead to increased rates of mutation. This concept is discussed further in Chapter 21.

Among the best studied of the human DNA repair diseases is xeroderma pigmentosum (FIGURE 17.31), a rare autosomal recessive condition that includes abnormal skin pigmentation and acute sensitivity to sunlight. Persons who have this disease also have a strong predisposition to skin cancer, with an incidence from 1000 to 2000 times that found in unaffected people.

Sunlight includes a strong UV component; so exposure to sunlight produces pyrimidine dimers in the DNA of skin cells. Although human cells lack photolyase (the enzyme that repairs pyrimidine dimers in bacteria), most pyrimidine dimers in humans can be corrected by nucleotide-excision repair. However, the cells of most people with xeroderma pigmentosum are defective in nucleotide-excision repair, and many of their pyrimidine dimers go uncorrected and may lead to cancer.

Xeroderma pigmentosum can result from defects in several different genes; studies have identified at least seven different xeroderma pigmentosum complementation groups, meaning that at least seven genes are required for nucleotide-excision repair in humans. Recent molecular research has led to the identification of genetic defects of nucleotide-excision repair associated with these complementation groups. Some persons with xeroderma pigmentosum have mutations in a gene encoding the protein that recognizes and binds to damaged DNA; others have mutations in a gene encoding helicase. Still others have defects in the genes that play a role in cutting the damaged strand on the 5′ or 3′ sides of the pyrimidine dimer. Some persons have a slightly different form of the disease (xeroderma pigmentosum variant) owing to mutations in the gene encoding polymerase η, the DNA polymerase that bypasses pyrimidine dimers by inserting AA.

Two other genetic diseases due to defects in nucleotide-excision repair are Cockayne syndrome and trichothiodystrophy (also known as brittle-hair syndrome). Persons who have either of these diseases do not have an increased risk of cancer but do exhibit multiple developmental and neurological problems. Both diseases result from mutations in some of the same genes that cause xeroderma pigmentosum. Several of the genes taking part in nucleotide-excision repair produce proteins that also play a role in recombination and the initiation of transcription. These other functions may account for the developmental symptoms seen in Cockayne syndrome and trichothiodystrophy.

Another genetic disease caused by faulty DNA repair is an inherited form of colon cancer called hereditary nonpolyposis colon cancer (HNPCC). This cancer is one of the most common hereditary cancers, accounting for about 15% of colon cancers. Research indicate that HNPCC arises from mutations in the proteins that carry out mismatch repair (see Figure 17.27).

Li-Fraumeni syndrome is caused by mutations in a gene called *p53*, which plays an important role in regulating the cell cycle. The product encoded by the *p53* gene can halt cell division until damage to DNA has been repaired; it can also directly stimulate DNA repair. The *p53* gene product may actually cause cells with damaged DNA to self-destruct (undergo apoptosis, or controlled cell death; see Chapter 21), preventing their mutated genetic instructions from being passed on. Patients who have Li-Fraumeni syndrome exhibit multiple independent cancers in different tissues. Some additional genetic diseases associated with defective DNA repair are summarized in Table 17.6.

Concepts

Defects in DNA repair are the underlying cause of several genetic diseases. Many of these diseases are characterized by a predisposition to cancer.

www.whfreeman.com/pierce Additional information about xeroderma pigmentosum

Connecting Concepts Across Chapters

This chapter has been our first comprehensive look at mutations, but we have been considering and using mutations throughout the book.

Mutation is a fact of life. Our DNA is continually assaulted by spontaneously arising and environmentally

Table 17.6 Genetic diseases associated with defects in DNA repair systems

Disease	Symptoms	Genetic Defect
Xeroderma pigmentosum	Frecklelike spots on skin, sensitivity to sunlight, predisposition to skin cancer	Defects in nucleotide-excision repair
Cockayne syndrome	Dwarfism, sensitivity to sunlight, premature aging, deafness, mental retardation	Defects in nucleotide-excision repair
Trichothiodystrophy	Brittle hair, skin abnormalities, short stature, immature sexual development, characteristic facial features	Defects in nucleotide-excision repair
Hereditary nonpolyposis colon cancer	Predisposition to colon cancer	Defects in mismatch repair
Fanconi anemia	Increased skin pigmentation, abnormalities of skeleton, heart, and kidneys, predisposition to leukemia	Possibly defects in the repair of interstrand cross-links
Ataxia telangiectasia	Defective muscle coordination, dilation of blood vessels in skin and eyes, immune deficiencies, sensitivity to ionizing radiation, predisposition to cancer	Defects in DNA damage detection and response
Li-Fraumeni syndrome	Predisposition to cancer in many different tissues	Defects in DNA damage response

induced mutations. These mutations are the raw material of evolution and, in the long run, allow organisms to adapt to the environment, a topic that will be taken up in Chapter 23. In spite of their long-term contribution to species evolution, the vast majority of mutations are, in the short term, detrimental to cells. The fact that most are detrimental is evidenced by the number mechanisms that cells possess to reduce the generation of errors in DNA and to repair those that do arise. A dominant theme of this chapter is that cells go to great lengths to prevent mutations.

This chapter has incorporated information presented in a number of earlier chapters, which you might want to review for a better understanding of the processes and structures discussed in the current chapter. Chromosome mutations and transposable elements (which frequently cause mutations) are discussed in Chapters 9 and 11. Although the structural nature of these mutations is different from that of gene mutations, many fundamental aspects of the mutational process that were introduced in this chapter also apply to these other types of mutations. The study of gene mutations is fundamentally about changes in DNA structure; so the discussion of DNA structure in Chapter 10 is critical for understanding the nature of mutations and how they arise. Some mutations spontaneously arise from errors in replication, and many DNA repair mechanisms include some DNA synthesis; hence, the process of replication outlined in Chapter 12 also is important. The relation between the nucleotide sequences of DNA and the amino acid sequences of proteins, which is discussed in Chapter 15, is particularly relevant for understanding the phenotypic effects of mutations and the nature of intra- and intergenic suppressors. Some of the material covered on bacterial and viral genetics in Chapter 15 is helpful for understanding complementation and the Ames test.

The current chapter has provided information that is important for understanding material presented in future chapters. Mutation is the molecular basis of cancer; so the contents of the current chapter will be highly relevant to the discussion of cancer genetics in Chapter 21. The importance of the mutation process to evolution will be revisited in Chapter 23.

CONCEPTS SUMMARY

- Mutations are heritable changes in genetic information. They are important for the study of genetics and can be used to unravel other biological processes.
- Somatic mutations occur in somatic cells; germ-line mutations occur in cells that give rise to gametes. Gene mutations are genetic alterations that affect a single gene; chromosome mutations entail changes in the number or structure of chromosomes.
- The simplest type of mutation is a base substitution, a change in a single base pair of DNA. Transitions are base substitutions in which purines are replaced by purines or pyrimidines are replaced by pyrimidines. Transversions are base substitutions in which a purine replaces a pyrimidine or a pyrimidine replaces a purine.
- Insertions are the addition of nucleotides, and deletions are the removal of nucleotides; these mutations often change the reading frame of the gene.
- Expanding trinucleotide repeats are mutations in which the number of copies of a trinucleotide increases through time; they are responsible for several human genetic diseases.
- A missense mutation alters the coding sequence so that one amino acid substitutes for another. A nonsense mutation changes a codon that specifies an amino acid to a termination codon. A silent mutation produces a synonymous codon that specifies the same amino acid as the original sequence, whereas a neutral mutation alters the amino acid sequence but does not change the functioning of the protein. A suppressor mutation reverses the effect of a previous mutation at a different site and may be intragenic (within the same gene as the original mutation) or intergenic (within a different gene).
- Mutation rate is the frequency with which a particular mutation arises in a population, whereas mutation frequency is the incidence of a mutation in a population. Mutation rates are usually low and are influenced by both genetic and environmental factors.
- Some mutations occur spontaneously. These mutations include the mispairing of bases in replication and spontaneous depurination and deamination.
- Insertions and deletions may arise from strand slippage in replication or from unequal crossing over.
- Base analogs may become incorporated into DNA in replication and pair with the wrong base in subsequent replication events. Alkylating agents and hydroxylamine modify the chemical structure of bases and lead to mutations. Intercalating agents insert into the DNA molecule and cause single-nucleotide additions and deletions. Oxidative reactions alter the chemical structures of bases.
- Ionizing radiation is mutagenic, altering base structures and breaking phosphodiester bonds. Ultraviolet light produces pyrimidine dimers, which block replication. Bacteria use the SOS response to overcome replication blocks produced by pyrimidine dimers and other lesions in DNA, but the SOS response causes the occurrence of more replication errors. Pyrimidine dimers in eukaryotic cells can be bypassed by DNA polymerase η but may result in the placement of incorrect bases opposite the dimer.
- The analysis of reverse mutations provides information about the molecular nature of the original mutation.
- The Ames tests uses bacteria to assess the mutagenic potential of chemical substances.
- Most damage to DNA is corrected by DNA repair mechanisms. These mechanisms include mismatch repair, direct repair, base-excision repair, nucleotide-excision repair, and other repair pathways. Although the details of the different DNA repair mechanisms vary, most require two strands of DNA and exhibit some overlap in the types of damage repaired. Proofreading and mismatch repair correct errors that arise in replication. Direct-repair mechanisms change the altered nucleotides back into their original condition, whereas base-excision and nucleotide-excision repair mechanisms replace nucleotides around the damaged segment of the DNA.
- Defects in DNA repair are the underlying cause of several genetic diseases.

IMPORTANT TERMS

mutation (p. 000)
somatic mutation (p. 000)
germ-line mutation (p. 000)
gene mutation (p. 000)
base substitution (p. 000)
transition (p. 000)
transversion (p. 000)
insertion (p. 000)
deletion (p. 000)
frameshift mutation (p. 000)
in-frame insertion (p. 000)
in-frame deletion (p. 000)
expanding trinucleotide repeat (p. 000)
forward mutation (p. 000)
reverse mutation (reversion) (p. 000)
missense mutation (p. 000)
nonsense mutation (p. 000)
silent mutation (p. 000)
neutral mutation (p. 000)
loss-of-function mutation (p. 000)
gain-of-function mutation (p. 000)
conditional mutation (p. 000)
lethal mutation (p. 000)
suppressor mutation (p. 000)
intragenic suppressor mutation (p. 000)

intergenic suppressor mutation (p. 000)
mutation rate (p. 000)
mutation frequency (p. 000)
spontaneous mutation (p. 000)
induced mutation (p. 000)
incorporated error (p. 000)
replicated error (p. 000)
strand slippage (p. 000)
unequal crossing over (p. 000)
depurination (p. 000)
deamination (p. 000)
mutagen (p. 000)
base analog (p. 000)
intercalating agent (p. 000)
pyrimidine dimer (p. 000)
SOS system (p. 000)
Ames test (p. 000)
direct repair (p. 000)
base-excision repair (p. 000)
nucleotide-excision repair (p. 000)

Worked Problems

1. A codon that specifies the amino acid Asp undergoes a single-base substitution that yields a codon that specifies Ala. Refer to the genetic code in Figure 15.12 and give all possible DNA sequences for the original and the mutated codon. Is the mutation a transition or a transversion?

• Solution

There are two possible RNA codons for Asp: GAU and GAC. The DNA sequences that encode these codons will be complementary to the RNA codons: CTA and CTG. There are four possible RNA codons for Ala: GCU, GCC, GCA, and GCG, which correspond to DNA sequences CGA, CGG, CGT, and CGC. If we organize the original and mutated sequences as shown in the following table, it is easy to see what type of mutations may have occurred:

Possible original sequence for Asp	Possible mutated sequence for Ala
CTA	CGA
CTG	CGG
	CGT
	CGC

If the mutation is confined to a single-base substitution, then the only mutations possible are that CTA mutated to CGA or that GTG mutated to CGG. In both, there is a T→G transversion in the middle nucleotide of the codon.

2. A gene encodes a protein with the following amino acid sequence:

Met-Arg-Cys-Ile-Lys-Arg

A mutation of a single nucleotide alters the amino acid sequence to:

Met-Asp-Ala-Tyr-Lys-Gly-Glu-Ala-Pro-Val

A second single-nucleotide mutation occurs in the same gene and suppresses the effects of the first mutation (an intragenic suppressor). With the original mutation and the intragenic suppressor present, the protein has the following amino acid sequence:

Met-Asp-Gly-Ile-Lys-Arg

What is the nature and location of the first mutation and the intragenic suppressor mutation?

• Solution

The first mutation alters the reading frame, because all amino acids after Met are changed. Insertions and deletions affect the reading frame; so the original mutation consists of a single-nucleotide insertion or deletion in the second codon. The intragenic suppressor restores the reading frame; so the intragenic suppressor also is most likely a single-nucleotide insertion or deletion: if the first mutation is an insertion, the suppressor must be a deletion; if the first mutation is a deletion, then the suppressor must be an insertion. Notice that the protein produced by the suppressor still differs from the original protein at the second and third amino acids, but the suppressor's second amino acid is the same as that in the protein produced by the original mutation. Thus the suppressor mutation must have occurred in the third codon, because the suppressor does not alter the second amino acid.

3. The mutations produced by the following compounds are reversed by the substances shown. What conclusions can you make about the nature of the mutations originally produced by these compounds?

	Reversed by			
Mutations produced by compound	5-Bromouracil	EMS	Hydroxylamine	Acridine orange
(a) 1	Yes	Yes	No	No
(b) 2	Yes	Yes	Some	No
(c) 3	No	No	No	Yes
(d) 4	Yes	Yes	Yes	Yes

• Solution

The ability of various compounds to produce reverse mutations reveals important information about the nature of the original mutation.

(a) Mutations produced by compound 1 are reversed by 5-bromouracil, which produces both A·T→G·C and G·C→A·T transitions. This tells us that compound 1 produces single-base substitutions that *may* include the generation of either A·T or

G·C pairs. The mutations produced by compound 1 are also reversed by EMS, which, like 5-bromouracil, produces both A·T→G·C and G·C→A·T transitions; so no additional information is provided here. Hydroxylamine does not reverse the mutations produced by compound 1. Because hydroxylamine produces only C·G→T·A transitions, we know that compound 1 does not generate C·G base pairs. Acridine orange, an intercalating agent that produces frameshift mutations, also does not reverse the mutations, revealing that compound 1 produces only single-base-pair substitutions, not insertions or deletions. In summary, compound 1 appears to causes single-base substitutions that generate T·A but not G·C base pairs.

(b) Compound 2 generates mutations that are reversed by 5-bromouracil and EMS, indicating that it may produce G·C or A·T base pairs. Some of these mutations are reversed by hydroxylamine, which produces only C·G→T·A transitions. This indicates that some of the mutations produced by compound 2 are T·A base pairs. None of the mutations are reversed by acridine orange; so compound 2 does not induce insertions or deletions. In summary, compound 2 produces single-base substitutions that generate both G·C and A·T base pairs.

(c) Compound 3 produces mutations that are reversed only by acridine orange; so compound 3 appears to produce only insertions and deletions.

(d) Compound 4 is reversed by 5 bromouracil, EMS, hydroxylamine, and acridine orange, indicating that this compound produces single-base substitutions, which include both G·C and A·T base pairs, and insertions and deletions.

The New Genetics
MINING GENOMES

Molecular Evolution

This exercise introduces you to some of the basic principles of molecular evolution, the study of the ways in which molecules evolve, and the reconstruction of the evolutionary history of molecules and organisms. You will use several of the Internet tools most frequently used by contemporary molecular geneticists to analyze analogous sequences from related organisms.

COMPREHENSION QUESTIONS

* 1. What is the difference between somatic mutations and germ-line mutations?

* 2. What is the difference between a transition and a transversion? Which type of base substitution is usually more common? Why?

* 3. Briefly describe expanding trinucleotide repeats. How do they account for the phenomenon of anticipation?

4. What is the difference between a missense mutation and a nonsense mutation? A silent mutation and a neutral mutation?

5. Briefly describe two different ways that intragenic suppressors may reverse the effects of mutations.

* 6. How do intergenic suppressors work?

* 7. What is the difference between mutation frequency and mutation rate?

* 8. What is the cause of errors in DNA replication?

9. How do insertions and deletions arise?

*10. How do base analogs lead to mutations?

11. How do alkylating agents, nitrous acid, and hydroxylamine produce mutations?

12. What types of mutations are produced by ionizing and UV radiation?

*13. What is the SOS system and how does it lead to an increase in mutations?

14. What is the purpose of the Ames test? How are *his*$^{-}$ bacteria used in this test?

*15. List at least three different types of DNA repair and briefly explain how each is carried out.

16. What features do mismatch repair, base-excision repair, and nucleotide-excision repair have in common?

APPLICATION QUESTIONS AND PROBLEMS

*17. A codon that specifies the amino acid Gly undergoes a single-base substitution to become a nonsense mutation. In accord with the genetic code given in Figure 15.12, is this mutation a transition or a transversion? At which position of the codon does the mutation occur?

*18. **(a)** If a single transition occurs in a codon that specifies Phe, what amino acids could be specified by the mutated sequence? **(b)** If a single transversion occurs in a codon that specifies Phe, what amino acids could be specified by the mutated sequence?

(**c**) If a single transition occurs in a codon that specifies Leu, what amino acids could be specified by the mutated sequence?

(**d**) If a single transversion occurs in a codon that specifies Leu, what amino acids could be specified by the mutated sequence?

19. Hemoglobin is a complex protein that contains four polypeptide chains. The normal hemoglobin found in adults—called adult hemoglobin—consists of two α and two β polypeptide chains, which are encoded by different loci. Sickle-cell hemoglobin, which causes sickle-cell anemia, arises from a mutation in the β chain of adult hemoglobin. Adult hemoglobin and sickle-cell hemoglobin differ in a single amino acid: the sixth amino acid from one end in adult hemoglobin is glutamic acid, whereas sickle-cell hemoglobin has valine at this position. After consulting the genetic code provided in Figure 15.12, indicate the type and location of the mutation that gave rise to sickle-cell anemia.

*20. The following nucleotide sequence is found on the template strand of DNA. First, determine the amino acids of the protein encoded by this sequence by using the genetic code provided in Figure 15.12. Then, give the altered amino acid sequence of the protein that will be found in each of the following mutations.

Sequence
of DNA
template
↳ 3′–TAC TGG CCG TTA GTT GAT ATA ACT–5′
↱ 1 24
Nucleotide
number

(**a**) Mutant 1: A transition at nucleotide 11.

(**b**) Mutant 2: A transition at nucleotide 13.

(**c**) Mutant 3: A one-nucleotide deletion at nucleotide 7.

(**d**) Mutant 4: A T→A transversion at nucleotide 15.

(**e**) Mutant 5: An addition of TGG after nucleotide 6.

(**f**) Mutant 6: A transition at nucleotide 9.

21. A polypeptide has the following amino acid sequence:

Met-Ser-Pro-Arg-Leu-Glu-Gly

The amino acid sequence of this polypeptide was determined in a series of mutants listed in parts *a* through *e*. For each mutant, indicate the type of change that occurred in the DNA (single-base substitution, insertion, deletion) and the phenotypic effect of the mutation (nonsense mutation, missense mutation, frameshift, etc.).

(**a**) Mutant 1: Met-Ser-Ser-Arg-Leu-Glu-Gly

(**b**) Mutant 2: Met-Ser-Pro

(**c**) Mutant 3: Met-Ser-Pro-Asp-Trp-Arg-Asp-Lys

(**d**) Mutant 4: Met-Ser-Pro-Glu-Gly

(**e**) Mutant 5: Met-Ser-Pro-Arg-Leu-Leu-Glu-Gly

*22. A gene encodes a protein with the following amino acid sequence:

Met-Trp-His-Val-Ala-Ser-Phe.

A mutation occurs in the gene. The mutant protein has the following amino acid sequence:

Met-Trp-His-Met-Ala-Ser-Phe.

An intragenic suppressor restores the amino acid sequence to that of the original the protein:

Met-Trp-His-Arg-Ala-Ser-Phe.

Give at least one example of base changes that could produce the original mutation and the intragenic suppressor? (Consult the genetic code in Figure 15.12.)

23. A gene encodes a protein with the following amino acid sequence:

Met-Lys-Ser-Pro-Ala-Thr-Pro

A nonsense mutation from a single-base-pair substitution occurs in this gene, resulting in a protein with the amino acid sequence Met-Lys. An intergenic suppressor mutation allows the gene to produce the full-length protein. With the original mutation and the intergenic suppressor present, the gene now produces a protein with the following amino acid sequence:

Met-Lys-Cys-Pro-Ala-Thr-Pro

Give the location and nature of the original mutation and the intergenic suppressor.

*24. Can nonsense mutations be reversed by hydroxylamine? Why or why not?

25. XG syndrome is a rare genetic disease that is due to an autosomal dominant gene. A complete census of a small European country reveals that 77,536 babies were born in 2000, of whom 3 had XG syndrome. In the same year, this country had a population of 5,964,321 people, and there were 35 living persons with XG syndrome. What are the mutation rate and mutation frequency of XG syndrome for this country?

*26. The following nucleotide sequence is found in a short stretch of DNA:

5′–ATGT–3′
3′–TACA–5′

If this sequence is treated with hydroxylamine, what sequences will result after replication?

27. The following nucleotide sequence is found in a short stretch of DNA:

5′–AG–3′
3′–TC–5′

(**a**) Give all the mutant sequences that may result from spontaneous depurination in this stretch of DNA.

(**b**) Give all the mutant sequences that may result from spontaneous deamination in this stretch of DNA.

28. In many eukaryotic organisms, a significant proportion of cytosine bases are naturally methylated to 5-methylcytosine. Through evolutionary time, the proportion of AT base pairs in the DNA of these organisms increases. Can you suggest a possible mechanism by which this increase occurs?

*29. A chemist synthesizes four new chemical compounds in the laboratory and names them PFI1, PFI2, PFI3, and PFI4. He gives the PFI compounds to a geneticist friend and asks her to determine their mutagenic potential. The geneticist finds that all four are highly mutagenic. She also tests the capacity of mutations produced by the PFI compounds to be reversed by other known mutagens and obtains the following results. What conclusions can you make about the nature of the mutations produced by these compounds?

	Reversed by			
Mutations produced by	**2-Amino-purine**	**Nitrous-acid**	**Hydroxy-lamine**	**Acridine orange**
PFI1	Yes	Yes	Some	No
PFI2	No	No	No	No
PFI3	Yes	Yes	No	No
PFI4	No	No	No	Yes

*30. A plant breeder wants to isolate mutants in tomatoes that are defective in DNA repair. However, this breeder does not have the expertise or equipment to study enzymes in DNA repair systems. How could the breeder identify tomato plants that are deficient in DNA repair? What are the traits to look for?

31. A genetics instructor designs a laboratory experiment to study the effects of UV radiation on mutation in bacteria. In the experiment, the students expose bacteria plated on petri plates to UV light for different lengths of time, place the plates in an incubator for 48 hours, and then count the number of colonies that appear on each plate. The plates that have received more UV radiation should have more pyrimidine dimers, which block replication; thus, fewer colonies should appear on the plates exposed to UV light for longer periods of time. Before the students carry out the experiment, the instructor warns them that, while the bacteria are in the incubator, the students must not open the incubator door unless the room is darkened. Why should the bacteria not be exposed to light?

CHALLENGE QUESTIONS

32. *Ochre* and *amber* are two types of nonsense mutations. Before the genetic code was worked out, Sydney Brenner, Anthony O. Stretton, and Samuel Kaplan applied different types of mutagens to bacteriophages in an attempt to determine the bases present in the codons responsible for *amber* and *ochre* mutations. They knew that *ochre* and *amber* mutants were suppressed by different types of mutations, demonstrating that each was a different termination codon. They obtained the following results.

(1) A single-base substitution could convert an *ochre* mutation into an *amber* mutation.

(2) Hydroxylamine induced both *ochre* and *amber* mutations in wild-type phages.

(3) 2-Aminopurine caused *ochre* to mutate to *amber*.

(4) Hydroxylamine did not cause *ochre* to mutate to *amber*.

These data do not allow the complete nucleotide sequence of the *amber* and *ochre* codons to be worked out, but they do provide some information about the bases found in the nonsense mutations.

(a) What conclusions about the bases found in the codons of *amber* and *ochre* mutations can be made from these observations?

(b) Of the three nonsense codons (UAA, UAG, UGA), which represents the *ochre* mutation.

33. To determine whether radiation associated with the atomic bombings of Hiroshima and Nagasaki produced recessive germ-line mutations, scientists examined the sex ratio of the children of the survivors of the blasts. Can you explain why an increase in germ-line mutations might be expected to alter the sex ratio?

34. The results of several studies provide evidence that DNA repair is rapid in genes that are undergoing transcription and that some proteins that play a role in transcription also participate in DNA repair. How are transcription and DNA repair related? Why might a gene that is being transcribed be repaired faster than a gene that is not being transcribed?

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18 Recombinant DNA Technology

Disembarkation of the Spanish at Veracruz by Mexican artist Diego Rivera. Anthropologists have suggested that Europeans first transmitted tuberculosis to the Native Americans. The polymerase chain reaction—a technique for amplifying very small amounts of DNA—has now demonstrated the presence of the bacterium that causes tuberculosis in a 1000 year old mummy from Peru, demonstrating that the disease was present in South America long before Europeans arrived. (Diego Rivera, *Disembarkation of the Spanish at Veracruz,* 1951. Schalkwijk/Art Resource.)

PCR and the Arrival of Tuberculosis in America

In the early 1600s, soon after Europeans arrived in the New World, devastating epidemics of tuberculosis ravaged many tribes of Native Americans. Anthropologists long argued that this disease was absent from the New World before 1492 and that Europeans first transmitted tuberculosis to the Native Americans. With no prior exposure to the disease and little natural immunity, the indigenous people would, it was argued, have been highly susceptible to tuberculosis.

A few anthropologists challenged this conventional view. On the basis of tuberculosis-like lesions found in a few skeletons and mummified remains that pre-date European contact, they suggested that the disease was present in Native Americans before European contact. On the other hand, many diseases and even bacteria that gain access to the body after death can produce similar marks, and the origin of tuberculosis in America remained controversial.

In 1994, pathologist Arthur C. Aufderheide and molecular biologist Wilmar Salo teamed up to resolve this controversy. Aufderheide obtained access to the remains of a woman who died and was naturally mummified in Peru about 1000 years ago, hundreds of years before Europeans arrived in South America. He removed samples of the woman's right lung and a lymph node and sent them to Salo, who used the newly developed polymerase chain reaction (PCR, which selectively amplifies sequences of DNA) to search the samples for DNA from *Mycobacterium tuberculosis*, the bacterium that causes tuberculosis. When applied to DNA from modern-day *Mycobacterium tuberculosis*, this technique produces copies of a 97-bp fragment of DNA. Salo detected an *identical* 97-bp piece of DNA by applying PCR to DNA samples from the ancient lung and lymph tissue, demonstrating unambiguously the presence of tuberculosis in the tissue of this 1000-year-old woman.

Although Europeans were not the *first* to transmit tuberculosis to Native Americans, they were still the most likely cause of the tuberculosis epidemics that accompanied their arrival in the Americas. European settlement was highly disruptive to many indigenous societies, often causing mass displacements of people and radically altering their traditional life styles. Stressful conditions, accompanied by crowding and malnutrition on reservations, probably lowered the resistance of many Native Americans and contributed to the spread of tuberculosis.

This story of the discovery of tuberculosis in the remains of a 1000-year-old woman illustrates the power of PCR, one of the techniques of molecular biology discussed in this chapter. We begin the chapter with a discussion of recombinant DNA technology and some of its effects. We then examine a number of methods used to isolate, study, alter, and recombine DNA sequences and place them back into cells. Finally, we explore some of the applications of recombinant DNA technology.

In reading this chapter, it will be helpful to understand two things. First, working at the molecular level is quite different from working with whole organisms: different approaches are needed, because the molecular objects of study cannot be seen directly. Second, there are a number of different approaches for isolating DNA sequences, amplifying them, and inserting them into bacteria, each approach with its own strengths and weaknesses. The optimal method depends on the starting materials, how much is known about the sequences to be isolated, and what the final objective is.

www.whfreeman.com/pierce More information about tuberculosis

Basic Concepts of Recombinant DNA Technology

In 1973, a group of scientists produced the first organisms with recombinant DNA molecules. Stanley Cohen at Stanford University and Herbert Boyer at the University of California School of Medicine at San Francisco and their colleagues inserted a piece of DNA from one plasmid into another, creating an entirely new, recombinant DNA molecule. They then introduced the recombinant plasmid into *E. coli* cells. Within a short time, they used the same methods to stitch together genes from two different types of bacteria, as well as to transfer genes from a frog to a bacterium. They called the hybrid DNA molecules *chimeras,* after the mythological Chimera, a creature with the head of a lion, the body of a goat, and the tail of a serpent. These experiments ushered in one of the most momentous revolutions in the history of science.

Recombinant DNA technology is a set of molecular techniques for locating, isolating, altering, and studying DNA segments. The term *recombinant* is used because frequently the goal is to combine DNA from two distinct sources. Genes from two different bacteria might be joined, for example, or a human gene might be inserted into a viral chromosome. Commonly called **genetic engineering,** recombinant DNA technology now encompasses an array of molecular techniques that can be used to analyze, alter, and recombine virtually any DNA sequences.

The Impact of Recombinant DNA Technology

Recombinant DNA technology has drastically altered the way that genes are studied. Previously, information about the structure and organization of genes was gained by examining their phenotypic effects, but the new technology makes it possible to read the nucleotide sequences themselves. Previously, geneticists had to wait for the appearance of random or induced mutations to analyze the effects of genetic differences; now they can create mutations at precisely defined spots and see how they alter the phenotype.

Recombinant DNA technology has provided new information about the structure and function of genes and has altered many fundamental concepts of genetics. For example, whereas the genetic code was once thought to be entirely universal, we now know that nonuniversal codons exist in mitochondrial DNA. Previously, we thought that the organization of eukaryotic genes was like that of prokaryotes, but we now know that many eukaryotic genes are interrupted by introns. Much of what we know today about replication, transcription, translation, RNA processing, and gene regulation has been learned through the use of recombinant DNA techniques. These techniques are also used in many other fields, including biochemistry, microbiology, developmental biology, neurobiology, evolution, and ecology.

Recombinant DNA technology is also used to create a number of commercial products, including drugs, hormones, enzymes, and crops (FIGURE 18.1). An entirely new industry—**biotechnology**—has grown up around the use of these techniques to develop new products. In medicine, recombinant DNA techniques are used to probe the nature

18.1 Recombinant DNA technology has been used to create genetically modified crops. Genetically engineered corn, which produces a toxin that kills insect pests, now comprises over 30% of all corn grown in the United States. (Chris Knapton/Photo Researchers.)

of cancer, diagnose genetic and infectious diseases, produce drugs, and treat hereditary disorders.

Concepts

Recombinant DNA technology is a set of methods used to locate, analyze, alter, study, and recombine DNA sequences. It is used to probe the structure and function of genes, address questions in many areas of biology, create commercial products, and diagnose and treat diseases.

www.whfreeman.com/pierce Information on genetic engineering and the biotechnology industry

Working at the Molecular Level

The manipulation of genes presents a serious challenge, often requiring strategies that may not, at first, seem obvious. The basic problem is that genes are minute and there are thousands of them in every cell. Even when viewed with the most powerful microscope, DNA appears as a tiny thread—individual nucleotides cannot be seen, and no physical features mark the beginning or the end of a gene.

To illustrate the problem, let's consider a typical situation faced by a molecular geneticist. Suppose we wanted to isolate a particular human gene, place it inside bacterial cells, and use the bacteria to produce large quantities of the encoded human protein. The first and most formidable problem is to find the desired gene. A haploid human genome consists of 3.3 billion base pairs of DNA. Let's assume that the gene that we want to isolate is 3000 bp long. Our target gene occupies only one-millionth of the genome; so searching for our gene in the huge expanse of genomic DNA is more difficult than looking for a needle in the proverbial haystack. But, even if we are able to locate the gene, how are we to separate it from the rest of the DNA? No forceps are small enough to pick up a single piece of DNA, and no mechanical scissors precise enough to snip out an individual gene.

If we did succeed in locating and isolating the desired gene, we would next need to insert it into a bacterial cell. Linear fragments of DNA are quickly degraded by bacteria; so the gene must be inserted in a stable form. It must also be able to successfully replicate or it will not be passed on when the cell divides.

If we succeed in transferring our gene to bacteria in a stable form, we still must ensure that the gene is properly transcribed and translated. Gene expression is a complex process requiring a number of DNA sequence elements, some of which lie outside the gene itself (Chapters 13 through 16). All of these elements must be present in their proper orientations and positions for the protein to be produced.

Finally, the methods used to isolate and transfer genes are inefficient and, of a million cells that are subjected to these procedures, only *one* cell might successfully take up and express the human gene. So we must search through many bacterial cells to find the one containing the recombinant DNA. We are back to the problem of the needle in the haystack.

Although these problems might seem insurmountable, molecular techniques have been developed to overcome all of them, and human genes are routinely transferred to bacterial cells, where the genes are expressed.

Concepts

Recombinant DNA technology requires special methods because individual genes make up a tiny fraction of the cellular DNA and they cannot be seen.

Recombinant DNA Techniques

In the sections that follow, we will examine some of the following techniques of recombinant DNA technology and see how they are used to create recombinant DNA molecules:

1. Methods for locating specific DNA sequences
2. Techniques for cutting DNA at precise locations
3. Procedures for amplifying a particular DNA sequence billions of times, producing enough copies of a DNA sequence to carry out further manipulations
4. Methods for mutating and joining DNA fragments to produce desired sequences
5. Procedures for transferring DNA sequences into recipient cells

Cutting and Joining DNA Fragments

The key development that made recombinant DNA technology possible was the discovery in the late 1960s of **restriction enzymes** (also called **restriction endonucleases**) that recognize and make double-stranded cuts in the sugar–phosphate backbone of DNA molecules at specific nucleotide sequences. These enzymes are produced naturally by bacteria, where they are used in defense against viruses. In bacteria, restriction enzymes recognize particular sequences in viral DNA and then cut it up. A bacterium protects its own DNA from a restriction enzyme by modifying the recognition sequence, usually by adding methyl groups to its DNA.

Three types of restriction enzymes have been isolated from bacteria (Table 18.1). Type I restriction enzymes recognize specific sequences in the DNA but cut the DNA at random sites that may be some distance (1000 bp or more) from the recognition sequence. Type III restriction enzymes recognize specific sequences and cut the DNA at nearby sites, usually about 25 bp away. Type II restriction enzymes recognize specific sequences and cut the DNA within the recognition sequence. Virtually all work on recombinant DNA is done with type II restriction enzymes; discussions of restriction enzymes throughout this book, refers to type II enzymes.

More than 800 different restriction enzymes that recognize and cut DNA at more than 100 different sequences have been isolated from bacteria. Many of these enzymes are commercially available; examples of some commonly used restriction enzymes are given in Table 18.2. Each restriction enzyme is referred to by a short abbreviation that signifies its bacterial origin.

The sequences recognized by restriction enzymes are usually from 4 to 8 bp long; most enzymes recognize a sequence of 4 or 6 bp. Most recognition sequences are palindromic—sequences that read the same forward and backward. Notice in Table 18.2 that the sequence on the bottom strand is the same as the sequence on the top strand, only reversed. All type II restriction enzymes recognize palindromic sequences.

Some of the enzymes make staggered cuts in the DNA. For example, *Hin*dIII recognizes the following sequence:

```
  ↓
5′–AAGCTT–3′
3′–TTCGAA–5′
        ↑
```

*Hin*dIII cuts the sugar–phosphate backbone of each strand at the point indicated by the arrow, generating fragments with short, single-stranded overhanging ends:

```
5′–A        AGCTT–3′
3′–TTCGA        A–5′
```

Such ends are called **cohesive ends** or sticky ends, because they are complementary to each other and can spontaneously pair to connect the fragments. Thus DNA fragments can be "glued" together: any two fragments cleaved by the same enzyme will have complementary ends and will pair (Figure 18.2). When their cohesive ends have paired, two DNA fragments can be joined together permanently by the enzyme DNA ligase, which seals nicks between the sugar–phosphate groups of the fragments.

Not all restriction enzymes produce staggered cuts and sticky ends. *Pvu*II cuts in the middle of its recognition site, producing blunt-ended fragments:

```
      ↓
5′–CAGCTG–3′
3′–GTCGAC–5′
      ↑
      |
      ↓
5′–CAG CTG–3′
3′–GTC GAC–5′
```

Fragments with blunt ends must be joined together in other ways, which will be discussed later.

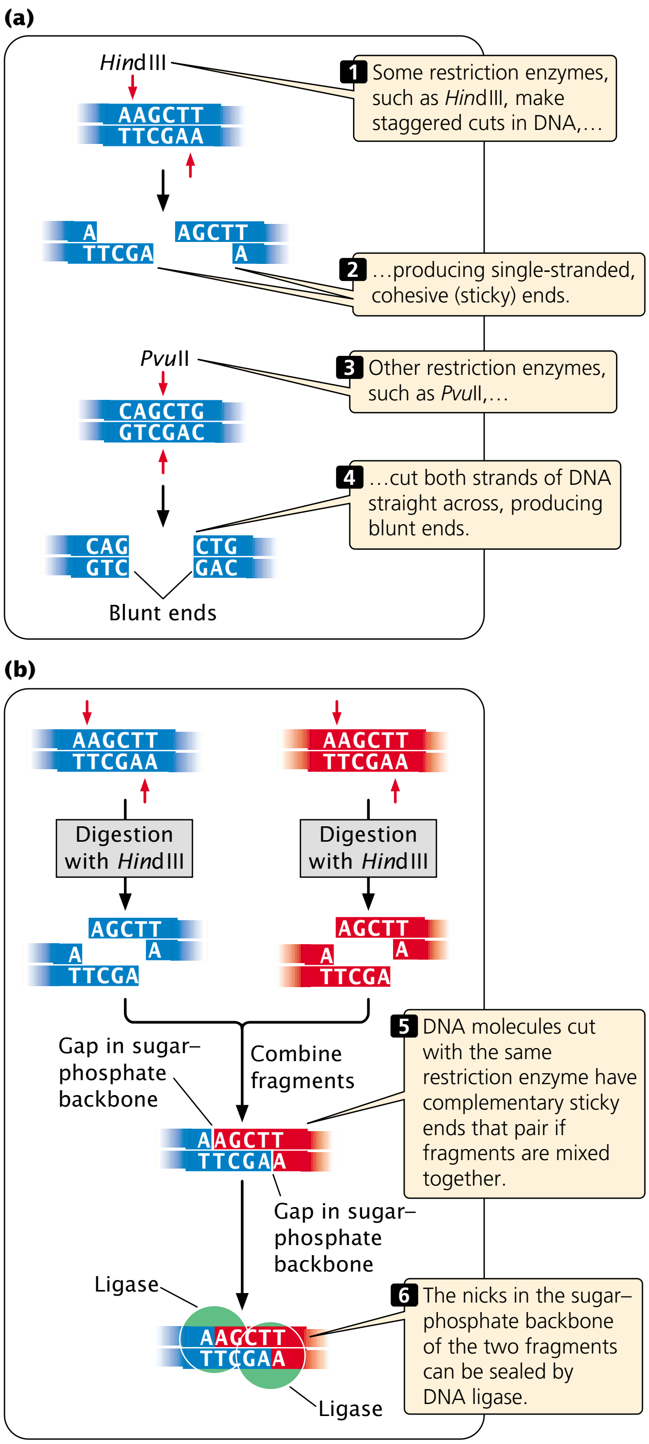
Table 18.1 Types of restriction enzymes

Type	Activity of Enzyme	ATP Required	Cleavage Site
I	Cleavage and methylation	Yes	Random sites distant from recognition site
II	Cleavage only	No	Within recognition site
III	Cleavage and methylation	Yes	Random sites near recognition site

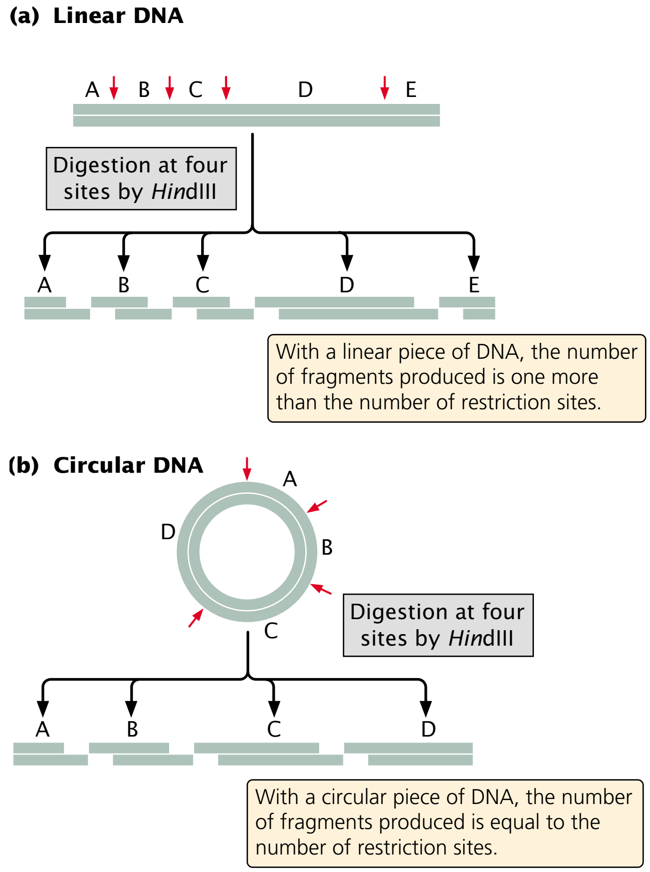
Table 18.2 Characteristics of some common type II restriction enzymes used in recombinant DNA technology

Enzyme	Microorganism from Which Enzyme Is Isolated	Recognition Sequence	Type of Fragment End Produced
*Bam*HI	*Bacillus amyloliquefaciens*	5′–G↓GATCC–3′ 3′–CCTAG↑G–3′	Cohesive
*Cof*I	*Clostridium formicoaceticum*	5′–GCG↓C–3′ 3′–C↑GCG–5′	Cohesive
*Dra*I	*Deinococcus radiophilus*	5′–TTT↓AAA–3′ 3′–AAA↑TTT–5′	Blunt
*Eco*RI	*Escherichia coli*	5′–G↓AATTC–3′ 3′–CTTAA↑G–5′	Cohesive
*Eco*RII	*Escherichia coli*	5′–↓CCAGG–3′ 3′–GGTCC↑–5′	Cohesive
*Hae*III	*Haemophilus aegyptius*	5′–GG↓CC–3′ 3′–CC↑GG–5′	Blunt
*Hind*III	*Haemophilus influenczae*	5′–A↓AGCTT–3′ 3′–TTCGA↑A–5′	Cohesive
*Hpa*II	*Haemophilus parainfluenzae*	5′–C↓CGG–3′ 3′–GGC↑C–5′	Cohesive
*Not*I	*Nocardia otitidis-caviarum*	5′–GC↓GGCCGC–3′ 3′–CGCCGG↑CG–5′	Cohesive
*Pst*I	*Providencia stuartii*	5′–CTGCA↓G–3′ 3′–G↑ACGTC–5′	Cohesive
*Pvu*II	*Proteus vulgaris*	5′–CAG↓CTG–3′ 3′–GTC↑GAC–5′	Blunt
*Sma*I	*Serratia marcescens*	5′–CCC↓GGG–3′ 3′–GGG↑CCC–5′	Blunt

Note: The first three letters of the abbreviation for each restriction enzyme refer to the bacterial species from which the enzyme was isolated (e.g., *Eco* refers to *E. coli*). A fourth letter may refer to the strain of bacteria from which the enzyme was isolated (the "R" in *Eco*RI indicates that this enzyme was isolated from the RY13 strain of *E. coli*). Roman numerals that follow the letters allow different enzymes from the same species to be identified. For convenience, molecular geneticists have come up with idiosyncratic pronunciations of the names: *Eco*RI is pronounced "echo-R-one," *Hind*III is "hin-D-three," and *Hae*III is "hay-three." These common pronunciations obey no formal rules and simply have to be learned.



18.2 Restriction enzymes make double-stranded cuts in the sugar–phosphate backbone of DNA, producing cohesive, or sticky, ends.



18.3 The number of restriction sites is related to the number of fragments produced when DNA is cut by a restriction enzyme.

The sequences recognized by a restriction enzyme occur randomly within genomic DNA. Consequently, there is a relation between the length of the recognition sequence and its frequency of occurrence: there are fewer long recognition sequences than short sequences because the probability of all the bases being in the required order is less.

Restriction enzymes are the workhorses of recombinant DNA technology and are used whenever DNA fragments must be cut or joined. In a typical restriction reaction, a concentrated solution of purified DNA is placed in a small tube with a buffer solution and a small amount of restriction enzyme. The reaction mixture is then heated at the optimal temperature for the enzyme, usually 37°C. Within a few hours, the enzyme cuts all the restriction sites in the DNA, producing a set of DNA fragments (FIGURE 18.3).

Concepts

Type II restriction enzymes cut DNA at specific base sequences. Some restriction enzymes make staggered cuts, producing DNA fragments with cohesive ends; others cut both strands straight across, producing blunt-ended fragments. There are fewer long recognition sequences in DNA than short sequences.

www.whfreeman.com/pierce Information on specific restriction enzymes

Viewing DNA Fragments

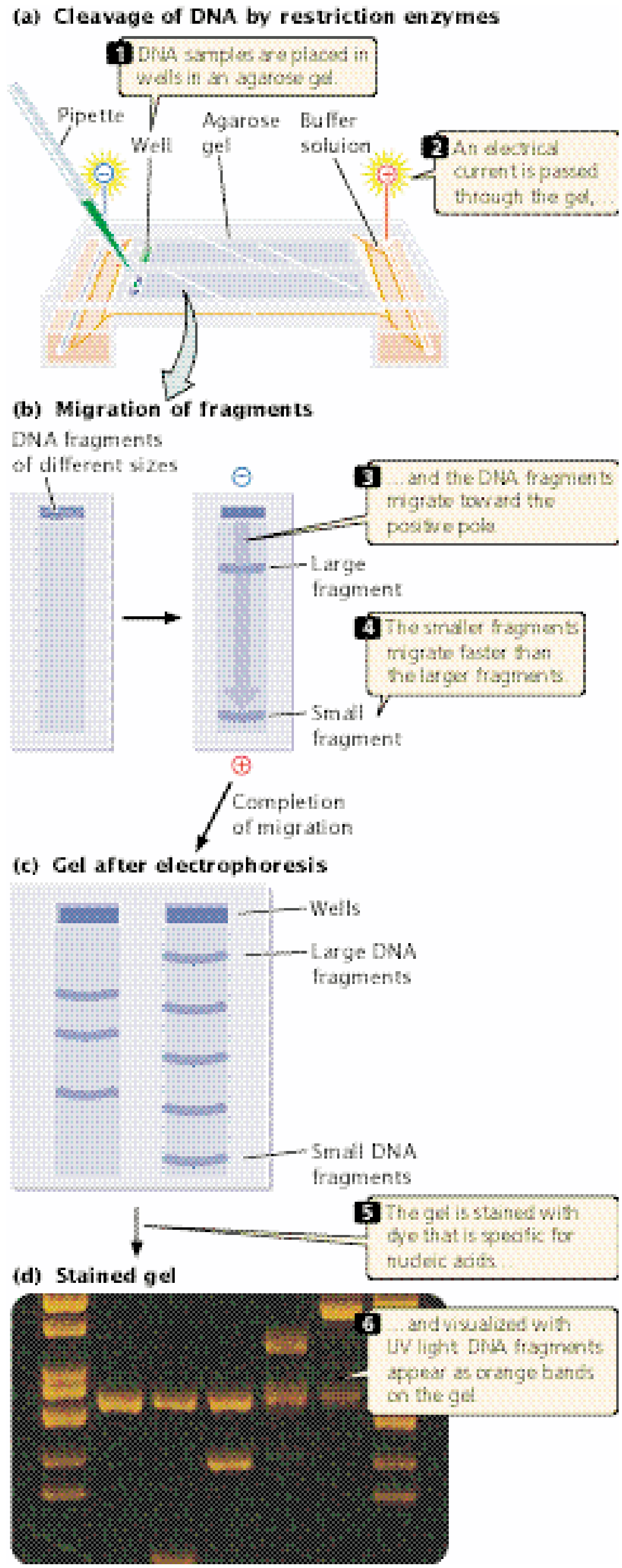
After the completion of a restriction reaction, a number of questions arise. Did the restriction enzyme cut the DNA? How many times was the DNA cut? What are the sizes of the resulting fragments? Gel electrophoresis provides us with a means of answering these questions.

Electrophoresis is a standard biochemical technique for separating molecules on the basis of their size and electrical charge. There are a number of different types of electrophoresis; to separate DNA molecules, **gel electrophoresis** is used. A porous gel is often made from agarose (a polysaccharide isolated from seaweed), which is melted in a buffer solution and poured into a plastic mold. As it cools, the agarose solidifies, making a gel that looks something like stiff gelatin.

Small indentions called wells are made at one end of the gel to hold solutions of DNA fragments (FIGURE 18.4a), and an electrical current is passed through the gel. Because the phosphate of each DNA nucleotide carries a negative charge, the DNA fragments migrate toward the positive end of the gel (FIGURE 18.4b). In this migration, the gel acts as a sieve; as the DNA molecules migrate toward the positive pole, they move through the pores between the gel particles. Small DNA fragments migrate more rapidly than do large ones and, with time, the fragments separate on the basis of their size. The distance that each fragment migrates depends on its size. Typically, DNA fragments of known length (a marker sample) are placed in another well. By comparing the migration distance of the unknown fragments with the distance traveled by the marker fragments, one can determine the approximate size of the unknown fragments.

After electrophoresis, the DNA fragments are separated according to size (FIGURE 18.4c). However, the DNA fragments are still too small to see; so the problem of visualizing the DNA needs to be addressed. Visualization can be accomplished in several ways. The simplest procedure is to stain the gel with a dye specific for nucleic acids, such as ethidium bromide, which wedges itself tightly (intercalates) between the bases of DNA. When exposed to UV light, ethidium bromide fluoresces bright orange; so copies of each DNA fragment appear as a brilliant orange band (FIGURE 18.4d). The original concentrated sample of purified DNA contained millions of copies of a DNA molecule, and thus each band represents millions of copies of identical DNA fragments.

Alternatively, DNA fragments can be visualized by adding a radioactive or chemical label to the DNA before it is placed in the gel. Nucleotides with radioactively labeled phosphate (^{32}P) can be used as the substrate for DNA synthesis and will be incorporated into the newly synthesized DNA strand. In another method called **end labeling,** the bacteriophage enzyme polynucleotide kinase is used to



18.4 **Gel electrophoresis can be used to separate DNA molecules on the basis of their size and electrical charge.** (Photo courtesy of Carol Eng.)

transfer a single ^{32}P to the 5′ end of each DNA strand. Radioactively labeled DNA can be detected with a technique called **autoradiography** (see Figure 10.4), in which a piece of X-ray film is placed on top of the gel. Radiation from the labeled DNA exposes the film, just as light exposes photographic film in a camera. The developed autoradiograph gives a picture of the fragments in the gel; each DNA fragment appears as a dark band on the film. Chemical labels can be detected by adding antibodies or other substances that carry a dye and will attach to the relevant DNA, which can be visualized directly.

Gel electrophoresis is used widely in recombinant DNA technology; it is often employed when there is a need to determine the number or size of DNA fragments or to isolate DNA fragments by size. For example, to determine the number and location of *Bam*HI restriction sites in a plasmid, we might cut the plasmid by using the *Bam*HI restriction enzyme and place the products of the restriction reaction in a well of an agarose gel. In another well of the same gel, we would place a set of control fragments of known size. After applying an electrical current to the gel for an hour or more, we would stain the gel with ethidium bromide and place it over a UV light. The appearance of three orange bands on the gel would indicate that the circular plasmid had been cut three times and that there are three *Bam*HI restriction sites in the plasmid. A comparison of the migration distance of the plasmid fragments with the migration distance of the standard fragments would reveal the sizes of the fragments and the distances between the *Bam*HI recognition sites.

Concepts

DNA fragments can be separated, and their sizes can be determined with the use of gel electrophoresis. The fragments can be viewed by using a dye that is specific for nucleic acids or by labeling the fragments with a radioactive or chemical tag.

Locating DNA Fragments with Southern Blotting and Probes

If a relatively small piece of DNA, such as a plasmid, is cut by a restriction enzyme, the few fragments produced can be seen as distinct bands on an electrophoretic gel. In contrast, if genomic DNA from a cell is cut by a restriction enzyme, a large number of fragments of different sizes are produced. A restriction enzyme that recognizes a four-base sequence would theoretically cut about once every 256 bp. The human genome, with 3.3 billion base pairs, would generate more than 12 million fragments when cut by this restriction enzyme. Separated by electrophoresis and stained, this large set of fragments would appear as a continuous smear on the gel because of the presence of so many fragments of differing size. Usually, one is interested in only a few of these fragments, perhaps those carrying a specific gene. How does one locate the desired fragments in such a large pool of DNA?

One approach is to use a **probe,** which is a DNA or RNA molecule with a base sequence complementary to a sequence in the gene of interest. The bases on a probe will pair only with the bases on a complementary sequence and, if suitably tagged with an identifying label, the probe can be used to locate a specific gene or other DNA sequence.

To use a probe, one first cuts the DNA into fragments by using one or more restriction enzymes and then separates the fragments with gel electrophoresis (FIGURE 18.5). Next, the separated fragments must be denatured and transferred to a thinner solid medium (such as nitrocellulose or nylon membrane) to prevent diffusion. **Southern blotting** (named after Edwin M. Southern) is one technique used to transfer the denatured, single-stranded fragments from a gel to a thin solid medium.

After the single-stranded DNA fragments have been transferred, the membrane is placed in a hybridization solution of a radioactively or chemically labeled probe (see Figure 18.5). The probe will bind to any DNA fragments on the membrane that bear complementary sequences. The membrane is then washed to remove any unbound probe; bound probe is detected by autoradiography or another method for chemically labeled probes.

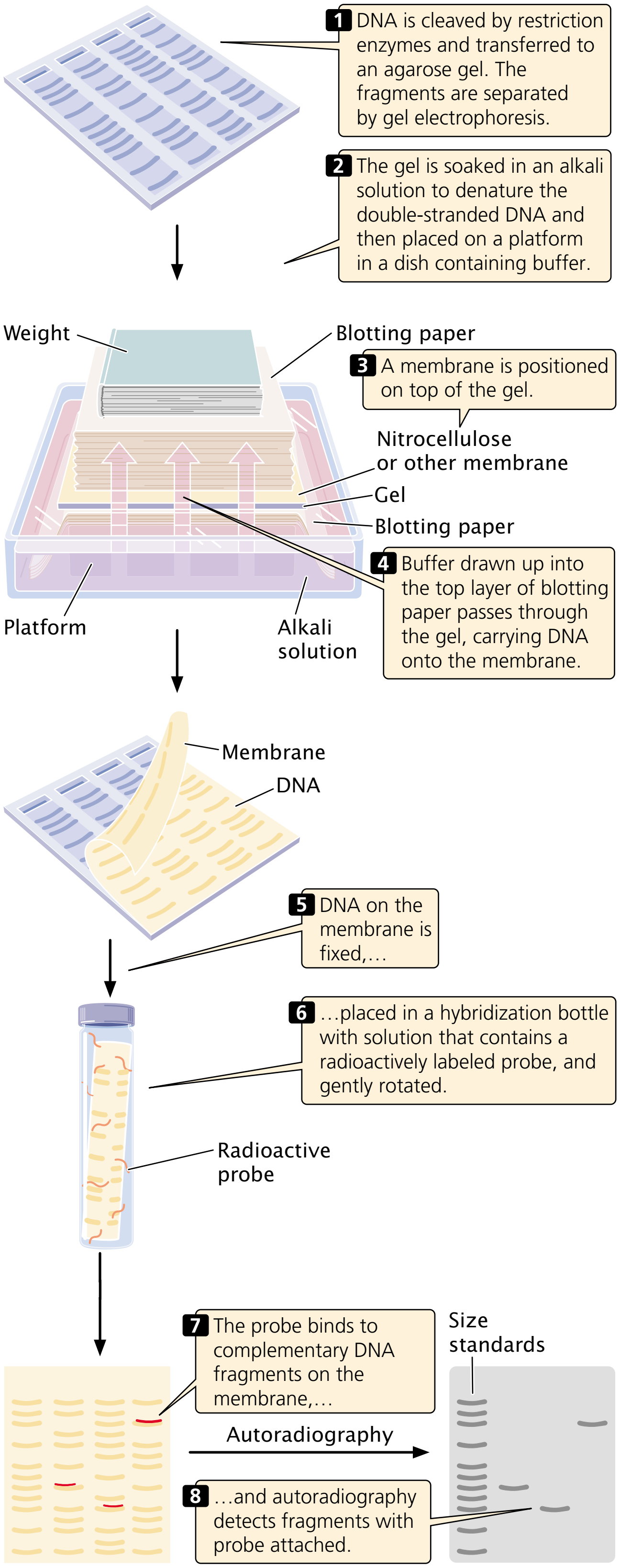
RNA can be transferred from a gel to a solid support by a related procedure called **Northern blotting** (not named after anyone but capitalized to match Southern). The hybridization of a probe can reveal the size of a particular mRNA molecule, its relative abundance, or the tissues in which the mRNA is transcribed. Here, the probe is usually an antibody, used to determine the size of a particular protein and the pattern of the protein's expression. **Western blotting** is the transfer of protein from a gel to a membrane.

Concepts

Labeled probes, which are sequences of RNA or DNA that are complementary to the sequence of interest, can be used to locate individual genes or DNA sequences. Southern blotting can be used to transfer DNA fragments from a gel to a membrane such as nitrocellulose.

Cloning Genes

Many recombinant DNA methods require numerous copies of a specific DNA fragment. One way to obtain these copies is to place the fragment in a bacterial cell and allow the cell to replicate the DNA. This procedure is termed **gene cloning,** because identical copies (clones) of the original piece of DNA are produced.



18.5 Southern blotting and hybridization with probes can be used to locate a few specific fragments in a large pool of DNA.

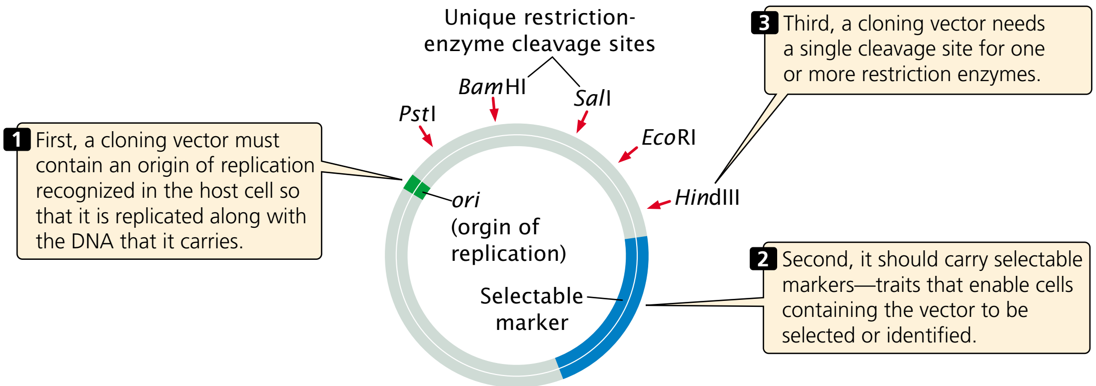
Cloning vectors A **cloning vector** is a stable, replicating DNA molecule to which a foreign DNA fragment can be attached for introduction into a cell. An effective cloning vector has three important characteristics (FIGURE 18.6): (1) an origin of replication, which ensures that the vector is replicated within the cell; (2) selectable markers, which enable any cells containing the vector to be selected or identified; (3) one or more unique restriction sites into which a DNA fragment can be inserted. The restriction sites used for cloning must be unique; if a vector is cut at multiple recognition sites, generating several pieces of DNA, there will be no way to get the pieces back together in the correct order. Three types of cloning vectors are commonly used for cloning genes in bacteria: plasmids, bacteriophages, and cosmids.

Plasmid vectors Plasmids are circular DNA molecules that exist naturally in bacteria (see p. 000 in Chapter 15). They contain origins of replication and are therefore able to replicate independently of the bacterial chromosome. The plasmids typically used in cloning have been constructed from the larger, naturally occurring bacterial plasmids.

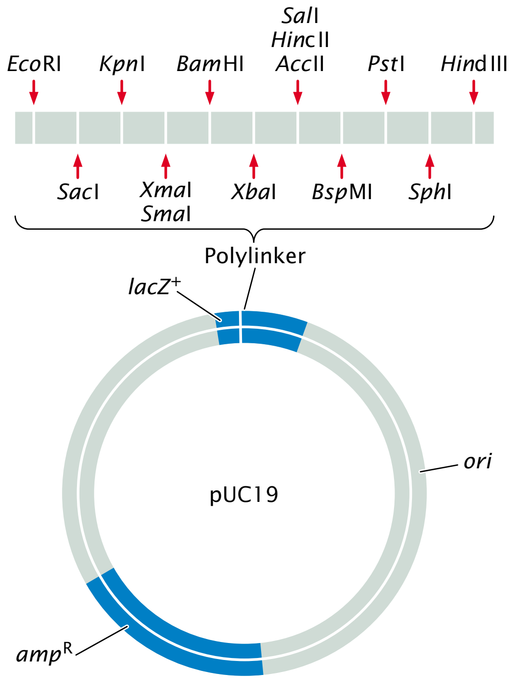
The pUC19 plasmid is a typical cloning vector (FIGURE 18.7). It has an origin of replication and two selectable markers—an ampicillin-resistance gene and a *lacZ* gene. Ampicillin is an antibiotic that normally kills bacterial cells, but any bacterium that contains a pUC19 plasmid will be resistant to this antibiotic. The *lacZ* gene encodes the enzyme β-galactosidase, which normally cleaves lactose to produce glucose and galactose (see p. 000 in Chapter 16). The enzyme will also cleave a chemical called X-gal, producing a blue substance; when X-gal is placed in the medium, any bacterial colonies that contain intact pUC19 plasmids will turn blue and can be easily identified. (In these experiments, the bacterium's own β-galactosidase gene has been inactivated, and so only bacteria with the plasmid turn blue.) The pUC19 plasmid also possesses a number of different unique restrictions sites grouped together (a polylinker) that allow DNA fragments to be inserted into the plasmid.

The easiest method for inserting a foreign DNA fragment into a plasmid is to use restriction cloning, in which the foreign DNA and the plasmid are cut by the same restriction enzyme. Restriction cloning produces complementary sticky ends on the foreign DNA and the plasmid (FIGURE 18.8a). The DNA and plasmid are then mixed together; some of the foreign DNA fragments will pair with the cut ends of the plasmid. DNA ligase seals the nicks in the sugar–phosphate backbone, creating a recombinant plasmid that contains the foreign DNA fragment.

Although simple, restriction cloning has several disadvantages. First, restriction cloning requires that a single restriction site in the plasmid matches sites on both ends of the foreign sequence to be cloned. If this arrangement of restriction sites is not available, this relatively straightfor-



18.6 Three characteristics of an idealized cloning vector. An origin of replication, one or more selectable markers, and one or more unique restriction sites.



18.7 The pUC19 plasmid is a typical cloning vector. It contains a cluster of unique restriction sites, an origin of replication, and two selectable markers—an ampicillin-resistance gene and a *lacZ* gene.

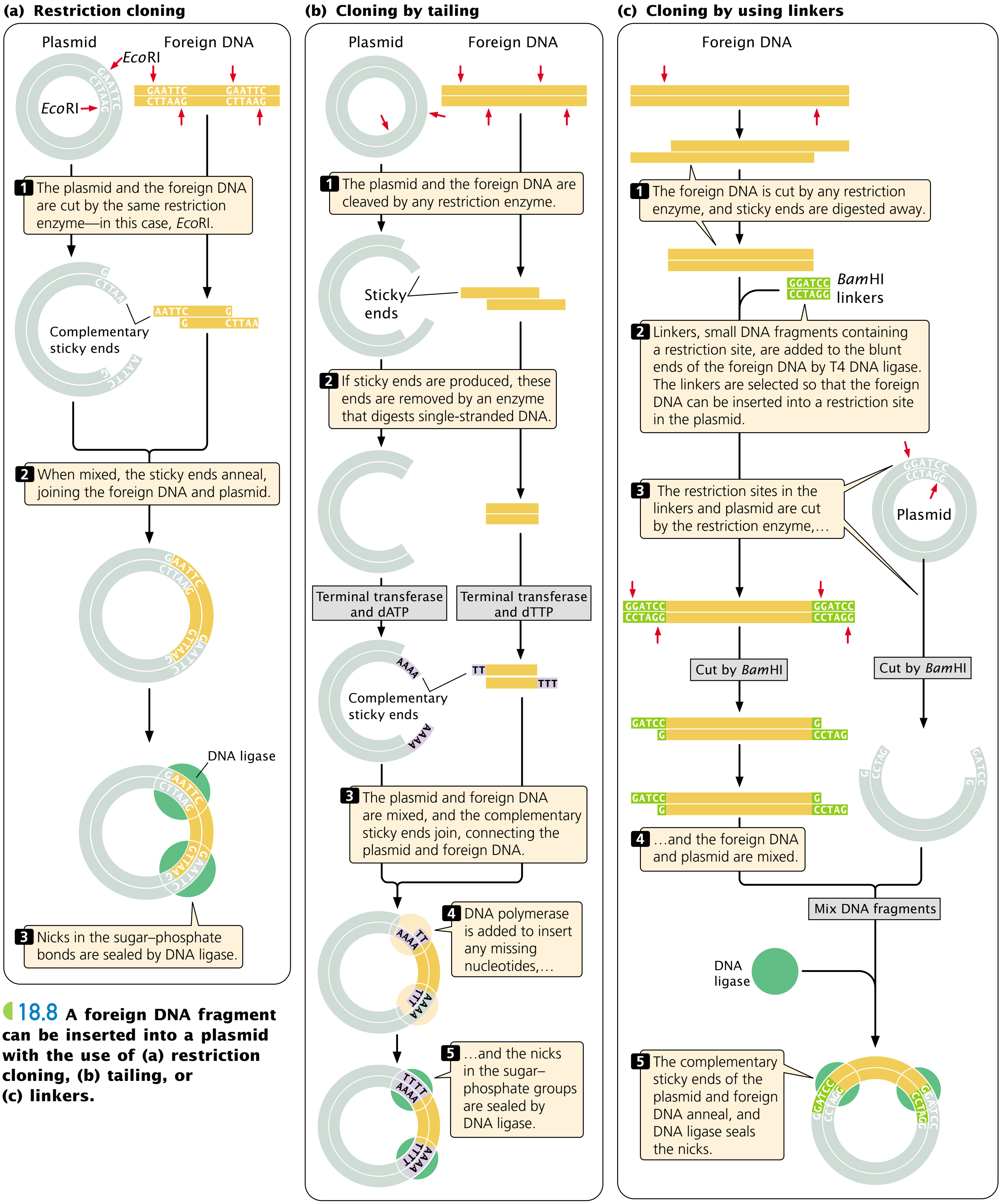
ward method cannot be used. Second, this technique often leads to undesirable products. The sticky ends of the plasmid are complementary to each other; so the two ends of the plasmid will often simply reanneal, reproducing the intact plasmid. Alternatively, the two complementary ends of the cleaved foreign DNA may anneal; or several pieces of foreign DNA or several plasmids may join. However, these undesirable products do not constitute a serious problem if an efficient method is used for screening bacterial cells for the presence of a recombinant plasmid.

Another method for inserting DNA into a plasmid (a method that gets around the problem of undesired products) is tailing (FIGURE 18.8b). In this procedure, complementary sticky ends are created on blunt-ended pieces of DNA. The plasmid and the foreign DNA are first cut by any restriction enzyme. If the restriction enzyme produces sticky ends, these ends are removed by an enzyme that digests single-stranded DNA. Alternatively, the plasmid and foreign DNA can be cut by a restriction enzyme that produces blunt ends.

Once the plasmid and the foreign DNA have blunt ends, single-stranded sticky ends are added by an enzyme called terminal transferase, which adds any available nucleotides to the 3′ end of DNA in a template-independent reaction. For example, terminal transferase and deoxyadenosine triphosphate (dATP) might be mixed with the plasmid DNA, creating poly(A) single-stranded tails on the 3′ ends of the plasmid. Terminal transferase and deoxythymidine triphosphate (dTTP) could be mixed with the blunt-ended foreign DNA fragments, creating poly(T) single-stranded tails on their 3′ ends. The poly(A) tail of the plasmid would be complementary to the poly(T) tail of the foreign DNA, allowing them to anneal and connecting the plasmid and foreign DNA together. DNA polymerase can be used to fill in any missing nucleotides, and DNA ligase can be used to seal the nicks in the sugar–phosphate backbone.

One advantage of tailing is that it prevents the production of the undesired products created by restriction cloning: the single-stranded ends of the plasmid are complementary only to the single-stranded ends of the foreign DNA. Another advantage is that identical restriction sites are not required in plasmid and foreign DNA; any restriction site can be used for cleavage. But tailing has several disadvantages of its own. First, it destroys the restriction site used to cut the original molecule, preventing later cleavage by the same restriction enzyme to retrieve the foreign DNA. Second, the new nucleotides (the complementary tails) introduced at the junctions between plasmid and foreign DNA sometimes interfere with the function of the cloned DNA.

A third method of inserting fragments into plasmids is to use the enzyme T4 ligase, which is capable of connecting any two pieces of blunt-ended DNA. Like tailing, this



18.8 A foreign DNA fragment can be inserted into a plasmid with the use of (a) restriction cloning, (b) tailing, or (c) linkers.

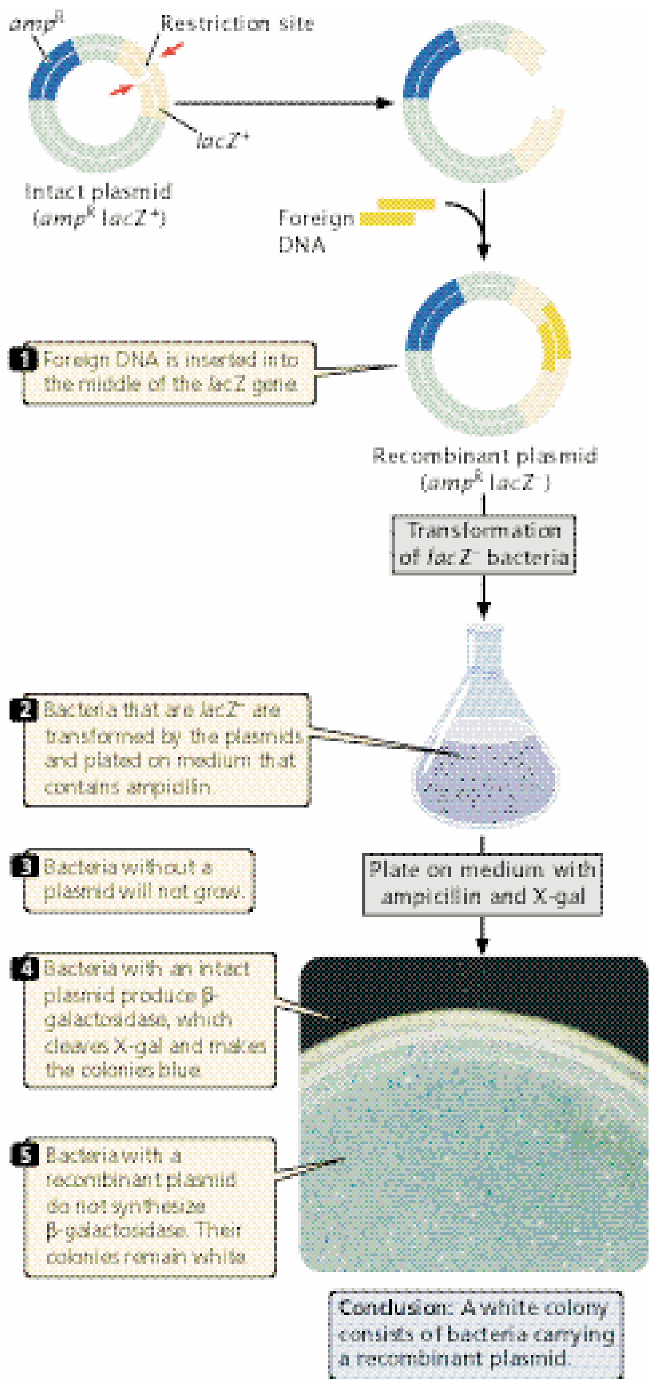
method requires no specific restriction sites and has great versatility; its chief drawback is that it creates a number of undesired products.

A fourth method, and one commonly used today, is the use of linkers to add complementary ends to DNA molecules (FIGURE 18.8c). Linkers are small, synthetic DNA fragments that contain one or more restriction sites. The foreign DNA of interest is cut by any restriction enzyme; if sticky ends are created, they are digested to produce blunt ends. The linkers are then attached to the blunt ends by T4 ligase and are then cut by a restriction enzyme, generating sticky ends that are complementary to sticky ends on the plasmid, which have been generated by using the same restriction enzyme to cut the plasmid. Mixing the plasmid and foreign DNA leads to the formation of recombinant DNA that can be stabilized by ligase. The great advantage of using linkers is that a particular restriction site can be added at almost any desired location; so any two pieces of DNA can be cut and joined.

Transformation When a gene has been placed inside a plasmid, the plasmid must be introduced into bacterial cells. This task is usually accomplished by *transformation,* which is the capacity of bacterial cells to take up DNA from the external environment (see Chapter 8). Some types of cells undergo transformation naturally; others must be treated chemically or physically before they will undergo transformation. Inside the cell, the plasmids replicate and multiply.

The use of selective markers Cells bearing recombinant plasmids can be detected by using the selectable markers on the plasmid. One type of selectable marker commonly used with plasmids is a copy of the *lacZ* gene (FIGURE 18.9). The *lacZ* gene contains a series of unique restriction sites into which may be inserted a fragment of DNA to be cloned. In the absence of an inserted fragment, the *lacZ* gene is active and produces β-galactosidase. When foreign DNA is inserted into the restriction site, it disrupts the *lacZ* gene, and β-galactosidase is not produced. The plasmid also usually contains a second selectable marker, which may be a gene that confers resistance to an antibiotic such as ampicillin.

Bacteria that are *lacZ*$^-$ are transformed by the plasmids and plated on medium that contains ampicillin. Only cells that have been successfully transformed and contain a plasmid with the ampicillin-resistance gene will survive and grow. Some of these cells will contain an intact plasmid, whereas others possess a recombinant plasmid. The medium also contains the chemical X-gal. Bacterial cells with an intact original plasmid—without an inserted fragment—have a functional *lacZ* gene and can synthesize β-galactosidase, which cleaves X-gal and turns the bacteria blue. Bacterial cells with a recombinant plasmid, however, have a β-galactosidase gene that is disrupted by the inserted DNA; they do not synthesize β-galactosidase and remain white. Thus, the color of the colony allows quick determination of whether a recombinant or intact plasmid is present in the cell.



18.9 **The *lacZ* gene can be used to screen bacteria containing recombinant plasmids.** A special plasmid carries a copy of the *lacZ* gene and an ampicillin resistance gene. (Photo: Cytographics/Visuals Unlimited.)

Plasmids make ideal cloning vectors but can hold only DNA less than about 15 kb in size. When large DNA fragments are inserted into a plasmid vector, the plasmid becomes unstable. Cloning DNA fragments that are longer than 15 kb requires the use of different cloning vectors.

Concepts

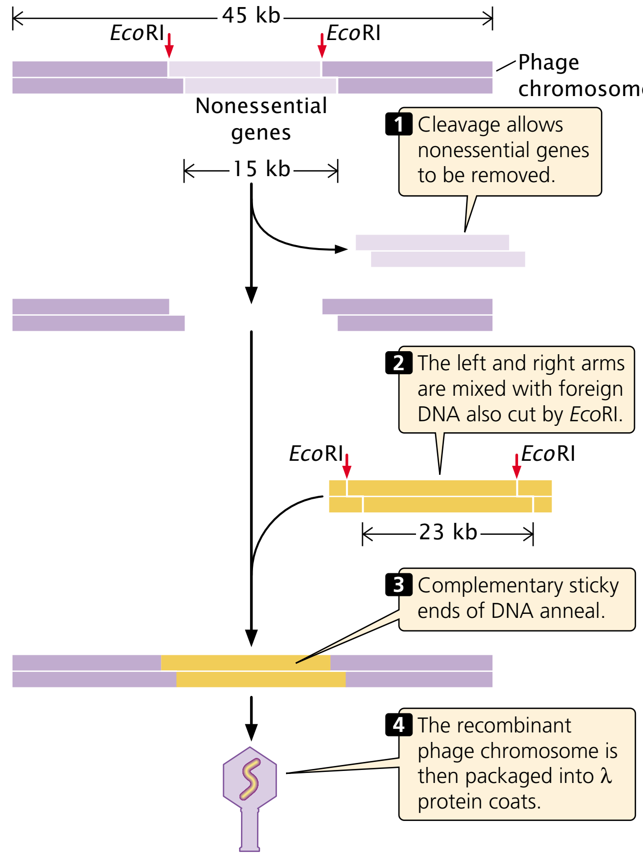
DNA fragments can be inserted into cloning vectors, stable pieces of DNA that will replicate within a cell. Cloning vectors must have an origin of replication, one or more unique restriction sites, and selectable markers. Plasmids are commonly used as cloning vectors.

Bacteriophage vectors Bacteriophages offer a number of advantages as cloning vectors. The most widely used bacteriophage vector is bacteriophage λ, which infects *E. coli.* One of its chief advantages is the high efficiency with which it transfers DNA into bacteria cells. A second advantage is that about a third of the λ genome is not essential for infection and reproduction; without these genes, a λ particle will still faithfully inject its DNA into a bacterial cell and reproduce. These nonessential genes, which comprise about 15 kb, can be replaced by as much as about 23 kb of foreign DNA. A third advantage is that DNA will not be packaged into a λ coat unless it is 40 to 50 kb long; so fragments of foreign DNA are not likely to be transferred by the vector unless they are inserted into the λ genome, which ensures that the foreign DNA fragment will be replicated after it enters the cell.

The essential genes of the phage λ genome are located in a cluster. Strains of phage λ, called replacement vectors, have been engineered with unique *Eco*RI sites on either side of the nonessential genes (FIGURE 18.10) so that, by using *Eco*RI, the nonessential genes can be removed. Foreign DNA cut with *Eco*RI will have sticky ends that are complementary to those on the ends of the essential λ genes, to which it can be connected by ligase. The λ chromosome possesses short, single-stranded ends called *cos* sites that are required for packaging λ DNA into a phage head. The recombinant phage chromosomes can then be packaged into protein coats and added to *E. coli*. The phages inject their recombinant DNA into the cell, where it will be replicated. Only DNA fragments of the proper size and containing essential genes will be packaged into the phage coats, providing an automatic selection system for recombinant vectors.

Cosmid vectors Although only about 23 kb of DNA can be cloned in λ vectors, DNA fragments as large as about 44 kb can be cloned in cosmids, which combine the properties of plasmids and phage vectors.

Cosmids are small plasmids that carry phage λ *cos* sites; they can be packaged into viral coats and transferred to bacteria by viral infection. Because all viral genes except the *cos* sites are missing, a cosmid can carry more than twice as much foreign DNA as can a phage vector. Cosmid vectors



18.10 **Phage λ is an effective cloning vector.**

have the following components: (1) a plasmid origin of replication (*ori*); (2) one or more unique restriction sites; (3) one or more selectable markers; and (4) *cos* sites to allow the packaging of DNA into phage heads.

Foreign DNA is inserted into cosmids in the same way that DNA is introduced into plasmids: the cosmid and foreign DNA are both cut by a restriction enzyme that produces complementary (sticky) ends, and they are joined by DNA ligase. Recombinant cosmids are incorporated into the coats, and the phage particles are used to infect bacterial cells, where the cosmid replicates as a plasmid. Table 18.3 compares the properties of plasmids, phage λ vectors, and cosmids.

Concepts

Bacteriophage vectors not only hold more DNA than do plasmids, but also transfer foreign DNA into bacterial cells at a relatively high rate. A cosmid vector consists of a plasmid with *cos* sites, which allow DNA to be packaged into phage protein coats. Cosmids hold more DNA than do bacteriophage vectors.

Table 18.3 Comparison of plasmids, phage lambda vectors, and cosmids

Cloning Vector	Size of DNA That Can Be Cloned	Method of Propagation	Introduction to Bacteria
Plasmid	As large as 15 kb	Plasmid replication	Transformation
Phage lambda	As large as 23 kb	Phage reproduction	Phage infection
Cosmid	As large as 44 kb	Plasmid reproduction	Phage infection

Note: 1 kb = 1000 bp

Expression vectors Sometimes the goal in gene cloning is not just to replicate the gene, but also to produce the protein that it encodes. One of the first commercial products produced by recombinant DNA technology was the protein insulin. The gene for human insulin was isolated and inserted into bacteria, which were then multiplied and used to synthesize human insulin. However, the successful expression of a human gene in a bacterial cell is not a straightforward matter. Although the universality of the genetic code allows human genes to specify the same protein in both human and bacterial cells, the sequences that regulate transcription and translation are quite different in bacteria and eukaryotes.

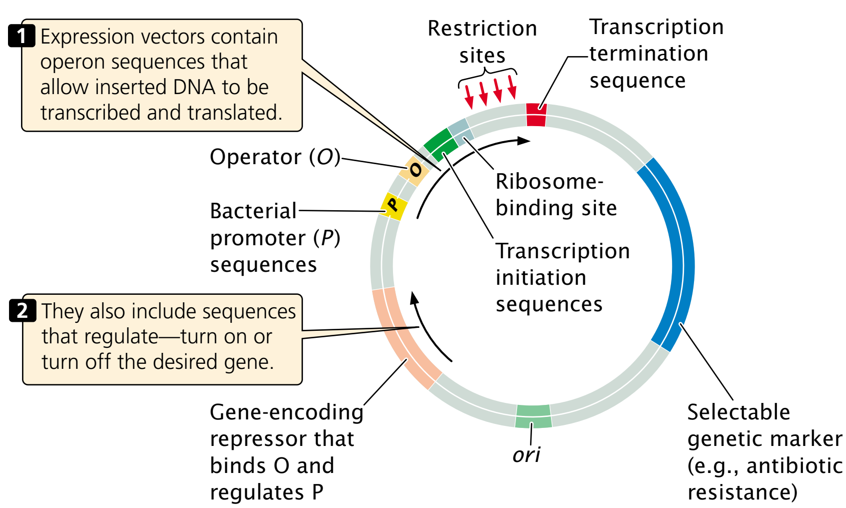
To ensure transcription and translation, a foreign gene is usually inserted into an **expression vector,** which, in addition to the usual origin of replication, restriction sites, and selectable markers, contains sequences required for transcription and translation in bacterial cells (FIGURE 18.11). These additional sequences may include:

1. A bacterial promoter, such as the *lac* promoter. The promoter precedes a restriction site where foreign DNA is to be inserted, allowing transcription of the foreign sequence to be regulated by adding substances that induce the promoter.
2. A DNA sequence that, when transcribed into RNA, produces a prokaryotic ribosome binding site.
3. Prokaryotic transcription initiation and termination sequences.
4. Sequences that control transcription initiation, such as regulator genes and operators.

The bacterial promoter and ribosome-binding site are usually placed upstream of the restriction site, which allows the foreign DNA to be inserted just downstream of the initiation codon. When the plasmid is placed in a bacterial cell, RNA polymerase binds to the promoter and transcribes the foreign DNA. Bacterial ribosomes attach to the ribosome-binding site on the RNA and translate the sequence into a foreign protein.

Concepts

An expression vector contains a promoter, a ribosome-binding site, and other sequences that allow a cloned gene to be transcribed and translated in bacteria.



18.11 To ensure transcription and translation, a foreign gene may be inserted into an expression vector—in this example, an *E. coli* expression vector.

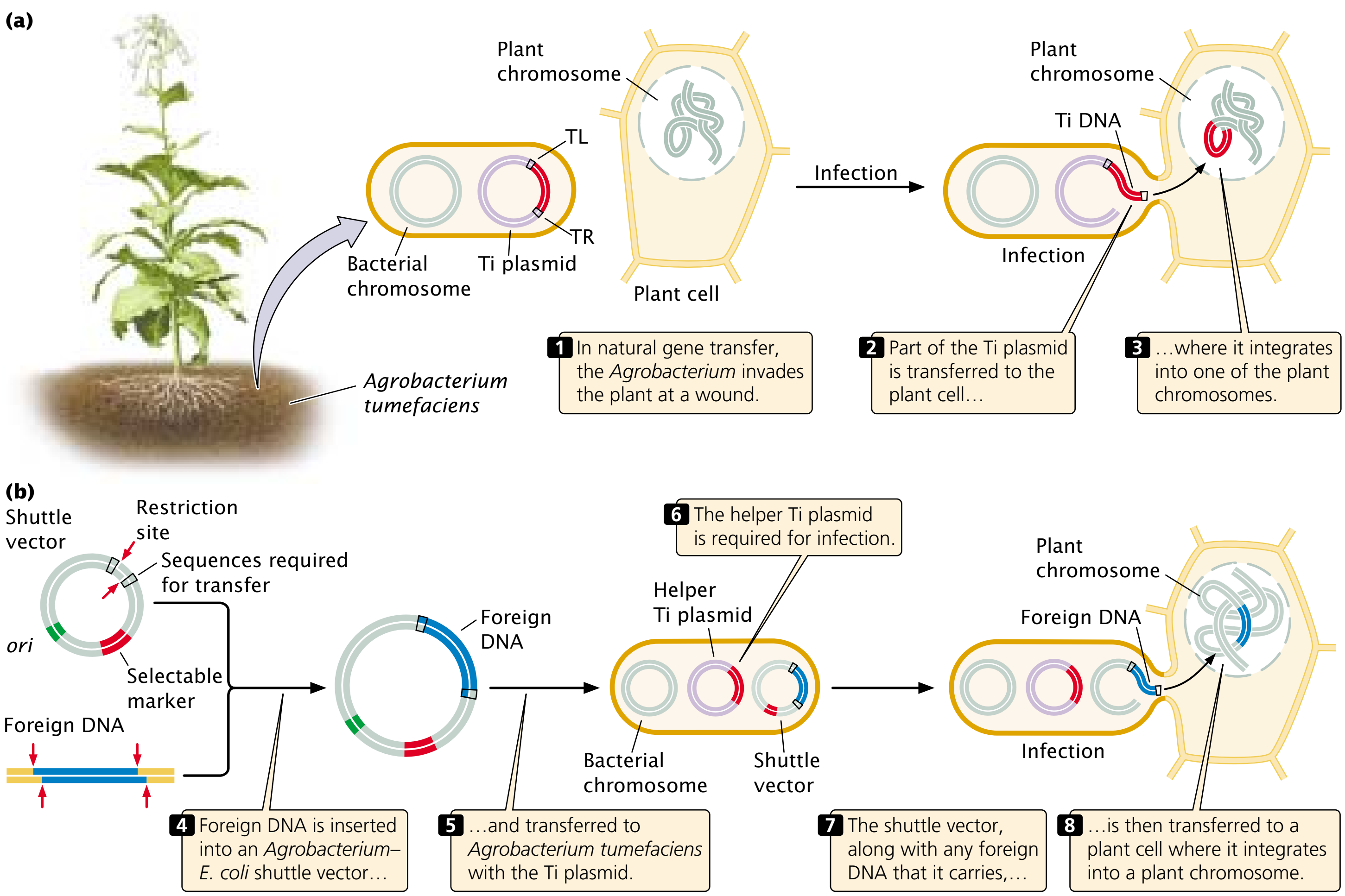
Cloning vectors for eukaryotes The vectors discussed so far allow genes to be cloned in bacterial cells. Other cloning vectors have been developed for transferring genes into eukaryotic cells. Special plasmids, for example, have been developed for cloning in yeast, and retroviral vectors have been developed for cloning in mammals.

Shuttle vectors are used to shuttle genes back and forth between two hosts. For example, plasmids have been engineered that allow gene sequences to be cloned and manipulated in bacteria and then transferred to yeast cells for study. For this reason, they must contain replication origins and selectable markers that work in both hosts.

Yeast artificial chromosomes (YACs) are DNA molecules with a yeast origin of replication, a pair of telomeres, and a centromere. Mitotic spindle fibers attach to the centromere, and YACs segregate in the same way as yeast chromosomes; the telomeres ensure that YACs remain stable within the cell; and the origin of replication allows YACs to be replicated. YACs are particularly useful because they can carry DNA fragments as large as 600 kb, and some special YACs can carry inserts of more than 1000 kb. YACs have been modified so that they can be used in eukaryotic organisms other than yeast (see introduction to Chapter 11). **Bacterial artificial chromosomes** (BACs), constructed from F factors (see Chapter 8), are used to clone large fragments ranging in length from 100 to 500 kb in bacteria.

The soil bacterium *Agrobacterium tumefaciens,* which invades plants through wounds and induces crown galls (tumors), has been used to transfer genes to plants. This bacterium contains a large plasmid called the **Ti plasmid,** part of which is transferred to a plant cell when *A. tumefaciens* infects a plant. In the plant, the Ti plasmid DNA integrates into one of the plant chromosomes where it is transcribed and translated to produce several enzymes that help support the bacterium (FIGURE 18.12a). Transfer of the DNA segment from the Ti plasmid to a plant chromosome requires two 25-bp sequences that flank the Ti DNA, as well as several genes located in the Ti plasmid.

Geneticists have engineered an *Agrobacterium–E. coli* shuttle vector that contains the flanking sequences required



18.12 The Ti plasmid can be used to transfer genes into plants.

to transfer DNA, a selectable marker, and restriction sites into which foreign DNA can be inserted (FIGURE 18.12b). When placed in *A. tumefaciens* with the Ti plasmid, the shuttle vector will transfer the foreign DNA that it carries into a plant cell, where it will integrate into a plant chromosome. This vector has been used to transfer genes that confer economically significant attributes such as resistances to herbicides, plant viruses, and insect pests.

Concepts

Special cloning vectors are used for introducing genes into eukaryotes; they include shuttle vectors, which can reproduce in two different hosts, yeast and bacterial artificial chromosomes, which hold DNA fragments hundreds of thousands of base pairs in length, and the Ti plasmid, which transfers genes to plants.

Finding Genes

In our consideration of gene cloning, we've glossed over a problem of major significance: How do we find the DNA sequence to be cloned in the first place? In fact, this problem is frequently the most significant one in cloning, because there are often millions or billions of base pairs of DNA in a cell. A discussion of how to solve this problem has been purposely delayed until now because, paradoxically, one must often clone a gene to find it.

This approach—to clone first and search later—is called "shotgun cloning," because it is like hunting with a shotgun: one sprays one's shots widely in the general direction of the quarry, knowing that there is a good chance that one or more of the pellets will hit the intended target. In shotgun cloning, one first clones a large number of DNA fragments, knowing that one or more contains the DNA of interest, and then searches for the fragment of interest among the clones.

A collection of clones containing all the DNA fragments from one source is called a **DNA library.** For example, we might isolate genomic DNA from human cells, break it into fragments, and clone all of them in bacterial cells or phages. The set of bacterial colonies or phages containing these fragments is a human **genomic library,** containing all the DNA sequences found in the human genome.

Creating a genomic library To create a genomic library, cells are collected and disrupted, which causes them to release their DNA and other cellular contents into an aqueous solution. There are several methods for isolating the DNA from the other cellular contents. In one method, phenol (an organic solvent that does not mix well with water) is added to the mixture, which is then shaken. The proteins from the cell associate with phenol, whereas the DNA and RNA remain in the aqueous solution, which is removed with the use of a pipette. The nucleic acids are then precipitated from this solution by adding cold alcohol. RNA can be removed by adding an enzyme that degrades RNA but not DNA.

When DNA has been extracted, it is cut into fragments by using a restriction enzyme to digest it for a limited amount of time only (a partial digestion) so that only *some* of the restriction sites in each DNA molecule are cut. Because which sites are cut is random, different DNA molecules will be cut in different places, and a set of overlapping fragments will be produced (FIGURE 18.13). The fragments are then joined to plasmid, phage, or cosmid vectors, which can be transferred to bacteria. This technique produces a set of bacterial cells or phage particles containing the overlapping genomic fragments. A few of the clones contain the entire gene of interest, a few contain parts of the gene, but most contain fragments that have no part of the gene of interest.

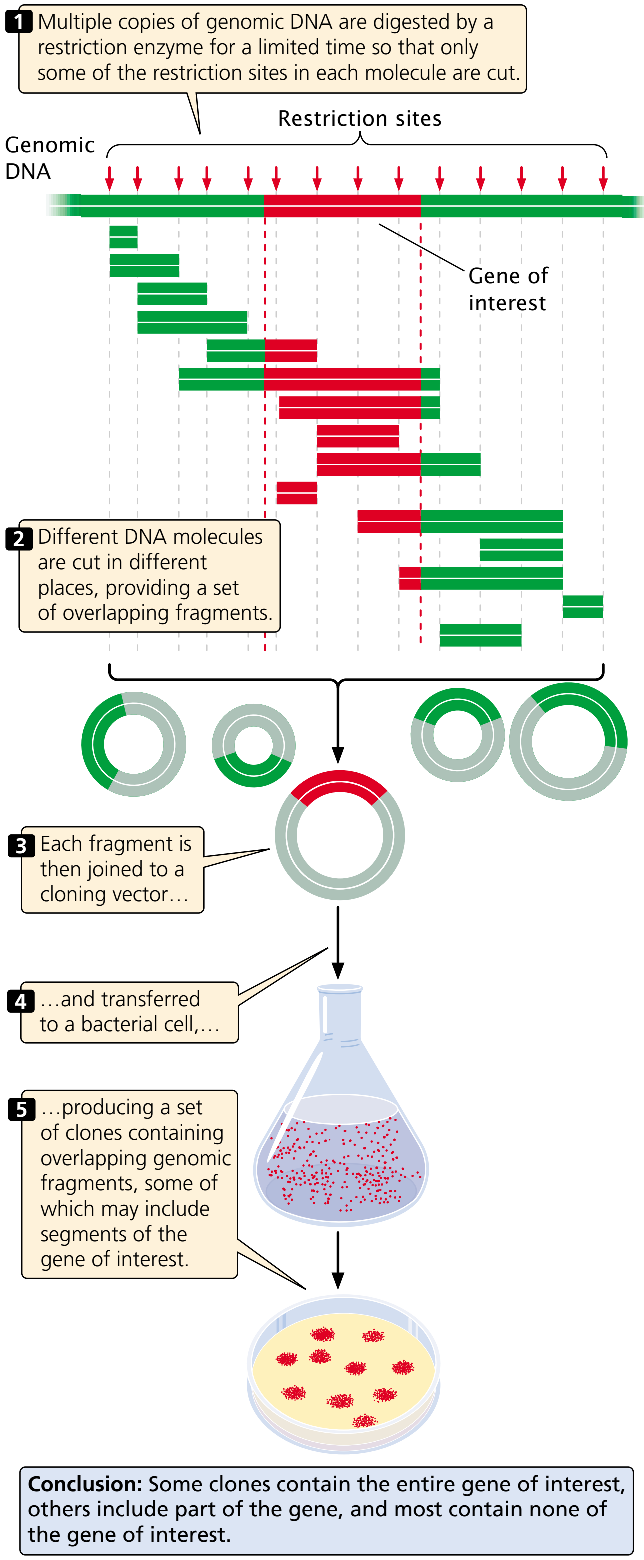
A genomic library must contain a large number of clones to ensure that all DNA sequences in the genome are represented in the library. A library of the human genome formed by using cosmids, each carrying a random DNA fragment from 35,000 to 44,000 bp long, would require about 350,000 cosmid clones to provide a 99% chance that every sequence is included in the library.

Creating a cDNA library An alternative to creating a genomic library is to create a library consisting only of those DNA sequences that are transcribed into mRNA (called a **cDNA library** because all the DNA in this library is *complementary* to mRNA). Much of eukaryotic DNA consists of repetitive (and other DNA) sequences that are not transcribed into mRNA (see p. 000 in Chapter 11), and the sequences are not represented in a cDNA library.

A cDNA library has two additional advantages. First, it is enriched with fragments from actively transcribed genes. Second, introns do not interrupt the cloned sequences; introns would pose a problem when the goal is to produce a eukaryotic protein in bacteria, because most bacteria have no means of removing the introns.

The disadvantage of a cDNA library is that it contains only sequences that are present in mature mRNA. Introns and any other sequences that are altered after transcription are not present; sequences, such as promoters and enhancers, that are not transcribed into RNA also are not present in a cDNA library. It is also important to note that the cDNA library represents only those gene sequences expressed in the tissue from which the RNA was isolated. Furthermore, the frequency of a particular DNA sequence in a cDNA library depends on the abundance of the corresponding mRNA in the given tissue. In contrast, almost all genes are present at the same frequency in a genomic DNA library.

To create a cDNA library, messenger RNA must first be separated from other types of cellular RNA (tRNA, rRNA, snRNA, etc.). Most eukaryotic mRNAs possess a string of adenine nucleotides at the 3′ end, and this poly(A) tail provides a convenient hook for separating eukaryotic mRNA from the other types. Total cellular RNA is isolated from cells and poured through a column packed



18.13 A genomic library contains all of the DNA sequences found in an organism's genome.

with short fragments of DNA consisting entirely of thymine nucleotides [oligo(dT) chains; FIGURE 18.14a]. As the RNA moves through the column, the poly(A) tails of mRNA molecules pair with the oligo(dT) chains and are retained in the column, whereas the rest of the RNA passes through. The mRNA can then be washed from the column by adding a buffer that breaks the hydrogen bonds between poly(A) tails and oligo(dT) chains.

The mRNA molecules are then copied into cDNA by reverse transcription. Short oligo(dT) primers are added to the mRNA. A primer pairs with the poly(A) tail at the 3′ end of the mRNA, providing a 3′-OH group for the initiation of DNA synthesis (FIGURE 18.14b). Reverse transcriptase, an enzyme isolated from retroviruses (see p. 000 in Chapter 8), synthesizes single-stranded complementary DNA from the RNA template by adding DNA nucleotides to the 3′-OH group of the primer.

The resulting RNA–DNA hybrid molecule is then converted into a double-stranded cDNA molecule by one of several methods. One common method is to treat the RNA–DNA hybrid with RNase to partly digest the RNA strand. Partial digestion leaves gaps in the RNA–DNA hybrid, allowing DNA polymerase to synthesize a second DNA strand by using the short undigested RNA pieces as primers and the first DNA strand as a template. DNA polymerase eventually displaces all the RNA fragments, replacing them with DNA nucleotides, and nicks in the sugar–phosphate backbone are sealed by DNA ligase.

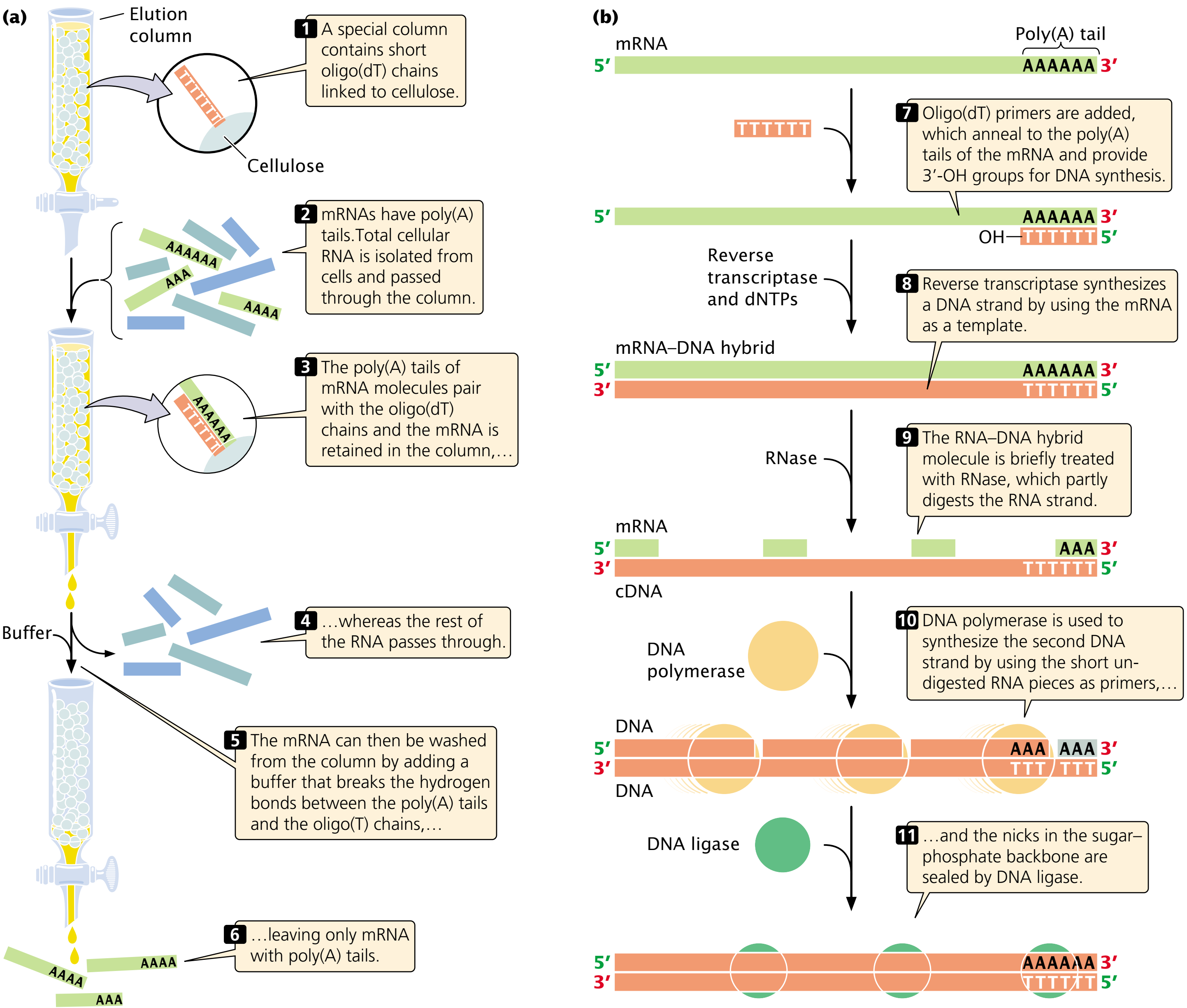
Concepts

One method of finding a gene is to create and screen a DNA library. A genomic library is created by cutting genomic DNA into overlapping fragments and cloning each fragment in a separate bacterial cell. A cDNA library is created from mRNA that is converted into cDNA and cloned in bacteria.

Screening DNA libraries Creating a genomic or cDNA library is relatively easy compared with screening the library to find clones that contain the gene of interest. The screening procedure used depends on what is known about the gene.

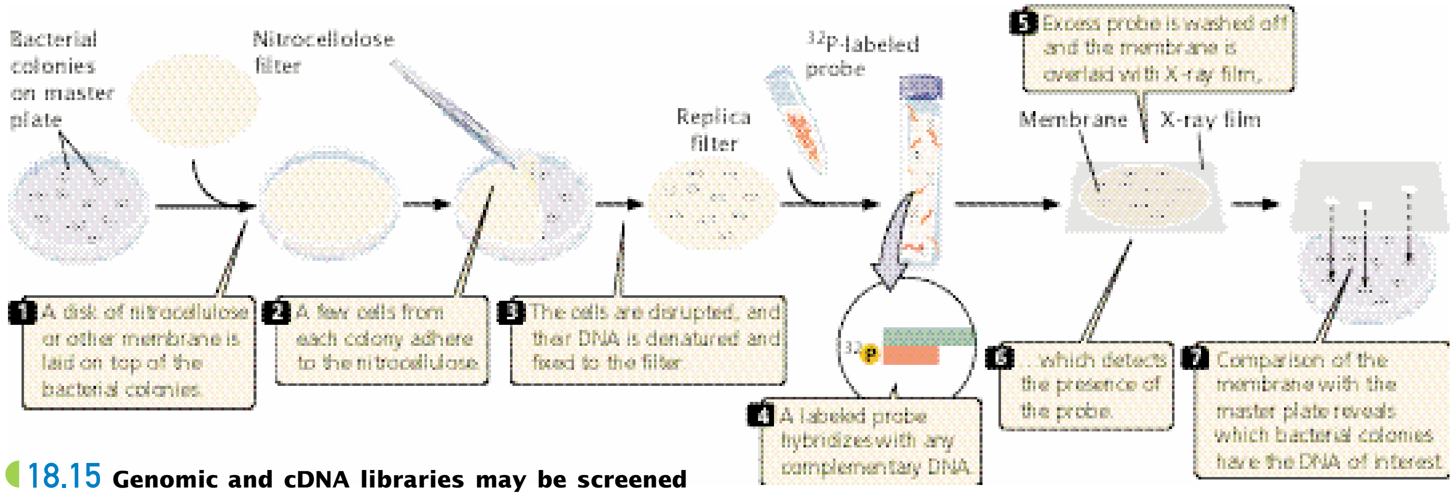
The first step in screening is to plate out the clones of the library. If a plasmid or cosmid vector was used to construct the library, the cells are diluted and plated so that each bacterium grows into a distinct colony. If a phage vector was used, the phages are allowed to infect a lawn of bacteria on a petri plate. Each plaque or bacterial colony contains a single, cloned DNA fragment that must be screened for the gene of interest.

One common way to screen libraries is with probes. We've seen how probes can be used to find specific fragments of DNA on an electrophoretic gel (see Figure 18.5). In a similar way, probes can be used to find cloned fragments of DNA in bacteria or phages. To use a probe, replicas of the plated colonies or plaques in the library must first be made. FIGURE 18.15 illustrates this procedure for a cosmid library.



18.14 A cDNA library contains only those DNA sequences that are transcribed into mRNA.

How is a probe obtained when the gene has not yet been isolated? One option is to use a similar gene from another organism as the probe. For example, if we wanted to screen a human genomic library for the growth-hormone gene and the gene had already been isolated from rats, we could use a purified rat gene sequence as the probe to find the human gene for growth hormone. Successful hybridization does not require perfect complementarity between the probe and the target sequence; so a related sequence can often be used as a probe. The temperature and salt concentration of the hybridization reaction can be adjusted to regulate the degree of complementarity required for pairing to take place. Alternatively, synthetic probes can be created if the protein produced by the gene has been isolated and its amino acid sequence has been determined. With the use of the genetic code and the amino acid sequence of the protein, possible nucleotide sequences of a small region of the gene can be deduced. Although only one sequence in the gene encodes a particular protein, the presence of synonymous codons means that the same protein could be produced by several different DNA sequences, and it is impossible to know which is correct. To overcome this problem, a mixture of all the possible DNA sequences is used as a probe. To minimize the number of sequences required in the mixture, a region of the protein is selected with relatively little degeneracy in its codons (FIGURE 18.16).



18.15 Genomic and cDNA libraries may be screened with a probe to find the gene of interest.

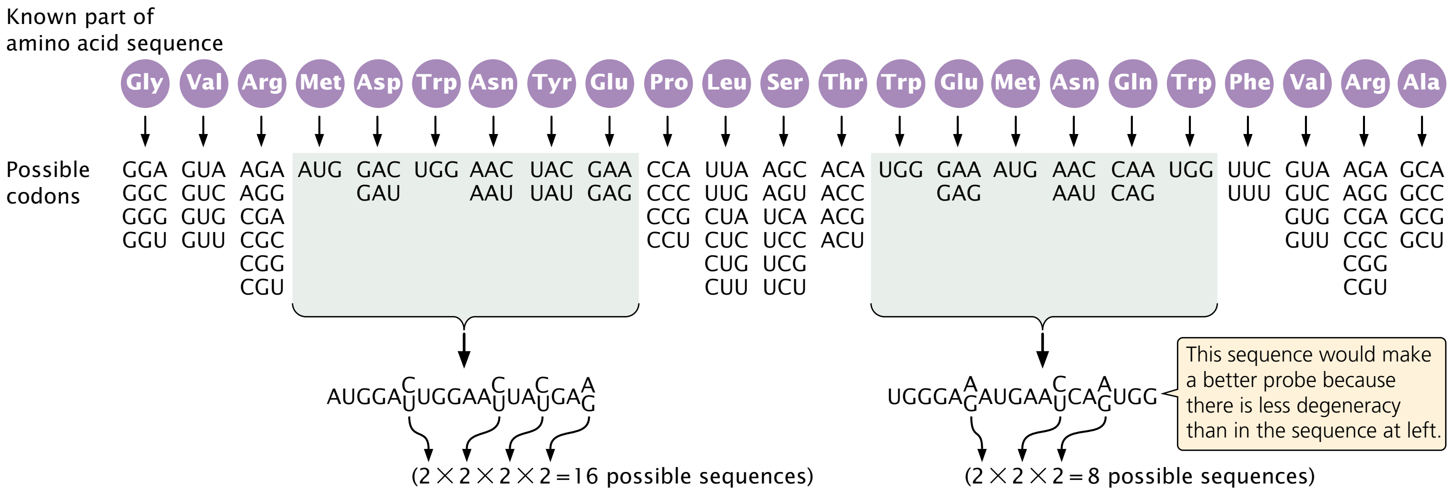
When part of the DNA sequence of the gene has been determined, a set of DNA probes can be synthesized chemically by using an automated machine known as an oligonucleotide synthesizer. The resulting probes can be used to screen a library for a gene of interest.

Yet another method of screening a library is to look for the protein product of a gene. This method requires that the DNA library be cloned in an expression vector. The clones can be tested for the presence of the protein by using an antibody that recognizes the protein or by using a chemical test for the protein product. This method depends on the existence of a test for the protein produced by the gene.

Almost any method used to screen a library will identify several clones, some of which will be false positives that do not contain the gene of interest; several screening methods may be needed to determine which clones actually contain the gene.

Concepts

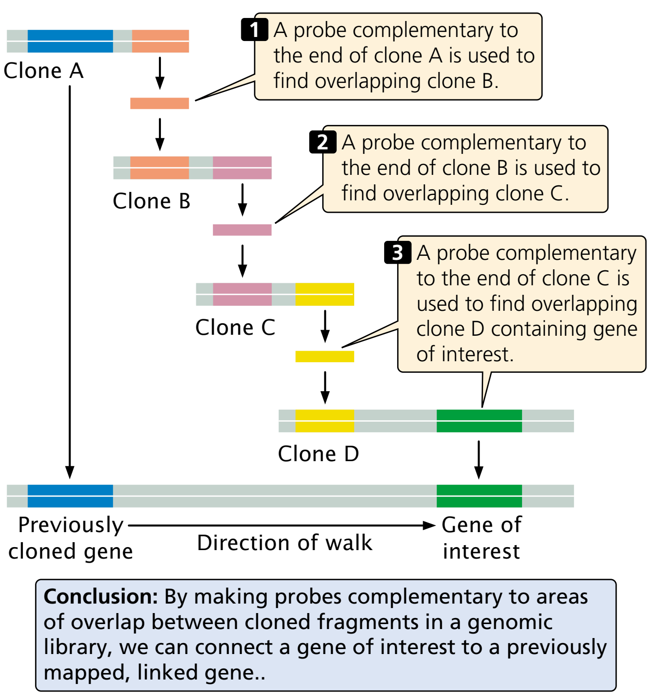
A DNA library can be screened for a specific gene by using complementary probes that hybridize to the gene. Alternatively, the library can be cloned into an expression vector, and the gene can be located by examining the clones for the protein product of the gene.



18.16 A synthetic probe can be designed on the basis of the genetic code and the known amino acid sequence of the protein encoded by the gene of interest. Because of ambiguity in the code, the same protein can be encoded by several different DNA sequences, and probes consisting of all the possible DNA sequences must be synthesized. To minimize the number of sequences that must be synthesized, a region of the gene with minimal degeneracy is picked.

Chromosome walking For many genes with important functions, no associated protein product is yet known. The biochemical bases of many human genetic diseases, for example, are still unknown. How could these genes be isolated? One approach is to first determine the general location of the gene on the chromosome by using recombination frequencies derived from crosses or pedigrees (see p. 000 in Chapter 7). After the gene has been placed on a chromosome map, neighboring genes that have already been cloned can be identified. With the use of a technique called **chromosome walking** (FIGURE 18.17), it is possible to move from these neighboring genes to the new gene of interest.

The basis of chromosome walking is the fact that a genomic library consists of a set of *overlapping* DNA fragments (see Figure 18.13). We start with a cloned gene or DNA sequence that is close to the new gene of interest so that the "walk" will be as short as possible. One end of the clone of a neighboring gene (clone A in Figure 18.17) is used to make a complementary probe. This probe is used to screen the genomic library to find a second clone (clone B) that overlaps with the first and extends in the direction of the gene of interest. This second clone is isolated and purified and a probe is prepared from its end. The second probe is used to screen the library for a third clone (clone C) that overlaps with the second. In this way, one can walk systematically toward the gene of interest, one clone at a time. A number of important human genes and genes of other organisms have been found in this way.



18.17 **In chromosome walking, neighboring genes are used to locate a gene of interest.**

Concepts

In chromosome walking, a gene is first mapped in relation to a previously cloned gene. A probe made from one end of the cloned gene is used to find an overlapping clone, which is then used to find another overlapping clone. In this way, it is possible to walk down the chromosome to the gene of interest.

Connecting Concepts

Cloning Strategies

All gene-cloning experiments have four basic steps:

1. Isolation of a DNA fragment
2. Joining of the fragment to a cloning vector
3. Introduction of the cloning vector, along with the inserted DNA fragment, into host cells
4. Identification of cells containing the recombinant DNA molecule

We've now considered a number of different methods for carrying out these four steps. There is no single procedure for cloning a gene but rather a variety of methods, each with its strengths and weaknesses. The particular combination of methods chosen for a cloning experiment is termed the **cloning strategy.**

In developing a cloning strategy, a number of factors must be taken into consideration. These factors include how much is known about the gene to be cloned, the size and nature of the gene, and the ultimate purpose of the cloning experiment. The procedure for cloning a small, well-characterized DNA fragment for sequencing would be very different from that for cloning a large, poorly known gene for the commercial production of a protein.

The first step in gene cloning is to find the particular gene or DNA fragment of interest. There are two basic approaches. In one approach, a DNA library can be constructed from genomic or cDNA, and the library can be screened to find the gene of interest. In the other approach, the gene can first be isolated and then cloned. Which approach is used depends largely on what is already known about the gene. Has the gene been mapped? Is there a probe available for screening? Is the amino acid sequence of a protein encoded by the gene known?

If one chooses to make and screen a DNA library, the next problem is to select the best source of DNA. Will it be a genomic library or a cDNA library? If the purpose is to clone the gene in an expression vector and produce a protein, then a cDNA library is ideal. Using a cDNA library means that introns (which bacteria cannot splice out) will be excluded, and fewer colonies will need to be screened. If, on the other hand, the purpose is to examine the regulatory sequences or the introns within a gene, then a genomic library is required.

The next important decision in developing a cloning strategy is to select the cloning vector. The choice depends on a number of factors:

- *The length of the sequence to be cloned.* For a sequence only a few thousand base pairs in length, a plasmid may be the best choice; if one wants to clone a gene that is 35 kb or longer, a cosmid will be required.
- *The organism in which the gene will be cloned.* Some vectors are specific for *E. coli,* whereas others are specific for other bacteria or for eukaryotic cells.
- *The selection methods used to find cells containing a plasmid with the inserted gene.* One may need a vector with selectable markers so that cells containing the gene can be identified.
- *The need for the inserted gene to be expressed.* If the protein product is desired, it may be necessary to use an expression vector that contains a promoter and other sequences that ensure transcription and translation of the inserted gene.
- *The need for efficiency of transfer to host cells.* If selection methods can be used to screen a large number of cells, then a low rate of transfer may be adequate; but, if screening is less efficient or is costly, a higher rate of transfer may be desirable.

A cloning strategy must also take into consideration the best method for joining the DNA fragment and cloning vector. Important points here include the simplicity and ease of the method, the need to retain restriction sites so that the foreign gene can later be retrieved from the vector, and whether the gene sequence must be joined to a promoter and other regulatory sequences to ensure transcription.

The method chosen for moving the vector into the host cell is usually dictated by the type of vector; plasmids are transferred to bacterial cells by transformation, whereas phage vectors and cosmids are transferred by viral infection.

The procedure for screening cells to find those with recombinant molecules depends on how much is known about the cloned fragment, the efficiency of transfer, and the cloning vector used. Considerations used in a developing a cloning strategy are summarized in Table 18.4.

Table 18.4 Considerations in developing a cloning strategy

Step in Gene Cloning	Considerations
1. Isolation of DNA fragment	a. The purpose of cloning (is expression required?). Is the entire sequence needed? b. What is known about the gene and the protein (if any) that it encodes? c. The size of the gene. d. Is the chromosomal location of the gene known? e. Size of the genome from which the gene is isolated.
2. Joining DNA fragment to vector	a. Type of cloning vector used. i. The size of the gene. ii. The organism into which the gene will be cloned. iii. The need for a selection mechanism. iv. Whether expression is required. v. Efficiency of transfer to host cell required. vi. The purpose of cloning. b. Method of joining the gene to vector. i. Simplicity of method. ii. Availability of restriction sites. iii. The need to retrieve the fragment from the vector. iv. Whether expression is required. v. The purpose of cloning.
3. Transfer of recombinant vector to host cell	a. Type of cloning vector used.
4. Identification of cells carrying recombinant molecule	a. Known information about the gene. b. Type of cloning vector used. c. Efficiency of transfer. d. Purpose of cloning.

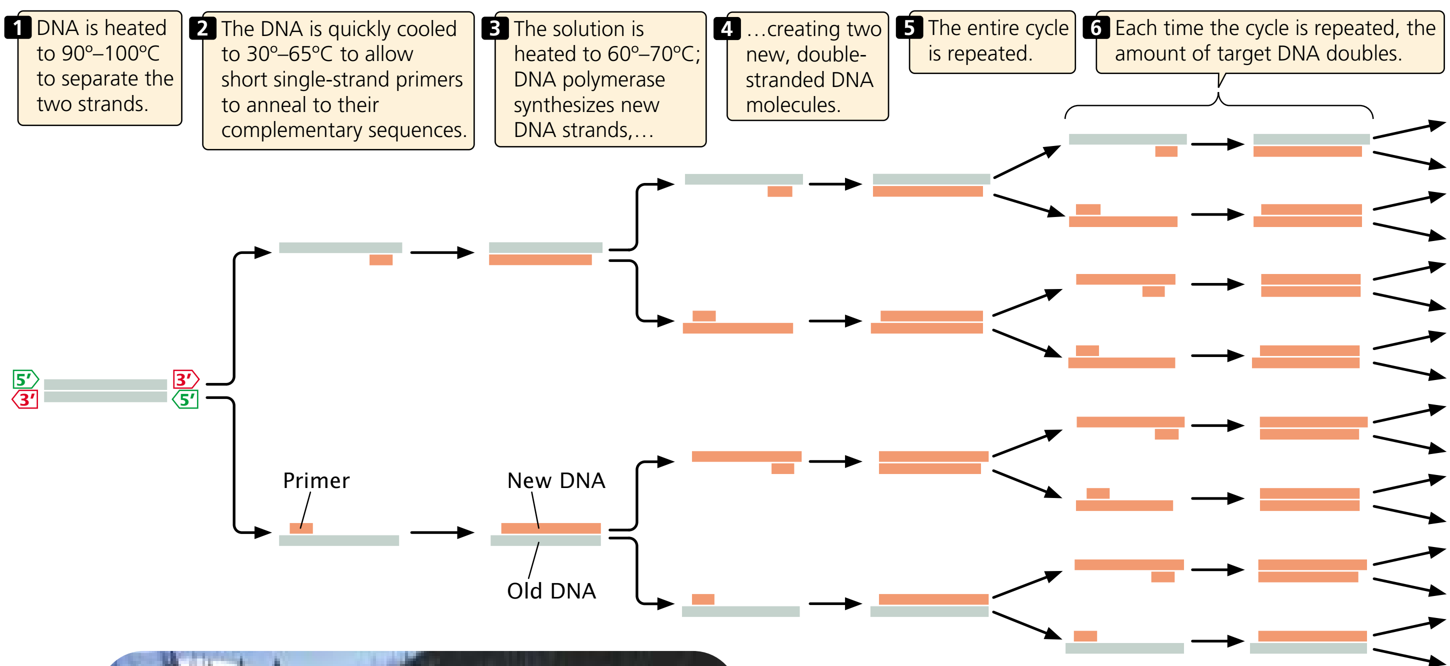
Using the Polymerase Chain Reaction to Amplify DNA

A major problem in working at the molecular level is that each gene is a tiny fraction of the total cellular DNA. Because each gene is rare, it must be isolated and amplified before it can be studied. Before mid-1980, the only procedure available for amplifying DNA was gene cloning—placing the gene in a bacterial cell and multiplying the bacteria. Cloning is labor intensive and requires at least several days to grow the bacteria. In 1983, Kary Mullis of the Cetus Corporation conceptualized a new technique for amplifying DNA in a test tube. The **polymerase chain reaction** allows DNA fragments to be amplified a billionfold within just a few hours. It can be used with extremely small amounts of original DNA, even a single molecule. The polymerase chain reaction has revolutionized molecular biology and is now one of the most widely used of all molecular techniques.

The basis of PCR is replication catalyzed by a DNA polymerase enzyme, which has two essential requirements: (1) a single-stranded DNA template from which a new DNA strand can be copied and (2) a primer with a 3′-OH group to which new nucleotides can be added.

Because a DNA molecule consists of two nucleotide strands, each of which can serve as a template to produce a new molecule of DNA, the amount of DNA doubles with each replication event. The starting point of DNA synthesis on the template is determined by the choice of primers. The primers used in PCR are short fragments of DNA, typically from 17 to 25 nucleotides long, that are complementary to known sequences on the template. A different primer is used for each strand.

To carry out PCR, one begins with a solution that includes the target DNA (the DNA to be amplified), DNA polymerase, all four deoxyribonucleoside triphosphates (dNTPs—the substrates for DNA polymerase), primers that are complementary to short sequences on each strand of the target DNA, and magnesium ions and other salts that are necessary for the reaction to proceed. A typical polymerase chain reaction includes three steps (FIGURE 18.18).



18.18 **The polymerase chain reaction is used to amplify even very small samples of DNA.** The photograph is of hot springs in Yellowstone National Park, habitat of *Thermus aquaticus*, the source of Taq enzyme used in PCR. (Photo: Fritz Pölking/Bruce Coleman.)

Table 18.5 Number of copies of DNA fragment in PCR amplification

Number of PCR Cycles (*n*)	Number of Double-Stranded Copies of Original DNA (2^n)
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1,024
20	1,048,576
30	1,073,741,824

In step 1, a starting solution of DNA is heated to between 90° and 100°C to break the hydrogen bonds between the two nucleotide strands and thus produce the necessary single-stranded templates. The reaction mixture is held at this temperature for only a minute or two. In step 2, the DNA solution is cooled quickly to between 30° and 65°C and held at this temperature for a minute or less. During this short interval, the DNA strands will not have a chance to reanneal, but the primers will be able to attach to their complementary sequences on the template strands. In step 3, the solution is heated to between 60° and 70°C, the temperature at which DNA polymerase can synthesize new DNA strands by adding nucleotides to the primers. Within a few minutes, two new double-stranded DNA molecules are produced for each original molecule of target DNA.

The whole cycle is then repeated. With each cycle, the amount of target DNA doubles; so the target DNA increases geometrically. One molecule of DNA increases to more than 1000 molecules in 10 PCR cycles, to more than 1 million molecules in 20 cycles, and to more than 1 billion molecules in 30 cycles (Table 18.5). Each cycle is completed within a few minutes; so a large amplification of DNA can be achieved within a few hours.

Two key innovations facilitated the use of PCR in the laboratory. The first was the discovery of a DNA polymerase that is stable at the high temperatures used in step 1 of PCR. The DNA polymerase from *E. coli* that was originally used in PCR denatures at 90°C. For this reason, fresh enzyme had to be added to the reaction mixture during *each* cycle, slowing the process considerably. This obstacle was overcome when DNA polymerase was isolated from the bacterium *Thermus aquaticus,* which lives in the boiling springs of Yellowstone National Park (see Figure 18.18). This enzyme, dubbed ***Taq* polymerase,** is remarkably stable at high temperatures and is not denatured during the strand-separation step of PCR; so it can be added to the reaction mixture at the beginning of the PCR process and will continue to function through many cycles.

The second key innovation was the development of automated thermal cyclers—machines that bring about the rapid temperature changes necessary for the different steps of PCR. Originally, tubes containing reaction mixtures were moved by hand among water baths set at the different temperatures required for the three steps of each cycle. In automated thermal cyclers, the reaction tubes are placed in a metal block that changes temperature rapidly according to a computer program.

The polymerase chain reaction is now often used in place of gene cloning, but it does have several limitations. First, the use of PCR requires prior knowledge of at least part of the sequence of the target DNA to allow construction of the primers. Therefore PCR cannot be used to amplify a gene that has not been at least partly sequenced. Second, the capacity of PCR to amplify extremely small amounts of DNA makes contamination a significant problem. Minute amounts of DNA from the skin of laboratory workers and even in small particles in the air can enter a reaction tube and be amplified along with the target DNA. Careful laboratory technique and the use of controls are necessary to circumvent this problem.

A third limitation of PCR is accuracy. Unlike other DNA polymerases, *Taq* polymerase does not have the capacity to proofread (see p. 000 in Chapter 12) and, under standard PCR conditions, it incorporates an incorrect nucleotide about once every 20,000 bp. DNA polymerases with proofreading capacity usually incorporate an incorrect nucleotide only about once every billion base pairs. For many applications, the error rate produced by PCR is not a problem, because only a few DNA molecules of the billions produced will contain an error. However, for other applications such as the cloning of PCR products, the relatively high error rate of PCR can pose significant problems. New heat-stable DNA polymerases *with* proofreading capacity have been isolated, giving more accurate PCR results.

A fourth limitation of PCR is that the size of the fragments that can be amplified by standard *Taq* polymerase is usually less than 2000 bp. By using a combination of *Taq* polymerase and a DNA polymerase with proofreading capacity and by modifying the reaction conditions, investigators have been successful in extending PCR amplification to larger fragments. In spite of its limitations, PCR is used routinely in a wide array of molecular applications.

Concepts

The polymerase chain reaction is an enzymatic, in vitro method for rapidly amplifying DNA. In this process, DNA is heated to separate the two strands, short primers attach to the target DNA, and DNA polymerase synthesizes new DNA strands from the primers. Each cycle of PCR doubles the amount of DNA.

Analyzing DNA Sequences

In addition to cloning and amplifying DNA, molecular techniques are used to analyze DNA molecules through a determination of their sequences and an investigation of their functions.

DNA sequencing A powerful technique to emerge from recombinant DNA technology is the ability to quickly sequence DNA molecules. *DNA sequencing* is the determination of the sequence of bases in DNA. Sequencing allows the genetic information in DNA to be read, providing an enormous amount of information about gene structure and function. Details of DNA sequencing will be covered in Chapter 19.

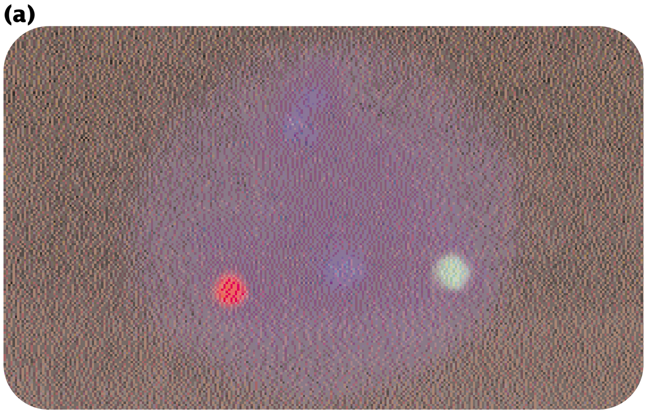
In situ hybridization DNA probes can be used to determine the chromosomal location of a gene or the cellular location of an mRNA in a process called **in situ hybridization.** The name is derived from the fact that DNA or RNA is visualized while it is in the cell (in situ). This technique requires that the cells be fixed and the chromosomes be spread on a microscope slide. The chromosomes are then briefly exposed to a solution with high pH, which disrupts the pairing of the DNA bases, making them accessible to probes. A labeled probe, which binds to any complementary DNA sequences, is added. Excess probe is washed off, and the location of the bound probe is detected. Originally, probes were radioactively labeled and detected with autoradiography, but now many probes carry attached fluorescent dyes that can be seen directly with the microscope (FIGURE 18.19a) Several probes with different colored dyes can be used simultaneously to investigate different sequences or chromosomes.

In situ hybridization can also be used to determine the tissue distribution of specific mRNA molecules, serving as a source of insight into how gene expression differs among cell types (FIGURE 18.19b). A labeled DNA or RNA probe complementary to a specific mRNA molecule is added to tissue, and the location of the probe is determined by using either autoradiography or fluorescent tags.

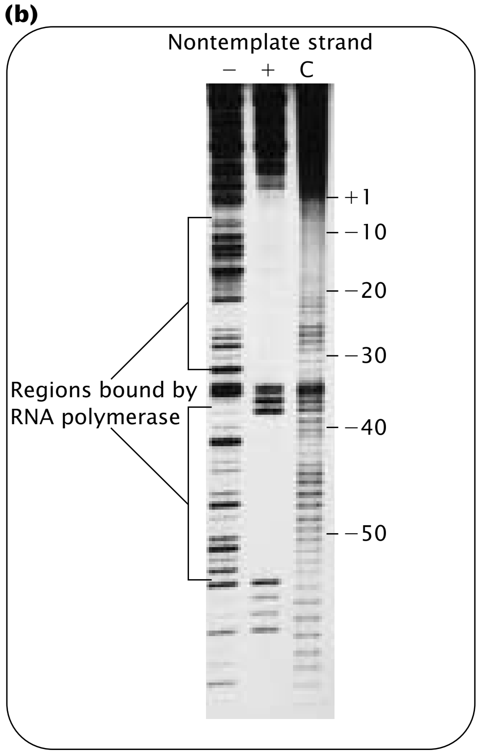
DNA footprinting Many important DNA sequences serve as binding sites for proteins; for example, consensus sequences in promoters are often binding sites for transcription factors (see p. 000 in Chapter 11). A technique called **DNA footprinting** can be used to determine which DNA sequences are bound by such proteins.

In a typical DNA-footprinting experiment, purified DNA fragments are labeled at one end with a radioactive isotope of phosphorus, ^{32}P. An enzyme or chemical that

(a)



(b)



18.19 In in situ hybridization, DNA probes are used to determine the cellular or chromosomal location of a gene. (a) Fluorescent probes are used to mark the locations of specific gene sequences on chromosomes. (b) In situ hybridization can also be used to detect specific mRNA sequences in tissues. Here, the distribution of mRNA produced by the *ftz* gene (which has a role in controlling development) in a young *Drosophila* embryo is detected by a radioactive probe and autoradiography. [Autoradiograph from E. Hafen, A. Kuriowa, and W. J. Gehring, *Cell* 37:833–841.]

makes cuts in DNA is used to cleave the DNA randomly into subfragments, which are then denatured and separated by gel electrophoresis. The positions of the subfragments are visualized with autoradiography. This procedure is carried out both in the presence and in the absence of a particular DNA-binding protein. When the protein is absent, cleavage is random along the DNA, producing a continuous "ladder" of bands on the autoradiograph (FIGURE 18.20). When the protein is present, it binds to specific nucleotides and protects their phosphodiester bonds from cleavage. Therefore, there is no cleavage in the area protected by the protein, and no labeled fragments terminating in the binding site appear on the autoradiograph. Their omission leaves a gap, or "footprint", on the ladder of bands (see Figure 18.20), and the position of the footprint identifies those nucleotides bound tightly by the protein.

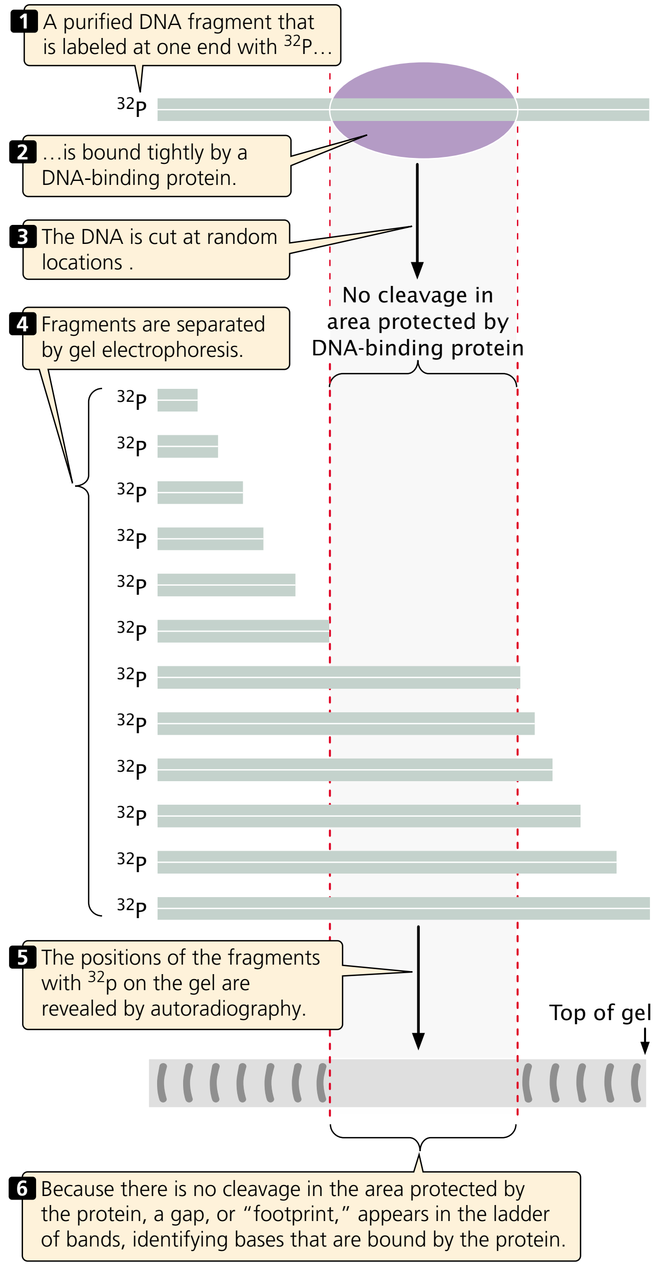
Concepts

In situ hybridization can be used to visualize the chromosomal location of a gene or to determine the tissue distribution of an mRNA transcribed from a specific gene. DNA footprinting is used to determine the sequences to which DNA-binding proteins attach.

Mutagenesis A powerful way to study gene function is to create mutations at specific locations in a process called **site-directed mutagenesis** and then to study the effects of these mutations on the organism.

A number of different strategies have been developed for site-directed mutagenesis. One strategy is to cut out a short sequence of nucleotides with restriction enzymes and replace it with a short, synthetic oligonulceotide that contains the desired mutated sequence (FIGURE 18.21). The success of this method depends on the availability of restriction sites flanking the sequence to be altered.

If appropriate restriction sites are not available, **oligonucleotide-directed mutagenesis** can be used (FIGURE 18.22). In this method, a single-stranded oligonucleotide is produced that differs from the target sequence by one or a few bases. Because they differ in only a few bases, the target DNA and the oligonucleotide will pair under the appropriate conditions. When successfully paired with the target DNA, the oligonucleotide can act as a primer to initiate DNA synthesis, which produces a double-stranded molecule with a mismatch in the primer region. When this DNA is transferred to bacterial cells, the mismatched bases will be repaired by bacterial enzymes. About half of the time the normal bases will be changed into mutant bases, and about half of the time the mutant bases will be changed into normal bases. The bacteria are then screened for the presence of the mutant gene.

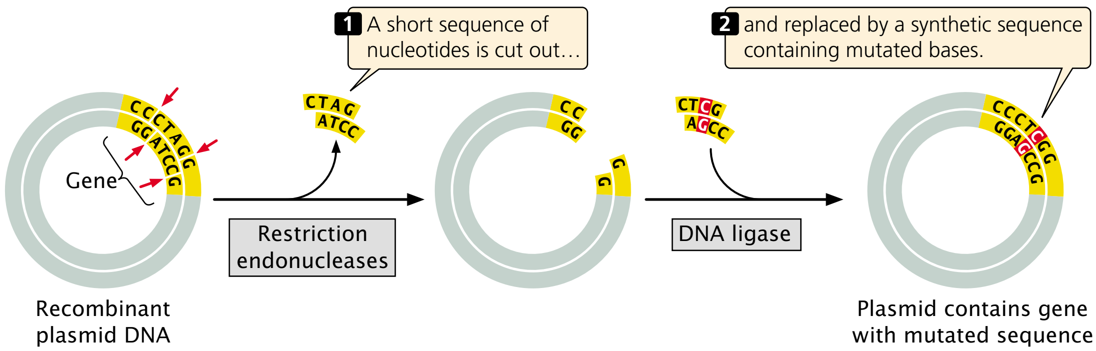


18.20 DNA footprinting can be used to determine which DNA sequences are bound by binding proteins.

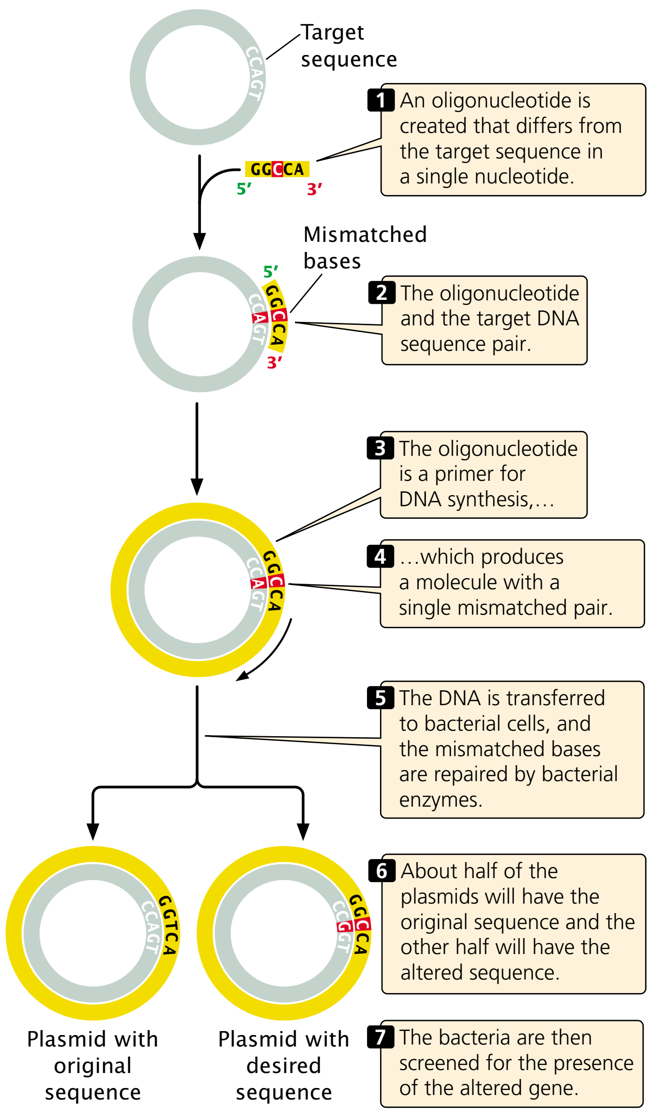
Concepts

Particular mutations can be introduced at specific sites within a gene by means of site-directed mutagenesis.

Transgenic animals The oocytes of mice and other mammals are large enough that DNA can be injected into them directly. Immediately after penetration by sperm,



18.21 In site-directed mutagenesis, restriction enzymes cut out a short sequence of nucleotides that is then replaced by a synthetic mutated DNA sequence.



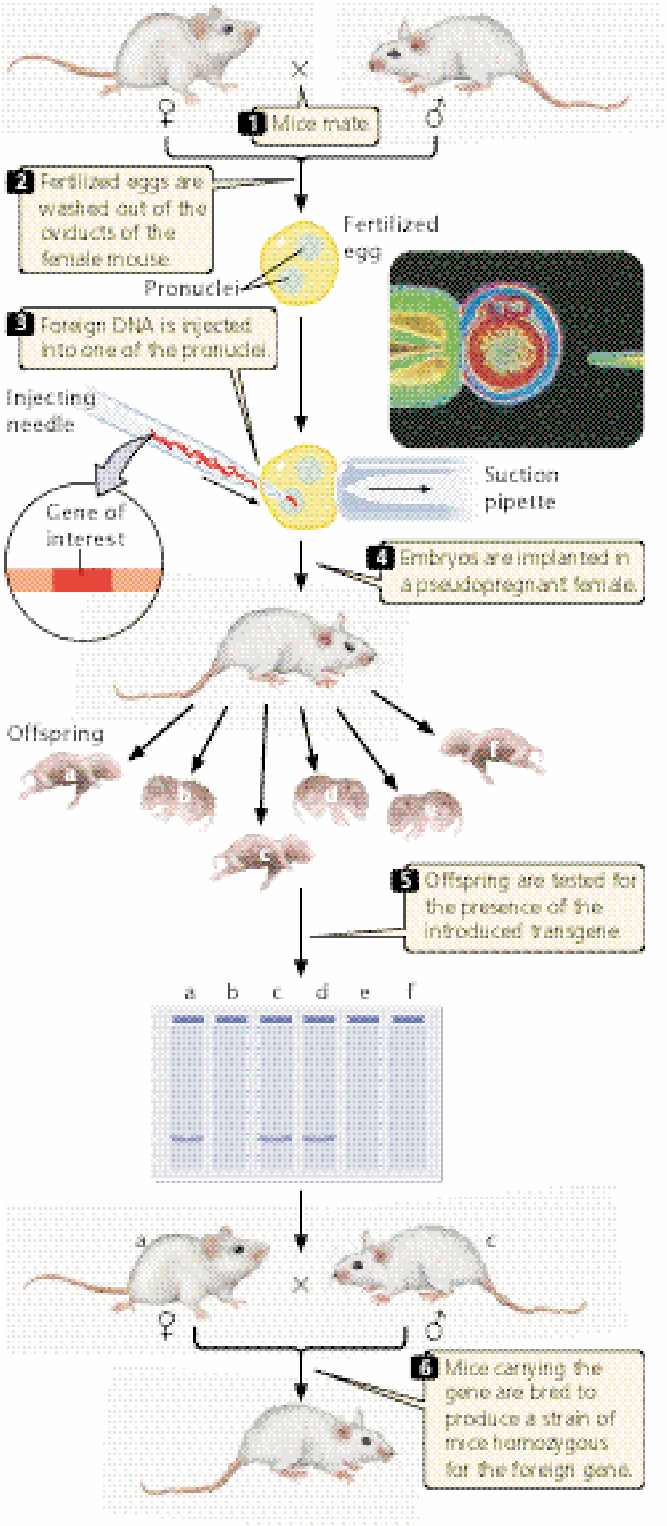
18.22 Oligonucleotide-directed mutagenesis is used to study gene function when appropriate restriction sites are not available.

a fertilized mouse egg contains two pronuclei, one from the sperm and one from the egg; these pronuclei later fuse to form the nucleus of the embryo. Mechanical devices can manipulate extremely fine, hollow glass needles to inject DNA directly into one of the pronuclei of a fertilized egg (FIGURE 18.23). Typically, a few hundred copies of cloned, linear DNA are injected into a pronucleus, and, in a few of the injected eggs, copies of the cloned DNA integrate randomly into one of the chromosomes through a process called nonhomologous recombination. After injection, the embryos are implanted in a pseudopregnant female—a surrogate mother that has been physiologically prepared for pregnancy by mating with a vasectomized male.

Only about 10% to 30% of the eggs survive and, of those that do survive, only a few have a copy of the cloned DNA stably integrated into a chromosome. Nevertheless, if several hundred embryos are injected and implanted, there is a good chance that one or more mice whose chromosomes contain the foreign DNA will be born. Moreover, because the DNA was injected at the one-cell stage of the embryo, these mice usually carry the cloned DNA in every cell of their bodies, including their reproductive cells, and will therefore pass the foreign DNA on to their progeny. Through interbreeding, a strain of mice that is homozygous for the foreign gene can be created. Animals that have been permanently altered in this way are said to be *transgenic*, and the foreign DNA that they carry is called a **transgene.**

Transgenic mice have proved useful in the study of gene function. For example, proof that the *SRY* gene (see p. 000 in Chapter 4) is the male-determining gene in mice was obtained by injecting a copy of the *SRY* gene into XX embryos and observing that these mice developed as males. In addition, a number of transgenic mouse strains that serve as experimental models for human genetic diseases have been created by injecting mutated copies of genes into mouse embryos.

www.whfreeman.com/pierce More information on transgenic animals

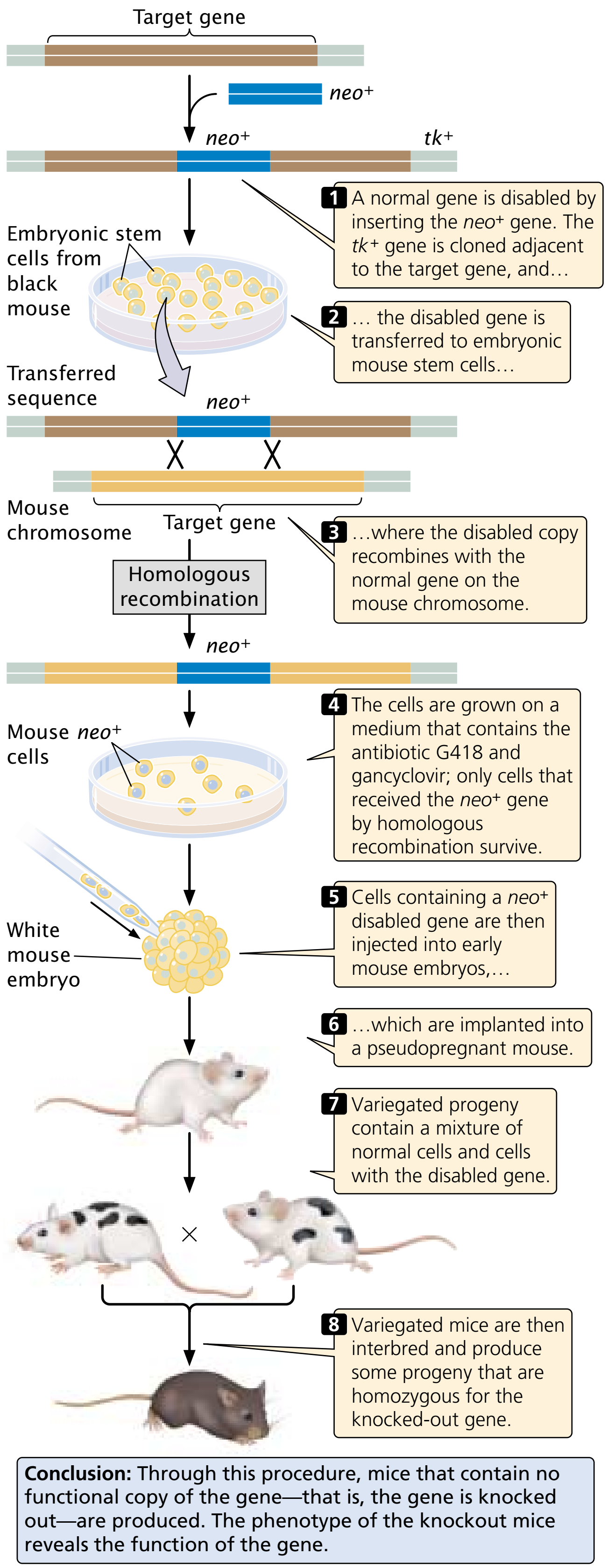


18.23 Transgenic animals have genomes that have been permanently altered through recombinant DNA technology. In this photograph a mouse embryo (red and blue) is being injected with DNA. (Photo: Jon Gordon/Phototake)

Knockout mice A particularly useful variant of the transgenic approach is to produce mice in which a normal gene has been disabled. The phenotypes of these animals, called **knockout mice,** help geneticists to determine the function of a gene. The creation of these knockout mice begins when a normal gene is cloned in bacteria and then "knocked out", or disabled. There are a number of ways to disable a gene, but a common method is to insert a gene called *neo,* which confers resistance to the antibiotic G418, into the middle of the target gene (FIGURE 18.24). The insertion of *neo* both disrupts (knocks out) the target gene and provides a convenient marker for finding copies of the disabled gene. In addition, a second gene, usually the herpes simplex viral thymidine kinase (*tk*) gene, is cloned adjacent to the disrupted gene. The disabled gene is then transferred to cultured embryonic mouse cells, where it may exchange places with the normal chromosomal copy through homologous recombination.

After the disabled gene has been transferred to the embryonic cells, the cells are screened by adding the antibiotic G418 to the medium. Only cells with the disabled gene containing the *neo* insert will survive. Because the frequency of nonhomologous recombination is higher than that of homologous recombination and because the intact target gene is replaced by the disabled copy only through homologous recombination, a means to select for the rarer homologous recombinants is required. The presence of the viral *tk* gene makes the cells sensitive to gancyclovir. Thus, transfected cells that grow on medium containing G418 and gancyclovir will contain the *neo* gene (disabled target gene) but not the adjacent *tk* gene. These cells contain the desired homologous recombinants. The nonhomologous recombinants (random insertions) will contain both the *neo* and the *tk* genes, and these transfected cells will die on the selection medium owing to the presence of gancyclovir. The surviving cells are injected into an early-stage mouse embryo, which is then implanted into a pseudopregnant mouse. Cells in the embryo carrying the disabled gene and normal embryonic cells carrying the wild-type gene will develop together, producing a chimera—a mouse that is a genetic mixture of the two cell types. The chimeric mice can be identified easily if the injected embryonic cells came from a black mouse and the embryos into which they are injected came from a white mouse; the resulting chimeras will have variegated black and white fur. The chimeras can then be interbred to produce some progeny that are homozygous for the knockout gene. The effects of disabling a particular gene can be observed in these homozygous mice.

Although they are a recent innovation, knockout mice have become important subjects for research in a number of fields. They have been used to study genes implicated in immune function, development, ethology, and human behavior.



18.24 Knockout mice possess a genome in which a gene has been disabled.

Concepts

Transgenic mice are produced by the injection of cloned DNA into the pronucleus of a fertilized egg, followed by implantation of the egg into a female mouse. In knockout mice, the injected DNA contains a mutation that disables a gene. Inside the mouse embryo, the disabled copy of the gene can exchange with the normal copy of the gene through homologous recombination.

www.whfreeman.com/pierce A catalog of knockout mice

Applications of Recombinant DNA Technology

In addition to providing valuable new information about the nature and function of genes, recombinant DNA technology has many practical applications. These applications include the production of pharmaceuticals and other chemicals, specialized bacteria, agriculturally important plants, and genetically engineered farm animals. The technology is also used extensively in medical testing and, in a few cases, is even being used to correct human genetic defects. Hundreds of firms now specialize in developing products through genetic engineering, and many large multinational corporations have invested enormous sums of money in recombinant DNA research. Recombinant DNA technology is also frequently used in criminal investigations and for the identification of human remains. For example, the remains of people who died in the collapse of the World Trade Center on September 11, 2001, are being identified through DNA comparisons with the use of recombinant DNA technology.

Pharmaceuticals

The first commercial products to be developed by using recombinant DNA technology were pharmaceuticals used in the treatment of human diseases and disorders. In 1979, the Eli Lilly corporation began selling human insulin produced with the use of recombinant DNA technology. Before this time, all the insulin used in the treatment of diabetics was isolated from the pancreases of farm animals slaughtered for meat. Although this source of insulin worked well for many diabetics, it was not human insulin, and some people suffered allergic reactions to the foreign protein. The human insulin gene was inserted into plasmids and transferred to bacteria that then produced human insulin. Pharmaceuticals produced through recombinant DNA technology include human growth hormone (for children with growth deficiencies), clotting factors (for hemophiliacs), and tissue plasminogen activator (used to dissolve blood clots in heart-attack patients).

Specialized Bacteria

Bacteria play an important role in many industrial processes, including the production of ethanol from plant material, the leaching of minerals from ore, and the treatment of sewage and other wastes. The bacteria engaged in these processes are being modified by genetic engineering so that they work more efficiently. New strains of technologically useful bacteria are being developed that will break down toxic chemicals and pollutants, enhance oil recovery, increase nitrogen uptake by plants, and inhibit the growth of pathogenic bacteria and fungi.

Agricultural Products

Recombinant DNA technology has had a major effect on agriculture, where it is now used to create crop plants and domestic animals with valuable traits. For many years, plant pathologists had recognized that plants infected with mild strains of viruses are resistant to infection by virulent strains. Using this knowledge, geneticists have created viral resistance in plants by transferring genes for viral proteins to the plant cells. A genetically engineered squash, called Freedom II, carries genes from the watermelon mosaic virus 2 and the zucchini yellow mosaic virus that protect the squash against viral infections.

Another objective has been to genetically engineer pest resistance into plants to reduce dependence on chemical pesticides. A protein toxin from the bacterium *Bacillus thuringiensis* selectively kills the larvae of certain insect pests but is harmless to wildlife, humans, and many other insects. The toxin gene has been isolated from the bacteria, linked to active promoters, and transferred into corn, tomato, potato, and cotton plants. The gene produces the insecticidal toxin in the plants, and caterpillars that feed on the plant die.

Recombinant DNA technology has also permitted the development of herbicide resistance in plants. A major problem in agriculture is the control of weeds, which compete with crop plants for water, sunlight, and nutrients. Although herbicides are effective at killing weeds, they can also damage the crop plants. Genes that provide resistance to broad-spectrum herbicides have been transferred into tomato, soybean, cotton, oilseed rape, and other commercially important crops. When the fields containing these crops are sprayed with herbicides, the weeds are killed but the genetically engineered plants are unaffected. In 1999, more than 21 million hectares (1 hectare = 2.471 acres) of genetically engineered soybeans and 11 million hectares of genetically engineered corn was grown throughout the world.

Recombinant DNA techniques are also applied to domestic animals. For example, the gene for growth hormone was isolated from cattle and cloned in *E. coli;* these bacteria produce large quantities of bovine growth hormone, which is administered to dairy cattle to increase milk production. Transgenic animals are being developed to carry genes that encode pharmaceutical products. For example, a gene for human clotting factor VIII has been linked to the regulatory region of the sheep gene for β-lactoglobulin, a milk protein. The fused gene was injected in sheep embryos, creating transgenic sheep that produce in their milk the human clotting factor, which is used to treat hemophiliacs. A similar procedure was used to transfer a gene for α_1-antitrypsin, a protein used to treat patients with hereditary emphysema, into sheep. Female sheep bearing this gene produce as much as 15 grams of α_1-antitrypsin in each liter of their milk, generating $100,000 worth of α_1-antitrypsin per year for each sheep.

The genetic engineering of agricultural products is controversial. One area of concern focuses on the potential effects of releasing novel organisms produced by genetic engineering into the environment. There are many examples in which nonnative organisms released into a new environment have caused ecological disruption because they are free of predators and other natural control mechanisms. Genetic engineering normally transfers only small sequences of DNA, relative to the large genetic differences that often exist between species, but even small genetic differences may alter ecologically important traits that might affect the ecosystem.

Another area of concern is that transgenic organisms may hybridize with native organisms and transfer their genetically engineered traits. For example, herbicide resistance engineered into crop plants might be transferred to weeds, which would then be resistant to the herbicides that are now used for their control. The results of some studies have demonstrated gene transfer between engineered plants and native plants, but the extent and effect of this transfer are uncertain. Other concerns focus on health-safety issues associated with the presence of engineered products in natural foods; some critics have advocated required labeling of all genetically engineered foods that contain transgenic DNA or protein. Such labeling is required in countries of the European Union but not in the United States.

On the other hand, the use of genetically engineered crops and domestic animals has potential benefits. Genetically engineered crops that are pest resistant have the potential to reduce the use of environmentally harmful chemicals, and research findings indicate that lower amounts of pesticides are used in the United States as a result of the adoption of transgenic plants. Transgenic crops also increase yields, providing more food per acre, which reduces the amount of land that must be used for agriculture.

Concepts

Recombinant DNA technology is used to create a wide range of commercial products, including pharmaceuticals, specialized bacteria, genetically engineered crops, and transgenic domestic animals.

www.whfreeman.com/pierce More information about the use of recombinant DNA and biotechnology in agriculture

Oligonucleotide Drugs

A recent application of DNA technology has been the development of oligonucleotide drugs, which are short sequences of synthetic DNA or RNA molecules that can be used to treat diseases. Antisense oligonucleotides are complementary to undesirable RNAs, such as viral RNA. When added to a cell, these antisense DNAs bind to the viral mRNA and inhibit its translation.

Single-stranded DNA oligonucleotides bind tightly to other DNA sequences, forming a triplex DNA molecule (FIGURE 18.25). The formation of triplex DNA interferes with the binding of RNA polymerase and other proteins required for transcription. Other oligonucleotides are ribozymes, RNA molecules that function as enzymes (see Chapter 13). These compounds bind to specific mRNA molecules and cleave them into fragments, destroying their ability to encode proteins. Several oligonucleotide drugs are already being tested for the treatment of AIDS and cancer.

Concepts

Oligonucleotide drugs are short pieces of DNA or RNA that prevent the expression of particular genes.

Genetic Testing

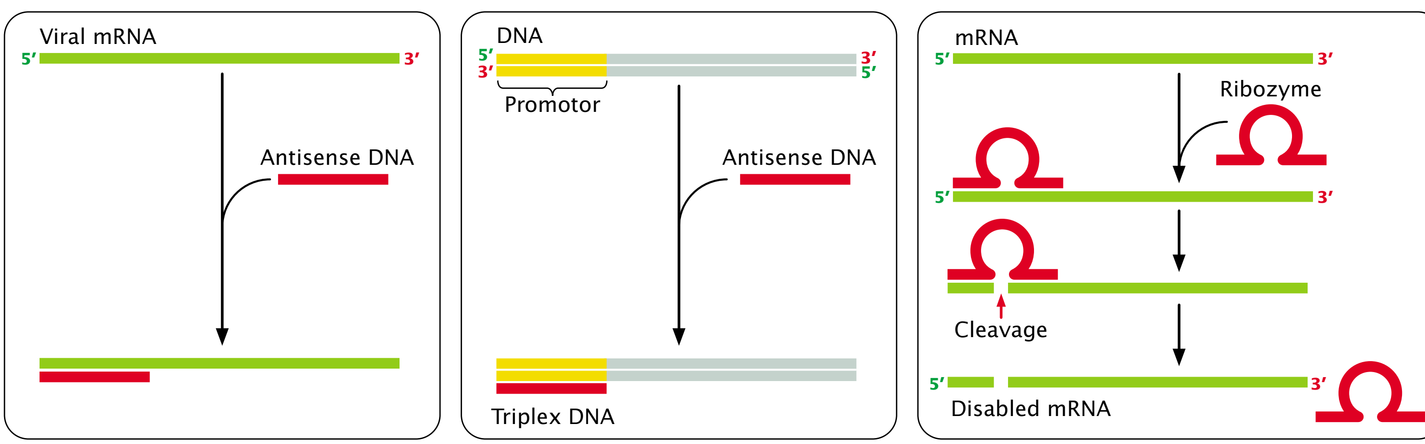
The identification and cloning of many important disease-causing human genes has allowed the development of probes for detecting disease-causing mutations. Prenatal testing is already available for several hundred genetic disorders (see Chapter 4). Additionally, presymptomatic genetic tests for adults and children are available for an increasing number of disorders.

The growing availability of genetic tests raises a number of ethical and social issues. For example, is it ethical to test for genetic diseases for which there is no cure or treatment? Huntington disease, an autosomal dominant disorder that appears in middle age, causes slow physical and mental deterioration and eventually death. No effective treatment is currently available. If one parent is affected, a child has a 50% chance of inheriting the gene for Huntington disease and eventually getting the disorder. Tests are now available that make it possible to determine whether a person carries the Huntington-disease gene, but is it beneficial to tell a young person that he or she has the Huntington-disease gene and will get the disease later in life?

Although learning that you do not have the gene might provide great peace of mind, learning that you *do* have it might lead to despair and depression. Many people at risk for Huntington disease want predictive testing, saying that the uncertainty of not knowing is more debilitating than the certain knowledge that they will get it and, in fact, a number of medical centers now offer predictive testing for Huntington disease. A few people who learned that they have the gene have committed suicide, and others had to be hospitalized for depression, but the results of several studies indicate that most people who undergo predictive testing for Huntington disease are able to cope with the information.

Other ethical and legal questions concern the confidentiality of test results. Who should have access to the results of genetic testing? Should insurance companies be allowed to use results from such tests to deny coverage to healthy people who are at risk for genetic diseases? Should relatives who also might be at risk be informed of the results of genetic testing?

Other concerns focus on whether the cost of genetic testing justifies the benefits. In some cases, genetic tests provide clear benefits because early identification allows for



Antisense DNA may bind to the 5′ end of viral mRNA and prevent the binding of the ribosome; so no viral protein is produced.

Antisense DNA may bind to a promoter (forming triplex DNA) and prevent the binding of RNA polymerase; so mRNA is not produced.

Antisense RNA in the form of a ribozyme may bind and cleave mRNA, destroying it.

18.25 Oligonucleotide drugs are short sequences of DNA or RNA that can be used to treat diseases.

better treatment. For example, when phenylketonuria (an autosomal recessive disorder that can cause mental retardation) is identified in infants, the administration of a special diet can prevent mental retardation. Because of this obvious benefit and the low cost of testing for this disorder, all states in the United States and many other countries require newborns to be tested for PKU.

Predictive testing for colorectal cancer and breast cancer also may be beneficial for at-risk people, because finding these cancers early improves the chances for successful treatment. Patients with genes that predispose to cancer may require more aggressive treatment than do patients with sporadically arising cancers. In these diseases, genetic testing provides clear benefits.

Another set of concerns are related to the accuracy of genetic tests. For many genetic diseases, the only predictive tests available are those that identify a *predisposing* mutation in DNA, but many genetic diseases may be caused by dozens or hundreds of different mutations. Probes that detect common mutations can be developed, but they won't detect rare mutations and will give a false negative result. Short of sequencing the entire gene—which is expensive and time consuming—there is no way to identify all predisposed persons. These questions and concerns are currently the focus of intense debate by ethicists, physicians, scientists, and patients.

www.whfreeman.com/pierce More information on genetic testing

Gene Therapy

Perhaps the ultimate application of recombinant DNA technology is **gene therapy,** the direct transfer of genes into humans to treat disease. When the first recombinant DNA experiments with bacteria were announced, many researchers recognized the potential for using this new technology in the treatment of patients with genetic diseases. But, before recombinant DNA could be used on humans, a number of difficult obstacles had to be overcome. The genes responsible for particular genetic diseases needed to be located and cloned, and special vectors had to be developed that would reliably and efficiently deliver genes to human cells.

In 1990, gene therapy became reality. W. French Anderson and his colleagues at the U.S. National Institutes of Health (NIH) transferred a functional gene for adenosine deaminase to a young girl with severe combined immunodeficiency disease, an autosomal recessive condition that produces impaired immune function.

Today, thousands of patients have received gene therapy, and many clinical trials are underway. Gene therapy is being used to treat genetic diseases, cancer, heart disease, and even some infectious diseases such as AIDS. All of these

Table 18.6 Vectors used in gene therapy

Vector	Advantages	Disadvantages
Retrovirus	Efficient transfer	Transfers DNA only to dividing cells, inserts randomly; risk of producing wild-type viruses
Adenovirus	Transfers to nondividing cells	Causes immune reaction
Adeno-associated virus	Does not cause immune reaction	Holds small amount of DNA; hard to produce
Herpes virus	Can insert into cells of nervous system; does not cause immune reaction	Hard to produce in large quantities
Lentivirus	Can accommodate large genes	Safety concerns
Liposomes and other lipid-coated vectors	No replication; does not stimulate immune reaction	Low efficiency
Direct injection	No replication; directed toward specific tissues	Low efficiency; does not work well with in some tissues
Pressure treatment	Safe, because tissues are treated outside the body and then transplanted into the patient	Most efficient with small DNA molecules
Gene gun (DNA coated on small gold particles and shot into tissue)	No vector required	Low efficiency

Source: After E. Marshall, Gene therapy's growing pains, *Science* 269(1995):1050–1055.

therapies depend on an introduced gene's ability to produce a therapeutic protein. A number of different methods for transferring genes into human cells are currently under development. Commonly used vectors include genetically modified retroviruses, adenoviruses, and adeno-associated viruses (Table 18.6). One method of gene transfer is to remove cells (such as white blood cells) from a patient's body, add viruses containing recombinant genes, and then reintroduce the cells back into the patient's body. In other cases, vectors are injected directly into the body.

In spite of the growing number of clinical trials for gene therapy, significant problems remain in transferring foreign genes into human cells, getting them expressed, and limiting immune responses to the gene products and the vectors used to transfer the genes to the cells. There are also heightened concerns about safety, especially after the death in 1999 of a patient participating in a gene-therapy trial who had a fatal immune reaction after he was injected with a viral vector carrying a gene to treat his metabolic disorder. Despite this setback, gene-therapy research has moved ahead. Unequivocal results demonstrating positive benefits from gene therapy for a severe combined immunodeficiency disease and for head and neck cancer were announced in 2000.

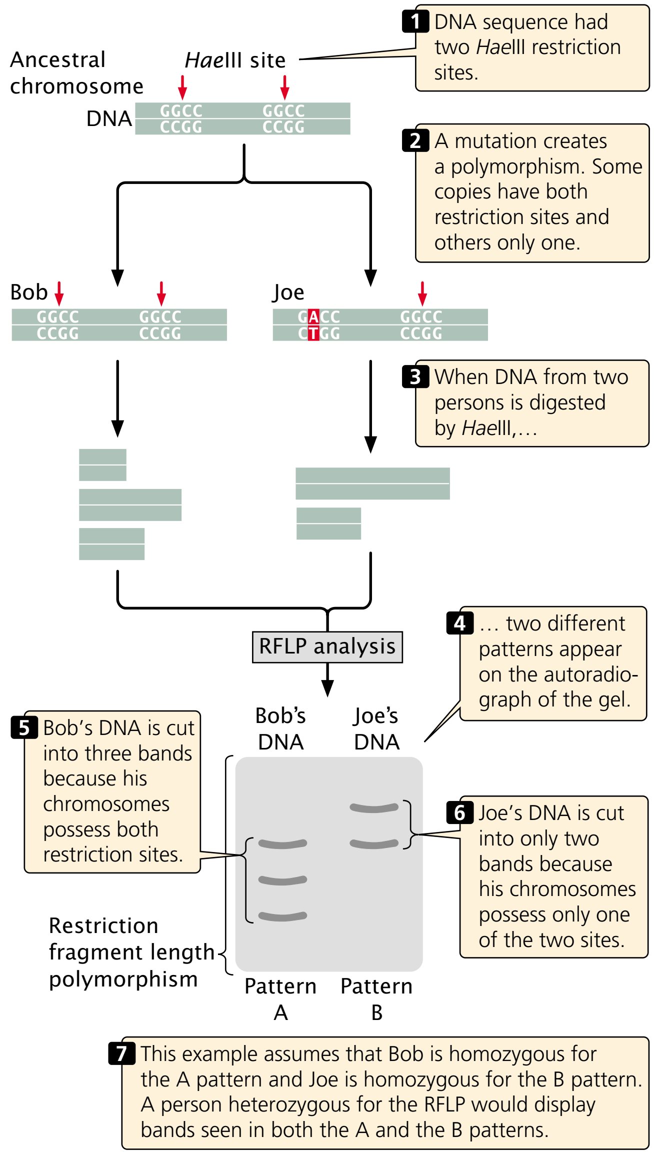
Gene therapy conducted to date has targeted only nonreproductive, somatic cells. Correcting a genetic defect in these cells (termed *somatic gene therapy*) may provide positive benefits to patients but will not affect the genes of future generations. Gene therapy that alters reproductive, or germ-line, cells (termed *germ-line gene therapy*) is technically possible but raises a number of significant ethical issues, because it has the capacity to alter the gene pool of future generations.

Concepts

Gene therapy is the direct transfer of genes into humans to treat disease. Gene therapy was first successfully implemented in 1990 and is now being used to treat genetic diseases, cancer, and infectious diseases.

Gene Mapping

A significant contribution of recombinant DNA technology has been to provide numerous genetic markers that can be used in gene mapping. One group of markers used in gene mapping comprises **restriction fragment length polymorphisms** (RFLPs, pronounced **rif**flips). RFLPs are variations (polymorphisms) in the patterns of fragments produced when DNA molecules are cut with the same restriction enzyme. If DNA from two persons is cut with the same restriction enzyme and different patterns of fragments are produced (FIGURE 18.26), these persons must possess differences in their DNA sequences. These differences are inherited and can be used in mapping, similar to the way in which allelic differences are used to map conventional genes.



18.26 **Restriction fragment length polymorphisms are genetic markers that can be used in mapping.**

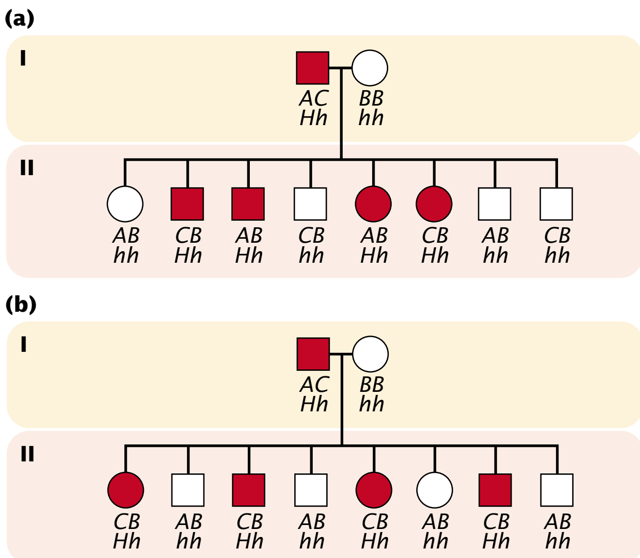
Traditionally, gene mapping has relied on the use of genetic differences that produce easily observable phenotypic differences. Unfortunately, because most traits are influenced by multiple genes and the environment, the number of traits with a simple genetic basis suitable for use in mapping is limited. RFLPs provide a large number of genetic markers that can be used in mapping.

To illustrate mapping with RFLPs, let's again consider Huntington disease. As mentioned earlier, this disease is caused by an autosomal dominant gene but, until recently, the chromosomal location of the gene was unknown. A team of scientists led by James Gusella (see introduction to Chapter 5) set out to determine the location of the Huntington gene, in the hope that, when the gene was found, its biochemical basis could be determined and possible treatments might be suggested. DNA was collected from members of the largest known family with Huntington disease, who live near Lake Maracaibo in Venezuela.

The basic strategy employed in the search for the Huntington-disease gene and a number of other human disease-causing genes is to look for coinheritance of the disease-causing gene and an RFLP with a known chromosomal location. If the disease gene and the RFLP have been inherited together, they must be physically linked.

This approach is summarized in FIGURE 18.27, which illustrates the coinheritance of two traits: (1) the presence or absence of Huntington disease and (2) the type of restriction pattern produced (pattern A or C). In the family shown, the father is heterozygous for Huntington disease (*Hh*) and is also heterozygous for a restriction pattern (*AC*). From the father, each child inherits either a Huntington-disease allele (*H*) or a normal allele (*h*); any child inheriting the Huntington-disease allele develops the disease, because it is an autosomal dominant disorder. The child also inherits one of the two RFLP alleles from the father, either *A* or *C*, which produces the corresponding RFLP pattern. In FIGURE 18.27a, there is no correspondence between the inheritance of the RFLP pattern and the inheritance of the disease: children who have inherited Huntington disease (and therefore the *H* allele) from their father are equally likely to have inherited the A or C RFLP pattern. Because, in this case, the *H* allele and the RFLP alleles segregate randomly, we know that they are not closely linked.

FIGURE 18.27b, on the other hand, shows that every child who inherits the C pattern from the father also inherits Huntington disease (and therefore the *H* allele), because the locus for the RFLP is closely linked to the locus for the disease-causing gene. The chromosomal location of the RFLP provides a general indication of the disease-causing locus. An examination of the cosegregation of other RFLPs from the same region can precisely determine the location of the gene. Actual RFLP patterns and part of the Huntington-disease gene are shown in FIGURE 18.28.



18.27 **Restriction fragment length polymorphisms can be used to detect linkage.** In this hypothetical pedigree, the father and half of the children are affected (red circles and squares) with Huntington disease, an autosomal dominant disease. The father is heterozygous (*Hh*) and will pass the chromosome with the Huntington gene to approximately half of his offspring. The father is also heterozygous for RFLP alleles *A* and *C*; each child receives one of these two alleles from the father. The mother is homozygous for RFLP allele *B*, so all children receive the *B* allele from her. (a) In this case, there is no correspondence between the inheritance of the RFLP allele and inheritance of the disease—children with the disease are just as likely to carry the *A* allele as they are the *C* allele. Thus the disease gene and RFLP alleles segregate independently and are not closely linked. (b) In this case, there is a close correspondence between the inheritance of the RFLP alleles and the presence of the disease—every child who inherits the *C* allele from the father also has the disease. This correspondence indicates that the RFLP is closely linked to the Huntington gene.

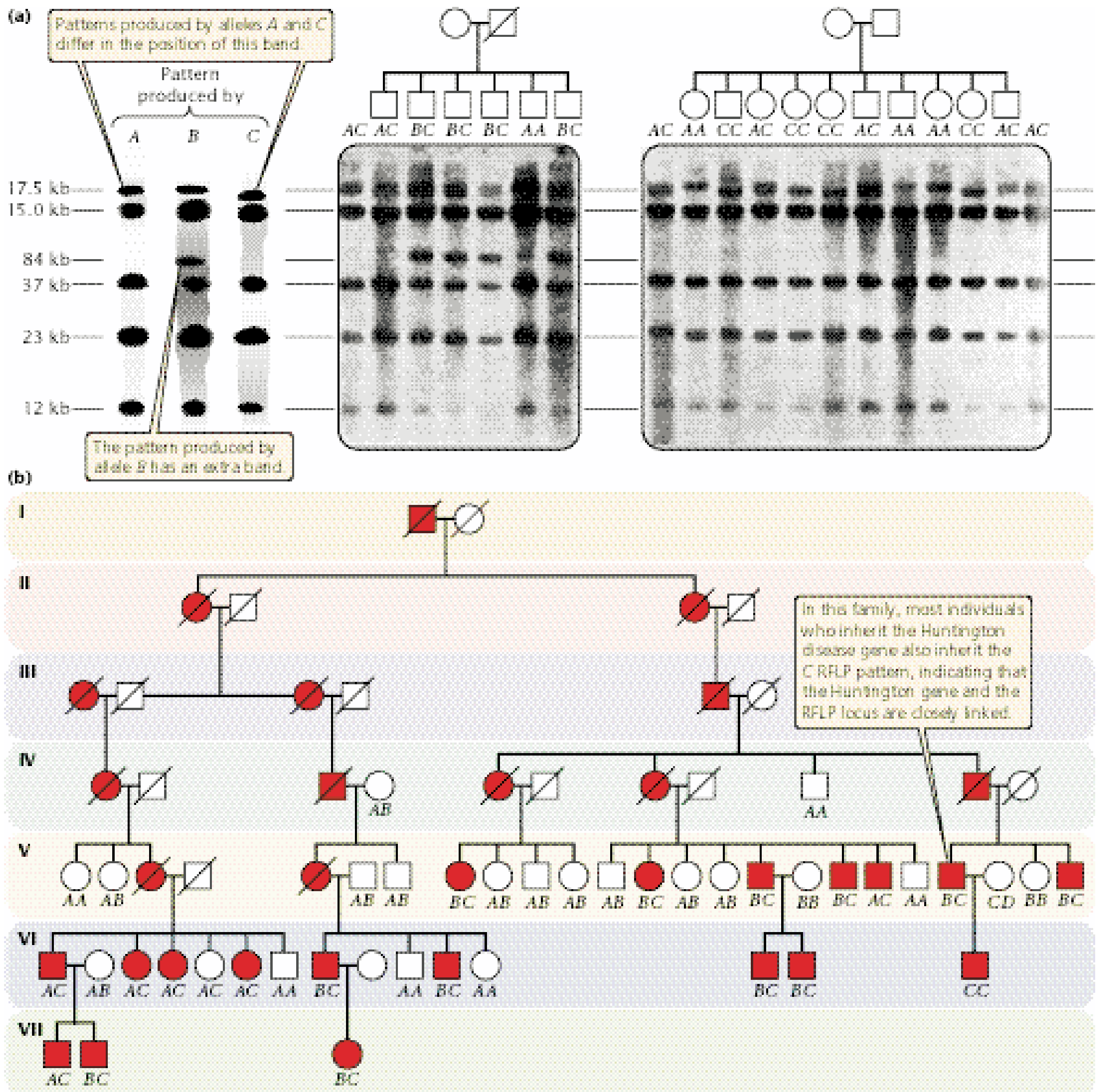
DNA Fingerprinting

Restriction fragment length polymorphisms are often found in noncoding regions of DNA and are therefore frequently quite variable in humans. Two randomly chosen people will differ at many RFLPs and, if enough RFLPs are examined, no two people (with the exception of identical twins) will be exactly the same. The use of DNA sequences to identify a person, called **DNA fingerprinting,** is a powerful tool for criminal investigations and other forensic applications.

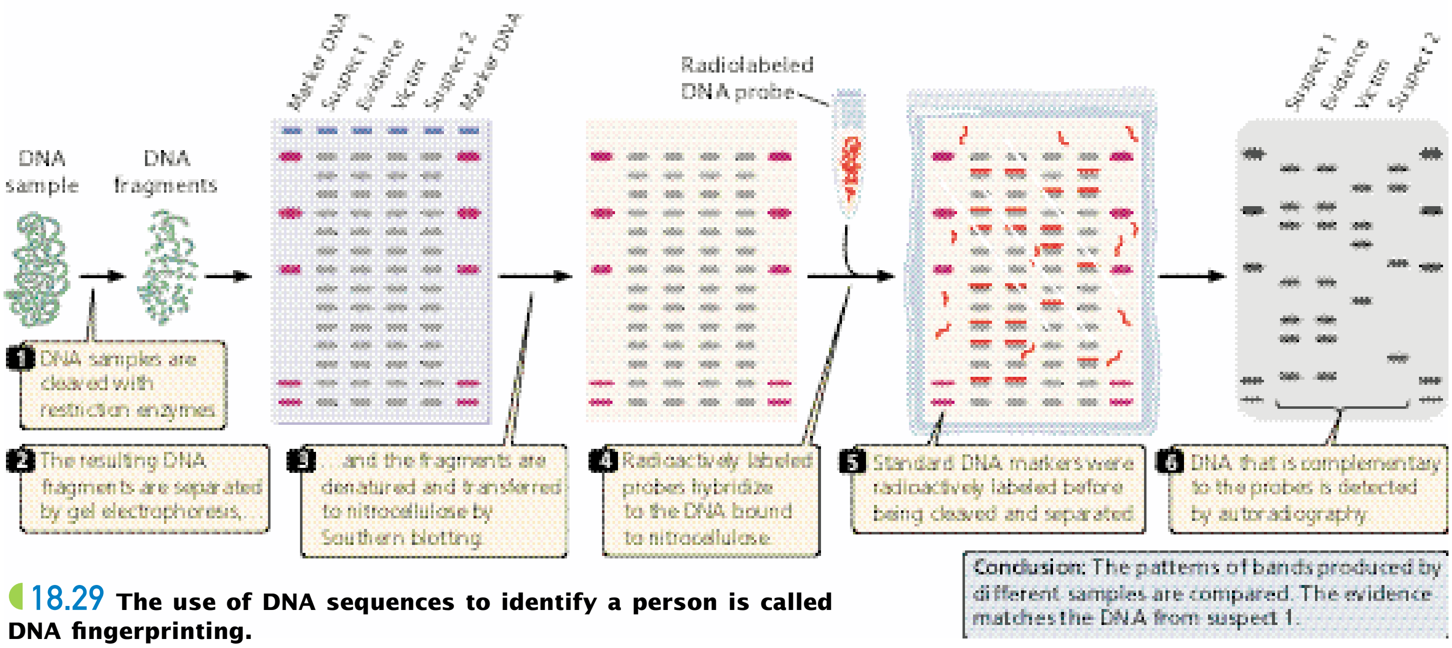
In a typical application, DNA fingerprinting might be used to confirm that a suspect was present at the scene of a crime (FIGURE 18.29). A sample of DNA from blood, semen, hair, or other body tissue is collected from the crime scene. If the sample is very small, PCR can be used to amplify it so that enough DNA is available for testing. Additional DNA samples are collected from one or more suspects.

Each DNA sample is cut with one or more restriction enzymes, and the resulting DNA fragments are separated by gel electrophoresis. The fragments in the gel are denatured and transferred to nitrocellulose paper by Southern blotting. One or more radioactive probes is then hybridized to the nitrocellulose and detected by autoradiography. The pattern of bands produced by DNA from the sample collected at the crime scene is then compared with the patterns produced by DNA from the suspects.

The probes used in DNA fingerprinting detect highly variable regions of the genome; so the chances of DNA from two people producing exactly the same banding pattern is low. When several probes are used in the analysis, the probability that two people have the same set of patterns becomes vanishingly small (unless they are identical twins). A match between the sample from the crime scene and one



18.28 Restriction fragment length polymorphisms were used to map the Huntington-disease gene to chromosome 8. (a) Autoradiograph showing different banding patterns revealed by cutting the DNA with *Hin*dIII and using a probe to chromosome 8. The RFLP *A* allele produces five bands. The *C* allele also produces five bands, but the first band is just below the first band produced by the *A* allele; *AC* heterozygotes have both bands, which are very close together. The *B* allele has an extra band representing a 4.9-kb fragment. (b) Partial pedigree of large family from Lake Maracaibo. Red symbols represent family members with Huntington disease; the RFLP genotypes are indicated below each person represented in the pedigree. Notice that persons with the disease carry the *C* allele, indicating that the sequences on chromosome 8 revealed by the probe are closely linked to the Huntington-disease gene.



18.29 The use of DNA sequences to identify a person is called DNA fingerprinting.

from the suspect can provide evidence that the suspect was present at the scene of the crime.

The probes most commonly used in DNA fingerprinting are complementary to short sequences repeated in tandem that are widely found in the human genome (see p. 000 in Chapter 11). People vary greatly in the number of copies of these repeats; thus, these polymorphisms are termed **variable number of tandem repeats** (VNTRs).

Since its introduction in the 1980s, DNA fingerprinting has helped convict a number of suspects in murder and rape cases. Suspects in other cases have been proved innocent when their DNA failed to match that from the crime scenes. Initially there was some controversy over calculating the odds of a match (the probability that two people could have the same pattern) and concerns about quality control (such as the accidental contamination of samples and the reproducibility of results) in laboratories where DNA analysis is done. In spite of the controversy, DNA fingerprinting has become an important tool in forensic investigations.

DNA fingerprinting has also been used to provide information about the relationships and sources of other organisms. For example, DNA fingerprinting was used to determine that several samples of anthrax mailed to different people in 2001 were all from the same source.

Concepts

RFLPs are variations in the pattern of fragments produced by restriction enzymes, which reveal variations in DNA sequences. They are used extensively in gene mapping. DNA fingerprinting detects genetic differences among people by using probes for highly variable regions of chromosomes.

www.whfreeman.com/pierce More information on DNA fingerprinting

Concerns About Recombinant DNA Technology

In 1971, as researchers were planning some of the first gene-cloning experiments, in which they planned to transfer genes from tumor viruses to *E. coli,* several scientists raised concerns about the safety of such experiments. *E. coli* is present in the human intestinal tract, and these scientists questioned whether it might be possible for recombinant bacteria to escape from the laboratory and infect people, eventually transferring tumor-causing genes to people. The risks were thought to be small, but the real hazards were quite unknown.

When the first experiments using recombinant DNA were performed in 1973, concerns about risks associated with recombinant technology were heightened. Although no hazard had been demonstrated, a number of potential dangers could be envisioned. In July 1974, leading molecular biologists published a letter in *Science* urging scientists to stop conducting certain types of potentially hazardous recombinant DNA experiments until their risks could be evaluated. In February 1975, a group of more than 100 molecular biologists met and agreed that some restrictions on recombinant DNA research were warranted. They formulated a series of recommendations concerning the types of recombinant DNA experiments that should be prohibited.

The National Institutes of Health then appointed a committee to develop guidelines for recombinant DNA research. Different types of cloning experiments were considered to have different degrees of risk, and more precautions were required for the more "risky" experiments. The

Recombinant DNA Advisory Committee was established to oversee the safety of this work in the United States, and similar committees were established in Europe.

After years of experience with recombinant DNA experiments, the initial concerns about risks were turned out to be largely unfounded, and the NIH guidelines have now been significantly relaxed. Current controversy about recombinant DNA technology revolves largely around the release of genetically modified organisms into the environment and the application of recombinant DNA technology to humans.

Connecting Concepts Across Chapters

This chapter has focused on recombinant DNA technology, a set of methods to isolate, study, and manipulate DNA sequences. Before the development of this technology, geneticists were forced to study genes by examining the phenotypes produced by the genes under study. The power of recombinant DNA technology is that it allows geneticists to read and alter genetic information directly, leading to an entirely new approach to the study of heredity in which genes are studied by altering DNA sequences and observing the associated change in phenotype.

A major theme of this chapter has been that working at the molecular level requires special approaches because DNA and other molecules are too small to see and manipulate directly. A number of recombinant DNA techniques are available and can be mixed and matched in different combinations or strategies; the particular set of methods used depends both on the sequences being manipulated and on the ultimate goal of the researcher.

Mastering the information in this chapter requires an understanding of material presented in many of the preceding chapters, particularly those on molecular genetics. A detailed understanding of DNA structure (Chapter 10), replication (Chapter 12), and the genetic code (Chapter 15) are essential for grasping the details of recombinant DNA technology. Knowledge of bacterial and viral genetics (Chapter 8) is helpful, because much of gene cloning takes place in bacteria, and plasmids and viruses are commonly used as cloning vectors. Knowledge of gene regulation (Chapter 16) is useful for understanding expression vectors and recombinant DNA applications where proteins are produced.

The information presented in this chapter will complement and enhance much of the material presented in the remaining chapters of the book. Chapter 19 deals with the use of recombinant DNA technology to compare the organization, content, and expression of genomes of different organisms.

CONCEPTS SUMMARY

- Recombinant DNA technology is a set of molecular techniques for locating, cutting, joining, analyzing, and altering DNA sequences and for inserting the sequences into a cell.
- Restriction endonucleases are enzymes that make double-stranded cuts in DNA at specific base sequences.
- DNA fragments can be separated with the use of gel electrophoresis and visualized by staining the gel with a dye that is specific for nucleic acids or by labeling the fragments with a radioactive or chemical tag.
- Individual genes can be studied by transferring DNA fragments from a gel to nitrocellulose or nylon and applying complementary probes.
- Gene cloning refers to placing a gene or a DNA fragment into a bacterial cell, where it will be multiplied as the cell divides.
- Plasmids, small circular pieces of DNA, are often used as vectors to ensure that a cloned gene is stable and replicated within the recipient cells.
- Bacteriophage λ offers several advantages over plasmids: it can hold larger fragments of foreign DNA and transfers DNA to cells with higher efficiency.
- Cosmids, which combine properties of plasmids and phage vectors, hold even larger amounts of foreign DNA. Yeast and bacterial artificial chromosomes can accommodate large inserts more than 100,000 bp in length.
- Expression vectors contain promoters, ribosome-binding sites, and other sequences necessary for foreign DNA to be transcribed and translated.
- Genes can be isolated by creating a DNA library, a set of bacterial colonies or viral plaques that each contain a different cloned fragment of DNA. A genomic library contains the entire genome of an organism, cloned as a set of overlapping fragments; a cDNA library contains DNA fragments complementary to all the different mRNAs in a cell.
- DNA libraries can be screened with probes complementary to particular genes or DNA fragments in the library can be cloned into an expression vector and screened by looking for the associated protein product.
- Genes can also be located by chromosome walking, in which a neighboring gene is used to make a probe; a genomic library is screened with this probe to find a clone that overlaps the gene. A probe is made from the end of this clone, and the probe is used to screen the library for a second clone that overlaps the first. The process is continued until the gene of interest is reached.
- The cloning strategy depends on the purpose of the cloning experiment, what is known about the gene, the size of the

gene to be cloned, the size of the genome from which it is isolated, and the organism into which it will be cloned.

- The polymerase chain reaction is a method for amplifying DNA enzymatically without cloning. A solution containing DNA is heated, so that the two DNA strands separate, and then quickly cooled, allowing primers to attach to the template DNA. The solution is then heated again, and DNA polymerase synthesizes new strands from the primers. Each time the cycle is repeated, the amount of DNA doubles.
- In situ hybridization can be used to determine the chromosomal location of a gene and the distribution of the mRNA produced by a gene. DNA footprinting reveals the nucleotides that are covered by DNA-binding proteins. Site-directed mutagenesis can be used to produce mutations at specific sites in DNA, allowing genes to be tailored for a particular purpose. Transgenic animals, produced by injecting DNA into fertilized eggs, contain foreign DNA that is integrated into a chromosome. Knockout mice are transgenic mice that have a normal gene disabled.
- Recombinant DNA technology has many applications, including not only the production of pharmaceuticals and other biological substances in bacteria but also the creation of bacteria that are genetically engineered for economically or medically important tasks. It is also being used in agriculture to transfer particular traits, such as disease and pest resistance, to crop plants. Transgenic domestic animals can be produced with desirable traits. Oligonucleotide drugs—short nucleotide sequences for treating diseases—are another application of recombinant DNA technology.
- In gene therapy, diseases are being treated by altering the genes of human cells.
- Restriction fragment length polymorphisms and variable number tandem repeats facilitate gene mapping by making available numerous genetic markers and are being used to identify people by their DNA sequences (DNA fingerprinting).

IMPORTANT TERMS

recombinant DNA technology (p. 000)
genetic engineering (p. 000)
biotechnology (p. 000)
restriction enzyme (p. 000)
restriction endonuclease (p. 000)
cohesive end (p. 000)
gel electrophoresis (p. 000)
end labeling (p. 000)
autoradiography (p. 000)
probe (p. 000)
Southern blotting (p. 000)
Northern blotting (p. 000)
Western blotting (p. 000)
gene cloning (p. 000)
cloning vector (p. 000)
cosmid (p. 000)
expression vector (p. 000)
shuttle vector (p. 000)
yeast artificial chromosome (YAC) (p. 000)
bacterial artificial chromosome (BAC) (p. 000)
Ti plasmid (p. 000)
DNA library (p. 000)
genomic library (p. 000)
cDNA library (p. 000)
chromosome walking (p. 000)
cloning strategy (p. 000)
polymerase chain reaction (PCR) (p. 000)
Taq polymerase (p. 000)
in situ hybridization (p. 000)
DNA footprinting (p. 000)
site-directed mutagenesis (p. 000)
oligonucleotide-directed mutagenesis (p. 000)
transgene (p. 000)
knockout mice (p. 000)
gene therapy (p. 000)
restriction fragment length polymorphism (RFLP) (p. 000)
DNA fingerprinting (p. 000)
variable number of tandem repeats (VNTRs) (p. 000)

Worked Problems

1. A molecule of double-stranded DNA that is 5 million base pairs long has a base composition that is 62% G + C. How many times, on average, are the following restriction sites likely to be present in this DNA molecule?

(a) *Bam*HI (recognition sequence = GGATCC)
(b) *Hind*III (recognitions sequence = AAGCTT)
(c) *Hpa*II (recognition sequence = CCGG)

• Solution

The percentages of G and C are equal in double-stranded DNA; so, if G + C = 62%, then %G = %C = 62%/2 = 31%. The percentage of A + T = (100% − G + C) = 48%, and %A = %T = 48%/2 = 24%. To determine the probability of finding a particular base sequence, we use the multiplicative rule, multiplying together the probably of finding each base at a particular site.

(a) The probability of finding the sequence GGATCC = $0.31 \times 0.31 \times 0.24 \times 0.24 \times 0.31 \times 0.31 = 0.00053$. To determine the average number of recognition sequences in a 5-million-base-pair piece of DNA, we multiply 5,000,000 bp × 0.00053 = 2659.5 recognition sequences.

(b) The number of AAGCTT recognition sequences is $0.24 \times 0.24 \times 0.31 \times 0.31 \times 0.24 \times 0.24 \times 5{,}000{,}000 = 1594$ recognition sequences.

(c) The number of CCGG recognition sequences is $0.31 \times 0.31 \times 0.31 \times 0.31 \times 5{,}000{,}000 = 46{,}176$ recognition sequences.

2. A protein has the following amino acid sequence:

Met-Leu-Arg-Ser-Arg-Met-Tyr-Trp-Asp-His-Glu-Thr

You wish to make a set of probes to screen a cDNA library for the sequence that encodes this protein. Your probes should be at least 18 nucleotides in length.

(a) Which amino acids in the protein should be used so that the smallest number of probes is required? (Consult the genetic code in Figure 15.12.)

(b) How many different sequences must be synthesized to be certain that you will find the correct cDNA sequence that specifies the protein?

• **Solution**

We first write out all the codons that can specify all the amino acids in the protein, using the genetic code in Figure 15.12 (see table below).

(a) The 18-bp region encoding amino acids 6 through 11 should be used, because this region has the fewest number of possible codons.

(b) For amino acids 6 through 11, there is one possible codon for Met, two for Tyr, one for Trp, two for Asp, two for His, and two for Glu. Thus $1 \times 2 \times 1 \times 2 \times 2 \times 2 = 16$ possible sequences must be synthesized to locate the gene.

1	2	3	4	5	6	7	8	9	10	11	12
Met	Leu	Arg	Ser	Arg	Met	Tyr	Trp	Asp	His	Glu	Thr
AUG	UUA	CGU	UCU	CGU	AUG	UAU	UGG	GAU	CAU	GAA	ACU
	UUG	CGC	UCC	CGC		UAC		GAC	CAC	GAG	ACC
	CUU	CGA	UCA	CGA							ACA
	CUC	CGG	UCG	CGG							ACG
	CUA		AGU	AGA							
	CUG		AGC	AGG							

The New Genetics
MINING GENOMES

Recombinant DNA Project

This exercise casts you in the role of research geneticist. Your job is to plan a project to clone a specific gene into a plasmid vector, including the selection of the restriction enzymes and vector that you will use. You will utilize the Web sites of some of the major suppliers of biotchnology reagents in the process.

COMPREHENSION QUESTIONS

1. List some of the effects and applications of recombinant DNA technology.
2. What common feature is seen in the sequences recognized by type II restriction enzymes?
3. What role do restriction enzymes play in bacteria? How do bacteria protect their own DNA from the action of restriction enzymes?
*4. Explain how gel electrophoresis is used to separate DNA fragments of different lengths.
*5. After DNA fragments are separated by gel electrophoresis, how can they be visualized?
6. What is the purpose of Southern blotting? How is it carried out?
*7. What are the differences between Southern, Northern, and Western blotting?
*8. Give three important characteristics of cloning vectors.
9. Briefly describe four different methods for inserting foreign DNA into plasmids, giving the strengths and weaknesses of each.
10. How are plasmids transferred into bacterial cells?
*11. Briefly explain how an antibiotic-resistance gene and the *lacZ* gene can be used as markers to determine which cells contain a particular plasmid.
12. How are genes inserted into bacteriophage λ vectors? What advantages do λ vectors have over plasmids?
*13. What is a cosmid? What are the advantages of using cosmids as gene vectors?
14. What are yeast artificial chromosomes and shuttle vectors? When are these cloning vectors used?
*15. How does a genomic library differ from a cDNA library? How is each created?
16. How are probes used to screen DNA libraries? Explain how a synthetic probe can be prepared when the protein product of a gene is known.
17. Explain how chromosome walking can be used to find a gene.
18. Discuss some of the considerations that must go into developing an appropriate cloning strategy.

*19. Briefly explain how the polymerase chain reaction is used to amplify a specific DNA sequence. What are some of the limitations of PCR?

*20. Briefly explain in situ hybridization, giving some applications of this technique.

21. What is DNA footprinting?

22. Briefly explain how site-directed mutagenesis is carried out.

*23. What are knockout mice, how are they produced, and for what are they used?

24. Describe how RFLPs can be used in gene mapping.

*25. What is DNA fingerprinting? What types of sequences are examined in DNA fingerprinting?

26. What is gene therapy?

27. As the first recombinant DNA experiments were being carried out, there was concern among some scientists about this research. What were these concerns and how were they addressed?

APPLICATION QUESTIONS AND PROBLEMS

*28. Suppose that a geneticist discovers a new restriction enzyme in the bacterium *Aeromonas ranidae*. This restriction enzyme is the first to be isolated from this bacterial species. Using the standard convention for abbreviating restriction enzymes, give this new restriction enzyme a name (for help, see footnote to Table 18.2).

29. How often, on average, would you expect a type II restriction endonuclease to cut a DNA molecule if the recognition sequence for the enzyme had 5 bp? (Assume that the four types of bases are equally likely to be found in the DNA and that the bases in a recognition sequence are independent.) How often would the endonuclease cut the DNA if the recognition sequence had 8 bp?

*30. A microbiologist discovers a new type II restriction endonuclease. When DNA is digested by this enzyme, fragments that average 1,048,500 bp in length are produced. What is the most likely number of base pairs in the recognition sequence of this enzyme?

31. Will restriction sites for an enzyme that has 4 bp in its restriction site be closer together, farther apart, or similarly spaced, on average, compared with those of an enzyme that has 6 bp in its restriction site? Explain your reasoning.

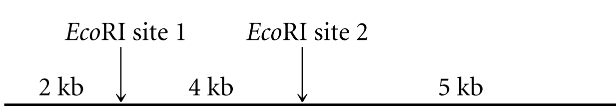
*32. About 30% of the base pairs in a human DNA molecule are AT. If the human genome has 3 billion base pairs of DNA, about how many times will the following restriction sites be present?

(a) *Bam*HI (restriction site = 5′–GGATCC–3′)

(b) *Eco*RI (restriction site = 5′–GAATTC–3′)

(c) *Hae*III (restriction site = 5′–GGCC–3′)

*33. Restriction mapping of a linear piece reveals the following *Eco*RI restriction sites.



(a) This piece of DNA is cut by *Eco*RI, the resulting fragments are separated by gel electrophoresis, and the gel is stained with ethidium bromide. Draw a picture of the bands that will appear on the gel.

(b) If a mutation that alters *Eco*RI site 1 occurs in this piece of DNA, how will the banding pattern on the gel differ from the one that you drew in part *a*?

(c) If mutations that alter *Eco*RI sites 1 and 2 occur in this piece of DNA, how will the banding pattern on the gel differ from the one that you drew in part *a*?

(d) If a 1000-bp insertion occurred between the two restriction sites, how would the banding pattern on the gel differ from the one that you drew in part *a*?

(e) If a 500-bp deletion occurred between the two restriction sites, how would the banding pattern on the gel differ from the one that you drew in part *a*?

*34. Which vectors (plasmid, phage λ, cosmid) can be used to clone a continuous fragment of DNA with the following lengths?

(a) 4 kb.

(b) 20 kb.

(c) 35 kb.

35. A geneticist uses a plasmid for cloning that has a gene that confers resistance to penicillin and the *lacZ* gene. The geneticist inserts a piece of foreign DNA into a restriction site that is located within the *lacZ* gene and transforms bacteria with the plasmid. Explain how the geneticist can identify bacteria that contain a copy of a plasmid with the foreign DNA.

*36. Suppose that you have just graduated from college and have started working at a biotechnology firm. Your first job assignment is to clone the pig gene for the hormone prolactin. Assume that the pig gene for prolactin has not yet been isolated, sequenced, or mapped; however, the mouse gene for prolactin has been cloned and the amino acid sequence of mouse prolactin is known. Briefly explain two different strategies that you might use to find and clone the pig gene for prolactin.

37. A genetic engineer wants to isolate a gene from a scorpion that encodes the deadly toxin found in its stinger, with the

ultimate purpose of transferring this gene to bacteria and producing the toxin for use as a commercial pesticide. Isolating the gene requires a DNA library. Should the genetic engineer create a genomic library or a cDNA library? Explain your reasoning.

*38. A protein has the following amino acid sequence:

Met-Tyr-Asn-Val-Arg-Val-Tyr-Lys-
Ala-Lys-Trp-Leu-Ile-His-Thr-Pro

You wish to make a set of probes to screen a cDNA library for the sequence that encodes this protein. Your probes should be at least 18 nucleotides in length.

(a) Which amino acids in the protein should be used to construct the probes so that the least degeneracy results? (Consult the genetic code in Table 15.12.)

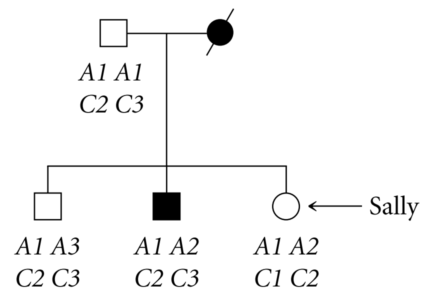
(b) How many different probes must be synthesized to be certain that you will find the correct cDNA sequence that specifies the protein?

*39. A gene in mice is discovered that is similar to a gene in yeast. How might it be determined whether this gene is essential for development in mice?

*40. A hypothetical disorder called G syndrome is an autosomal dominant disease characterized by visual, skeletal, and cardiovascular defects. The disorder appears in middle age. Because the symptoms of the disorder are variable, the disorder is difficult to diagnose. Early diagnosis is important, however, because the cardiovascular symptoms can be treated if the disorder is recognized early. The gene for G syndrome is known to reside on chromosome 7, and it is closely linked to two RFLPs on the same chromosome, one at the *A* locus and one at the *C* locus. The genes at the *G*, *A*, and *C* loci are very close together, and there is little crossing over between them. The following RFLP alleles are found at the *A* and *C* loci:

A locus: *A1, A2, A3, A4*
C locus: *C1, C2, C3*

Sally, shown in the following pedigree, is concerned that she might have G syndrome. Her deceased mother had G syndrome, and she has a brother with the disorder. A geneticist genotypes Sally and her immediate family for the *A* and *C* loci and obtains the genotypes shown on the pedigree.



(a) Assume that there is no crossing over between the *A*, *C*, and *G* loci. Does Sally carry the gene that causes G syndrome? Explain why or why not?
(b) Draw the arrangement of the *A*, *C*, and *G* alleles on the chromosomes for all members of the family.

CHALLENGE QUESTIONS

41. Suppose that you are hired by a biotechnology firm to produce a giant strain of fruit flies, by using recombinant DNA technology, so that genetics students will not be forced to strain their eyes when looking at tiny flies. You go to the library and learn that growth in fruit flies is normally inhibited by a hormone called shorty substance P (SSP). You decide that you can produce giant fruit flies if you can somehow turn off the production of SSP. SSP is synthesized from a compound called XSP in a single-step reaction catalyzed by the enzyme *runtase*:

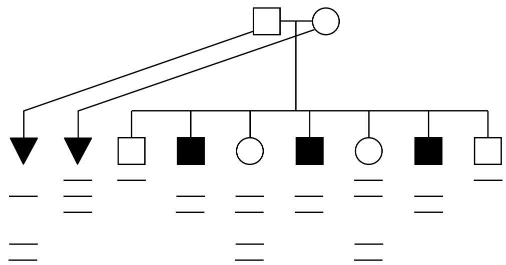
$$\text{XSP} \xrightarrow[\text{runtase}]{} \text{SSP}$$

A researcher has already isolated cDNA for runtase and has sequenced it, but the location of the runtase gene in the *Drosophila* genome is unknown.

In attempting to devise a strategy for turning off the production of SSP and producing giant flies by using standard recombinant DNA techniques, you discover that deleting, inactivating, or otherwise mutating this DNA sequence in *Drosophila* turns out to be extremely difficult. Therefore you must restrict your genetic engineering to gene augmentation (adding new genes to cells). Describe the methods that you will use to turn off SSP and produce giant flies by using recombinant DNA technology.

42. A rare form of polydactyly (extra fingers and toes) in humans is due to an X-linked recessive gene, whose chromosomal location is unknown. Suppose a geneticist studies the family whose pedigree is shown here. She isolates DNA from each member of this family, cuts the DNA with a restriction enzyme, separates the resulting fragments by gel electrophoresis, and transfers the DNA to nitrocellulose by Southern blotting. She then hybridizes the nitrocellulose with a cloned DNA sequence that comes from the X chromosome. The pattern of bands

that appear on the autoradiograph is shown below each person in the pedigree.



(a) For each person in the pedigree, give his or her genotype for RFLPs revealed by the probe. (Remember that males are hemizygous for X-linked genes, and females can be homozygous or heterozygous.)

(b) Is there evidence for close linkage between the probe sequence and the X-linked gene for polydactyly? Explain your reasoning.

(c) How many of the daughters in the pedigree are likely to be carriers of X-linked polydactyly? Explain your reasoning.

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19 Genomics

A young girl with leprosy, a disease caused by the bacteria *Mycobacterium leprae.* Leprosy causes characteristic patches of skin with a loss of sensation, as seen on the face of this young patient; if untreated, leprosy may lead to nerve damage and disfigurement. Genomic studies reveal the genome of *M. leprae* has undergone extensive gene loss, mutation, and rearrangement over evolutionary time. (WHO/OMS.)

The Decaying Genome of *Mycobacterium leprae*

Leprosy, one of the most feared diseases of history, was well known in ancient times and is still a major public health problem today; from 2 million to 3 million people are affected worldwide, and approximately 650,000 new cases are reported each year. In its severest form, leprosy causes paralysis, blindness, and disfigurement. Although human genes play some role in susceptibility to leprosy, the disease is caused by the bacterium *Mycobacterium leprae*, which infects cells of the nervous system and causes nerve damage, sensory loss, and disfigurement. In 1873, Armauer Hansen observed these bacteria in tissue samples taken from people with leprosy, but to this day no one has successfully cultured the bacterium in laboratory media, severely restricting the study of the disease agent.

In 2001, scientists in Britain and France determined the sequence of the entire genome of *M. leprae.* Comparing its genome with that of its close relative *M. tuberculosis* (the pathogen that causes tuberculosis) and other mycobacteria has been a source of important insight into the unique properties of this pathogen.

The genome of *M. leprae* is 3,268,203 bp in size, 1 million base pairs smaller than the genomes of other mycobacteria. In most bacterial genomes, the vast majority of the DNA encodes proteins—there is little noncoding DNA between genes. In contrast, only 50% of the DNA of *M. leprae* encodes proteins (Table 19.1), and *M. leprae* has 2300 fewer genes than *M. tuberculosis.* An incredible 27% of *M. leprae*'s genome consists of pseudogenes—nonfunctional copies of genes that have been inactivated by mutations. *M. leprae* has 1116 pseudogenes, whereas its close relative, *M. tuberculosis,* has just 6.

Table 19.1 Comparison of the genomes of *Mycobacterium leprae,* which causes leprosy, and *Mycobacterium tuberculosis,* which causes tuberculosis

Characteristics	*M. leprae*	*M. tuberculosis*
Genome size (bp)	3,268,203	4,411,532
Percentage of genome that encodes proteins	49.5%	90.8%
Protein-encoding genes (bp)	1604	3959
Pseudogenes (bp)	1116	6
Gene density (bp/gene)	2037	1114
Average length of gene (bp)	1011	1012

Source: S. T. Cole et al., Massive gene decay in the leprosy bacillus, *Nature* 409 (2001), p. 1007.

The reduced DNA content, fewer functional genes, and the large number of pseudogenes suggest that, evolutionarily, the genome of *M. leprae* has undergone massive decay through time, losing DNA and acquiring mutations that have inactivated many of its genes. Furthermore, the genome of *M. leprae* has undergone extensive rearrangement; comparison with the genome of *M. tuberculosis* has identified at least 65 gene segments that are arranged in different order and distribution.

The mechanisms responsible for gene decay and genomic rearrangement in *M. leprae* are not known, although the loss of proofreading ability in the bacterium's DNA polymerase III (the enzyme responsible for most bacterial DNA replication, see Chapter 12) may contribute to a high rate of mutation and the large number of pseudogenes. Because the leprosy bacterium resides in a highly specialized habitat (human nerve cells), it may have lost the need for many enzymatic functions found in other bacteria. When a function is no longer required for survival, genes encoding that function usually accumulate mutations and deletions.

Regardless of the mechanism for gene inactivation and loss, this genomic decay helps explain some of the bacterium's unique properties. Genes for many metabolic enzymes and structural proteins have been lost, which may explain why the bacterium cannot be cultured on synthetic media containing traditional carbon sources; it may also account for the bacterium's slow growth, with a doubling time of 14 days, compared with a doubling time of 20 minutes for *E. coli.*

A comparison of *M. leprae*'s genome with those of other related bacteria has identified a few unique genes that may contribute to its pathogenesis. The study of these genes has opened the door to an improved understanding of leprosy, better diagnostic tests, and the development of new drugs for the disease.

The information gleaned from sequencing the genome of *M. leprae* illustrates the power of genomics, which is the focus of this chapter. **Genomics** is the field of genetics that attempts to understand the content, organization, function, and evolution of genetic information contained in whole genomes. Genomics consists of two complementary fields: structural genomics and functional genomics. **Structural genomics** determines the organization and sequence of the genetic information contained within a genome, and **functional genomics** characterizes the function of sequences elucidated by structural genomics. A third area, **comparative genomics,** compares the gene content, function, and organization of genomes of different organisms.

The field of genomics is at the cutting edge of modern biology; information resulting from research in this field has made significant contributions to human health, agriculture, and numerous other areas. It has also provided gene sequences necessary for producing medically important proteins through recombinant DNA technology. Comparisons of genome sequences from different organisms are leading to a better understanding of evolution and the history of life.

We begin this chapter by examining genetic and physical maps and methods for sequencing entire genomes. Next, we explore functional genomics—how genes are identified in genomic sequences and how their functions are defined. Some of the genomes that have been sequenced are then examined in detail. We end the chapter by briefly considering the future of genomics.

Concepts

The field of genomics comprises structural genomics, which focuses on the content and organization of genomic information, and functional genomics, which attempts to understand the function of information in genomes. Comparative genomics compares the content and organization of genomes of different organisms. Genomics makes important contributions to human health, agriculture, biotechnology, and our understanding of evolution.

www.whfreeman.com/pierce Internet sources of information on leprosy

Structural Genomics

Structural genomics is concerned with sequencing and understanding the content of genomes. Often, one of the first steps in characterizing a genome is to prepare genetic and physical maps of its chromosomes. These maps provide information about the relative locations of genes, molecular markers, and chromosome segments, which are often essential for positioning chromosome segments and aligning stretches of sequenced DNA into a whole-genome sequence.

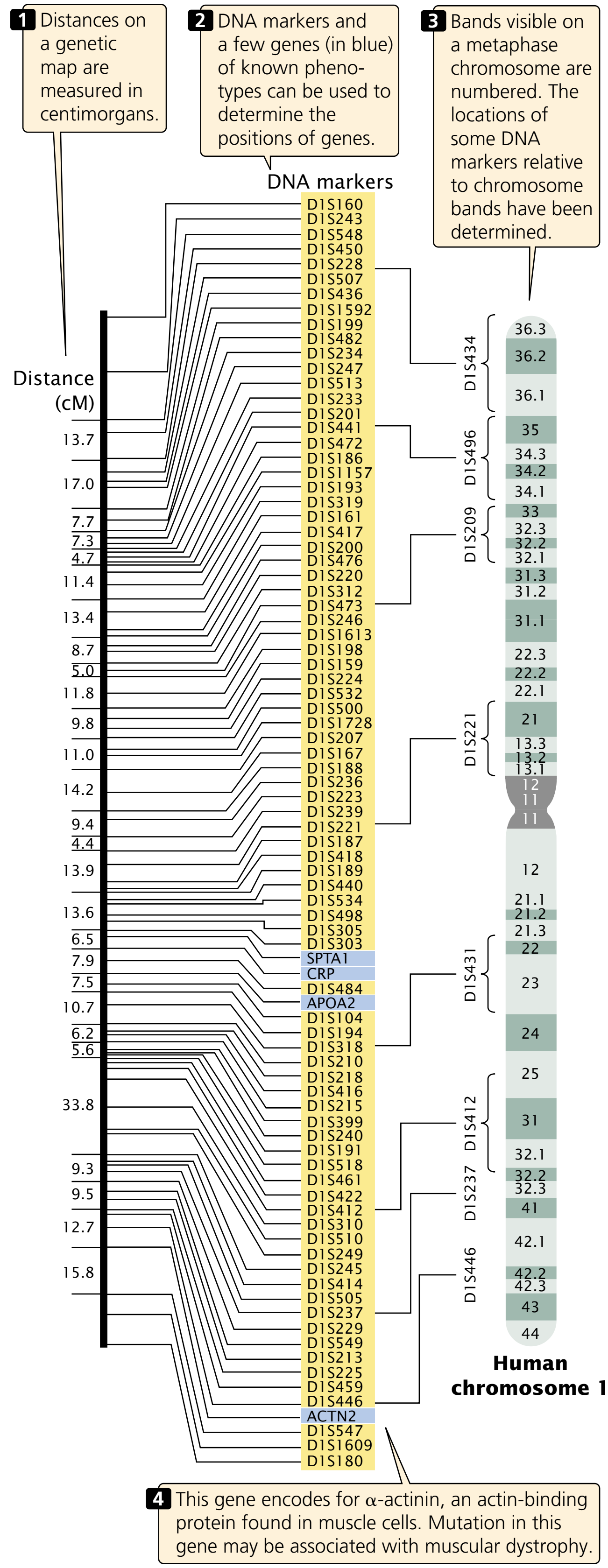
Genetic Maps

Everyone has used a map at one time or another. Maps are indispensable for finding a new friend's house, the way to an unfamiliar city in your state, or the location of a country on the globe. Each of these examples requires a map with a different scale. For finding a friend's house, you would probably use a city street map; for finding your way to an unknown city, you might pick up a state highway map; for finding a country such as Kazakhstan, you would need a world atlas. Similarly, navigating a genome requires maps of different types and scales.

Genetic maps (also called linkage maps) provide a rough approximation of the locations of genes relative to the locations of other known genes (FIGURE 19.1). These maps are based on the *genetic* function of recombination (hence the name genetic map). The basic principles of constructing genetic maps are discussed in detail in Chapter 7. In short, individuals heterozygous at two or more genetic loci are crossed, and the frequency of recombination between loci is determined by examining the progeny. If the recombination frequency between two loci is 50%, then the loci are located on different chromosomes or are far apart on the same chromosome. If the recombination frequency is less than 50%, the loci are located close together on the same chromosome (they belong to the same linkage group). For linked genes, the rate of recombination is proportional to the physical distance between the loci. Distances on genetic maps are measured in percent recombination (centimorgans, cM) or map units. Data from multiple two-point or three-point crosses can be integrated into linkage maps for whole chromosomes.

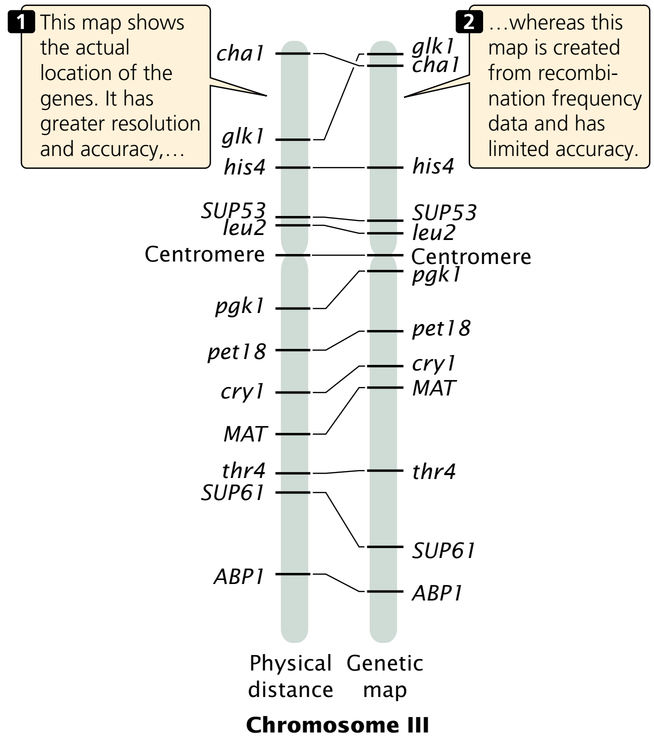
For many years, genes could be detected only by observing their influence on a trait (the phenotype), and construction of genetic maps was limited by the availability of single-locus traits that could be examined for evidence of recombination. Eventually, this limitation was overcome by the development of molecular techniques such as restriction fragment length polymorphisms, the polymerase chain reaction, and DNA sequencing (see Chapter 18) that are able to provide molecular markers that can be used to construct and refine genetic maps.

Genetic maps have several limitations, the first of which is resolution or detail. The human genome includes 3.4 billion base pairs of DNA and has a total genetic distance of about 4000 cM, an average of 850,000 bp/cM.



19.1 Genetic maps are based on rates of recombination. Shown here is a genetic map of human chromosome 1.

Even if a marker occurred every centimorgan (which is unrealistic), the resolution in regard to the physical structure of the DNA would still be quite low. In other words, the detail of the map is very limited. A second problem with genetic maps is that they do not always accurately correspond to physical distances between genes. Genetic maps are based on rates of crossing over, which vary somewhat from one part of a chromosome to another; so the distances on a genetic map are only approximations of real physical distances along a chromosome. FIGURE 19.2 compares the genetic map of chromosome III of yeast with a physical map determined by DNA sequencing. There are some discrepancies between the distances and even among the positions of some genes. In spite of these limitations, genetic maps have been critical to the development of physical maps and the sequencing of whole genomes.

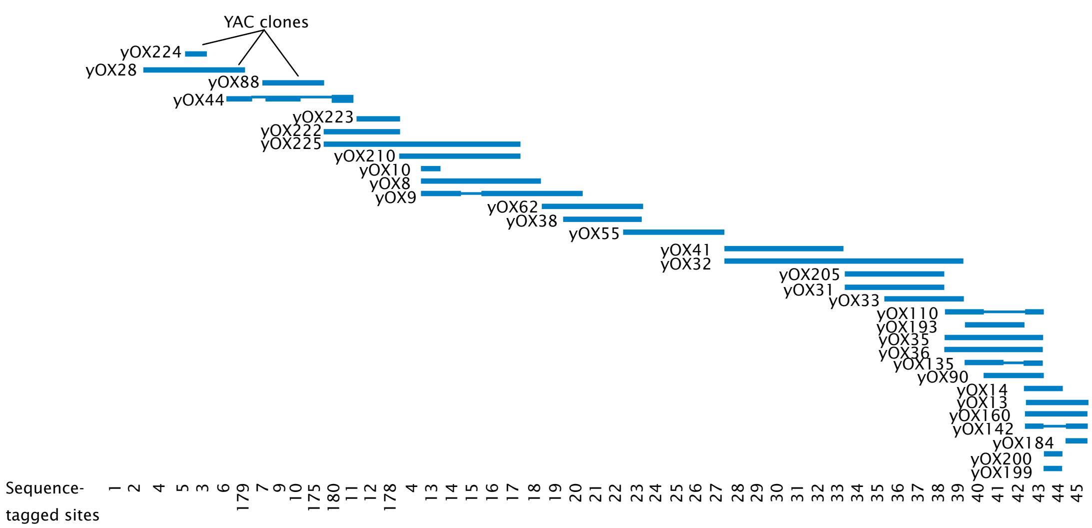


19.2 Genetic and physical maps may differ in relative distances and even in the position of genes on a chromosome. Genetic and physical maps of yeast chromosome III reveal such differences.

Physical Maps

Physical maps are based on the direct analysis of DNA, and they place genes in relation to distances measured in number of base pairs, kilobases, or megabases (FIGURE 19.3). A common type of physical map is one that connects isolated pieces of genomic DNA that have been cloned in bacteria or yeast. Physical maps generally have higher resolution and are more accurate than genetic maps. A physical map is analogous to a neighborhood map that shows the location of every house along a street, whereas a genetic map is analogous to a highway map that shows the locations of major towns and cities.

A number of techniques exist for creating physical maps, including restriction mapping, which determines the positions of restriction sites on DNA; sequence-tagged site (STS) mapping, which locates the positions of short unique sequences of DNA on a chromosome; fluorescent in situ hybridization (FISH), by which markers can be visually mapped to locations on chromosomes (see Figure 7.22); and DNA sequencing.

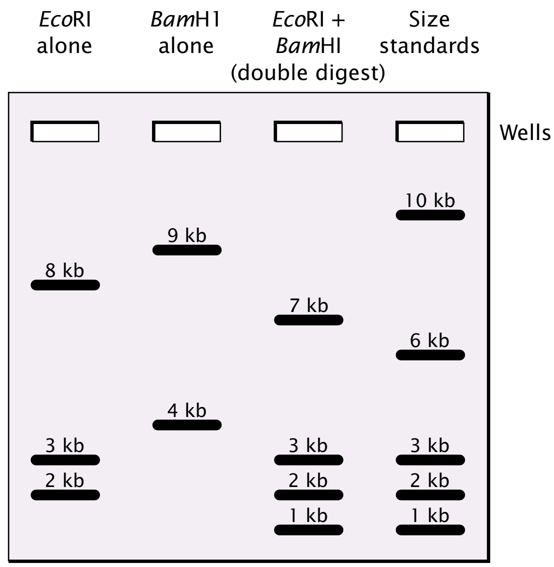


19.3 Physical maps are often used to order cloned DNA fragments. A part of a physical map of a set of overlapping YAC clones from one end of the human Y chromosome.

Concepts

Both genetic and physical maps provide information about the relative positions and distances between genes, molecular markers, and chromosome segments. Genetic maps are based on rates of recombination and are measured in percent recombination, or centimorgans. Physical maps are based on the physical distances and are measured in base pairs.

Restriction mapping determines the relative positions of restriction sites on a piece of DNA. When a piece of DNA is cut with a restriction enzyme and the fragments are separated by gel electrophoresis, the number of restriction sites in the DNA and the distances between them can be determined by the number and positions of bands on the gel (p. 000 in Chapter 18), but this information does not tell us the order or the precise location of the restriction sites. To map restriction sites, a sample of the DNA is cut with one restriction enzyme, and another sample is cut with a different restriction enzyme. A third sample is cut with both restriction enzymes together (a double digest). The DNA fragments produced by these restriction digests are then separated by gel electrophoresis, and their sizes are compared. Overlap in size of fragments produced by the digests can be used to position the restriction sites on the original DNA molecule.



19.4 Restriction sites can be mapped by comparing DNA fragments produced by digestion by restriction enzymes used alone and in various combinations. A sample of a linear piece of DNA was first digested with *Eco*RI alone. Another sample was digested by *Bam*HI alone, and finally a third sample was digested by both *Eco*RI and *Bam*HI. The resulting fragments were separated by gel electrophoresis and stained with ethidium bromide.

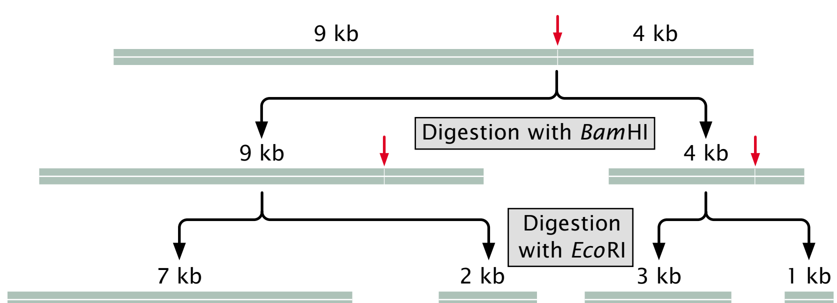
Worked Problem

One sample of a linear 13,000-bp (13-kb) DNA fragment is cut with the restriction enzyme *Eco*RI; a second sample of the same DNA is cut with *Bam*HI; and a third sample is cut with *both Eco*RI and *Bam*HI together. The resulting fragments are separated and sized by gel electrophoresis (Figure 19.4). Determine the positions of the *Eco*RI and *Bam*HI restriction sites on the original 13-kb fragment.

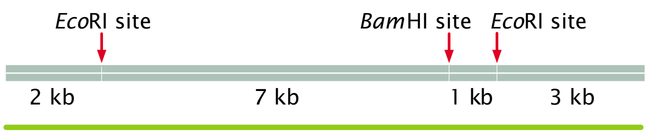
• Solution

Using the sizes of the fragments produced from the three digests in Figure 19.4, we can order the positions of the restriction sites on the original 13-kb piece of DNA. First, note that digestion with *Eco*RI alone produced 8-kb, 3-kb, and 2-kb fragments, indicating that there are two *Eco*RI restriction sites in the original linear piece of DNA. Digestion with *Bam*HI produced 9-kb and 4-kb fragments, indicating that there is only one *Bam*HI site. The *Bam*HI restriction site must be 9 kb from one end and 4 kb from the other end.

The double digest produced four pieces of DNA: 7-kb, 3-kb, 2-kb, and 1-kb fragments. Neither of the fragments generated by *Bam*HI alone is present in the double digest, and so *Eco*RI must have cut both of the *Bam*HI fragments. Consider the 9-kb fragment. How could this fragment be cut by *Eco*RI to produce the fragments found in the double digest? Two of the fragments produced by the double digest, the 7-kb and 2-kb fragments, add up to 9 kb, the length of one fragment produced by digestion by *Bam*HI alone. Similarly, the 3-kb fragment and the 1-kb fragment of the double digest add up to 4 kb, the length of the other fragment produced by *Bam*HI alone. Therefore, *Eco*RI cut the first *Bam*HI fragment into 7-kb and 2-kb fragments and cut the second *Bam*HI fragment into 3-kb and 1-kb fragments:



We now know that the *Bam*HI site lies between these two *Eco*RI sites. Considering the four fragments produced by the double digest, there are several possible arrangements by which a *Bam*HI site could fit in between the two *Eco*RI sites. To determine which of the arrangements is correct, compare the results of the *Eco*RI digestion with the double digest. When the original 13-kb DNA fragment was cut by *Eco*RI alone, the three fragments produced were 8 kb, 3 kb, and 2 kb in length. The 2-kb and 3-kb bands are also present in the double digest, indicating that these fragments do not contain a *Bam*HI site. The 8-kb fragment present in the *Eco*RI digest disappears in the double digest and is replaced by the 7-kb fragment and the 1-kb fragment, indicating that the 8-kb fragment has the *Bam*HI site. Thus the 7-kb and 1-kb fragments must lie next to each other, and the 2-kb and 3-kb fragments are on the ends. Thus, the correction arrangement of the restriction sites is:



In the example in the Worked Problem, we can map the restriction sites in our heads or with a few simple sketches. Most restriction mapping is done with several restriction enzymes, used alone and in various combinations, producing many restriction fragments. With long pieces of DNA (greater than 30 kb), computer programs are used to determine the restriction maps, and restriction mapping may be facilitated by tagging one end of a large DNA fragment with radioactivity or by identifying the end with the use of a probe.

Physical maps, such as restriction maps of DNA fragments or even whole chromosomes, are often created for genomic analysis. These lengthy maps are often put together by combining maps of shorter, overlapping genomic fragments.

Concepts

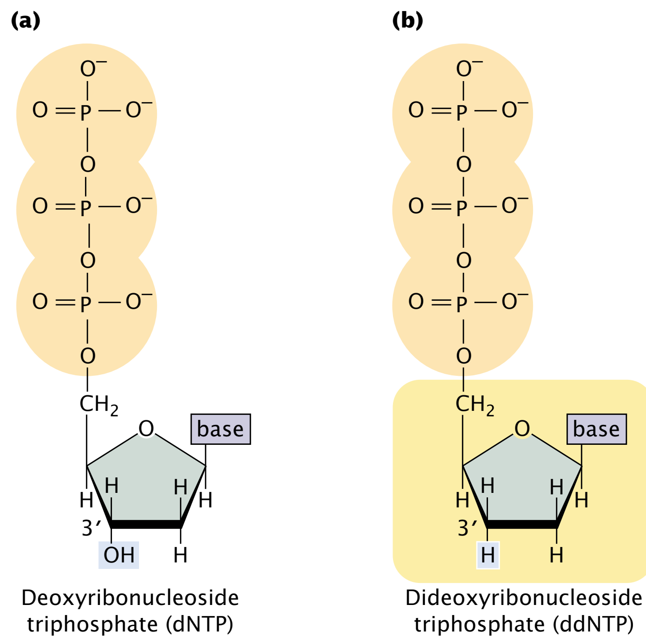
The locations of restriction sites can be mapped by cutting DNA with several restriction enzymes, first with each restriction enzyme alone and then with combinations of restriction enzymes.

DNA-Sequencing Methods

The most detailed physical maps are based on direct DNA sequence information. The first methods for quickly sequencing DNA were developed between 1975 and 1977. Frederick Sanger and his colleagues created the dideoxy sequencing method based on the elongation of DNA; Allan Maxam and Walter Gilbert developed a second method based on the chemical degradation of DNA. The Sanger method quickly became the standard procedure for sequencing any purified fragment of DNA.

The Sanger, or dideoxy, method of DNA sequencing is based on the process of replication. The fragment to be sequenced is used as a template to make a series of new DNA molecules. In the process, replication is sometimes (but not always) terminated when a specific base is encountered, producing DNA strands of different length, each of which ends in the same base.

The method relies on the use of a special substrate for DNA synthesis. Normally, DNA is synthesized from deoxyribonucleoside triphosphates (dNTPs), which have an OH group on the 3′-carbon atom (FIGURE 19.5a). In DNA synthesis, two phosphate groups on the 5′-carbon atom of a dNTP are removed, and a phosphodiester bond is formed between the remaining 5′-phosphate group of the dNTP and the 3′-OH group of the last nucleotide on the growing DNA chain (see p. 000 in Chapter 12). In the Sanger method, a special nucleotide, called a **dideoxyribonucleoside triphosphate** (ddNTP; FIGURE 19.5b), is used as substrate. The ddNTPs are identical with dNTPs, except that they lack a 3′-OH group. Like dNTPs, ddNTPs possess three phosphate groups on their 5′ ends, and so they are incorporated into a growing DNA chain. When a ddNTP has been incorporated into a DNA chain, however, no more nucleotides can be added, because there is



19.5 The dideoxy sequencing reaction requires a special substrate for DNA synthesis. (a) Structure of deoxyribonucleoside triphosphate, the normal substrate for DNA synthesis. (b) Structure of dideoxyribonucleoside triphosphate, which lacks an OH group on the 3′ carbon atom.

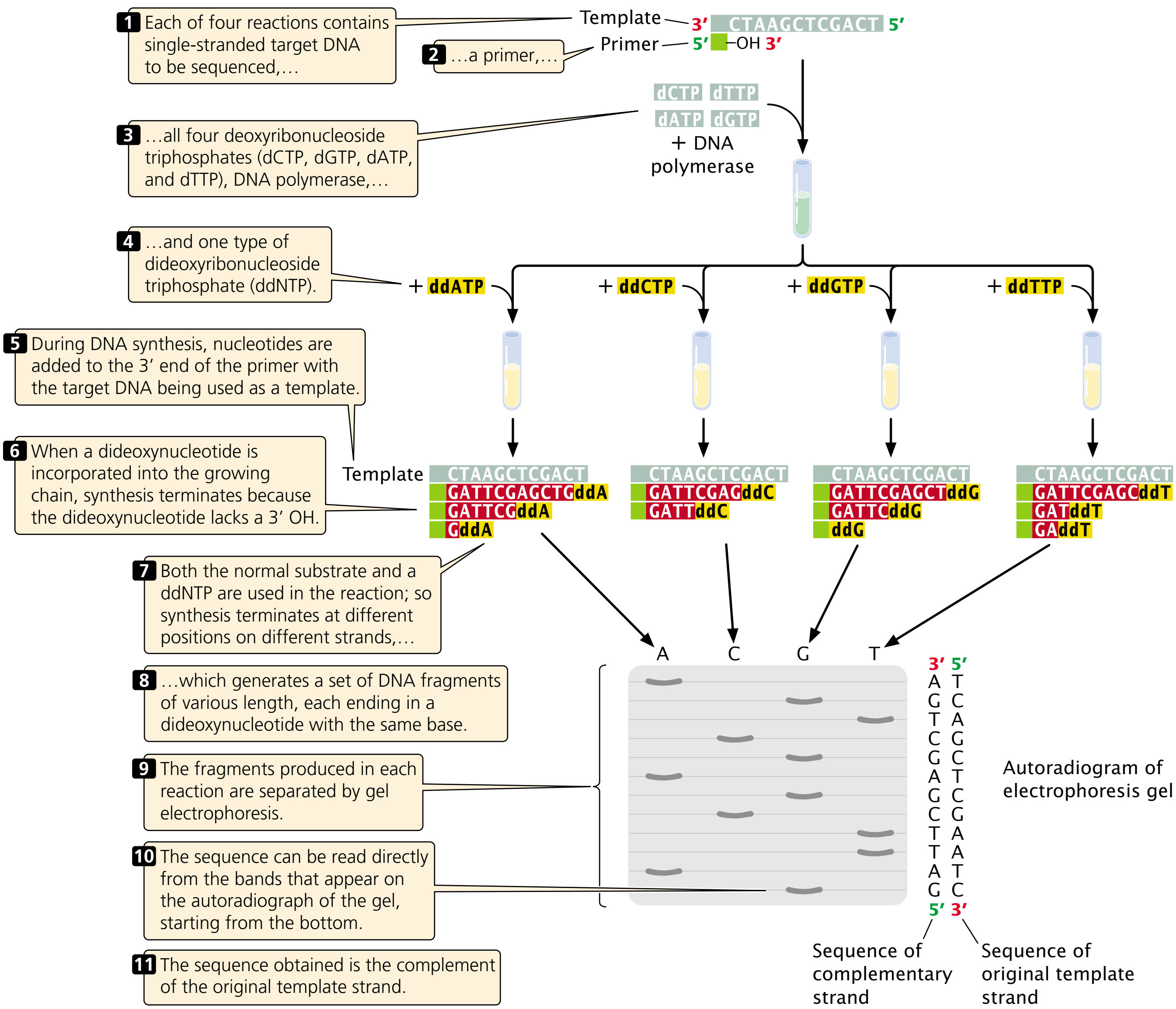
no 3′-OH group to form a phosphodiester bond with an incoming nucleotide. Thus, ddNTPs terminate DNA synthesis.

A single DNA molecule cannot be sequenced; so any DNA fragment to be sequenced must first be amplified by PCR or by cloning in bacteria. Copies of the target DNA are isolated and split into four parts (FIGURE 19.6). Each part is placed in a different tube, to which are added:

1. many copies of a primer that is complementary to one end of the target DNA strand;
2. all four deoxyribonucleoside triphosphates (dCTP, dATP, dGTP, and dTTP), the normal precursors of DNA synthesis;
3. a small amount of *one* of the four types of dideoxyribonucleoside triphosphates (ddCTP, ddATP, ddGTP, *or* ddTTP), which will terminate DNA synthesis as soon as it is incorporated into any growing chain (each of the four tubes received a different ddNTP); and
4. DNA polymerase.

Either the primer or one of the dNTPs is radioactively or chemically labeled so that newly produced DNA can be detected.

Within each of the four tubes, the DNA polymerase enzyme carries out DNA synthesis. Let's consider the reaction in one of the four tubes; the one that received ddATP. Within this tube, each of the single strands of target DNA



19.6 **The dideoxy method of DNA sequencing is based on the termination of DNA synthesis.**

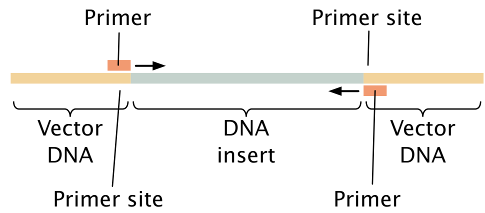
serves as template for DNA synthesis. The primer pairs to its complementary sequence at one end of each template strand, providing a 3′-OH group for the initiation of DNA synthesis. DNA polymerase elongates a new strand of DNA from this primer, by using the target DNA strand as a template. Wherever DNA polymerase encounters a T on the template strand, it uses at random either a dATP or a ddATP to introduce an A in the newly synthesized strand. Because there is more dATP than ddATP in the reaction mixture, dATP is incorporated most often, allowing DNA synthesis to continue. Occasionally, however, ddATP is incorporated into the strand and synthesis terminates. The incorporation of ddA into the new strand occurs randomly at different positions in different copies, producing a set of DNA chains of different length (12, 7, and 2 nucleotides long in the example illustrated in Figure 19.6), each ending in a nucleotide with adenine.

Equivalent reactions take place in the other three tubes. In the tube that received ddCTP, all the chains terminate in a nucleotide with cytosine; in the tube that received ddGTP, all the chains terminate in a nucleotide with guanine; and, in the tube that received ddTTP, all the chains terminate in a nucleotide with thymine. After the completion of the polymerization reactions, all the DNA in the tubes is denatured, and the single-strand products of each reaction are separated by gel electrophoresis.

The contents of the four tubes are separated side by side on an acrylamide gel so that DNA strands differing in length by only a single nucleotide can be distinguished. After electrophoresis, the locations of the DNA strands in the gel are revealed by autoradiography. The shortest strands, which terminated at positions early in the DNA sequence, migrate quickly and end up near the bottom of the gel; longer fragments, which terminated late in the sequence, migrate more slowly and end up near the top of the gel.

Reading the DNA sequence is simple and the shortest part of the procedure. In Figure 19.6, you can see that the band closest to the bottom of the gel is from the tube that contained the ddGTP reaction, which means that the first nucleotide synthesized had guanine (G). The next band up is from the tube that contained ddATP; so the next nucleotide in the sequence is adenine (A), and so forth. In this way, the sequence is read from the bottom to the top of the gel, with the nucleotides near the bottom corresponding to the 5′ end of the newly synthesized DNA strand and those near the top corresponding to the 3′ end. Keep in mind that the sequence obtained is not that of the target DNA but that of its *complement.*

You may have wondered how the primers used in dideoxy sequencing are constructed, because the sequence of the target DNA may not be known ahead of time. The trick is to insert a sequence that will be recognized by the primer into the target DNA. This is often done by first cloning the target DNA in a vector that contains sequences



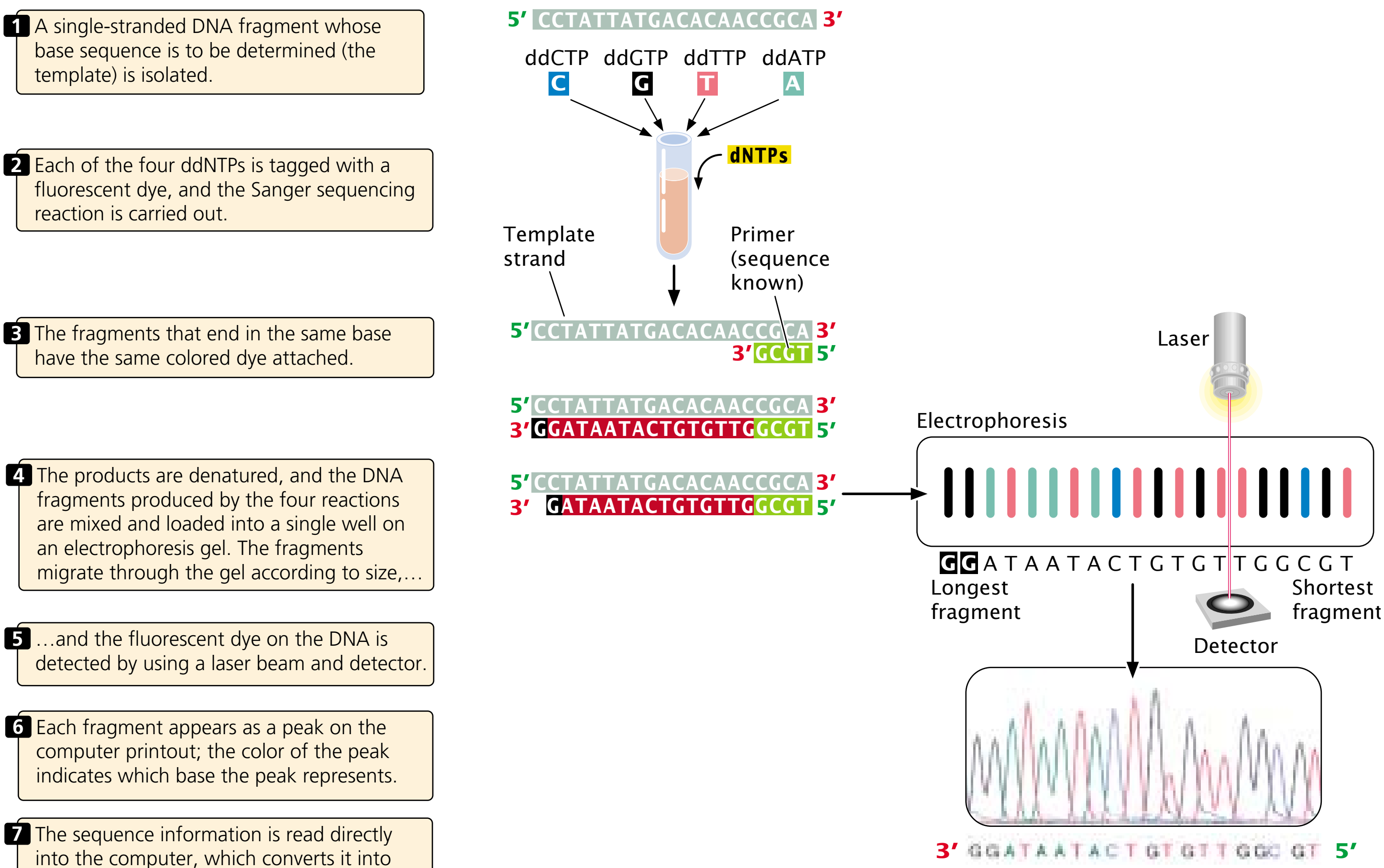
19.7 Sites recognized by sequencing primers are added to the target DNA by cloning the DNA in a vector that contains universal sequencing primer sites on either side of the site where the target DNA will be inserted.

recognized by a common primer (called universal sequencing primer sites) on either side of the site where the target DNA will be inserted. The target DNA is then isolated from the vector and will contain universal sequencing primer sites at each end (FIGURE 19.7).

Sequencing is often carried out by automated machines that use fluorescent dyes and laser scanners to sequence thousands of base pairs in a few hours (FIGURE 19.8). The dideoxy reaction is also used here, but the ddNTPs used in the reaction are labeled with a fluorescent dye, and a different colored dye is used for each type of dideoxynucleotide. For example, a red dye might be used for nucleotides with thymine, a green dye for those with adenine, a black dye for those with guanine, and a blue dye for those with cytosine. In this case, the four sequencing reactions can take place in the same test tube and can be placed in the same well during electrophoresis, given that each ddNTP is distinctively marked. The most recently developed sequencing machines carry out electrophoresis in gel-containing capillary tubes. The different-sized fragments produced by the sequencing reaction separate within a tube and migrate past a laser beam and detector. As the fragments pass the laser, their fluorescent dyes are activated and the resulting fluorescence is detected by an optical scanner. Each colored dye emits fluorescence of a characteristic wavelength, which is read by the optical scanner. The information is fed into a computer for interpretation, and the results are printed out as a set of peaks on a graph (See Figure 19.8). Automated sequencing machines may contain 96 or more capillary tubes, allowing from 50,000 to 60,000 bp of sequence to be read in a few hours.

Concepts

DNA can be rapidly sequenced by the dideoxy method, in which ddNTPs are used to terminate DNA synthesis at specific bases. Automated sequencing methods allow tens of thousands of base pairs to be read in just a few hours.



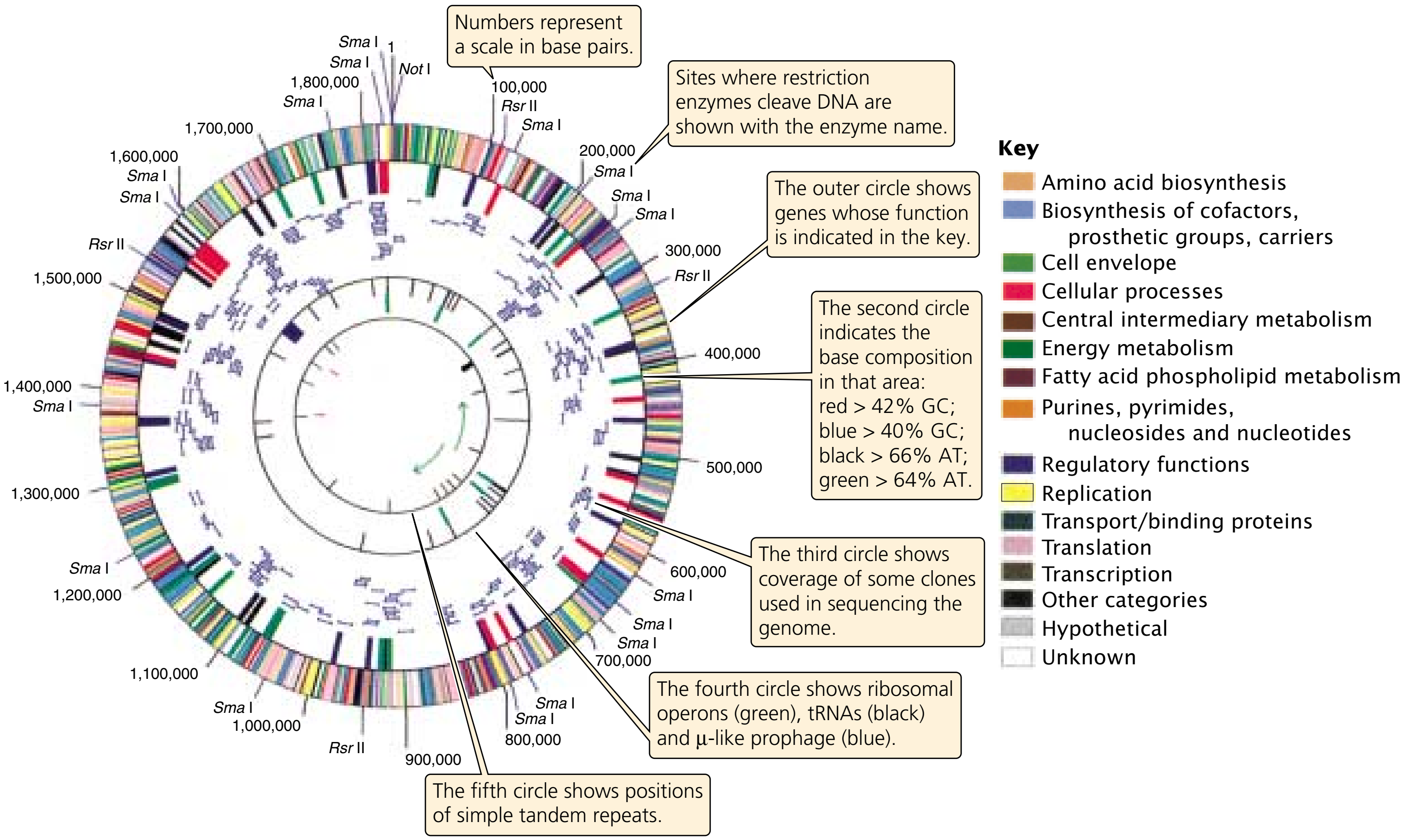
19.8 **The dideoxy sequencing method can be automated.**

Sequencing an Entire Genome

The ultimate goal of structural genomics is to determine the ordered nucleotide sequences of entire genomes of organisms. The main obstacle to this task is the immense size of most genomes. Bacterial genomes are usually at least several million base pairs long; many eukaryotic genomes are billions of base pairs long and are distributed among dozens of chromosomes. In addition, for technical reasons, it is not possible to begin sequencing at one end of a chromosome and continue straight through to the other end; only small fragments of DNA—usually from 500 to 700 nucleotides—can be sequenced at one time. Therefore, determining the sequence for an entire genome requires that the DNA be broken into thousands or millions of smaller fragments that can then be sequenced. The difficulty lies in putting these short sequences back together in the correct order. As we will see, two different approaches have been used to assemble the short, sequenced fragments into a complete genome.

The first genomes to be sequenced were small genomes of some viruses. The genome of bacteriophage λ, consisting of 49,000 bp, was completed in 1982. In 1995, the first genome of a living organism (*Haemophilus influenzae*) was sequenced by Craig Venter and Claire Fraser of the Institute for Genomic Research (TIGR) and Hamilton Smith of Johns Hopkins University. This bacterium has a relatively small genome of 1.8 million base pairs (FIGURE 19.9). By 1996, the genome the first eukaryotic organism (yeast) had been determined, followed by the genome of *Eschericia coli* (1997), *Caenorhabditis elegans* (1998), and *Drosophila melanogaster* (2000). The first draft of the human genome was completed in June 2000.

Map-based sequencing The first method for assembling short, sequenced fragments into a whole-genome sequence, called a **map-based approach,** requires the initial creation of detailed genetic and physical maps of the genome, which provide known locations of genetic markers (restriction sites, other genes, or known DNA sequences) at regularly spaced intervals along each chromosome. These markers can later be used to help align the short, sequenced fragments into their correct order.

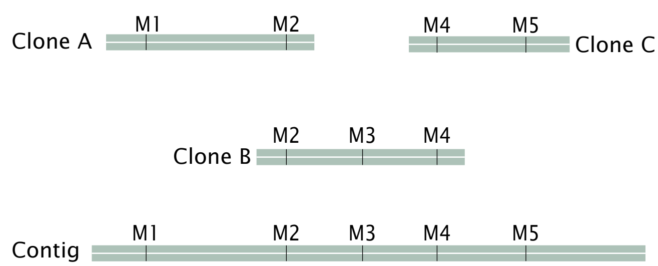


19.9 The bacterium *Haemophilas influenzae* was the first free-living organism to be sequenced. (From R.D. Fleischman et al., 1993, *Science* 269:469; Scan courtesy of TIGR.)

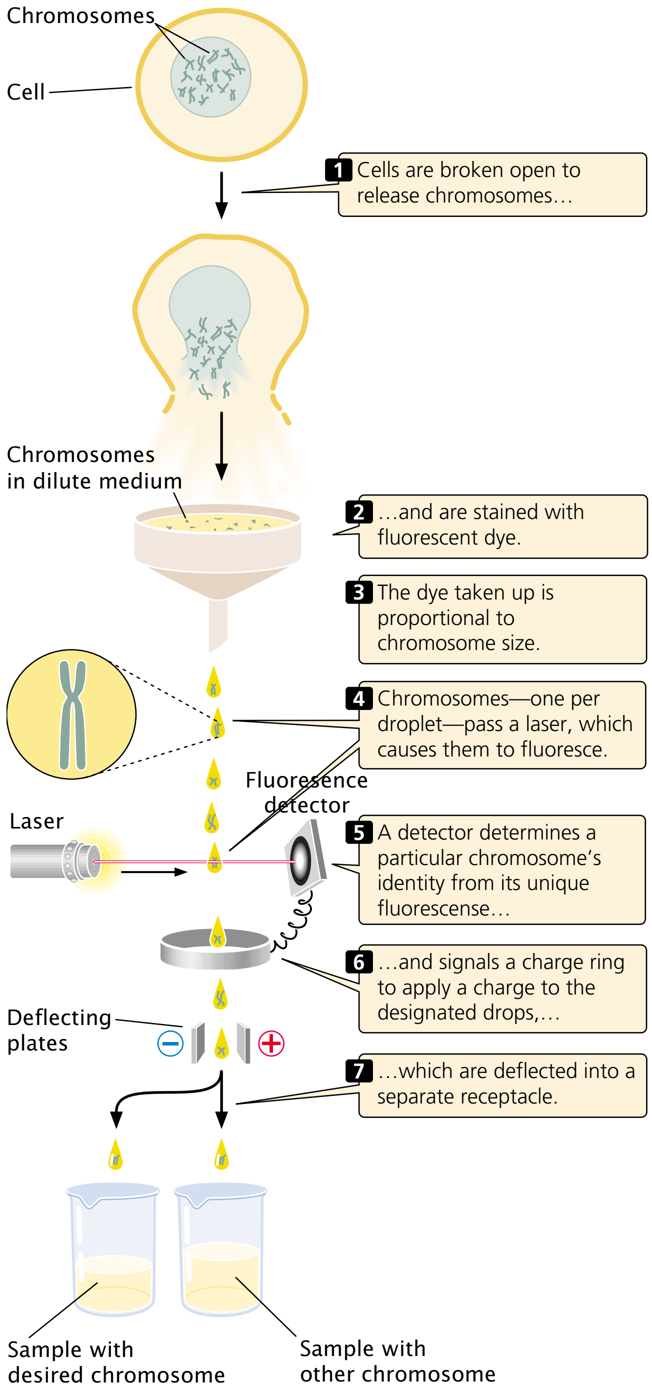
After the genetic and physical maps are available, chromosomes or large pieces of chromosomes are separated by pulsed-field gel electrophoresis (PFGE) or by flow cytometry. In pulsed field gel electrophoresis (which is similar to standard gel electrophoresis), large molecules of DNA or whole chromosomes are separated in a gel by periodically alternating the orientation of an electrical current. In flow cytometry, chromosomes are sorted optically by size (FIGURE 19.10).

Each chromosome (or sometimes the entire genome) is then cut up by partial digestion with restriction enzymes (FIGURE 19.11). Partial digestion means that the restriction enzymes are allowed to act only for a limited time so that not all restriction sites in every DNA molecule are cut. Thus partial digestion produces a set of large overlapping DNA fragments, which are then cloned by using cosmids, yeast artificial chromosomes (YACs), or bacterial artificial chromosomes (BACs).

Next, these large-insert clones are put together in their correct order on the chromosome (see Figure 19.11). This assembly can be done in several ways. One method relies on the presence of a high-density map of genetic markers. A complementary DNA probe is made for each genetic marker, and a library of the large-insert clones is screened with the probe, which will hybridize to any colony containing a clone with the marker. The library is then screened for neighboring markers. Because the clones are much larger than the markers used as probes, some clones will have more than one marker. For example, clone A might have markers M1 and M2, clone B markers M2, M3, and M4, and clone C markers M4 and M5. Such a result would indicate that these clones contain areas of overlap, as shown here.

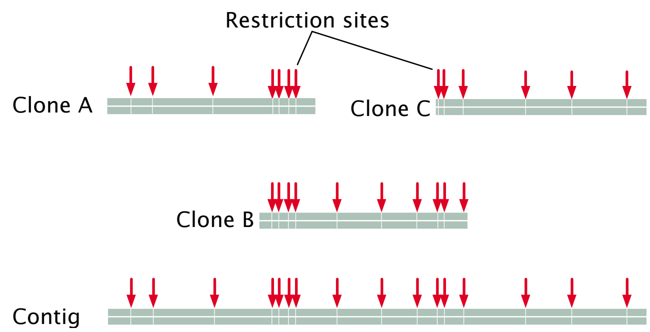


A set of two or more overlapping DNA fragments that form a contiguous stretch of DNA is called a **contig.** This approach was used in 1993 to create a contig of the human Y chromosome consisting of 196 overlapping YAC clones (see Figure 19.3).



19.10 Flow cytometry is used to separate individual chromosomes.

It is also possible to determine the order of clones without the use of preexisting genetic maps. For example, each clone can be cut with a series of restriction enzymes, and the resulting fragments are then separated by gel electrophoresis. This method generates a unique set of restriction fragments, called a fingerprint, for each clone. The restriction patterns for the clones are stored in a database. A computer program is then used to examine the restriction patterns of all the clones and look for areas of overlap. The overlap is then used to arrange the clones in order, as shown here:



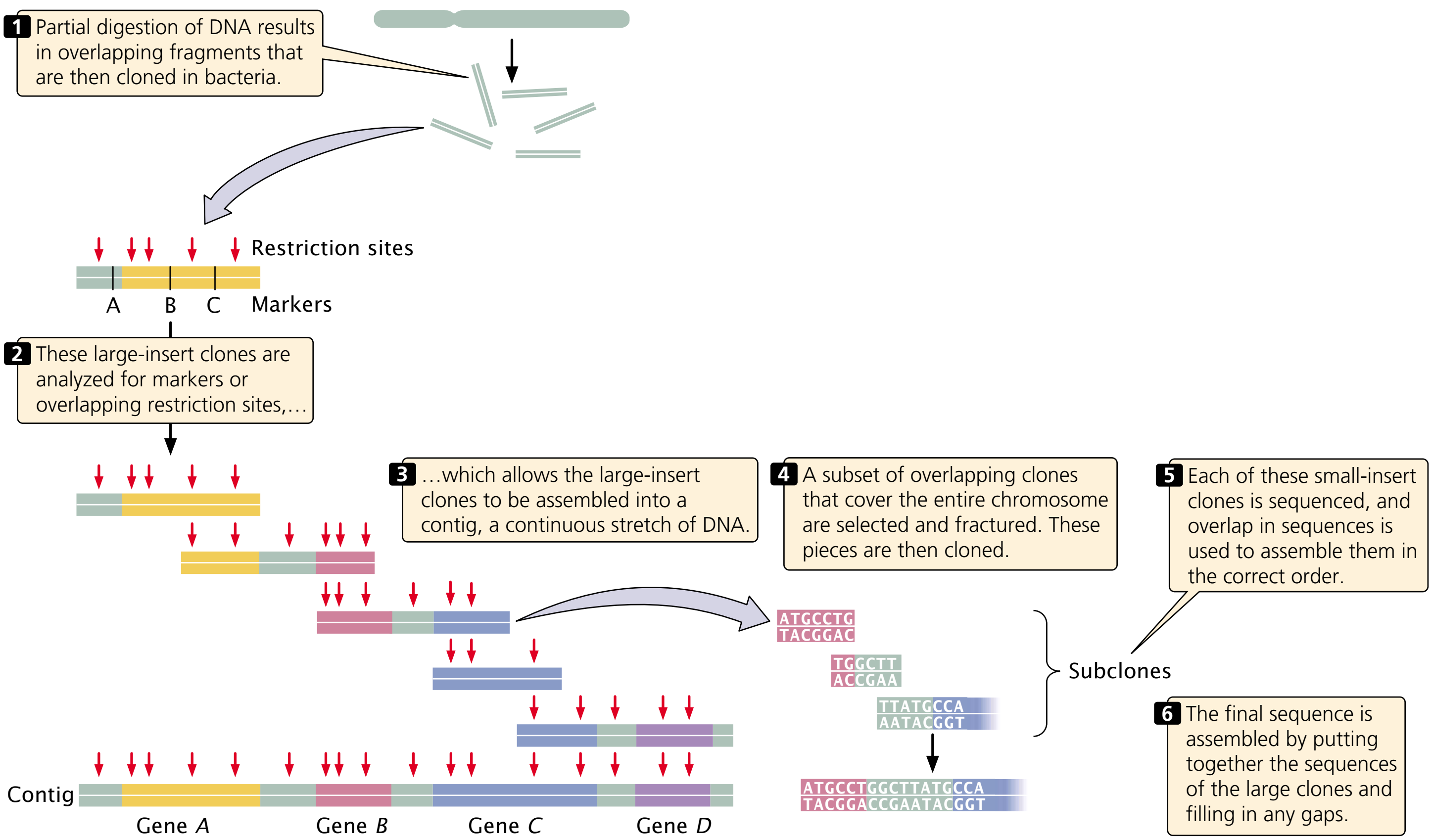
Other genetic markers can be used to help position contigs along the chromosome.

When the large-insert clones have been assembled into the correct order on the chromosome, a subset of overlapping clones that efficiently cover the entire chromosome can be chosen for sequencing. Each of the selected large-insert clones is fractured into smaller overlapping fragments, which are themselves cloned with the use of phages or cosmids (Figure 19.11). These smaller clones (called small-insert clones) are then sequenced. The sequences of the small-insert clones are examined for overlap, which allows them to be correctly assembled to give the sequence of the larger insert clones. Enough overlapping small-insert clones are usually sequenced to ensure that the entire genome is sequenced several times. Finally, the whole genome is assembled by putting together the sequences of all overlapping contigs (Figure 19.11). Often, gaps still exist in the genome map that must be filled in by using other methods.

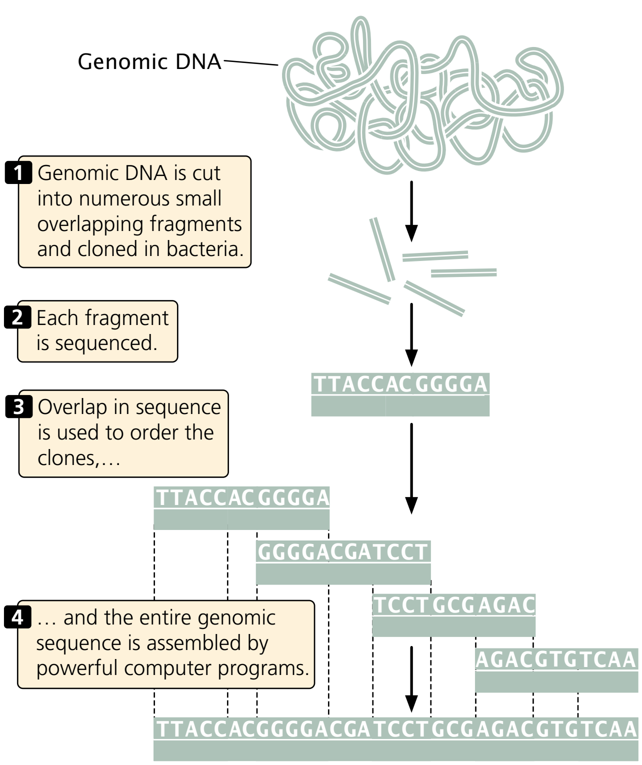
Whole-genome shotgun sequencing The second approach to genome sequencing does not map and assemble the large-insert clones. In this approach, called **whole-genome shotgun sequencing** (Figure 19.12), small-insert clones are prepared directly from genomic DNA and sequenced. Powerful computer programs then assemble the entire genome by examining overlap among the small-insert clones. The requirement for overlap means that most of the genome will be sequenced multiple (often from 10 to 15) times.

Concepts

Sequencing a genome requires breaking it up into small overlapping fragments whose DNA sequences can be determined in a sequencing reaction. The sequences can be ordered into the final genome sequence by a map-based approach (large fragments are ordered with the use of genetic and physical maps) or by whole-genome shotgun sequencing (overlap between the sequences of small fragments is compared by computers).



19.11 Map-based approaches to whole-genome sequencing rely on detailed genetic and physical maps to align sequenced fragments.

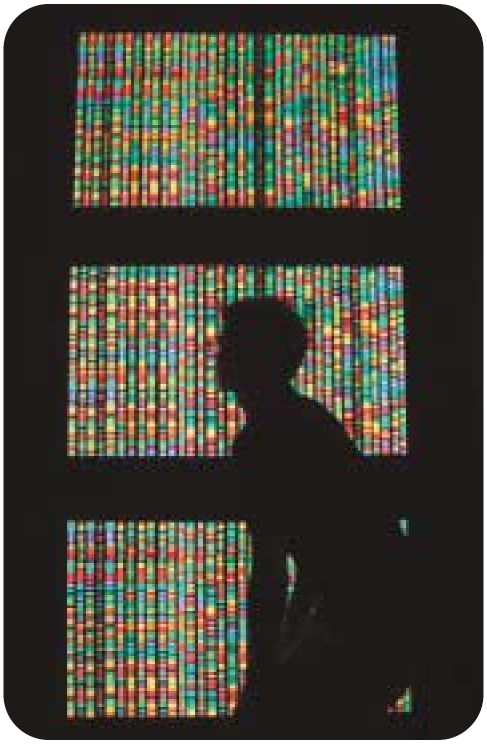


19.12 Whole-genome shotgun sequencing utilizes sequence overlap to align sequenced fragments.

www.whfreeman.com/pierce More about current developments in DNA sequencing

The Human Genome Project

By 1980, methods for mapping and sequencing DNA fragments had been sufficiently developed that geneticists began seriously proposing that the entire human genome could be sequenced. An international collaboration was planned to undertake the Human Genome Project (FIGURE 19.13); initial estimates suggested that 15 years and $3 billion would be required to accomplish the task. As a part of the effort, the genomes of several model organisms, including *Escherichia coli*, *Saccharomyces cerevisiae* (yeast), *Drosophila melanogaster* (fruit fly), *Arabidopsis thaliani* (a plant), and *Caenorhabditis elegans* (a nematode) were to be sequenced as well. The genomes of these model organisms were sequenced to help develop methods that could then be applied to the sequencing of the human genome and to



19.13 **The Human Genome Project has produced an initial draft of the sequence for the human genome.** (Mario Tama/Getty.)

provide sequenced genomes with which to compare the organization and structure of the human genome.

The Human Genome Project officially got underway in October 1990. Initial efforts focused on developing new and automated methods for cloning and sequencing DNA and on generating detailed physical and genetic maps of the human genome. The methods described earlier for mapping, sequencing, and assembling DNA fragments were pivotal in these early stages of the project. By 1993, large-scale physical maps were completed for all 24 pairs of human chromosomes. At the same time, automated sequencing techniques (FIGURE 19.14) had been developed that made large-scale sequencing feasible.

The initial effort to sequence the genome was a public project consisting of the international collaboration of 20 research groups and hundreds of individual researchers who formed the International Human Genome Sequencing Consortium. In 1998, Craig Venter announced that he would lead a company called Celera Genomics in a private effort to sequence the human genome.

The public and private efforts moved forward simultaneously but used different approaches. The Human Genome Consortium used a map-based approach; many copies of the human genome were cut up into fragments of about 150,000 bp each, which were inserted into bacterial artificial chromosomes. Yeast artificial chromosomes and cosmids had been used in early stages of the project but did not prove to be as stable as the BAC clones, although YAC clones were instrumental in putting together some of the larger contigs. Restriction fingerprints were used to assemble the BAC clones into contigs, which were positioned on the chromosomes by using genetic markers and probes. The individual BAC clones were sheared into smaller overlapping fragments and sequenced, and the whole genome was assembled by putting together the sequence of the BAC clones.

Celera Genomics used a whole-genome shotgun approach to determine the human genome sequence, although the genetic and physical maps produced by the public effort helped Celera assemble the final sequence. In this approach, small-insert clones were prepared directly from genomic DNA and then sequenced. The overlapping of DNA sequences among these small-insert clones was then used to assemble the entire genome.

Both public and private sequencing projects announced the completion of a rough draft that included most of the sequence of the human genome in the summer of 2000, 5 years ahead of schedule. Analysis of this sequence was published 6 months later.

The availability of the complete sequence of the human genome is proving to be of enormous benefit. It is greatly facilitating the identification and isolation of genes that contribute to many human diseases and is providing probes that can be used in genetic testing, diagnosis, and drug development. The sequence is also providing important information about many basic cellular processes. Comparisons of the human genome with those of other organisms are adding to our understanding of evolution and the history of life.

19.14 **Automated sequencers and powerful computers allowed a rough draft of the human genome sequence to be completed in just 10 years.** (Whitehead/MIT Genome Center, 2001; from *NATURE* 409: 860–921.)

The New Genetics
ETHICS • SCIENCE • TECHNOLOGY

Mapping the Human Genome—Where It Leads, What It Means

Arthur L. Caplan and Kelly A. Carroll

In June 2000, scientists from the Human Genome Project and Celera Genomics stood at a podium with President Bill Clinton to announce a stunning achievement—they had successfully constructed a sequence of the entire human genome. Soon this process of identifying and sequencing each and every human gene became characterized as "mapping the human genome." As with maps of the physical world, the map of the human genome provides a picture of locations, terrains, and structures. But, like explorers, scientists must continue to decipher what each location on the map can tell us about diseases, human health, and biology. The map accelerates this process because it allows researchers to identify key structural dimensions of the gene that they are exploring and reminds them where they have been and where they have yet to explore.

What does the map of the human genome depict? When researchers discuss the sequencing of the genome, they are describing the identification of the patterns and order of the 3 billion human DNA base pairs. Although such identification provides valuable information about overall structure and the evolution of humans in relation to other organisms, researchers really want the key information encoded in just 2% of this enormous map—the information that encodes most of the proteins of which you and I are composed. Proteins stand as the link between genes and pharmaceutical drug development, they show which genes are being expressed at any given moment, and they provide information about gene function.

Knowing our genes will lead to a greater understanding and radically improved treatment of many diseases. However, sequencing the entire human genome, in conjunction with sequencing of various nonhuman genomes under the same project, has raised fundamental questions about what it means to be human. After all, fruit flies possess about one-third the number of genes possessed by humans, and an ear of corn has approximately the same number of genes as a human. In addition, the overall DNA sequence of a chimpanzee is about 99% the same as the human genome sequence. As the genomes of other species become available, the similarities to the human genome in both structure and sequence pattern will continue to be identified. At a basic level, the discovery of so many commonalities

Dr. Craig Venter (Celera Genomics), President Clinton, and Dr. Francis Collins (NIH). (Ron Edmonds/AP)

and links and ancestral trees with other species adds credence to principles of evolution and Darwinism.

Some of the most expected developments and potential benefits of the Human Genome Project directly affect human health; researchers, practicing physicians, and the general public eagerly await the development of targeted pharmaceutical agents and more specific diagnostic tests. Pharmacogenomics is at the intersection of genetics and pharmacology; It is the study of how one's genetic makeup will affect one's response to various drugs. In the future, medicine will potentially be safer, cheaper, and more disease specific, all while causing fewer side effects and acting more effectively, the first time around.

There are, however, some hard ethical questions that follow in the wake of new genetic knowledge. Patients will have to undergo genetic testing in order to match drugs to their genetic makeup. Who will have access to these results—just the health care practitioner? Or will the patient's insurance company, employer, school, or family have access? Although the tests may have been administered for one case, will information derived from them be used for other purposes, such as for the identification of other conditions or future diseases or even in research studies?

How should researchers conduct studies in pharmacogenomics? Often, they need to study subjects by some kind of identifiable trait that they believe will assist in separating groups of drugs, and in turn they separate people into populations. The order of almost all the DNA base pairs (99.9 %) is exactly the same in all humans, leaving a small window of difference. The potential for the stigmatization of individuals and groups of people based on race and ethnicity is inherent in genomic research and analysis. As scientists continue drug development, they must be careful not to further such ideas, especially because studies of nuclear DNA indicate that there is often more genetic variation within ethnic groups or cultures than between ethnic groups or cultures.

These are just a few of the ethical issues arising out of one development of the Human Genome Project. The potential applications of genome research are staggering, and the mapping is just the beginning.

Concepts

The Human Genome Project is an effort to sequence the entire human genome. Begun in 1990, a rough draft of the sequence was completed by two competing teams, an international consortium of publicly supported investigators and a private company, both of which finished a rough draft of the genome sequence in 2000.

www.whfreeman.com/pierce Information about the Human Genome Project and numerous links to it

Single-Nucleotide Polymorphisms

In addition to the DNA sequence of an entire genome, several other types of data are useful for genomic projects and have been the focus of sequencing efforts. One consists of **single-nucleotide polymorphisms** (SNPs, pronounced "snips"), which are single-base-pair differences in DNA sequence between individual members of a species. Arising through mutation, SNPs are inherited as allelic variants (just like alleles that produce phenotypic differences, such as blood types), although SNPs do not usually produce a phenotypic difference. Single-nucleotide polymorphisms are numerous and are present throughout genomes. In a comparison of the same chromosome from two different people, a SNP can be found approximately every 1000 bp.

Because of their variability and widespread occurrence throughout the genome, SNPs are valuable as markers in linkage studies. For example, human SNPs are being cataloged and mapped for use in identifying genes that contribute to disease. When a SNP is physically close to a disease-causing locus, it will tend to be inherited along with the disease-causing allele. Thus the SNP marks the location of a genetic locus that causes the disease. A SNP can also be useful for determining family relationships—most SNPs are unique within a population, having arisen only once by mutation. Thus the presence of the same SNP in two persons often indicates that they have a common ancestor.

Expressed-Sequence Tags

Another type of data identified by sequencing projects consists of databases of **expressed-sequence tags** (ESTs). In most eukaryotic organisms, only a small percentage of the DNA actually encodes proteins; in humans, less than 2% of human DNA encodes the amino acids of proteins. If only protein-encoding genes are of interest, it is often more efficient to examine RNA than the entire DNA genomic sequence. RNA can be examined by using ESTs—markers associated with DNA sequences that are expressed as RNA. Expressed-sequence tags are obtained by isolating RNA from a cell and subjecting it to reverse transcription, producing a set of cDNA fragments that correspond to RNA molecules from the cell. Short stretches of these cDNA fragments are then sequenced, and the sequence obtained (called a tag) provides a marker that identifies the DNA fragment. Expressed-sequence tags can be used to find active genes in a particular tissue or at a particular point in development.

Concepts

In addition to the genomic-sequence data, genomic projects are collecting databases of nucleotides that vary among individuals (single-nucleotide polymorphisms, SNPs) and markers associated with transcribed sequences (expressed-sequence tags, ETSs).

Bioinformatics

By the time this book is published, complete genome sequences will have been determined for more than 100 different organisms, with many additional projects underway. These studies are producing tremendous quantities of sequence data. GenBank, one of the major databases of DNA sequence information, now contains more than 19 billion base pairs of sequence, and this number increases in size every month. Cataloging, storing, retrieving, and analyzing this huge data set are a major challenge of modern genetics. **Bioinformatics** is an emerging field consisting of molecular biology and computer science that centers on developing databases, computer-search algorithms, gene-prediction software, and other analytical tools that are used to make sense of DNA, RNA, and protein sequence data. Bioinformatics develops and applies these tools to "mine the data," extracting the useful information from sequencing projects.

Before being sequenced, most genomes contain few genes whose locations have already been determined, which, coupled with the enormous amount of DNA in a genome and the complexities of gene structure, makes finding genes a difficult task. Computer programs have been developed to look for specific sequences in DNA that are associated with certain genes. For example, protein-encoding genes are characterized by an **open reading frame** (ORF), which includes a start codon and a stop codon in the same reading frame. Specific sequences mark the splice sites at the beginning and end of introns; other specific sequences are present in promoters immediately upstream of start codons. Still other sequences are associated with particular functions in certain classes of proteins. Computer programs have been developed that scan the DNA for these sequences and identify genes on the basis of their presence and position. Some of these programs are capable of examining databases of EST and protein sequences to see if there is evidence that a potential gene is expressed.

It is important to recognize that the programs that have been developed to identify genes on the basis of DNA sequence are not perfect. Therefore, the numbers of genes reported in most genome projects are estimates. The presence of multiple introns, alternative splicing, multiple copies of some genes, and much noncoding DNA between genes makes accurate identification and counting of genes difficult.

www.whfreeman.com/pierce Information on ESTs, SNPs, and bioinformatics

Functional Genomics

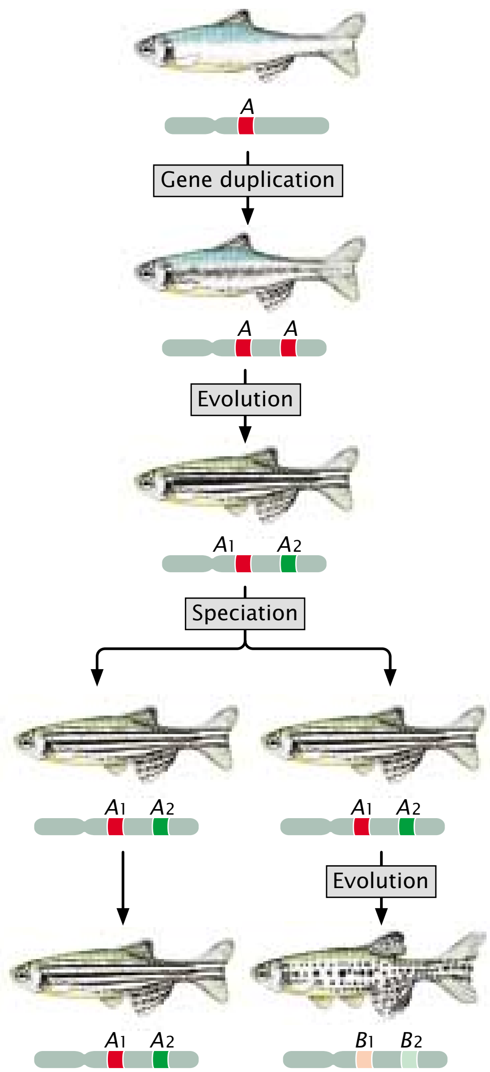
A genomic sequence is, by itself, of limited use. It would be like having a huge set of encyclopedias without being able to read—you could recognize the different letters but the text would be meaningless. Functional genomics is, in essence, probing genome sequences for meaning—identifying genes, recognizing their organization, and understanding their function. The goals of functional genomics include identifying all the RNA molecules transcribed from a genome (the **transcriptome**) and all the proteins encoded by the genome (the **proteome**). Functional genomics exploits both bioinformatics and laboratory-based experimental approaches in its search to define the function of DNA sequences.

Chapter 18 considered several methods for identifying genes and assessing their functions, including in situ hybridization, DNA footprinting, experimental mutagenesis, and the use of transgenic animals and knockouts. These methods can be applied to individual genes and can provide important information about the locations and functions of genetic information. In this section, we will focus primarily on methods that rely on knowing the sequences of other genes or that can be applied to large numbers of genes simultaneously.

Predicting Function from Sequence

The nucleotide sequence of a gene can be used to predict the amino acid sequence of the protein that it encodes. The protein can then be synthesized or isolated and its properties studied to determine its function. However, this biochemical approach to understanding gene function is both time consuming and expensive. A major goal of functional genomics has been to develop computational methods that allow gene function to be identified from DNA sequence alone, bypassing the laborious process of isolating and characterizing individual proteins.

Homology searches One computational method (often the first employed) for determining gene function is to conduct a homology search, which relies on comparing DNA and protein sequences from the same and different organisms. Genes that are evolutionarily related are said to be **homologous.** Homologous genes found in different species that evolved from the same gene in a common ancestor are called **orthologs** (Figure 19.15). For example, both mouse and human genomes contain a gene that encodes the alpha subunit of hemoglobin; the mouse and human alpha-hemoglobin genes are said to be orthologs, because both genes evolved from an alpha-hemoglobin gene in a mammalian ancestor common to mice and humans. Homologous genes in the same organism (arising by duplication of a single gene in the evolutionary past) are called **paralogs** (see Figure 19.15). Within the human genome is a gene that



Conclusion: Genes A1 and A2 are paralogs.
Genes B1 and B2 are paralogs.
Genes A1 and B1 are orthologs.
Genes A2 and B2 are orthologs.

19.15 **Homologous sequences are evolutionarily related.** Orthologs are homologous sequences found in different species; paralogs are homologous genes in the same species that arise from gene duplication.

encodes the alpha subunit of hemoglobin and another homologous gene that encodes the beta subunit of hemoglobin. These two genes arose because an ancestral gene underwent duplication and the resulting two genes diverged through evolutionary time, giving rise to the alpha- and beta-subunit genes; these two genes are paralogs. Homologous genes (both orthologs and paralogs) often have the same or related functions; so, after a function has been assigned to a particular gene, it can provide a clue to the function of a homologous gene.

Databases containing genes and proteins found in a wide array of organisms are available for homology searches. Powerful computer programs have been developed for scanning these databases to look for particular sequences. A commonly used homology search program is BLAST (Basic Local Alignment Search Tool). Suppose a geneticist sequences a genome and locates a gene that encodes a protein of unknown function. A homology search conducted on databases containing the DNA or protein sequences of other organisms may identify one or more orthologous sequences. If a function is known for one of these sequences, that function may provide information about the function of the newly discovered protein.

In a similar way, computer programs can search a single genome for paralogs. Eukaryotic organisms often contain families of genes that have arisen by duplication of a single gene. If a paralog is found and its function has been previously assigned, this function can provide information about a possible function of the unknown gene. However, paralogs often evolve new functions; so information about their functions must be used cautiously. Of the genes newly identified through genomic-sequencing projects, 50% are significantly similar to orthologs and paralogs whose function has already been described. The 50% of newly identified genes that *cannot* be assigned a function on the basis of homology searches will undoubtedly decrease in number as functions are assigned to more and more genes and as more genomes are sequenced.

Other sequence comparisons Complex proteins often contain regions that have specific shapes or functions called **protein domains.** For example, certain DNA-binding proteins attach to DNA in the same way; these proteins have in common a domain that provides the DNA-binding function. Each protein domain has an arrangement of amino acids common to that domain. There are probably a limited, though large, number of protein domains, which have mixed and matched through evolutionary time to yield the protein diversity seen in present-day organisms.

Many protein domains have been characterized, and their molecular functions have been determined. The sequence from a newly identified gene can be scanned against a database of known domains. If the gene sequence encodes one or more domains whose functions have been previously determined, the function of the domain can provide important information about a possible function of the new gene.

Another computational method for predicting protein function is a **phylogenetic profile.** In this method, the presence-and-absence pattern of a particular protein is examined across a set of organisms whose genomes have been sequenced. If two proteins are either both present or both absent in all genomes surveyed, the two proteins may be functionally related. For example, the two proteins might function as consecutive steps in a biochemical pathway. The idea is that the two proteins depend on each other and will evolve together. One protein cannot function without the other, and they will either both be present or both be absent.

Consider the following proteins in four bacterial species (FIGURE 19.16a):

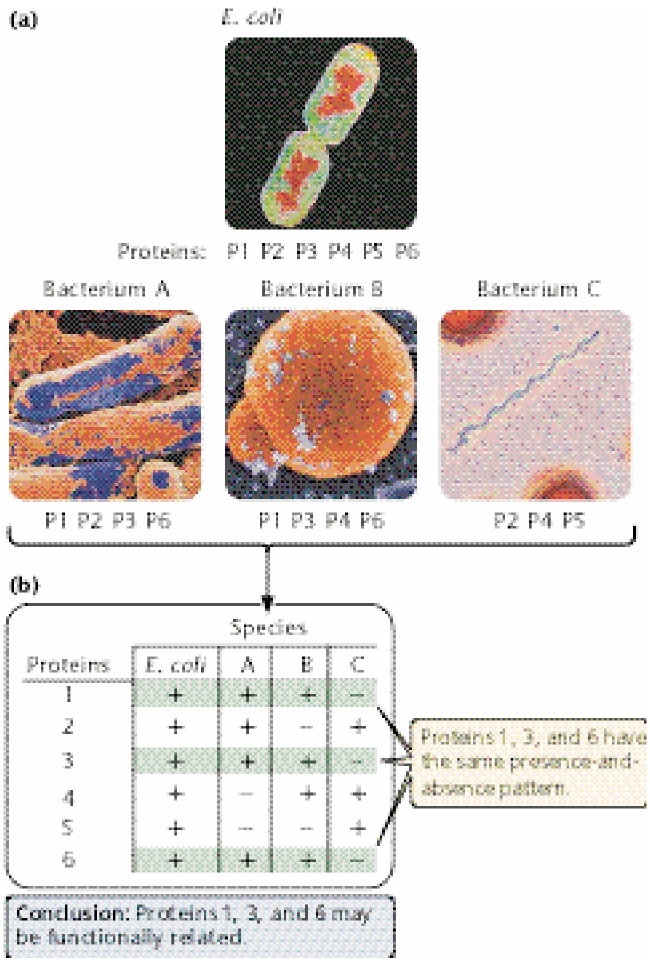
E. coli:	protein 1, protein 2, protein 3, protein 4, protein 5, protein 6
Species A:	protein 1, protein 2, protein 3, protein 6
Species B:	protein 1, protein 3, protein 4, protein 6
Species C:	protein 2, protein 4, protein 5

We can create a phylogenetic profile by constructing a table comparing the presence (+) or absence (−) of the proteins in the four bacterial species (FIGURE 19.16b). The phylogenetic profile reveals that proteins 1, 3, and 6 are either all present or all absent in all species; so these proteins might be functionally related.

Examining **fusion patterns** among proteins is another method for predicting functional relations; this technique is sometimes called the Rosetta Stone method. Functionally related, separate proteins in one organism sometimes exist as a single, fused protein in another organism. Thus, the presence of a fused A + B protein in one species suggests that separate proteins A and B in another organism may be functionally related.

Yet another method for determining the function of an unknown gene is **gene neighbor analysis** (FIGURE 19.17). Genes that encode functionally related proteins are often closely linked in bacteria. For example, if two genes are consistently linked in the genomes of several bacteria, they might be functionally related. Functionally related genes are sometimes also linked in eukaryotes; examples are the *hox* genes, which play an important role in embryonic development (Chapter 21).

It is important to recognize that functions suggested by computational methods such as homology searches, phylogenetic profiling, fusion proteins, and neighbor analysis do not define a protein's function; rather these computational methods provide hints about possible

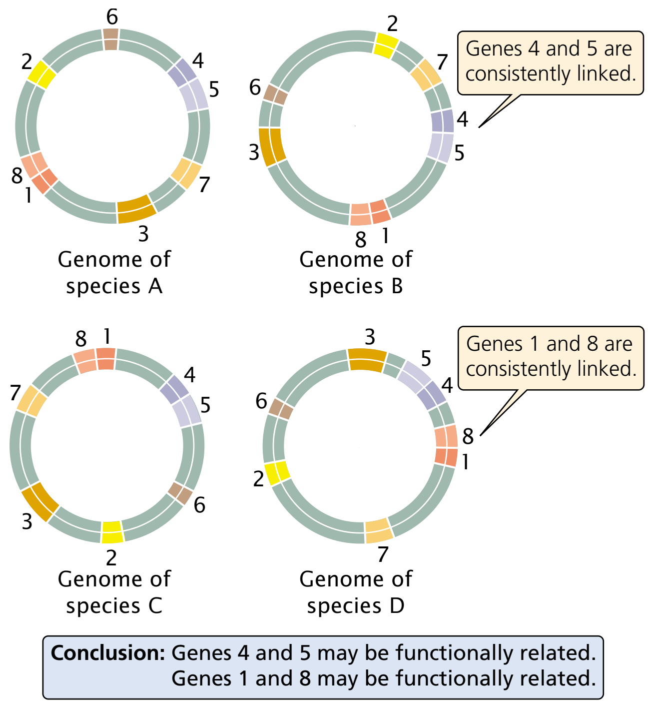


19.16 Phylogenetic profiling can be used to infer protein function. (Micrographs from: top, CNRI/SPL/Photo Researchers; middle left and center, Gary Gaugler/Visuals unlimited; middle right, M. Abbey/Visuals unlimited.)

functions that can be pursued through detailed analyses of the biochemistry and cellular location of the protein. Nevertheless, these computational methods and others like them have proved to be invaluable in determining the functions of genes revealed in genomic studies.

Concepts

Genes can be identified by computer programs that look for characteristic features of genes, such as start and stop codons in the same reading frame, sequences that mark the beginning and the end of introns, and sequences found within promoters. Clues to the functions of genes can be obtained by homology searches, comparing protein domains, phylogenetic profiling, protein fusion patterns, and gene neighbor analysis.



19.17 The gene neighbor method infers gene function on the basis of the linkage arrangements of the genes. Genes that are consistently linked in different genomes may be functionally related.

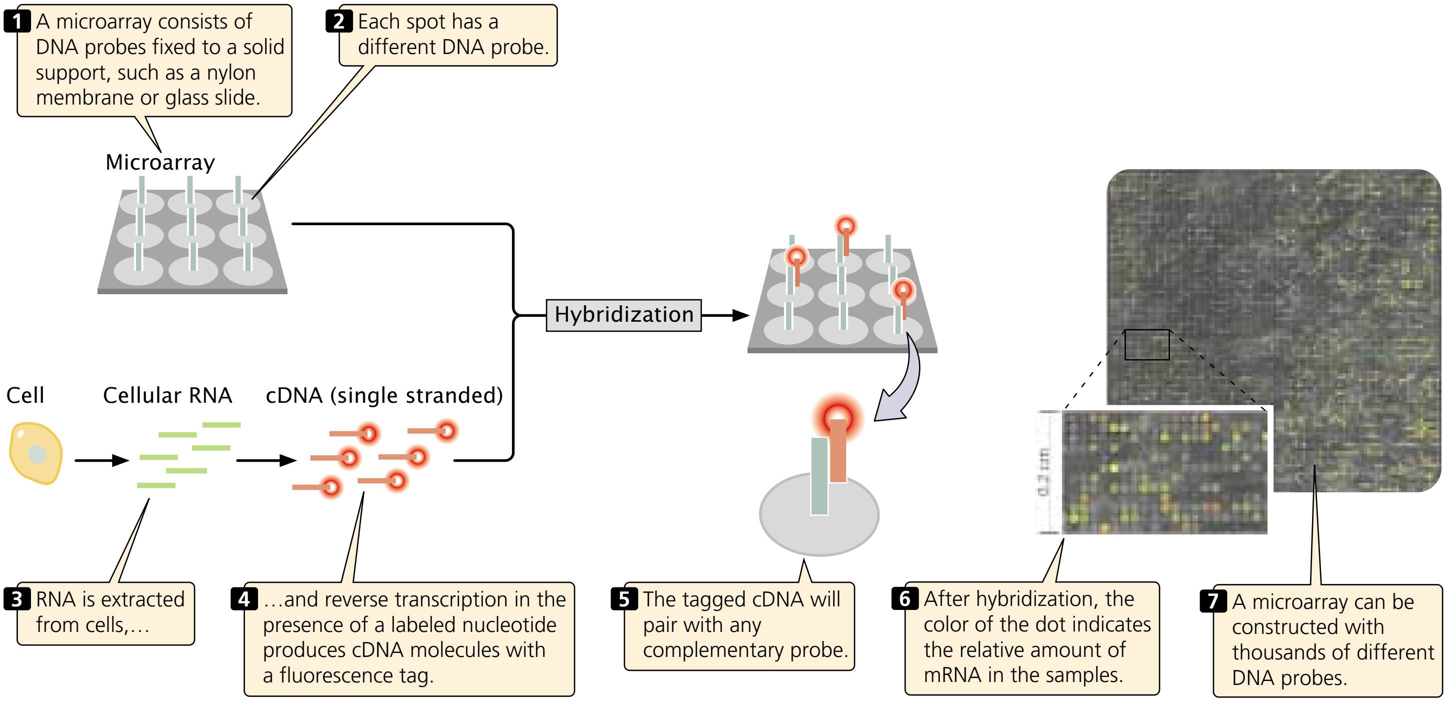
Gene Expression and Microarrays

Many important clues about gene function come from knowing when and where the genes are expressed. The development of microarrays has allowed the expression of thousand of genes to be monitored simultaneously.

Microarrays rely on nucleic acid hybridization (see Chapter 18), in which a known DNA fragment is used as a probe to find complementary sequences (FIGURE 19.18). The probe is usually fixed to some type of solid support, such as a nylon filter or a glass slide. A solution containing a mixture of DNA or RNA is applied to the solid support; any nucleic acid that is complementary to the probe will bind to it. Nucleic acids in the mixture are labeled with a radioactive or fluorescent tag so that molecules bound to the probe can be easily detected.

In a microarray (also called a gene chip), numerous known DNA fragments are fixed to a solid support in an orderly pattern or array, usually as a series of dots. These DNA fragments (the probes) usually correspond to known genes.

When the microarray has been constructed, mRNA, DNA, or cDNA isolated from experimental cells is labeled with fluorescent nucleotides and applied to the array. Any of the DNA or RNA molecules that are complementary to probes on the array will hybridize with them and emit fluo-



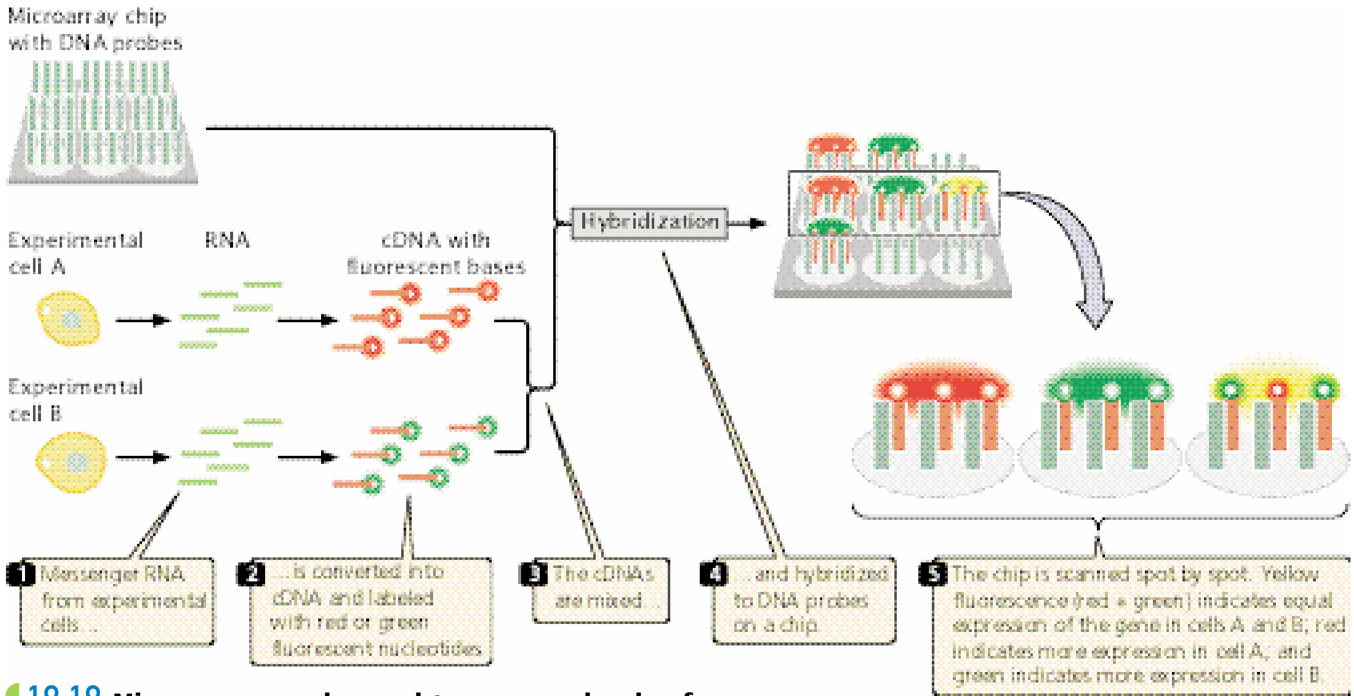
19.18 **Microarrays are used to simultaneously detect the expression of many genes.** (D. Lockhart and E. Winzeler, 2000, *Nature* 405:827.)

rescence, which can be detected by an automated scanner. An array containing tens of thousands of probes can be applied to a glass slide or silicon wafer just a few square centimeters in size.

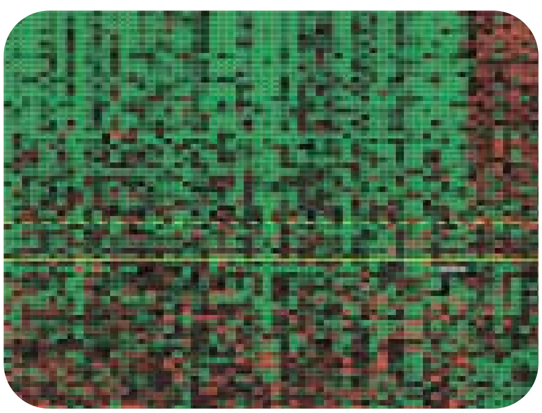
One type of DNA chip is illustrated in FIGURE 19.19. For this chip, mRNA from experimental cells is converted into cDNA and labeled with red fluorescent nucleotides. MessengerRNA from control cells is converted into cDNA and labeled with green fluorescent nucleotides. The labeled cDNAs are mixed and hybridized to the DNA chip, which contains DNA probes from different genes. Hybridization of the red (experimental) and green (control) cDNAs is proportional to the relative amounts of mRNA in the samples. The fluorescence of each spot is assessed with microscopic scanning and appears as a single color. Red indicates the overexpression of a gene in the experimental cells relative to that in the control cells (more red-labeled cDNA hybridizes), whereas green indicates the underexpression of a gene in the experimental cells relative to that in the control cells (more green-labeled cDNA hybridizes). Yellow indicates equal expression in experimental and control cells (equal hybridization of red- and green-labeled cDNAs), and no color indicates no expression in either experimental or control cells. Microarrays that allow the detection of specific alleles, SNPs, and even particular proteins also have been created.

Microarrays allow the expression of thousands of genes to be monitored simultaneously, enabling scientists to study which genes are active in particular tissues. They can also be used to investigate how gene expression changes in the course of biological processes such as development or disease progression. In one study, researchers examined gene expression to predict the long-term outcome for women who had undergone treatment for breast cancer. Breast cancer affects 1 of 10 women in the United States, and half of those with the disease die from it. Current treatment depends on a number of factors, including a woman's age, the size of the tumor, the characteristics of tumor cells, and whether the cancer has already spread to nearby lymph nodes. Many women whose cancer has not spread are treated by removal of the tumor and radiation therapy, yet the cancer later reappears in some of the women thus treated. These women might benefit from more-aggressive treatment when the cancer is first detected.

Using microarrays, researchers examined the expression patterns of 25,000 genes from primary tumors of 78 young women who had breast cancer. In 34 of these patients, the cancer later spread to other sites; the other 44 patients remained free of breast cancer for 5 years after their initial diagnoses. The researchers identified a subset of 70 genes whose expression patterns in the initial tumors accurately predicted whether the cancer would later spread (FIGURE 19.20). This degree of prediction was much higher than that of traditional predictive measures, which are based on the size and histology of the tumor. These results, though preliminary and confined to a small sample of cancer patients, suggest that gene-expression data



19.19 Microarrays can be used to compare levels of gene expression in different types of cells.



19.20 Microarrays can be used to examine gene expression associated with disease progression. Shown here are expression patterns of 70 genes in the initial tumors from patients whose cancer later spread to other sites and from other patients who remained free of breast cancer for 5 years after their initial diagnosis. Red indicates higher gene expression; green indicates lower gene expression; black indicates no change in gene expression; and gray indicates no data available. Each row represents the primary tumor from a patient and each column represents a different gene. Tumors below the solid yellow line came primarily from patients in which the cancer spread to distant sites within 5 years of diagnosis; tumors above the solid line came primarily from patients who remained cancer free for at least 5 years. (L.J. Van't Veer, 2002, *Nature* 405:532.)

obtained from microarrays can be a powerful tool in determining the nature of cancer treatment.

Concepts

Microarrays, consisting of DNA probes attached to a solid support, can be used to determine which RNA and DNA sequences are present in a mixture of nucleic acids. They are capable of determining which RNA molecules are being synthesized and thus can be used to examine changes in gene expression.

www.whfreeman.com/pierce More on microarrays

Genomewide Mutagenesis

One of the best methods for determining the function of a gene is to examine the phenotypes of individual organisms that possess a mutation in the gene. Traditionally, genes encoding naturally occurring variations in a phenotype were mapped, the causative genes were isolated, and their products were studied. But this procedure was limited by the number of naturally occurring mutations and the difficulty of mapping genes with a limited number of chromosomal markers. The number of naturally occurring mutations can be increased by exposure to mutagenic agents, and the accuracy of mapping is increased dramatically by the availability of mapped molecular markers, such as RFLPs, microsatellites, STSs, ETSs, and SNPs. These two methods—random inducement of mutations on a genomewide basis and mapping with molecular markers—are coupled and automated in a **mutagenesis screen.**

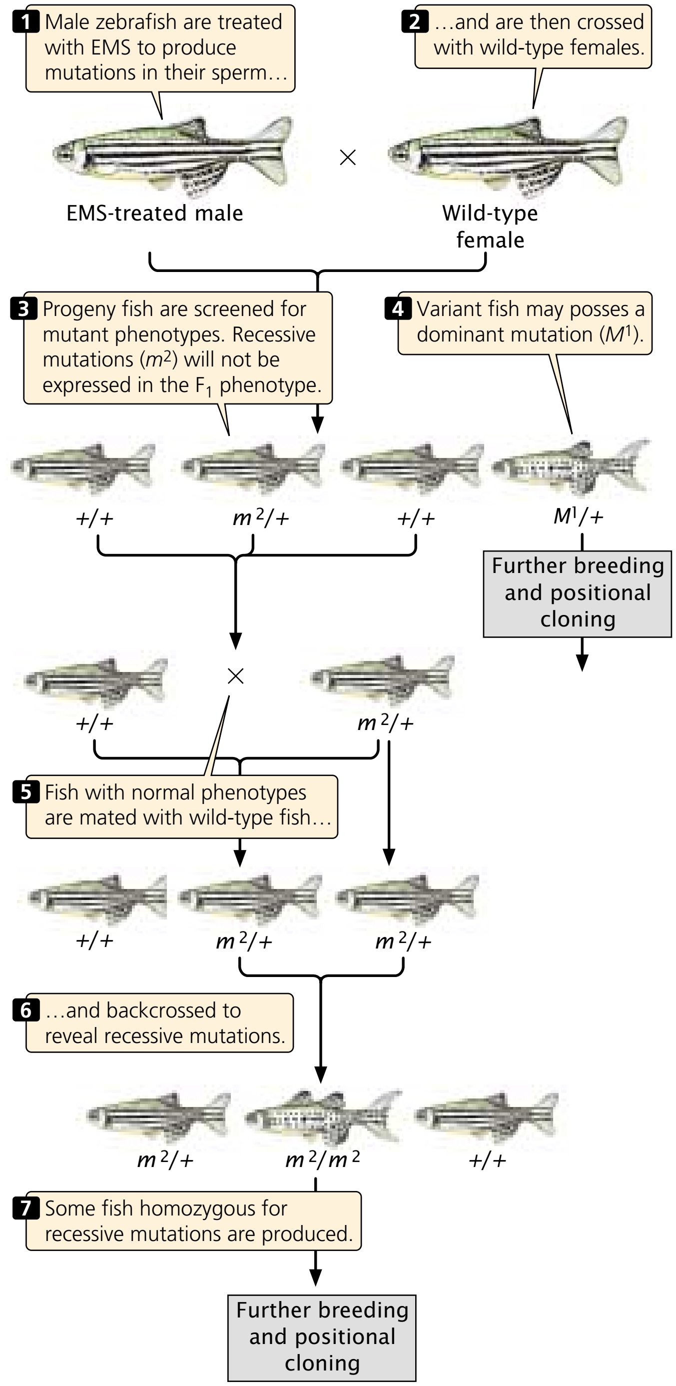
Mutagenesis screens can be used to search for specific genes encoding a particular function or trait. For example, mutagenesis screens of mice are being used to identify genes having roles in cardiovascular function. When genes that affect cardiovascular function are located in mice, homology searches are carried out to determine if similar genes exist in humans. These genes can then be studied to better understand cardiac disease in humans.

To conduct a mutagenesis screen, random mutations are induced in a population of organisms, creating new phenotypes. The mutations are induced by exposing the organisms to radiation, a chemical mutagen (Chapter 17), or transposable elements (DNA sequences that insert randomly into the DNA; Chapter 20). The procedure for a typical mutagenesis screen is illustrated in FIGURE 19.21. Here, male zebra fish are treated with ethylmethylsulfonate, or EMS, a chemical that induces mutations in their sperm. The treated males are mated with wild-type female fish. The offspring are heterozygous for mutations induced by EMS and are screened for any variant phenotypes that might be the product of dominant mutations expressed in these heterozygous fish.

Recessive mutations will not be expressed in the F_1 progeny but can be revealed with further breeding. The F_1 offspring are mated with wild-type fish, and the offspring from this cross are then backcrossed with their male parents, producing fish that are homozygous for recessive mutations. The offspring of the backcross are then screened for variant phenotypes.

The fish with variant phenotypes undergo further breeding experiments to verify that their variant phenotype is, in fact, due to a single-gene mutation. After the genetic nature of an abnormal phenotype has been verified, the gene that causes the phenotype can be located by **positional cloning.** The first step in positional cloning is to demonstrate linkage between the trait and one or more already mapped genetic markers. The progeny of genetic crosses that include the mutant phenotype are examined for a large number of molecular markers that cover the entire genome. The cosegregation of markers and the mutant phenotype provides evidence of linkage, indicating that the marker and the gene encoding the mutant phenotype are physically linked on the same chromosome. Cosegregating markers provide information about the general chromosome region in which the gene is located.

The next step is to localize the mutated gene to a smaller region of the chromosome, which is usually done by examining a linkage map of the chromosome region to identify other molecular markers in close proximity to the gene of interest. The gene causing the mutant phenotype is then mapped in relation to these markers. Next is the creation of a physical map, which requires a set of overlapping clones from the area of interest. A physical map of these overlapping clones that includes information about the molecular markers allows the identification of one or more clones that contain the gene of interest. These clones



19.21 **Genes affecting a particular characteristic or function can be identified by a genomewide mutagenesis screen.** In this illustration, M^1 represents a dominant mutation and m^2 represents a recessive mutation.

are then sequenced to find potential candidate genes that might encode the mutant phenotype. Candidate genes are evaluated by studying their expression patterns, protein products, and homology to genes of known function. This information might suggest that one or more of the candidate genes is likely to be the cause of the phenotype. The candidate genes can be examined for the presence of mutations in the gene sequences carried by those individuals having a mutant phenotype. Further proof that a particular gene causes the phenotype can be obtained by mutating a specific gene and observing the phenotype in the offspring.

Concepts

Genomewide mutagenesis screening coupled with positional cloning can be used to identify genes that affect a specific characteristic or function.

Comparative Genomics

Genome-sequencing projects provide detailed information about gene content and organization in different species and even in different members of the same species, allowing inferences about how genes function and genomes evolve. They also provide important information about evolutionary relationships among organisms and about factors that influence the speed and direction of evolution.

Prokaryotic Genomes

A large number of bacterial genomes have now been sequenced (Table 19.2). Most prokaryotic genomes consist of a single circular chromosome, but there are exceptions, such as *Vibrio cholerae* (the bacterium that causes cholera; see introduction to Chapter 8), which has two circular chromosomes, and *Borrelia burgdorferi,* which has one large linear chromosome and 21 smaller chromosomes.

The total amount of DNA in prokaryotic genomes ranges from more than 7 million base pairs in *Mesorhizo-*

Table 19.2 Characteristics of some completely sequenced representative prokaryotic genomes

Species	Size (Millions of Base Pairs)	Number of Predicted Genes	G + C (%)
Archaea			
Archaeoglobus fulgidus	2.18	2407	49
Methanobacterium thermoautotrophicum	1.75	1869	50
Methanococcus jannaschii	1.66	1715	32
Thermoplasma acidophilum	1.56	1478	46
Eubacteria			
Bacillus subtilis	4.21	4100	44
Bordetella parapertussis	4.75	*	69
Buchnera species	0.64	564	27
Campylobacter jejuni	1.64	1654	31
Escherichia coli	4.64	4289	51
Haemophilus influenzae	1.83	1709	39
Mesorhizobium loti	7.04	6752	63
Mycobacterium tuberculosis	4.41	3918	66
Mycoplasma genitalium	0.58	480	32
Staphylococcus aureus	2.88	2697	33
Treponema pallidum	1.14	1031	53
Ureaplasma urealyticum	0.75	611	26
Vibrio cholerae	4.03	3828	48

Source: Data from the Genome Atlas of the Center for Biological Sequence Analysis, http://www.cbs.dtu.dk/services/GenomeAtlas/
* Data not available.

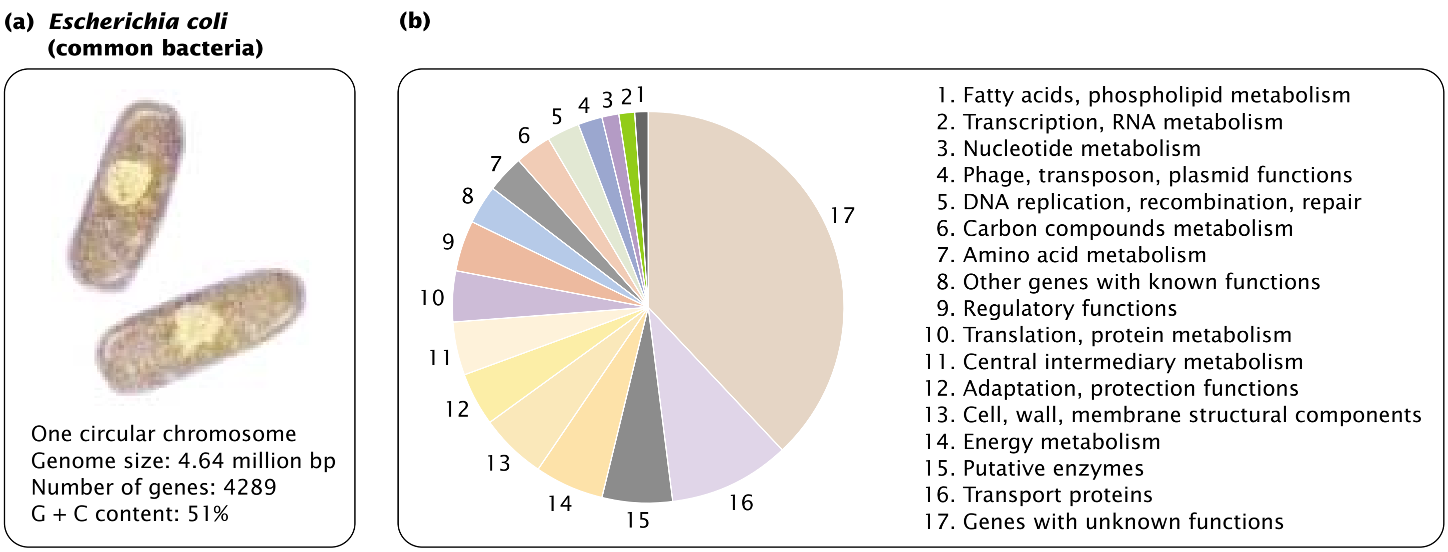
bium loti to only 580,000 bp in *Mycoplasma genitalium.* *Escherichia coli,* the most widely used bacterium for genetic studies, has 4.6 million base pairs (FIGURE 19.22a). The number of genes is usually from 1000 to 2000, but some species have as many as 6700, and others as few as 480. The density of genes is rather constant across all species, with about 1 gene for every 1000 bp. Thus bacteria with larger genomes usually have more genes.

Only about half of the genes identified in prokaryotic genomes can be assigned a function. Almost a quarter of the genes have no significant sequence similarity to any other known genes in bacteria, suggesting that there is considerable genetic diversity among bacteria. The number of genes that encode biological functions such as transcription and translation tends to be similar among species, even when their genomes differ greatly in size. This similarity suggests that these functions are encoded by a basic set of proteins that does not vary among species. On the other hand, the number of genes taking part in biosynthesis, energy metabolism, transport, and regulatory functions varies greatly among species and tends to be higher in larger genomes. The functions of predicted genes (i.e., genes identified by computer programs) and known genes in *E. coli* are presented in FIGURE 19.22b. A substantial part of the "extra" DNA found in the larger bacterial genomes is made up of paralogous genes that have arisen by duplication.

The G + C content (percentage of bases that consist of guanine or cytosine) of prokaryotic genomes varies widely, from 26% to 69%. This more-than-twofold difference in G + C content affects the frequency of particular amino acids in the proteins produced by different bacterial species. For example, glycine, alanine, proline, and arginine are encoded by codons that have G and C nucleotides; so these amino acids are incorporated into proteins with higher frequency in organisms whose genomes have a high G + C content. On the other hand, isoleucine, phenylalanine, tyrosine, and methionine are encoded by codons that tend to have A and T (U in RNA) nucleotides; so these amino acids are found more frequently in proteins encoded by species whose genome has a low G + C content. Which synonymous codons are used is also affected by the G + C content; some synonymous codons have more G and C nucleotides than do others, and these codons tend to be used more frequently in those species with high G + C content.

The results of genomic studies of prokaryotic species support the conclusion that archaea and eubacteria are evolutionarily unique (see Chapter 2). The results also reveal that both closely and distantly related bacterial species periodically exchange genetic information over evolutionary time, a process called **horizontal gene exchange.** Such exchange may take place through bacterial uptake of DNA in the environment (transformation), through the exchange of plasmids, and through viral vectors (see Chapter 8). Horizontal gene exchange has been recognized for some time, but analyses of many microbial genomes now indicate that it is more extensive than was previously recognized. For example, an analysis of two eubacteria species demonstrated that from 20% to 25% of their genes were more similar to genes from archaea than to those from other eubacterial species.

www.whfreeman.com/pierce Information on prokaryotic and other genomes



19.22 **Genomic characteristics of the bacterium *E. coli.*** (a) Genome size, number of genes, and G + C content. (b) Percentages of genes affecting various known and unknown functions.

Table 19.3 Characteristics of Some Eukaryotic Genomes That Have Been Completely Sequenced

Species	Genome Size (Millions of Base Pairs)	Number of Predicted Genes	Number of Protein-Domain Families
Saccharomyces cerevisiae (yeast)	12	6,144	851
Arabidopsis thaliana (plant)	125	25,706	1012
Caenorhabditis elegans (roundworm)	100	18,266	1014
Drosophila melanogaster (fruit fly)	180	13,338	1035
Homo sapiens (human)	3400	~32,000	1262

Source: Number of genes and protein-domain families from International Human Genome Sequencing Consortium, Initial sequencing and analysis of the human genome, *Nature* 409 (2001), Table 23.

Eukaryotic Genomes

The genomes of only a few eukaryotic organisms have been completely sequenced, but some tentative statements can be made about the content and organization of eukaryotic genetic information from these organisms.

The genomes of eukaryotic organisms (Table 19.3) are larger than those of prokaryotes, and, in general, multicellular eukaryotes have more DNA than do simple, single-celled eukaryotes such as yeast (see p. 000 in Chapter 11). There is no close relation, however, between genome size and complexity among the multicellular eukaryotes. For example, the roundworm *Caenorhabditis elegans* is structurally more complex than the plant *Arabidopsis* but has considerably less DNA. Eukaryotic genomes also contain more genes than do prokaryotes, and the genomes of multicellular eukaryotes have more genes than do the genomes of single-celled eukaryotes. The *number* of genes among multicellular eukaryotes also is not obviously related to phenotypic complexity: humans have more genes than do invertebrates but only twice as many as fruit flies and only slightly more than the plant *Arabidopsis*. Eukaryotic genomes contain multiple copies of many genes, indicating that gene duplication has been an important process in genome evolution.

A substantial part of the genomes of multicellular organisms consists of moderately and highly repetitive sequences (see Chapter 11), and the percentage of repetitive sequences is usually higher in those species with larger genomes (Table 19.4). Most of these repetitive sequences appear to have arisen through transposition. This is particularly evident in the human genome, where 45% of the DNA is derived from transposable elements, many of which are defective and no longer able to move. The majority of DNA in multicellular organisms is noncoding, and many genes are interrupted by introns. In the more complex eukaryotes, both the number and the length of the introns are greater.

In spite of only a modest increase in gene number, vertebrates have considerably more protein diversity than do invertebrates. The human genome does not encode many new protein domains; there are 1262 domains in humans compared with 1035 in fruit flies (see Table 19.3). However, the existing domains in humans are assembled into more combinations, leading to many more types of proteins. For example, the human genome contains almost two times as many arrangements of protein domains as worms or flies contain and almost six times as many as yeast contains. Humans, worms, and flies have many of the same families of genes in common, but each family in the human genome has a greater number of different genes, suggesting that gene duplication has been an important process in vertebrate evolution.

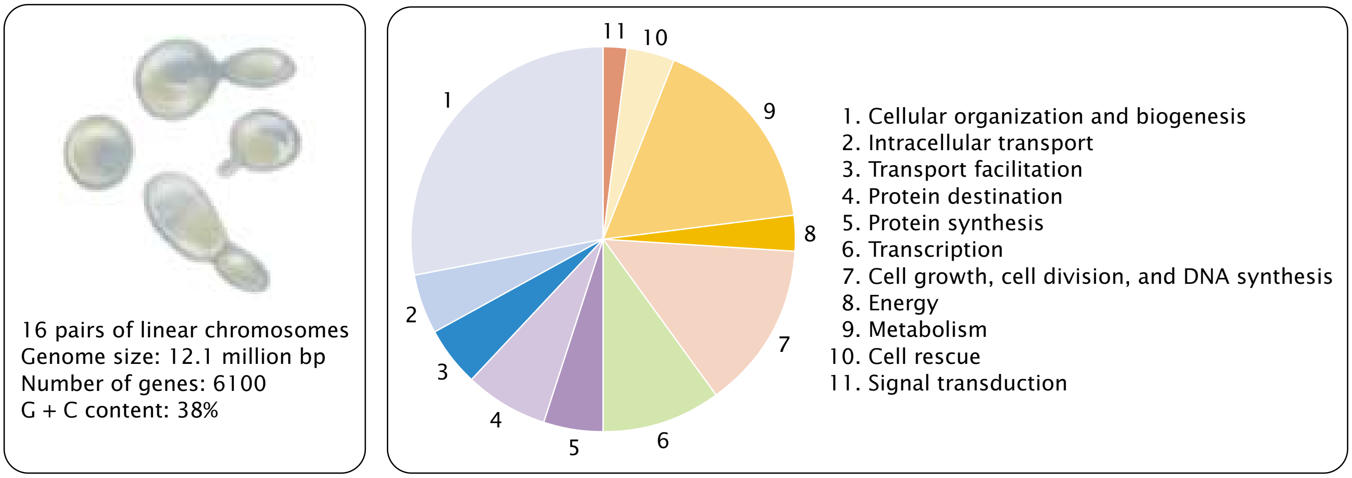
Table 19.4 Percentage of genome consisting of interspersed repeats derived from transposable elements

Organism	Percentage of Genome
Plant (*Arabidopsis thaliana*)	10.5
Worm (*Caenorhabditis elegans*)	6.5
Fly (*Drosophila melanogaster*)	3.1
Human (*Homo sapiens*)	44.4

Concepts

Comparative genomics compares the content and organization of whole genomic sequences from different organisms. Prokaryotic genomes are small, usually ranging from 1 million to 3 million base pairs of DNA, with several thousand genes. Among multicellular eukaryotic organisms, there is no clear relation between organismal complexity and amount of DNA or gene number. A substantial part of the genome in eukaryotic organisms consists of repetitive DNA, much of which is derived from transposable elements.

(a) *Saccharomyces cerevisiae* (yeast) **(b)**



19.23 Genomic characteristics of yeast, *Saccharomyces cerevisiae.* (a) Number of chromosomes, genome size, number of genes, and G + C content. (b) Percentages of genes affecting various known and unknown functions.

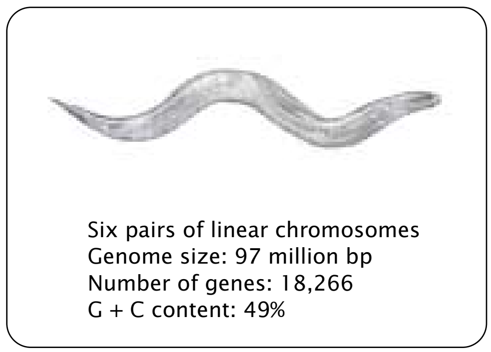
Yeast genome As mentioned earlier, *Saccharomyces cerevisiae* (yeast) was the first eukaryotic genome to be completely sequenced. Its genome consists of 12.1 million base pairs of DNA and 6100 potential genes, of which about 5900 encode proteins (FIGURE 19.23a), giving a gene density of about one gene for every 2000 bp of DNA. The distribution of gene functions in yeast is displayed in FIGURE 19.23b. The yeast genome contains considerable redundancy; there are a number of blocks of repeated sequences in the genome, and 30% of the genes exist in two or more copies.

Worm genome *Caenorhabditis elegans,* a roundworm, has a genome consisting of 97 million base pairs of DNA (FIGURE 19.24). More than 18,000 protein-encoding genes have been identified in the *C. elegans* genome, of which more than 40% are homologous with genes found in other organisms. There is one gene for about every 5000 bp of DNA, and gene density is more uniform across chromosomes than it is in most eukaryotes.

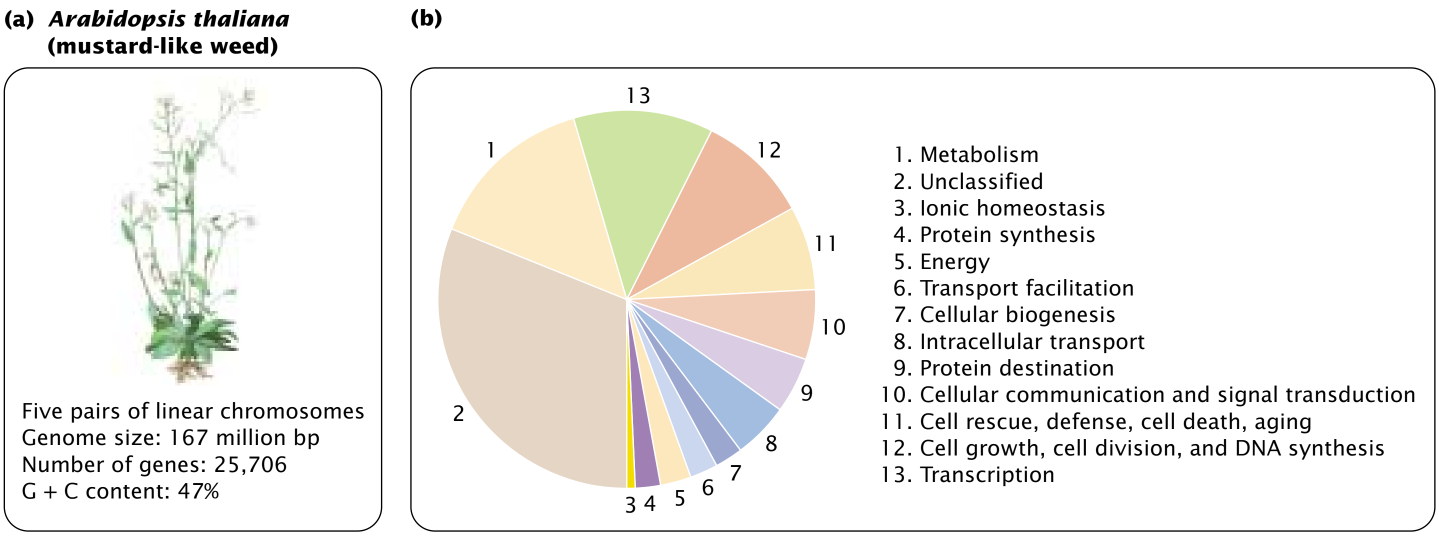
Plant genome The genome of *Arabidopsis thaliana,* a small mustardlike plant, consists of 167 million base pairs of DNA (FIGURE 19.25a), encoding 25,706 predicted genes. Although *Arabidopsis* has many proteins in common with yeast, worm, fly, and humans, it has roughly 150 protein families not seen in other eukaryotes, including structural proteins, transcription factors, enzymes, and proteins of unknown function. FIGURE 19.25b shows the distribution of gene functions in *Arabidopsis.*

Gene duplication has played an important role in the evolution of *Arabidopsis,* with 60% of its genome consisting of duplicated segments. Seventeen percent of the genes exist in tandem arrays, which are multiple copies of the same gene positioned one after another. One of the processes that produce tandem arrays of duplicated genes is unequal crossing over (see p. 000 in Chapter 9). A number of large duplicated regions, encompassing hundreds of thousands or millions of base pairs of DNA also are present. The large extent of duplication in the *Arabidopsis* genome suggests that this species had a tetraploid (4N) ancestor (see Chapter 9) and that all genes were duplicated in the past, followed by extensive gene rearrangement and divergence. Thus, at least two different mechanisms seem to have led to the large number of duplications seen in the *Arabidopsis* genome: (1) duplication of the whole genome through polyploidy; and

***Caenorhabditis elegans* (round worm)**



19.24 Genomic characteristics of the roundworm, *Caenorhabditis elegans.*



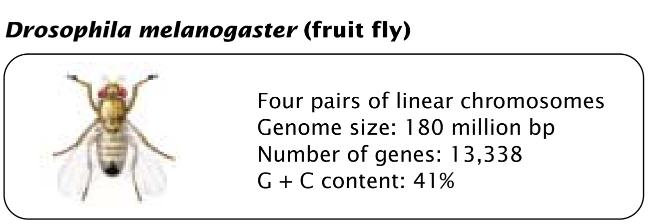
19.25 Genomic characteristics of the mustard plant, *Arabidopsis thaliana.* (a) Number of chromosomes, genome size, number of genes, and G + C content. (b) Percentages of genes affecting various known and unknown functions.

(2) duplication of individual genes arrayed in tandem through unequal crossing over.

Transposable elements are common in the *Arabidopsis* genome and make up about 10% of the genome but are much less frequent than in the human genome and in some other plant genomes. Most of these transposable elements are not transcribed, and many are concentrated in the regions surrounding the centromere.

Although *Arabidopsis, C. elegans,* and *Drosophila* have similar numbers of proteins, the *Arabidopsis* genome has more genes. This difference can be explained by the large number of duplicated copies of genes found in the *Arabidopsis* genome.

Fly genome *Drosophila melanogaster,* the fruit fly, has a genome of 180 million base pairs of DNA located on four chromosomes (FIGURE 19.26). A third of its genome is made up of heterochromatin, which contains few genes. This extensive heterochromatin, consisting mainly of short simple repeats, made sequencing the genome of *Drosophila* difficult (because the repeats lead to much overlap in sequence among cloned fragments, making it difficult to assemble the clones in the correct order). *Drosophila* has more than 13,000 predicted genes. There are 14,113 RNA transcripts produced from these genes, with some genes encoding multiple transcripts through alternative splicing. *Drosophila* genes average four exons per gene, although this number is probably an underestimate. The average RNA molecule encoded by a gene is 3058 nucleotides in length.

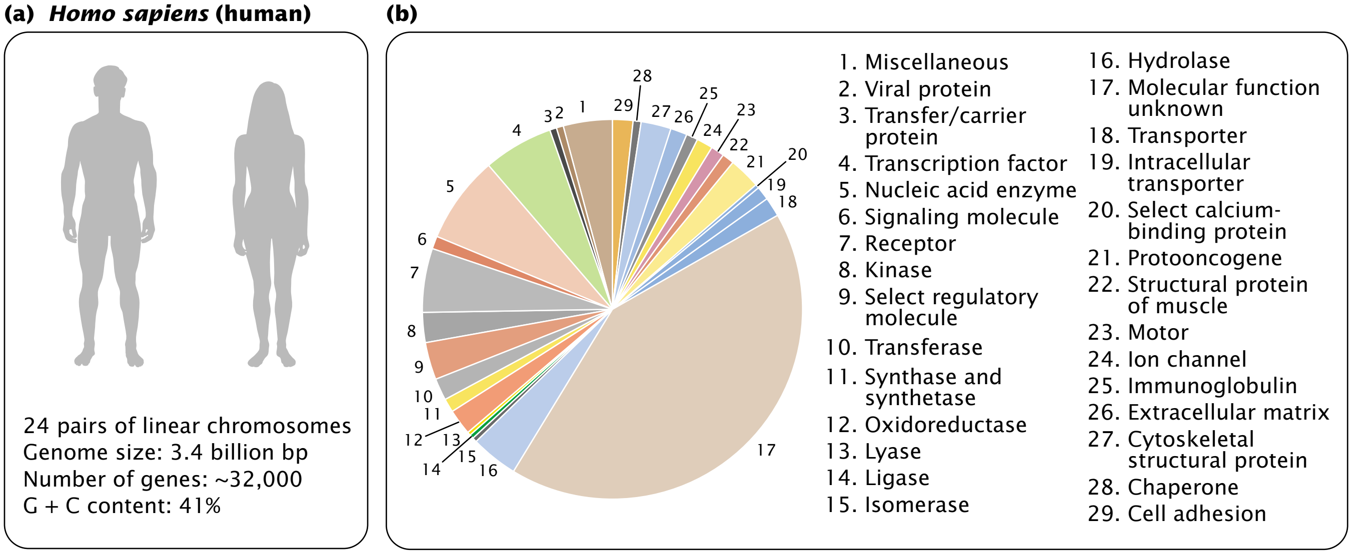


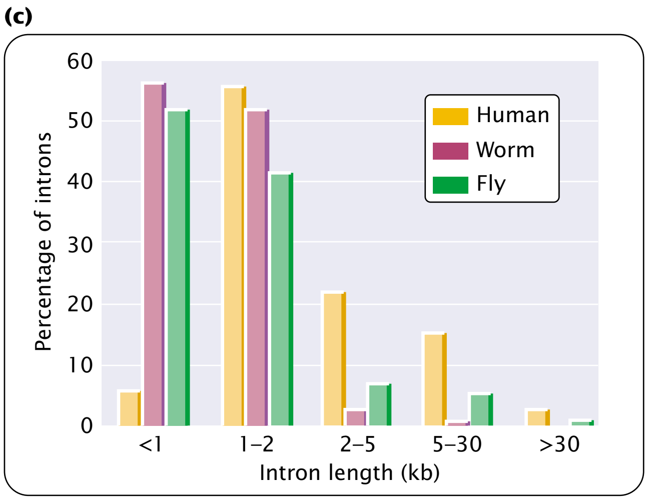
19.26 Genomic characteristics of *Drosophila melanogaster.*

Human genome The human genome is 3.4 billion base pairs in length (FIGURE 19.27a). Only about 25% of the DNA is transcribed into RNA, and less than 2% actually encodes proteins (FIGURE 19.27b). Active genes are often separated by vast deserts of noncoding DNA, much of which consists of repeated sequences derived from transposable elements.

The average gene in the human genome is approximately 27,000 bp in length, with about 9 exons. (Table 19.5). (One exceptional gene has 234 exons.) The introns of human genes are much longer, and there are more of them than in other genomes (FIGURE 19.27c) The human genome does not encode substantially more protein domains (see Table 19.3), but the domains are combined in more ways to produce a relatively diverse proteome. Gene functions encoded by the human genome are presented in Figure 19.27b. A single gene often encodes multiple proteins through alternative splicing; each gene may encode, on the average, two or three different mRNAs, meaning that the human genome, with approximately 32,000 genes, might encode as many as 96,000 proteins.

Gene density varies among human chromosomes; chromosomes 17, 19, and 22 have the highest density and





19.27 Genomic characteristics of *Homo sapiens*. (a) Number of chromosomes, genome size, number of genes, and G + C content. (b) Percentages of genes affecting various known and unknown functions. (c) Intron length of genes in humans, worm, and fly.

Table 19.5 Average characteristics of genes in the human genome

Characteristic	Average
Number of exons	8.8
Size of internal exon	145 bp
Size of intron	3365 bp
Size of 5′ untranslated region	300 bp
Size of 3′ untranslated region	770 bp
Size of coding region	1340 bp
Total length of gene	27,000 bp

chromosomes X, 4, 18, 13, and Y have the lowest density. Some proteins encoded by the human genome that are not found in other animals include those affecting immune function; neural development, structure and function; intercellular and intracellular signaling pathways in development; hemostasis; and apoptosis.

Transposable elements are much more common in the human genome than in worm, plant, and fruit-fly genomes (Table 19.4). The density of transposable elements varies, depending on chromosome location. In one region of the X chromosome, 89% of the DNA is made up of transposable elements, whereas other regions are largely devoid of these elements. There are variety of types of transposable elements in the human genome, including LINEs, SINEs, retrotransposons, and DNA transposons (see Chapter 11). Most appear to be evolutionarily old and are defective, containing mutations and deletions so that they are no longer capable of transposition.

www.whfreeman.com/pierce Information on the human genome and other genome-sequencing projects, including animal, plant, protozoan, fungal, and bacterial genomes

The Future of Genomics

The genomes of numerous organisms are in the process of being sequenced. These sequencing efforts, combined with the large amount of known DNA sequence that now exists, provide information that is tremendously useful for agriculture, human health, and biotechnology. The complete genome sequences of the mouse and the chimpanzee will serve as important sources of insight into the function and evolution of the human genome, inasmuch as these organisms are related to humans and are often used in studies of human health. Having complete genome sequences of crop plants and domestic animals will make it easier to identify genes that affect yield, disease and pest resistance, and other agriculturally important traits, which can then be manipulated by traditional breeding or genetic engineering to produce greater quantities and more-nutritious foods.

In the future, whole or partial genomic sequence information will be used in individual patient care. Currently, newborn babies are screened for a few treatable genetic diseases, such as phenylketonuria, which can be identified with the use of simple biochemical tests. In the future, newborns may be screened for a large number of variations in genetic sequence that confer high risk to treatable diseases, such as coronary artery disease, hypertension, asthma, and certain types of cancer. For those persons who are identified as genetically at risk, preventive treatment may be started early. In what has been called "personalized medicine," a person's DNA sequence may be used to predict responses to different treatment regimes, and drug therapy may then be fine-tuned to a person's genetic background. Genetic testing of both patients and pathogens will allow faster and more-precise diagnoses of many diseases.

Along with the many potential benefits of having complete sequence information are concerns about the misuse of this information. With the knowledge gained from genomic sequencing, many more genes for diseases, disorders, and behavioral and physical traits will be identified, increasing the number of genetic tests that can be performed to make predictions about the future phenotype and health of a person. There is concern that information from genetic testing might be used to discriminate against people who are carriers of disease-causing genes or who might be at risk for some future disease. Questions arise about who owns a person's genome sequence. Should employers and insurance companies have access to this information? What about relatives, who have similar genomes and who might also be at risk for some of the same diseases? There are also questions about the use of this information to select for specific traits in future offspring. All of these concerns are legitimate and must be addressed if we are to use the information from genome sequencing responsibly.

www.whfreeman.com/pierce Ethical issues associated with the Human Genome Project and genomics in general

Connecting Concepts Across Chapters

Genomics, the focus of this chapter, uses many of the techniques described in Chapter 18 for studying individual genes and applies them to the entire genome. What is different about genomics is the tremendous amount of information that is produced by using these techniques, requiring special computational tools. Although the details of many of these methods are beyond the scope of this book, an understanding of the underlying principles of genomics and the general trends emerging from the results of genomic studies is important to a student in a general genetics course. Genomics holds great potential for understanding biological processes and for applications in health, agriculture, and biotechnology. It will undoubtedly be one of the most important areas of future genetic research.

A surprising result to emerge from the study of genomics is the finding that organisms that differ greatly in phenotype and complexity may possess many similar genes and, in fact, may not differ greatly in the total number of genes that they possess. This finding suggests that differences in phenotype are often due more to differing patterns of gene expression than to differences in the protein-coding information of their genomes.

Much of what has already been covered in this book is relevant to the study of genomics. Information on gene mapping (Chapter 7), DNA structure (Chapter 10), chromosome organization (Chapter 11), transcription (Chapter 13), protein synthesis (Chapter 15), and recombinant DNA (Chapter 18) is particularly critical for understanding the concepts presented in this chapter. Comprehension of some of the topics covered in subsequent chapters will be facilitated by an understanding of the information in this chapter; such topics include organelle DNA in Chapter 20 and evolutionary genetics in Chapter 23.

CONCEPTS SUMMARY

- Genomics is the field of genetics that attempts to understand the content, organization, and function of genetic information contained in whole genomes.
- Structural genomics concerns the organization and sequence of the genome. Functional genomics studies the biological function of genomic information. Comparative genomics compares the genomic information in different organisms.
- Genetic maps position genes relative to other genes by determining rates of recombination and are measured in percent recombination. Physical maps are based on the physical distances between genes and are measured in base pairs.
- The location of sites recognized by restriction enzymes can be determined by cutting the DNA with each restriction enzyme separately and in combinations and then comparing the restriction fragments produced.
- DNA sequencing determines the base sequence of nucleotides along a stretch of DNA. The Sanger (dideoxy) method uses

special substrates for DNA synthesis (dideoxynucleoside triphosphates, ddNTPs) that terminate synthesis after they are incorporated into the newly made DNA. Four reactions, each with a different ddNTP, are set up. In each reaction, DNA fragments of varying length are produced, all of which terminate in nucleotides with the same base. The products of the four reactions are separated by gel electrophoresis, and the sequence of the DNA synthesized is read from the pattern of bands on the gel.

- Sequencing a whole genome requires breaking the genome into small overlapping fragments whose DNA sequence can be determined in sequencing reactions. The individual sequences can be ordered into a whole genome sequence with the use of a map-based approach, in which fragments are assembled in order by using previously created genetic and physical maps, or with the use of a whole-genome shotgun approach, in which overlap between fragments is used to assemble them into a whole-genome sequence.
- The Human Genome Project is an effort to determine the entire sequence of the human genome. The project began officially in 1990; rough drafts of the human genome sequence were completed in 2000.
- Single-nucleotide polymorphisms are single-base differences in DNA between individuals and are valuable as markers in linkage studies.
- Expressed-sequence tags are markers associated with expressed (transcribed) DNA sequences. RNA from a cell is subjected to reverse transcription, producing cDNA molecules. A short stretch of the cDNA is then sequenced, which provides a marker that tags (identifies) the DNA fragment. Expressed-sequence tags can be used to find the genes expressed in a genome.
- Bioinformatics is a synthesis of molecular biology and computer science that develops tools to store, retrieve, and analyze DNA, cDNA, and protein sequence data.
- A transcriptome is the set of all RNA molecules transcribed from a genome; a proteome is the set of all the proteins encoded by the genome.
- Computer programs can identify genes by looking for characteristic features of genes within a sequence.
- Homologous genes are evolutionarily related. Orthologs are homologous sequences found in different organisms, whereas paralogs are homologous sequences found in the same organism. Gene function may be determined by looking for homologous sequences (both orthologs and paralogs) whose function has been previously determined.
- Functions of unknown genes may be inferred by searching databases for protein domains in genes that have been previously characterized.
- The functions of unknown genes can be inferred by using methods that compare DNA sequences, including phylogenetic profiling, protein fusion patterns, and linkage arrangements of genes in different organisms.
- A microarray consists of DNA fragments fixed in an orderly pattern to a solid support, such as a nylon filter or glass slide. When a solution containing a mixture of DNA or RNA is applied to the array, any nucleic acid that is complementary to the probe being used will bind to the probe. Microarrays can be used to monitor the expression of thousands of genes simultaneously.
- Genes affecting a particular function or trait can be identified through whole-genome mutagenesis screens. In this process, a group of organisms is screened for abnormal phenotypes subsequent to mutagenesis, and the mutated genes causing the abnormal phenotypes are identified by positional cloning.
- The genomes of many prokaryotic organisms have been determined. Most species have between 1 million and 3 million base pairs of DNA and from 1000 to 2000 genes. Compared with that of eukaryotic genomes, the density of genes in prokaryotic genomes is relatively uniform, with about one gene per 1000 bp. There is relatively little noncoding DNA between prokaryotic genes. Horizontal gene transfer (the movement of genes between different species) has been an important evolutionary process in prokaryotes.
- Eukaryotic genomes are larger and more variable in size than prokaryotic genomes. There is no clear relation between organismal complexity and the amount of DNA or number of genes among multicellular organisms. Much of the genomes of eukaryotic organisms consist of repetitive DNA. Transposable elements are very common in most eukaryotic genomes.
- Genomics is making important contributions to human health, agriculture, biotechnology, and our understanding of evolution.

IMPORTANT TERMS

genomics (p. 000)
structural genomics (p. 000)
functional genomics (p. 000)
comparative genomics (p. 000)
genetic map (p. 000)
physical map (p. 000)
restriction mapping (p. 000)
DNA sequencing (p. 000)
dideoxyribonucleoside triphosphate (ddNTP) (p. 000)
map-based sequencing (p. 000)
contig (p. 000)
whole-genome shotgun sequencing (p. 000)
single-nucleotide polymorphism (SNP) (p. 000)
expressed-sequence tag (EST) (p. 000)
bioinformatics (p. 000)
open reading frame (p. 000)
transcriptome (p. 000)
proteome (p. 000)

homologous genes (p. 000)
orthologous genes (p. 000)
paralogous genes (p. 000)
protein domain (p. 000)
phylogenetic profile (p. 000)
fusion pattern (p. 000)
gene neighbor analysis (p. 000)
microarray (p. 000)
mutagenesis screen (p. 000)
positional cloning (p. 000)
horizontal gene exchange (p. 000)

Worked Problems

1. A linear piece of DNA that is 30 kb long is first cut with *Bam*HI, then with *Hpa*II, and finally with both *Bam*HI and *Hpa*II together. Fragments of the following size were obtained from this reaction.

*Bam*HI: 20-kb, 6-kb, and 4-kb fragments

*Hpa*II: 21-kb and 9-kb fragments

*Bam*HI and *Hpa*II: 20-kb, 5-kb, 4-kb, and 1-kb fragments

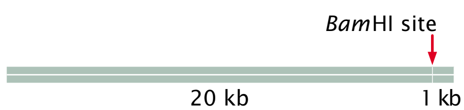
Draw a restriction map of the 30-kb piece of DNA, indicating the locations of the *Bam*HI and *Hpa*II restriction sites.

• Solution

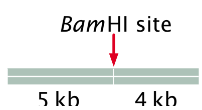
This problem can be solved correctly through a variety of approaches; this solution applies one possible approach.

When cut by *Bam*HI alone, the linear piece of DNA is cleaved into three fragments; so there must be two *Bam*HI restriction sites. When cut with *Hpa*II alone, a clone of the same piece of DNA is cleaved into only two fragments; so there is a single *Hpa*II site.

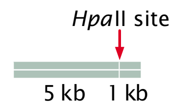
Let's begin to determine the location of these sites by examining the *Hpa*II fragments. Notice that the 21-kb fragment produced when the DNA is cut by *Hpa*II is not present in the fragments produced when the DNA is cut by *Bam*HI and *Hpa*II together (the double digest); this result indicates that the 21-kb *Hpa*II fragment has within it a *Bam*HI site. If we examine the fragments produced by the double digest, we see that the 20-kb and 1-kb fragments sum to 21 kb; so a *Bam*HI site must be 20 kb from one end of the fragment and 1 kb from the other end.



Similarly, we see that the 9-kb *Hpa*II fragment does not appear in the double digest and that the 5-kb and 4-kb fragments in the double digest add up to 9 kb; so another *Bam*HI site must be 5 kb from one end of this fragment and 4 kb from the other end.

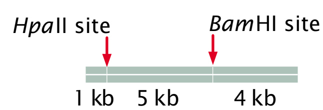


Now, let's examine the fragments produced when the DNA is cut by *Bam*HI alone. The 20-kb and 4-kb fragments are also present in the double digest; so neither of these fragments contains an *Hpa*II site. The 6-kb fragment, however, is not present in the double digest, and the 5-kb and 1-kb fragments in the double digest sum to 6 kb; so this fragment contains an *Hpa*II site that is 5 kb from one end and 1 kb from the other end.



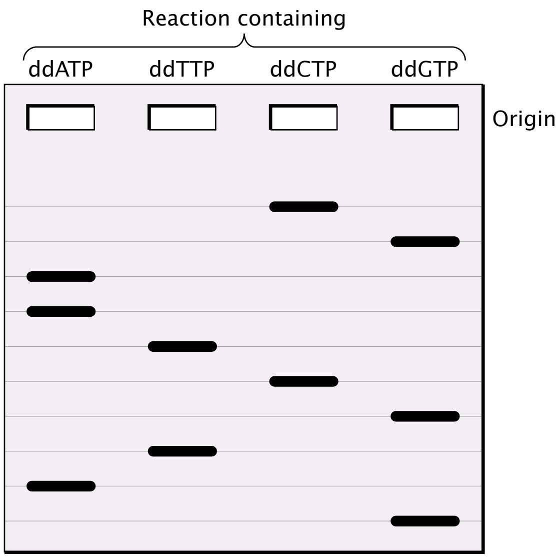
We have accounted for all the restriction sites, but we must still determine the order of the sites on the original 30-kb fragment.

Notice that the 5-kb fragment must be adjacent to both the 1-kb and 4-kb fragments; so it must be in between these two fragments.



We have also established that the 1-kb and 20-kb fragments are adjacent; because the 5-kb fragment is on one side, the 20-kb fragment must be on the other, completing the restriction map:

2. You are given the following DNA fragment to sequence: 5′–GCTTAGCATC–3′. You first clone the fragment in bacterial cells to produce sufficient DNA for sequencing. You isolate the DNA from the bacterial cells and carry out the dideoxy sequencing method. You then separate the products of the polymerization reactions by gel electrophoresis. Draw the bands that should appear on the gel from the four sequencing reactions.



• Solution

In the dideoxy sequencing reaction, the original fragment is used as a template for the synthesis of a new DNA strand; and it is the sequence of the new strand that is actually determined. The first task, therefore, is to write out the sequence of the newly synthesized fragment, which will be complementary and antiparallel to the original fragment. The sequence of the newly synthesized strand, written $5' \rightarrow 3'$ is: $5'$–GATGCTAAGC–$3'$. Bands representing this sequence will appear on the gel, with the bands representing nucleotides near the $5'$ end of the molecule at the bottom of the gel.

The New Genetics

MINING GENOMES

Genome Analysis and Comparative Genomics

Recent developments in genomics are revolutionizing our understanding of life, evolution, and medicine. This exercise allows you to explore and compare completed genomes at the Comprehensive Microbial Resources site at the Institute for Genomic Research. You will also explore the Human Genome and the Mouse Genome by using tools at the National Center for Biotechnology Information (NCBI).

COMPREHENSION QUESTIONS

1. (a) What is genomics and how does structural genomics differ from functional genomics?
   (b) What is comparative genomics?
*2. What is the difference between a genetic map and a physical map? Which generally has higher resolution and accuracy and why?
3. What is the purpose of the dideoxynucleoside triphosphate in the dideoxy sequencing reaction?
*4. What is the difference between a map-based approach to sequencing a whole genome and a whole-genome shotgun approach?
5. How are DNA fragments ordered into a contig by using restriction sites?
*6. Describe the different approaches to sequencing the human genome that were taken by the international collaboration and Celera Genomics.
7. (a) What is an expressed-sequence tag (EST)?
   (b) How are ESTs created?
   (c) How are ESTs used in genomics studies?
8. What is a single-nucleotide polymorphism (SNP), and how are SNPs used in genomic studies?
9. How are genes recognized within genomic sequences?
*10. What are homologous sequences? What is the difference between orthologs and paralogs?
11. Describe several different methods for inferring the function of a gene by examining its DNA sequence.
12. What is a microarray and how can it be used to obtain information about gene function?
*13. Briefly outline how a mutagenesis screen is carried out.
14. Eukaryotic genomes are typically much larger than prokaryotic genomes. What accounts for the increased amount of DNA seen in eukaryotic genomes?
15. What is one consequence of differences in the G + C content of different genomes?
*16. What is horizontal gene exchange? How might it take place between different species of bacteria?
17. DNA content varies considerably among different multicellular organisms. Is this variation closely related to

the number of genes and the complexity of the organism? If not, what accounts for the differences?

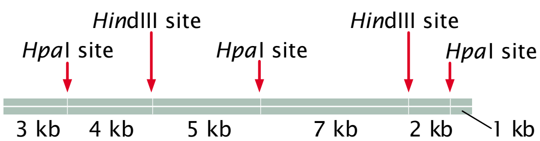
*18. More than half of the genome of *Arabidopsis thaliana* consists of duplicated sequences. What mechanisms are thought to have been responsible for these extensive duplications?

19. The human genome does not encode substantially more protein domains than do invertebrate genomes, and yet it encodes many more proteins. How are more proteins encoded when the number of domains does not differ substantially?

20. What are some of the ethical concerns arising out of the information produced by the Human Genome Project?

APPLICATION QUESTIONS AND PROBLEMS

*21. A 22-kb piece of DNA has the following restriction sites.



A batch of this DNA is first fully digested by *Hpa*I alone, then another batch is fully digested by *Hin*dIII alone, and finally a third batch is fully digested by both *Hpa*I and *Hin*dIII together. The fragments resulting from each of the three digestions are placed in separate wells of an agarose gel, separated by gel electrophoresis, and stained by ethidium bromide. Draw the bands as they would appear on the gel.

*22. A piece of DNA that is 14 kb long is cut first by *Eco*RI alone, then by *Sma*I alone, and finally by both *Eco*RI and *Sma*I together. The following results are obtained.

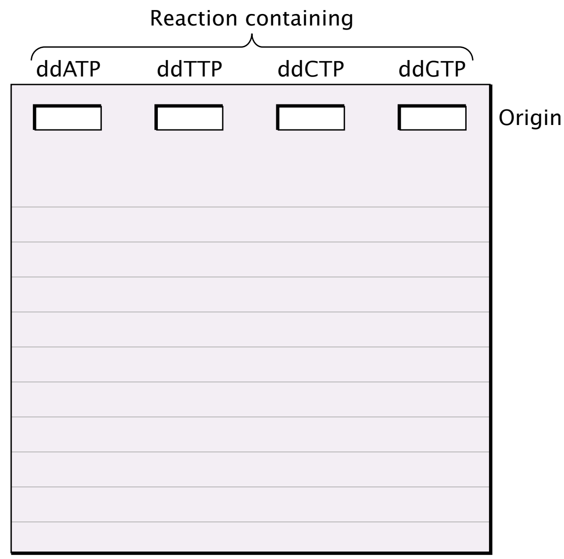
Digestion by *Eco*RI alone	Digestion by *Sma*I alone	Digestion by both *Eco*RI and *Sma*I
3-kb fragment	7-kb fragment	2-kb fragment
5-kb fragment	7-kb fragment	3-kb fragment
6-kb fragment		4-kb fragment
		5-kb fragment

Draw a map of the *Eco*RI and *Sma*I restriction sites on this 14-kb piece of DNA, indicating the relative positions of the restriction sites and the distances between them.

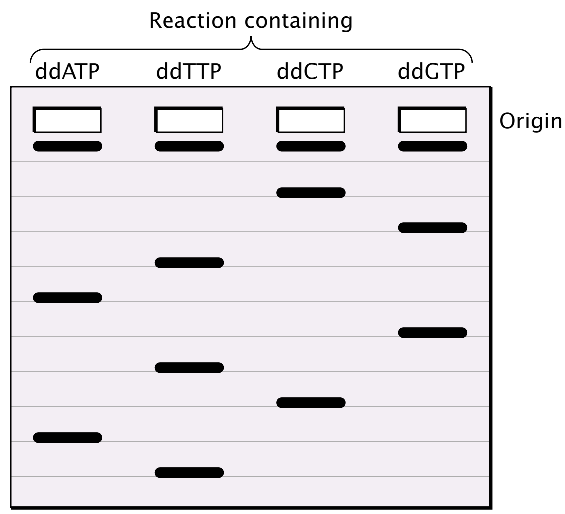
23. Suppose that you want to sequence the following DNA fragment:

Fragment to be sequenced:
5′–TCCCGGGAAA-primer site–3′

You first clone the fragment in bacterial cells to produce sufficient DNA for sequencing. You isolate the DNA from the bacterial cells and carry out the dideoxy sequencing method. You then separate the products of the polymerization reactions by gel electrophoresis. Draw the bands that should appear on the gel from the four sequencing reactions.



*24. Suppose that you are given a short fragment of DNA to sequence. You clone the fragment, isolate the cloned DNA fragments, and set up a series of four dideoxy reactions. You then separate the products of the reactions by gel electrophoresis and obtain the following banding pattern:

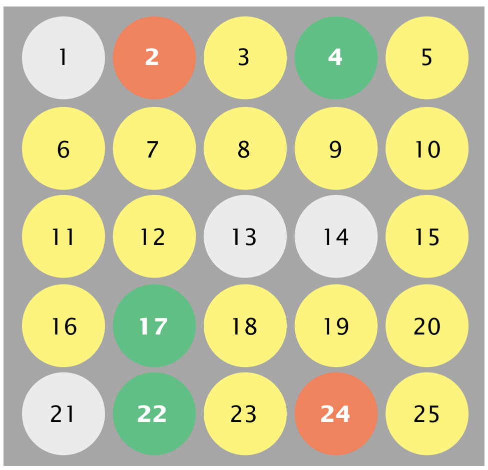


Write out the base sequence of the original fragment that you were given.

Original sequence: 5′–________________–3′

25. Microarrays can be used to determine the levels of gene expression. In one type of microarray, hybridization of the red (experimental) and green (control) cDNAs is proportional to the relative amounts of mRNA in the samples. Red indicates the overexpression of a gene and green indicates the underexpression of a gene in the experimental cells relative to the control cells, yellow indicates equal expression in experimental and control cells, and no color indicates no expression in either experimental or control cells.

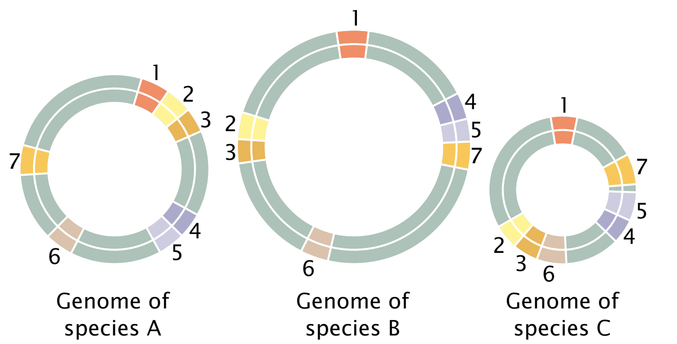
In one experiment, mRNA from a strain of antibiotic-resistant bacteria (experimental cells) is converted into cDNA and labeled with red fluorescent nucleotides; mRNA from a nonresistant strain of the same bacteria (control cells) is converted into cDNA and labeled with green fluorescent nucleotides. The cDNAs from the resistant and nonresistant cells are mixed and hybridized to a chip containing spots of DNA from genes 1 through 25. The results are shown in the adjoining illustration. What conclusions can you make about which genes might be implicated in antibiotic resistance in these bacteria? How might this information be used to design new antibiotics that are less vulnerable to resistance?



*26. Genes for the following proteins are found in five different species whose genomes have been completely sequenced. On the basis of the presence-and-absence patterns of these proteins in the genomes of the five species, which proteins are most likely to be functionally related? (Hint: Create a table listing the presence or absence of each protein in the five species.)

Species	Proteins
A	P1, P2, P3, P4, P5
B	P1, P2, P3, P5
C	P2, P4
D	P3, P5
E	P1, P3, P4, P5

27. The physical locations of several genes determined from genomic sequences are shown here for three bacterial species. On the basis of this information, which genes might be functionally related?



28. The presence (+) or absence (−) of six sequence-tagged sites (STSs) in each of five bacterial artificial chromosome (BAC) clones (A–E) is indicated in the following table. Using these markers, put the BAC clones in their correct order and indicate the locations of the STS sites within them.

	STSs					
BAC clone	**1**	**2**	**3**	**4**	**5**	**6**
A	+	−	−	−	+	−
B	−	−	−	+	−	+
C	−	+	+	−	−	−
D	−	−	+	−	+	−
E	+	−	−	+	−	−

29. How does the density of genes found on chromosome 22 compare with the density of genes found on chromosome 21, two similar-sized chromosomes? How does the number of genes on chromosome 22 compare with the number found on the Y chromosome?

To answer these questions, go to the Ensembl Web site:

http://www.ensembl.org/

Under the heading *Ensembl Species,* click *Human.* On the left-hand side of the next page are pictures of the human chromosomes. Click on chromosome 22. You will be shown a picture of this chromosome and a histogram illustrating the density of total genes (uncolored bars) and known genes (colored bars). The number of known and novel (uncharacterized) genes is given in the upper right-hand side of the page.

Now go to chromosome 21 by pulling down the Change Chromosome menu and selecting chromosome 21. Examine the density and total number of genes for chromosome 21. Now do the same for the Y chromosome.

(**a**) Which chromosome has the highest density and greatest number of genes? Which has the fewest?

(b) Examine in more detail the genes at the tip of the short arm of the Y chromosome by clicking on the top bar in the histogram of genes. A more detailed view will be shown. What known genes are found in this region? How many novel genes are there in this region?

*30. Some researchers have proposed creating an entirely new, free-living organism with a minimal genome, the smallest set of genes that allows for replication of the organism in a particular environment. This organism could be used to design and create, from "scratch," novel organisms that might perform specific tasks such as the breakdown of toxic materials in the environment.

(a) How might the minimal genome required for life be determined?

(b) What, if any, social and ethical concerns might be associated with the creation of novel organisms by constructing an entirely new organism with a minimal genome?

31. What are some of the major differences between the ways in which genetic information is organized in the genomes of prokaryotes versus eukaryotes?

32. How do the following genomic features of prokaryotic organisms compare with those of eukaryotic organisms? How do they compare among eukaryotes?

(a) Genome size

(b) Number of genes

(c) Gene density (bp/gene)

(d) G + C content

(e) Number of exons

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20 Organelle DNA

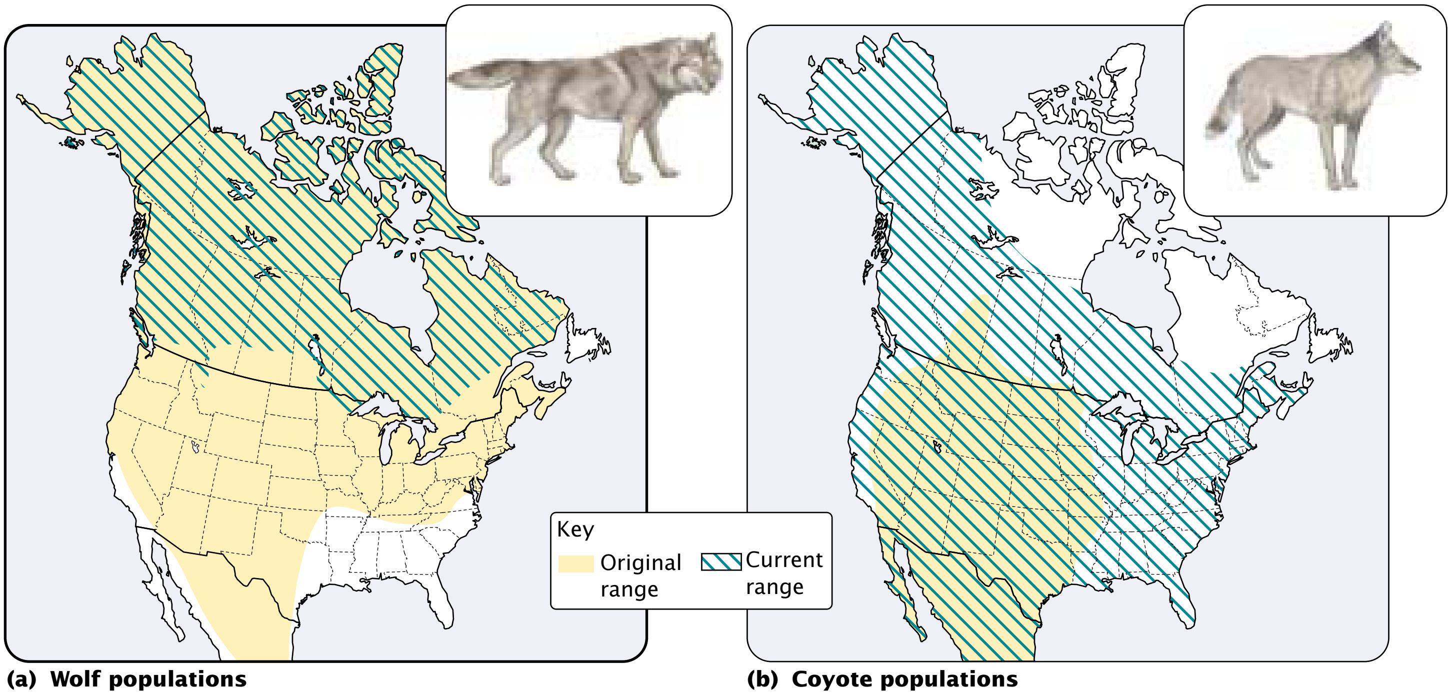
(John Shaw/Bruce Coleman.)

Coyote Genes in Declining Wolves

North America is home to two wild canids: the gray wolf (*Canis lupus*) and the coyote (*Canis latrans*). Before European settlement, gray wolves ranged across much of North America, occupying forest, plains, desert, and tundra habitat (FIGURE 20.1). Coyotes had a more limited distribution, being confined primarily to plains and deserts. With the expansion of European settlement and the development of intensive agriculture in the eighteenth and nineteenth centuries, the distribution of wolves and coyotes changed dramatically (see Figure 20.1). Wolf populations in North America declined precipitously owing to habitat destruction and deliberate extermination. In contrast, coyote populations expanded, probably because competition from wolves was eliminated and because coyotes were better able to adapt to human disruption of the ecosystem. Today coyotes are found throughout most of North America.

Habitat alternation and changes in their distributions have increased interactions between wolves and coyotes in recent times, providing more opportunities for hybridization between the two species. In captivity, wolves and coyotes will interbreed and produce fertile hybrids. Large coyotes in New England and southeastern Canada may be the result of hybridization between wolves and coyotes in these areas. To what extent is hybridization occurring between coyotes and wolves in nature?

To answer this question, Niles Lehman and his colleagues studied DNA in the mitochondria of wolves and coyotes. **Mitochondrial DNA** (mtDNA) can be helpful in determining hybridization between animals for two reasons: (1) in animals, it is inherited only from the female



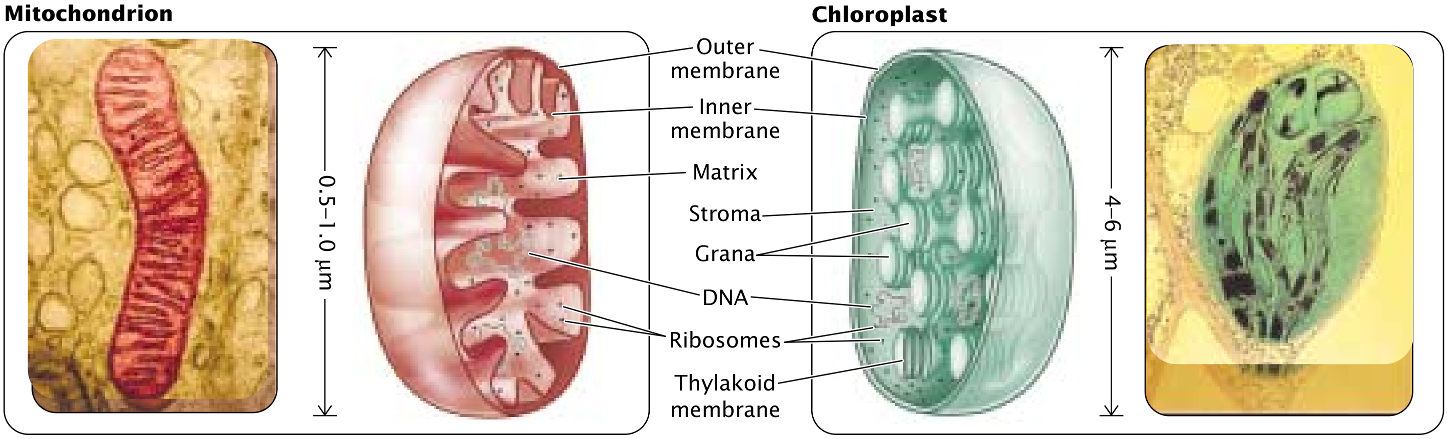
20.1 Original and current ranges of the gray wolf *(Canis lupus)* and the coyote *(Canis latrans).* Originally, the gray wolf occupied most of North America, but its current range is restricted to northern Minnesota, Canada, and Alaska. The coyote originally occupied the plains and desert habitat in the midwestern United States and Mexico. Today, the coyote is found throughout most of North America. The results of studies of mitochondrial DNA reveal that hybridization is occurring between wolves and coyotes.

parent and (2) it evolves rapidly. Lehman and his colleagues gathered tissue and blood samples from more than 500 gray wolves and coyotes in North America, extracted mtDNA from the samples, and analyzed restriction fragment length polymorphisms (see Chapter 18) in the mtDNA. The results of his study revealed two major clusters of mtDNA among the canids: one consisting of wolf mtDNA and another of coyote-like mtDNA. Surprisingly, the coyote-like mtDNA cluster included several samples that had been obtained from wolves, indicating that some wolves possessed coyote-like mtDNA. The wolves with coyote-like mtDNA were all from the U.S.–Canadian border area, which has recently been invaded by coyotes. No wolves from Alaska or northern Canada possessed coyote-like mtDNA. All the coyotes had only coyote-like mtDNA.

These results indicate that unidirectional hybridization has taken place between coyotes and wolves: coyote mtDNA has entered wolf populations, but wolf mtDNA has not entered coyote populations. The fact that in animals mtDNA is inherited only from the female parent implies that female coyotes are mating successfully with male wolves and the wolf–coyote hybrids are backcrossing with wolves, introducing coyote genes into wolf populations.

These findings have important implications for the future of gray wolves in North America. Hybridization between wolves and coyotes threatens to erode the genetic integrity of wolves. As human activities encroach on areas occupied by wolves, wolves and coyotes will be forced into closer contact and there will be more hybridization; the wolf genome (both mitochondrial and nuclear) will become increasingly diluted by coyote DNA. Current efforts to reintroduce wolves into former territories, which are now occupied by coyotes, may lead to further hybridization, ultimately harming, rather than helping, wolf populations.

DNA sequences found in mitochondria and other organelles possess unique properties that make these sequences useful in the fields of conservation biology, evolution, and genetic diseases. Uniparental inheritance exhibited by genes found in mitochondria and chloroplasts was discussed in Chapter 5; the present chapter examines molecular aspects of organelle DNA. We begin by briefly considering the structures of mitochondria and chloroplasts, the inheritance of traits encoded by their genes, and their evolutionary origin. We then examine the general characteristics of mtDNA, followed by a discussion of the organization and function of different types of mitochondrial genomes.



20.2 Comparison of the structures of mitochondria and chloroplasts. (Left, Don Fawcett/Visuals Unlimited; (right) Biophoto Associates; Photo Researchers.)

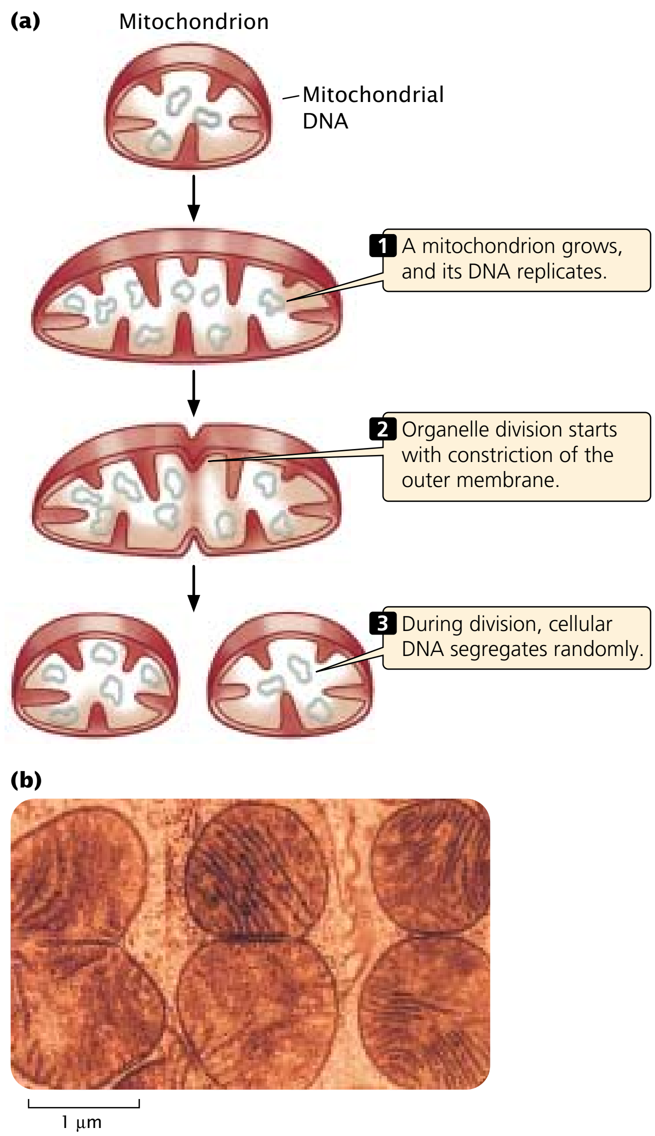
Finally, we turn to **chloroplast DNA** (cpDNA), examining its characteristics, organization, and function.

The Biology of Mitochondria and Chloroplasts

Mitochondria and chloroplasts are membrane-bounded organelles located in the cytoplasm of eukaryotic cells (FIGURE 20.2). Mitochondria are present in almost all eukaryotic cells, whereas chloroplasts are found in multicellular plants and some algae. Both organelles generate ATP, the universal energy carrier of cells.

Mitochondrion and Chloroplast Structure

Mitochondria are from 0.5 to 1.0 micrometer (µm) in diameter, about the size of a typical bacterium; chloroplasts are typically from about 4 to 6 µm in diameter. Both are surrounded by two membranes that enclose a region (called the matrix in mitochondria and the stroma in chloroplasts) that contains enzymes, ribosomes, RNA, and DNA. In mitochondria, the inner membrane is highly folded; embedded within it are the enzymes that catalyze electron transport and oxidative phosphorylation. Chloroplasts have a third membrane, called the thylakoid membrane, which is highly folded and stacked to form aggregates called grana. This membrane bears the pigments and enzymes required for photophosphorylation. New mitochondria and chloroplasts arise by the division of existing organelles (FIGURE 20.3). Mitochondria



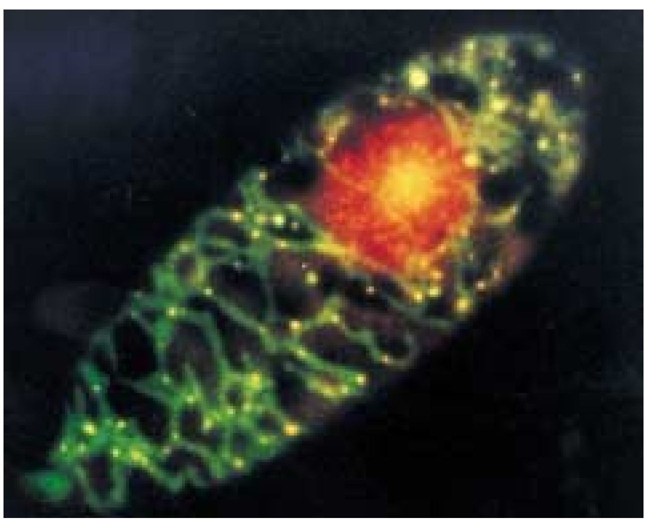
20.3 New mitochondria arise by division of existing mitochondria. (a) DNA molecules within the mitochondria segregate randomly in organelle division. (b) Electron micrograph of a dividing mitochondrion from a liver cell. (T. Kanaseki and D. Fawcett/Visuals Unlimited.)

and chloroplasts possess DNA that encodes polypeptides used by the organelle, as well as rRNAs and tRNAs needed for the translation of these proteins.

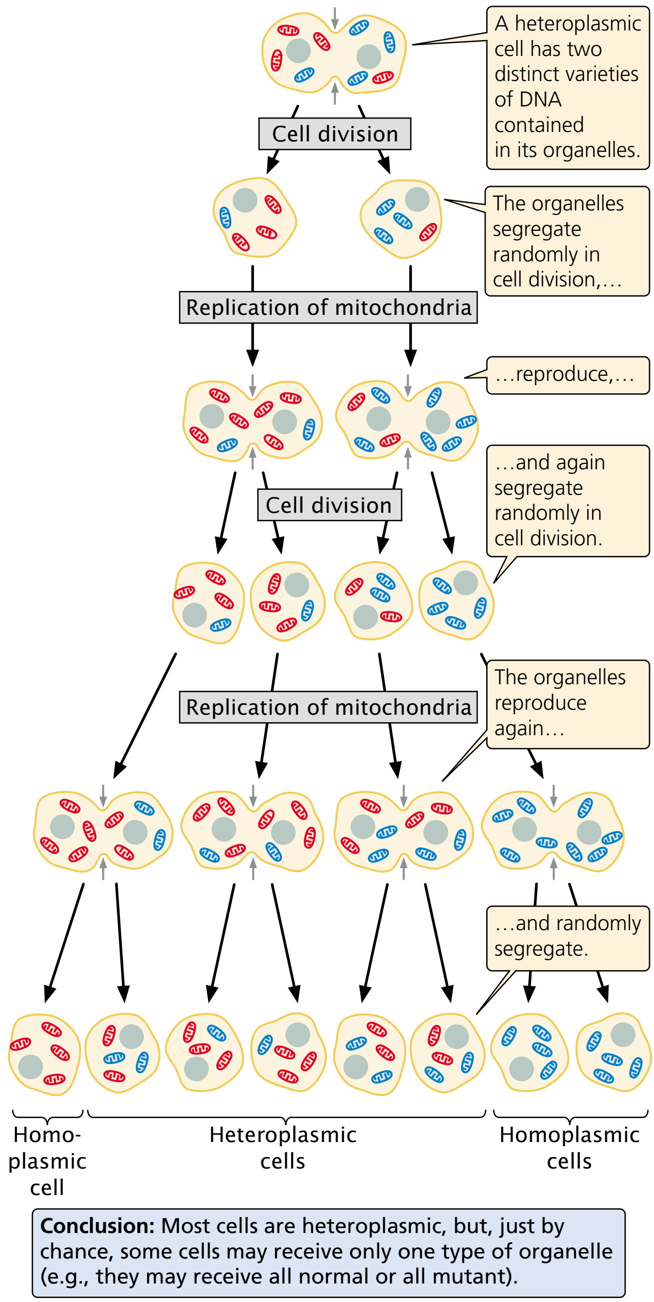
The Genetics of Organelle-Encoded Traits

Mitochondria and chloroplasts are present in the cytoplasm and are usually inherited from a single parent. Thus traits encoded by mtDNA and cpDNA exhibit uniparental inheritance. In animals, mtDNA is inherited almost exclusively from the female parent, although occasional male transmission of mtDNA has been documented. Paternal inheritance of organelles is common in gymnosperms and occurs occasionally in angiosperms as well. Some plants even exhibit biparental inheritance of mtDNA and cpDNA.

Individual cells may contain from dozens to hundreds of organelles, each with numerous copies of the organelle genome; so each cell typically possesses from hundreds to thousands of copies of mitochondrial and chloroplast genomes (FIGURE 20.4). A mutation arising within one organellar DNA molecule generates a mixture of mutant and wild-type DNA sequences within that cell. The occurrence of two distinct varieties of DNA within the cytoplasm of a single cell is termed **heteroplasmy.** When a heteroplasmic cell divides, the organelles segregate randomly into the two progeny cells in a process called **replicative segregation** (FIGURE 20.5), and chance determines the proportion of mutant organelles in each cell. Although most progeny cells will inherit a mixture of mutant and normal organelles, just by chance some cells may receive organelles with only mutant or only wild-type sequences; this situation is known as **homoplasmy.**



20.4 **Individual cells may contain many mitochondria, each with several copies of the mitochondrial genome.** Shown is a cell of *Euglena gracilis*, stained so that the nucleus appears red, mitochondria green, and mtDNA yellow. (From Y. Huyashi and K. Veda, *Journal of Cell Sciences* 93, 1989, 565.)



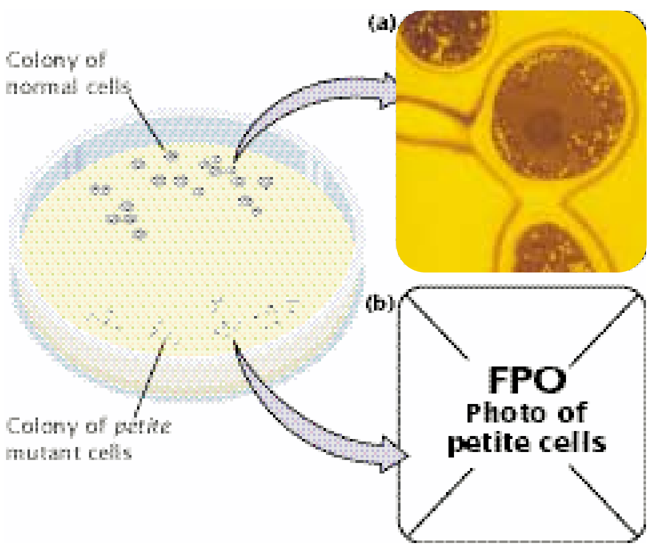
20.5 **Organelles in a heteroplasmic cell divide randomly into the progeny cells.** This diagram illustrates replicative segregation in mitosis; the same process also takes place in meiosis.

When replicative segregation occurs in somatic cells, it may create phenotypic variation within a single organism; different cells of the organism may possess different proportions of mutant and wild-type sequences, resulting in different degrees of phenotypic expression among tissues. When replicative segregation occurs in the germ cells of a heteroplasmic cytoplasmic donor, the offspring may show quite different phenotypes.

The disease known as myoclonic epilepsy and ragged-red fiber disease syndrome (MERRF) is caused by a mutation in an mtDNA gene. A 20-year-old person who carried this mutation in 85% of his mtDNAs displayed a normal phenotype, whereas a cousin who had the mutation in 96% of his mtDNAs was severely affected. In diseases caused by mutations in mtDNA, the severity of the disease is frequently related to the proportion of mutant mtDNA sequences inherited at birth.

A number of traits encoded by organellar DNA have been studied. One of the first to be examined in detail was the phenotype produced by *petite* mutations in yeast (FIGURE 20.6). In the late 1940s, Boris Ephrussi and his colleagues noticed that, when grown on solid medium, some colonies of yeast were much smaller than normal. Examination of these *petite* colonies revealed that growth rates of the cells within the colonies were greatly reduced. The results of biochemical studies demonstrated that *petite* mutants were unable to carry out aerobic respiration; they obtained all of their energy from anaerobic respiration (glycolysis), which is much less efficient than aerobic respiration and results in the smaller colony size.

Some *petite* mutations are inherited from both parents and are defects in nuclear DNA. However, most *petite* mutations are inherited from only a single parent; such mutants possess large deletions in mtDNA or, in some cases, are missing mtDNA entirely. Because much of their mtDNA encodes enzymes that catalyze aerobic respiration, the *petite* mutants are unable to carry out aerobic respiration and therefore cannot produce normal quantities of ATP, which inhibits their growth.



20.6 The *petite* mutants have large deletions in their mtDNA and are unable to carry out oxidative phosphorylation. Electron micrograph of mitochondria in (a) a normal yeast cell and (b) a *petite* mutant. (Part a, David M. Phillips/Visuals Unlimited; part b, .)

Another known mtDNA mutation occurs in *Neurospora*. Isolated by Mary Mitchell in 1952, *poky* mutants grow slowly, display cytoplasmic inheritance, and have abnormal amounts of cytochromes. Cytochromes are protein components of the electron-transport chain of the mitochondria and play an integral role in the production of ATP. Most organisms have three primary types of cytochromes: cytochrome *a*, cytochrome *b*, and cytochrome *c*. *Poky* mutants have cytochrome *c* but no cytochrome *a* or *b*. Like *petite* mutants, *poky* mutants are defective in ATP synthesis and therefore grow more slowly than normal wild-type cells.

In recent years, a number of genetic diseases that result from mutations in mtDNA have been identified in humans. Leber hereditary optic neuropathy (LHON), which typically leads to sudden loss of vision in middle age, results from mutations in the mtDNA genes that encode electron-transport proteins. Another disease caused by mitochondrial mutations is neurogenic muscle weakness, ataxia, and tetinitis pigmentosa (NARP), which is characterized by seizures, dementia, and developmental delay. Other mitochondrial diseases include Kearns-Sayre syndrome (KSS) and chronic external opthalmoplegia (CEOP), both of which result in paralysis of the eye muscles, droopy eyelids, and, in severe cases, vision loss, deafness, and dementia. All of these diseases exhibit cytoplasmic inheritance and variable expression (see Chapter 5).

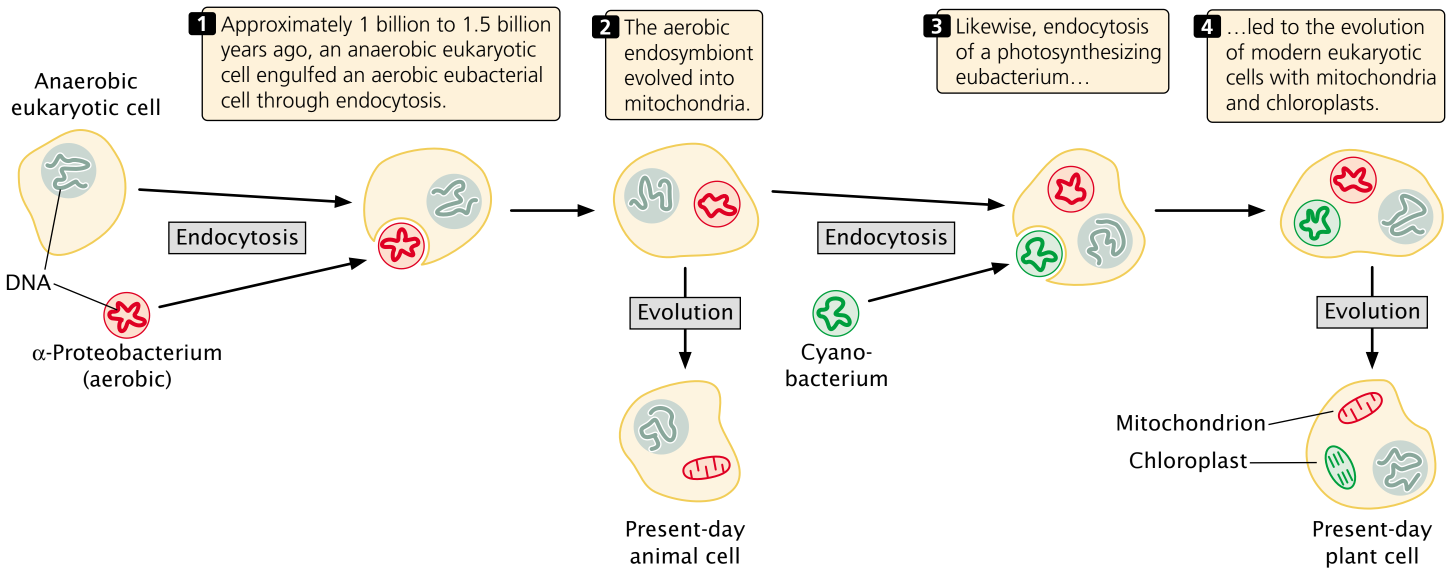
A trait in plants that is produced by mutations in mitochondrial genes is cytoplasmic male sterility, a mutant phenotype found in more than 140 different plant species and inherited only from the maternal parent. These mutations inhibit pollen development but do not affect female fertility.

A number of cpDNA mutants also have been discovered. One of the first to be recognized was leaf variegation in the *Mirabilis jalapaa,* which was studied by Carl Correns in 1909 (see p. 000 in Chapter 5). In the green alga *Chlamydomonas,* streptomycin-resistant mutations occur in cpDNA, and a number of mutants exhibiting altered pigmentation and growth in higher plants have been traced to defects in cpDNA.

Concepts

In most organisms, genes encoded by mtDNA and cpDNA are inherited entirely from a single parent. A gamete may contain more than one distinct type of mtDNA or cpDNA; in these cases, random segregation of the organelle DNA may produce phenotypic variation within a single organism or it may produce different degrees of phenotypic expression among progeny of a cross.

www.whfreeman.com/pierce More information about mitochondria and chloroplasts



20.7 **The endosymbiotic theory proposes that mitochondria and chloroplasts in eukaryotic cells arose from eubacteria.** An original ancestral cell gave rise to prokaryotic and eukaryotic cells.

The Endosymbiotic Theory

Chloroplasts and mitochondria are in many ways similar to bacteria. This resemblance is not superficial; indeed there is compelling evidence that these organelles evolved from eubacteria (see p. 000 in Chapter 2). The **endosymbiotic theory** (FIGURE 20.7) proposes that mitochondria and chloroplasts were once free-living bacteria that became internal inhabitants (endosymbionts) of early eukaryotic cells. According to this theory, between 1 billion and 1.5 billion years ago, a large, anaerobic eukaryotic cell engulfed an aerobic eubacterium, one that possessed the enzymes necessary for oxidative phosphorylation. The eubacterium provided the formerly anaerobic cell with the capacity for oxidative phosphorylation and allowed it to produce more ATP for each organic molecule digested. With time, the endosymbiont became an integral part of the eukaryotic host cell, and its descendants evolved into present-day mitochondria. Sometime later, a similar relation arose between photosynthesizing eubacteria and eukaryotic cells, leading to the evolution of chloroplasts.

A great deal of evidence supports the idea that mitochondria and chloroplasts originated as eubacterial cells. Many modern, single-celled eukaryotes (protists) are hosts to endosymbiotic bacteria. Mitochondria and chloroplasts are similar in size to present-day eubacteria and possess their own DNA, which has many characteristics in common with eubacterial DNA. Mitochondria and chloroplasts possess ribosomes, some of which are similar in size and structure to eubacterial ribosomes. Finally, antibiotics that inhibit protein synthesis in eubacteria but do not affect protein synthesis in eukaryotic cells also inhibit protein synthesis in these organelles.

The strongest evidence for the endosymbiotic theory comes from the study of DNA sequences in organellar DNA. Ribosomal RNA and protein-encoding gene sequences in mitochondria and chloroplasts have been found to be more closely related to sequences in the genes of eubacteria than they are to those found in the eukaryotic nucleus. Mitochondrial DNA sequences are most similar to sequences found in a group of eubacteria called the α-proteobacteria, suggesting that the original bacterial endosymbiont came from this group. Chloroplast DNA sequences are most closely related to sequences found in cyanobacteria, a group of photosynthesizing eubacteria. All of this evidence indicates that mitochondria and chloroplasts are more closely related to eubacterial cells than they are to the eukaryotic cells in which they are now found.

Concepts

Mitochondria and chloroplasts are membrane-bounded organelles of eukaryotic cells that generally possess their own DNA. The well-supported endosymbiotic theory proposes that these organelles began as free-living eubacteria that developed stable endosymbiotic relations with early eukaryotic cells.

Mitochondrial DNA

In animals and most fungi, the mitochondrial genome consists of a single, highly coiled, circular DNA molecule Plant mitochondrial genomes often exist as a complex collection of multiple circular DNA molecules. Each mitochondrion

Table 20.1 Sizes of mitochondrial genomes in selected organisms

Organism	Size of mtDNA in Nucleotide Pairs
Ascaris summ (nematode worm)	14,284
Drosophila melanogaster (fruit fly)	19,517
Lumbricus terrestis (earthworm)	14,998
Xenopus laevis (frog)	17,553
Mus musculus (house mouse)	16,295
Canis familiaris (dog)	16,728
Homo sapiens (human)	16,569
Pichia canadensis (fungus)	27,694
Podospora anserina (fungus)	100,314
Schizosaccharomyces pome (fungus)	19,431
Saccharomyces cerevisiae (fungus)	85,779
Chlamydomonas reinhardtii (green alga)	15,758
Paramecium aurelia (protist)	40,469
Reclinomonas americana (protist)	69,034
Arabidopsis thaliana (plant)	166,924
Brassica hirta (plant)	208,000
Cucumis melo (plant)	2,400,000

contains multiple copies of the mitochondrial genome, and a cell may contain many mitochondria. A typical rat liver cell, for example, has from 5 to 10 mtDNA molecules in each of about 1000 mitochondria; so each cell possesses from 5000 to 10,000 copies of the mitochondrial genome, and mtDNA constitutes about 1% of the total cellular DNA in a rat liver cell. Like eubacterial chromosomes, mtDNA lacks the histone proteins normally associated with eukaryotic nuclear DNA. The guanine–cytosine (GC) content of mtDNA is often sufficiently different from that of nuclear DNA that mtDNA can be separated from nuclear DNA by density gradient centrifugation.

Mitochondrial genomes are small compared with nuclear genomes and vary greatly in size among different organisms (Table 20.1). Most of this size variation is in noncoding sequences such as introns and intergenic regions.

Gene Structure and Organization of mtDNA

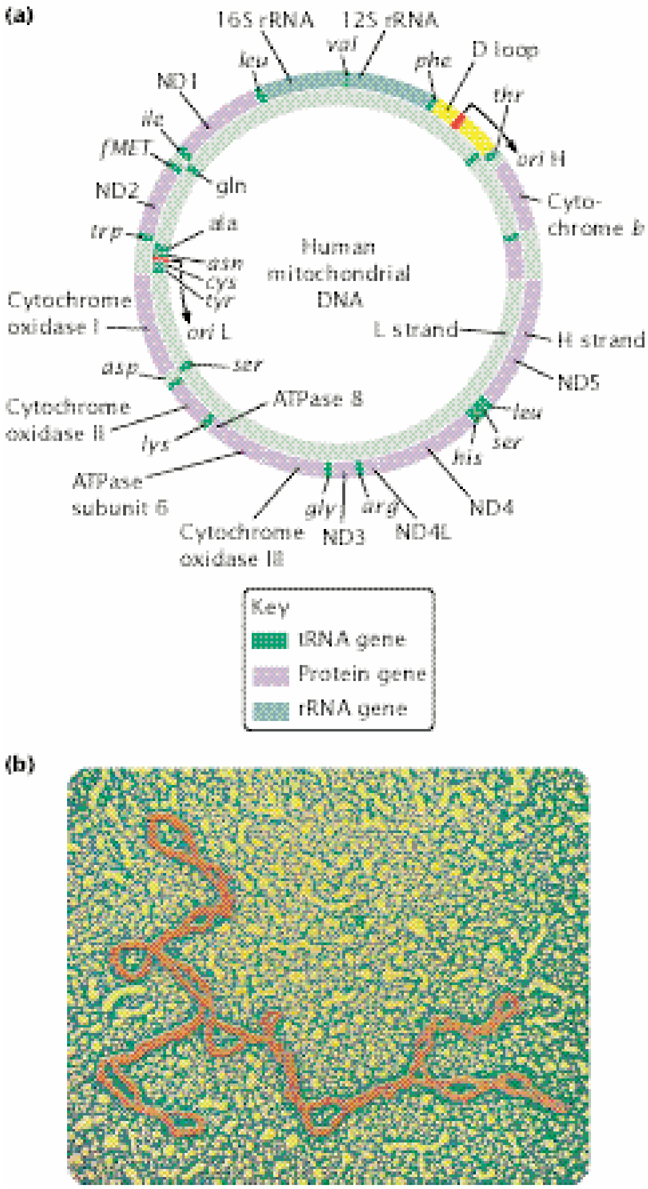
The nucleotide sequence of the mitochondrial genome has been determined for a variety of different organisms, including protists, fungi, plants, and animals. The genes for many of the structural proteins and enzymes found in mitochondria are actually encoded by *nuclear* DNA, translated on cytoplasmic ribosomes, and then transported into the mitochondria; the mitochondrial genome typically encodes only a few rRNA and tRNA molecules needed for mitochondrial protein synthesis. The organization of these mitochondrial genes and how they are expressed is extremely diverse across organisms.

Ancestral and derived mitochondrial genomes Mitochondrial genomes can be divided in two basic types—ancestral genomes and derived genomes—although there is much variation within each type and the mtDNA of some organisms does not fit well into either category. Ancestral mitochondrial genomes are found in some plants and protists and retain many characteristics of their eubacterial ancestors. These mitochondrial genomes contain more genes than do derived genomes, have rRNA genes that encode eubacterial-like ribosomes, and have a complete or almost complete set of tRNA genes. They possess few introns and little noncoding DNA between genes, generally use universal codons, and have their genes organized into clusters similar to those found in eubacteria.

Derived mitochondrial genomes, in contrast, are usually smaller than ancestral genomes and contain fewer genes. Their rRNA genes and ribosomes differ substantially from those found in typical eubacteria. The DNA sequences found in derived mitochondrial genomes differ more from typical eubacterial sequences than do ancestral genomes, and they contain nonuniversal codons. Most animal and fungal mitochondrial genomes fit into this category.

Human mtDNA Human mtDNA is a circular molecule encompassing 16,569 bp that encode two rRNAs, 22 tRNAs, and 13 proteins. The two nucleotide strands of the molecule differ in their base composition: the heavy (H) strand has more guanine nucleotides, whereas the light (L) strand has more cytosine nucleotides. The H strand is the template for both rRNAs, 14 of the 22 tRNAs, and 12 of the 13 proteins, whereas the L strand serves as template for only 8 of the tRNAs and one protein.

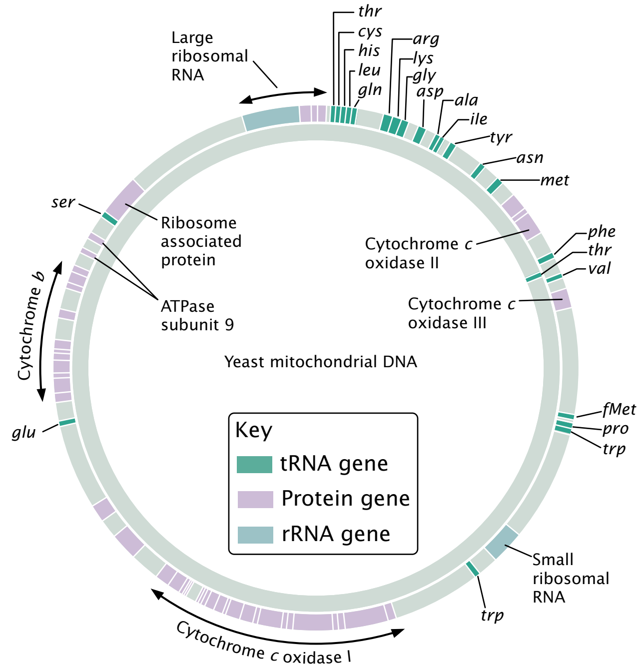
The origin of replication for the H strand is within a region known as the **D loop** (FIGURE 20.8), which also contains promoters for both the H and L strands. Human mtDNA is highly economical in its organization: there are few noncoding nucleotides between the genes; almost all the mRNA is translated (there are no 5′ and 3′ untranslated regions); and there are no introns. Each strand has only a single promoter; so transcription produces two very large RNA precursors that are later cleaved into individual RNA molecules. Many of the genes that encode polypeptides even lack a complete termination codon, ending in either U or UA; the addition of a poly(A) tail to the 3′ end of the mRNA provides a UAA termination codon that halts translation. Human mtDNA also contains very little repetitive DNA. The one region of the human mtDNA that does contain some noncoding nucleotides is the D loop.



20.8 The human mitochondrial genome, consisting of 16,569 bp, is highly economic in its organization. (a) The outer circle represents the heavy (H) strand, and the inner circle represents the light (L) strand. The origins of replication for the H and L strands are *ori* H and *ori* L, respectively. (b) Electron micrograph of isolated mtDNA. (Part b, CNRI/Photo Researchers.)

www.whfreeman.com/pierce Information on genes of the human mitochondrial genome

Yeast mtDNA The organization of yeast mtDNA is quite different from that of human mtDNA. Although the yeast mitochondrial genome with 78,000 bp is nearly five times as large, it encodes only six additional genes, for a total of 2 rRNAs, 25 tRNAs, and 16 polypeptides (FIGURE 20.9). Most of the extra DNA in the yeast mitochondrial genome



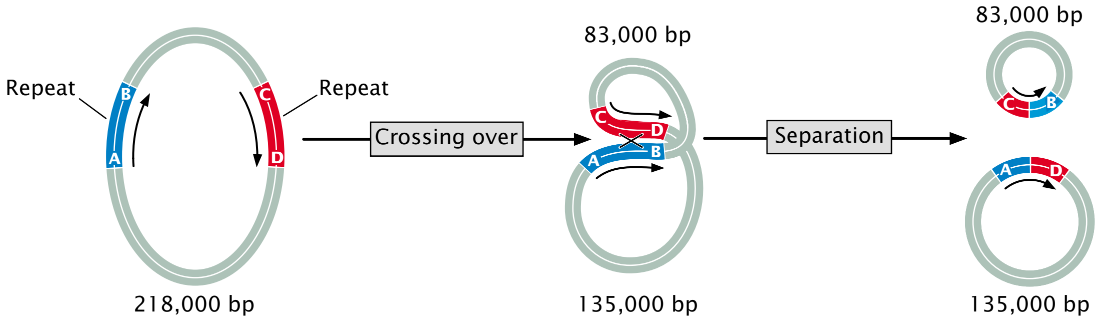
20.9 The yeast mitochondrial genome, consisting of 78,000 bp, contains much noncoding DNA.

consists of noncoding sequences. Yeast mitochondrial genes are separated by long intergenic spacer regions that have no known functions. The genes encoding polypeptides often include regions that encode 5′ and 3′ untranslated regions of the mRNA; there are also short repetitive sequences and some duplications.

www.whfreeman.com/pierce Information on the Fungal Mitochondrial Genome Project (FMGP), whose goals are to sequence and analyze complete mitochondrial genomes from all major groups of fungi

Flowering plant mtDNA Flowering plants (angiosperms) have the largest and most complex mitochondrial genomes known; their mitochondrial genomes range in size from 186,000 bp in white mustard to 2,400,000 bp in muskmelon. Even closely related plant species may differ greatly in the sizes of their mtDNA.

Part of the extensive size variation in the mtDNA of flowering plants can be explained by the presence of large direct repeats, which constitute large parts of the mitochondrial genome. Crossing over between these repeats can generate multiple circular chromosomes of different sizes. The mitochondrial genome in turnip, for example, consists of a "master circle" consisting of 218,000 bp that has direct repeats (FIGURE 20.10). Homologous recombination between the repeats can generate two smaller circles of 135,000 bp and 83,000 bp. Other species contain several direct repeats, providing possibilities for complex crossing-over events that may increase or decrease the number and sizes of the circles.



20.10 Size variation in plant mtDNA can be generated through recombination between direct repeats. In turnips, the mitochondrial genome consists of a "master circle" of 218,000 bp, which has direct repeats that are separated by 135,000 bp on one side and 83,000 bp on the other. Crossing over between the direct repeats produces two smaller circles of 135,000 bp and 83,000 nucleotide pairs.

Nonuniversal Codons in mtDNA

In the vast majority of bacterial and eukaryotic DNA, the same codons specify the same amino acids (see p. 000 in Chapter 15). However, there are exceptions to this universal code, and many of these exceptions are in mtDNA (Table 20.2). There is not a "mitochondrial code"; rather, exceptions to the universal code exist in mitochondria, and these exceptions often differ among organisms. For example, AGA specifies arginine in the universal code, but AGA codes for serine in *Drosophila* mtDNA and is a stop codon in mammalian mtDNA.

Concepts

The mitochondrial genome consists of circular DNA with no associated histone proteins. The size and structure of mtDNA differ greatly among organisms. Human mtDNA exhibits extreme economy, but mtDNAs found in yeast and flowering plants contain many noncoding nucleotides and repetitive sequences. Mitochondrial DNA in most flowering plants is large and typically has one or more large direct repeats that can recombine to generate smaller or larger molecules.

Replication, Transcription, and Translation of mtDNA

Mitochondrial DNA does not replicate in the orderly, regulated manner of nuclear DNA. Mitochondrial DNA is synthesized throughout the cell cycle and is not coordinated with the synthesis of nuclear DNA. Which mtDNA molecules are replicated at any particular moment appears to be random; within the same mitochondrion, some molecules are replicated two or three times, whereas others are not replicated at all. Furthermore, the two strands in human mtDNA may not replicate synchronously. Mitochondrial DNA is replicated by a special DNA polymerase called DNA polymerase γ (gamma). Presumably, helicases and topoisomerases are required for mitochondrial DNA replication, just as they are in eubacterial and nuclear DNA replication.

The processes of transcription and translation of mitochondrial genes exhibit extensive variation among different organisms. In human mtDNA, eubacterial-like operons are absent, and there are two promoters, one for each nucleotide strand, within the D loop. Transcription of the two strands proceeds in opposite directions, generating two giant precursor RNAs that are then cleaved to yield individual rRNAs, tRNAs, and mRNAs. As the tRNAs are transcribed, they fold up into three-dimensional configurations. These configurations are recognized and cut out by enzymes. The tRNA

Table 20.2 Nonuniversal codons found in mtDNA

		mtDNA		
Codon	**Universal Code**	**Vertebrate**	***Drosophila***	**Yeast**
UGA	Stop	Tryptophan	Tryptophan	Tryptophan
AUA	Isoleucine	Methionine	Methionine	Methionine
AGA	Arginine	Stop	Serine	Arginine

Source: After T. D. Fox, *Annual Review of Genetics* 21 (1987), p. 69.

genes generally flank the protein and rRNA genes; so cleavage of the tRNAs releases mRNAs and rRNAs. In the mitochondrial genomes of fungi, plants, and protists, there are multiple promoters, although genes are occasionally arranged and transcribed in operons.

Most mRNA molecules produced by the transcription of mtDNA are not capped at their 5′ ends, unlike mRNA transcribed from nuclear genes (See Figure 14.6). Poly(A) tails are added to the 3′ end of some mRNAs encoded by animal mtDNA, but poly(A) tails are missing from those encoded by mtDNA in fungi, plants, and protists. The poly(A) tails added to animal mitochondrial mRNAs are shorter than those attached to nuclear-encoded mRNA and are probably added by an entirely different mechanism.

Some of the genes in yeast and plant mitochondrial DNA contain introns, many of which are self-splicing. RNA encoded by some mitochondrial genomes undergoes extensive editing (see p. 000 in Chapter 14).

Translation in mitochondria has some similarities to eubacterial translation, but there are also important differences. In mitochondria, protein synthesis is initiated at AUG start codons by *N*-formylmethionine, just as in eubacterial initiation of translation. Mitochondrial translation also employs elongation factors similar to those seen in eubacteria, and the same antibiotics that inhibit translation in eubacteria also inhibit translation in mitochondria. However, mitochondrial ribosomes are variable in structure and are often different from those seen in both eubacterial and eukaryotic cells. Additionally, the initiation of translation in mitochondria must be different from that of both eubacterial and eukaryotic cells, because animal mitochondrial mRNA contains no Shine-Dalgarno ribosome-binding site and no 5′ cap. (A Shine-Dalgarno sequence has been observed in mitochondrial mRNA of the protozoan *Reclinomonas americana,* which has a very primitive, eubacterial-like mitochondrion.)

There is also much diversity in the tRNAs encoded by various mitochondrial genomes. Human mtDNA encodes 22 of the 32 tRNAs required for translation in the cytoplasm. (Only 32 are required in cytoplasmic translation because wobble at the third position of the codon allows tRNAs to pair with more than one codon; see p. 000 in Chapter 15.) In human mitochondrial translation, there is even more wobble than in cytoplasmic translation; many mitochondrial tRNAs will recognize any of the four nucleotides in the third position of the codon, permitting translation to take place with even fewer tRNAs. The increased wobble also means that any change in a DNA nucleotide at the third position of the codon will be a silent mutation (see p. 000 in Chapter 17) and will not alter the amino acid sequence of the protein. Thus more of the changes that occur in mtDNA are silent and accumulate over time, contributing to a higher rate of evolution. In some organisms, fewer than 22 tRNAs are encoded by mtDNA; in these organisms, nuclear-encoded tRNAs are imported from the cytoplasm to help carry out translation. In yet other organisms, the mitochondrial genome encodes a complete set of all 32 tRNAs.

Concepts

The processes of replication, transcription, and translation vary widely among mitochondrial genomes and exhibit a curious mix of eubacterial, eukaryotic, and unique characteristics.

Evolution of mtDNA

As already mentioned, comparisons of DNA sequences in mitochondrial genomes with homologous sequences in other organisms strongly support a common eubacterial origin for all mtDNA. Nevertheless, patterns of evolution seen in mtDNA vary greatly among different groups of organisms.

The sequences of vertebrate mtDNA exhibit an accelerated rate of change: mammalian mtDNA, for example, typically evolves from 5 to 10 times as fast as mammalian nuclear DNA. The gene content and organization of vertebrate mitochondrial genomes, however, is relatively constant. In contrast, sequences of plant mtDNA evolve slowly at a rate only one-tenth that of the nuclear genome, but their gene content and organization change rapidly. The reason for these basic differences in rates of evolution is not yet known.

One possible reason for the accelerated rate of evolution seen in vertebrate mtDNA is a high mutation rate in mtDNA, which would allow DNA sequences to change quickly. Increased errors associated with replication, the absence of DNA repair functions, and the frequent replication of mtDNA may increase the number of mutations. The large amount of wobble in mitochondrial translation may also allow mutations to accumulate over time, as discussed earlier. The use of mtDNA in evolutionary studies will be described in more detail in Chapter 23.

Concepts

All mtDNA appears to have evolved from a common eubacterial ancestor, but the patterns of evolution seen in different mitochondrial genomes varies greatly. Vertebrate mtDNA exhibits rapid change in sequence but little change in gene content and organization, whereas the mtDNA of plants exhibits little change in sequence but much variation in gene content and organization.

www.whfreeman.com/pierce Data on mitochondrial genomes that have been completely sequenced, and more information on human diseases and disorders caused by defects in mitochondria

Chloroplast DNA

Geneticists have long recognized that many traits associated with chloroplasts exhibit cytoplasmic inheritance, indicating that these traits are not encoded by nuclear genes. In 1963, chloroplasts were shown to have their own DNA (FIGURE 20.11).

Among different plants, the chloroplast genome ranges in size from 80,000 to 600,000 bp, but most chloroplast genomes range from 120,000 to 160,000 bp (Table 20.3). Chloroplast DNA is usually contained on a single, double-stranded DNA molecule that is circular, is highly coiled, and lacks associated histone proteins. As in mtDNA, multiple copies of the chloroplast genome are found in each chloroplast, and there are multiple organelles per cell; so there are several hundred to several thousand copies of cpDNA in a typical plant cell.

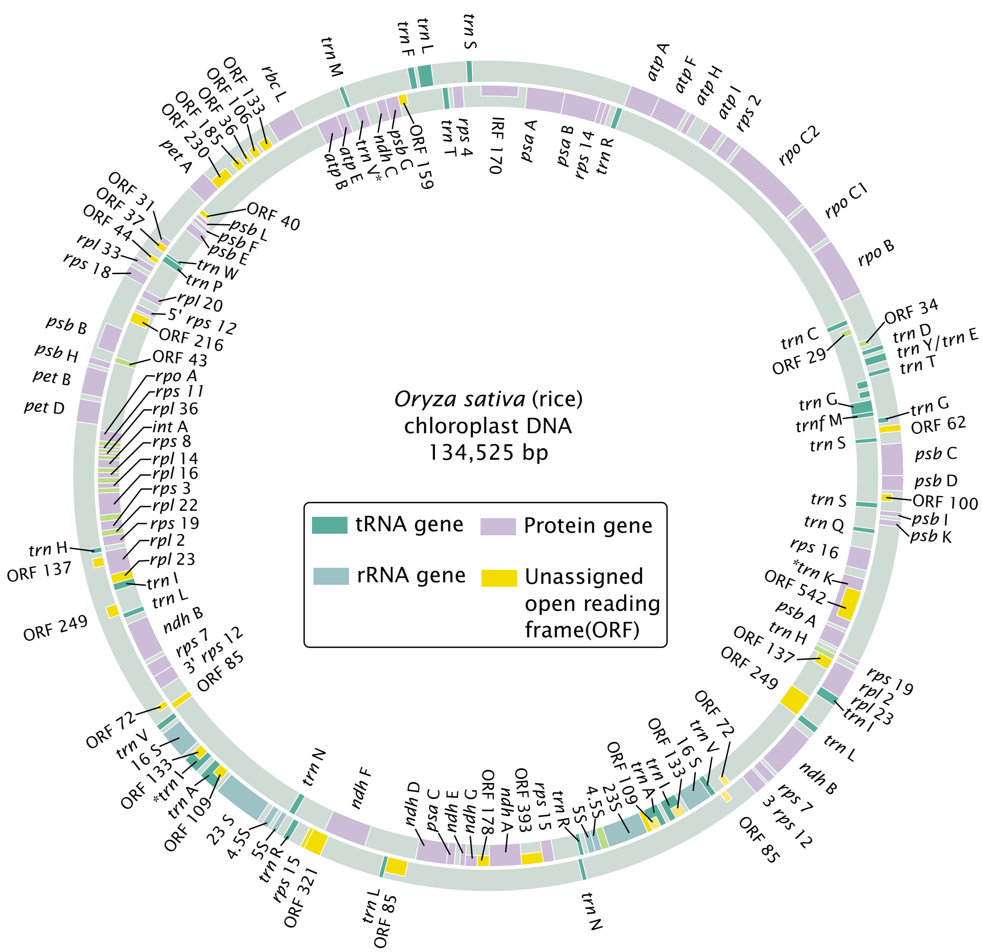
Gene Structure and Organization of cpDNA

The chloroplast genomes from a number of plant and algal species have been sequenced, and cpDNA is now recognized to be basically eubacterial in its organization: the order of some groups of genes is the same as that observed in *E. coli*, and many chloroplast genes are organized into operon-like clusters.

Table 20.3 Size of the chloroplast genomes in selected organisms

Organism	**Size of cpDNA in Nucleotide Pairs**
Euglena gracilis (protist)	143,172
Porphyra purpurea (red alga)	191,028
Chlorella vulgaris (green alga)	150,613
Marchantia polymorpha (liverwort)	121,024
Nicotiana tabacum (tobacco)	155,939
Zea mays (corn)	140,387
Pinus thunbergii (black pine)	119,707

Among vascular plants, chloroplast chromosomes are similar in gene content and gene order. A typical chloroplast genome encodes 4 rRNA genes, from 30 to 35 tRNA genes, a number of ribosomal proteins, many proteins engaged in



20.11 **Chloroplast DNA of rice.**

photosynthesis, and several proteins having roles in nonphotosynthesis processes. A key protein encoded by cpDNA is ribulose-1,5-bisphosphate carboxylase-oxygenase (abbreviated RuBisCO), which participates in carbon fixation of photosynthesis. RuBisCO makes up about 50% of the protein found in green plants and is therefore considered the most abundant protein on earth. It is a complex protein consisting of eight identical large subunits and eight identical small subunits. The large subunit is encoded by chloroplast DNA, whereas the small subunit is encoded by nuclear DNA.

The circular chloroplast genome has genes on both of its strands. Some chloroplast genes have been identified on the basis of a start and stop codon in the same reading frame, but no protein products have yet been isolated for these genes. These sequences are referred to as *open reading frames.* A prominent feature of most chloroplast genomes is the presence of a large inverted repeat. In rice, this repeat includes genes for 23S rRNA, 4.5S rRNA, and 5S rRNA, as well as several genes for tRNAs and proteins (see Figure 20.11). In some plants, these repeats include the majority of the genome, whereas, in others, the repeats are absent entirely. Much of cpDNA consists of noncoding sequences, and introns are found in many chloroplast genes. Finally, many of the sequences in cpDNA are quite similar to those found in equivalent eubacterial genes.

Concepts

Most chloroplast genomes consist of a single, circular DNA molecule not complexed with histone proteins. Although there is considerable size variation, the cpDNAs found in most vascular plants are about 150,000 bp. Genes are scattered in the circular chloroplast genome, and many contain introns. Most cpDNAs contain a large inverted repeat.

Replication, Transcription, and Translation of cpDNA

Little is known about the process of replication of cpDNA. The results of studies viewing cpDNA replication with electron microscopy suggest that replication begins within two D loops and spreads outward to form a theta-like structure. After an initial round of replication, DNA synthesis may switch to a rolling-circle-type mechanism (see Figure 12.5).

The transcription and translation of chloroplast genes are similar in many respects to these processes in eubacteria. For example, promoters found in cpDNA are virtually identical with those found in eubacteria and possess sequences similar to the -10 and -35 consensus sequences of eubacterial promoters. The same antibiotics that inhibit protein synthesis in eubacteria (as well as in mitochondria) inhibit protein synthesis in chloroplasts, indicating that protein synthesis in eubacteria and chloroplasts is similar. Chloroplast translation is initiated by *N*-formylmethionine, just as it is in eubacteria.

Most genes in cpDNA are transcribed in groups; only a few genes have their own promoters and are transcribed as separate mRNA molecules. The RNA polymerase that transcribes cpDNA is more similar to eubacterial RNA polymerase than to any of the RNA polymerases that transcribe eukaryotic nuclear genes. Like eubacterial mRNAs, chloroplast mRNAs are not capped at the 5′ ends, and poly(A) tails are not added to the 3′ ends. However, introns are removed from some RNA molecules after transcription, and the 5′ and 3′ ends may undergo some additional processing before the molecules are translated. Like eubacterial mRNAs, many chloroplast mRNAs have a Shine-Dalgarno sequence in the 5′ untranslated region, which may serve as a ribosome-binding site.

Chloroplasts, like eubacteria, contain 70S ribosomes that consist of two subunits, a large 50S subunit and a smaller 30S subunit. The small subunit includes a single RNA molecule that is 16S in size, similar to that found in the small subunit of eubacterial ribosomes. The larger 50S subunit includes three rRNA molecules: a 23S rRNA, a 5S rRNA, and a 4.5 rRNA. In eubacterial ribosomes, the large subunit possesses only two rRNA molecules, which are 23S and 5S in size. The 4.5S rRNA molecule found in the large subunit of chloroplast ribosomes is homologous to the 3′ end of the 23S rRNA found in eubacteria; so the structure of the chloroplast ribosome is very similar to that of ribosomes found in eubacteria.

Initiation factors, elongation factors, and termination factors function in chloroplast translation and eubacterial translation in similar ways. Most chloroplast chromosomes encode from 30 to 35 different tRNAs, suggesting that the expanded wobble seen in mitochondria does not exist in chloroplast translation. Only universal codons have been found in cpDNA, and translation in chloroplast starts with *N*-formylmethionine as the first amino acid.

Evolution of cpDNA

The DNA sequences of chloroplasts are very similar to those found in cyanobacteria; so chloroplast genomes clearly have a eubacterial ancestry. Overall, cpDNA sequences evolve slowly compared with sequences in nuclear DNA and some mtDNA. For most chloroplast genomes, size and gene organization are similar, although there are some notable exceptions.

Concepts

Many aspects of the transcription and translation of cpDNA are similar to those of eubacteria. Chloroplast DNA sequences are most similar to DNA sequences in cyanobacteria which supports the endosymbiotic theory. Most cpDNA evolves slowly in sequence and structure.

www.whfreeman.com/pierce Information on chloroplast genomes that have been sequenced

Connecting Concepts

Genome Comparisons

A theme running through the preceding discussions of mitochondrial and chloroplast genomes has been a comparison of these genomes with those found in eubacterial and eukaryotic cells (Table 20.4). The endosymbiotic theory indicates that mitochondria and chloroplasts evolved from eubacterial ancestors, and one might therefore assume that mtDNA and cpDNA would be similar to DNA found in eubacterial cells. The actual situation is more complex: mitochondrial DNA and chloroplast DNA possess a mix of eubacterial, eukaryotic, and unique characteristics.

The mitochondrial and chloroplast genomes are similar to those of eubacterial cells in that they are relatively small, lack histone proteins, and are usually on circular DNA molecules. Gene organization and the expression of organelle genomes, however, display some similarities to eubacterial genomes and some similarities to eukaryotic genomes. Introns are present in some organelle genomes but are absent from others. Pre-mRNA introns (see p. 000 in Chapter 14 for a discussion of different types of introns) are absent from mitochondrial and chloroplast genes, as they are from eubacterial genes. Group II introns are present in some organelle and eubacterial genomes but are absent from eukaryotic nuclear genomes. Group I introns are common in some mtDNA and in most cpDNA, and these introns are also found in eubacterial, archaeal, and eukaryotic genomes.

Polycistronic mRNA, which is common in eubacteria but uncommon in eukaryotes, is also found in mitochondria and especially chloroplasts. Human mtDNA, which has little noncoding DNA between genes and little repetitive DNA, is similar in organization to that of typical eubacterial chromosomes, but other mitochondrial and chloroplast genomes possess long noncoding sequences between genes.

Antibiotics that inhibit eubacterial translation also inhibit organelle translation, and the 5′ cap, which is added to eukaryotic mRNA after transcription, is absent from organelle mRNA. A 3′ poly(A) tail, characteristic of most nuclear mRNAs, is present only in some animal mitochondrial mRNA, and it appears to be fundamentally different from that found in nuclear mRNAs. Shine-Dalgarno sequences, the ribosome-binding sites characteristic of eubacterial DNA, are present in some cpDNA but are absent in mtDNA. Finally, some mitochondrial genomes use nonuniversal codons and have extended wobble, which is rare in both eubacterial and eukaryotic DNA.

What conclusions can we draw from these comparisons? Clearly, the genomes of mitochondria and chloroplasts are not typical of the nuclear genomes of the eukaryotic cells in which they reside. In sequence, organelle DNA is most similar to eubacterial DNA, but many aspects of organization and expression in organelle genomes are unique. It is important to remember that the endosymbiotic theory does not propose that mitochondria and chloroplasts are eubacterial in nature but that they arose

Table 20.4 Comparison of nuclear eukaryotic, eubacterial, mitochondrial, and chloroplast genomes

Characteristic	Eukaryotic Genome	Eubacterial Genome	Mitochondrial Genome	Chloroplast Genome
Genome consists of double-stranded DNA	Yes	Yes	Yes	Yes
Circular	No	Yes	Most	Yes
Histone proteins	Yes	No	No	No
Size	Large	Small	Small	Small
Single molecule per genome	No	Yes	Yes in animals No in some plants	Yes
Pre-mRNA introns	Common	Absent	Absent	Absent
Group I introns	Present	Present	Present	Present
Group II introns	Absent	Present	Present	Present
Polycistronic mRNA	Uncommon	Common	Present	Common
5′ cap added to mRNA	Yes	No	No	No
3′ poly(A) tail added to mRNA	Yes	No	Some in animals	No
Shine-Dalgarno sequence in 5′untranslated region of mRNA	No	Yes	Rare	Some
Nonuniversal codons	Rare	Rare	Yes	No
Extended wobble	No	No	Yes	No
Translation inhibited by tetracycline	No	Yes	Yes	Yes

from eubacterial ancestors more than a billion years ago. Through time, the genomes of the endosymbiont have undergone considerable evolutionary change and have evolved characteristics that distinguish them from contemporary eubacterial and eukaryotic genomes.

The Intergenomic Exchange of Genetic Information

Many proteins found in modern mitochondria and chloroplasts are encoded by nuclear genes, which suggests that much of the original genetic material in the endosymbiont has probably been transferred to the nucleus. This assumption is supported by the observation that some DNA sequences normally found in mtDNA have been detected in nuclear DNA of some strains of yeast and maize. Likewise, chloroplast sequences have been found in the nuclear DNA of spinach Furthermore, the sequences of nuclear genes that encode organelle proteins are most similar to their eubacterial counterparts.

There is also evidence that genetic material has moved from chloroplasts to mitochondria. For example, DNA fragments from the 16S rRNA gene and two tRNA genes that are normally encoded by cpDNA have been found in the mtDNA of maize. Sequences from the gene that encodes the large subunit of RuBisCO, which is normally encoded by cpDNA, are duplicated in maize mtDNA. And there is even evidence that some nuclear genes have moved into mitochondrial genomes. The exchange of genetic material between the nuclear, mitochondrial, and chloroplast genomes has given rise to the term "promiscuous DNA" to describe this phenomenon. The mechanism by which this exchange takes place is not entirely clear.

Mitochondrial DNA and Aging in Humans

Symptoms of many human genetic diseases caused by defects in mtDNA first appear in middle age or later and increase in severity as people age. One hypothesis to explain the late onset and progressive worsening of mitochondrial diseases is related to the decline in oxidative phosphorylation with aging.

Oxidative phosphorylation is the process that generates ATP, the primary carrier of energy in the cell. This process takes place on the inner membrane of the mitochondrion and requires a number of different proteins, some encoded by mtDNA and others encoded by nuclear genes. Oxidative phosphorylation normally declines with age and, if it falls below some critical threshold, tissues do not make enough ATP to sustain vital functions and disease symptoms appear. Most people start life with an excess capacity for oxidative phosphorylation; this capacity decreases with age, but most people reach old age or die before the critical threshold is passed. Persons born with mitochondrial diseases carry mutations in their mtDNA that lower their oxidative phosphorylation capacity. At birth, their capacity may be sufficient to support their ATP needs but, as their oxidative phosphorylation capacity declines with age, they cross the critical threshold and begin to experience symptoms. These symptoms usually first appear in tissues that are most critically dependent on mitochondrial energy: the central nervous system, heart and skeletal muscle, pancreatic islets, kidneys, and the liver.

Why does oxidative phosphorylation capacity decline with age? One possible explanation is that damage to mtDNA accumulates with age; deletions and base substitutions in mtDNA increase with age. For example, a common 5000-bp deletion in mtDNA is absent in normal heart muscle cells before the age of 40, but afterward this deletion is present with increasing frequency. The same deletion is found at a low frequency in normal brain tissue before age 75 but is found in 11% to 12% of mtDNAs in the basal ganglia by age 80. People with mtDNA genetic diseases may age prematurely because they begin life with damaged mtDNA.

The mechanism of age-related increases in mtDNA damage is not yet known. Oxygen radicals, highly reactive compounds that are natural by-products of oxidative phosphorylation, are known to damage DNA (see p. 000 in Chapter 17). Because mtDNA is physically close to the enzymes taking part in oxidative phosphorylation, it may be more prone to oxidative damage than nuclear DNA. When mtDNA has been damaged, the cell's capacity to produce ATP drops. To produce sufficient ATP to meet the cell's energy needs, even more oxidative phosphorylation must occur, which in turn may stimulate further production of oxygen radicals, leading to a vicious cycle.

Significantly elevated levels of mtDNA defects have been observed in some patients with late-onset degenerative diseases, such as diabetes mellitus, ischemic heart disease, Parkinson disease, Alzheimer disease, and Huntington disease. All of these diseases appear in middle to old age and have symptoms associated with tissues that critically depend on oxidative phosphorylation for ATP production. However, because Huntington disease and some cases of Alzheimer disease are inherited as autosomal dominant conditions, mtDNA defects cannot be the primary cause of these diseases, although they may contribute to their progression.

Connecting Concepts Across Chapters

This chapter is about the unique properties of organelle DNA, which is part of the cytoplasm and usually exhibits uniparental inheritance. A unifying theme has been that mitochondria and chloroplasts evolved from free-living eubacteria that entered into an endosymbiotic relation with the eukaryotic cells in which they are found. Endosymbiosis helps to explain many of the characteristics of mitochondrial DNA and chloroplast DNA, which resemble eubacterial DNA more than they do nuclear eukaryotic DNA. However, not all aspects of mtDNA and cpDNA are similar to eubacterial DNA; organelle DNA has a number of properties that are unique.

Another prominent theme that runs through this chapter is that cpDNA and mtDNA display a bewildering diversity of variation in size and organization. The reason for this variation is unknown, but the variation makes summarizing mitochondrial and chloroplast genomes difficult.

Traits encoded by mitochondrial and chloroplast genes are inherited in a very different manner from those encoded by nuclear genes. Because organelle DNA is located in the cytoplasm, the traits that it encodes exhibit cytoplasmic inheritance and are typically inherited from a single parent, most often the mother. Many traits encoded by mtDNA and cpDNA exhibit phenotypic variation among progeny of a single cross and even among cells and tissues within an individual organism; the latter occurs when there are two or more genetic variants in a single cell and random segregation of the organelles in cell division produces cells with different proportions of the two types of DNA.

Understanding the inheritance of mitochondrial- and chloroplast-encoded traits builds on earlier discussions of uniparental inheritance in Chapter 5 and biparental inheritance (with which it is contrasted) in Chapter 3. Material in the present chapter is closely linked to information on DNA structure and organization found in Chapters 10 and 11 and to discussions of replication, transcription, RNA processing, and translation found in Chapters 12 through 15. Molecular techniques described in this chapter are covered more thoroughly in Chapter 18. The use of mtDNA in evolutionary studies will be discussed in more detail in Chapter 23.

CONCEPTS SUMMARY

- Mitochondria and chloroplasts are eukaryotic organelles that possess their own DNA. Traits encoded by mtDNA and cpDNA exhibit cytoplasmic inheritance and usually are inherited from a single parent, most often the mother. Random segregation of organelles in cell division may produce phenotypic variation among cells within a single individual and among the offspring of a single female.
- The endosymbiotic theory proposes that mitochondria and chloroplasts originated as free-living prokaryotic (specifically eubacterial) organisms that entered into a beneficial association with eukaryotic cells. Similarities in the gene sequences of organelle and eubacterial DNA support a eubacterial origin for mitochondrial and chloroplast DNA.
- The mitochondrial genome usually consists of a single, circular DNA molecule that lacks histone proteins, although plants may have multiple circular molecules. Mitochondrial DNA varies in size among different groups of organisms; most of this variation is due to noncoding DNA. Each cell contains many copies of mtDNA.
- The organization of genes in the mitochondrial genome differs among organisms. Ancestral mitochondrial genomes typically have characteristics of eubacterial genomes, including eubacterial-like ribosomes, a complete or almost complete set of tRNA genes, few introns, little noncoding DNA between genes, genes organized into eubacterial-like clusters, and the use of only universal codons. Derived mitochondrial genomes are smaller and contain fewer genes. Their rRNA genes and ribosomes differ from those found in eubacteria, and they use some nonuniversal codons.
- Human mtDNA is highly economical, with few noncoding nucleotides. Fungal and plant mtDNAs contain much noncoding DNA between genes, introns within genes, and extensive 5′ and 3′ untranslated regions. Most plant mitochondrial genomes contain one or more large direct repeats, which may recombine to produce smaller or larger DNA molecules.
- Mitochondrial DNA is synthesized throughout the cell cycle, and its synthesis is not coordinated with the replication of nuclear DNA.
- The transcription of mitochondrial genes varies among different organisms. Messenger RNAs produced by the transcription of mtDNA are not capped at their 5′ ends; poly(A) tails are added to the 3′ ends of some animal mRNAs, but these tails are different from the poly(A) tails found on nuclear-encoded mRNAs.
- Antibiotics that inhibit eubacterial ribosomes also inhibit mitochondrial ribosomes. Protein synthesis in mitochondria is initiated at AUG start codons by *N*-formylmethionine and employs eubacterial-like elongation factors. Many mitochondrial genomes encode a limited number of tRNAs, with relaxed codon–anticodon pairing rules and extended wobble.
- Comparisons of mtDNA sequences suggest that mitochondria evolved from a eubacterial ancestor. Vertebrate mtDNA exhibits rapid change in sequence but little change in gene content and organization. Plant mtDNA exhibits little change in sequence but much variation in gene content and organization.
- Chloroplast genomes consist of a single, circular DNA molecule that varies little in size and lacks histone proteins. Each plant cell contains multiple copies of cpDNA.
- Most chloroplast chromosomes possess large inverted repeats; some chloroplast genes contain introns.
- Transcription and translation are similar in chloroplasts and eubacteria: most chloroplast genes are transcribed as polycistronic units, their mRNAs are not capped, no poly(A)

tails are added, and they possess a Shine-Dalgarno ribosome-binding sequence.

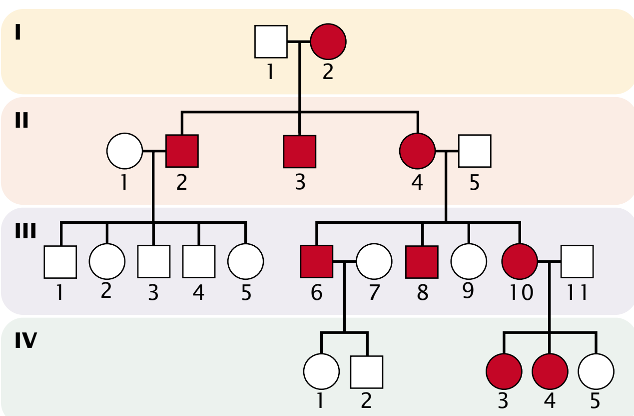
- Chloroplast DNA sequences are most similar to those in cyanobacteria. Chloroplast DNA sequences tend to evolve slowly.
- Through evolutionary time, many mitochondrial and chloroplast genes have moved to nuclear chromosomes. In some plants, there is evidence that copies of chloroplast genes have moved to the mitochondrial genome.

IMPORTANT TERMS

mitochondrial DNA (mtDNA) (p. 000)
chloroplast DNA (cpDNA) (p. 000)
heteroplasmy (p. 000)
replicative segregation (p. 000)
homoplasmy (p. 000)
endosymbiotic theory (p. 000)
D loop (p. 000)

Worked Problems

1. A physician examines a young man who has a progressive muscle disorder and visual abnormalities. A number of the patient's relatives have the same condition, as shown in the adjoining pedigree. The degree of expression of the trait is highly variable among members of the family: some are only slightly affected, whereas others developed severe symptoms at an early age. The physician concludes that this disorder is due to a mutation in the mitochondrial genome. Do you agree with the physician's conclusion? Why or why not? Could the disorder be due to a mutation in a nuclear gene? Explain your reasoning.



• Solution

The conclusion that the disorder is caused by a mutation in the mitochondrial genome is supported by the pedigree and the observation of variable expression in affected members of the same family. The disorder is passed only from affected mothers to offspring; when fathers are affected, none of their children have the trait (as seen in the children of II-2 and III-6). This outcome is expected of traits determined by mutations in mtDNA, because mitochondria are in the cytoplasm and usually inherited only from a single (in humans, the maternal) parent.

The facts that some offspring of affected mothers do not show the trait (III-9 and IV-5) and that expression varies from one person to another suggest that affected persons are heteroplasmic, with both mutant and wild-type mitochondria. Random segregation of mitochondria in meiosis may produce gametes having different proportions of mutant and wild-type sequences, resulting in different degrees of phenotypic expression among the offspring. Most likely, symptoms of the disorder develop when some minimum proportion of the mitochondria are mutant. Just by chance, some of the gametes produced by an affected mother contain few mutant mitochondria and result in offspring that lack the disorder.

Another possible explanation for the disorder is that it results from an autosomal dominant gene. When an affected (heterozygous) person mates with an unaffected (homozygous) person, about half of the offspring are expected to have the trait, but just by chance some affected parents will have no affected offspring. It is possible that individuals II-2 and III-6 in the pedigree just happened to be male and their sex is unrelated to the mode of transmission. The variable expression could be explained by variable expressivity (see p. 000 in Chapter 3).

2. Suppose that a new organelle is discovered in an obscure group of protists. This organelle contains a small DNA genome and some scientists are arguing that, like chloroplasts and mitochondria, this organelle originated as a free-living eubacterium that entered into an endosymbiotic relation with the protist. Outline a research plan to determine if the new organelle evolved from a free-living eubacterium. What kinds of data would you collect and what predictions would you make if the theory is correct?

• Solution

We could examine the structure, organization, and sequences of the organelle genome. If the organelle shows only characteristics of eukaryotic DNA, then it most likely has a eukaryotic origin but, if it displays some characteristics of eubacterial DNA, then this finding supports the theory of a eubacterial origin. However, on the basis of our knowledge of mitochondrial and chloroplast genomes, we should not expect the organelle genome to be entirely eubacterial in its characteristics.

We could start by examining the overall characteristics of the organelle DNA. If it has a eubacterial origin, we might expect that the organelle genome will consist of a circular molecule and will lack histone proteins. We might then sequence the organelle DNA to determine its gene content and organization. The presence of any group II introns would suggest a eubacterial origin, because these introns have been found only in eubacterial genomes and genomes derived from eubacteria. The presence of any pre-mRNA introns, on the other hand, would suggest a eukaryotic origin, because these introns have been found only in nuclear eukaryotic genomes. If the organelle genome has a eubacterial origin, we might expect to see polycistronic mRNA, the absence of a 5′ cap, and inhibition of translation by those antibiotics that typically inhibit eubacterial translation.

Finally, we could compare the DNA sequences found in the organelle genome with homologous sequences from eubacteria and eukaryotic genomes. If the theory of an endosymbiotic origin is correct, then the organelle sequences should be most similar to homologous sequences found in eubacteria.

The New Genetics
MINING GENOMES

Evolutionary Analysis with the Use of Mitochondrial Genomes

Phylogenetic trees are graphic representations of the relationships between different organisms. Traditionally based on morphological, physiological, and behavioral data, evolutionary analysis has been substantially changed by the use of molecular information. In this exercise, you will pose and evaluate a question relating to human evolution. You will use mitochondrial DNA sequences and the tools available at the Biology Workbench, managed by the San Diego Supercomputing Center at the University of California, San Diego.

COMPREHENSION QUESTIONS

* 1. Briefly describe the general structures of mtDNA and cpDNA. How are they similar? How do they differ? How do their structures compare with the structures of eubacterial and eukaryotic (nuclear) DNA.

2. Explain why many traits encoded by mtDNA and cpDNA exhibit considerable variation in their expression, even among members of the same family.

* 3. What is the endosymbiotic theory? How does it help to explain some of the characteristics of mitochondria and chloroplasts?

4. What evidence supports the endosymbiotic theory?

5. How are genes organized in the mitochondrial genome? How does this organization differ between ancestral and derived mitochondrial genomes?

* 6. What are nonuniversal codons? Where are they found?

7. How does replication of mtDNA differ from replication of nuclear DNA in eukaryotic cells.

* 8. The human mitochondrial genome encodes only 22 tRNAs, whereas at least 32 tRNAs are required for cytoplasmic translation. Why are fewer tRNAs needed in mitochondria?

9. What are some possible explanations for an accelerated rate of evolution in the sequences of vertebrate mtDNA.

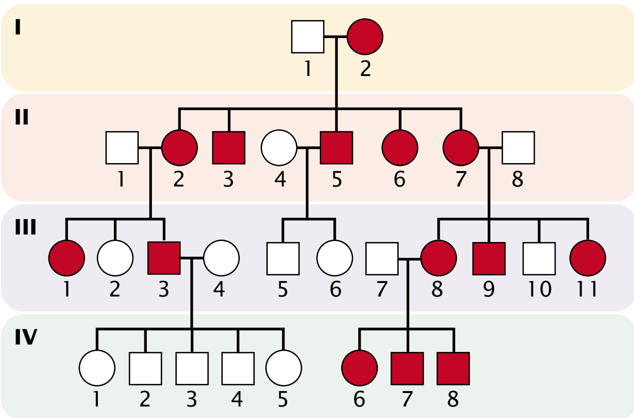
*10. Briefly describe the organization of genes on the chloroplast genome.

11. What is meant by the term “promiscuous DNA”?

APPLICATION QUESTIONS AND PROBLEMS

12. A wheat plant that is light green in color is found growing in a field. Biochemical analysis reveals that chloroplasts in this plant produce only 50% of the chlorophyll normally found in wheat chloroplasts. Propose a set of crosses to determine whether the light-green phenotype is caused by a mutation in a nuclear gene or a chloroplast gene.

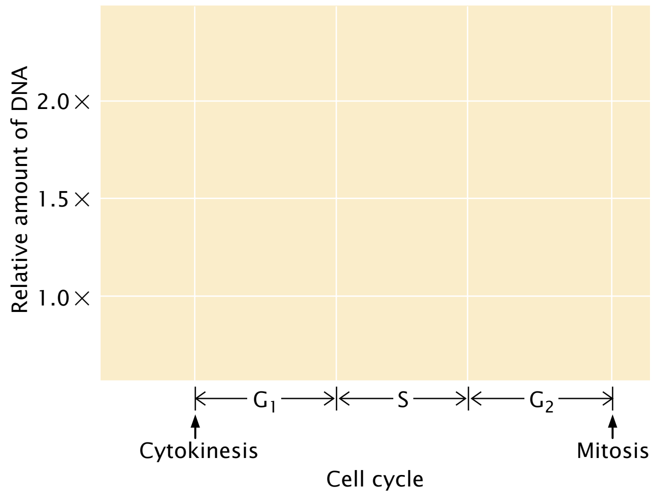
*13. A rare neurological disease is found in the family illustrated in the following pedigree. What is the most likely mode of inheritance for this disorder? Explain your reasoning.



14. In a particular strain of *Neurospora*, a *poky* mutation exhibits biparental inheritance, whereas *poky* mutations in other strains are inherited only from the maternal parent. Explain these results.

15. Antibiotics such as chloramphenicol, tetracycline, and erythromycin inhibit protein synthesis in eubacteria but have no effect on protein synthesis encoded by nuclear genes. Cycloheximide inhibits protein synthesis encoded by nuclear genes but has no effect on eubacterial protein synthesis. How might these compounds be used to determine which proteins are encoded by the mitochondrial and chloroplast genomes?

16. A scientist collects cells at various points in the cell cycle and isolates DNA from them. Using density gradient centrifugation, she separates the nuclear and mtDNA. She then measures the amount of mtDNA and nuclear DNA present at different points in the cell cycle. On the following graph, draw a line to represent the relative amounts of nuclear DNA that you expect her to find per cell throughout the cell cycle. Then, draw a dotted line on the same graph to indicate the relative amount of mtDNA that you would expect to see at different points throughout the cell cycle.



17. The introduction to Chapter 1 described how bones found in 1979 outside Ekaterinburg, Russia, were shown to be those of Tsar Nicholas and his family, who were executed in 1918 by a Bolshevik firing squad in the Russian Revolution. To prove that the skeletons were those of the royal family, mtDNA was extracted from the bone samples, amplified by PCR, and compared with mtDNA from living relatives of the tsar's family. Why was DNA from the mitochondria analyzed instead of nuclear DNA? What are some of the advantages of using mtDNA for this type of study?

18. From Figure 20.8, determine as best you can the percentage of human mtDNA that is coding (transcribed into RNA) and the percentage that is noncoding (not transcribed).

CHALLENGE QUESTIONS

19. Mitochondrial DNA sequences have been detected in the nuclear genomes of many organisms, and cpDNA sequences are sometimes found in the mitochondrial genome. Propose a mechanism for how such "promiscuous DNA" might move between nuclear, mitochondrial, and chloroplast genomes.

20. Steven A. Frank and Laurence D. Hurst argued that a cytoplasmically inherited mutation in humans that has severe effects in males but no effect in females will not be eliminated from a population by natural selection, because only females pass on mtDNA. Using this argument, explain why males with Leber hereditary optic neuropathy are more severely affected than females.

21. Several families have been described that exhibit vision problems, muscle weakness, and deafness. This disorder is inherited as an autosomal dominant trait and the disease-causing gene has been mapped to chromosome 10 in the nucleus. Analysis of the mtDNA from affected persons in these families reveals that large numbers of their mitochondrial genomes possess deletions of varying length. Different members of the same family and even different mitochondria from the same person possess deletions of different sizes; so the underlying defect appears to be a tendency for the mtDNA of affected persons to have deletions. Propose an explanation for how a mutation in a nuclear gene might lead to deletions in mtDNA.

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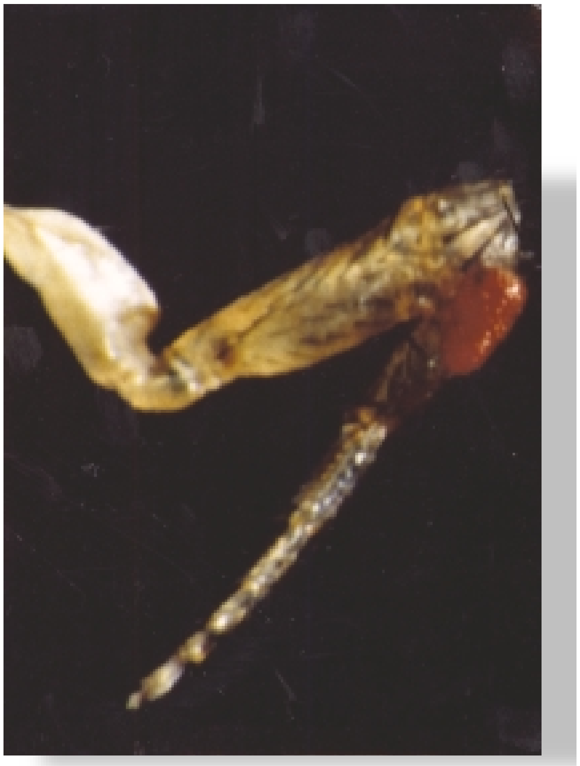
21 Advanced Topics in Genetics: Developmental Genetics, Immunogenetics, and Cancer Genetics



Flies have been genetically engineered to have extra eyes on their legs, wings, and elsewhere. Through genetic engineering, the *eyeless* gene can be expressed in cells of body parts where eyes do not normally appear.

Flies with Extra Eyes

We can all imagine situations where an extra set of eyes might come in handy: eyeing members of the opposite sex while still paying attention to the professor during lecture, looking both ways at the same time before crossing the street; or watching your backside in a barroom brawl. However useful extra eyes might be, creating them at selected locations is no simple matter. An eye is, after all, an exceedingly complex structure, consisting of photoreceptors, lens, nerves, and other tissues. It would be very unlikely for all of these structures to develop at a site where eyes don't normally exist. Nevertheless, in 1995, a group of geneticists succeeded in genetically engineering fruit flies with extra eyes on their wings, legs, and antennae. How was this amazing feat accomplished?

The story of creating flies with extra eyes began in 1915, when Mildred Hoge discovered a mutant fruit fly with small eyes due to a recessive mutation in a gene called *eyeless*. The product of the normal allele of the *eyeless* locus is required for proper development of the fruit-fly eye.

In 1993, Walter Gehring and his collaborators were investigating *Drosophila* genes that encode transcription factors (see Chapter 13). One of these genes mapped to the same location as that of the *eyeless* gene and, in fact, turned out to *be* the *eyeless* gene. To see what effect *eyeless* might have on development, Gehring's group genetically engineered cells that expressed the *eyeless* gene in parts of the fly where the gene is not normally expressed. When these flies hatched, they had huge eyes on their wings, antennae, and legs. These structures were not just tissue that resembled eyes; they were complete eyes with a cornea, cone cells, and photoreceptors that responded to light, although the flies could not use them to see, because they were not connected to the nervous system. The *eyeless* gene appears to be one of the long-sought master control switches of development: its protein activates a set of other genes that are responsible for making a complete eye.

The *eyeless* gene has counterparts in mice and humans that affect the development of mammalian eyes. There is a striking similarity between the *eyeless* gene of *Drosophila* and the *Small eye* gene that exists in mice. In mice, a mutation in one copy of *Small eye* causes small eyes; a mouse that is homozygous for the *Small eye* mutation has no eyes. There is also a similarity between the *eyeless* gene in *Drosophila* and the *Aniridia* gene in humans; a mutation in *Aniridia* produces a severely malformed human eye. Similarities in the sequences of *eyeless, Small eye,* and *Aniridia* suggest that all three genes evolved from a common ancestral sequence. This possibility is surprising, because the eyes of insects and mammals were thought to have evolved independently. Similarities among *eyeless, Small eye,* and *Aniridia* suggest that a common pathway underlies eye development in flies, mice, and humans.

This chapter focuses on three specialized topics in genetics: developmental genetics, immunogenetics, and cancer genetics. We begin with a discussion of the genetic control of the early development of *Drosophila* embryos, one of the best-understood developmental systems. We then turn to the genetics of the immune system in vertebrates. This system is capable of generating proteins that recognize virtually any foreign substance in the body. The generation of this huge diversity of proteins relies on a special type of genetic recombination unique to the immune system. Last, we consider the genetic basis of cancer and how mutations in particular types of genes contribute to the growth of tumors in humans.

Developmental Genetics

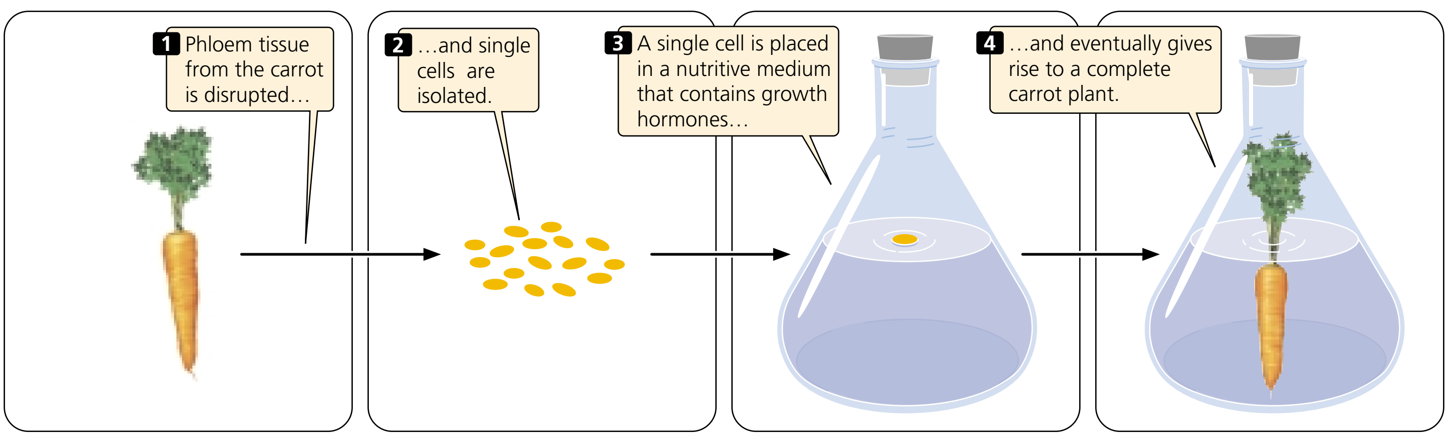
Every multicellular organism begins life as a unicellular, fertilized egg. This single-celled zygote undergoes repeated cell divisions, eventually producing millions or trillions of cells that constitute a complete adult organism. Initially, each cell in the embryo is **totipotent**—it has the potential to develop into any cell type. Many cells in plants and fungi remain totipotent, but animal cells usually become committed to developing into specific types of cells after just a few early embryonic divisions. This commitment often comes well before a cell begins to exhibit any characteristics of a particular cell type; once the cell becomes committed, it cannot reverse its fate and develop into a different cell type. A cell becomes committed by a process called **determination,** the mechanism of which is still unknown.

For many years, the work of developmental biologists was limited to describing the changes that take place in the course of development, because techniques for probing the intracellular processes behind these changes were unavailable. But, in recent years, powerful genetic and molecular techniques have had a tremendous influence on the study of development. In a few model systems such as *Drosophila,* the molecular mechanisms underlying developmental change are now beginning to be understood.

Cloning Experiments

If all cells in a multicellular organism are derived from the same original cell, how do different cells types arise? One possibility is that, throughout development, genes might be selectively lost or altered, causing different cell types to have different genomes. Alternatively, each cell might contain the same genetic information, but different genes might be expressed in each cell type. Early cloning experiments helped to answer this question.

In the 1950s, Frederick Steward developed methods for cloning plants. He disrupted phloem tissue from the root of a carrot, separating and isolating individual cells. He then placed individual cells in a sterile medium that contained nutrients. Steward was successful in getting the cells to grow and divide, and eventually he obtained whole edible carrots from single cells (FIGURE 21.1). Because all parts of the plant were regenerated from a specialized phloem cell,



21.1 **Many plants can be cloned from isolated single cells.** Thus none of the original genetic material is lost during development.

Steward concluded that each phloem cell contained the genetic potential for a whole plant; none of the original genetic material was lost during determination.

The results of later studies demonstrated that most animal cells also retain a complete set of genetic information during development. In 1952, Robert Briggs and Thomas King removed the nuclei from unfertilized oocytes of the frog *Rana pipiens*. They then isolated nuclei from frog blastulas (an early embryonic stage) and injected these nuclei individually into the oocytes. The eggs were then pricked with a needle to stimulate them to divide. Although most were damaged in the process, a few eggs developed into complete tadpoles that eventually metamorphosed into frogs.

In the late 1960s, John Gurdon used these methods to successfully clone a few frogs with nuclei isolated from intestinal cells of tadpoles. This accomplishment suggested that the differentiated intestinal cells carried the genetic information necessary to encode traits found in all other cells. However, Gurdon's successful clonings may have resulted from the presence of a few undifferentiated stem cells in the intestinal tissue, which were inadvertently used as the nuclei donors.

In 1997, researchers at the Roslin Institute of Scotland announced that they had successfully cloned a sheep by using the genetic material from a differentiated cell of an adult animal. To perform this experiment, they fused an udder cell from a white-faced Finn Dorset ewe with an enucleated egg cell and stimulated the egg electrically to initiate development. After growing it in the laboratory for a week, they implanted the embryo into a Scottish black-faced surrogate mother. Dolly, the first mammal cloned from an adult cell, was born on July 5, 1996 (FIGURE 21.2). Since the cloning of Dolly, other sheep, mice, and calves have been cloned from differentiated adult cells.

21.2 In 1996, researchers at the Roslin Institute of Scotland successfully cloned a sheep named Dolly. They used the genetic material from a differentiated cell of an adult animal.

These cloning experiments demonstrated that genetic material is not lost or permanently altered during development—development must require the selective expression of genes. But how do cells regulate their gene expression in a coordinated manner to give rise to a complex, multicellular organism? Research has now begun to provide some answers to this important question.

Concepts

The ability to clone plants and animals from single specialized cells demonstrates that genes are not lost or permanently altered during development.

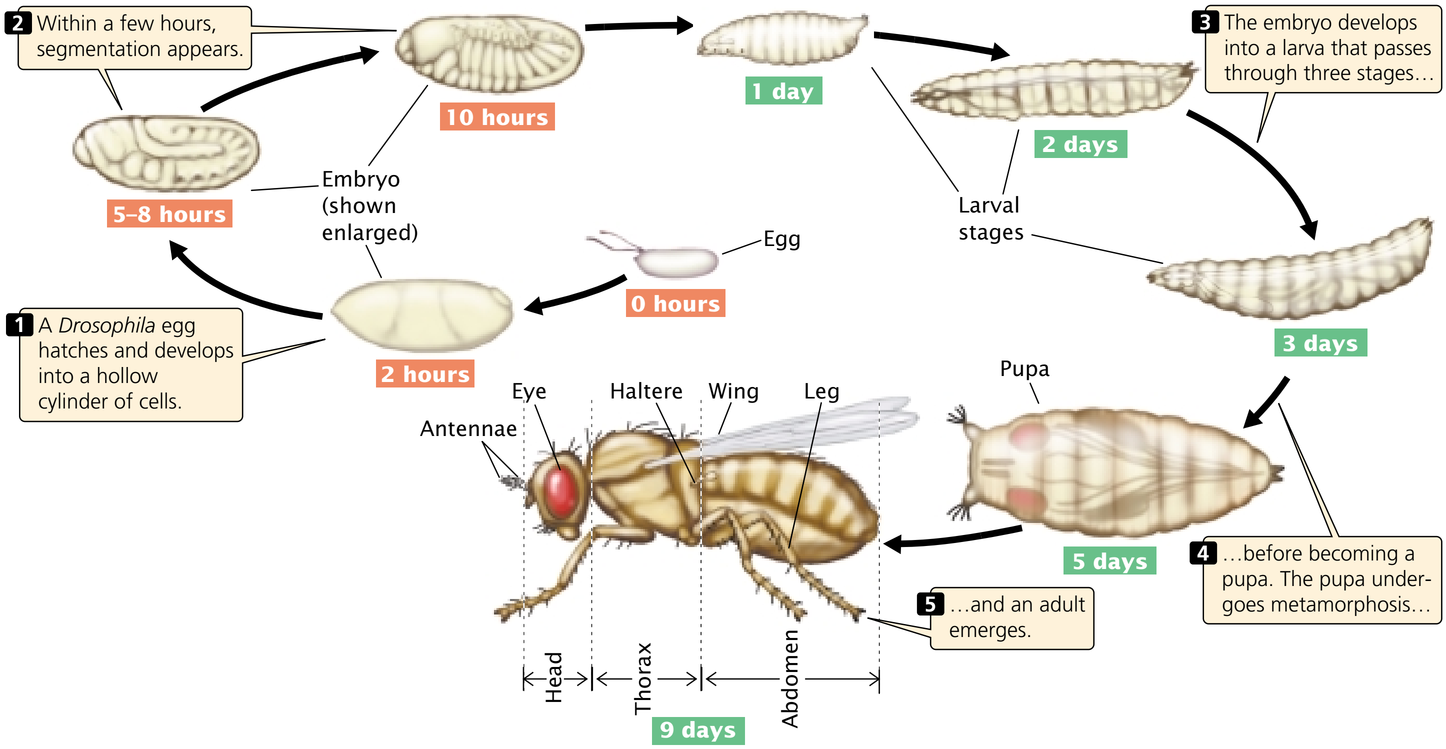
www.whfreeman.com/pierce Information about cloning, nuclear transfer research, and about the ethics of cloning

The Genetics of Pattern Formation in *Drosophila*

One of the best-studied systems for the genetic control of pattern formation is the early embryonic development of *Drosophila melanogaster*. Geneticists have isolated a large number of mutations in fruit flies that influence all aspects of their development, and these mutations have been subjected to molecular analysis, providing much information about how genes control early development in *Drosophila*.

The development of the fruit fly An adult fruit fly possesses three basic body parts: head, thorax, and abdomen (FIGURE 21.3). The thorax consists of three segments: the first thoracic segment carries a pair of legs; the second thoracic segment carries a pair of legs and a pair of wings; and the third thoracic segment carries a pair of legs and the halteres (rudiments of the second pair of wings found in most other insects). The abdomen contains nine segments.

When a *Drosophila* egg has been fertilized, its diploid nucleus (FIGURE 21.4a) immediately divides nine times without division of the cytoplasm, creating a single, multinucleate cell (FIGURE 21.4b). These nuclei are scattered throughout the cytoplasm but later migrate toward the periphery of the embryo and divide several more times (FIGURE 21.4c). Next, the cell membrane grows inward and around each nucleus, creating a layer of approximately 6000 cells at the outer surface of the embryo (FIGURE 21.4d). Four nuclei at one end of the embryo develop into pole cells, which eventually give rise to germ cells. The early embryo then undergoes further development in three distinct stages: (1) the anterior–posterior axis and the dorsal–ventral axis of the embryo are established (FIGURE 21.5a); (2) the number and orientation of the body segments are determined (FIGURE 21.5b); and (3) the identity of each individual



21.3 The fruit fly, *Drosophila melanogaster,* passes through three larval stages and a pupa before developing into an adult fly. The three major body parts of the adult are head, thorax, and abdomen.

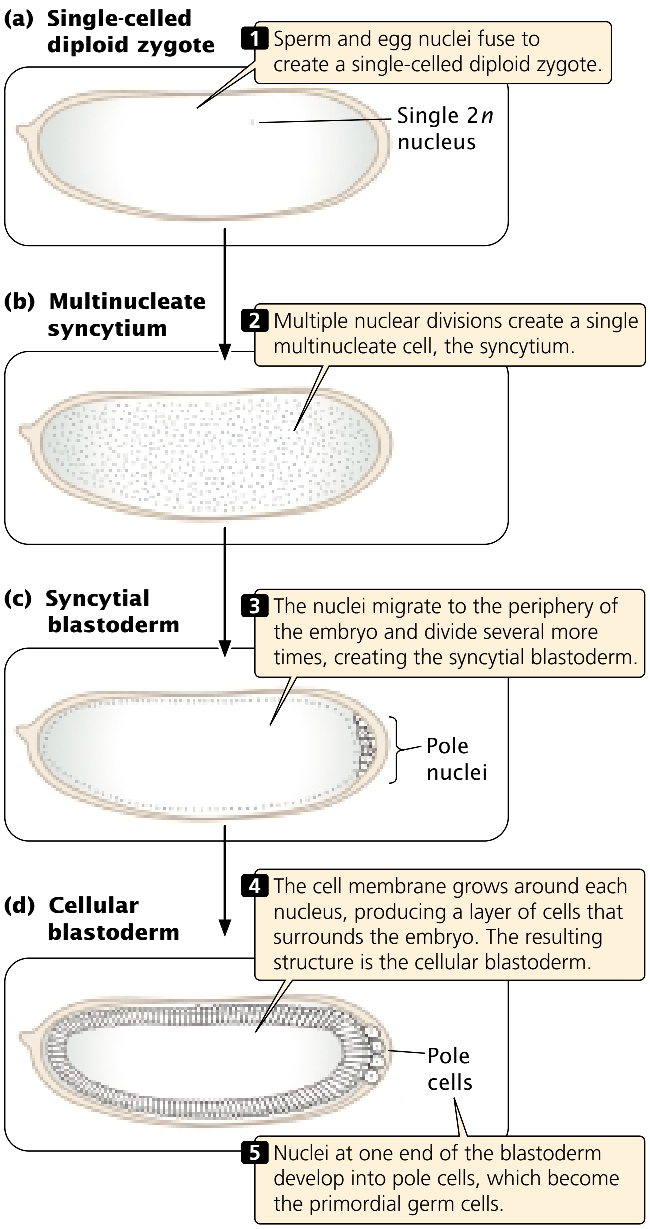
segment is established (FIGURE 21.5c). Different sets of genes control each of these three stages (Table 21.1).

Egg-polarity genes The **egg-polarity genes** play a crucial role in establishing the two main axes of development in fruit flies. You can think of these axes as the longitude and latitude of development: any location in the *Drosophila* embryo can be defined in relation to these two axes. There are two sets of egg-polarity genes: one set determines the anterior–posterior axis and the other determines the dorsal–ventral axis. These genes work by setting up concentration gradients of morphogens within the developing embryo. A **morphogen** is a protein whose concentration gradient affects the developmental fate of the surrounding region.

The egg-polarity genes are transcribed into mRNAs during egg formation in the maternal parent, and these mRNAs become incorporated into the cytoplasm of the egg. After fertilization, the mRNAs are translated into proteins that play an important role in determining the anterior–posterior and dorsal–ventral axes of the embryo. Because the mRNAs of the polarity genes are produced by the female parent and influence the phenotype of their offspring, the traits encoded by them are examples of genetic maternal effects (see p. 000 in Chapter 4).

Egg-polarity genes function by producing proteins that become asymmetrically distributed in the cytoplasm, giving the egg polarity, or direction. This asymmetrical distribution may take place in a couple of ways. The mRNA may be localized to particular regions of the egg cell, leading to an abundance of the protein in those regions when the mRNA is translated. Alternatively, the mRNA may be randomly distributed, but the protein that it encodes may become asymmetrically distributed, either by a transport system that delivers it to particular regions of the cell or by its removal from particular regions by selective degradation.

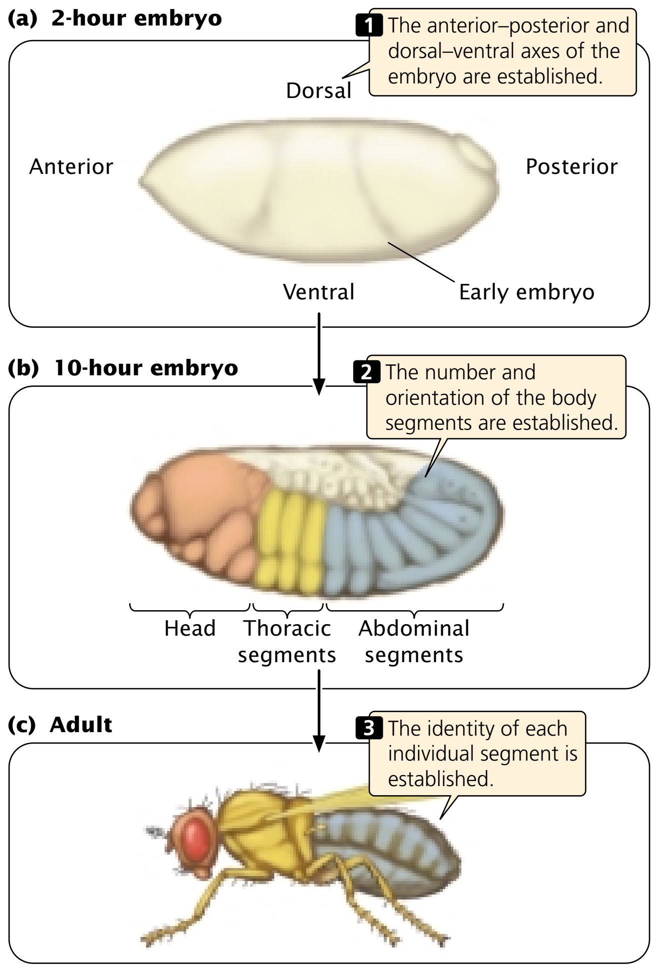
Determination of the dorsal–ventral axis The dorsal–ventral axis defines the back (dorsum) and belly (ventrum) of a fly (see Figure 21.5). At least 12 different genes determine this axis, one of the most important being a gene called *dorsal*. The *dorsal* gene is transcribed and translated in the maternal ovary, and the resulting mRNA and protein are transferred to the egg during oogenesis. In a newly laid egg, mRNA and protein encoded by the *dorsal* gene are uniformly distributed throughout the cytoplasm but, after the nuclei migrate to the periphery of the embryo (see Figure 21.4c), Dorsal protein becomes redistributed. Along one side of the embryo, Dorsal protein remains in the cytoplasm; this side will become the dorsal surface. Along the other side, Dorsal protein is taken up into the nuclei; this side will become the ventral surface. At this point, there is a smooth gradient of increasing nuclear Dorsal concentration from the dorsal to the ventral side (FIGURE 21.6).



21.4 Early development of a *Drosophila* embryo.

The nuclear uptake of Dorsal protein is thought to be governed by a protein called Cactus, which binds to Dorsal protein and traps it in the cytoplasm. The presence of yet another protein, called Toll, can alter Dorsal, allowing it to dissociate from Cactus and move into the nucleus. Together, Cactus and Toll regulate the nuclear distribution of Dorsal protein, which in turn determines the dorsal–ventral axis of the embryo.

Inside the nucleus, Dorsal protein acts as a transcription factor, binding to regulatory sites on the DNA and activating or repressing the expression of other genes (Table 21.2). High nuclear concentration of Dorsal protein (as on the ventral side of the embryo) activates a gene called *twist,* which causes mesoderm to develop. Low concentrations of

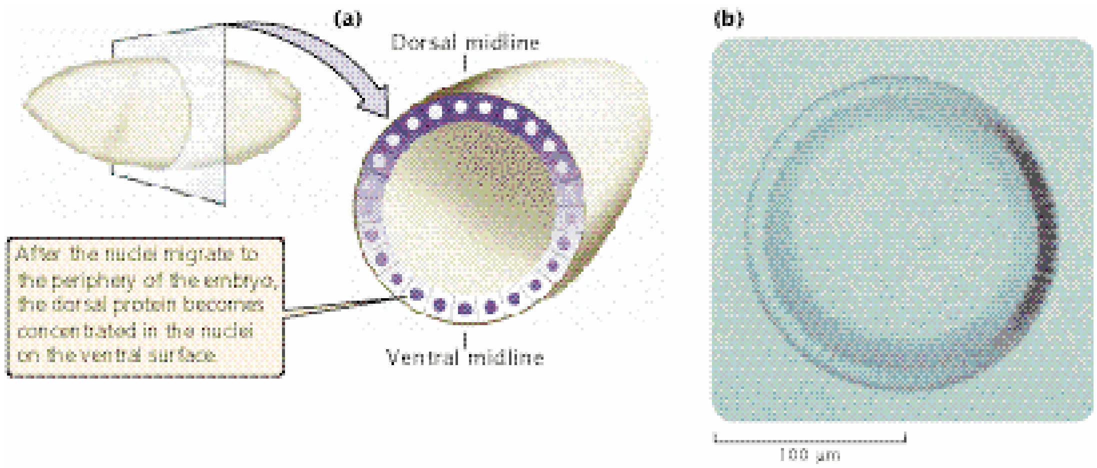


21.5 In an early *Drosophila* embryo, the major body axes are established, the number and orientation of the body segments are determined, and the identity of each individual segment is established. Different sets of genes control each of these three stages.

Dorsal protein (as in cells on the dorsal side of the embryo), activates a gene called *decapentaplegic,* which specifies dorsal structures. In this way, the ventral and dorsal sides of the embryo are determined.

Table 21.1 Stages in the early development of fruit flies and the genes that control each stage

Developmental Stage	Genes
Establishment of main body axes	Egg polarity genes
Determination of number and polarity of body segments	Segmentation genes
Establishment of identity of each segment	Homeotic genes



21.6 **Dorsal protein in the nuclei helps to determine the dorsal–ventral axis of the *Drosophila* embryo.** (a) Relative concentrations of Dorsal protein in the cytoplasm and nuclei of cells in the early *Drosophila* embryo. (b) Micrograph of a cross section of the embryo showing the Dorsal protein, darkly stained, in the nuclei along the ventral surface.

Determination of the anterior–posterior axis Establishing the anterior–posterior axis of the embryo is a crucial step in early development. We will consider several genes in this pathway (Table 21.3). One important gene is *bicoid,* which is first transcribed in the ovary of an adult female during oogenesis. *Bicoid* mRNA becomes incorporated into the cytoplasm of the egg and, as it is passes into the egg, *bicoid* mRNA becomes anchored to the anterior end of the egg by part of its 3′ end. This anchoring causes *bicoid* mRNA to become concentrated at the anterior end (FIGURE 21.7a). (A number of other genes that are active in the ovary are required for proper localization of *bicoid* mRNA in the egg.) When the egg has been laid, *bicoid* mRNA is translated into Bicoid protein. Because most of the mRNA is at the anterior end of the egg, Bicoid protein is synthesized there and forms a concentration gradient along the anterior–posterior axis of the embryo, with a high concentration at the anterior end and a low concentration at posterior end. This gradient is maintained by the continuous synthesis of Bicoid protein and its short half-life.

The high concentration of Bicoid protein at the anterior end induces the development of anterior structures such as the head of the fruit fly. Bicoid—like Dorsal—is a morphogen. It stimulates the development of anterior structures by binding to regulatory sequences in the DNA and influencing the expression of other genes. One of the most important of the genes stimulated by Bicoid protein is *hunchback,* which is required for the development of the head and thoracic structures of the fruit fly.

The development of the anterior–posterior axis is also greatly influenced by a gene called *nanos,* an egg-polarity

Table 21.2 Key genes that control development of the dorsal–ventral axis in fruit flies and their action

Gene	Where Expressed	Action of Gene Product
dorsal	Ovary	Affects expression of genes such as *twist* and *decapentaplegic*
cactus	Ovary	Traps Dorsal protein in cytoplasm
toll	Ovary	Alters Dorsal protein, allowing it to dissociate from Cactus protein and move into nuclei of ventral cells
twist	Embryo	Takes part in development of mesodermal tissues
decapentaplegic	Embryo	Takes part in development of gut structures

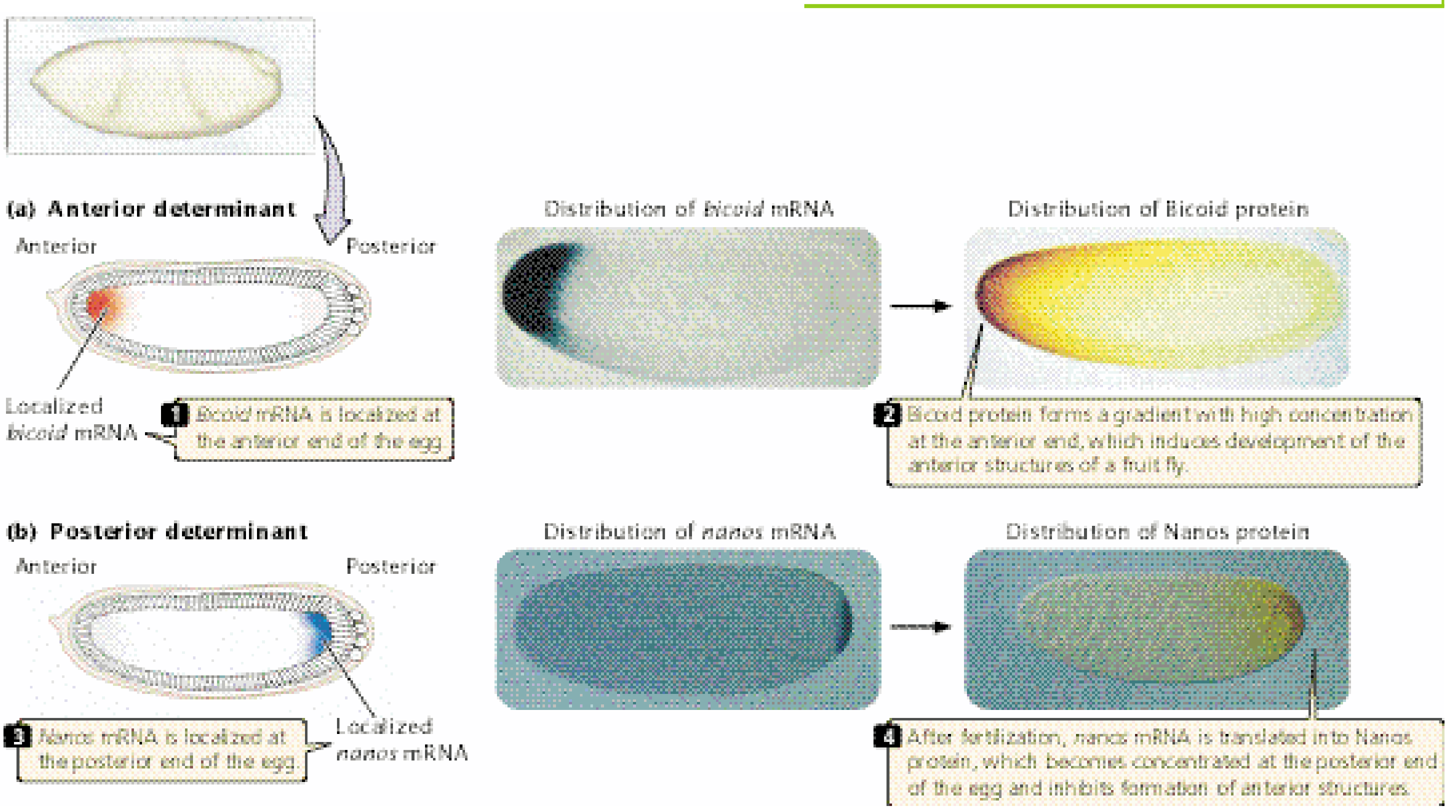
Table 21.3 Some key genes that determine the anterior–posterior axis in fruit flies

Gene	Where Expressed	Action
bicoid	Ovary	Regulates expression of genes responsible for anterior structures; stimulates *hunchback*
nanos	Ovary	Regulates expression of genes responsible for posterior structures; inhibits translation of *hunchback* mRNA
hunchback	Embryo	Regulates transcription of genes responsible for anterior structures

gene that acts at the posterior end of the axis. The *nanos* gene is transcribed in the adult female, and the resulting mRNA becomes localized at the posterior end of the egg (FIGURE 21.7b). After fertilization, *nanos* mRNA is translated into Nanos protein, which diffuses slowly toward the anterior end. The Nanos protein gradient is opposite that of Bicoid protein: Nanos is most concentrated at the posterior end of the embryo and is least concentrated at the anterior end. Nanos protein inhibits the formation of anterior structures by repressing the translation of *hunchback* mRNA. The synthesis of the Hunchback protein is therefore stimulated at the anterior end of the embryo by Bicoid protein and is repressed at the posterior end by Nanos protein. This combined stimulation and repression results in a Hunchback protein concentration gradient along the anterior–posterior axis that, in turn, affects the expression of other genes and helps determine the anterior and posterior structures.

Concepts

The major axes of development in early fruit-fly embryos are established as a result of initial differences in the distribution of specific mRNAs and proteins encoded by genes in the female parent (genetic maternal effect). These differences in distribution establish concentration gradients of morphogens, which cause different genes to be activated in different parts of the embryo.



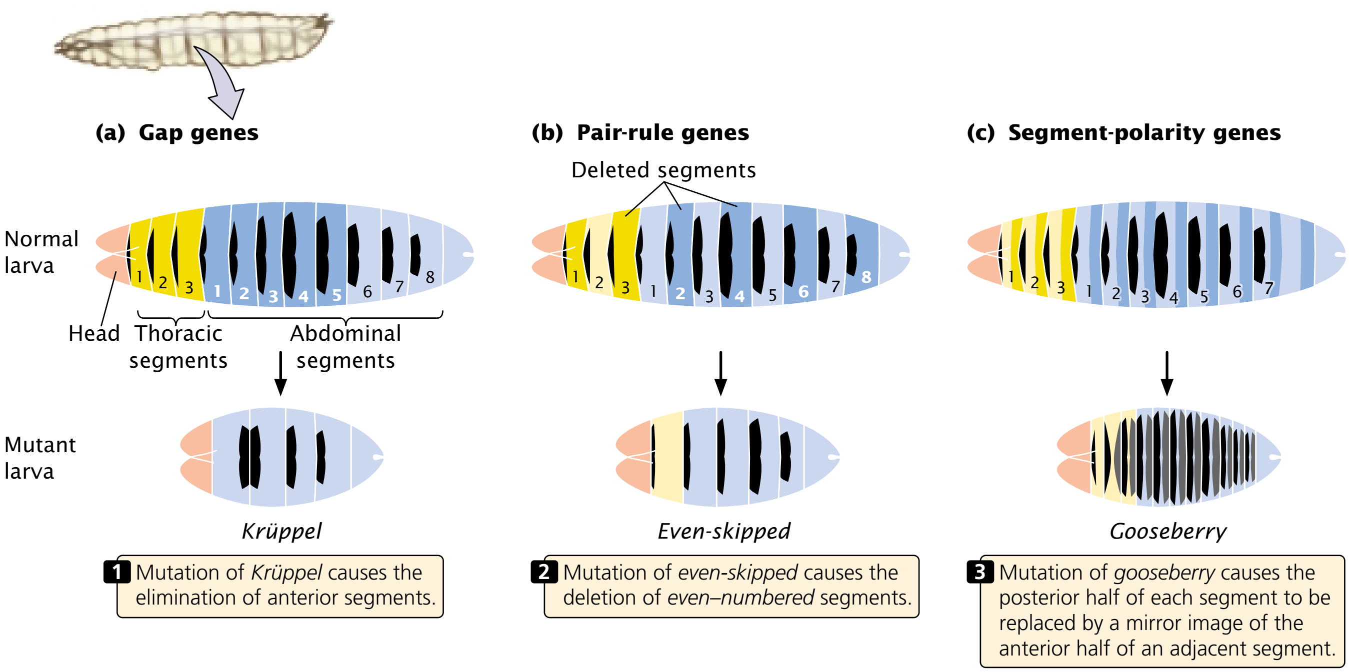
21.7 The anterior–posterior axis in a *Drosophila* embryo is determined by concentrations of Bicoid and Nanos proteins.

Table 21.4 Segmentation genes and the effects of mutations in them

Class of Gene	Effect of Mutations	Examples of Genes
Gap genes	Delete adjacent segments	*hunchback, Krüppel, knirps, giant, tailless*
Pair-rule genes	Delete same part of pattern in every other segment	*runt, hairy, fushi tarazu, even paired, odd paired, skipped, sloppy, paired, odd skipped*
Segment-polarity genes	Affect polarity of segment; part of segment replaced by mirror image of part of another segment	*engrailed, wingless, gooseberry, cubitus interruptus, patched, hedgehog, disheveled, costal-2, fused*

Segmentation genes Like all insects, the fruit fly has a segmented body plan. When the basic dorsal–ventral and anterior–posterior axes of the fruit-fly embryo have been established, **segmentation genes** control the differentiation of the embryo into individual segments. These genes affect the number and organization of the segments, and mutations in them usually disrupt whole sets of segments. The approximately 25 segmentation genes in *Drosophila* are transcribed after fertilization; so they don't exhibit a genetic maternal effect, and their expression is regulated by the Bicoid and Nanos protein gradients.

The segmentation genes fall into three groups as shown in Table 21.4 and FIGURE 21.8. **Gap genes** define large sections of the embryo; mutations in these genes eliminate whole groups of adjacent segments. Mutations in the *Krüppel* gene, for example, cause the absence of several adjacent segments. **Pair-rule genes** define regional sections of the embryo and affect alternate segments. Mutations in the *even-skipped* gene cause the deletion of even-numbered segments, whereas mutations in the *fushi tarazu* gene cause the absence of odd-numbered segments. **Segment-polarity genes** affect the organization of segments. Mutations in



21.8 Segmentation genes control the differentiation of the *Drosophila* embryo into individual segments. The gap genes affect large sections of the embryo. The pair-rule genes affect alternate segments. The segment-polarity genes affect the polarity of segments.

these genes cause part of each segment to be deleted and replaced by a mirror image of part or all of an adjacent segment. For example, mutations in the *gooseberry* gene cause the posterior half of each segment to be replaced by the anterior half of an adjacent segment.

The gap genes, pair-rule genes, and segment-polarity genes act sequentially, affecting progressively smaller regions of the embryo. First, the egg-polarity genes activate or repress the gap genes, which divide the embryo into broad regions. The gap genes, in turn, regulate the pair-rule genes, which affect the development of pairs of segments. Finally, the pair-rule genes influence the segment-polarity genes, which guide the development of individual segments.

Concepts

When the major axes of the fruit-fly embryo have been established, segmentation genes determine the number, orientation, and basic organization of the body segments.

Homeotic genes After the segmentation genes have established the number and orientation of the segments, **homeotic genes** become active and determine the *identity* of individual segments. Eyes normally arise only on the head segment, whereas legs develop only on the thoracic segments. The products of homeotic genes activate other genes that encode these segment-specific characteristics. Mutations in the homeotic genes cause body parts to appear in the wrong segments.

Homeotic mutations were first identified in 1894, when William Bateson noticed that floral parts of plants occasionally appeared in the wrong place: he found, for example, flowers in which stamens grew in the normal place of petals. In the late 1940s, Edward Lewis began to study homeotic mutations in *Drosophila,* which caused bizarre rearrangements of body parts. Mutations in the *Antennapedia* gene, for example, cause legs to develop on the head of a fly in place of the antenna (FIGURE 21.9).

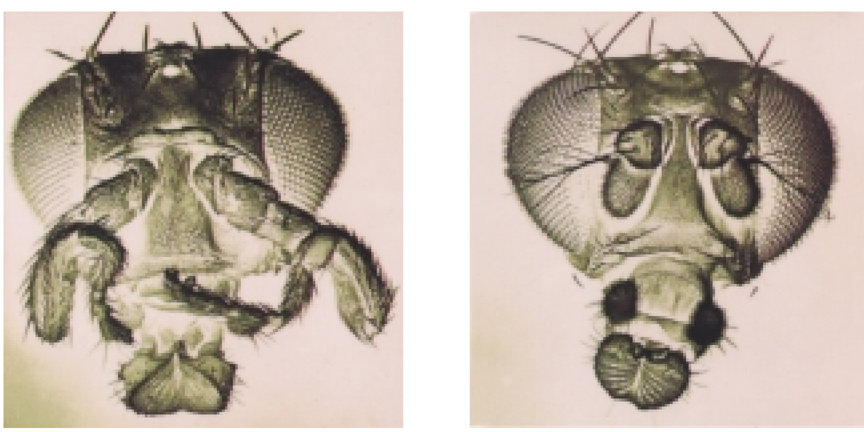
Homeotic genes create addresses for the cells of particular segments, telling the cells where they are within the regions defined by the segmentation genes. When a homeotic gene is mutated, the address is wrong and cells in the segment develop as though they were somewhere else in the embryo.

Homeotic genes are expressed after fertilization and are activated by specific concentrations of the proteins produced by the gap, pair-rule, and segment-polarity genes. The homeotic gene *Ultrabithorax (Ubx),* for example, is activated when the concentration of Hunchback protein (a product of a gap gene) is within certain values. These concentrations exist only in the middle region of the embryo; so *Ubx* is expressed only in these segments.

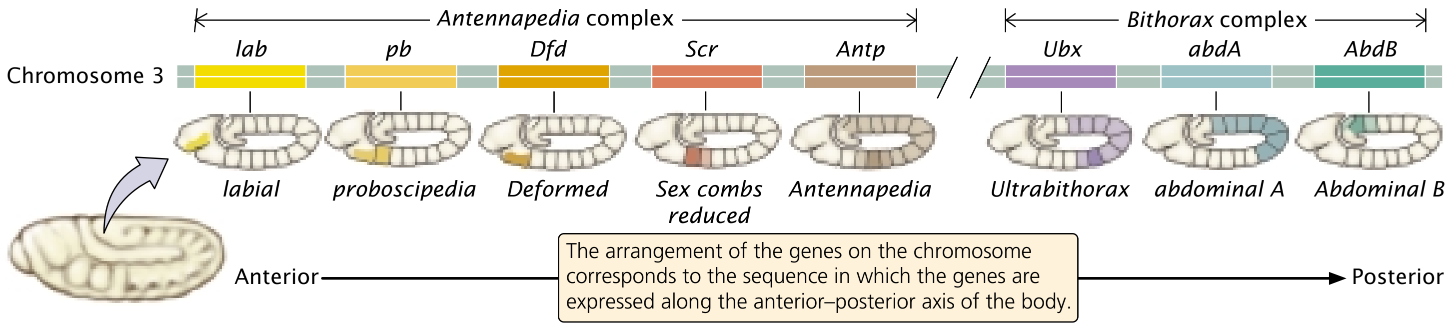
The homeotic genes encode regulatory proteins that bind to DNA; each gene contains a subset of nucleotides, called a **homeobox,** that are similar in all homeotic genes. The homeobox consists of 180 nucleotides and encodes 60 amino acids that serve as a DNA-binding domain; this domain is related to the helix-turn-helix motif (See Figure 16.2a). Homeoboxes are also present in segmentation genes and other genes that play a role in spatial development.

There are two major clusters of homeotic genes in *Drosophila*. One cluster, the ***Antennapedia* complex,** affects the development of the adult fly's head and anterior thoracic segments. The other cluster consists of the ***bithorax* complex** and includes genes that influence the adult fly's posterior thoracic and abdominal segments. Together, the *bithorax* and *Antennapedia* genes are termed the **homeotic complex** (HOM-C). In *Drosophila,* the *bithorax* complex contains three

(a) **(b)**



21.9 The homeotic mutation *Antennapedia* substitutes legs for the antenna of a fruit fly. (a) Normal, wild-type antenna. (b) *Antennapedia* mutant.



21.10 Homeotic genes, which determine the identity of individual segments in *Drosophila*, are present in two complexes. The *Antennapedia* complex has five genes, and the *bithorax* complex has three genes.

genes, and the *Antennapedia* complex has five; they are all located on the same chromosome (FIGURE 21.10). In addition to these eight genes, HOM-C contains many sequences that regulate the homeotic genes.

Remarkably, the order of the genes in the HOM-C is the same as the order in which the genes are expressed along the anterior–posterior axis of the body. The genes that are expressed in the more anterior segments are found at the one end of the complex, whereas those expressed in the more posterior end of the embryo are found at the other end of complex (See Figure 21.10). The reason for this correlation is unknown.

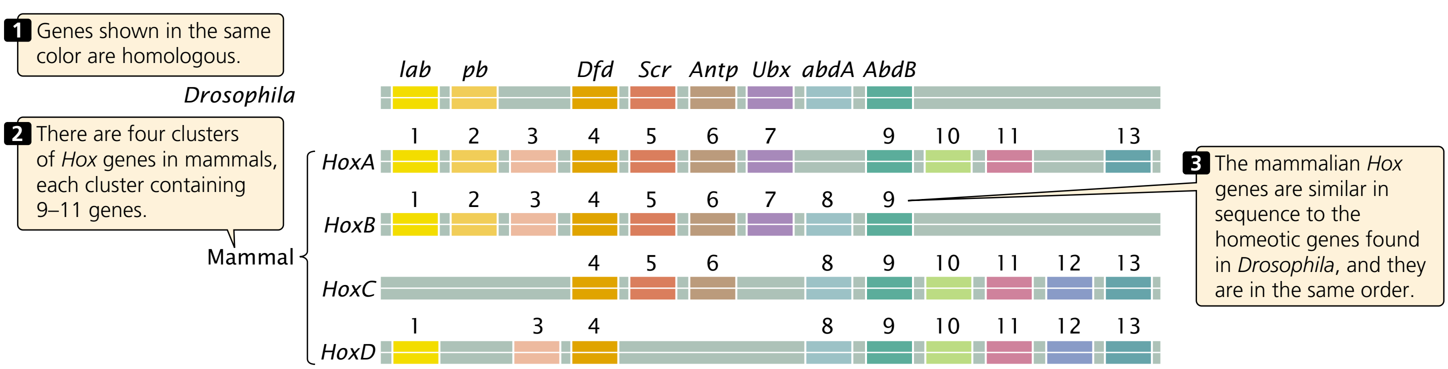
Concepts

Homeotic genes help determine the identity of individual segments in *Drosophila* embryos by producing DNA-binding proteins that activate other genes. Each homeotic gene contains a consensus sequence called a homeobox, which encodes the DNA-binding domain.

Homeobox Genes in Other Organisms

After homeotic genes in *Drosophila* had been isolated and cloned, molecular geneticists set out to determine if similar genes exist in other animals; probes complementary to the homeobox of *Drosophila* genes were used to search for homologous genes that might play a role in the development of other animals. The search was hugely successful: homeobox-containing (***Hox***) **genes** have been found in all animals studied so far, including nematodes, beetles, sea urchins, frogs, birds, and mammals. They have even been discovered in fungi and plants, indicating that *Hox* genes arose early in the evolution of eukaryotes.

In vertebrates, there are four clusters of *Hox* genes, each of which contains from 9 to 11 genes. Interestingly, the *Hox* genes of other organisms exhibit the same relation between order on the chromosome and order of their expression along the anterior–posterior axis of the embryo as that of *Drosophila* (FIGURE 21.11). Mammalian *Hox* genes, like those in *Drosophila,* encode transcription factors that help determine the identity of body regions along an anterior–posterior axis.



21.11 Homeotic genes in mammals are similar to those found in *Drosophila.* The complexes are arranged so that genes with similar sequences lie in the same column. See Figure 21.10 for the full names of the *Drosophila* genes.

Concepts

Homeobox-containing genes are found in many organisms, in which they regulate development.

www.whfreeman.com/pierce More information about *Hox* genes

Connecting Concepts

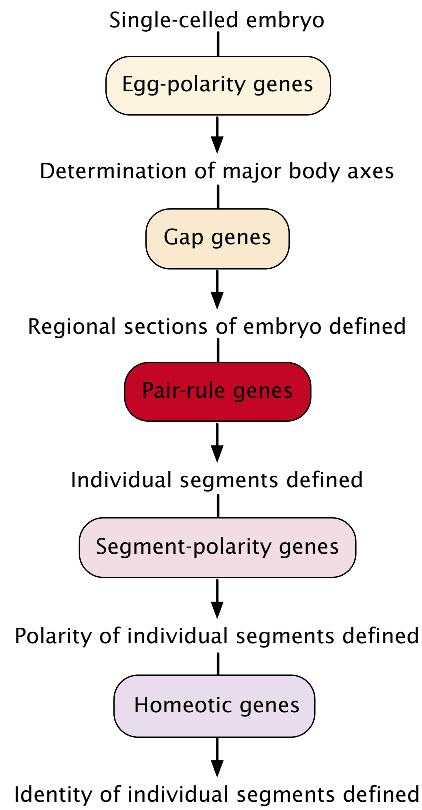
The Control of Development

Development is a complex process consisting of numerous events that must take place in a highly specific sequence. The results of studies in fruit flies and other organisms reveal that this process is regulated by a large number of genes. In *Drosophila,* the dorsal–ventral axis and the anterior–posterior axis are established by maternal genes; these genes encode mRNAs and proteins that are localized to specific regions within the egg and cause specific genes to be expressed in different regions of the embryo. The proteins of these genes then stimulate other genes, which in turn stimulate yet other genes in a cascade of control. As might be expected, most of the gene products in the cascade are regulatory proteins, which bind to DNA and activate other genes.

In the course of development, successively smaller regions of the embryo are determined (FIGURE 21.12). In *Drosophila,* first, the major axes and regions of the embryo are established by egg polarity genes. Next, patterns within each region are determined by the action of segmentation genes: the gap genes define large sections; the pair-rule genes define regional sections of the embryo and affect alternate segments; and the segment-polarity genes affect individual segments. Finally, the homeotic genes provide each segment with a unique identity. Initial gradients in proteins and mRNA stimulate localized gene expression, which produces more finely located gradients that stimulate even more localized gene expression. Developmental regulation thus becomes more and more narrowly defined.

The processes by which limbs, organs, and tissues form (called morphogenesis) are less well understood, although this pattern of generalized-to-localized gene expression is encountered frequently.

www.whfreeman.com/pierce *Drosophila* development, images of fruit-fly anatomy and development, images of mammalian embryos, and many resources on development biology



21.12 A cascade of gene regulation establishes the polarity and identity of individual segments of *Drosophila.* In development, successively smaller regions of the embryo are determined.

Programmed Cell Death in Development

Cell death is an integral part of multicellular life. Cells in many tissues have a limited life span, and they die and are replaced continually by new cells. Cell death shapes many body parts during development: it is responsible for the disappearance of a tadpole's tail during metamorphosis and causes the removal of tissue between the digits to produce the human hand. Cell death is also used to eliminate dangerous cells that have escaped normal controls (see next section on cancer).

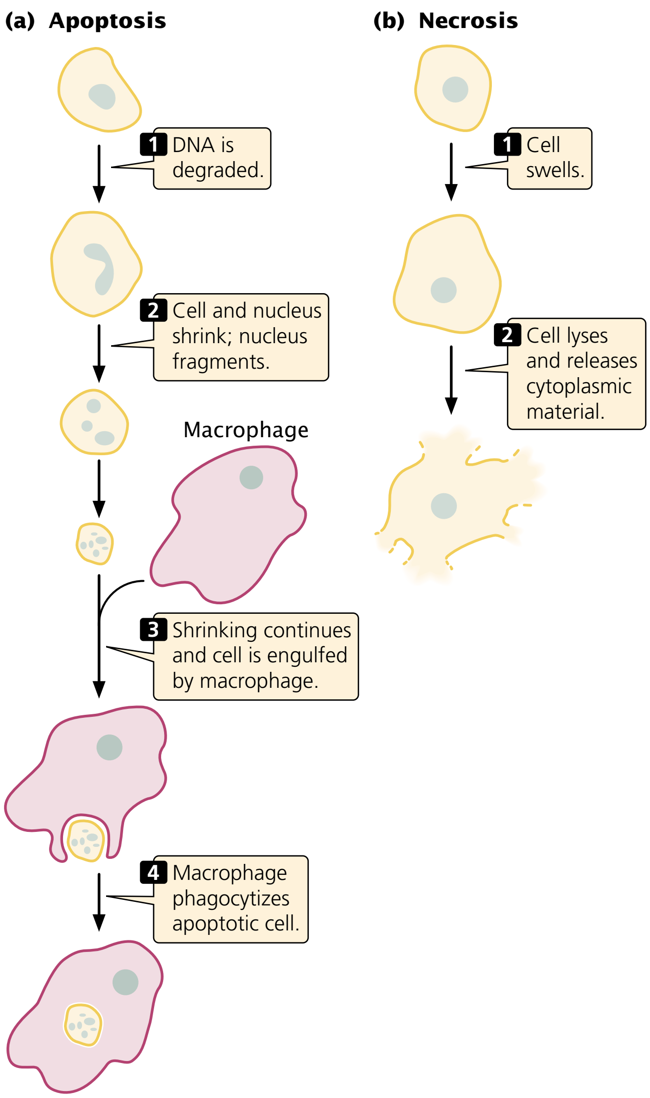
Cell death in animals is often initiated by the cell itself in a kind of cellular suicide termed **apoptosis.** In this process, a cell's DNA is degraded, its nucleus and cytoplasm shrink, and the cell undergoes phagocytosis by other cells without any leakage of its contents (FIGURE 21.13a). Cells that are injured, on the other hand, die in a relatively uncontrolled manner called necrosis. In this process, a cell swells and bursts, spilling its contents over neighboring cells and eliciting an inflammatory response (FIGURE 21.13b). Apoptosis is essential to embryogenesis; most multicellular animals cannot complete development if the process is inhibited.

Surprisingly, most cells are programmed to undergo apoptosis and will survive only if the internal death program is continually held in check. The process of apoptosis is highly regulated and depends on numerous signals inside and outside the cell. Geneticists have identified a number of genes having roles in various stages of the regulation of apoptosis. Some of these genes encode enzymes called **caspases,** which cleave other proteins at specific sites (after aspartic acid). Each caspase is synthesized as a large, inactive precursor (a procaspase) that is activated by cleavage, often by another caspase. When one caspase is activated, it cleaves other procaspases that trigger even more caspase activity. The resulting cascade of caspase activity eventually cleaves proteins essential to cell function, such as those supporting the nuclear membrane and cytoskeleton. Caspases also cleave a protein that normally keeps an enzyme that degrades DNA (DNAse) in an inactive form. Cleavage of this protein activates DNAse and leads to the breakdown of cellular DNA, which eventually leads to cell death.

Procaspases and other proteins required for cell death are continuously produced by healthy cells, so the potential for cell suicide is always present. A number of different signals can trigger apoptosis; for instance, infection by a virus can activate immune cells to secrete substances onto an infected cell, causing that cell to undergo apoptosis. This process is believed to be a defense mechanism designed to prevent the reproduction and spread of viruses. Similarly, DNA damage can induce apoptosis and thus prevent the replication of mutated sequences. Damage to mitochondria and the accumulation of a misfolded protein in the endoplasmic reticulum also stimulate programmed cell death.

Apoptosis in animal development is still poorly understood but is believed to be controlled through cell–cell signaling. The cell death that causes the disappearance of a tadpole's tail, for example, is triggered by thyroxin, a hormone produced by the thyroid gland that increases in concentration during metamorphosis. The elimination of cells between developing fingers in humans is thought to result from localized signals from nearby cells.

The symptoms of many diseases and disorders are caused by apoptosis or, in some cases, its absence. In neurodegenerative diseases such as Parkinson disease and Alzheimer disease, symptoms are caused by a loss of neurons through apoptosis. In heart attacks and stroke, some cells die through necrosis, but many others undergo apoptosis. Cancer is often stimulated by mutations in genes that regulate apoptosis, leading to a failure of apoptosis that would normally eliminate cancer cells.



21.13 Programmed cell death by apoptosis is distinct from uncontrolled cell death through necrosis.

Concepts

Cells are capable of apoptosis (programmed cell death), a highly regulated process that depends on enzymes called caspases. Apoptosis plays an important role in animal development and is implicated in a number of diseases.

www.whfreeman.com/pierce Additional information on apoptosis

Evo-Devo: The Study of Evolution and Development

"Ontogeny recapitulates phylogeny" is a familiar phrase that was coined in the 1860s by German zoologist Ernst Haeckel to describe his belief—now considered wrong—that organisms repeat their evolutionary history during development. According to Haeckel's theory, a human embryo passes through fish, amphibian, reptilian, and mammalian stages before developing human traits.

Although ontogeny does not recapitulate phylogeny, many evolutionary biologists today are turning to the study of development for a better understanding of the processes and patterns of evolution. Sometimes called "evo-devo," the study of evolution through the analysis of development is revealing that the same genes often shape developmental pathways in distantly related organisms. In humans and insects, for example, the same gene controls the development of eyes, despite the fact that insect and mammalian eyes are thought to have evolved independently. Similarly, biologists once thought that segmentation in vertebrates and invertebrates was only superficially similar, but we now know that, in both *Drosophila* and amphioxus (a marine organism closely related to vertebrates), a gene called *engrailed* divides the embryo into specific segments. A gene called *distalless,* which creates the legs of a fruit fly, has also been found to also play a role in the development of crustacean branched appendages. This same gene also stimulates body outgrowths of many other organisms, from polycheate worms to starfish.

Similar genes may be part of a developmental pathway common to two different species but have quite different effects. For example, a *Hox* gene called *AbdB* helps define the posterior end of a *Drosophila* embryo; a similar group of genes in birds divides the wing into three segments. In another example, the *sog* gene in fruit flies stimulates cells to assume a ventral orientation in the embryo, but the expression of a similar gene called *chordin* in vertebrates causes cells to assume dorsal orientation, exactly the opposite of the situation in fruit flies.

The theme emerging from these studies is that a small, common set of genes may underlie many basic developmental processes in many different organisms. Although Haeckel's euphonious phrase "ontogeny recapitulates phylogeny" was incorrect, evo-devo is proving that development can reveal much about the process of evolution.

Immunogenetics

A basic assumption of developmental biology is that every somatic cell carries an identical set of genetic information and that no genes are lost during development. Although this assumption holds for most cells, there are some important exceptions, one of which concerns genes that encode immune function in vertebrates.

The immune system provides protection against infection by specific bacteria, viruses, fungi, and parasites. The focus of an immune response is an **antigen,** defined as any molecule that elicits an immune reaction. Although any molecule can be an antigen, most are proteins. The immune system is remarkable in its ability to recognize an almost unlimited number of potential antigens.

The body is full of proteins, so it is essential that the immune system be able to distinguish between self-antigens and foreign antigens. Occasionally, the ability to make this distinction breaks down, and the body produces an immune reaction to its own antigens, resulting in an **autoimmune disease** (Table 21.5).

The Organization of the Immune System

The immune system contains a number of different components and uses several mechanisms to provide protection against pathogens, but most immune responses can be grouped into two major classes: humoral immunity and cellular immunity. Although it is convenient to think of these classes as separate systems, they interact and influence each other significantly.

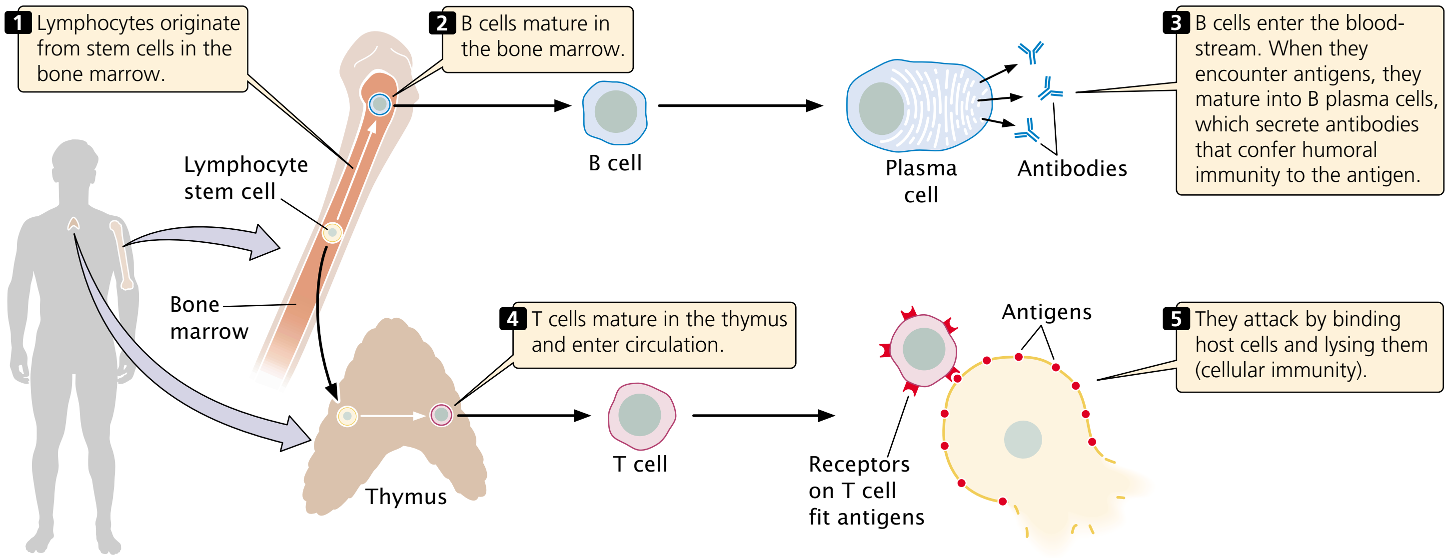
Humoral immunity centers on the production of antibodies by special lymphocytes called **B cells** (FIGURE 21.14), which mature in the bone marrow. **Antibodies** are proteins that circulate in the blood and other body fluids, binding to specific antigens and marking them for destruction by phagocytic cells. Antibodies also activate a set of proteins called complement that help to lyse cells and attract macrophages.

Cellular immunity is conferred by **T cells** (see Figure 21.14), which are specialized lymphocytes that mature in the thymus and respond only to antigens found on the surfaces of the body's own cells. After a pathogen such as a virus has infected a host cell, some viral antigens appear on the cell surface. Proteins, called **T-cell receptors,** on the surfaces of T cells bind to these antigens and mark the infected cell for destruction. T-cell receptors must simultaneously bind a foreign antigen and a self-antigen called a **major histocompatibility complex** (MHC) **antigen** on the cell surface. Not all T cells attack cells having foreign antigens; some help regulate immune responses, providing communication among different components of the immune system.

How can the immune system recognize an almost unlimited number of foreign antigens? Remarkably, each mature lymphocyte is genetically programmed to attack

Table 21.5 Examples of autoimmune diseases

Disease	Tissues Attacked
Graves disease, Hashimoto thyroiditis	Thyroid gland
Rheumatic fever	Heart muscle
Systematic lupus erythematosus	Joints, skin, and other organs
Rheumatoid arthritis	Joints
Insulin-dependent diabetes mellitus	Insulin-producing cells in pancreas
Multiple sclerosis	Myelin sheath around nerve cells



21.14 Immune responses are divided into humoral immunity, in which antibodies are produced by B cells, and cellular immunity produced by T cells.

one and only one specific antigen: each mature B cell produces antibodies against a single antigen, and each T cell is capable of attaching to only one type of foreign antigen.

If each lymphocyte is specific for only one type of antigen, how does an immune response develop? The **theory of clonal selection** proposes that initially there is a large pool of millions of different lymphocytes, each capable of binding only one antigen (FIGURE 21.15); so millions of different foreign antigens can be detected. To illustrate clonal selection, let's imagine that a foreign protein enters the body. Only a few lymphocytes in the pool will be specific for this particular foreign antigen. When one of these lymphocytes encounters the foreign antigen and binds to it, that lymphocyte is stimulated to divide. The lymphocyte proliferates rapidly, producing a large population of genetically identical cells—a clone—each of which is specific for that particular antigen.

This initial proliferation of antigen-specific B and T cells is known as a **primary immune response** (see Figure 21.15); in most cases, the primary response destroys the foreign antigen. Subsequent to the primary immune response, most of the lymphocytes in the clone die, but a few continue to circulate in the body. These **memory cells** may remain in circulation for years or even for the rest of one's life. Should the same antigen reappear at some time in the future, memory cells specific to that antigen become activated and quickly give rise to another clone of cells capable of binding the antigen. The rise of this second clone is termed a **secondary immune response** (see Figure 21.15). The ability to quickly produce a second clone of antigen-specific cells permits the long-lasting immunity that often follows recovery from a disease. For example, people who have chicken pox usually have life-long immunity to the disease. The secondary immune response is also the basis for vaccination, which stimulates a primary immune response to an antigen and results in memory cells that can quickly produce a secondary response if that same antigen appears in the future.

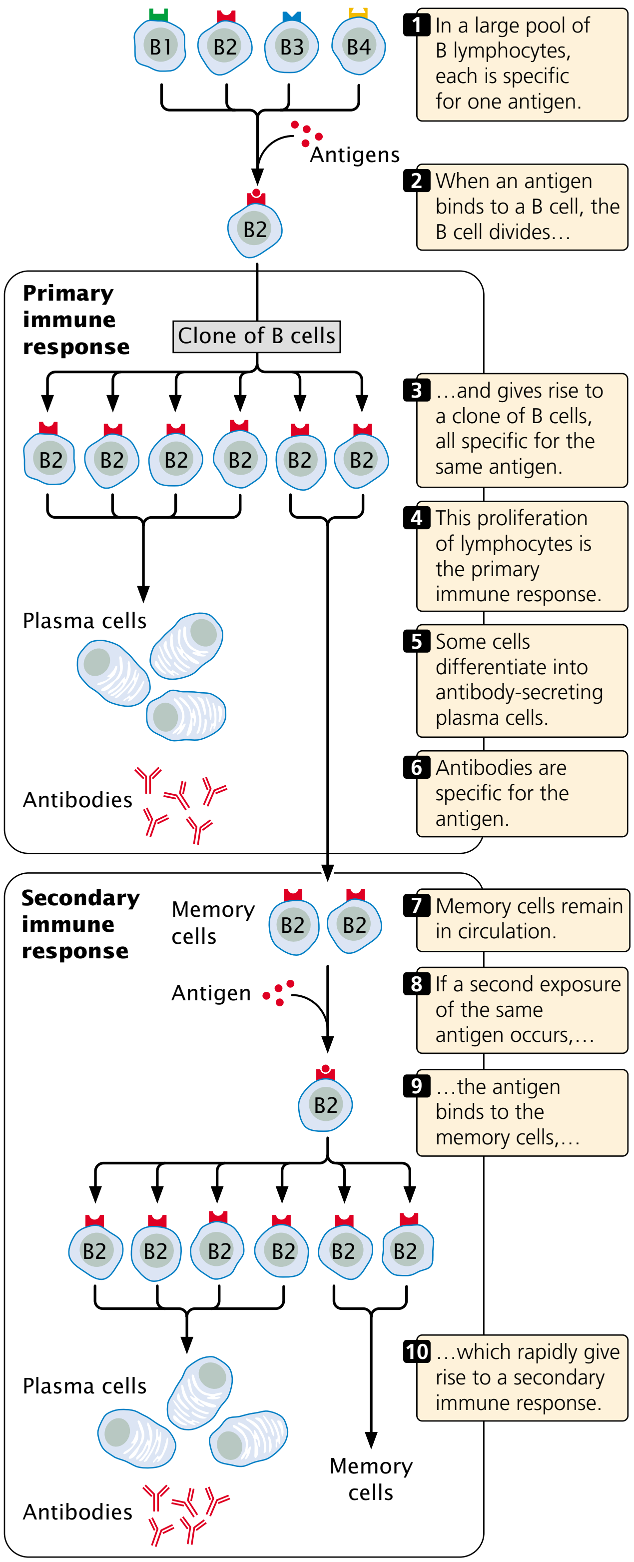
Three sets of proteins are required for immune responses: antibodies, T-cell receptors, and the major histocompatibility antigens. The next section explores how the enormous diversity in these proteins is generated.

Concepts

Each B cell and T cell of the immune system is genetically capable of binding one type of foreign antigen. When a lymphocyte binds to an antigen, the lymphocyte undergoes repeated division, giving rise to a clone of genetically identical lymphocytes (the primary response), all of which are specific for that same antigen. Memory cells remain in circulation for long periods of time; if the antigen reappears, the memory cells undergo rapid proliferation and generate a secondary immune response.

Immunoglobulin Structure

The principal products of the humoral immune response are antibodies—also called immunoglobulins. Each immunoglobulin (Ig) molecule consists of four polypeptide chains—two identical light chains and two identical heavy chains—which form a Y-shaped structure (FIGURE 21.16). Disulfide bonds link the two heavy chains in the stem of the Y

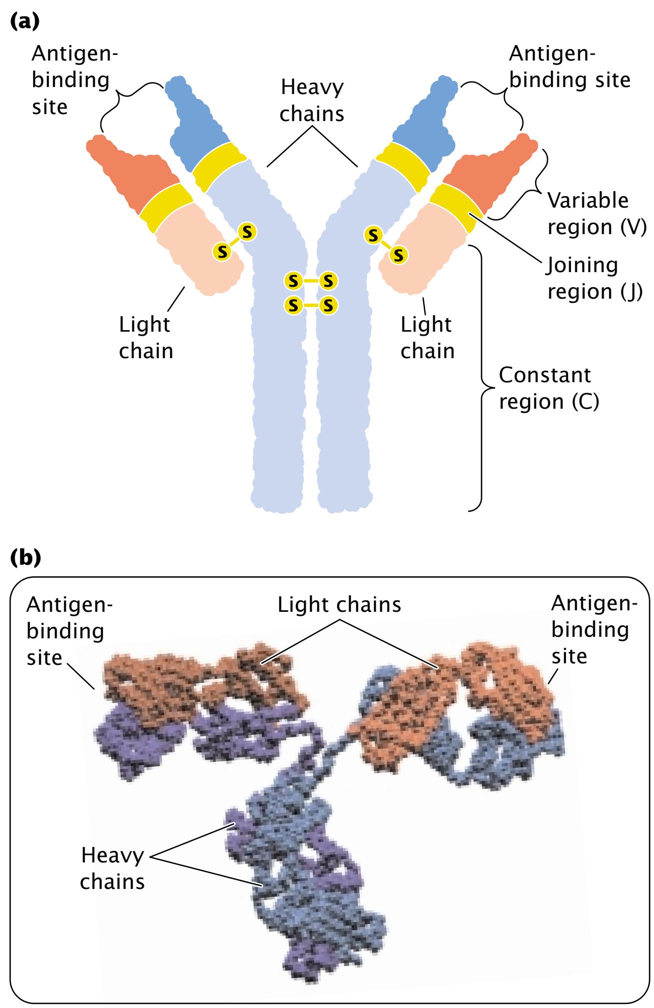


21.15 An immune response to a specific antigen is produced through clonal selection.

and attach a light chain to a heavy chain in each arm of the Y. Binding sites for antigens are at the ends of the two arms.

The light chains of an immunoglobulin come in two basic types, called kappa chains and lambda chains. An immunoglobulin molecule can have two kappa chains or two lambda chains, but it cannot have one of each type. Both the light and the heavy chain has a variable region at one end and a constant region at the other end; the variable regions of different immunoglobulin molecules vary in amino acid sequence, whereas the constant regions of different immunoglobulins are similar in sequence. The variable regions of both light and heavy chains make up the antigen-binding region and specify the type of antigen that the antibody can bind.

Mammals have five basic classes of immunoglobulins, known as IgM, IgD, IgE, IgG, and IgA. Each class is defined



21.16 Each immunoglobulin molecule consists of four polypeptide chains—two light chains and two heavy chains—that combine to form a Y-shaped structure. (a) Structure of an immunoglobulin. (b) Folded, space-filling model.

by the type of heavy chain found in the immunoglobulin. The different classes of antibodies have different functions or they appear at different times during an immune response or both. For example, in a primary response, all B cells initially make IgM but, as the immune response develops, they switch to producing a combination of IgM and IgD. Later, the B cells may switch to one of the other immunoglobulin classes.

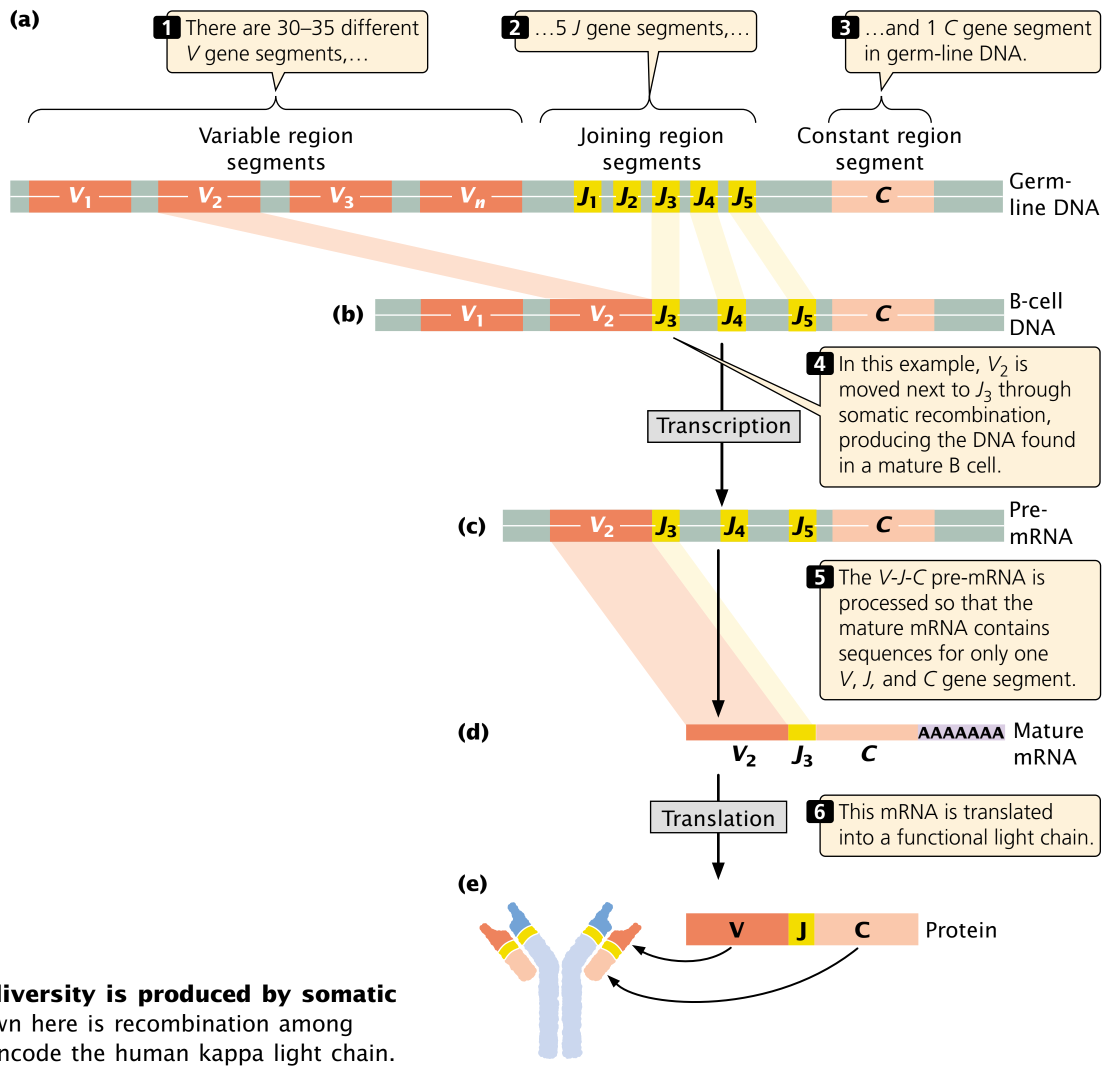
The Generation of Antibody Diversity

The immune system is capable of making antibodies against virtually any antigen that might be encountered in one's lifetime: each human is capable of making about 10^{15} different antibody molecules. Antibodies are proteins; so the amino acid sequences of all 10^{15} potential antibodies must be encoded in the human genome. However, there are fewer than 1×10^5 genes in the human genome and, in fact, only 3×10^9 total base pairs; so how can this huge diversity of antibodies be encoded?

The answer lies in the fact that antibody genes are composed of segments. There are a number of copies of each type of segment, each differing slightly from the others. In the maturation of a lymphocyte, the segments are joined to create an immunoglobulin gene. The particular copy of each segment used is random and, because there are multiple copies of each type, there are many possible combinations of the segments. A limited number of segments can therefore encode a huge diversity of antibodies.

To illustrate this process of antibody assembly, let's consider the immunoglobulin light chains. Kappa and lambda chains are encoded by separate genes on different chromosomes. Each gene is composed of three types of segments: *V*, for variable; *J*, for joining; and *C*, for constant. The *V* segments encode most of the variable region of the light chains, the *C* segment encodes the constant region of the chain, and the *J* segments encode a short set of nucleotides that join the *V* segment and the *C* segments together.

The number of *V*, *J*, and *C* segments differs among species. For the human kappa gene, there are from 30 to 35



21.17 Antibody diversity is produced by somatic recombination. Shown here is recombination among gene segments that encode the human kappa light chain.

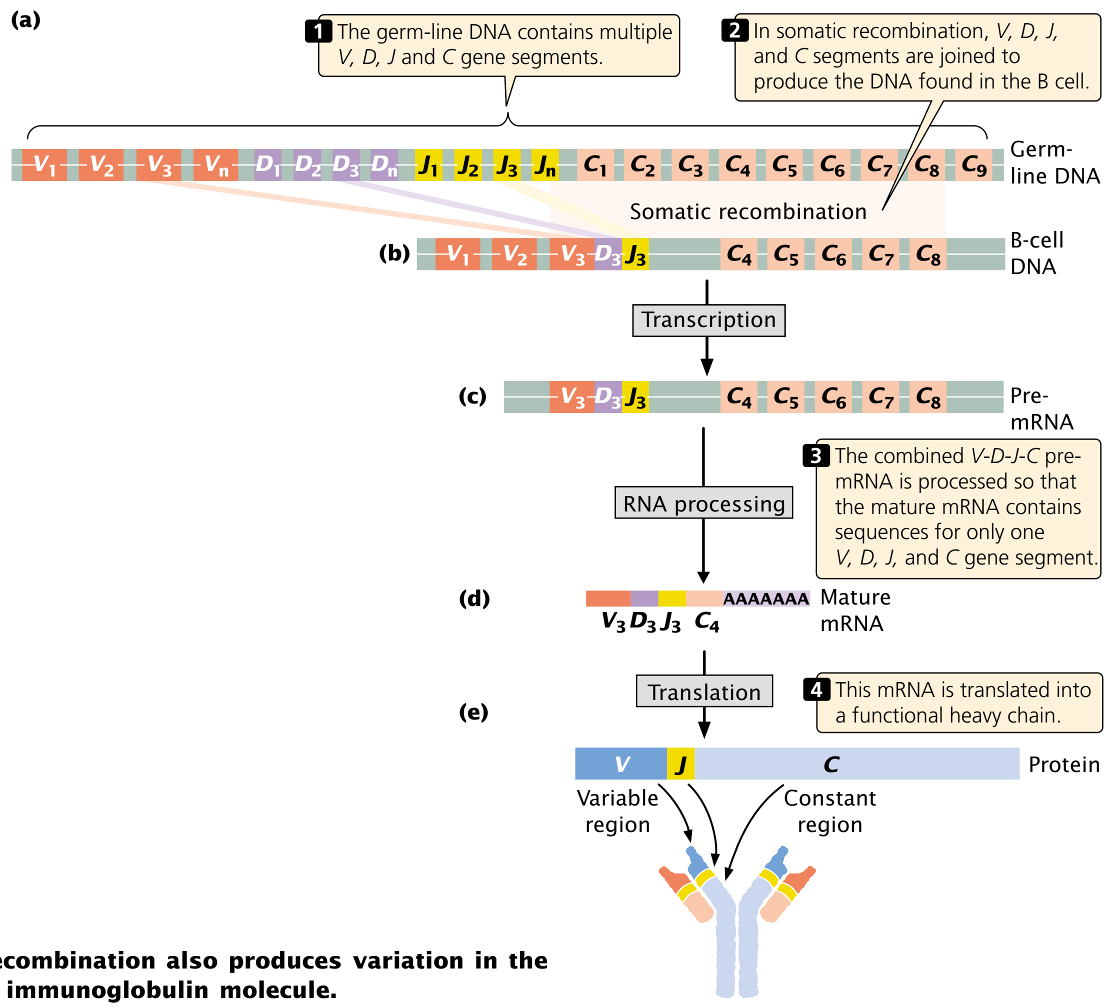
different functional *V* gene segments, 5 different *J* genes, and a single *C* gene segment, all of which are present in the germ-line DNA (FIGURE 21.17a). The *V* gene segments, which are about 400 bp in length, are located on the same chromosome and are separated from one another by about 7000 bp. The *J* gene segments are about 30 bp in length and all together encompass about 1400 bp.

Initially, an immature lymphocyte inherits all of the *V* gene segments and all of the *J* gene segments present in the germ line. In the maturation of the lymphocyte, **somatic recombination** within a single chromosome moves one of the *V* genes to a position next to one of the *J* gene segments (FIGURE 21.17b). In Figure 21.17b, V_2 (the second of approximately 35 different *V* gene segments) undergoes somatic recombination, which places it next to J_3 (the third of 5 *J* gene segments); the intervening segments are lost.

After somatic recombination has taken place, the combined *V-J-C* gene is transcribed and processed (FIGURE 21.17c and d). The mature mRNA that results contains only sequences for a single *V*, *J*, and *C* segment; this mRNA is translated into a functional light chain (FIGURE 21.17e). In this way, each mature human B cell produces a unique type of kappa light chain, and different B cells produce slightly different kappa chains, depending on the combination of *V* and *J* segments that are joined.

The gene that encodes the lambda light chain is organized in a similar way but differs from the kappa gene in the number of copies of the different segments. In the human gene for the lambda light chain, there are from 29 to 33 different functional *V* gene segments and 4 or 5 different functional *J* and *C* gene segments (each *C* gene segment is attached to a different *J* segment). Somatic recombination takes place among the segments in the same way as that in the kappa gene, generating many possible combinations of lambda light chains.

The gene that encodes the immunoglobulin heavy chain is arranged in *V*, *J*, and *C* segments, but this gene also possesses *D* (for diversity) segments. Somatic recombination taking place in lymphocyte maturation joins one *D* gene segment to one *J* gene segment, and then a *V* gene segment is joined to this combined *D-J* gene segment (FIGURE 21.18a and b). Transcription and RNA processing of this gene produces a mRNA that encodes only one



21.18 Somatic recombination also produces variation in the heavy chain of the immunoglobulin molecule.

particular type of heavy ch36ain (FIGURE 21.18c–e). Thus, many different types of light and heavy chains are possible.

Somatic recombination is brought about by RAG1 and RAG2 proteins, which generate double-strand breaks at specific nucleotide sequences called recombination signal sequences that flank the *V, D, J,* and *C* gene segments. DNA repair proteins then process and join the ends of particular segments together (FIGURE 21.19).

In addition to somatic recombination, other mechanisms add to antibody diversity. First, each type of light chain can potentially combine with each type of heavy chain to make a functional immunoglobulin molecule, increasing the amount of possible variation in antibodies. Second, the recombination process that joins *V, J, D,* and *C* gene segments in the developing B cell is imprecise, and a few random nucleotides are frequently lost or gained at the junctions of the recombining segments. This **junctional diversity** greatly enhances variation among antibodies. Third, a high rate of mutation, called **somatic hypermutation** (the cause of which is unknown), is characteristic of the immunoglobulin genes.

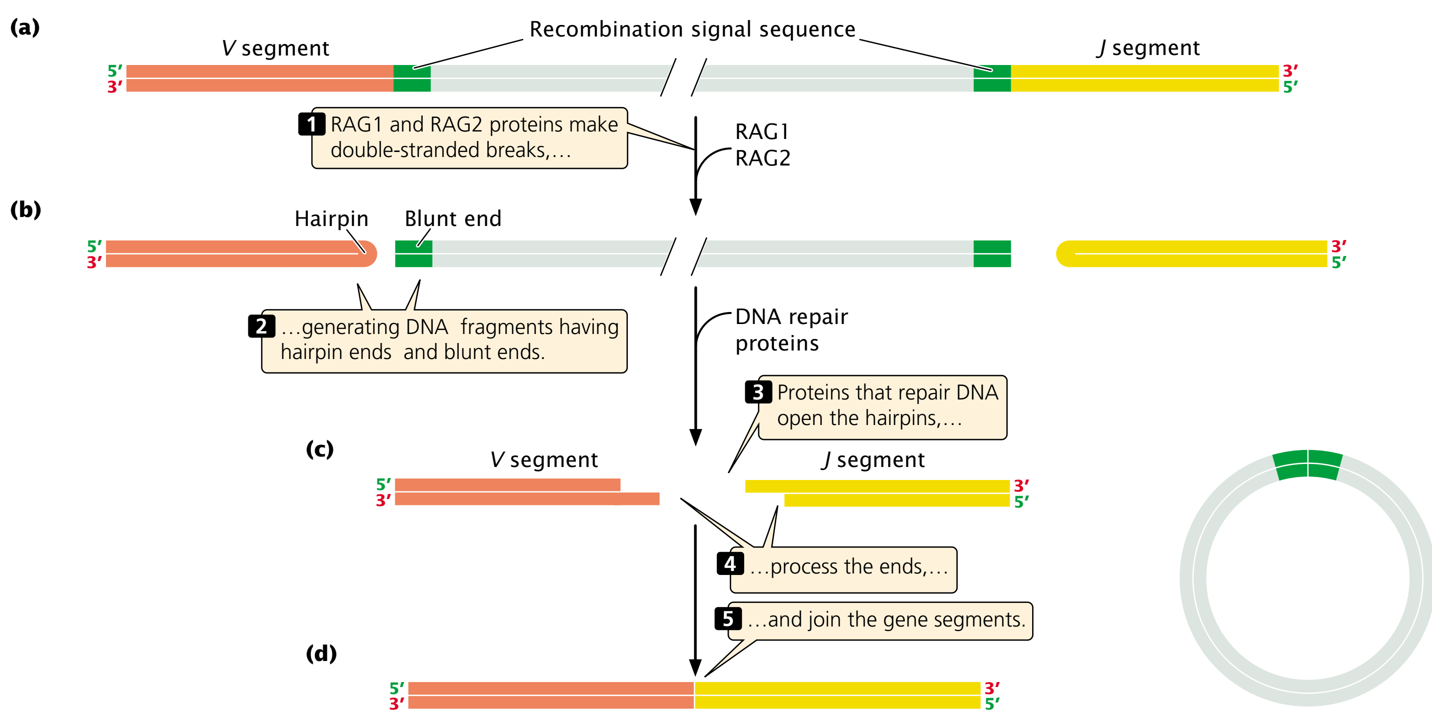
Concepts

The genes encoding the antibody chains are organized in segments, and germ-line DNA contains multiple versions of each segment. The many possible combinations of *V, J,* and *D* segments permit an immense variety of different antibodies to be generated. This diversity is augmented by the different combinations of light and heavy chains, the random addition and deletion of nucleotides at the junctions of the segments, and the high mutation rates in the immunological genes.

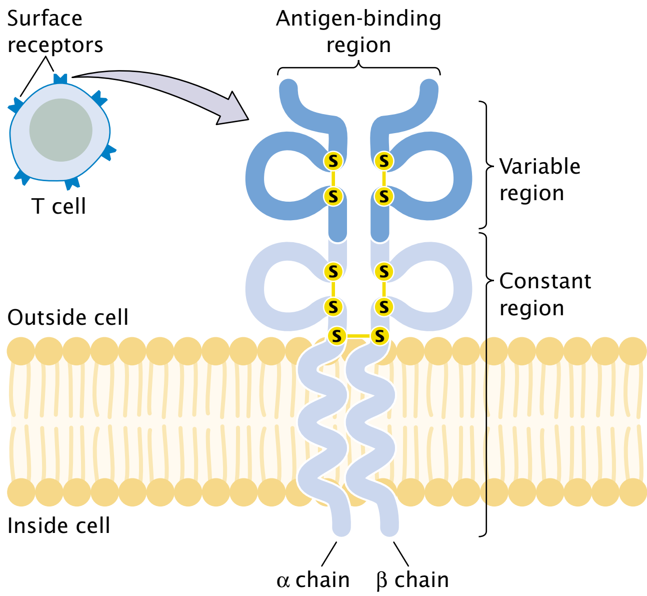
T-Cell-Receptor Diversity

Like B cells, each mature T cell has genetically determined specificity for one type of antigen that is mediated through the cell's receptors. T-cell receptors are structurally similar to immunoglobulins (FIGURE 21.20) and are located on the cell surface; most T-cell receptors are composed of one alpha and one beta polypeptide chain held together by disulfide bonds. One end of each chain is embedded in the cell membrane; the other end projects away from the cell and binds antigens. Like the immunoglobulin chains, each chain of the T-cell receptor possesses a constant region and a variable region (see Figure 21.20); the variable regions of the two chains provide the antigen-binding site.

The genes that encode the alpha and beta chains of the T-cell receptor are organized much like those that encode the heavy and light chains of immunoglobulins: each gene is made up of segments that undergo somatic recombination before the gene is transcribed. For example, the human gene for the alpha chain initially consists of 44 to 46 *V* gene segments, 50 *J* gene segments, and a single *C* gene segment. The organization of the gene for the beta chain is similar, except that it also contains *D* segments. Random combination of alpha and beta chains and junctional diversity takes place, but there is no evidence for somatic hypermutation in T-cell-receptor genes.



21.19 Somatic recombination is brought about by RAG1 and RAG2 proteins and DNA repair proteins.



21.20 A T-cell receptor is composed of two polypeptide chains, each having a variable and constant region. Most T-cell receptors are composed of alpha (α) and beta (β) polypeptide chains held together by disulfide bonds. One end of each chain traverses the cell membrane; the other end projects away from the cell and binds antigens.

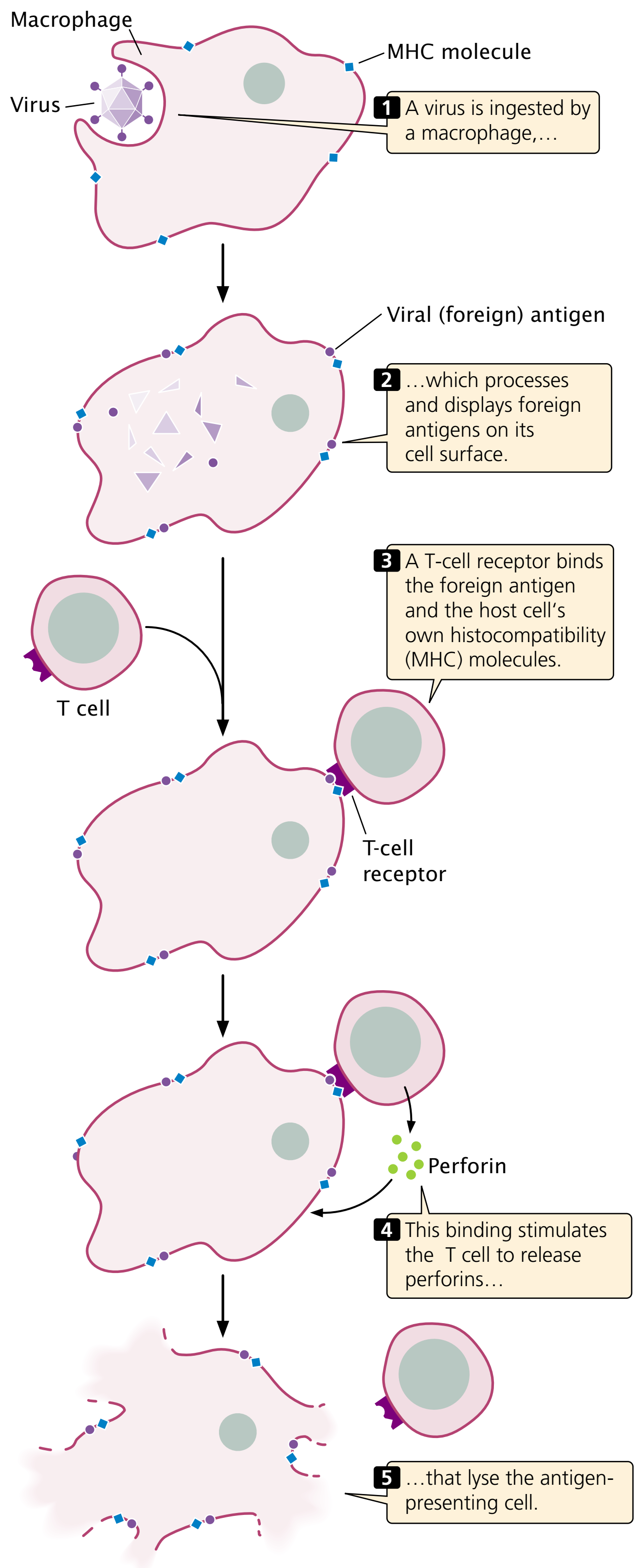
Concepts

Like the genes that encode antibodies, the genes for the T-cell-receptor chains consist of segments that undergo somatic recombination, generating an enormous diversity of antigen-binding sites.

Major Histocompatibility Complex Genes

When tissues are transferred from one species to another or even from one individual member to another within a species, the transplanted tissues are usually rejected by the host animal. The results of early studies demonstrated that this graft rejection is due to an immune response that occurs when antigens on the surface of the grafted tissue are detected and attacked by T cells in the host organism. The antigens that elicit graft rejection are referred to as histocompatibility antigens, and they are encoded by a cluster of genes called the major histocompatibility complex.

T cells are activated only when the T-cell receptor simultaneously binds *both* a foreign antigen *and* the host cell's own histocompatibility antigen. The reason for this requirement is not clear; it may reserve T cells for action against pathogens that have invaded cells. When a foreign body, such as a virus, is ingested by a macrophage or other cell, partly digested pieces of the foreign body containing antigens are displayed on the macrophage's surface (FIGURE 21.21). Through their T-cell receptors, T cells bind to both



21.21 T cells are activated by binding to a foreign antigen and a histocompatibility antigen on the surface of a self-cell.

the histocompatibility protein and the foreign antigen and secrete substances that either destroy the antigen-containing cell or activate other B and T cells or both.

The MHC genes are among the most variable genes known: there are more than 100 different alleles for some MHC loci. Because each person possesses five or more MHC loci and because many alleles are possible at each locus, no two people (with the exception of identical twins) produce the same set of histocompatibility antigens. The variation in histocompatibility antigens provides each of us with a unique identity for our own cells, which allows our immune systems to distinguish self from nonself. This variation is also the cause of rejection in organ transplants.

Concepts

The MHC genes encode proteins that provide identity to the cells of each individual organism. To bring about an immune response, a T-cell receptor must simultaneously bind both a histocompatibility (self) antigen and a specific foreign antigen.

www.whfreeman.com/pierce Additional information about the genetics of the immune system

Genes and Organ Transplants

For a person with a seriously impaired organ, a transplant operation may offer the only hope of survival. Successful transplantation requires more than the skills of a surgeon; it also requires a genetic match between the patient and the person donating the organ.

The fate of transplanted tissue depends largely on the type of antigens present on the surface of its cells. Because foreign tissues are usually rejected by the host, the successful transplantation of tissues between different persons is very difficult. Tissue rejection can be partly inhibited by drugs that interfere with cellular immunity. Unfortunately, this treatment can create serious problems for transplant patients, because they may have difficulty fighting off common pathogens and thus may die of infection. The only other option for controlling the immune reaction is to carefully match the donor and the recipient, maximizing the genetic similarities.

The tissue antigens that elicit the strongest immune reaction are the very ones used by the immune system to mark its own cells, those encoded by the major histocompatibility complex. The MHC spans a region of more than 3 million base pairs on human chromosome 6 and has many alleles, providing different MHC antigens on the cells of different people and allowing the immune system to recognize foreign cells.

The severity of an immune rejection of a transplanted organ depends on the number of mismatched MHC antigens on the cells of the transplanted tissue. The ABO red-blood-cell antigens also are important because they elicit a strong immune reaction. The ideal donor is the patient's own identical twin, who will have exactly the same MHC and ABO antigens. Unfortunately, most patients don't have an identical twin. The next best donor is a sibling with the same major MHC and ABO antigens. If a sibling is not available, donors from the general population are considered. An attempt is made to match as many of the MHC antigens of the donor and recipient as possible, and immunosuppressive drugs are used to control rejection that occurs because of the mismatches. The long-term success of transplants depends on the closeness of the match. Survival rates after kidney transplants (the most successful of the major organ transplants) increase from 63% with zero or one MHC match to 90% with four matches.

Cancer Genetics

Cancer kills one of every five Americans, and cancer treatments cost billions of dollars every year. Cancer is not a single disease; rather, it is a heterogeneous group of disorders characterized by the presence of cells that do not respond to the normal controls on division. Cancer cells divide rapidly and continuously, creating tumors that crowd out normal cells and eventually rob healthy tissues of nutrients. The cells of an advanced tumor can separate from the tumor and travel to distant sites in the body, where they may take up residence and develop into new tumors. The most common cancers in the United States are those of the prostate, breast, lung, colon and rectum, and blood (Table 21.6).

The Nature of Cancer

Normal cells grow, divide, mature, and die in response to a complex set of internal and external signals. A normal cell receives both stimulatory and inhibitory signals, and its growth and division are regulated by a delicate balance between these opposing forces. In a cancer cell, one or more of the signals has been disrupted, which causes the cell to proliferate at an abnormally high rate. As they lose their response to the normal controls, cancer cells gradually lose their regular shape and boundaries, eventually forming a distinct mass of abnormal cells—a tumor. If the cells of the tumor remain localized, the tumor is said to be benign; if the cells invade other tissues, the tumor is said to be **malignant.** Cells that travel to other sites in the body, where they establish secondary tumors, have undergone **metastasis.**

Cancer As a Genetic Disease

Cancer arises as a result of fundamental defects in the regulation of cell division, and its study therefore has significance not only for public health, but also for our basic understanding of cell biology. Through the years, a large number of theories have been put forth to explain cancer, but we now recognize that most, if not all, cancers arise from defects in DNA.

Table 21.6 Estimated incidences of various cancers and cancer mortality in the United States in 2002

Type of Cancer	Estimated New Cases per Year	Estimated Deaths per Year
Prostate	189,000	30,200
Breast	205,000	40,000
Lung and bronchus	169,400	154,900
Colon and rectum	148,300	56,600
Lymphoma	60,900	25,800
Bladder	56,500	12,600
Melanoma	53,600	7,400
Uterus	39,300	6,600
Leukemias	30,800	21,700
Oral cavity and pharynx	28,900	7,400
Pancreas	30,300	29,700
Ovary	23,300	13,900
Stomach	21,600	12,400
Brain and nervous system	17,000	13,100
Liver	16,600	14,100
Uterine cervix	13,000	4,100
Cancers of soft tissues including heart	8,300	3,900
All cancers	1,268,000	553,400

Source: American Cancer Society, *Cancer Facts and Figures, 2002* (Atlanta: American Cancer Society, 2001), p. 80

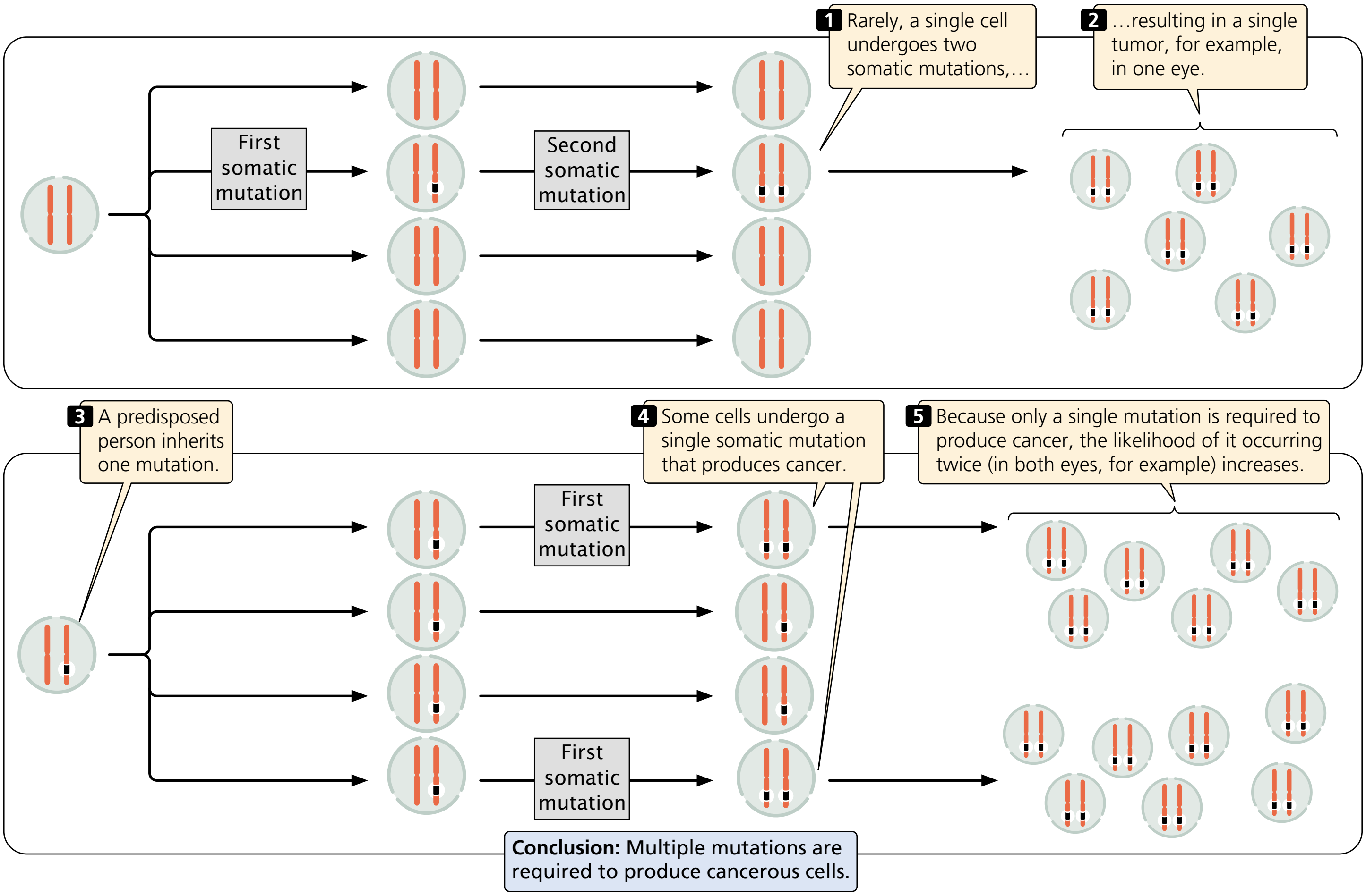
Early observations suggested that cancer might result from genetic damage. First, it was recognized that many agents such as ionizing radiation and chemicals that cause mutations also cause cancer (are carcinogens). Second, some cancers are consistently associated with particular chromosome abnormalities. About 90% of people with chronic myeloid leukemia, for example, have a reciprocal translocation between chromosome 22 and chromosome 9 (see Figure 9.34). Third, some specific types of cancers tend to run in families. Retinoblastoma, a rare childhood cancer of the retina, appears with high frequency in a few families and is inherited as an autosomal dominant trait, suggesting that a single gene is responsible for these cases of the disease.

Although these observations hinted that genes play some role in cancer, the theory of cancer as a genetic disease had several significant problems. If cancer is inherited, every cell in the body should receive the cancer-causing gene, and therefore every cell should become cancerous. In those types of cancer that run in families, however, tumors typically appear only in certain tissues and often only when the person reaches an advanced age. Finally, many cancers do not run in families at all and, even in regard to those cancers that generally do, isolated cases crop up in families with no history of the disease.

In 1971, Alfred Knudson proposed a model to explain the genetic basis of cancer. Knudson was studying retinoblastoma, a cancer that usually develops in only one eye but occasionally appears in both. Knudson found that, when retinoblastoma appears in both eyes, onset is at an early age, and affected children often have close relatives who also have retinoblastoma.

Knudson proposed that retinoblastoma results from two separate genetic defects, both of which are necessary for cancer to develop (FIGURE 21.22). He suggested that, in the cases in which the disease affects just one eye, a single cell in one eye undergoes two successive mutations. Because the chance of these two mutations occurring in a single cell is remote, retinoblastoma is rare and typically develops in only one eye. For bilateral cases, Knudson proposed that the child inherited one of the two mutations required for the cancer, and so every cell contains this initial mutation. In these cases, all that is required for cancer to develop is for one eye cell to undergo the second mutation. Because each eye possesses millions of cells, there is a high probability that the second mutation will occur in at least one cell of each eye, producing tumors in both eyes at an early age.

Knudson's hypothesis suggests that cancer is the result of a multistep process that requires several mutations. If one or more of the required mutations is inherited, fewer



21.22 Alfred Knudson proposed that retinoblastoma results from two separate genetic defects, both of which are necessary for cancer to develop.

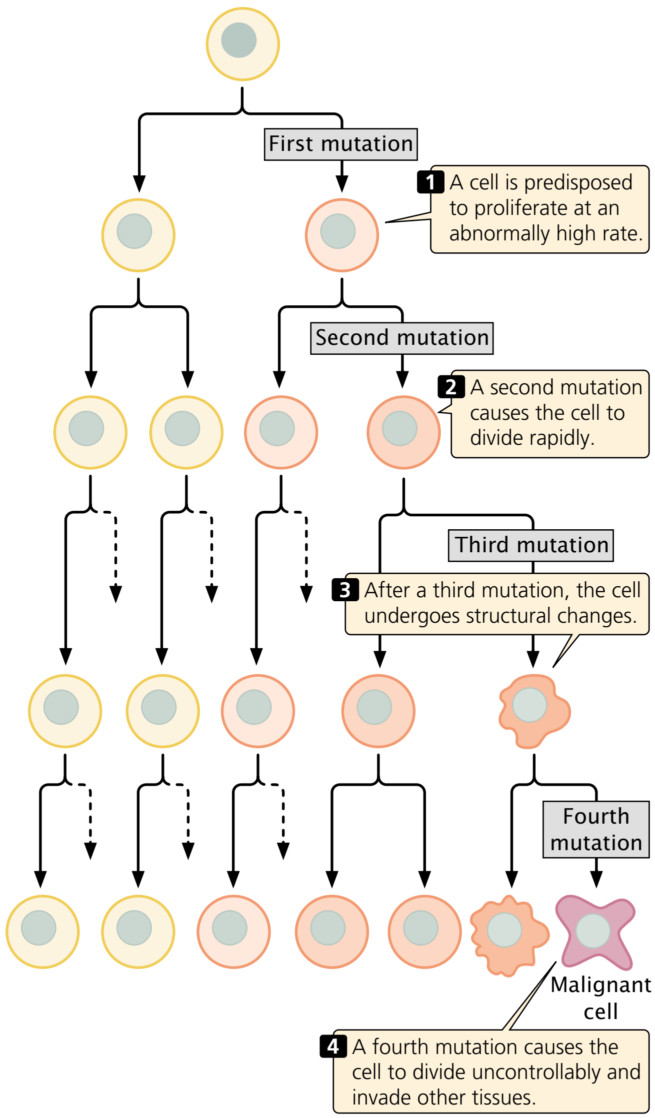
additional mutations are required to produce cancer, and the cancer will tend to run in families. The idea that cancer results from multiple mutations turns out to be correct for most cancers.

Knudson's genetic theory for cancer has been confirmed by the identification of genes that, when mutated, cause cancer. Today, we recognize that cancer is fundamentally a genetic disease, although few cancers are actually inherited. Most tumors arise from somatic mutations that accumulate during our life span, either through spontaneous mutation or in response to environmental mutagens.

The clonal evolution of tumors Cancer begins when a single cell undergoes a mutation that causes the cell to divide at an abnormally rapid rate. The cell proliferates, giving rise to a clone of cells, each of which carries the same mutation. Because the cells of the clone divide more rapidly than normal, they soon outgrow other cells. Additional mutations that arise in the clone may further enhance the ability of those cells to proliferate, and cells carrying both mutations soon become dominant in the clone. Eventually, they may be overtaken by cells that contain yet more mutations that enhance proliferation. In this process, called **clonal evolution,** the tumor cells acquire more mutations that allow them to become increasingly more aggressive in their proliferative properties (FIGURE 21.23).

The rate of clonal evolution depends on the frequency with which new mutations arise. Any genetic defect that allows more mutations to arise will accelerate cancer progression. Genes that regulate DNA repair are often found to have been mutated in the cells of advanced cancers, and inherited disorders of DNA repair are usually characterized by increased incidences of cancer. Because DNA repair mechanisms normally eliminate many of the mutations that arise, without DNA repair, mutations are more likely to persist in all genes, including those that regulate cell division. Xeroderma pigmentosum, for example, is a rare disorder caused by a defect in DNA repair (see p. 000 in Chapter 17). People with this condition have elevated rates of skin cancer when exposed to sunlight (which induces mutation).

Mutations in genes that affect chromosome segregation also may contribute to the clonal evolution of tumors. Many cancer cells are aneuploid, and it is clear that chromosome mutations contribute to cancer progression by



21.23 Through clonal evolution, tumor cells acquire multiple mutations that allow them to become increasingly aggressive and proliferative.

duplicating some genes (those on extra chromosomes) and eliminating others (those on deleted chromosomes). Cellular defects that interfere with chromosome separation increase aneuploidy and therefore may accelerate cancer progression.

Concepts

Cancer is fundamentally a genetic disease. Mutations in several genes are usually required to produce cancer. If one of these mutations is inherited, fewer somatic mutations are necessary for cancer to develop, and the person may have a predisposition to cancer. Clonal evolution is the accumulation of mutations in a clone of cells.

The role of environment in cancer Although cancer is fundamentally a genetic disease, most cancers are not inherited, and there is little doubt that many cancers are influenced by environmental factors. The role of environmental factors in cancer is suggested by differences in the incidence of specific cancers throughout the world (Table 21.7). The results of studies show that migrant populations typically take on the cancer incidence of their host country. For example, the overall rates of cancer are considerably lower in Japan than in Hawaii. However, within a single generation after migration to Hawaii, Japanese people develop cancer at rates similar to those of native Hawaiians. Smoking is a good example of an environmental factor that is strongly associated with cancer. Other environmental factors such as chemicals, ultraviolet light, ionizing radiation, and viruses are known carcinogens and are associated with variation in the incidence of many cancers.

Genes That Contribute to Cancer

The signals that regulate cell division fall into two basic types: molecules that stimulate cell division and those that inhibit it. These control mechanisms are similar to the accelerator and brake of an automobile. In normal cells (but hopefully not your car), both accelerators and brakes are applied at the same time, causing cell division to proceed at the proper speed.

Because cell division is affected by both accelerators and brakes, cancer can arise from mutations in either type of signal, and there are several fundamentally different routes to cancer (FIGURE 21.24). A stimulatory gene can be made hyperactive or active at inappropriate times, analogously to having the accelerator of an automobile stuck in the floored position. Mutations in stimulatory genes are usually dominant, because a mutation in a single copy of the gene is usually sufficient to produce a stimulatory effect. Dominant-acting stimulatory genes that cause cancer are termed **oncogenes.** Cell division may also be stimulated when inhibitory genes are made *inactive,* analogously to having a defective brake in an automobile. Mutated inhibitory genes generally have recessive effects, because both copies must be mutated to remove all inhibition. Inhibitory genes in cancer are termed **tumor-suppressor genes.**

Although oncogenes or mutated tumor-suppressor genes or both are required to produce cancer, mutations in DNA repair genes can increase the likelihood of acquiring mutations in these genes. Having mutated DNA repair genes is analogous to having a lousy car mechanic who does not make the necessary repairs to a broken accelerator or brake.

Oncogenes and tumor-suppressor genes Oncogenes were the first cancer-causing genes to be identified. In 1910, Peyton Rous described a virus that caused connective-tissue tumors (sarcomas) in chickens; this virus became known as the Rous sarcoma virus. A number of other cancer-causing

Table 21.7 Examples of geographic variation in the incidence of cancer

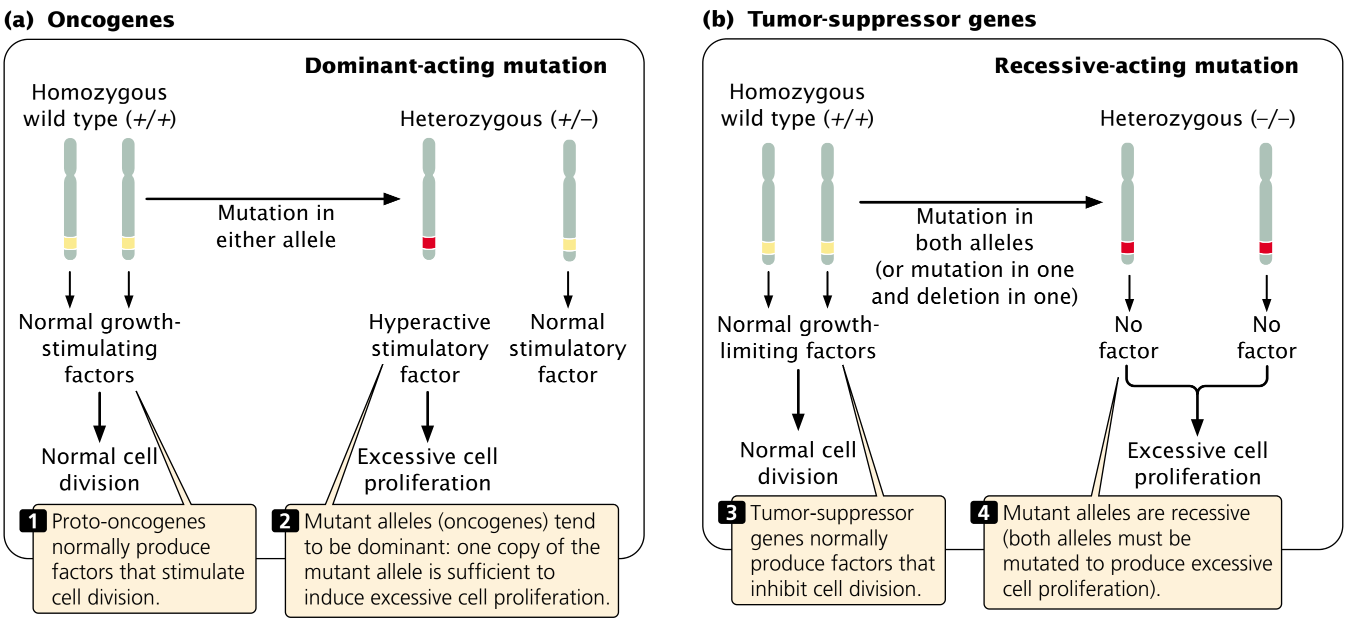
Type of Cancer	Location	Incidence Rate*
Lip	Canada (Newfoundland)	15.1
	Brazil (Fortaleza)	1.2
Nasopharynx	Hong Kong	30.0
	United States (Utah)	0.5
Colon	United States (Iowa)	30.1
	India (Bombay)	3.4
Lung	United States (New Orleans, African Americans)	110.0
	Costa Rica	17.8
Prostate	United States (Utah)	70.2
	China (Shanghai)	1.8
Bladder	United States (Connecticut, Whites)	25.2
	Philippines (Rizal)	2.8
All cancer	Switzerland (Basel)	383.3
	Kuwait	76.3

Source: C. Muir et al., *Cancer incidence in Five Continents*, vol. 5 (Lyon: International Agency for Research on Cancer, 1987), Table 12-2.

*The incidence rate is the age-standardized rate in males per 100,000 population.

viruses were subsequently isolated from various animal tissues. These viruses were generally assumed to carry a cancer-causing gene that was transferred to the host cell. The first oncogene, called *src*, was isolated from the Rous sarcoma virus in 1970.

In 1975, Michael Bishop, Harold Varmus, and their colleagues began to use probes for viral oncogenes to search for related sequences in normal cells. They discovered that the genomes of all normal cells carry DNA sequences that are closely related to viral oncogenes. These cellular genes are called **proto-oncogenes.** They are responsible for basic cellular functions in normal cells but, when mutated, they become oncogenes that contribute to the development of cancer. When a virus infects a cell, a proto-oncogene may



21.24 Both oncogenes and tumor-suppressor genes contribute to cancer but differ in their modes of action and dominance.

Table 21.8 Some oncogenes and functions of their corresponding proto-oncogenes

Oncogene	Cellular Location	Function of Proto-oncogene
sis	Secreted	Growth factor
erbB	Cell membrane	Part of growth-factor receptor
erbA	Cytoplasm	Thyroid hormone receptor
src	Cell membrane	Protein tyrosine kinase
ras	Cell membrane	GTP binding and GTPase
myc	Nucleus	Transcription factor
fos	Nucleus	Transcription factor
jun	Nucleus	Transcription factor
bcl-1	Nucleus	Cell cycle

become incorporated into the viral genome through recombination. Within the viral genome, the proto-oncogene may mutate to an oncogene that, when inserted back into a cell, causes rapid cell division and cancer. Because the proto-oncogenes are more likely to undergo mutation or recombination within a virus, viral infection is often associated with the cancer.

Proto-oncogenes can be converted into oncogenes in viruses by several different ways. The sequence of the proto-oncogene may be altered or truncated as it is being incorporated into the viral genome. This mutated copy of the gene may then produce an altered protein that causes uncontrolled cell proliferation. Alternatively, through recombination, a proto-oncogene may end up next to a viral promoter or enhancer, which then causes the gene to be overexpressed. Finally, sometimes the function of a proto-oncogene in the host cell may be altered when a virus inserts its own DNA into the gene, disrupting its normal function.

Many oncogenes have been identified by experiments in which selected fragments of DNA are added to cells in culture. Some of the cells take up the DNA and, if these cells become cancerous, then the DNA fragment that was added to the culture must contain an oncogene. The fragments can then be sequenced, and the oncogene can be identified. More than 70 oncogenes have now been discovered (Table 21.8).

Tumor-suppressor genes are more difficult than oncogenes to identify because they *inhibit* cancer and are recessive; both alleles must be mutated before the inhibition of cell division is removed. Because it is the *failure* of their function that promotes cell proliferation, tumor-suppressor genes cannot be identified by adding them to cells and looking for cancer.

One of the first tumor-suppressor genes to be identified was the retinoblastoma gene. In 1985, Raymond White and Webster Cavenne showed that large segments of chromosome 13 were missing in cells of retinoblastoma tumors, and later the tumor-suppressor gene was isolated from these segments. A number of tumor-suppressor genes have now been discovered in this way (Table 21.9).

Genes controlling the cell cycle Genes that control the cell cycle often serve as proto-oncogenes or tumor-suppressor genes. Let's briefly revisit the regulation of the cell cycle, which was discussed in Chapter 2. The cell cycle is regulated by cyclins, whose concentration oscillates during the cell cycle, and cyclin-dependent kinases (CDKs), which have a relatively constant concentration. Cyclins bind to CDKs, producing activated protein kinases that initiate key events in the cell cycle. Genes that encode cyclins and factors that inhibit or stimulate the formation of activated CDKs are often oncogenes and tumor-suppressor genes, respectively. Mutated cyclin genes have been associated with cancers of the immune system, breast, stomach, and esophagus; genes, such as *p16* and *p21*, that encode inhibitors of CDKs are mutated or missing in many cancer cells.

Some proto-oncogenes and suppressor genes have roles in apoptosis. Cells have the ability to assess themselves and, when they are abnormal or damaged, they normally undergo apoptosis (see p. 000). Cancer cells frequently have chromosome mutations, DNA damage, and other cellular anomalies that would normally stimulate apoptosis and

Table 21.9 Some tumor-suppressor genes and their functions

Gene	Cellular Location	Function
NF1	Cytoplasm	GTPase activator
p53	Nucleus	Transcription factor, regulates apoptosis
RB	Nucleus	Transcription factor
WT-1	Nucleus	Transcription factor

Source: J. Marx, Learning how to suppress cancer, *Science* 261(1993):1385.

prevent their proliferation. Often these cells have mutations in genes that regulate apoptosis, and therefore they do not undergo programmed cell death. The ability of a cell to initiate apoptosis in response to DNA damage, for example, depends on a gene called *p53,* which is inactivate in many human cancers.

www.whfreeman.com/pierce Additional information about *p53* and its role in cancer

DNA repair genes Cancer arises from the accumulation of multiple mutations in a single cell. Some cancer cells have normal rates of mutation, and multiple mutations accumulate because each mutation gives the cell a replicative advantage over cells without the mutations. Other cancer cells may have higher-than-normal rates of mutation in all of their genes, which leads to more frequent mutation of oncogenes and tumor-suppressor genes. What might be the source of these high rates of mutation in some cancer cells?

Two processes control the rate at which mutations arise within a cell: (1) the rate at which errors arise during and after replication; and (2) the efficiency with which these errors are corrected. The error rate during replication is controlled by the fidelity of DNA polymerases and other proteins in the replication process (see Chapter 12). However, defects in genes encoding replication proteins have not been strongly linked to cancer.

The mutation rate is also strongly affected by whether errors are corrected by DNA repair systems (see p. 000 in Chapter 17). Defects in genes that encode components of these repair systems have been consistently associated with a number of cancers. People with xeroderma pigmentosum, for example, are defective in nucleotide-excision repair, an important cellular repair system that normally corrects DNA damage caused by a number of mutagens, including ultraviolet light. Likewise, about 13% of colorectal, endometrial, and stomach cancers have cells that are defective in mismatch repair, another major repair system in the cell.

Some types of colon cancer are inherited as an autosomal dominant trait. In families with this condition, a person can inherit one mutated and one normal allele of a gene that controls mismatch repair. The normal allele provides sufficient levels of the protein for mismatch repair to function, but it is highly likely that this normal allele will become mutated or lost in at least a few cells. If it does so, there is no mismatch repair, and these cells undergo higher-than-normal rates of mutation, leading to defects in oncogenes and tumor-suppressor genes that cause the cells to proliferate.

www.whfreeman.com/pierce Additional information on DNA repair

Genes affecting chromosome segregation Most advanced tumors contain cells that exhibit a variety of chromosome anomalies, including extra chromosomes, missing chromosomes, and chromosome rearrangements. Aneuploidy in somatic cells usually arises when chromosomes do not segregate properly in mitosis. Normal cells have a checkpoint that monitors the proper assembly of the mitotic spindle; if chromosomes are not properly attached to the microtubules at metaphase, the onset of anaphase is blocked. Some aneuploid cancer cells contain mutant alleles for genes that encode proteins having roles in this checkpoint; in these cells, anaphase is entered into despite the improper or lack of assembly of the spindle, and chromosome abnormalities result.

The tumor-suppressor gene *p53,* in addition to controlling apoptosis, plays a role in the duplication of the centrosome, which is required for proper formation of the spindle and for chromosome segregation. Normally, the centrosome duplicates once per cell cycle. If *p53* is mutated or missing, however, the centrosome may undergo extra duplications, resulting in the unequal segregation of chromosomes. In this way, mutation of the *p53* gene may generate chromosome mutations that contribute to cancer. The *p53* gene is also a tumor-suppressor gene that prevents cell division when the DNA is damaged.

Sequences that regulate telomerase Another factor that may contribute to the progression of cancer is the inappropriate activation of an enzyme called telomerase. Telomeres are special sequences at the ends of eukaryotic chromosomes (see p. 000 in Chapter 11). In DNA replication in somatic cells, DNA polymerases require a 3′-OH group to add new nucleotides. For this reason, the ends of chromosomes cannot be replicated, and telomeres become shorter with each cell division. This shortening eventually leads to the destruction of the chromosome and cell death; so somatic cells are capable of a limited number of cell divisions.

In germ cells, telomerase replicates the chromosome ends (see p. 000 in Chapter 12), thereby maintaining the telomeres, but this enzyme is not normally expressed in somatic cells. In many tumor cells, however, sequences that regulate the expression of the telomerase gene are mutated so that the enzyme *is* expressed, and the cell is capable of unlimited cell division. Although the expression of telomerase appears to contribute to the development of many cancers, its precise role in tumor progression is still being investigated.

Genes that promote vascularization and the spread of tumors A final set of factors that contribute to the progression of cancer includes genes that affect the growth and spread of tumors. Oxygen and nutrients, which are essential to the survival and growth of tumors, are supplied by blood vessels, and the growth of new blood vessels (angiogenesis) is important to tumor progression. Angiogenesis is stimulated by growth factors and others proteins encoded by genes whose expression is carefully regulated in normal cells. In tumor cells, genes encoding these proteins are often overexpressed compared with normal cells, and inhibitors of angiogenesis-promoting factors may be inactivated or underexpressed. At least one inherited cancer syndrome—van Hippel-Lindau disease, in which people develop multiple types of tumors—is caused by the mutation of a gene that affects angiogenesis.

In the development of many cancers, the primary tumor gives rise to cells that spread to distant sites, producing secondary tumors. This process of metastasis is the cause of death in 90% of human cancer cases; it is influenced by cellular changes induced by somatic mutation. By using microarrays to measure levels of gene expression (see Chapter 19), researchers have identified several genes that are transcribed at a significantly higher rate in metastatic cells compared with nonmetastatic cells. These genes encode components of the extracellular matrix and the cytoskeleton, which are thought to affect the migration of cells. Other genes that affect metastasis include adhesion proteins that help hold cells together.

Concepts

Oncogenes are dominant in their action and stimulate cell proliferation. Tumor-suppressor genes are recessive in their action and inhibit cell proliferation. Defects in DNA repair genes allow a higher-than-normal rate of mutation in oncogenes and tumor suppressor genes. Mutations in genes that control chromosome segregation allow chromosome mutations to accumulate, which may then contribute to cancer progression. Mutations that allow telomerase to be expressed in somatic cells and that affect vascularization and metastasis also may contribute to cancer progression.

www.whfreeman.com/pierce General information about cancer, the genetics of cancer, and telomerase

The Molecular Genetics of Colorectal Cancer

Mutations that contribute to colorectal cancer have received extensive study, and this cancer is an excellent example of how cancer often arises through the accumulation of successive genetic defects.

Colorectal cancers arise in the cells lining the colon and rectum. More than 135,000 new cases of colorectal cancer are diagnosed in the United States each year, where this cancer is responsible for more than 56,000 deaths annually. If detected early, colorectal cancer can be treated successfully; consequently, there has been much interest in identifying the molecular events responsible for the initial stages of colorectal cancer.

Colorectal cancer is thought to originate as benign tumors called adenomatous polyps. Initially, these polyps are microscopic, but in time they enlarge, and the cells of the polyp acquire the abnormal characteristics of cancer cells. In the later stages of the disease, the tumor may invade the muscle layer surrounding the gut and metastasize. The progression of the disease is slow; from 10 to 35 years may be required for a benign tumor to develop into a malignant tumor.

Most cases of colorectal cancer are sporadic, developing in people with no family history of the disease, but a few families display a clear genetic predisposition to this disease. In one form of hereditary colon cancer, known as familial adenomatous polyposis coli, hundreds or thousands of polyps develop in the colon and rectum; if these polyps are not removed, one or more almost invariably becomes malignant.

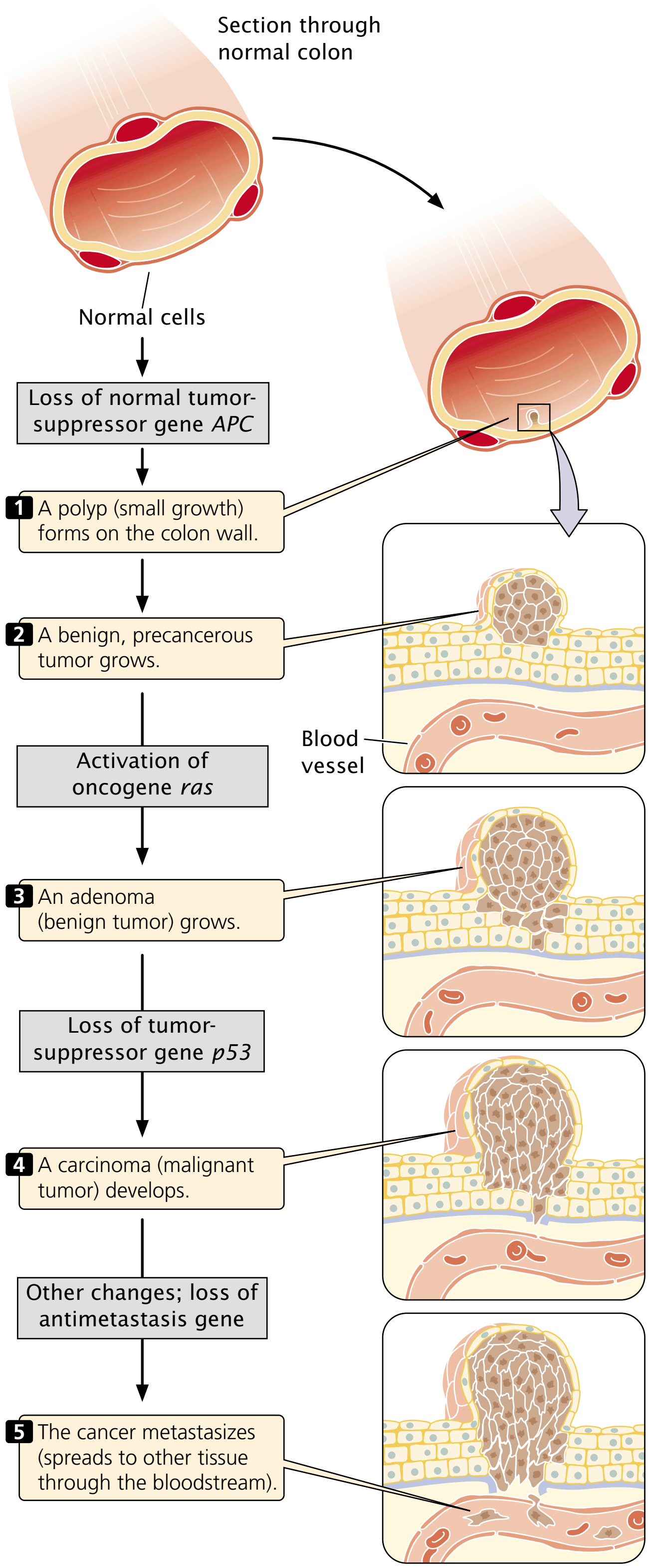
Because polyps and tumors of the colon and rectum can be easily observed and removed with a colonoscope (a fiber optic instrument that is used to view the interior of the rectum and colon), much is known about the progression of colorectal cancer, and some of the genes responsible for its clonal evolution have been identified. About 75% of colorectal cancers have mutations in tumor-suppressor gene *p53,* and many also have a mutation in the *ras* proto-oncogene. Families with adenomatous polyposis coli carry a defect in a gene called *APC,* and mutations in *APC* are found in the cells of tumors that arise sporadically (in persons without a family history). Additional genes that are frequently mutated in colorectal cancer include the oncogenes *myc* and *neu* and the tumor-suppressor gene *HNPCC.*

Mutations in these genes are responsible for the different steps of colorectal cancer progression. One of the earliest steps is a mutation that inactivates the *APC* gene, which increases the rate of cell division, leading to polyp formation (FIGURE 21.25). A person with familial adenomatous polyposis coli inherits one defective copy of the *APC* gene, and defects in this gene are associated with the numerous polyps that appear in those who have this disorder. Mutations in *APC* are also found in the polyps that develop in people who do not have adenomatous polyposis coli.

Mutations of the *ras* oncogene usually occur later, in larger polyps comprising cells that have acquired some genetic mutations. The protein produced by the normal *ras* proto-oncogene sits inside the cell membrane. From there it relays signals from growth factors that stimulate cell division. When *ras* is mutated, the protein that it encodes continually relays a stimulatory signal for cell division, even when growth factor is absent.

Mutations in *p53* and other genes appear still later in tumor progression; these mutations are rare in polyps but common in malignant cells. Because *p53* prevents the replication of cells with genetic damage and controls proper chromosome segregation, mutations in *p53* may allow a cell to rapidly acquire further gene and chromosome mutations, which then contribute to further proliferation and invasion into surrounding tissues.

The sequence of steps just outlined is not the only route to colorectal cancer, and the mutations need not occur in the order presented here, but this sequence is a common pathway by which colon and rectal cells become cancerous.



21.25 Mutations in multiple genes contribute to the progression of colorectal cancer.

Connecting Concepts Across Chapters

This chapter has focused on three specialized but important topics: the genetics of development, the immune system, and cancer. In addition to their relevance to genetics, these topics have obvious medical importance and all are the subject of intense research.

The results of early experiments demonstrated that genes are not usually lost or permanently altered in the course of development; rather, development proceeds through the regulation of gene expression. The basic question for development is how are different sets of genes expressed in different parts of the embryo? Our study of pattern formation in *Drosophila* revealed that many genes take part and that they are regulated in a highly sequential manner. The process is initiated by maternally produced mRNA and proteins that become localized to particular regions of the egg. Sets of genes are successively activated, each set controlling the expression of other sets, so that successively smaller regions of the embryo are determined.

The immune system also is encoded by a complex set of genes whose products interact closely. Unlike those in pattern development, genes encoding antibodies and T-cell receptors are permanently altered in lymphocyte maturation. Lymphocytes violate the general principle that all cells contain the same set of genetic information.

Cancer also is influenced by complex interactions among multiple genes. Paradoxically, cancer is fundamentally a genetic disease, but most cancers are not inherited, because cancer usually requires somatic mutations at multiple genes. Even for those cancers for which a predisposition is clearly inherited, additional somatic mutations are required for cancer to arise. These mutations, each rare, accumulate because they provide the cell with a growth advantage.

This chapter has synthesized much of the information provided in preceding chapters. Gene regulation (Chapter 16) is the basis of development, the understanding of which also requires knowledge of genetic maternal effects (Chapter 5), transcription (Chapter 13), and translation (Chapter 15). The rearrangement of segments in genes of the immune system builds on our understanding of recombination (Chapter 12) and RNA processing (Chapter 14). Chromosome and gene mutations (Chapters 9 and 17) are essential to understanding cancer progression. Many oncogenes and tumor-suppressor genes control the cell cycle (Chapter 2), and predisposition to some cancers may be inherited as single-gene traits (Chapter 3). Cancer may also entail mutations in DNA repair genes (Chapter 17), genes affecting chromosome segregation (Chapter 2), and the regulation of telomerase (Chapter 12). Recombinant DNA techniques (Chapter 18) have contributed tremendously to our understanding of all of these processes.

The New Genetics
ETHICS • SCIENCE • TECHNOLOGY

Breast Cancer

Ron Green

Scientists in a medical genetics program of a major university medical school enroll a 54-year-old woman with metastatic breast cancer into a research protocol. The patient reports that several of her maternal relatives have breast cancer, but no pathological specimens from affected relatives are available for verifying the diagnoses. The patient dies before research studies are completed.

The patient has two daughters who are identical twins. Shortly after her mother's death, one of the twins requests access to her mother's test results. She explains that she wants this information because it will help her learn whether she carries the same mutation that might have contributed to her mother's disease. She has been informed that laboratories will not test for a mutation in a person before the identification of a known cancer-causing mutation in another family member affected by breast cancer. She wishes to learn whether she has a mutation because she is considering a prophylactic mastectomy to reduce her risk of developing breast cancer.

The research team learns from the head of the Institutional Review Board that there are no legal obstacles to this request, because the legal rights of the mother are not being compromised—she is deceased. The deceased are not considered research subjects under existing federal regulations.

On discovering her sister's intentions to request her mother's results, the second twin objects to the release of their mother's genetic information and says that she does not want to know whether she has inherited a greater risk of developing breast cancer. However, she went on to say, there is no way that the information could be kept from her, because she would inevitably learn of her sister's decision to have surgery. What should the research team do?

The case raises questions that are common in genetic research and clinical care today. Because of the nature and complexity of the questions raised, many of them require interdisciplinary examination. The GenEthics Consortium (GEC) was formed to bring scientists, bioethicists, lawyers, genetic counselors, and consumers together to discuss ethical issues emerging from research associated with the Human Genome Project.

In the late 1990s, the GEC convened to consider this particular case. In the course of their discussion, some members expressed the opinion that the clinical setting should focus on meeting the needs of the individual person, and the research setting should focus on gathering evidence to support a hypothesis. Others disagreed, arguing that the results of genetic testing in research settings often have clinical implications for subjects.

No explicit guidance was given by the mother,—should researchers release this information to a child who requests it? Some argued that the fact that the mother's DNA was being tested for mutations that are markers for breast cancer implies that the mother was not opposed to this type of testing. Some also stated that we can safely presume that the mother expected to share that information with her husband and children.

But whether children have a right to access the mother's test results was not the only issue that merited discussion. What should clinicians and researchers do when family members disagree about whether such information should be made available? Some felt that the sister who first came to the researchers with a request for her mother's genetic information has a privileged position. Others objected strenuously, saying that one's right to information in complex cases such as this one cannot merely be a matter of who gets to the doctor first.

Still other issues arose in the course of this discussion, most of which fell under the umbrella question of whether the risks of breast cancer for the twins could really be determined with precision. Persons in high-risk families with hereditary breast cancer and cancer-causing mutations face a much greater than average lifetime risk. However, in the absence of a thorough family history, the presence of a certain mutation and breast cancer in the mother alone do not provide the basis for assuming the existence of "hereditary" breast cancer in the family.

A second major area of uncertainty was highlighted when genetic professionals from different institutions disagreed about the statistics regarding risk reduction from prophylactic mastectomy. Some argued that the weighing of benefits and harms must include a comparison of the protection of one twin from the risk of premature death with the risk of psychological distress for the second twin. The research team would be morally justified in choosing the action that is most likely to reduce risk of death by cancer. However, others disagreed on the basis of inconclusive scientific evidence that mastectomy is a valid risk-reduction strategy.

Although there was no agreement on how genetic professionals should respond to this situation, there *was* a sense that, at this stage of genetic research, individual professionals might ethically and responsibly come to different conclusions about what they would do when faced with competing requests of this sort.

CONCEPTS SUMMARY

- Each multicellular organism begins as a single cell that has the potential to develop into any cell type. As development proceeds, cells become committed to particular fates. The results of early cloning experiments demonstrated that this process arises from differential gene expression.
- In the early *Drosophila* embryo, determination is effected through a cascade of gene control.
- The dorsal–ventral and anterior–posterior axes of the *Drosophila* embryo are established by egg-polarity genes. These genes are expressed in the female parent and produce RNA and proteins that are deposited in the egg cytoplasm. Initial differences in the distribution of these molecules regulate gene expression in various parts of the embryo. The dorsal–ventral axis is defined by a concentration gradient of the Dorsal protein, and the anterior–posterior axis is defined by concentration gradients of Bicoid and Nanos proteins.
- After the establishment of the major axes of development, three types of segmentation genes act sequentially to determine the number and organization of the embryonic segments in *Drosophila*. The gap genes establish large sections of the embryo, the pair-rule genes affect alternate segments, and the segment-polarity genes affect the organization of individual segments.
- Homeotic genes then define the identity of individual *Drosophila* segments. All these genes contain a consensus sequence called a homeobox that encodes a DNA-binding domain; the products of homeotic genes are DNA-binding proteins that regulate the expression of other genes. Genes with homeoboxes are found in many other organisms.
- Apoptosis, or programmed cell death, plays an important role in the development of many animals. In apoptosis, DNA is degraded, the nucleus and cytoplasm shrink, and the cell undergoes phagocytosis by other cells. Apoptosis is a highly regulated process that depends on caspases—proteins that cleave proteins. Each caspase is originally synthesized as an inactive precursor that must be activated, often through cleavage by another caspase.
- The immune system is the primary defense network in vertebrates. In humoral immunity, B cells produce antibodies that bind foreign antigens; in cellular immunity, T cells attack cells carrying foreign antigens.
- Each B and T cell is capable of binding only one type of foreign antigen. There are vast numbers of different types of B and T cells, and any potential antigen can be bound. When a lymphocyte binds to an antigen, the lymphocyte divides and gives rise to a clone of cells, each specific for that same antigen. This process is a primary immune response. A few memory cells remain in circulation for long periods of time. If the same antigen is encountered again, memory cells can proliferate rapidly and generate a secondary immune response.
- Immunoglobulins (antibodies) consists of two light chains and two heavy chains, each containing variable and constant regions. Light chains are of two basic types: kappa and lambda chains. The genes that encode the immunoglobulin chains consist of several types of gene segments; germ-line DNA contains multiple copies of these gene segments, which differ slightly in sequence. In B-cell maturation, somatic recombination randomly brings together one version of each segment to produce a single complete gene. Many combinations of the different segments are possible. The potential for diversity of antibodies is further increased by the random addition and deletion of nucleotides at the junctions of the segments. A high mutation rate also increases the potential diversity of antibodies.
- T-cell receptors are composed of alpha and beta chains. The germ-line genes for these proteins consist of segments with multiple varying copies. Somatic recombination allows many different types of T-cell receptors in different cells. Junctional diversity also adds to T-cell receptor variability.
- The major histocompatibility complex encodes a number of histocompatibility antigens. Each T cell simultaneously binds a foreign antigen and a host MHC antigen. The MHC antigen allows the immune system to distinguish self from nonself. Each locus for the MHC contains many alleles.
- Cancer is fundamentally a genetic disorder, arising from somatic mutations in multiple genes that affect cell division and proliferation. If one or more mutations is inherited, then fewer additional mutations are required for cancer to develop.
- A mutation that allows a cell to divide rapidly provides the cell with a growth advantage; this cell gives rise to a clone of cells with the same mutation. Within this clone, other mutations occur that provide additional growth advantages, and cells with these additional mutations become dominant in the clone. In this way, the clone evolves. Environmental factors play an important role in the development of many cancers by increasing the rate of somatic mutations.
- Several types of genes contribute to cancer progression. Oncogenes are dominant mutated copies of genes that normally stimulate cell division. Tumor-suppressor genes normally inhibit cell division; recessive mutations in these genes may contribute to cancer. Oncogenes and tumor-suppressor genes often control the cell cycle or regulate apoptosis.
- Defects in DNA repair genes and genes that control chromosome segregation often increase the overall mutation rate of other genes, leading to defects in proto-oncogenes and tumor-suppressor genes that may contribute to cancer progression.
- Mutations in sequences that regulate telomerase, an enzyme that replicates the ends of chromosomes, are often associated with cancer. Telomerase allows cells to divide indefinitely but is not usually expressed in somatic cells. Mutations in tumor cells allow telomerase to be expressed.
- Tumor progression is also affected by mutations in genes that promote vascularization and the spread of tumors.

- Colorectal cancer offers a model system for understanding tumor progression in humans. Initial mutations stimulate cell division, leading to a small benign polyp. Additional mutations allow the polyp to enlarge, invade the muscle layer of the gut, and eventually spread to other sites. Mutations in particular genes affect different stages of this progression.

IMPORTANT TERMS

totipotent (p. 000)
determination (p. 000)
egg polarity gene (p. 000)
morphogen (p. 000)
segmentation gene (p. 000)
gap gene (p. 000)
pair-rule gene (p. 000)
segment-polarity gene (p. 000)
homeotic gene (p. 000)
homeobox (p. 000)
Antennapedia complex (p. 000)
bithorax complex (p. 000)
homeotic complex (p. 000)
Hox gene (p. 000)
apoptosis (p. 000)
caspase (p. 000)
antigen (p. 000)
autoimmune disease (p. 000)
humoral immunity (p. 000)
B cell (p. 000)
antibody (p. 000)
cellular immunity (p. 000)
T cell (p. 000)
T-cell receptor (p. 000)
major histocompatibility complex (MHC) antigen (p. 000)
theory of clonal selection (p. 000)
primary immune response (p. 000)
memory cell (p. 000)
secondary immune response (p. 000)
somatic recombination (p. 000)
junctional diversity (p. 000)
somatic hypermutation (p. 000)
malignant tumor (p. 000)
metastasis (p. 000)
clonal evolution (p. 000)
oncogene (p. 000)
tumor-suppressor gene (p. 000)
proto-oncogene (p. 000)

Worked Problems

1. If a fertilized *Drosophila* egg is punctured at the anterior end and a small amount of cytoplasm is allowed to leak out, what will be the most likely effect on the development of the fly embryo?

• Solution

The egg-polarity genes determine the major axes of development in the *Drosophila* embryo. One of these genes is *bicoid,* which is transcribed in the maternal parent. As *bicoid* mRNA passes into the egg, the mRNA becomes anchored to the anterior end of the egg. After the egg is laid, *bicoid* mRNA is translated into Bicoid protein, which forms a concentration gradient along the anterior–posterior axis of the embryo. The high concentration of Bicoid protein at the anterior end induces the development of anterior structures such as the head of the fruit fly. If the anterior end of the egg is punctured, cytoplasm containing high concentrations of Bicoid protein will leak out, reducing the concentration of Bicoid protein at the anterior end. The result will be that the embryo fails to develop head and thoracic structures at the anterior end.

2. In some cancer cells, a specific gene has become duplicated many times. Is this gene likely to be an oncogene or a tumor-suppressor gene? Explain your reasoning.

• Solution

The gene is likely to be an oncogene. Oncogenes stimulate cell proliferation and act in a dominant manner. Therefore, extra copies of an oncogene will result in cell proliferation and cancer. Tumor-suppressor genes, on the other hand, suppress cell proliferation and act in a recessive manner; a single copy of a tumor-suppressor gene is sufficient to prevent cell proliferation. Therefore extra copies of the suppressor gene will not lead to cancer.

3. The immunoglobulin molecules of a particular mammalian species has kappa and lambda light chains and heavy chains. The kappa gene consists of 250 *V* and 8 *J* segments. The lambda gene contains 200 *V* and 4 *J* segments. The gene for the heavy chain consists of 300 *V*, 8 *J*, and 4 *D* segments. Considering just somatic recombination and random combinations of light and heavy chains, how many different types of antibodies can be produced by this species?

• Solution

For the kappa light chain, there are $250 \times 8 = 2000$ combinations; for the lambda light chain, there are $200 \times 4 = 800$ combinations; so a total of 2800 different types of light chains are possible. For the heavy chains, there are $300 \times 8 \times 4 = 9600$ possible types. Any of the 2800 light chains can combine with any of the 9600 heavy chains; so there are $2800 \times 9600 = 26{,}880{,}000$ different types of antibodies possible from somatic recombination and random combination alone. Junctional diversity and somatic hypermutation would greatly increase this diversity.

COMPREHENSION QUESTIONS

* 1. What experiments suggested that genes are not lost or permanently altered in development?

2. Briefly explain how the Dorsal protein is redistributed in the formation of the *Drosophila* embryo and how this redistribution helps to establish the dorsal–ventral axis of the early embryo.

* 3. Briefly describe how the *bicoid* and *nanos* genes help to determine the anterior–posterior axis of the fruit fly.

* 4. List the three major classes of segmentation genes and outline the function of each.

5. What role do homeotic genes play in the development of fruit flies?

* 6. What is apoptosis and how is it regulated?

* 7. Explain how each of the following processes contributes to antibody diversity.
   (a) Somatic recombination.
   (b) Junctional diversity.
   (c) Hypermutation.

8. What is the function of the MHC antigens? Why are the genes that encode these antigens so variable?

* 9. Outline Knudson's multistage theory of cancer and describe how it helps to explain unilateral and bilateral cases of retinoblastoma.

10. Briefly explain how cancer arises through clonal evolution.

*11. What is the difference between an oncogene and a tumor-suppressor gene? Give some examples of the functions of proto-oncogenes and tumor suppressers in normal cells.

12. Why do mutations in genes that encode DNA repair enzymes and chromosome segregation often produce a predisposition to cancer?

*13. What role do telomeres and telomerase play in cancer progression?

APPLICATION QUESTIONS AND PROBLEMS

14. If telomeres are normally shortened after each round of replication in somatic cells, what prediction would you make about the length of telomeres in Dolly, the first cloned sheep?

*15. Give examples of genes that affect development in fruit flies by regulating gene expression at the level of (a) transcription and (b) translation.

16. What would be the most likely effect on development of puncturing the posterior end of a *Drosophila* egg, allowing a small amount of cytoplasm to leak out, and then injecting that cytoplasm into the anterior end of another egg?

*17. What would be the most likely result of injecting *bicoid* mRNA into the posterior end of a *Drosophila* embryo and inhibiting the translation of *nanos* mRNA?

18. What would be the most likely effect of inhibiting the translation of *hunchback* mRNA throughout the embryo?

*19. Molecular geneticists have performed experiments in which they altered the number of copies of the *bicoid* gene in flies, affecting the amount of Bicoid protein produced.
   (a) What would be the effect on development of an increased number of copies of the *bicoid* gene?
   (b) What would be the effect of a decreased number of copies of *bicoid*?
   Justify your answers.

20. What would be the most likely effect on fruit-fly development of a deletion in the *nanos* gene?

21. Give an example of a gene found in each of the categories of genes (egg-polarity, gap, pair-rule, and so forth) listed in Figure 21.12.

*22. In a particular species, the gene for the kappa light chain has 200 *V* gene segments and 4 *J* segments. In the gene for the lambda light chain, this species has 300 *V* segments and 6 *J* segments. Considering only the variability arising from somatic recombination, how many different types of light chains are possible?

23. In the fictional book *Chromosome 6* by Robin Cook, a biotechnology company genetically engineers individual bonobos (a type of chimpanzee) to serve as future organ donors for clients. The genes of the bonobos are altered so that no tissue rejection takes place when their organs are transplanted into a client. What genes would need to be altered for this scenario to work? Explain your answer.

*24. A couple has one child with bilateral retinoblastoma. The mother is free from cancer, but the father had unilateral retinoblastoma and he has a brother who has bilateral retinoblastoma.
   (a) If the couple has another child, what is the probability that this next child will have retinoblastoma?
   (b) If the next child has retinoblastoma, is it likely to be bilateral or unilateral?
   (c) Propose an explanation for why the father's case of retinoblastoma was unilateral, whereas his sons and brother's cases are bilateral.

25. Some cancers are consistently associated with the deletion of a particular part of a chromosome. Does the deleted region contain an oncogene or a tumor-suppressor gene? Explain why.

26. Cells in a tumor contain mutated copies of a particular gene that promotes tumor growth. Gene therapy can be used to introduce a normal copy of this gene into the tumor cells. Would you expect this therapy to be effective if the mutated gene were an oncogene? A tumor-suppressor gene? Explain your reasoning.

CHALLENGE QUESTIONS

27. As we have learned in this chapter, the Nanos protein inhibits the translation of *hunchback* mRNA, thus lowering the concentration of Hunchback protein at the posterior end of a fruit-fly embryo and stimulating the differentiation of posterior characteristics. The results of experiments have demonstrated that the action of Nanos on *hunchback* mRNA depends on the presence of an 11-base sequence that is located in the 3′ untranslated region of *hunchback* mRNA. This sequence has been termed the Nanos response element (NRE). There are two copies of NREs in the trailer of *hunchback* mRNA. If a copy of NRE is added to the 3′ untranslated region of another mRNA produced by a different gene, the mRNA now becomes repressed by Nanos. The repression is greater if several NREs are added. On the basis of these observations, propose a mechanism for how Nanos inhibits Hunchback translation.
28. Offer a possible explanation for the widespread distribution of *Hox* genes among animals.
29. Many cancer cells are immortal (will divide indefinitely) because they have mutations that allow telomerase to be expressed. How might this knowledge be used to design anticancer drugs?

SUGGESTED READINGS

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McKinnell, R. G., and M. A. Di Berardino. 1999. The biology of cloning: history and rationale. *Bioscience* 49:875–885.

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Pennisi, E., and G. Vogel. 2000. Clones: a hard act to follow. *Science* 288:1722–1727.

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Raff, M. 1998. Cell suicide for beginners. *Nature* 396:119–122.

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Gellert, M. 2002. V(D)J recombination: RAG proteins, repair factors, and regulation. *Annual Review of Biochemistry* 71:101–132.

A review of the mechanism of recombination that leads to antibody diversity.

Leder, P. 1982. The genetics of antibody diversity. *Scientific American* 247(5):102–115.

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Discusses the first successful attempt to convert normal human cells into cancer cells by artificially introducing telomerase-expressing genes, oncogenes, and tumor-suppressor genes into a cell.

22 Quantitative Genetics

Quantitative genetic methods are being used to improve and predict racing speed in thoroughbred horses.

Thoroughbred Winners Through Quantitative Genetics

For more than 300 years, thoroughbred horses have been raised for a single purpose—to win at the racetrack. The origin of these horses can be traced to a small group that was imported to England from North Africa and the Middle East in the 1600s. The population of racing horses remained small until the 1800s, when horse racing became increasing popular; today there are approximately half a million thoroughbred horses worldwide.

Breeding and racing thoroughbred horses is a multibillion-dollar industry that relies on the premise that a horse's speed is inherited. Speed is not, however, a simple genetic characteristic such as seed shape in peas. Numerous genes and nongenetic factors such as diet, training, and the jockey who rides the horse all contribute to a horse's success or failure in a race. The inheritance of racing speed in thoroughbreds is more complex than that of any of the characteristics that we have studied up to this point. Can the inheritance of a complex characteristic such as racing speed be studied? Is it possible to predict the speed of a horse on the basis of its pedigree? The answers are yes—at least in part—but these questions cannot be addressed with the methods that we used for simple genetic characteristics. Instead, we must use statistical procedures that have been developed for analyzing complex characteristics. The genetic analysis of complex characteristics such as racing speed of thoroughbreds is known as **quantitative genetics.**

Although the mathematical methods for analyzing complex characteristics may seem to be imposing at first, most people can intuitively grasp the underlying logic of quantitative genetics. We all recognize family resemblance: we talk about inheriting our father's height or our mother's intelligence. Family resemblance lies at the heart of the statistical methods used in quantitative genetics. When genes influence a characteristic, related individuals resemble one another more than unrelated individuals. Closely related individuals (such as siblings) should resemble one another more than distantly related individuals (such as cousins). Comparing individuals with different degrees of relatedness, then, provides information about the extent to which genes influence a characteristic.

This type of analysis has been applied to the inheritance of racing speed in thoroughbreds. In 1988, Patrick Cunningham and his colleagues examined records of more than 30,000 3-year-old horses that raced between 1961 and 1985. They reasoned that, if genes influence racing success, a horse's racing success should be more similar to that of its parents than to that of unrelated horses. Similarly, the racing speeds of half-brothers and half-sisters should be more similar than the speeds of unrelated horses are. When Cunningham and his colleagues statistically analyzed the racing records for thoroughbreds, they found that a considerable amount of variation in track performance was due to genetic differences—racing speed is heritable. With the use of statistics, it is possible to estimate, with some degree of accuracy, the track performance of a horse from the performance of its relatives.

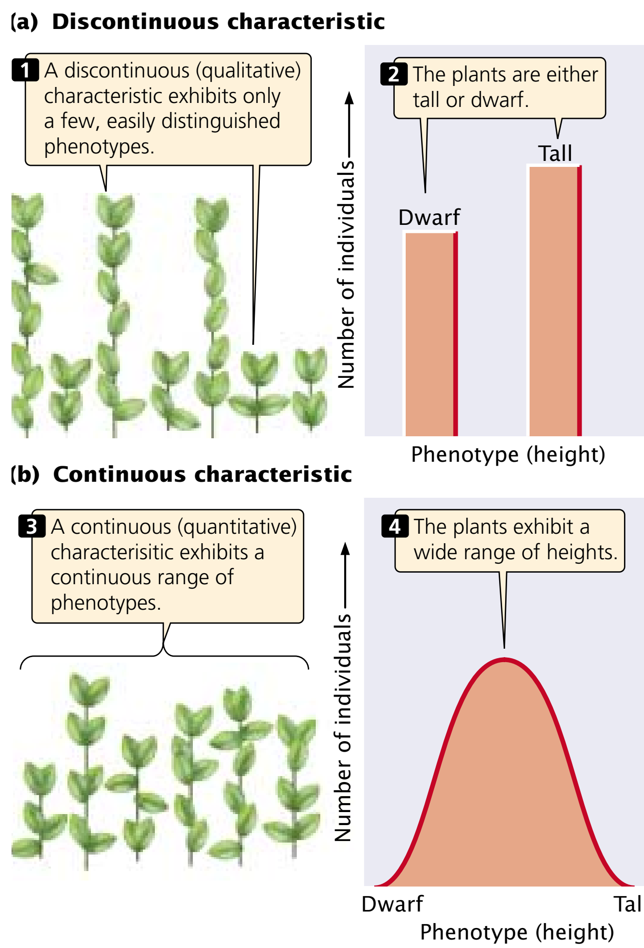
This chapter is about the genetic analysis of complex characteristics such as racing speed. We begin by considering the differences between quantitative and qualitative characteristics and why the expression of some characteristics varies continuously. We'll see how quantitative characteristics are often influenced by many genes, each of which has a small effect on the phenotype. Next, we will examine statistical procedures for describing and analyzing quantitative characteristics. We will consider the question of how much of phenotypic variation can be attributed to genetic and environmental influences and will conclude looking at the effects of selection on quantitative characteristics. It's important to recognize that the methods of quantitative genetics are not designed to identify individual genes and genotypes. Rather, the focus is on statistical predictions based on groups of individuals.

www.whfreeman.com/pierce More information on horse genetics, including the genetic basis of coat color, gene mapping, chromosomes, and genetic disorders

Quantitative Characteristics

Qualitative, or discontinuous, characteristics possess only a few distinct phenotypes (FIGURE 22.1a); these characteristics are the types studied by Mendel and have been the focus of our attention thus far. However, many characteristics vary continuously along a scale of measurement with many overlapping phenotypes (FIGURE 22.1b). They are referred to as *continuous characteristics;* they are also called *quantitative characteristics* because any individual's phenotype must be described with a quantitative measurement. Quantitative characteristics might include height, weight, and blood pressure in humans, growth rate in mice, seed weight in plants, and milk production in cattle.

Quantitative characteristics arise from two phenomena. First, many are polygenic—they are influenced by



22.1 Discontinuous and continuous characteristics differ in the number of phenotypes exhibited.

genes at many loci. If many loci take part, many genotypes are possible, each producing a slightly different phenotype. Second, quantitative characteristics often arise when environmental factors affect the phenotype, because environmental differences result in a single genotype producing a range of phenotypes. Most continuously varying characteristics are *both* polygenic *and* influenced by environmental factors, and these characteristics are said to be multifactorial.

The Relation Between Genotype and Phenotype

For many discontinuous characteristics, there is a relatively straightforward relation between genotype and phenotype. Each genotype produces a single phenotype, and most phenotypes are encoded by a single genotype. Dominance and epistasis may allow two or three genotypes to produce the same phenotype, but the relation remains relatively simple. This simple relation between genotype and phenotype allowed Mendel to decipher the basic rules of inheritance from his crosses with pea plants; it also permits us both to predict the outcome of genetic crosses and to assign genotypes to individuals.

For quantitative characteristics, the relation between genotype and phenotype is often more complex. If the characteristic is polygenic, many different genotypes are possible, several of which may produce the same phenotype. For instance, consider a plant whose height is determined by three loci (A, B, and C), each of which has two alleles. Assume that one allele at each locus (A^+, B^+, and C^+) encodes a plant hormone that causes the plant to grow 1 cm above its baseline height of 10 cm. The second allele at each locus (A^-, B^-, and C^-) encodes no plant hormone and does not contribute to additional height. Considering only the two alleles at a single locus, 3 genotypes are possible (A^+A^+, A^+A^-, and A^-A^-). If all three loci are taken into account, there are a total of $3^3 = 27$ possible multilocus genotypes ($A^+A^+B^+B^+C^+C^+$, $A^+A^-B^+B^+C^+C^+$, etc.). Although there are 27 genotypes, they produce only seven phenotypes (10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, and 16 cm in height). Some of the genotypes produce the same phenotype (Table 22.1); for example, genotypes $A^+A^-B^-B^-C^-C^-$, $A^-A^-B^+B^-C^-C^-$, and $A^-A^-B^-B^-C^+C^-$ all have one gene that encodes plant hormone. These genotypes produce one dose of the hormone and a plant that is 11 cm tall. Even in this simple example of only three loci, the relation between genotype and phenotype is quite complex. The more loci encoding a characteristic, the greater the complexity.

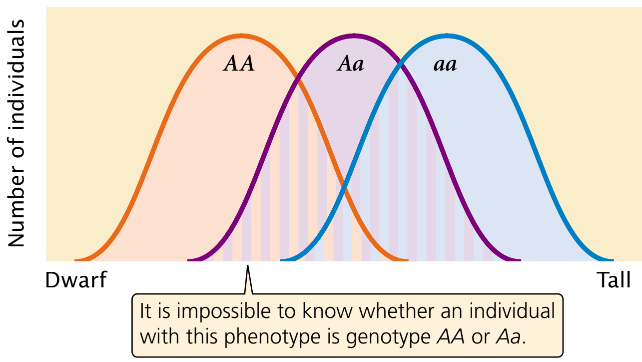
The influence of environment on a characteristic also can complicate the relation between genotype and phenotype. Because of environmental effects, the same genotype may produce a range of potential phenotypes (the norm of reaction; see p. 00 in Chapter 5). The phenotypic ranges of different genotypes may overlap, making it difficult to know whether individuals differ in phenotype because of genetic or environmental differences (FIGURE 22.2).

Table 22.1 Hypothetical example of plant height determined by pairs of alleles at each of three loci

Genotype	Doses of Plant Hormone	Height (cm)
$A^-A^-B^-B^-C^-C^-$	0	10
$A^+A^-B^-B^-C^-C^-$	1	11
$A^-A^-B^+B^-C^-C^-$		
$A^-A^-B^-B^-C^-C^+$		
$A^+A^+B^-B^-C^-C^-$	2	12
$A^-A^-B^+B^+C^-C^-$		
$A^-A^-B^-B^-C^+C^+$		
$A^+A^-B^+B^-C^-C^-$		
$A^+A^-B^-B^-C^+C^-$		
$A^-A^-B^+B^-C^+C^-$		
$A^+A^+B^+B^-C^-C^-$	3	13
$A^+A^+B^-B^-C^+C^-$		
$A^+A^-B^+B^+C^-C^-$		
$A^-A^-B^+B^+C^+C^-$		
$A^+A^-B^-B^-C^+C^+$		
$A^-A^-B^+B^-C^+C^+$		
$A^+A^-B^+B^-C^+C^-$		
$A^+A^+B^+B^+C^-C^-$	4	14
$A^+A^+B^+B^-C^+C^-$		
$A^+A^-B^+B^+C^+C^-$		
$A^-A^-B^+B^+C^+C^+$		
$A^+A^+B^-B^-C^+C^+$		
$A^-A^-B^+B^-C^+C^+$		
$A^+A^+B^+B^+C^+C^-$	5	15
$A^+A^-B^+B^+C^+C^+$		
$A^+A^+B^+B^-C^+C^+$		
$A^+A^+B^+B^+C^+C^+$	6	16

Note: Each + allele contributes 1 cm in height above a baseline of 10 cm.

In summary, the simple relation between genotype and phenotype that exists for many qualitative (discontinuous) characteristics is absent in quantitative characteristics, and it is impossible to assign a genotype to an individual on the basis of its phenotype alone. The methods used for analyzing qualitative characteristics (examining the phenotypic ratios of progeny from a genetic cross) will not work with quantitative characteristics. Our goal remains the same: we wish to make predictions about the phenotypes of offspring produced in a genetic cross. We may also want to know how much of the variation in a characteristic results from genetic differences and how much results from environmental differences. To answer these questions, we must turn to statistical methods that allow us to make predictions about the inheritance of phenotypes in the absence of information about the underlying genotypes.



22.2 For a quantitative characteristic, each genotype may produce a range of possible phenotypes. In this hypothetical example, the phenotypes produced by genotypes *AA*, *Aa*, and *aa* overlap.

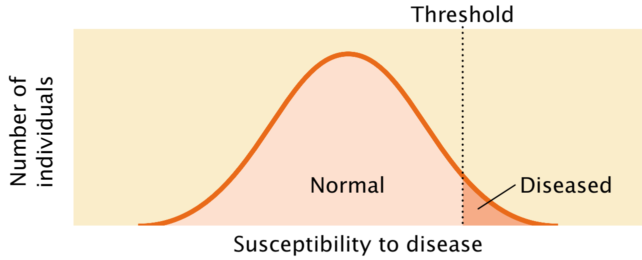
www.whfreeman.com/pierce Information about some current research in quantitative genetics

Types of Quantitative Characteristics

Before we look more closely at polygenic characteristics and relevant statistical methods, we need to more clearly define what is meant by a quantitative characteristic. Thus far, we have considered only quantitative characteristics that vary continuously in a population. A *continuous characteristic* can theoretically assume any value between two extremes; the number of phenotypes is limited only by our ability to precisely measure the phenotype. Human height is a continuous characteristic because, within certain limits, people can theoretically have any height. Although the number of phenotypes possible with a continuous characteristic is infinite, we often group similar phenotypes together for convenience; we may say that two people are both 5 feet 11 inches tall, but careful measurement may show that one is slightly taller than the other.

Some characteristics are not continuous but are nevertheless considered quantitative because they are determined by multiple genetic and environmental factors. **Meristic characteristics,** for instance, are measured in whole numbers. An example is litter size: a female mouse may have 4, 5, or 6 pups but not 4.13 pups. A meristic characteristic has a limited number of distinct phenotypes, but the underlying determination of the characteristic may still be quantitative. These characteristics must therefore be analyzed with the same techniques that we use to study continuous quantitative characteristics.

Another type of quantitative characteristic is a **threshold characteristic,** which is simply present or absent. Although threshold characteristics exhibit only two phenotypes, they are considered quantitative because they, too, are determined by multiple genetic and environmental factors. The expression of the characteristic depends on an underlying susceptibility (usually referred to as liability or risk) that



22.3 Threshold characteristics display only two possible phenotypes—the trait is either present or absent—but they are quantitative because the underlying susceptibility to the characteristic varies continuously. When the susceptibility exceeds a threshold value, the characteristic is expressed.

varies continuously. When the susceptibility is larger than a threshold value, a specific trait is expressed (FIGURE 22.3). Diseases are often threshold characteristics because many factors, both genetic and environmental, contribute to disease susceptibility. If enough of the susceptibility factors are present, the disease develops; otherwise, it is absent. Although we focus on the genetics of continuous characteristics in this chapter, the same principles apply to many meristic and threshold characteristics.

It is important to point out that just because a characteristic can be measured on a continuous scale does not mean that it exhibits quantitative variation. One of the characteristics studied by Mendel was height of the pea plant, which can be described by measuring the length of the plant's stem. However, Mendel's particular plants exhibited only two distinct phenotypes (some were tall and others short), and these differences were determined by alleles at a single locus. The differences that Mendel studied were therefore discontinuous in nature.

Concepts

Characteristics whose phenotypes vary continuously are called quantitative characteristics. For most quantitative characteristics, the relation between genotype and phenotype is complex. Some characteristics whose phenotypes do not vary continuously also are considered quantitative because they are influenced by multiple genes and environmental factors.

Polygenic Inheritance

The rediscovery of Mendel's work in 1900 provided a cohesive theory of inheritance, but the characteristics that Mendel studied were all discontinuous. Questions soon arose about the inheritance of continuously varying characteristics. These characteristics had already been the focus of a group of biologists and statisticians, led by Francis Galton, who were known as biometricians. They examined the inheritance of quantitative characteristics such as human height and intelligence by using statistical procedures. The results of these studies showed that quantitative characteristics are inherited, although the mechanism of inheritance was as yet unknown. After Mendel's work was rediscovered, a bitter dispute broke out about whether Mendel's principles applied to quantitative characteristics. Some biometricians argued that the inheritance of quantitative characteristics could not be explained by Mendelian principles, whereas others felt that Mendel's principles acting on numerous genes (polygenes) could adequately account for the inheritance of quantitative characteristics.

This conflict began to be resolved by the work of Wilhelm Johannsen, who showed that continuous variation in the weight of beans was influenced by both genetic and environmental factors. George Udny Yule, a mathematician, proposed in 1906 that several genes acting together could produce continuous characteristics. This hypothesis was later confirmed by Herman Nilsson-Ehle, working on wheat and tobacco, and by Edward East, working on corn. The argument was finally laid to rest in 1918, when Ronald Fisher demonstrated that the inheritance of quantitative characteristics could indeed be explained by the cumulative effects of many genes, each following Mendel's rules.

Kernel Color in Wheat

To illustrate how multiple genes acting on a characteristic can produce a continuous range of phenotypes, let us examine one of the first demonstrations of polygenic inheritance. Nilsson-Ehle studied kernel color in wheat and found that the intensity of red pigmentation was determined by three unlinked loci, each of which had two alleles.

Nilsson-Ehle obtained several homozygous varieties of wheat that differed in color. Like Mendel, he performed crosses between these homozygous varieties and studied the ratios of phenotypes in the progeny. In one experiment, he crossed a variety of wheat that possessed white kernels with a variety that possessed purple (very dark red) kernels and obtained the following results:

P plants with white kernels $\times$ plants with purple kernels

$\downarrow$

F_1 plants with red kernels

$\downarrow$

F_2
$^1/_{16}$ plants with purple kernels
$^4/_{16}$ plants with dark-red kernels
$^6/_{16}$ plants with red kernels
$^4/_{16}$ plants with light-red kernels
$^1/_{16}$ plants with white kernels

Nilsson-Ehle interpreted this phenotypic ratio as the result of segregation of alleles at two loci. (Although he found alleles at three loci that affected kernel color, the two varieties used in this cross differed only at two of the loci.) He proposed that there were two alleles at each locus: one that produced red pigment and another that produced no pigment. We'll designate the alleles that encoded pigment A^+ and B^+ and the alleles that encoded no pigment A^- and B^-. Nilsson-Ehle recognized that the effects of the genes were additive. Each gene seemed to contribute equally to color; so the overall phenotype could be determined by adding the effects of all the genes, as shown in this table.

Genotype	Doses of pigment	Phenotype
$A^+A^+B^+B^+$	4	purple
$A^+A^+B^+B^-$ $A^+A^-B^+B^+$	3	dark red
$A^+A^+B^-B^-$ $A^-A^-B^+B^+$ $A^+A^-B^+B^-$	2	red
$A^+A^-B^-B^-$ $A^-A^-B^+B^-$	1	light red
$A^-A^-B^-B^-$	0	white

Notice that the purple and white phenotypes are each encoded by a single genotype, but other phenotypes may result from several different genotypes.

From these results, we see that five phenotypes are possible when alleles at two loci influence the phenotype and the effects of the genes are additive. When alleles at more than two loci influence the phenotype, more phenotypes are possible, and this would make the color appear to vary continuously between white and purple. If environmental factors had influenced the characteristic, individuals of the same genotype would vary somewhat in color, making it even more difficult to distinguish between discrete phenotypic classes. Luckily, environment played little role in determining kernel color in Nilsson-Ehle's crosses, and only a few loci encoded color; so Nilsson-Ehle was able to distinguish among the different phenotypic classes. This ability allowed him to see the Mendelian nature of the characteristic.

Let's now see how Mendel's principles explain the ratio obtained by Nilsson-Ehle in his F_2 progeny. Remember that Nilsson-Ehle crossed a homozygous purple variety ($A^+A^+B^+B^+$) with the homozygous white variety ($A^-A^-B^-B^-$), producing F_1 progeny that were heterozygous at both loci ($A^+A^-B^+B^-$). All the F_1 plants possessed two pigment-producing alleles that allowed two doses of color to make red kernels. The types and proportions of progeny expected in the F_2 can be found by applying Mendel's principles of segregation and independent assortment.

Let's first examine the effects of each locus separately. At the first locus, two heterozygous F_1s are crossed ($A^+A^- \times A^+A^-$). As we learned in Chapter 3, when two heterozygotes are crossed, we expect progeny in the proportions

$^1/_4\ A^+A^+$, $^1/_2\ A^+A^-$, and $^1/_4\ A^-A^-$. At the second locus, two heterozygotes also are crossed, and again we expect progeny in the proportions $^1/_4\ B^+B^+$, $^1/_2\ B^+B^-$, and $^1/_4\ B^-B^-$.

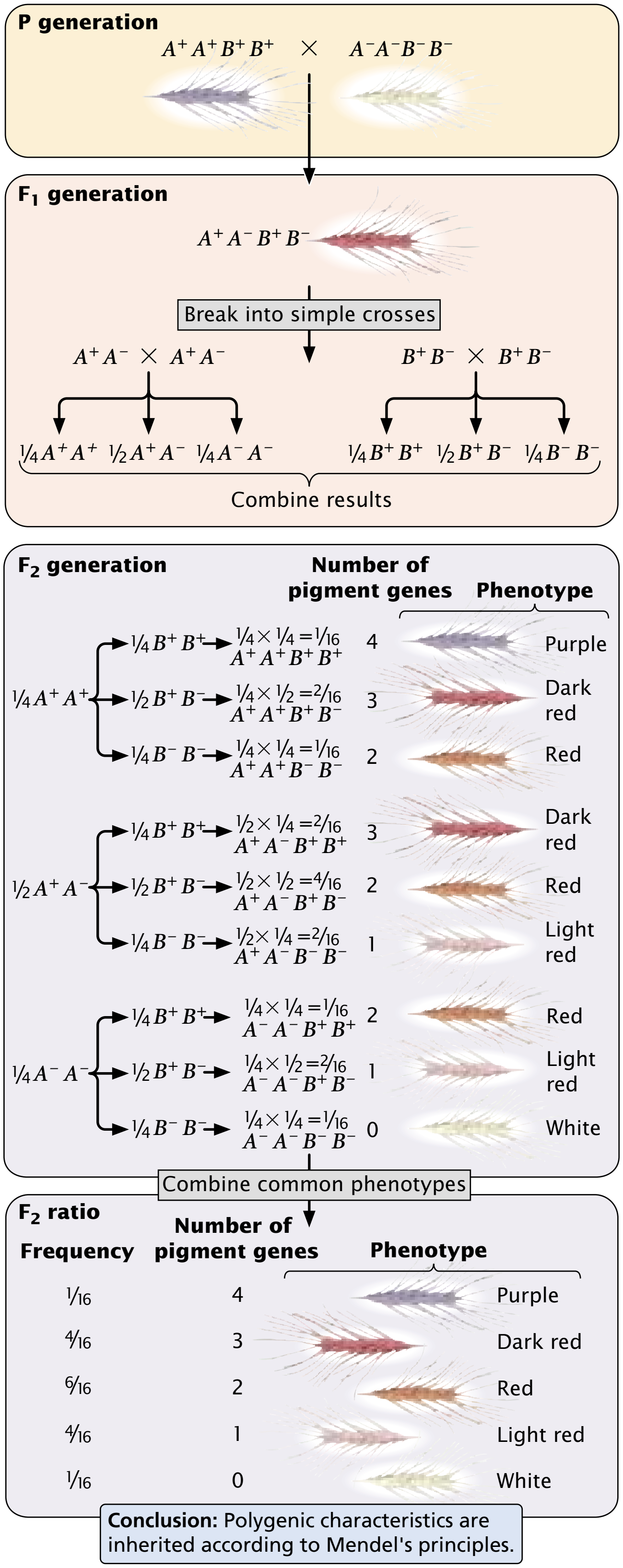
To obtain the probability of combinations of genes at both loci, we must use the multiplication rule of probability (p. 000 in Chapter 3), which is based on Mendel's principle of independent assortment. The expected proportion of F_2 progeny with genotype $A^+A^+B^+B^+$ is the product of the probability of obtaining genotype A^+A^+ ($^1/_4$) and the probability of obtaining genotype B^+B^+ ($^1/_4$), or $^1/_4 \times {}^1/_4 = {}^1/_{16}$ (FIGURE 22.4). The probabilities of each of the phenotypes can then be obtained by adding the probabilities of all the genotypes that produce that phenotype. For example, the red phenotype is produced by three genotypes:

Genotype	Probability
$A^+A^+B^-B^-$	$^1/_{16}$
$A^-A^-B^+B^+$	$^1/_{16}$
$A^+A^-B^+B^-$	$^1/_4$

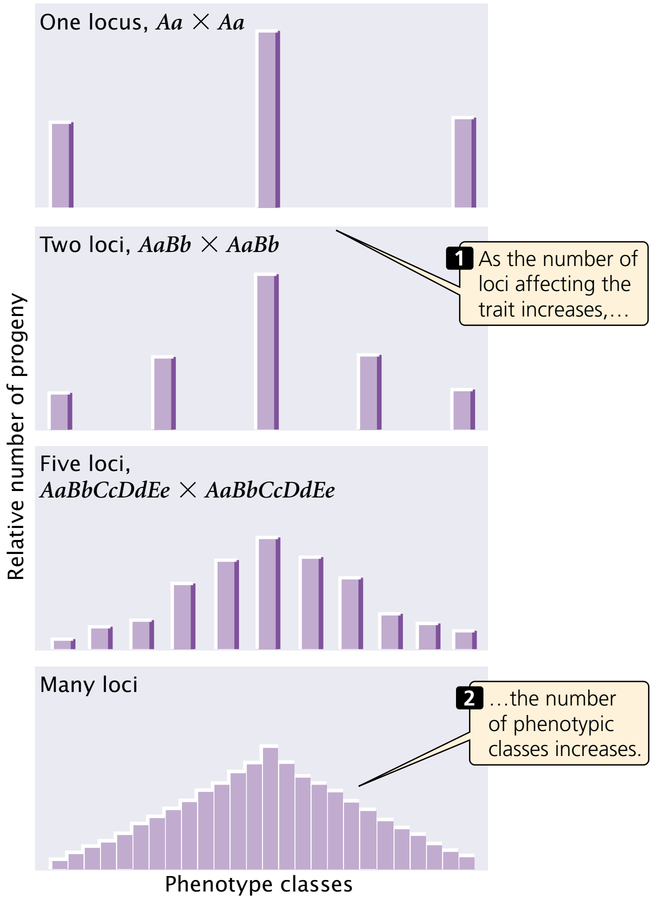
Thus, the overall probability of obtaining red kernels in the F_2 progeny is $^1/_{16} + {}^1/_{16} + {}^1/_4 = {}^6/_{16}$. Figure 22.4 shows that the phenotypic ratio expected in the F_2 is $^1/_{16}$ purple, $^4/_{16}$ dark red, $^6/_{16}$ red, $^4/_{16}$ light red, and $^1/_{16}$ white. This phenotypic ratio is precisely what Nilsson-Ehle observed in his F_2 progeny, demonstrating that the inheritance of a continuously varying characteristic such as kernel color is indeed according to Mendel's basic principles.

Nilsson-Ehle's crosses demonstrated that the difference between the inheritance of genes influencing quantitative characteristics and the inheritance of genes influencing discontinuous characteristics is in the *number* of loci that determine the characteristic. When multiple loci affect a character, more genotypes are possible; so the relation between the genotype and the phenotype is less obvious. As the number of loci affecting a character increases, the number of phenotypic classes in the F_2 increases (FIGURE 22.5).

Several conditions of Nilsson-Ehle's crosses greatly simplified the polygenic inheritance of kernel color and made it possible for him to recognize the Mendelian nature of the characteristic. First, genes affecting color segregated at only two or three loci. If genes at many loci had been segregating, he would have had difficulty in distinguishing the phenotypic classes. Second, the genes affecting kernel color had strictly additive effects, making the relation between genotype and phenotype simple. Third, environment played almost no role in the phenotype; had environmental factors



22.4 Nilsson-Ehle demonstrated that kernel color in wheat is inherited according to Mendelian principles. He crossed two varieties of wheat that differed in pairs of alleles at two loci affecting kernel color. A purple strain ($A^+A^+B^+B^+$) was crossed with a white strain ($A^-A^-B^-B^-$), and the F_1 was intercrossed to produce F_2 progeny. The ratio of phenotypes in the F_2 can be determined by breaking the dihybrid cross into two simple single-locus crosses and combining the results with the multiplication rule.



22.5 The results of crossing individuals heterozygous for different numbers of loci affecting a characteristic.

modified the phenotypes, distinguishing between the five phenotypic classes would have been difficult. Finally, the loci that Nilsson-Ehle studied were not linked; so the genes assorted independently. Nilsson-Ehle was fortunate—for many polygenic characteristics, these simplifying conditions are not present and Mendelian inheritance of these characteristics is not obvious.

Determining Gene Number for a Polygenic Characteristic

When two individuals homozygous for different alleles at a single locus are crossed ($A^1A^1 \times A^2A^2$) and the resulting F_1 are interbred ($A^1A^2 \times A^1A^2$), one-fourth of the F_2 should be homozygous like each of the original parents. If the original parents are homozygous for different alleles at *two* loci, as are those in Nilsson-Ehle's crosses, then $^1/_4 \times {}^1/_4 = {}^1/_{16}$ of the F_2 should resemble one of the original homozygous parents. Generally, $(^1/_4)^n$ will be the number of individuals in the F_2 progeny that should resemble each of the original homozygous parents, where n equals the number of loci with a segregating pair of alleles that affects the characteristic. This equation provides us with a possible means of determining the number of loci influencing a quantitative characteristic.

To illustrate the use of this equation, assume that we cross two different homozygous varieties of pea plants that differ in height by 16 cm, interbreed the F_1, and find that approximately $^1/_{256}$ of the F_2 are similar to one of the original homozygous parental varieties. This outcome would suggest that 4 loci with segregating pairs of alleles ($^1/_{256} = {}^1/_4{}^4$) are responsible for the height difference between the two varieties. Because the two homozygous strains differ in height by 16 cm and there are 4 loci each with two alleles (8 alleles in all), each of the alleles contributes 16 cm/8 = 2 cm in height.

This method for determining the number of loci affecting phenotypic differences requires the use of homozygous strains, which may be difficult to obtain in some organisms. It also assumes that all the genes influencing the characteristic have equal effects, that their effects are additive, and that the loci are unlinked. For many polygenic characteristics, these assumptions are not valid, so this method of determining the number of genes affecting a characteristic has limited application.

Concepts

The principles that determine the inheritance of quantitative characteristics are the same as the principles that determine the inheritance of discontinuous characteristics, but more genes take part in the determination of quantitative characteristics.

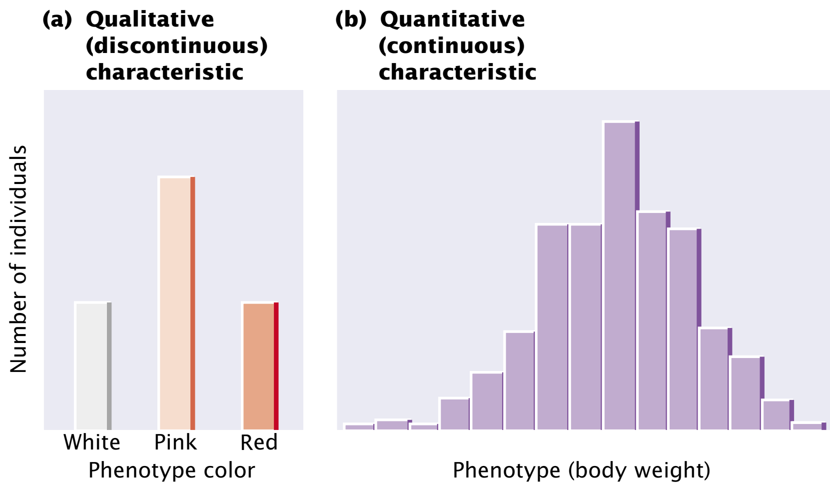
Statistical Methods for Analyzing Quantitative Characteristics

Because quantitative characteristics are described by a measurement and are influenced by multiple factors, their inheritance must be analyzed statistically. This section will explain the basic concepts of statistics that are used to analyze quantitative characteristics.

Distributions

Understanding the genetic basis of any characteristic begins with a description of the numbers and kinds of phenotypes present in a group of individuals. Phenotypic variation in a group, such as the progeny of a cross, can be conveniently represented by a **frequency distribution,** which is a graph of the frequencies (numbers or proportions) of the different phenotypes (FIGURE 22.6). In a typical frequency distribution, the phenotypic classes are plotted on the horizontal (x) axis and the numbers (or proportions) of individuals in each class on the vertical (y) axis. Unlike qualitative characteristics (FIGURE 22.6a), quantitative characteristics often exhibit many phenotypes, so a frequency distribution is a concise method of summarizing them all (FIGURE 22.6b).

Connecting the points of a frequency distribution with a line creates a curve that is characteristic of the distribution (FIGURE 22.7). Many quantitative characteristics exhibit a



22.6 A frequency distribution is a graph that displays the number or proportion of different phenotypes. Phenotypic values are plotted on the horizontal axis and the numbers (or proportions) of individuals in each class are plotted on the vertical axis.

symmetrical (bell-shaped) curve called a **normal distribution** (FIGURE 22.7a). Normal distributions arise when a large number of independent factors contribute to a measurement. Quantitative characteristics are frequently affected by numerous genes and environmental factors; so their phenotypes often exhibit normal distributions. Two other common types of distributions (skewed and bimodal) are illustrated in FIGURE 22.7b and c.

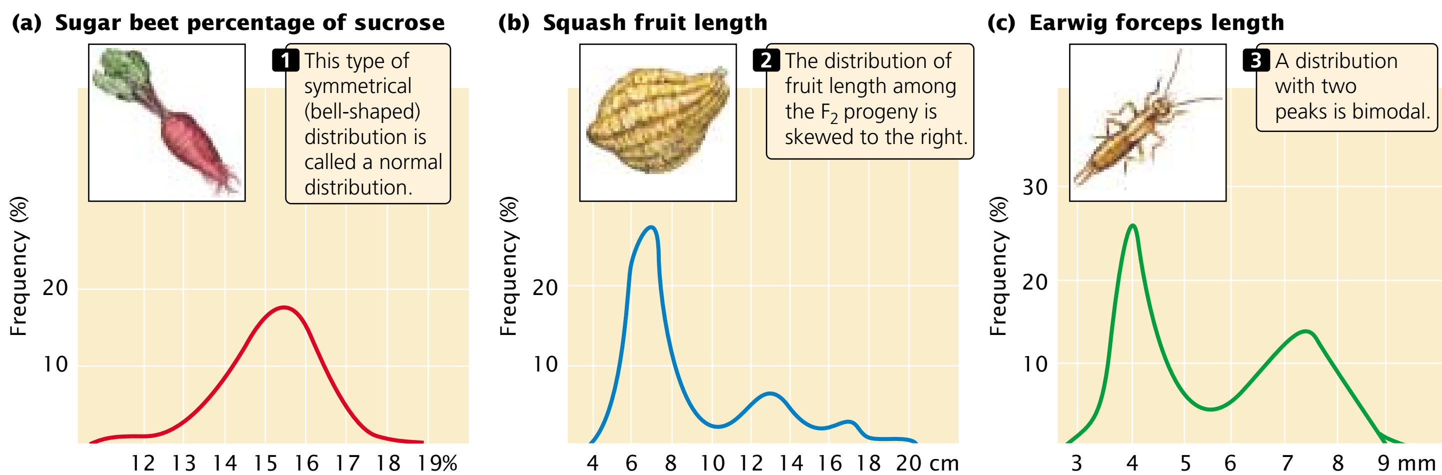
Samples and Populations

Biologists frequently need to describe the distribution of phenotypes exhibited by some group of individuals. We might want to describe the height of students at the University of Texas (UT), but there are more than 40,000 students at UT, and measuring every one of them would be impractical. Scientists are constantly confronted with this problem: the group of interest, called the **population,** is too large for a complete census. One solution is to measure a smaller collection of individuals, called a **sample,** and use measurements made on the sample to describe the population.

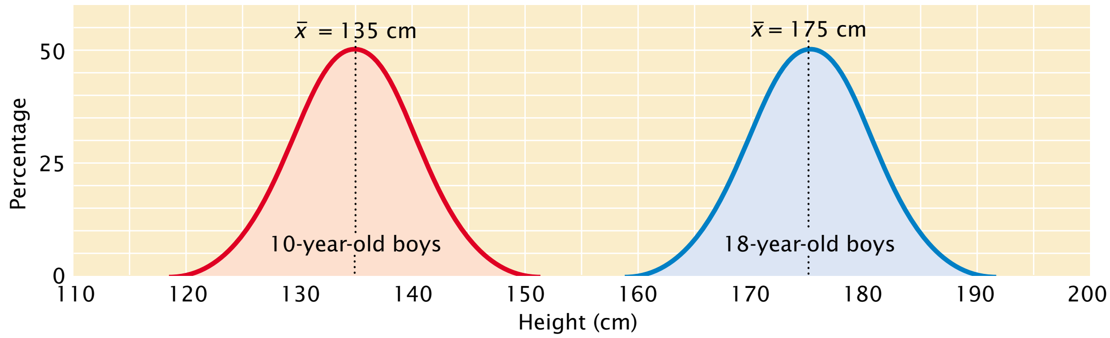
To provide an accurate description of the population, a good sample must have several characteristics. First, it must be representative of the whole population. If our sample consisted entirely of members of the UT basketball team, for instance, we would probably overestimate the true height of the students. One way to ensure that a sample is representative of the population is to select the members of the sample randomly. Second, the sample must be large enough that chance differences between individuals in the sample and the overall population do not distort the estimate of the population measurements. If we measured only three students at UT and just by chance all three were short, we would underestimate the true height of the student population. Statistics can provide information about how much confidence to expect from estimates based on random samples.

Concepts

In statistics, the population is the group of interest; a sample is a subset of the population. The sample should be representative of the population and large enough to minimize chance differences between the population and the sample.



22.7 Distributions of phenotypes may assume several different shapes.



22.8 The mean provides information about the center of a distribution. Both distributions of heights of 10-year-old and 18 year-old boys are normal, but they have different locations along a continuum of height, which makes their means different.

The Mean

The **mean,** also called the average, provides information about the center of the distribution. If we measured the heights of 10-year-old and 18-year-old boys and plotted a frequency distribution for each group, we would find that both distributions are normal, but the two distributions would be centered at different heights, and this difference would be indicated in their different means (FIGURE 22.8).

If we represent a group of measurements as x_1, x_2, x_3, and so forth, then the mean ($\bar{x}$) is calculated by adding all the individual measurements and dividing by the total number of measurements in the sample (n):

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n} \quad (22.1)$$

A shorthand way to represent this formula is

$$\bar{x} = \frac{\Sigma x_i}{n} \quad (22.2)$$

or

$$\bar{x} = \frac{1}{n}\sum x_i \quad (22.3)$$

where the symbol Σ means "the summation of" and x_i represents individual x values.

The Variance and Standard Deviation

A statistic that provides key information about a distribution is the **variance,** which indicates the variability of a group of measurements (how spread out the distribution is). Distributions may have the same mean but different variances (FIGURE 22.9). The larger the variance, the greater the spread of measurements in a distribution about its mean.

The variance (s^2) is defined as the average squared deviation from the mean:

$$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1} \quad (22.4)$$

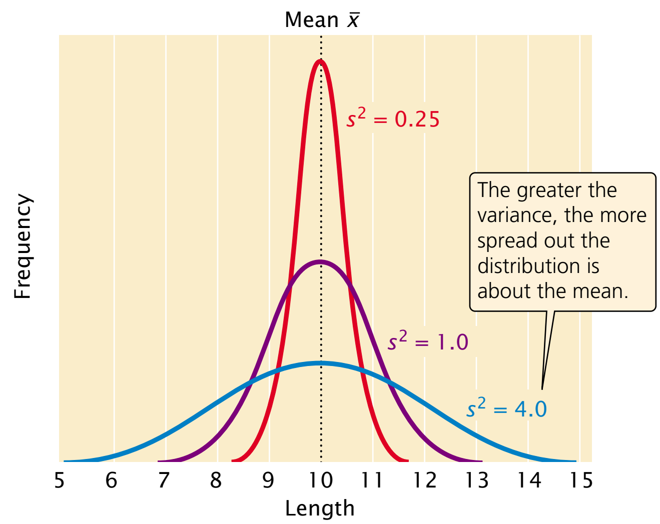
To calculate the variance, we (1) subtract the mean from each measurement and square the value obtained, (2) add all the squared deviations, and (3) divide this sum by the number of original measurements minus one.

Another statistic that is closely related to the variance is the **standard deviation** (s), which is defined as the square root of the variance:

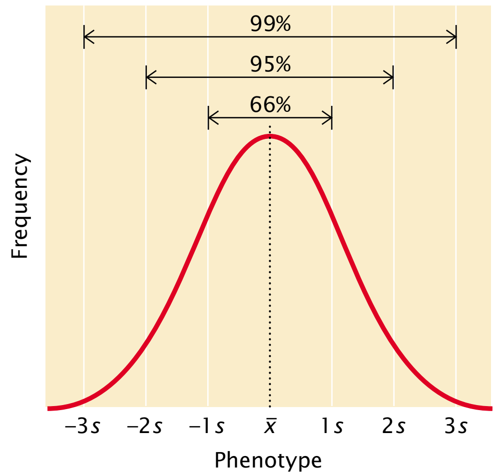
$$s^2 = \sqrt{s^2} \quad (22.5)$$

Whereas the variance is expressed in units squared, the standard deviation is in the same units as the original measurements; so the standard deviation is often preferred for describing the variability of a measurement.

A normal distribution is symmetrical; so the mean and standard deviation are sufficient to describe its shape. The mean plus or minus one standard deviation ($\bar{x} \pm s$) includes approximately 66% of the measurements in a normal distribution; the mean plus or minus two standard



22.9 The variance provides information about the variability of a group of phenotypes. Shown here are three distributions with the same mean but different variances.



22.10 The proportions of a normal distribution occupied by plus or minus one, two, and three standard deviations from the mean.

deviations ($\bar{x} \pm 2s$) includes approximately 95% of the measurements, and the mean plus or minus three standard deviations ($\bar{x} \pm 3s$) includes approximately 99% of the measurements (Figure 22.10). Thus, only 1% of a normally distributed population lies outside the range of $\bar{x} \pm 3s$.

Concepts

The mean and variance describe a distribution of measurements: the mean provides information about the location of the center of a distribution, and the variance provides information about its variability.

Worked Problem

The following table lists yearly amounts (in hundreds of pounds) of milk produced by 10 two-year-old Jersey cows. Calculate the mean, variance, and standard deviation of milk production for this sample of 10 cows.

Annual milk production (hundreds of pounds)
60
74
58
61
56
55
54
57
65
42

• Solution

The mean is calculated by using the following formula: $\bar{x} = \Sigma x_i / n$. The value of Σx_i is obtained by summing all the individual measurements, which equals 582; n is the total number of measurements, which equals 10; so $\bar{x} = (582/10) = 58.20$, or 58,200 pounds per year.

The variance is calculated by using the following formula:

$$s_x^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1}$$

so we need to determine the deviation of each individual measurement from the mean ($x_i - \bar{x}$), square each value, and sum the squared deviations from the mean.

Annual milk production (hundreds of pounds) x	Variance $x_i - \bar{x}$	Standard deviation $(x_i - \bar{x})^2$
60	1.80	3.24
74	15.80	249.64
58	−0.20	0.04
61	2.80	7.84
56	−2.20	4.84
55	−3.20	10.24
54	−4.20	17.64
57	−1.20	1.44
65	6.80	46.24
42	−16.20	262.44

$$\Sigma(x_i - \bar{x})^2 = 603.60$$

The variance is therefore:

$$s_x^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1} = \frac{603.60}{9} = 67.07$$

The standard deviation is the square root of the variance:

$$s_x = \sqrt{s_x^2} = \sqrt{67.07} = 8.19.$$

Correlation

The mean and the variance can be used to describe an individual characteristic, but geneticists are frequently interested in more than one characteristic. Often, two or more characteristics vary together. For instance, both the number and the weight of eggs produced by hens are important to the poultry industry. These two characteristics are not independent of each other. There is an inverse relation between egg number and weight: hens that lay more eggs produce smaller eggs. This kind of relation between two characteristics is called a **correlation.** When two characteristics are correlated, a change in one characteristic is likely to be associated with a change in the other.

Correlations between characteristics are measured by a **correlation coefficient** (designated r), which measures the strength of their association. Consider two characteristics, such as human height (x) and arm length (y). To determine how these characteristics are correlated, we first obtain the covariance (cov) of x and y:

$$\mathrm{cov}_{xy} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{n - 1} \qquad (22.6)$$

The covariance is computed by (1) taking an x value for an individual and subtracting it from the mean of x ($\bar{x}$); (2) taking the y value for the same individual and subtracting it from the mean of y ($\bar{y}$); (3) multiplying the results of these two subtractions; (4) adding the results for all the xy pairs; and (5) dividing this sum by $n - 1$ (where n equals the number of xy pairs).

The correlation coefficient (r) is obtained by dividing the covariance of x and y by the product of the standard deviations of x and y:

$$r = \frac{\mathrm{cov}_{xy}}{s_x s_y} \qquad (22.7)$$

A correlation coefficient can theoretically range from -1 to $+1$ (Figure 22.11). A positive value indicates that there is a direct association between the variables (Figure 22.11a); as one variable increases, the other variable also tends to increase. A positive correlation exists for human height and weight: tall people tend to weigh more. A negative correlation coefficient indicates that there is an inverse relation between the two variables (Figure 22.11b); as one variable increases, the other tends to decrease (as is the case for egg number and hen weight).

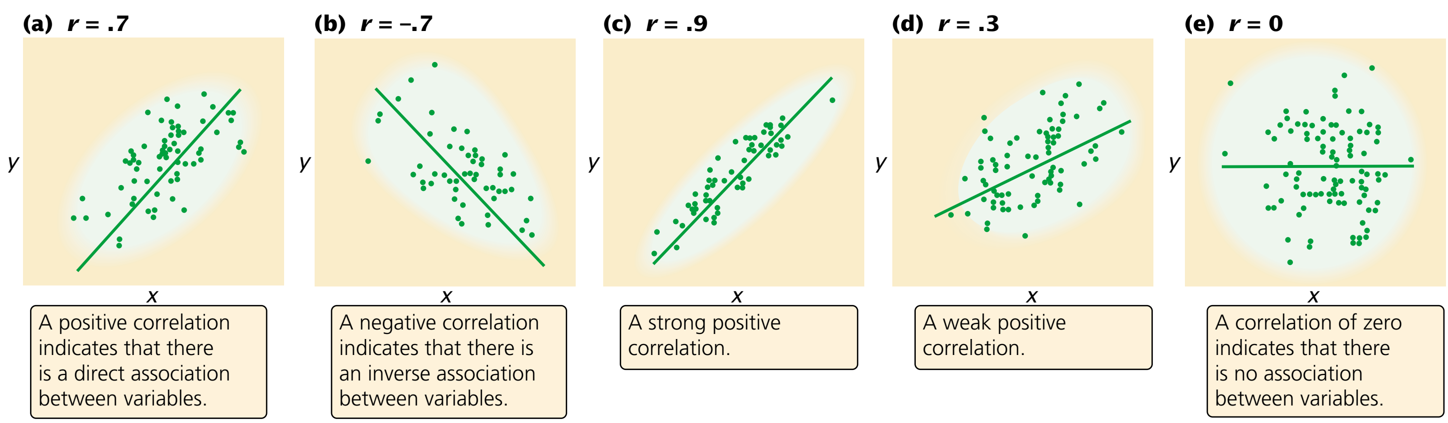
The absolute value of the correlation coefficient (the size of the coefficient, ignoring its sign) provides information about the strength of association between the variables. A coefficient of -1 or $+1$ indicates a perfect correlation between the variables, meaning that a change in x is always accompanied by a proportional change in y. Correlation coefficients close to -1 or close to $+1$ indicate a strong association between the variables—a change in x is almost always associated with a proportional increase in y, as seen in Figure 22.11c. On the other hand, a correlation coefficient closer to 0 indicates a weak correlation—a change in x is associated with a change in y but not always (Figure 22.11d). A correlation of 0 indicates that there is no association between variables (Figure 22.11e).

A correlation coefficient can be computed for two variables measured for the same individual, such as height (x) and weight (y). A correlation coefficient can also be computed for a single variable measured for pairs of individuals. For example, we can calculate for fish the correlation between the number of vertebrae of a parent (x) and the number of vertebrae of its offspring (y), as shown in Figure 22.12. This approach is often used in quantitative genetics.

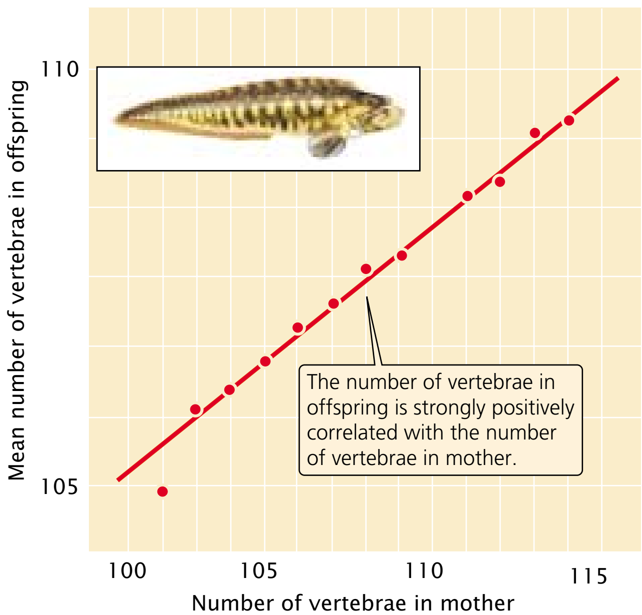
A correlation between two variables indicates only that the variables are associated; it does not imply a cause and effect relation. Correlation also does not mean that the values of two variables are the same; it means only that a change in one variable is associated with a proportional change in the other variable. For example, the x and y variables in the following list are almost perfectly correlated, with a correlation coefficient of .99.

	x value	y value
	12	123
	14	140
	10	110
	6	61
	3	32
Average:	9	90

A high correlation is found between these x and y variables; larger values of x are always associated with larger values of y. Note that the y values are about 10 times as large as the corresponding x values; so, although x and y are correlated, they are not identical. The distinction between correlation



22.11 The correlation coefficient describes the relation between two or more variables.



22.12 A correlation coefficient can be computed for a single variable measured for pairs of individuals. Here, the number of vertebrae in mothers and offspring of the fish Zoarces viviparus is compared.

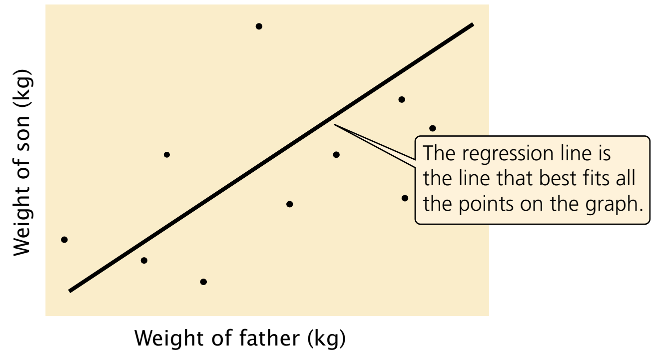
and identity becomes important when we consider the effects of heredity and environment on the correlation of characteristics.

Regression

Correlation provides information only about the strength and direction of association between variables, but often we want to know more than just whether two variables are associated; we want to be able to predict the value of one variable, given a value of the other.

A positive correlation exists between body weight of parents and body weight of their offspring; this correlation exists in part because genes influence body weight, and parents and children share the same genes. Because of this association between phenotypes of parent and offspring, we can predict the weight of an individual on the basis of the weights of its parents. This type of statistical prediction is called **regression.** This technique plays an important role in quantitative genetics because it allows us to predict characteristics of offspring from a given mating, even without knowledge of the genotypes that encode the characteristic.

Regression can be understood by plotting a series of x and y values. FIGURE 22.13 illustrates the relation between the weight of fathers (x) and the weight of their offspring (y). Each father–offspring pair is represented by a point on the graph. The overall relation between these two variables is depicted by the regression line, which is the line that best fits all the points on the graph (deviations of the points from the line are minimized). The regression line defines the relation between the x and y variables and can be represented by



22.13 A regression line defines the relation between two variables. Illustrated here is a regression of the weights of fathers against the weights of sons. Each father–offspring pair is represented by a point on the graph: the x value of a point is the father's weight and the y value of the point is the offspring's weight.

$$y = a + bx \tag{22.8}$$

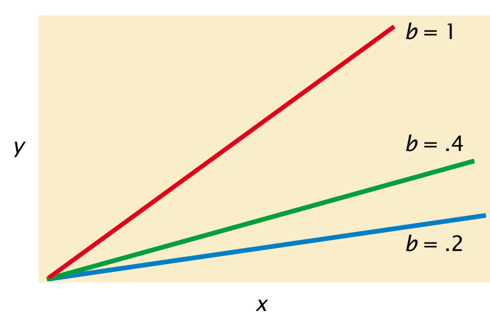
In Equation 22.8, x and y represent the x and y variables (in this case, the father's weight and the offspring's weight, respectively). The variable a is the y intercept of the line, which is the expected value of y when x is 0. Variable b is the slope of the regression line, also called the **regression coefficient;** it indicates how much y increases, on average, per increase in x.

Trying to position a regression line by eye is not only very difficult but also inaccurate when there are many points scattered over a wide area. Fortunately, the regression coefficient and y intercept can be obtained mathematically. The regression coefficient (b) can be computed from the covariance of x and y (cov_{xy}) and the variance of x (s_x^2) by

$$b = \frac{\text{cov}_{xy}}{s_x^2} \tag{22.9}$$

Several regression lines with different regression coefficients are illustrated in FIGURE 22.14.

After the regression coefficient has been calculated, the y intercept can be calculated by substituting the regression



22.14 The regression coefficient (*b*) represents the change in *y* per unit change in *x.* Shown here are regression lines with different regression coefficients.

coefficient and the mean values of x and y into the following equation:

$$a = \bar{y} - b\bar{x} \tag{22.10}$$

The regression equation ($y = a + bx$) can then be used to predict the value of any y given the value of x.

Concepts

A correlation coefficient measures the strength of association between two variables. The sign (positive or negative) indicates the direction of the correlation; the absolute value measures the strength of the association. Regression is used to predict the value of one variable on the basis of the value of a correlated variable.

Worked Problem

Body weights of 11 female fish and the numbers of eggs that they produce are given in the following table.

Weight (mg) x	Eggs (thousands) y
14	61
17	37
24	65
25	69
27	54
33	93
34	87
37	89
40	100
41	90
42	97

What are the correlation coefficient and the regression coefficient for body weight and egg number in these 11 fish?

• Solution

The computations needed to answer this question are given in the table below. To calculate the correlation and regression coefficients, we first obtain the sum of all the x_i values (Σx_i) and the sum of all the y_i values (Σy_i); these sums are shown in the last row of the table. We can calculate the means of the two variables by dividing the sums by the number of measurements, which is 11:

$$\bar{x} = \frac{\Sigma x_i}{n} = \frac{334}{11} = 30.36$$

$$\bar{y} = \frac{\Sigma y_i}{n} = \frac{842}{11} = 76.55$$

After the means have been calculated, the deviations of each value from the means are computed; these deviations are shown in columns B and E of the table. The deviations are then squared (columns C and F) and summed (last row of columns C and F). Next, the products of the deviation of the x values and the deviation of the y values $[(x_i - \bar{x})(y_i - \bar{y})]$ are calculated; these products are shown in column G, and their sum is shown in the last row of column G.

To calculate the covariance, we use Formula 22.6:

$$\text{cov}_{xy} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{n - 1} = \frac{1743.84}{10} = 174.38$$

To calculate the covariance and regression requires the variances and standard deviations of x and y:

$$s_x^2 = \frac{(x_i - \bar{x})^2}{n - 1} = \frac{932.55}{10} = 93.26$$

A Weight (mg) x	B $x_i - \bar{x}$	C $(x_i - \bar{x})^2$	D Eggs (thousands) y	E $y_i - \bar{y}$	F $(y_i - \bar{y})^2$	G $(x_i - \bar{x})(y_i - \bar{y})$
14	−16.36	267.65	61	−15.55	241.80	254.40
17	−13.36	178.49	37	−39.55	1564.20	528.39
24	−6.36	40.45	65	−11.55	133.40	73.46
25	−5.36	28.73	69	−7.55	57.00	40.47
27	−3.36	11.29	54	−22.55	508.50	75.77
33	2.64	6.97	93	16.45	270.60	43.43
34	3.64	13.25	87	10.45	109.20	38.04
37	6.64	44.09	89	12.45	155.00	82.67
40	9.64	92.93	100	23.45	549.90	226.06
41	10.64	113.21	90	13.45	180.90	143.11
42	11.64	135.49	97	20.45	418.20	238.04
$\Sigma x_i = 334$		$\Sigma(x - \bar{x})^2 = 932.55$	$\Sigma y_i = 842$		$\Sigma(y - \bar{y})^2 = 4188.70$	$\Sigma(x_i - \bar{x})(y_i - \bar{y}) = 1743.84$

Source: R. R. Sokal and F. J. Rohlf, *Biometry*, 2d ed. (San Francisco: W. H. Freeman and Company, 1981.)

$$s_x = \sqrt{s_x^2} = \sqrt{93.26} = 9.66$$

$$s_y^2 = \frac{(y_i - \bar{y})^2}{n - 1} = \frac{4188.70}{10} = 418.87$$

$$s_y = \sqrt{s_y^2} = \sqrt{418.87} = 20.47$$

We can now compute the correlation and regression coefficients.

Correlation coefficient:

$$r = \frac{\mathrm{cov}_{xy}}{s_x s_y} = \frac{174.38}{9.66 \times 20.47} = 0.88$$

Regression coefficient:

$$b = \frac{\mathrm{cov}_{xy}}{s_x^2} = \frac{174.38}{93.26} = 1.87$$

www.whfreeman.com/pierce Numerous links to Web sites on statistics

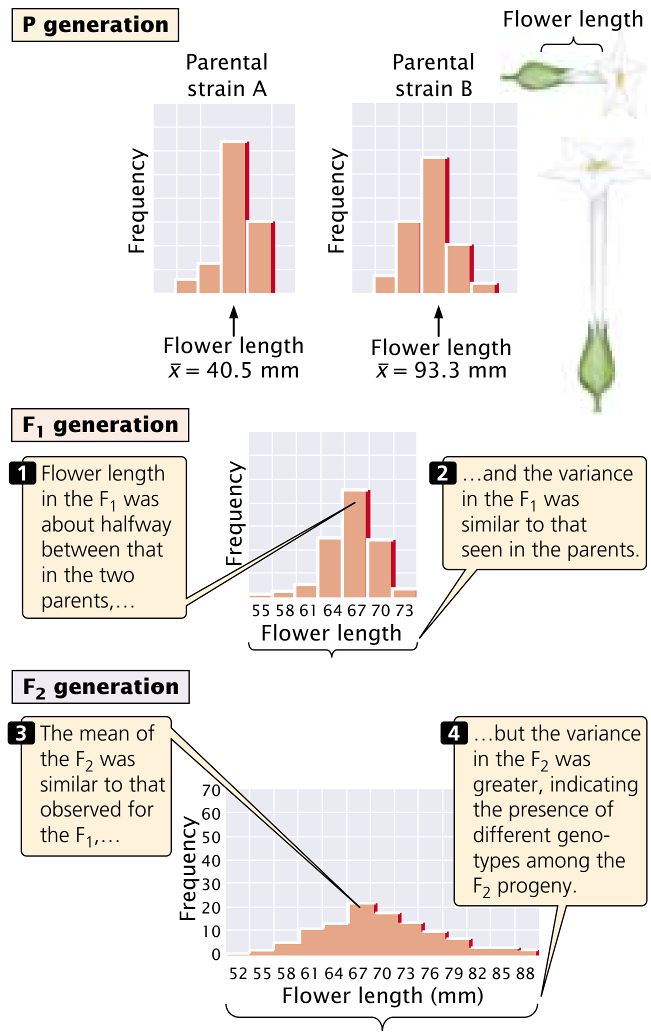
Applying Statistics to the Study of a Polygenic Characteristic

Edward East carried out one early statistical study of polygenic inheritance on the length of flowers in tobacco (*Nicotiana longiflora*). He obtained two varieties of tobacco that differed in flower length: one variety had a mean flower length of 40.5 mm, and the other had a mean flower length of 93.3 mm (FIGURE 22.15). These two varieties had been inbred for many generations and were homozygous at all loci contributing to flower length. Thus, there was no genetic variation in the original parental strains; the small differences in flower length within each strain were due to environmental effects on flower length.

When East crossed the two strains, he found that flower length in the F_1 was about halfway between that in the two parents (see Figure 22.15), as would be expected if the genes determining the differences in the two strains were additive in their effects. The variance of flower length in the F_1 was similar to that seen in the parents, because the F_1 were, like their parents, uniform in genotype (the F_1 were all heterozygous at the genes that differed between the two parental varieties).

East then interbred the F_1 to produce F_2 progeny. The mean flower length of the F_2 was similar to that of the F_1, but the variance of the F_2 was much greater (see Figure 22.15). This greater variability indicates that there were genetic differences within the F_2 progeny.

East selected some F_2 plants and interbred them to produce F_3 progeny. He found that flower length of the F_3 depended on flower length in the plants selected as their parents. This finding demonstrated that flower-length differences in the F_2 were partly genetic and thus were passed to the next generation. None of the 444 F_2 plants that East raised exhibited flower lengths similar to those of the two parental strains. This result suggested that more than four loci with pairs of alleles affected flower length in his varieties, because four allelic pairs are expected to produce 1 of 256 progeny ($\frac{1}{4}^4 = \frac{1}{256}$) having one or the other of the original parental phenotypes.



22.15 Edward East conducted an early statistical study of the inheritance of flower length in tobacco.

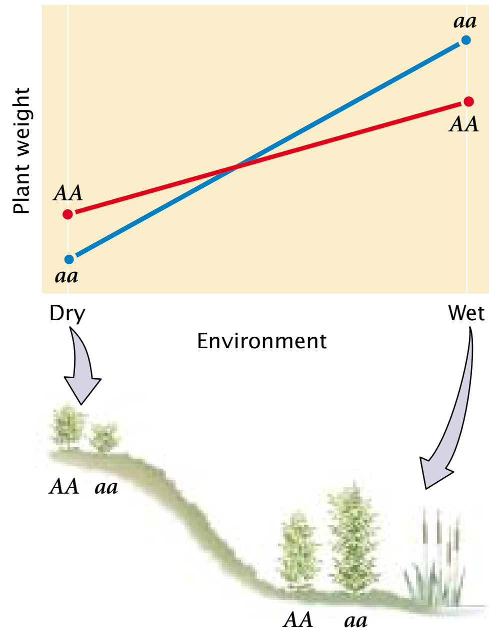
Heritability

In addition to being polygenic, quantitative characteristics are frequently influenced by environmental factors. It is often useful to know how much of the variation in a quantitative characteristic is due to genetic differences and how much is due to environmental differences. That proportion of the total phenotypic variation that is due to genetic differences is known as the **heritability.**

Consider a dairy farmer who owns several hundred milk cows. The farmer notices that some cows consistently produce more milk than others. The *nature* of these differences is important to the profitability of his dairy operation. If the differences in milk production are largely genetic in origin, then the farmer may be able to boost milk production by selectively breeding the cows that produce the most milk. On the other hand, if the differences are largely environmental in origin, selective breeding will have little effect on milk production, and the farmer might better boost milk production by adjusting the environmental factors associated with higher milk production. To determine the extent of genetic and environmental influences on variation in a characteristic, phenotypic variation in the characteristic must be partitioned into components attributable to different factors.

Phenotypic Variance

To determine how much of phenotypic differences in a population is due to genetic and environmental factors, we must first have some quantitative measure of the phenotype under consideration. Consider a population of wild plants that differ in size. We could collect a representative sample of plants from the population, weigh each plant in the sample, and calculate the mean and variance of plant weight. This **phenotypic variance** is represented by V_P.



22.16 Genetic–environmental interaction variance occurs when the effect of a gene depends on the specific environment in which it is found. In this example, the genotype affects plant weight, but the environmental conditions determine which genotype produces the heavier plant.

Components of phenotypic variance

Phenotypic variance, which represents the phenotypic differences among individual members of a group, can be attributed to several factors. First, some of the differences in phenotype may be due to differences in genotypes among individual members of the population. These differences are termed the **genetic variance** and are represented by V_G.

Second, some of the differences in phenotype may be due to environmental differences among the plants; these differences are termed the **environmental variance,** V_E. Environmental variance includes differences that can be attributed to specific environmental factors, such as the amount of light or water that the plant receives; it also includes random differences in development that cannot be attributed to any specific factor. Any variation in phenotype that is not inherited is, by definition, a part of the environmental variance.

Third, **genetic–environmental interaction variance** (V_{GE}) arises when the effect of a gene depends on the specific environment in which it is found. An example is shown in FIGURE 22.16. In a dry environment, genotype *AA* produces a plant that averages 12 g in weight, and genotype *aa* produces a smaller plant that averages 10 g. In a wet environment, genotype *aa* produces the larger plant, averaging 24 g in weight, whereas genotype *AA* produces a plant that averages 20 g. In this example, there are clearly differences in the two environments: both genotypes produce heavier plants in the wet environment. There are also differences in the weights of the two genotypes, but the relative performances of the genotypes depend on whether the plants are grown in a wet or dry environment. In this case, the influences on phenotype cannot be neatly allocated into genetic and environmental components, because the expression of the genotype depends on the environment in which the plant grows. The phenotypic variance must therefore include a component that accounts for the way in which genetic and environmental factors interact.

In summary, the total phenotypic variance can be apportioned into three components:

$$V_P = V_G + V_E + V_{GE} \tag{22.11}$$

Components of genetic variance

Genetic variance can be further subdivided into components consisting of different types of genetic effects. First, **additive genetic variance** (V_A) comprises the additive effects of genes on the phenotype, which can be summed to determine the overall effect on the phenotype. For example, suppose that, in a plant, allele A^1 contributes 2 g in weight and allele A^2 contributes 4 g. If the alleles are strictly additive, then the genotypes would have the following weights:

$$A^1A^1 = 2 + 2 = 4 \text{ g}$$

$$A^1A^2 = 2 + 4 = 6 \text{ g}$$

$$A^2A^2 = 4 + 4 = 8 \text{ g}$$

The genes that Nilsson-Ehle studied, which affected kernel color in wheat, were additive in this way.

Second, there is **dominance genetic variance** (V_D) when some genes have a dominance component. In this case, the alleles at a locus are not additive; rather, the effect of an allele depends on the identity of the other allele at that locus. Here, we cannot simply add the effects of the alleles together. Instead, we must add a component (V_D) to the genetic variance to account for the way that alleles interact.

Third, genes at different loci may interact in the same way that alleles at the same locus interact. When this genic interaction occurs, the effects of genes are not additive, and we must include a third component, called **genic interaction variance** (V_I), to the genetic variance:

$$V_G = V_A + V_D + V_I \tag{22.12}$$

Summary equation We can now integrate these components into one equation to represent all the potential contributions to the phenotypic variance:

$$V_P = V_A + V_D + V_I + V_E + V_{GE} \tag{22.13}$$

This equation provides us with a model that describes the potential causes of differences that we observe among individual phenotypes. It's important to note that this model deals strictly with the observable *differences* (variance) in phenotypes among individual members of a population; it says nothing about the absolute value of the characteristic or about the underlying genotypes that produce these differences.

Types of Heritability

The model of phenotypic variance that we've just developed can be used to address the question of how much of the phenotypic variance in a characteristic is due to genetic differences. **Broad-sense heritability** (H^2) represents the proportion of phenotypic variance that is due to genetic variance and is calculated by dividing the genetic variance by the phenotypic variance:

$$\text{broad-sense heritability} = H^2 = \frac{V_G}{V_P} \tag{22.14}$$

It is symbolized H^2 because it is a measure of variance, which is in units squared.

Broad-sense heritability can potentially range from 0 to 1. A value of 0 indicates that none of the phenotypic variance results from differences in genotype and all of the differences in phenotype result from environmental variation. A value of 1 indicates that all of the phenotypic variance results from differences in genotype. A heritability value between 0 and 1 indicates that both genetic and environmental factors influence the phenotypic variance.

Often, we are more interested in the proportion of the phenotypic variance that results from the additive genetic variance, because the additive genetic variance primarily determines the resemblance between parents and offspring. **Narrow-sense heritability** (h^2) is equal to the additive genetic variance divided by the phenotypic variance:

$$\text{narrow-sense heritability} = h^2 = \frac{V_A}{V_P} \tag{22.15}$$

The Calculation of Heritability

Having considered the components that contribute to phenotypic variance and having developed a general concept of heritability, we can ask, How does one go about estimating these different components and calculating heritability? There are several ways to measure the heritability of a characteristic. They include eliminating one or more variance components, comparing the resemblance of parents and offspring, comparing the phenotypic variances of individuals with different degrees of relatedness, and measuring the response to selection. The mathematical theory that underlies these calculations of heritability is complex and beyond the scope of this book. Nevertheless, we can develop a general understanding of how heritability is measured.

Heritability by elimination of variance components

One way of calculating the broad-sense heritability is to eliminate one of the variance components. We have seen that $V_P = V_G + V_E + V_{GE}$. If we eliminate all environmental variance ($V_E = 0$), then $V_{GE} = 0$ (because, if either V_G or V_E is zero, no genetic–environmental interaction can take place), and $V_P = V_G$. In theory, we might make V_E equal to 0 by ensuring that all individuals were raised in exactly the same environment but, in practice, it is virtually impossible. Instead, we could make V_G equal to 0 by raising genetically identical individuals, causing V_P to be equal to V_E. In a typical experiment, we might raise cloned or highly inbred, identically homozygous individuals in a defined environment and measure their phenotypic variance to estimate V_E. We could then raise a group of genetically variable individuals and measure their phenotypic variance (V_P). Using V_E calculated on the genetically identical individuals, we could obtain the genetic variance of the variable individuals by subtraction:

$$V_{G[\text{of genetically varying individuals}]} = V_{P[\text{of genetically varying individuals}]} - V_{E[\text{of genetically identical individuals}]} \tag{22.16}$$

The broad-sense heritability of the genetically variable individuals would then be calculated as follows:

$$H^2 = \frac{V_{G[\text{of genetically varying individuals}]}}{V_{P[\text{of genetically varying individuals}]}} \tag{22.17}$$

Sewall Wright used this method to estimate the heritability of white spotting in guinea pigs. He first measured the phenotypic variance for white spotting in a genetically variable population and found that $V_P = 573$. Then he inbred the guinea pigs for many generations so that they were essentially homozygous and genetically identical. When he measured their phenotypic variance in white spotting, he obtained V_P equal to 340. Because $V_G = 0$ in this group, their $V_P = V_E$. Wright assumed this value of environmental variance for the original (genetically variable) population and estimated their genetic variance:

$$V_P - V_E = V_G$$

$$573 - 340 = 233$$

He then estimated the broad-sense heritability from the genetic and phenotypic variance:

$$H^2 = \frac{V_G}{V_P}$$

$$H^2 = \frac{233}{573} = .41$$

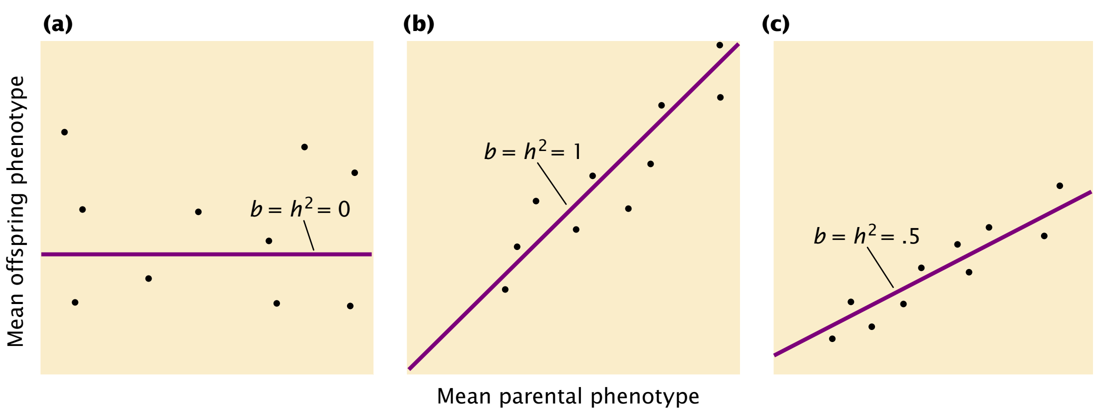
This value implies that 41% of the variation in spotting of guinea pigs in Wright's population was due to differences in genotype.

Estimating heritability by using this method assumes that the environmental variance of genetically identical individuals is the same as the environmental variance of the genetically variable individuals, which may not be true. Additionally, this approach can be applied only to organisms for which it is possible to create genetically identical individuals.

Heritability by parent–offspring regression Another method for estimating heritability is to compare the phenotypes of parents and offspring. When genetic differences are responsible for phenotypic variance, offspring should resemble their parents more than they resemble unrelated individuals, because offspring and parents have some genes in common that help determine their phenotype. Correlation and regression can be used to analyze the association of phenotypes in different individuals.

To calculate the narrow-sense heritability in this way, we first measure the characteristic on a series of parents and offspring. The data are arranged into families, and the mean parental phenotype is plotted against the mean offspring phenotype (FIGURE 22.17). Each data point in the graph represents one family; the value on the x (horizontal) axis is the mean phenotypic value of the parents in a family, and the value on the y (vertical) axis is the mean phenotypic value of the offspring for the family.

Let's assume that there is no narrow-sense heritability for the characteristic ($h^2 = 0$); genetic differences do not contribute to the phenotypic differences among individuals. In this case, offspring will be no more similar to their parents than they are to unrelated individuals, and the data points will be scattered randomly, generating a regression coefficient of zero (FIGURE 22.17a). Next, let's assume that all of the phenotypic differences are due to additive genetic differences ($h^2 = 1.0$). In this case, the mean phenotype of the offspring will be equal to the mean phenotype of the parents, and the regression coefficient will be 1 (FIGURE 22.17b). If genes and environment both contribute to the differences in phenotype, both heritability and the regression coefficient will lie between 0 and 1 (FIGURE 22.17c). The regression coefficient therefore provides information about the *magnitude* of the heritability.



22.17 The narrow-sense heritability (h^2) equals the regression coefficient (b) in a regression of the mean phenotype of the offspring on the mean phenotype of the parents. (a) There is no relation between the parental phenotype and the offspring phenotype. (b) The offspring phenotype is the same as the parental phenotypes. (c) Both genes and environment contribute to the differences in phenotype.

A complex mathematical proof (which we will not go into here) demonstrates that, in a regression of the mean phenotype of the offspring against the mean phenotype of the parents, narrow-sense heritability (h^2) equals the regression coefficient (b):

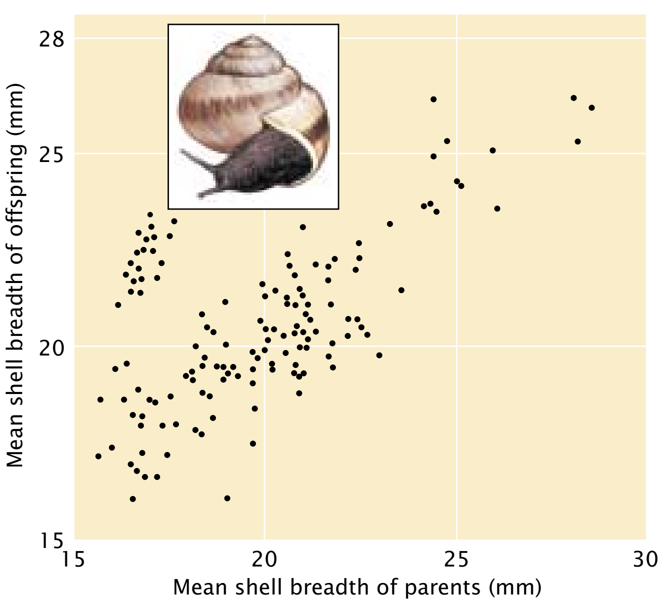
$$h^2 = b_{(\text{regression of mean offspring against mean of both parents})} \quad (22.18)$$

An example of calculating heritability by regression of the phenotypes of parents and offspring is illustrated in FIGURE 22.18. In a regression of the mean offspring phenotype against the phenotype of only *one* parent, the narrow-sense heritability equals twice the regression coefficient:

$$h^2 = 2b_{(\text{regression of mean offspring against mean of one parent})} \quad (22.19)$$

With only one parent, the heritability is twice the regression coefficient because only half the genes of the offspring come from one parent; thus, we must double the regression coefficient to obtain the full heritability.

Heritability and degrees of relatedness A third method for calculating heritability is to compare the phenotypes of individuals having different degrees of relatedness. This method is based on the concept that, the more closely related two individuals are, the more genes they have in common.



22.18 **The heritability of shell breadth in snails can be determined by regression of the phenotype of offspring against the mean phenotype of the parents.** The regression coefficient, which equals the heritability, is .70. (From L. M. Cook, 1965. *Evolution* 19:86–94.)

Monozygotic (identical) twins have 100% of their genes in common, whereas dizygotic (nonidentical) twins have, on average, 50% of their genes in common. If genes are important in determining variability in a characteristic, then monozygotic twins should be more similar in a particular characteristic than dizygotic twins. By using correlation to compare the phenotypes of monozygotic and dizygotic twins, we can estimate broad-sense heritability. A rough estimate of the broad-sense heritability can be obtained by taking twice the difference of the correlation coefficients for a quantitative characteristic in monozygotic and dizygotic twins:

$$H^2 = 2(r_{MZ} - r_{DZ}) \quad (22.20)$$

where r_{MZ} equals the correlation coefficient among monozygotic twins and r_{DZ} equals the correlation coefficient among dizygotic twins. This calculation assumes that the two individuals of a monozygotic twin pair experience environments that are no more similar to each other than those experienced by the two individuals of a dizygotic twin pair, which is often not the case, unless the twins have been reared apart.

Narrow-sense heritability can also be estimated by comparing the phenotypic variances for a characteristic in full sibs (who have both parents in common, as well as 50% of their genes on the average) and half sibs (who have only one parent in common and thus 25% of their genes on the average).

All estimates of heritability depend on the assumption that the environments of related individuals are not more similar than those of unrelated individuals. This assumption is difficult to meet in human studies, because related people are usually reared together. Heritability estimates for humans should therefore always be viewed with caution.

Concepts

Broad-sense heritability is the proportion of phenotypic variance that is due to genetic variance. Narrow-sense heritability is the proportion of phenotypic variance that is due to additive genetic variance. Heritability can be measured by eliminating one of the variance components, analyzing parent–offspring regression, or comparing individuals having different degrees of relatedness.

The Limitations of Heritability

Knowledge of heritability has great practical value, because it allows us to statistically predict the phenotypes of offspring on the basis of their parent's phenotype. It also provides useful information about how characteristics will respond to selection (see next section). In spite of its importance, heritability is frequently misunderstood. Heritability does not provide information about an individual's genes or the environmental factors that control the development of a characteristic, and it says nothing about the nature of differences between groups. This section outlines some limitations and common misconceptions concerning broad- and narrow-sense heritability.

Heritability does *not* indicate the degree to which a characteristic is genetically determined Heritability is the proportion of the phenotypic variance that is due to genetic variance; it says nothing about the degree to which genes determine a characteristic. Heritability indicates only the degree to which genes determine *variation* in a characteristic. The determination of a characteristic and the determination of variation in a characteristic are two very different things.

Consider polydactyly (the presence of extra digits) in rabbits, which can be caused either by environmental factors or by a dominant gene. Suppose we have a group of rabbits all homozygous for a gene that produces normal numbers of digits. None of the rabbits in this group carries a gene for polydactyly, but a few of the rabbits are polydactylous because of environmental factors. Broad-sense heritability for polydactyly in this group is zero, because there is no genetic variation for polydactyly; all of the variation is due to environmental factors. However, it would be incorrect for us to conclude that genes play no role in determining the number of digits in rabbits. Indeed, we know that there are specific genes that can produce extra digits. Heritability indicates nothing about whether genes control the development of a characteristic; it only provides information about causes of the *variation* in a characteristic within a defined group.

An individual does *not* have heritability Broad- and narrow-sense heritabilities are statistical values based on the genetic and phenotypic variances found in a *group* of individuals. It is impossible to calculate heritability for an individual, and heritability has no meaning for a specific individual. Suppose we calculate the narrow-sense heritability of adult body weight for the students in a biology class and obtain a value of .6. We could conclude that 60% of the variation in adult body weight among the students in this class is determined by additive genetic variation. We could not, however, conclude that 60% of any particular student's body weight is due to additive genes.

There is *no* universal heritability for a characteristic The value of heritability for a characteristic is specific for a given population in a given environment. Recall that broad-sense heritability is genetic variance divided by phenotypic variance. Genetic variance depends on which genes are present, which often differs between populations. In the example of polydactyly in rabbits, there were no genes for polydactyly in the group; so the heritability of the characteristic was zero. A different group of rabbits might contain many genes for polydactyly, and the heritability of the characteristic might be high.

Environmental differences may affect heritability, because V_P is composed of both genetic and environmental variance. When the environmental differences that affect a characteristic differ between two groups, the heritabilities for the two groups also will often differ.

Because heritability is specific to a defined population in a given environment, it is important not to extrapolate heritabilities from one population to another. For example, human height is determined by environmental factors (such as nutrition and health) and by genes. If we measured the heritability of height in a developed country, we might obtain a value of .8, indicating that the variation in height in this population is largely genetic. This population has a high heritability because most people have adequate nutrition and health care (V_E is low); so most of the phenotypic variation in height is genetically determined. It would be incorrect for us to assume that height has a high heritability in all human populations. In developing countries, there may be more variation in a range of environmental factors; some people may enjoy good nutrition and health, whereas others may have a diet deficient in protein and suffer from diseases that affect stature. If we measured the heritability of height in such a country, we would undoubtedly obtain a lower value than we observed in the developed country, because there is more environmental variation and the genetic variance in height constitutes a smaller proportion of the phenotypic variation, making the heritability lower. The important point to remember is that heritability must be calculated separately for each population and each environment.

Even when heritability is high, environmental factors may influence a characteristic High heritability does not mean that environmental factors cannot influence the expression of a characteristic. High heritability indicates only that the environmental variation to which the population is *currently* exposed is not responsible for variation in the characteristic. Let's look again at human height. In most developed countries, heritability of human height is high, indicating that genetic differences are responsible for most of the variation in height. It would be wrong for us to conclude that human height cannot be changed by alteration of the environment. Indeed, height decreased in several European cities during World War II owing to hunger and disease, and height can be increased dramatically by the administration of growth hormone to children. The absence of environmental variation in a characteristic does not mean that the characteristic will not respond to environmental change.

Heritabilities indicate nothing about the nature of population differences in a characteristic A common misconception about heritability is that it provides information about population differences in a characteristic. Heritability is specific for a given population in a given environment, so it cannot be used to draw conclusions about why populations differ in a characteristic.

Suppose we measured heritability for human height in two groups. One group is from a small town in a developed country, where everyone consumes a high-protein diet. Because there is little variation in the environmental factors that affect human height and there is some genetic variation, the heritability of height in this group is high. The second group comprises the inhabitants of a single village in a developing country. The consumption of protein by these people is only 25% of that consumed by those in the first group; so their average adult height is several centimeters less than that in the developed country. Again, there is little variation in the environmental factors that determine height in this group, because everyone in the village eats the same types of food and is exposed to same diseases. Because there is little environmental variation and there is some genetic variation, the heritability of height in this group also is high.

Thus, the heritability of height in both groups is high, and the average height in the two groups is considerably different. We might be tempted to conclude that the difference in height between the two groups is genetically based—that the people in the developed country are genetically taller than the people in the developing country. This conclusion is obviously wrong, however, because the differences in height are due largely to diet—an environmental factor. Heritability provides no information about the causes of differences between populations.

These limitations of heritability have often been ignored, particularly in arguments about possible social implications of genetic differences between humans. Soon after Mendel's principles of heredity were rediscovered, some geneticists began to claim that many human behavioral characteristics are determined entirely by genes. This claim led to debates about whether characteristics such as human intelligence are determined by genes or environment. Many of the early claims of genetically based human behavior were based on poor research; unfortunately, the results of these studies were often accepted at face value and led to a number of eugenic laws that discriminated against certain groups of people. Today, geneticists recognize that many behavioral characteristics are influenced by a complex interaction of genes and environment and that it is very difficult to separate genetic effects from those of the environment.

The results of a number of modern studies indicate that human intelligence as measured by IQ and other intelligence tests has a moderately high heritability (usually from .4 to .8). On the basis of this observation, some people have argued that intelligence is innate and that enhanced educational opportunities cannot boost intelligence. This argument is based on the misconception that, when heritability is high, changing the environment will not alter the characteristic. In addition, because heritabilities of intelligence range from .4 to .8, a considerable amount of the variance in intelligence originates from environmental differences.

Another argument based on a misconception about heritability is that ethnic differences in measures of intelligence are genetically based. Because the results of some genetic studies show that IQ has moderate heritability and because other studies find differences in the average IQ of ethnic groups, some people have suggested that ethnic differences in IQ are genetically based. As in the example of the effects of diet on nutrition, heritability provides no information about causes of differences among groups; it indicates only the degree to which phenotypic variance within a single group is genetically based. High heritability for a characteristic does not mean that phenotypic differences between ethnic groups are genetic. We should also remember that separating genetic and environmental effects in humans is very difficult; so heritability estimates themselves may be unreliable.

Concepts

Heritability provides information only about the degree to which *variation* in a characteristic is genetically determined. There is no universal heritability for a characteristic; heritability is specific for a given population in a specific environment. Environmental factors can potentially affect characteristics with high heritability, and heritability says nothing about the nature of population differences in a characteristic.

Locating Genes That Affect Quantitative Characteristics

The statistical methods described for use in analyzing quantitative characteristics can be used both to make predictions about the average phenotype expected in offspring and to estimate the overall contribution of genes to variation in the characteristic. These methods do not, however, allow us to identify and determine the influence of individual genes that affect quantitative characteristics. The genes that control polygenic characteristics are referred to as **quantitative trait loci** (QTLs). Although quantitative genetics has made important contributions to basic biology and to plant and animal breeding, the inability to identify QTLs and measure their individual effects has severely limited the application of quantitative genetic methods.

Mapping QTLs In recent years, numerous genetic markers have been identified and mapped with the use of recombinant DNA techniques, making it possible to identify QTLs by linkage analysis. The underlying idea is simple: if the inheritance of a genetic marker is associated consistently with the inheritance of a particular characteristic (such as increased height), then that marker must be linked to a QTL that affects height. The key is to have enough genetic markers so that QTLs can be detected throughout the genome. With the introduction of restriction fragment length polymorphisms and microsatellite variations (see pp. 000 and pp. 000 in Chapter 18), variable markers are now available for mapping QTLs in a number of different organisms (FIGURE 22.19).

A common procedure for mapping QTLs is to cross two homozygous strains that differ in alleles at many loci. The resulting F_1 progeny are then intercrossed or backcrossed to allow the genes to recombine through independent assortment and crossing over. Genes on different chromosomes and genes that are far apart on the same chromosome will recombine freely; genes that are closely linked will be inherited together. The offspring are measured for one or more quantitative characteristics; at the same time, they are genotyped for numerous genetic markers that span the genome. Any correlation between the inheritance of a particular marker allele and a quantitative phenotype indicates that a QTL is linked to that marker. If enough markers are used, it is theoretically possible to detect all the QTLs affecting a characteristic. This approach has been used to detect genes affecting various characteristics in several plant and animal species (Table 22.2).

Applications of QTL mapping The number of genes affecting a quantitative characteristic can be estimated by locating QTLs with genetic markers and adding up the number of QTLs detected. This method will always be an underestimate, because QTLs that are located close together on the same chromosome will be counted together, and those with small effects are likely to be missed.

QTL mapping also provides information about the magnitude of the effects that individual genes have on a quantitative characteristic. The polygenic model assumes that many genes affect a quantitative characteristic, that the effect of each gene is small, and that the effects of the genes are equal and additive. The results of studies of QTLs in a number of organisms now show that these assumptions are not always valid. Polygenes appear to vary widely in their effects. In many of the characteristics that have been studied, a few QTLs account for much of the phenotypic variation. In some instances, individual QTLs have been mapped that account for more than 20% of the variance in the characteristic.

22.19 The availability of molecular markers makes the mapping of QTLs possible in many organisms.

Table 22.2 Quantitative characteristics for which QTLs have been detected

Organism	Quantitative Characteristic	Number of QTLs Detected
Tomato	Soluble solids	7
	Fruit mass	13
	Fruit pH	9
	Growth	5
	Leaflet shape	9
	Height	9
Corn	Height	11
	Leaf length	7
	Tiller number	1
	Glume hardness	5
	Grain yield	18
	Number of ears	9
	Thermotolerance	6
Common bean	Number of nodules	4
Mung bean	Seed weight	4
Cow pea	Seed weight	2
Wheat	Preharvest sprout	4
Pig	Growth	2
	Length of small intestine	1
	Average back fat	1
	Abdominal fat	1
Mouse	Epilepsy	2
Rat	Hypertension	2

Source: After S. D. Tanksley, Mapping polygenes, *Annual Review of Genetics* 27 (1993):218.

Concepts

The availability of numerous genetic markers revealed by molecular methods make it possible to map individual genes that contribute to polygenic characteristics.

www.whfreeman.com/pierce More on QTLs

Response to Selection

Evolution is genetic change. Several different forces are potentially capable of producing evolution, and we will explore these forces and the process of evolution more fully in the next chapter. Here, we consider how one of these forces—natural selection—may bring about genetic change in a quantitative characteristic.

Charles Darwin proposed the idea of natural selection in his book *On the Origin of Species* in 1859. **Natural selection** arises through the differential reproduction of individuals with different genotypes, allowing individuals with certain genotypes to produce more offspring than others. Natural selection is one of the most important of the forces that brings about evolutionary change and can be summarized as follows:

Observation 1—Many more individuals are produced each generation than are capable of surviving long enough to reproduce.

Observation 2—There is much phenotypic variation within natural populations.

Observation 3—Some phenotypic variation is heritable. In the terminology of quantitative genetics, some of the phenotypic variation in these characteristics is due to genetic variation, and these characteristics have heritability.

Logical consequence—Individuals with certain characters (called adaptive traits) survive and reproduce better that others. Because the adaptive traits are heritable, offspring will tend to resemble their parents with regard to these traits, and there will be more individuals with these adaptive traits in the next generation. Thus, adaptive traits will tend to increase in the population through time.

In this way, organisms become genetically suited to their environments; as environments change, organisms change in ways that make them better able to survive and reproduce.

For thousands of years, humans have practiced a form of selection by promoting the reproduction of organisms with traits perceived as desirable. This form of selection is **artificial selection,** and it has produced the domestic plants and animals that make modern agriculture possible. The power of artificial selection, the first application of genetic principles by humans, is illustrated by the tremendous diversity of shapes, colors, and behaviors of modern domesticated dogs (FIGURE 22.20).

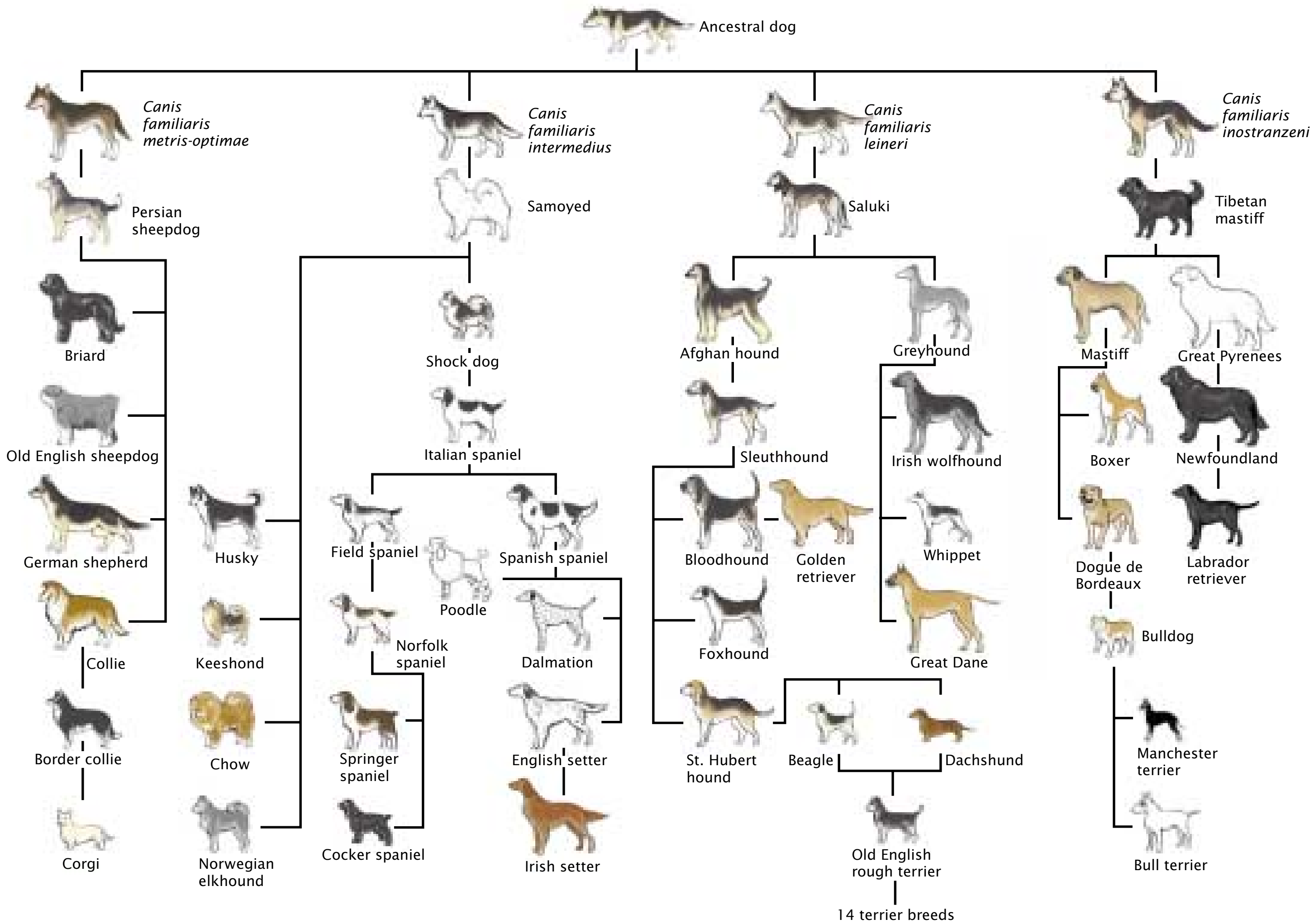
Predicting the Response to Selection

When a quantitative characteristic is subjected to natural or artificial selection, it will increase with the passage of time, provided there is genetic variation for that characteristic in the population. Suppose a dairy farmer breeds only those cows in his herd that have the highest milk production. If there is genetic variation in milk production, the mean milk production in the offspring of the selected cows should be higher than the mean milk production of the original herd. This increased production is due to the fact that the selected cows possess more genes for high milk production than does the average cow, and these genes are passed on to the offspring. The offspring of the selected cows possess a higher proportion of genes for greater milk yield and therefore produce more milk than the average cow in the initial herd.

The extent to which a characteristic subjected to selection changes in one generation is termed the **response to selection.** Suppose that the average cow in a dairy herd produces 80 liters of milk per week. A farmer selects for increased milk production by breeding the highest milk producers, and the progeny of these selected cows produce 100 liters of milk per week on average. The response to selection is calculated by subtracting the mean phenotype of the original population (80 liters) from the mean phenotype of the offspring (100 liters), obtaining a response to selection of $100 - 80 = 20$ liters per week.

The response to selection is determined primarily by two factors. First, it is affected by the narrow-sense heritability, which largely determines the degree of resemblance between parents and offspring. When the narrow-sense heritability is high, offspring will tend to resemble their parents; conversely, when the narrow-sense heritability is low, there will be little resemblance between parents and offspring.

The second factor that determines the response to selection is how much selection there is. If the farmer is very stringent in the choice of parents and breeds only the highest milk producers in the herd (say, the top 2 cows), then all the offspring will receive genes for high-quality milk production. If the farmer is less selective and breeds the top 20 milk producers in the herd, then the offspring will not carry as many superior genes for high milk production, and they will not, on average, produce as much milk as the offspring of the top 2 producers. The response to selection depends on the phenotypic difference of the individuals that are selected as parents; this phenotypic difference is measured by the **selection differential,** defined as the difference between the mean phenotype of the selected parents and the mean phenotype of the original population. If the aver-



22.20 Artificial selection has produced the tremendous diversity of shape, size, color, and behavior seen today among breeds of domestic dogs.

age milk production of the original herd is 80 liters and the farmer breeds cows with an average milk production of 120 liters, then the selection differential is $120 - 80 = 40$ liters.

The response to selection (R) depends on the narrow-sense heritability (h^2) and the selection differential (S):

$$R = h^2 \times S \tag{22.21}$$

This equation can be used to predict the magnitude of change in a characteristic when a given selection differential is applied. G. Clayton and his colleagues estimated the response to selection that would take place in abdominal bristle number of *Drosophila melanogaster*. Using several different methods, including parent–offspring regression, they first estimated the narrow-sense heritability of abdominal bristle number in one population of fruit flies to be .52. The mean number of bristles in the original population was 35.3. They selected individual flies with a mean bristle number of 40.6 and intercrossed them to produce the next generation. The selection differential was $40.6 - 35.3 = 5.3$; so they predicted a response to selection to be

$$R = .52 \times 5.3 = 2.8$$

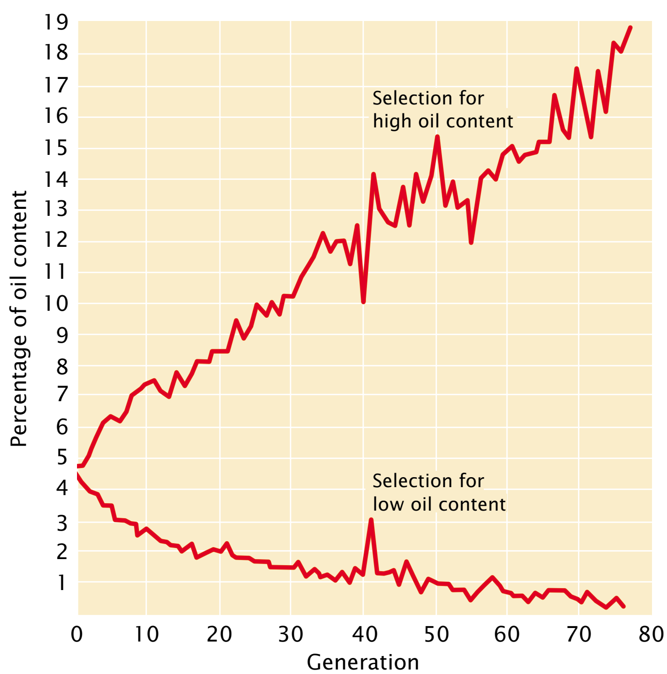
The response to selection of 2.8 represents the expected increase in the characteristic of the offspring above that of the original population. They therefore expected the average number of abdominal bristles in the offspring of their selected flies to be $35.3 + 2.8 = 38.1$. Indeed, they found an average bristle number of 37.9 in these flies.

Rearranging Equation 22.21 provides another way to calculate the narrow-sense heritability:

$$h^2 = \frac{R}{S} \qquad (22.22)$$

In this way, h^2 can be calculated by conducting a response-to-selection experiment. First, the selection differential is obtained by subtracting the population mean from the mean of selected parents. The selected parents are then interbred, and the mean phenotype of their offspring is measured. The difference between the mean of the offspring and that of the initial population is the response to selection, which can be used with the selection differential to estimate the heritability. Heritability determined by a response-to-selection experiment is usually termed the **realized heritability.** If certain assumptions are met, the realized heritability is identical with the narrow-sense heritability.

One of the longest selection experiments is a study of oil and protein content in corn seeds (FIGURE 22.21). This experiment began at the University of Illinois on 163 ears of corn with an oil content ranging from 4% to 6%. Corn plants having high oil content and those having low oil content were selected and interbreed. Response to selection for increased oil content (the upper line in Figure 22.21) reached about 20%, whereas response to selection for decreased oil content reached a lower limit near zero. Genetic analysis of the high- and low-oil-content strains revealed that at least 20 loci take part in determining oil content.



22.21 **In a long-term response-to-selection experiment, selection for oil content in corn increased oil content in one line to about 20%, while almost eliminating it altogether in another line.**

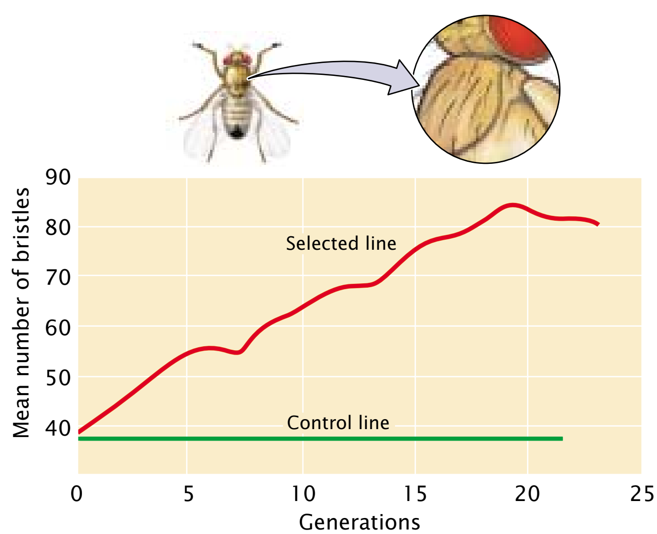
Concepts

The response to selection is influenced by narrow-sense heritability and the selection differential.

Limits to Selection Response

When a characteristic has been selected for many generations, the response eventually levels off, and the characteristic no longer responds to selection (FIGURE 22.22). A potential reason for this leveling off is that the genetic variation in the population may be exhausted; at some point, all individuals in the population have become homozygous for alleles that encode the selected trait. When there is no more additive genetic variation, heritability equals zero, and no further response to selection can occur.

The response to selection may level off even while some genetic variation remains in the population, however, because natural selection opposes further change in the characteristic. Response to selection for small body size in mice, for example, eventually levels off because the smallest animals are sterile and cannot pass on their genes for small body size. In this case, artificial selection for small size is opposed by natural selection for fertility, and the population can no longer respond to the artificial selection.



22.22 **The response of a population to selection often levels off at some point in time.** In a response-to-selection experiment for increased abdominal chaetae bristle number in female fruit flies, the number of bristles increased steadily for about 20 generations and then leveled off.

Correlated Responses

Two or more characteristics are often correlated. Human height and weight exhibit a positive correlation: tall people, on the average, weigh more than short people. This correlation is a **phenotypic correlation,** because the association is between two phenotypes of the same person. Phenotypic correlations may be due to environmental or genetic correlations. Environmental correlations refer to two or more characteristics that are influenced by the same environmental factor. Moisture availability, for example, may affect both the size of a plant and the number of seeds produced by the plant. Plants growing in environments with lots of water are large and produce many seeds, whereas plants growing in environments with limited water are small and have few seeds.

Alternatively, a phenotypic correlation may result from a **genetic correlation,** which means that the genes affecting two characteristics are associated. The primary genetic cause of phenotypic correlations is pleiotropy, which is due to the effect of one gene on two or more characteristics (see p. 000 in Chapter 5). In humans, for example, many body structures respond to growth hormone, and there are genes that affect the amount of growth hormone secreted by the pituitary gland. People with certain genes produce high levels of growth hormone, which increases both height and hand size. Others possess genes that produce lower levels of growth hormone, which leads to both short stature and small hands. Height and hand size are therefore phenotypically correlated in humans, and this correlation is due to a genetic correlation—the fact that both characteristics are affected by the same genes that control the amount of growth hormone. Genetically speaking, height and hand size are the same characteristic, because they are the phenotypic manifestation of a single set of genes. When two characteristics are influenced by the same genes they are genetically correlated.

Genetic correlations are quite common (Table 22.3) and may be positive or negative. A positive genetic correlation between two characteristics means that genes that cause an increase in one characteristic also produce an increase in the other characteristic. Thorax length and wing length in *Drosophila* are positively correlated because the genes that increase thorax length also increase wing length. A negative genetic correlation means that genes that cause an increase in one characteristic produce a decrease in the other characteristic. Milk yield and percentage of butterfat are negatively correlated in cattle: genes that cause higher milk production result in milk with a lower percentage of butterfat.

Genetic correlations are important in animal and plant breeding because they produce a correlated response to selection, which means that, when one characteristic is selected, genetically correlated characteristics also change. Correlated responses to selection occur because both characteristics are influenced by the same genes; selection for one characteristic causes a change in the genes affecting that characteristic, and these genes also affect the second characteristic, causing it to change at the same time. Correlated responses may well be undesirable and may limit the ability to alter a characteristic by selection. From 1944 to 1964, domestic turkeys were subjected to intense selection for growth rate and body size. At the same time, fertility, egg production, and egg hatchability all declined. These correlated responses were due to negative genetic correlations between body size and fertility; eventually, these genetic correlations limited the extent to which the growth rate of turkeys could respond to selection. Genetic correlations may also limit the ability of natural populations to respond to selection in the wild and adapt to their environments.

Table 22.3 Genetic correlations in various organisms

Organism	Characteristics	Genetic Correlation
Cattle	Milk yield and percentage of butterfat	−.38
Pig	Weight gain and back-fat thickness	.13
	Weight gain and efficiency	.69
Chicken	Body weight and egg weight	.42
	Body weight and egg production	−.17
	Egg weight and egg production	−.31
Mouse	Body weight and tail length	.29
Fruit fly	Abdominal bristle number and sternopleural bristle number	.41

Source: After D. S. Falconer, *Introduction to Quantitative Genetics* (London: Longman, 1981), p. 284.

Concepts

Genetic correlations result from pleiotropy. When two characteristics are genetically correlated, selection for one characteristic will produce a correlated response in the other characteristic.

www.whfreeman.com/pierce Information and links on the use of quantitative genetics in plant and animal breeding

Connecting Concepts Across Chapters

In this chapter, our perspective has shifted from individual genotypes (emphasized in transmission genetics) and the physical nature of the gene (emphasized in molecular genetics) to the genetic properties of groups of individuals. This shift will also be our perspective in Chapter 23, on population genetics.

Many of the most important characteristics in nature are those that display complex phenotypes and vary continuously. Body weight, reproductive output, susceptibility to diseases, and behavioral attributes often have continuous phenotypes. These types of characteristics are important in agriculture and are frequently significant in human health and evolution. An important theme of this chapter has been that such complex characteristics are inherited according to Mendelian principles, but more genes take part and environmental factors modify the phenotype. Because many factors influence the phenotypes of these complex characteristics, individual genes are difficult to identify, and we cannot predict precise phenotypic ratios among the offspring of a particular cross. Nevertheless, statistical procedures can be used to predict the average offspring phenotype and to assess the extent to which genetic and environmental factors are responsible for phenotypic differences in a characteristic.

Because the genes that influence quantitative characteristics are inherited according to Mendelian principles, the study of quantitative genetics requires a thorough understanding of the basic principles of heredity, which were covered in Chapters 3 through 7. Twin studies, which can be used to calculate heritability, are discussed in detail in Chapter 6; restriction fragment length polymorphisms and microsatellite variants, used to map quantitative trait loci, are explained in Chapter 18. The study of quantitative genetics depends on the genetic composition of populations and how that composition changes with time, which is the focus of Chapter 23.

CONCEPTS SUMMARY

- Quantitative genetics focuses on the inheritance of complex characteristics whose phenotype varies continuously. For many quantitative characteristics, the relation between genotype and phenotype is complex because many genes and environmental factors influence a characteristic.
- Quantitative characteristics also include meristic (counting) characteristics and threshold characteristics whose underlying genetic basis is influenced by multiple factors.
- Many quantitative characteristics are polygenic. The individual genes that influence a polygenic characteristic follow the same Mendelian principles that govern discontinuous characteristics, but, because many genes participate, the expected ratios of phenotypes are obscured.
- A population is the group of interest, and a sample is a subset of the population used to describe it.
- A frequency distribution, in which the phenotypes are represented on one axis and the number of individuals possessing the phenotype is represented on the other, is a convenient means of summarizing phenotypes found in a group of individuals.
- The mean and variance provide key information about a distribution: the mean gives the central location of the distribution, and the variance provides information about how the phenotype varies within a group.
- The correlation coefficient measures the direction and strength of association between two variables. Regression can be used to predict the value of one variable on the basis of the value of a correlated variable.
- Phenotypic variance in a characteristic can be divided into components that are due to additive genetic variance, dominance genetic variance, genic interaction variance, environmental variance, and genetic–environmental interaction variance.
- Broad-sense heritability is the proportion of the phenotypic variance that is due to genetic variance; narrow-sense heritability is the proportion of the phenotypic variance due to additive genetic variance.
- Broad-sense heritability can be estimated by eliminating the environmental variance component. Narrow-sense heritability can be estimated by comparing the phenotypes of parents and offspring or by comparing phenotypes of individuals with different degrees of relatedness, such as identical twins and nonidentical twins.
- Heritability provides information only about the degree to which variation in a characteristic results from genetic differences. It does not indicate the degree to which a characteristic is genetically determined. Heritability is based on the variances present within a group of individuals, and an individual does not have heritability. Heritability of a characteristic varies among populations and among environments. Even if heritability for a characteristic is high, the characteristic may still be altered by changes in the environment. Heritabilities provide no information about the nature of population differences in a characteristic.
- Quantitative trait loci are genes that control polygenic characteristics. QTLs can be mapped by examining the association between the inheritance of a quantitative characteristic and the inheritance of genetic markers. The mapping of numerous genetic markers with molecular techniques has made QTL mapping feasible for many organisms.
- When selection is applied to a quantitative characteristic, the characteristic will change if additive genetic variation for the characteristic is present. The amount that a quantitative

characteristic changes in a single generation when subjected to selection (the response to selection) is directly related to the selection differential and narrow-sense heritability. By applying a selection differential and measuring the response to selection, narrow-sense heritability can calculated.

- After selection has been applied to a quantitative characteristic for a number of generations, the response to selection may level off because no additive genetic variation in the characteristic remains. Alternatively, the response to selection may level off because of genetic correlations between the selected trait and other traits that affect fitness.
- A genetic correlation may be present when the same gene affects two or more characteristics (pleiotropy). Genetic correlations produce correlated responses to selection.

IMPORTANT TERMS

quantitative genetics (p. 000)
meristic characteristic (p. 000)
threshold characteristic (p. 000)
frequency distribution (p. 000)
normal distribution (p. 000)
population (p. 000)
sample (p. 000)
mean (p. 000)
variance (p. 000)
standard deviation (p. 000)
correlation (p. 000)
correlation coefficient (p. 000)
regression (p. 000)
regression coefficient (p. 000)
heritability (p. 000)
phenotypic variance (p. 000)
genetic variance (p. 000)
environmental variance (p. 000)
genetic–environmental interaction variance (p. 000)
additive genetic variance (p. 000)
dominance genetic variance (p. 000)
genic interaction variance (p. 000)
broad-sense heritability (p. 000)
narrow-sense heritability (p. 000)
quantitative trait locus (QTL) (p. 000)
natural selection (p. 000)
artificial selection (p. 000)
response to selection (p. 000)
selection differential (p. 000)
realized heritability (p. 000)
phenotypic correlation (p. 000)
genetic correlation (p. 000)

Worked Problems

1. Seed weight in a particular plant species is determined by pairs of alleles at two loci (a^+ a^- and b^+ b^-) that are additive and equal in their effects. Plants with genotype $a^-a^-b^-b^-$ have seeds that average 1 g in weight, whereas plants with genotype $a^+a^+b^+b^+$ have seeds that average 3.4 g in weight. A plant with genotype $a^-a^-b^-b^-$ is crossed with a plant of genotype $a^+a^+b^+b^+$.

(a) What is the predicted weight of seeds from the F_1 progeny of this cross?

(b) If the F_1 plants are intercrossed, what are the expected seed weights and proportions of the F_2 plants?

• Solution

The difference in average seed weight of the two parental genotypes is 3.4 g − 1 g = 2.4 g. These two genotypes differ in four genes; so, if the genes have equal and additive effects, each gene difference contributes an additional 24 g/4 = 6 g of weight to the 1-g weight of a plant with none of these contributing genes ($a^-a^-b^-b^-$).

The cross between the two homozygous genotypes produces the following F_1 and F_2 progeny:

P $a^-a^-b^-b^-$ (1 g) × $a^+a^+b^+b^+$ (3.4 g)

↓

F_1 $a^+a^-b^+b^-$ (2.2 g)

↓

F_2:

Genotype		Propability	Number of contributing genes	Average seed weight
$a^-a^-b^-b^-$	$1/4 \times 1/4 = 1/16$	$1/16$	0	$1\text{ g} + (0 \times 0.6\text{ g}) = 1\text{ g}$
$a^+a^-b^-b^-$	$1/2 \times 1/4 = 1/8$	$2/8 = 4/16$	1	$1\text{ g} + (1 \times 0.6\text{ g}) = 1.6\text{ g}$
$a^-a^-b^+b^-$	$1/4 \times 1/2 = 1/8$			
$a^+a^+b^-b^-$	$1/4 \times 1/4 = 1/16$	$2/16 + 1/4 = 6/16$	2	$1\text{ g} + (2 \times 0.6\text{ g}) = 2.2\text{ g}$
$a^-a^-b^+b^+$	$1/4 \times 1/4 = 1/16$			
$a^+a^-b^+b^-$	$1/2 \times 1/2 = 1/4$			
$a^+a^+b^+b^-$	$1/4 \times 1/2 = 1/8$	$2/8 = 4/16$	3	$1\text{ g} + (3 \times 0.6\text{ g}) = 2.8\text{ g}$
$a^+a^-b^+b^+$	$1/2 \times 1/4 = 1/8$			
$a^+a^+b^+b^+$	$1/4 \times 1/4 = 1/16$	$1/16$	4	$1\text{ g} + (4 \times 0.6\text{ g}) = 3.4\text{ g}$

(**a**) The F_1 are heterozygous at both loci ($a^+a^-b^+b^-$) and possess two genes that contribute an additional 0.6 g each to the 1-g weight of a plant with no contributing genes. Therefore, the seeds of the F_1 should average 1 g + 2(0.6 g) = 2.2 g.

(**b**) The F_2 will have the following phenotypes and proportions: $\frac{1}{16}$ 1 g; $\frac{4}{16}$ 1.6 g; $\frac{6}{16}$ 2.2 g; $\frac{4}{16}$ 2.8 g; and $\frac{1}{16}$ 3.4 g.

2. Phenotypic variation is analyzed for milk production in a herd of dairy cattle and the following variance components are obtained.

Additive genetic variance (V_A)	= .4
Dominance genetic variance (V_D)	= .1
Genic interaction variance (V_I)	= .2
Environmental variance (V_E)	= .5
Genetic-environmental interaction variance (V_{GE})	= .0

(**a**) What is the narrow-sense heritability of milk production?

(**b**) What is the broad-sense heritability of milk production?

• Solution

To determine the heritabilities, we first need to calculate V_P and V_G.

$$\begin{aligned} V_P &= V_A + V_D + V_I + V_E + V_{GE} \\ &= .4 + .1 + .2 + .5 + .5 \\ &= 1.2 \end{aligned}$$

$$\begin{aligned} V_G &= V_A + V_D + V_I \\ &= .7 \end{aligned}$$

(**a**) The narrow sense heritability is:

$$h^2 = \frac{V_A}{V_P} = \frac{0.4}{1.2} = .33$$

(**b**) The broad sense heritability is:

$$H^2 = \frac{V_G}{V_P} = \frac{0.7}{1.2} = .58$$

3. The heights of parents and their offspring are measured for 10 families:

Mean height of parents (cm)	Mean height of offspring (cm)
150	152
157	163
188	193
165	163
160	152
142	157
170	183
183	175
152	163
173	180

From these data, determine:

(**a**) the mean, variance, and standard deviation of height of parents and offspring;

(**b**) the correlation and regression coefficients for a regression of mean offspring height on mean parental height; and

(**c**) the narrow-sense heritability of height in these families.

(**d**) What conclusions can be drawn from the heritability value determined in part c?

• Solution

(**a**) The best way to begin is by constructing a table, as shown below. To calculate the means, we need to sum the values of x and y, which are shown in the last rows of columns A and D of the table.

For the mean of parental height,

$$\bar{x} = \frac{\Sigma x_i}{n} = \frac{1640}{10} = 164 \text{ cm}$$

A Mean height parents (cm) $\bar{x}$	B $x_i - \bar{x}$	C $(x_i - \bar{x})^2$	D Mean height offspring (cm) $\bar{y}$	E $y_i - \bar{y}$	F $(y_i - \bar{y})^2$	G $(x_i - \bar{x})(y_i - \bar{y})$
150	−14	196	152	−16.1	259.21	225.4
157	−7	49	163	−5.1	26.01	35.7
188	24	576	193	24.9	620.01	597.6
165	1	1	163	−5.1	26.01	−5.1
160	−4	16	152	−16.1	259.21	64.4
142	−22	484	157	−11.1	123.21	244.2
170	6	36	183	14.9	222.01	89.4
183	19	361	175	6.9	47.61	131.1
152	−12	144	163	−5.1	26.01	61.2
173	9	81	180	11.9	141.61	107.1
$\Sigma\bar{x}_i = 1640$		$\Sigma(x - \bar{x})^2 = 1944$	$\Sigma\bar{y}_i = 1681$		$\Sigma(y - \bar{y})^2 = 1750.9$	$\Sigma(x_i - \bar{x})(y_i - \bar{y}) = 1551$

For the mean of the offspring height,

$$\bar{y} = \frac{\Sigma y_i}{n} = \frac{1681}{10} = 168.1 \text{ cm}$$

For the variance, we subtract each x and y value from its mean (columns B and E) and square these differences (columns C and F). The sums of the these squared deviations are shown in the last row of columns C and F. For the regression, we need the covariance, which requires that we take the difference between each x value and its mean and multiply it by the difference between each y value and its mean $[(x_i - \bar{x})(y_i - \bar{y})$, column G] and then sum these products (last row of column G).

The variance is the sum of the squared deviations from the mean divided by $n - 1$, where n is the number of measurements:

$$s_x^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1} = \frac{1944}{9} = 216$$

$$s_y^2 = \frac{\Sigma(y_i - \bar{y})^2}{n - 1} = \frac{1750.9}{9} = 194.54$$

The standard deviation is the square root of the variance:

$$s_x = \sqrt{s_x^2} = \sqrt{216} = 14.70$$

$$s_y = \sqrt{s_y^2} = \sqrt{194.54} = 13.95$$

To calculate the correlation coefficient and regression coefficient, we need the covariance:

$$\text{cov}_{xy} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{n - 1} = \frac{1551}{9} = 172.33 \text{ cm}$$

The correlation coefficient is the covariance divided by the standard deviation of x and the standard deviation of y:

$$r = \frac{\text{cov}_{xy}}{s_x s_y} = \frac{172.33}{(14.70)(13.95)} = .84$$

The regression coefficient is the covariance divided by the variance of x:

$$r = \frac{\text{cov}_{xy}}{s_x^2} = \frac{172.33}{216} = .80$$

(b) In a regression of the mean phenotype of the offspring against the mean phenotype of the parents, the regression coefficient equals the narrow-sense heritability, which is .80.

(c) We conclude that 80% of the variance in height among the members of these families results from additive genetic variance.

4. A farmer is raising rabbits. The average body weight in his population of rabbits is 3 kg. The farmer selects the 10 largest rabbits in his population, whose average body weight is 4 kg, and interbreeds them. If the heritability of body weight in the rabbit population is .7, what is the expected body weight among offspring of the selected rabbits?

• Solution

The farmer has carried out a response-to-selection experiment, in which the response to selection will equal the selection differential times the narrow-sense heritability. The selection differential equals the difference in average weights of the selected rabbits and the entire population: 4 kg − 3 kg = 1 kg. The narrow-sense heritability is given as .7; so the expected response to selection is: $R = h^2 \times S = .7 \times 1 \text{ kb} = 0.7$ kg. This is the increase in weight that is expected in the offspring of the selected parents; so the average weight of the offspring is expected to be: 3 kg + 0.7 kg = 3.7 kg.

COMPREHENSION QUESTIONS

* 1. How does a quantitative characteristic differ from a discontinuous characteristic?

2. Briefly explain why the relation between genotype and phenotype is frequently complex for quantitative characteristics.

* 3. Why do polygenic characteristics have many phenotypes?

* 4. Explain the relation between a population and a sample. What characteristics should a sample have to be representative of the population?

5. What information do the mean and variance provide about a distribution?

6. How is the standard deviation related to the variance?

* 7. What information does the correlation coefficient provide about the association between two variables?

8. What is regression? How is it used?

*10. List all the components that contribute to the phenotypic variance and define each component.

*11. How do the broad-sense and narrow-sense heritabilities differ?

12. Briefly outline some of the ways that heritability can be calculated.

13. Briefly discuss common misunderstandings or misapplications of the concept of heritability.

14. Briefly explain how genes affecting a polygenic characteristic are located with the use of QTL mapping.

*15. How is the response to selection related to the narrow-sense heritability and the selection differential? What information does the response to selection provide?

16. Why does the response to selection often level off after many generations of selection?

*17. What is the difference between phenotypic and genetic correlations?

APPLICATION QUESTIONS AND PROBLEMS

*18. For each of the following characteristics, indicate whether it would be considered a discontinuous characteristic or a quantitative characteristic. Briefly justify your answer.

(a) Kernel color in a strain of wheat, in which two codominant alleles segregating at a single locus determine the color. Thus, there are three phenotypes present in this strain: white, light red, and medium red.

(b) Body weight in a family of Labrador retrievers. An autosomal recessive allele that causes dwarfism is present in this family. Two phenotypes are recognized: dwarf (less than 13 kg) and normal (greater than 13 kg).

(c) Presence or absence of leprosy; susceptibility to leprosy is determined by multiple genes and numerous environmental factors.

(d) Number of toes in guinea pigs, which is influenced by genes at many loci.

(e) Number of fingers in humans; extra (more than five) fingers are caused by the presence of an autosomal dominant allele.

*19. The following data are the numbers of digits per foot in 25 guinea pigs. Construct a frequency distribution for these data.

4, 4, 4, 5, 3, 4, 3, 4, 4, 5, 4, 4, 3, 2, 4, 4, 5, 6, 4, 4, 3, 4, 4, 4, 5

20. Ten male Harvard students were weighed in 1916. Their weights are given in the following table. Calculate the mean, variance, and standard deviation for these weights.

Weight (kg) of Harvard students (class of 1920)
51
69
69
57
61
57
75
105
69
63

*21. In Moab, Utah, temperature and rainfall exhibit a correlation of .67. Assuming that this correlation is significant (not due to chance), is there more rainfall, on the average, in the summer or in the winter at this location?

22. Among a population of tadpoles, the correlation coefficient for size at metamorphosis and time required for metamorphosis is −.74. On the basis of this correlation, what conclusions can you make about the relative sizes of tadpoles that metamorphose quickly and those that metamorphose more slowly?

*23. A researcher studying alcohol consumption in North American cities finds a significant, positive correlation between the number of Baptist preachers and alcohol consumption. Is it reasonable for the researcher to conclude that the Baptist preachers are consuming most of the alcohol? Why or why not?

24. Body weight and length were measured on six mosquito fish; these measurements are given in the following table. Calculate the correlation coefficient for weight and length in these fish.

Wet weight (g)	Length (mm)
115	18
130	19
210	22
110	17
140	20
185	21

*25. The heights of mothers and daughters are given in the following table.

(a) Calculate the correlation coefficient for the heights of the mothers and daughters.

(b) Using regression, predict the expected height of a daughter whose mother is 67 inches tall.

Height of mother (in)	Height of daughter (in)
64	66
65	66
66	68
64	65
63	65
63	62
59	62
62	64
61	63
60	62

*26. Assume that plant weight is determined by a pair of alleles at each of two independently assorting loci (*A* and *a*, *B* and *b*) that are additive in their effects. Further, assume that each allele represented by an uppercase letter contributes 4 g to weight and each allele represented by a lowercase letter contributes 1 g to weight.

(a) If a plant with genotype *AABB* is crossed with a plant with genotype *aabb*, what weights are expected in the F_1 progeny?

(b) What is the distribution of weight expected in the F_2 progeny?

27. Assume that three loci, each with two alleles (*A* and *a*, *B* and *b*, *C* and *c*), determine the differences in height between two homozygous strains of a plant. These genes are additive and equal in their effects on plant height. One strain

(*aabbcc*) is 10 cm in height. The other strain (*AABBCC*) is 22 cm in height. The two strains are crossed, and the resulting F_1 are interbred to produce F_2 progeny. Give the phenotypes and the expected proportions of the F_2 progeny.

*28. A farmer has two homozygous varieties of tomatoes. One variety, called *Little Pete*, has fruits that average only 2 cm in diameter. The other variety, *Big Boy*, has fruits that average a whopping 14 cm in diameter. The farmer crosses *Little Pete* and *Big Boy*; he then intercrosses the F_1 to produce F_2 progeny. He grows 2000 F_2 tomato plants and doesn't find any F_2 offspring that produce fruits as small as *Little Pete* or as large as *Big Boy*. If we assume that the differences in fruit size of these varieties are produced by genes with equal and additive effects, what conclusion can we make about the minimum number of loci with pairs of alleles determining the differences in fruit size of the two varieties?

29. Seed size in a plant is a polygenic characteristic. A grower crosses two pure-breeding varieties of the plant and measures seed size in the F_1 progeny. He then backcrosses the F_1 plants to one of the parental varieties and measures seed size in the backcross progeny. The grower finds that seed size in the backcross progeny has a higher variance than does seed size in the F_1 progeny. Explain why the backcross progeny are more variable.

*30. Phenotypic variation in tail length of mice has the following components:

Additive genetic variance (V_A)	= .5
Dominance genetic variance (V_D)	= .3
Genic interaction variance (V_I)	= .1
Environmental variance (V_E)	= .4
Genetic–environmental interaction variance (V_{GE})	= .0

(a) What is the narrow-sense heritability of tail length?

(b) What is the broad-sense heritability of tail length?

31. The narrow-sense heritability of ear length in Reno rabbits is .4. The phenotypic variance (V_P) is .8 and the environmental variance (V_E) is .2. What is the additive genetic variance (V_A) for ear length in these rabbits?

*32. Assume that human ear length is influenced by multiple genetic and environmental factors. Suppose you measured ear length on three groups of people, in which group A consists of five unrelated persons, group B consists of five siblings, and group C consists of five first cousins.

(a) Assuming that the environment for each group is similar, which group should have the highest phenotypic variance? Explain why.

(b) Is it realistic to assume that the environmental variance for each group is similar? Explain your answer.

33. A characteristic has a narrow-sense heritability of .6.

(a) If the dominance variance (V_D) increases and all other variance components remain the same, what will happen to the narrow-sense heritability? Will it increase, decrease, or remain the same? Explain.

(b) What will happen to the broad-sense heritability? Explain.

(c) If the environmental variance (V_E) increases and all other variance components remain the same, what will happen to the narrow-sense heritability? Explain.

(d) What will happen to the broad-sense heritability? Explain.

34. Flower color in the pea plants that Mendel studied is controlled by alleles at a single locus. A group of peas homozygous for purple flowers is grown in a garden. Careful study of the plants reveals that all their flowers are purple, but there is some variability in the intensity of the purple color. If heritability were estimated for this variation in flower color, what would it be. Explain your answer.

*35. A graduate student is studying a population of bluebonnets along a roadside. The plants in this population are genetically variable. She counts the seeds produced by 100 plants and measures the mean and variance of seed number. The variance is 20. Selecting one plant, the graduate student takes cuttings from it, and cultivates these cuttings in the greenhouse, eventually producing many genetically identical clones of the same plant. She then transplants these clones into the roadside population, allows them to grow for 1 year, and then counts the number of seeds produced by each of the cloned plants. The graduate student finds that the variance in seed number among these cloned plants is 5. From the phenotypic variance of the genetically variable and genetically identical plants, she calculates the broad-sense heritability.

(a) What is the broad-sense heritability of seed number for the roadside population of bluebonnets?

(b) What might cause this estimate of heritability to be inaccurate?

*36. The length of the middle joint of the right index finger was measured on 10 sets of parents and their adult offspring. The mean parental lengths and the mean offspring lengths for each family are listed in the following table. Calculate the regression coefficient for regression of mean offspring length against mean parental length and estimate the narrow-sense heritability for this characteristic.

Mean parental length (mm)	Mean offspring length (mm)
30	31
35	36
28	31
33	35
26	27
32	30
31	34
29	28
40	38
33	34

*37. Mr. Jones is a pig farmer. For many years, he has fed his pigs the food left over from the local university cafeteria, which is known to be low in protein, deficient in vitamins, and downright untasty. However, the food is free, and his pigs don't complain. One day a salesman from a feed company visits Mr. Jones. The salesman claims that his company sells a new, high-protein, vitamin-enriched feed that enhances weight gain in pigs. Although the food is expensive, the salesman claims that the increased weight gain of the pigs will more than pay for the cost of the feed, increasing Mr. Jones's profit. Mr. Jones responds that he took a genetics class when he went to the university and that he has conducted some genetic experiments on his pigs; specifically, he has calculated the narrow-sense heritability of weight gain for his pigs and found it to be .98. Mr. Jones says that this heritability value indicates that 98% of the variance in weight gain among his pigs is determined by genetic differences, and therefore the new pig feed can have little effect on the growth of his pigs. He concludes that the feed would be a waste of his money. The salesman doesn't dispute Mr. Jones' heritability estimate, but he still claims that the new feed can significantly increase weight gain in Mr. Jones' pigs. Who is correct and why?

*38. Joe is breeding cockroaches in his dorm room. He finds that the average wing length in his population of cockroaches is 4 cm. He picks six cockroaches that have the largest wings; the average wing length among these selected cockroaches is 10 cm. Joe interbreeds these selected cockroaches. From previous studies, he knows that the narrow-sense heritability for wing length in his population of cockroaches is .6.

(a) Calculate the selection differential and expected response to selection for wing length in these cockroaches.

(b) What should be the average wing length of the progeny of the selected cockroaches?

39. Three characteristics in beef cattle—body weight, fat content, and tenderness—are measured and the following variance components are estimated.

	Body weight	Fat content	Tenderness
V_A	22	45	12
V_D	10	25	5
V_I	3	8	2
V_E	42	64	8
V_{GE}	0	0	1

In this population, which characteristic would respond best to selection? Explain your reasoning.

*40. A rancher determines that the average amount of wool produced by a sheep in his flock is 22 kg per year. In an attempt to increase the wool production of his flock, the rancher picks five male and five female sheep with the greatest wool production; the average amount of wool produced per sheep by those selected is 30 kg. He interbreeds these selected sheep and finds that the average wool production among the progeny of the selected sheep is 28 kg. What is the narrow-sense heritability for wool production among the sheep in the rancher's flock?

41. The narrow-sense heritability of wing length in a population of *Drosophila melanogaster* is .8. The narrow-sense heritability of head width in the same population is .9. The genetic correlation between wing length and head width is −.86. If a geneticist selects for increased wing length in these flies, what will happen to head width?

CHALLENGE QUESTIONS

42. We have explored some of the difficulties in separating genetic and environmental components of human behavioral characteristics. Considering these difficulties and what you know about calculating heritability, propose an experimental design for accurately measuring the heritability of musical ability.

43. A student who has just learned about quantitative genetics says, "Heritability estimates are worthless! They don't tell you anything about the genes that affect a characteristic. They don't provide any information about the types of offspring to expect from a cross. Heritability estimates measured in one population can't be used for other populations; so they don't even give you any general information about how much of a characteristic is genetically determined. I can't see that heritabilities do anything other than make undergraduate students sweat during tests." How would you respond to this statement? Is the student correct? What good are heritabilities, and why do geneticists bother to calculate them?

44. A geneticist selects for increased size in a population of fruit flies that she is raising in her laboratory. She starts with the two largest males and the two largest females and uses them as the parents for the next generation. From the progeny produced by these selected parents, she selects the two largest males and the two largest females and mates them. She repeats this procedure each generation. The average weight of flies in the initial population was 1.1 mg. The flies respond to selection, and their body size steadily increases. After 20 generations of selection, the average weight is 2.3 mg. However, after about 20 generations, the response to selection in subsequent generations levels off, and the average size of the flies no longer increases. At this point, the geneticist takes a long vacation; while she is gone, the fruit flies in her population interbreed randomly. When

she returns from vacation, she finds that the average size of the flies in the population has decreased to 2.0 mg.

(a) Provide an explanation for why the response to selection leveled off after 20 generations.

(b) Why did the average size of the fruit flies decrease when selection was no longer applied during the geneticist's vacation?

45. Manic-depressive illness is a psychiatric disorder that has a strong hereditary basis, but the exact mode of inheritance is not known. Previous research has shown that siblings of patients with manic-depressive illness are more likely also to develop the disorder than are siblings of unaffected persons. A recent study demonstrated that the ratio of manic-depressive brothers to manic-depressive sisters is higher when the patient is male than when the patient is female. In other words, relatively more brothers of manic-depressive patients also have the disease when the patient is male than when the patient is female. What does this new observation suggest about the inheritance of manic-depressive illness?

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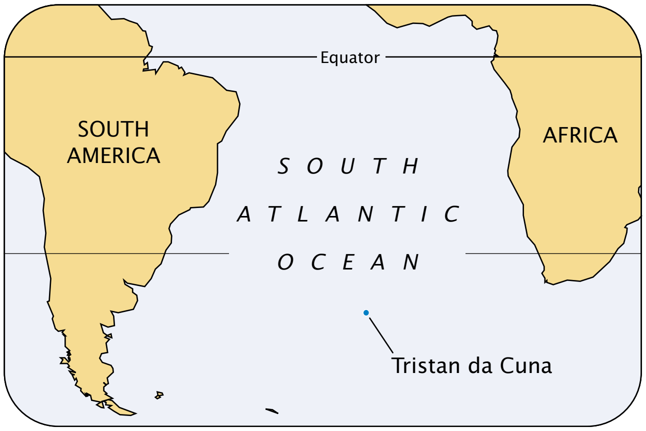
23 Population and Evolutionary Genetics

The inhabitants of the island of Tristan da Cuna have one of the highest incidences of asthma in the world due to the population's unique genetic history. (John Eckwall.)

The Genetic History of Tristan da Cuna

In the fall of 1993, geneticist Noé Zamel arrived at Tristan da Cuna, a small remote island in the South Atlantic (FIGURE 23.1). It had taken Zamel 9 days to make the trip from his home in Canada, first by plane from Toronto to South Africa and then aboard a small research vessel to the island. Because of its remote location, the people of Tristan da Cuna call their home "the loneliest island," but isolation was not what attracted Zamel to Tristan da Cuna. Zamel was looking for a gene that causes asthma, and the inhabitants of Tristan da Cuna have one of the world's highest incidences of hereditary asthma: more than half of the islanders display some symptoms of the disease.

The high frequency of asthma on Tristan da Cuna derives from the unique history of the island's gene pool. The population traces its origin to William Glass, a Scot who moved his family there in 1817. They were joined by some shipwrecked sailors and a few women who migrated from the island of St. Helena but, owing to its remote location and lack of a deep harbor, the island population remained largely isolated. The descendants of Glass and the other settlers intermarried, and slowly the island population increased in number; by 1855, about 100 people inhabited the island. However, Tristan da Cuna's population dropped markedly when, after William Glass's death in 1856, many islanders migrated to South America and South Africa. By 1857, only 33 people remained, and the population grew slowly afterward. It was reduced again in 1885 when a small



23.1 **Tristan de Cuna is a small island in the South Atlantic.**

boat carrying 15 men was capsized by a huge wave, drowning all on board. Many of the widows and their children left the island, and the population dropped from 106 to 59. In 1961, a volcanic eruption threatened the main village. Fortunately, all of the islanders were rescued and transported to England, where they spent 2 years before returning to Tristan da Cuna.

Today, just a little more than 300 people permanently inhabit the island. These islanders have many genes in common and, in fact, all the island's inhabitants are no less closely related than cousins. Because the founders of the colony were few in number and many were already related, many of the genes in today's population can be traced to just a few original settlers. The population has always been small, which also gives rise to inbreeding and allows chance factors to have a large effect on the frequencies of the alleles in the population. The abrupt population reductions in 1856 and 1885 eliminated some alleles from the population and elevated the frequencies of others. As will be discussed in this chapter, the events affecting these islanders (small number of founders, limited population size, inbreeding, and population reduction) affect the proportions of alleles in a population. All of these factors have contributed to the high proportion of alleles that cause asthma among the inhabitants of Tristan da Cuna.

Tristan da Cuna illustrates how the history of a population shapes its genetic makeup. *Population genetics* is the branch of genetics that studies the genetic makeup of *groups* of individuals and how a group's genetic composition changes with time. Population geneticists usually focus their attention on a **Mendelian population,** which is a group of interbreeding, sexually reproducing individuals that have a common set of genes, the **gene pool.** A population evolves through changes in its gene pool; so population genetics is therefore also the study of evolution. Population geneticists study the variation in alleles within and between groups and the evolutionary forces responsible for shaping the patterns of genetic variation found in nature. In this chapter, we will learn how the gene pool of a population is measured and what factors are responsible for shaping it. In the later part of the chapter, we will examine molecular studies of genetic variation and evolution.

Genetic Variation

An obvious and pervasive feature of life is variability. Consider a group of students in a typical college class, the members of which vary in eye color, hair color, skin pigmentation, height, weight, facial features, blood type, and susceptibility to numerous diseases and disorders. No two students in the class are likely to be even remotely similar in appearance (FIGURE 23.2a).

Humans are not unique in their extensive variability; almost all organisms exhibit variation in phenotype. For instance, lady beetles are highly variable in their patterns of spots (FIGURE 23.2b), mice vary in body size, snails have different numbers of stripes on their shells, and plants vary in their susceptibility to pests. Much of this phenotypic variation is hereditary. Recognition of the extent of phenotypic variation and its genetic basis led Charles Darwin to the idea of evolution through natural selection.

In fact, even more genetic variation exists in populations than is visible in the phenotype. Much variation exists at the molecular level owing to the redundancy of the genetic code, which allows different codons to specify the same amino acids. Thus two individuals can produce the same protein even if their DNA sequences are different. DNA sequences between the genes and introns within genes do not encode proteins; so much of the variation in these sequences also has little effect on the phenotype.

The amount of genetic variation within natural populations and the forces that limit and shape it are of primary interest to population geneticists. Genetic variation is the basis of all evolution, and the extent of genetic variation within a population affects its potential to adapt to environmental change.

An important, but frequently misunderstood, tool used in population genetics is the mathematical model. Let's take a moment to consider what a model is and how it can be used. A mathematical model usually describes a process in terms of an equation. Factors that may influence the process are represented by variables in the equation; the equation defines the way in which the variables influence the process. Most models are simplified representations of a process, because it is impossible to simultaneously consider all of the influencing factors; some must be ignored in order to examine the effects of others. At first, a model might consider only one or a few factors, but, after their effects are understood, the model can be improved by the addition of

23.2 All organisms exhibit genetic variation. (a) Extensive variation among humans. (b) Variation in spotting patterns of Asian lady beetles. (Part a, Paul Warner/AP)

more details. It is important to realize that even a simple model can be a source of valuable insight into how a process is influenced by key variables.

www.whfreeman.com/pierce More information on genetic diversity within the human species

Before we can explore the evolutionary processes that shape genetic variation, we must be able to describe the genetic structure of a population. The usual way of doing so is to enumerate the types and frequencies of genotypes and alleles in a population.

Calculation of Genotypic Frequencies

A frequency is simply a proportion or a percentage, usually expressed as a decimal fraction. For example, if 20% of the alleles at a particular locus in a population are *A*, we would say that the frequency of the *A* allele in the population is .20. For large populations, where it is not practical to determine the genes of all individuals, a sample of individuals from the population is usually taken and the genotypic and allelic frequencies are calculated for this sample (see Chapter 22 for a discussion of samples.) The genotypic and allelic frequencies of the sample are then used to represent the gene pool of the population.

To calculate a **genotypic frequency,** we simply add up the number of individuals possessing the genotype and divide by the total number of individuals in the sample (N). For a locus with three genotypes AA, Aa, and aa, the frequency (f) of each genotype is:

$$f(AA) = \frac{\text{number of } AA \text{ individuals}}{N} \qquad (23.1)$$

$$f(Aa) = \frac{\text{number of } Aa \text{ individuals}}{N}$$

$$f(aa) = \frac{\text{number of } aa \text{ individuals}}{N}$$

The sum of all the genotypic frequencies always equals 1.

Calculation of Allelic Frequencies

The gene pool of a population can also be described in terms of the allelic frequencies. There are always fewer alleles than genotypes; so the gene pool of a population can be described in fewer terms when the allelic frequencies are used. In a sexually reproducing population, the genotypes are only temporary assemblages of the alleles: the genotypes break down each generation when individual alleles are passed to the next generation through the gametes, and so it is the types and numbers of alleles, not genotypes, that have real continuity from one generation to the next and that make up the gene pool of a population.

Allelic frequencies can be calculated from (1) the numbers or (2) the frequencies of the genotypes. To calculate the **allelic frequency** from the numbers of genotypes, we count the number of copies of a particular allele present in a sample and divide by the total number of all alleles in the sample:

$$\text{frequency of an allele} = \frac{\text{number of copies of the allele}}{\text{number of copies of all alleles at the locus}} \tag{23.2}$$

For a locus with only two alleles (A and a), the frequencies of the alleles are usually represented by the symbols p and q, and can be calculated as follows:

$$p = f(A) = \frac{2n_{AA} + n_{Aa}}{2N} \tag{23.3}$$

$$q = f(a) = \frac{2n_{aa} + n_{Aa}}{2N}$$

where n_{AA}, n_{Aa}, and n_{aa} represent the numbers of AA, Aa, and aa individuals, and N represents the total number of individuals in the sample. We divide by $2N$ because each diploid individual has two alleles at a locus. The sum of the allelic frequencies always equals 1 ($p + q = 1$); so after p has been obtained, q can be determined by subtraction: $q = 1 - p$.

Alternatively, allelic frequencies can also be calculated from the genotypic frequencies. To do so, we add the frequency of the homozygote for each allele to half the frequency of the heterozygote (because half of the heterozygote's alleles are of each type):

$$p = f(A) = f(AA) + \tfrac{1}{2} f(Aa) \tag{23.4}$$

$$q = f(a) = f(aa) + \tfrac{1}{2} f(Aa)$$

We obtain the same values of p and q whether we calculate the allelic frequencies from the numbers of genotypes (Equation 23.3) or from the genotypic frequencies (Equation 23.4).

Loci with multiple alleles We can use the same principles to determine the frequencies of alleles for loci with more than two alleles. To calculate the allelic frequencies from the numbers of genotypes, we count up the number of copies of an allele by adding twice the number of homozygotes to the number of heterozygotes that possess the allele and divide this sum by twice the number of individuals in the sample. For a locus with three alleles (A^1, A^2, and A^3) and six genotypes (A^1A^1, A^1A^2, A^1A^3, A^2A^2, A^2A^3, and A^3A^3), the frequencies (p, q, and r) of the alleles are:

$$p = f(A^1) = \frac{2n_{A^1A^1} + n_{A^1A^2} + n_{A^1A^3}}{2N} \tag{23.5}$$

$$q = f(A^2) = \frac{2n_{A^2A^2} + n_{A^1A^2} + n_{A^2A^3}}{2N}$$

$$r = f(A^3) = \frac{2n_{A^3A^3} + n_{A^1A^3} + n_{A^2A^3}}{2N}$$

Alternatively, we can calculate the frequencies of multiple alleles from the genotypic frequencies by extending Equation 23.4. Once again, we add the frequency of the homozygote to half the frequency of each heterozygous genotype that possesses the allele:

$$p = f(A^1) = f(A^1A^1) + \tfrac{1}{2} f(A^1A^2) + \tfrac{1}{2} f(A^1A^3) \tag{23.6}$$

$$q = f(A^2) = f(A^2A^2) + \tfrac{1}{2} f(A^1A^2) + \tfrac{1}{2} f(A^2A^3)$$

$$r = f(A^3) = f(A^3A^3) + \tfrac{1}{2} f(A^1A^3) + \tfrac{1}{2} f(A^2A^3)$$

X-linked loci To calculate allelic frequencies for genes at X-linked loci, we apply these same principles. However, we must remember that a female possesses two X chromosomes and therefore has two X-linked alleles, whereas a male has only a single X chromosome and has one X-linked allele.

Suppose there are two alleles at an X-linked locus, X^A and X^a. Females may be either homozygous ($X^A X^A$ or X^aX^a) or heterozygous (X^AX^a). All males are hemizygous (X^AY or X^aY). To determine the frequency of the X^A allele (p), we first count the number of copies of X^A: we multiply the number of X^AX^A females by two and add the number of X^AX^a females and the number of X^AY males. We then divide the sum by the total number of alleles at the locus, which is twice the total number of females plus the number of males:

$$p = f(X^A) = \frac{2n_{X^AX^A} + n_{X^AX^a} + n_{X^AY}}{2n_{\text{females}} + n_{\text{males}}} \tag{23.7a}$$

Similarly, the frequency of the X^a allele is:

$$q = f(X^a) = \frac{2n_{X^aX^a} + n_{X^AX^a} + n_{X^aY}}{2n_{\text{females}} + n_{\text{males}}} \tag{23.7b}$$

The frequencies of X-linked alleles can also be calculated from genotypic frequencies by adding the frequency of the females that are homozygous for the allele, half the frequency of the females that are heterozygous for the allele, and the frequency of males hemizygous for the allele:

$$p = f(X^A) = f(X^AX^A) + {}^1\!/_2 f(X^AX^a) + f(X^AY) \quad (23.8)$$

$$q = f(X^a) = f(X^aX^a) + {}^1\!/_2 f(X^AX^a) + f(X^aY)$$

If you remember the logic behind all of these calculations, you can determine allelic frequencies for any set of genotypes, and it will not be necessary to memorize the formulas.

Concepts

Population genetics is concerned with the genetic composition of a population and how it changes with time. The gene pool of a population can be described by the frequencies of genotypes and alleles in the population.

Worked Problem

The human MN blood type antigens are determined by two codominant alleles, L^M and L^N (see p. 000 in Chapter 5). The MN blood types and corresponding genotypes of 398 Finns from Karjala are tabulated here.

Phenotype	Genotype	Number
MM	L^ML^M	182
MN	L^ML^N	172
NN	L^NL^N	44

Source: W. C. Boyd, *Genetics and the Races of Man* (Boston: Little, Brown, 1950.)

Calculate the allelic and genotypic frequencies at the MN locus for the Karjala population.

• **Solution**

The genotypic frequencies for the population are calculated with the following formula:

$$\text{genotypic frequency} = \frac{\text{number of individuals with genotype}}{\text{total number of individudals in sample}(N)}$$

$$f(L^ML^M) = \frac{\text{number of } L^ML^M \text{ individuals}}{N} = \frac{182}{398} = .457$$

$$f(L^ML^N) = \frac{\text{number of } L^ML^N \text{ individuals}}{N} = \frac{172}{398} = .432$$

$$f(L^NL^N) = \frac{\text{number of } L^NL^N \text{ individuals}}{N} = \frac{44}{398} = .111$$

The allelic frequencies can be calculated from either the numbers or the frequencies of the genotypes. To calculate allelic frequencies from numbers of genotypes, we add the number of copies of the allele and divide by the number of copies of all alleles at that locus.

$$\text{frequency of an allele} = \frac{\text{number of copies of the allele}}{\text{number of copies of all alleles}}$$

$$p = f(L^M) = \frac{(2n_{L^ML^M}) + (n_{L^ML^N})}{2N} = \frac{2(182) + 172}{2(398)}$$
$$= \frac{536}{796} = .673$$

$$q = f(L^N) = \frac{(2n_{L^NL^N}) + (n_{L^ML^N})}{2N} = \frac{2(44) + 172}{2(398)}$$
$$= \frac{260}{796} = .327$$

To calculate the allelic frequencies from genotypic frequencies, we add the frequency of the homozygote for that genotype to half the frequency of each heterozygote that contains that allele:

$$p = f(L^M) = f(L^ML^M) + {}^1\!/_2 f(L^ML^N) = .457 + {}^1\!/_2(.432)$$
$$= .673$$

$$p = f(L^N) = f(L^NL^N) + {}^1\!/_2 f(L^ML^N) = .111 + {}^1\!/_2(.432)$$
$$= .327$$

The Hardy-Weinberg Law

The primary goal of population genetics is to understand the processes that shape a population's gene pool. First, we must ask what effects reproduction and Mendelian principles have on the genotypic and allelic frequencies: How do the segregation of alleles in gamete formation and the combining of alleles in fertilization influence the gene pool? The answer to this question lies in the **Hardy-Weinberg law,** one of the most important principles of population genetics.

The Hardy-Weinberg law was formulated independently by both Godfrey H. Hardy and Wilhelm Weinberg in 1908. (Similar conclusions were reached by several other geneticists about the same time.) The law is actually a mathematical model that evaluates the effect of reproduction on the genotypic and allelic frequencies of a population. It makes several simplifying assumptions about the population and provides two key predictions if these assumptions are met. For an autosomal locus with two alleles, the Hardy-Weinberg law can be stated as follows:

> **Assumptions**—If a population is large, randomly mating, and not affected by mutation, migration, or natural selection, then:
>
> **Prediction 1**—the allelic frequencies of a population do not change; and

Prediction 2—the genotypic frequencies stabilize (will not change) after one generation in the proportions p^2 (the frequency of *AA*), $2pq$ (the frequency of *Aa*), and q^2 (the frequency of *aa*), where p equals the frequency of allele *A* and q equals the frequency of allele *a*.

The Hardy-Weinberg law indicates that, when the assumptions are met, reproduction alone does not alter allelic or genotypic frequencies and the allelic frequencies determine the frequencies of genotypes.

The statement that genotypic frequencies stabilize after one generation means that they may change in the first generation after random mating, because one generation of random mating is required to produce Hardy-Weinberg proportions of the genotypes. Afterward, the genotypic frequencies, like allelic frequencies, do not change as long as the population continues to meet the assumptions of the Hardy-Weinberg law. When genotypes are in the expected proportions of p^2, $2pq$, and q^2, the population is said to be in **Hardy-Weinberg equilibrium.**

Concepts

The Hardy-Weinberg law describes how reproduction and Mendelian principles affect the allelic and genotypic frequencies of a population.

Closer Examination of the Assumptions of the Hardy-Weinberg Law

Before we consider the implications of the Hardy-Weinberg law, we need to take a closer look at the three assumptions that it makes about a population. First, it assumes that the population is large. How big is "large"? Theoretically, the Hardy-Weinberg law requires that a population be infinitely large in size, but this requirement is obviously unrealistic. In practice, a population need only be large enough that chance deviations from expected ratios do not cause significant changes in allelic frequencies. Later in the chapter, we will examine the effects of small population size on allelic frequencies.

A second assumption of the Hardy-Weinberg law is that individuals in the population mate randomly, which means that each genotype mates in proportion to its frequency. For example, suppose that three genotypes are present in a population in the following proportions: $f(AA) = .6$, $f(Aa) = .3$, and $f(aa) = .1$. With random mating, the frequency of mating between two *AA* homozygotes ($AA \times AA$) will be equal to the multiplication of their frequencies: $.6 \times .6 = .36$, whereas the frequency of mating between two *aa* homozygotes ($aa \times aa$) will be only $.1 \times .1 = .01$.

A third assumption of the Hardy-Weinberg law is that the allelic frequencies of the population are not affected by natural selection, migration, and mutation. Although mutation occurs in every population, its rate is so low that it has little effect on the predictions of the Hardy-Weinberg law. Although natural selection and migration are significant factors in real populations, we must remember that the purpose of the Hardy-Weinberg law is to examine only the effect of reproduction on the gene pool. When this effect is known, the effects of other factors (such as migration and natural selection) can be examined.

A final point that should be mentioned is that the assumptions of the Hardy-Weinberg law apply to a *single* locus. No real population mates randomly for all traits; nor is a population completely free of natural selection for all traits. The Hardy-Weinberg law, however, does not require random mating and the absence of selection, migration, and mutation for all traits; it requires these conditions only for the locus under consideration. A population may be in Hardy-Weinberg equilibrium for one locus but not for others.

Implications of the Hardy-Weinberg Law

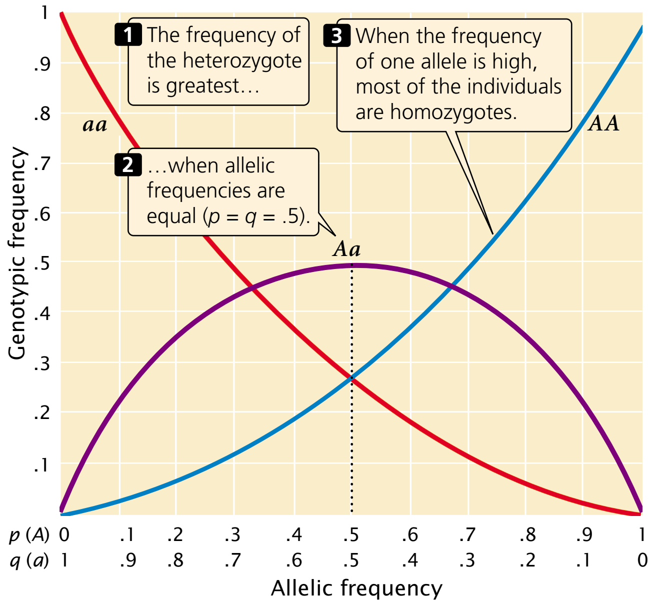
The Hardy-Weinberg law has several important implications for the genetic structure of a population. One implication is that a population cannot evolve if it meets the Hardy-Weinberg assumptions, because evolution consists of change in the allelic frequencies of a population. Therefore the Hardy-Weinberg law tells us that reproduction alone will not bring about evolution. Other processes such as natural selection, mutation, migration, or chance in small populations are required for populations to evolve.

A second important implication is that, when a population is in Hardy-Weinberg equilibrium, the genotypic frequencies are determined by the allelic frequencies. For a locus with two alleles, the frequency of the heterozygote is greatest when allelic frequencies are between .33 and .66 and is at a maximum when allelic frequencies are each .5 (Figure 23.3). The heterozygote frequency also never exceeds .5. Furthermore, when the frequency of one allele is low, homozygotes for that allele will be rare, and most of the copies of a rare allele will be present in heterozygotes. As you can see from Figure 23.3, when the frequency of allele *a* is .2, the frequency of the *aa* homozygote is only .04 (q^2), but the frequency of *Aa* heterozygotes is .32 ($2pq$); 80% of the *a* alleles are in heterozygotes.

A third implication of the Hardy-Weinberg law is that a single generation of random mating produces the equilibrium frequencies of p^2, $2pq$, and q^2. The fact that genotypes are in Hardy-Weinberg proportions does not prove that the population is free from natural selection, mutation, and migration. It means only that these forces have not acted since the last time random mating took place.

Extensions of the Hardy-Weinberg Law

The Hardy-Weinberg law can also be applied to multiple alleles and X-linked alleles. The genotypic frequencies expected under Hardy-Weinberg equilibrium will differ according to the situation.



23.3 When a population is in Hardy-Weinberg equilibrium, the proportions of genotypes are determined by frequencies of alleles.

Hardy-Weinberg expectations for loci with multiple alleles In general, the genotypic frequencies expected at equilibrium are the square of the allelic frequencies. For an autosomal locus with two alleles, these frequencies are $(p + q)^2 = p^2 + 2pq + q^2$. We can also use the square of the allelic frequencies to calculate the equilibrium frequencies for a locus with multiple alleles. An autosomal locus with three alleles, A^1, A^2, and A^3, has six genotypes: A^1A^1, A^1A^2, A^2A^2, A^1A^3, A^2A^3, and A^3A^3. According to the Hardy-Weinberg law, the frequencies of the genotypes at equilibrium depend on the frequencies of the alleles. If the frequencies of alleles A^1, A^2, and A^3 are p, q, and r, respectively, then the equilibrium genotypic frequencies will be the square of the allelic frequencies $(p + q + r)^2 = p^2 + 2pq + q^2 + 2pr + 2qr + r^2$, where:

$$p^2 = f(A^1A^1) \tag{23.9}$$

$$2pq = f(A^1A^2)$$

$$q^2 = f(A^2A^2)$$

$$2pr = f(A^1A^3)$$

$$2qr = f(A^2A^3)$$

$$r^2 = f(A^3A^3)$$

The square of the allelic frequencies can also be used to calculate the expected genotypic frequencies for loci with four or more alleles.

Hardy-Weinberg expectations for X-linked loci For an X-linked locus with two alleles, X^A and X^a, there are five possible genotypes: X^AX^A, X^AX^a, X^aX^a, X^AY, and X^aY. Females possess two X-linked alleles, and the expected proportions of the female genotypes can be calculated by using the square of the allelic frequencies. If the frequencies of X^A and X^a are p and q, respectively, then the equilibrium frequencies of the female genotypes are $(p + q)^2 = p^2$ (frequency of X^AX^A) + $2pq$ (frequency of X^AX^a) + q^2 (frequency of X^aX^a). Males have only a single X-linked allele, and so the frequencies of the male genotypes are p (frequency of X^AY) and q (frequency of X^aY). Notice that these expected frequencies are the proportions of the genotypes among males and females rather than the proportions among the entire population. Thus, p^2 is the expected proportion of females with the genotype X^AX^A; if females make up 50% of the population, then the expected proportion of this genotype in the entire population is $.5 \times p^2$.

The frequency of an X-linked recessive trait among males is q, whereas the frequency among females is q^2. When an X-linked allele is uncommon, the trait will therefore be much more frequent in males than in females. Consider hemophilia A, a clotting disorder caused by an X-linked recessive allele with a frequency (q) of approximately 1 in 10,000, or .0001. At Hardy-Weinberg equilibrium, this frequency will also be the frequency of the disease among males. The frequency of the disease among females, however, will be $q^2 = (.0001)^2 = .00000001$, which is only 1 in 10 million. Hemophilia is 1000 times as frequent in males as in females.

Testing for Hardy-Weinberg Proportions

If a population is in equilibrium, then it is randomly mating for the locus in question, and selection, migration, mutation, and small population size have not significantly influenced the genotypic frequencies since random mating last took place. To determine whether these conditions are met, the genotypic proportions expected under the Hardy-Weinberg law must be compared with the observed genotypic frequencies. To do so, we first calculate the allelic frequencies, then find the expected genotypic frequencies by using the square of the allelic frequencies, and finally compare the observed and expected genotypic frequencies by using a chi-square test.

Worked Problem

Jeffrey Mitton and his colleagues found three genotypes (R^2R^2, R^2R^3, and R^3R^3) at a locus encoding the enzyme peroxidase in ponderosa pine trees growing in Colorado. The observed numbers of these genotypes at Glacier Lake, Colorado, were:

Genotypes	Number observed
R^2R^2	135
R^2R^3	44
R^3R^3	11

Are the ponderosa pine trees at Glacier Lake, Colorado, in Hardy-Weinberg equilibrium at the peroxidase locus?

• Solution

If the frequency of the R^2 allele equals p and the frequency of the R^3 allele equals q, the frequency of the R^2 allele is:

$$p = f(R^2) = \frac{(2n_{R^2R^2}) + (n_{R^2R^3})}{2N} = \frac{135 + 44}{2(190)} = .826$$

The frequency of the R^3 allele is obtained by subtraction:

$$q = f(R^3) = 1 - p = .174$$

The frequencies of the genotypes expected under Hardy-Weinberg equilibrium are then calculated by using p^2, $2pq$, and q^2:

$$R^2R^2 = p^2 = (.826)^2 = .683$$

$$R^2R^3 = 2pq = 2(.826)(.174) = .287$$

$$R^3R^3 = q^2 = (.174)^2 = .03$$

Multiplying each of these expected genotypic frequencies by the total number of observed individuals in the sample (190), we obtain the *numbers* expected for each genotype:

$$R^2R^2 = .683 \times 190 = 129.7$$

$$R^2R^3 = .287 \times 190 = 54.5$$

$$R^3R^3 = .03 \times 190 = 5.7$$

Comparing these expected numbers with the observed numbers of each genotype, we see that there are more R^2R^2 homozygotes and fewer R^2R^3 heterozygotes and R^3R^3 homozygotes in the population than we expect at equilibrium.

A goodness-of-fit chi-square test is used to determine whether the differences between the observed and the expected numbers of each genotype are due to chance:

$$\begin{aligned}\chi^2 &= \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} \\ &= \frac{(135 - 129.7)^2}{129.7} + \frac{(44 - 54.5)^2}{54.5} + \frac{(11 - 5.7)^2}{5.7} \\ &= 0.22 + 2.02 + 4.93 = 7.17\end{aligned}$$

The calculated chi-square value is 7.17; to obtain the probability associated with this chi-square value, we determine the appropriate degrees of freedom.

Up to this point, the chi-square test for assessing Hardy-Weinberg equilibrium has been identical with the chi-square tests that we used in Chapter 3 to assess progeny ratios in a genetic cross, where the degrees of freedom were $n - 1$ and n equaled the number of expected genotypes. For the Hardy-Weinberg test, however, we must subtract an additional degree of freedom, because the expected numbers are based on the observed allelic frequencies; therefore, the observed numbers are not completely free to vary. In general, the degrees of freedom for a chi-square test of Hardy-Weinberg equilibrium equal the number of expected genotypic classes minus the number of associated alleles. For this particular Hardy-Weinberg test, the degrees of freedom are $3 - 2 = 1$.

Once we have calculated both the chi-square value and degrees of freedom, the probability associated with this value can be sought in a chi-square table (Table 3.4). With one degree of freedom, a chi-square value of 7.17 has a probability between .01 and .001. It is very unlikely that the peroxidase genotypes observed at Glacier Lake are in Hardy-Weinberg proportions.

Concepts

The observed number of genotypes in a population can be compared to the Hardy-Weinberg expected proportions by using a goodness of fit chi-square test.

Estimating Allelic Frequencies with the Hardy-Weinberg Law

A practical use of the Hardy-Weinberg law is that it allows us to calculate allelic frequencies when dominance is present. For example, cystic fibrosis is an autosomal recessive disorder characterized by respiratory infections, incomplete digestion, and abnormal sweating (see p. 000 in Chapter 6). Among North American Caucasians, the incidence of the disease is approximately 1 person in 2000. The formula for calculating allelic frequency (Equation 23.3) requires that we know the numbers of homozygotes and heterozygotes, but cystic fibrosis is a recessive disease, and so we cannot easily distinguish between homozygous normal persons and heterozygous carriers. Although molecular tests are available for identifying heterozygous carriers of the cystic fibrosis gene, the low frequency of the disease makes widespread screening impractical. In such situations, the Hardy-Weinberg law can be used to estimate the allelic frequencies.

If we assume that a population is in Hardy-Weinberg equilibrium with regard to this locus, then the frequency of the recessive genotype (aa) will be q^2, and the allelic frequency is the square root of the genotypic frequency:

$$q = \sqrt{f(aa)} \qquad (23.10)$$

The frequency of cystic fibrosis in North American Caucasians is approximately 1 in 2000, or .0005; so $q = \sqrt{0.0005} = .02$. Thus, about 2% of the alleles in the Caucasian population encode cystic fibrosis. We can calculate the frequency of the normal allele by subtracting: $p = 1 - q = 1 - .02 = .98$. After we have calculated p and q, we can use the Hardy-Weinberg law to determine the frequencies of homozygous normal people and heterozygous carriers of the gene:

$$f(AA) = p^2 = (.98)^2 = .960$$

$$f(Aa) = 2pq = 2(.02)(.98) = .0392$$

Thus about 4% (1 of 25) of Caucasians are heterozygous carriers of the allele that causes cystic fibrosis.

Concepts

Although allelic frequencies cannot be calculated directly for traits that exhibit dominance, the Hardy-Weinberg law can be used to estimate the allelic frequencies if the population is in Hardy-Weinberg equilibrium for that locus. The frequency of the recessive allele will be equal to the square root of the frequency of the recessive trait.

Nonrandom Mating

An assumption of the Hardy-Weinberg law is that mating is random with respect to genotype. Nonrandom mating affects the way in which alleles combine to form genotypes and alters the genotypic frequencies of a population.

We can distinguish between two types of nonrandom mating. **Positive assortative mating** refers to a tendency for like individuals to mate. For example, humans exhibit positive assortative mating for height: tall people mate preferentially with other tall people; short people mate preferentially with other short people. **Negative assortative mating** refers to a tendency for unlike individuals to mate. If people engaged in negative assortative mating for height, tall and short people would preferentially mate.

One form of nonrandom mating is **inbreeding,** which is preferential mating between related individuals. Inbreeding is actually positive assortative mating for relatedness, but it differs from other types of assortative mating because it affects all genes, not just those that determine the trait for which the mating preference occurs. Inbreeding causes a departure from the Hardy-Weinberg equilibrium frequencies of p^2, $2pq$, and q^2. More specifically, it leads to an increase in the proportion of homozygotes and a decrease in the proportion of heterozygotes in a population. **Outcrossing** is the avoidance of mating between related individuals.

Inbreeding is usually measured by the **inbreeding coefficient,** designated F, which is a measure of the probability that two alleles are "identical by descent." In a diploid organism, homozygous individual has two copies of the same allele. These two copies may be the same in *state,* which means that the two alleles are alike in structure and function but do not have a common origin. Alternatively, the two alleles in a homozygous individual may be the same because they are identical by *descent*—the copies are descended from a single allele that was present in an ancestor (FIGURE 23.4). If we go back far enough in time, many alleles are likely to be identical by descent but, for calculating the effects of inbreeding, we consider identity by descent by going back only a few generations.

Inbreeding coefficients can range from 0 to 1. A value of 0 indicates that mating in a large population is random; a value of 1 indicates that all alleles are identical by descent. Inbreeding coefficients can be calculated from analyses of pedigrees or they can be determined from the reduction in the heterozygosity of a population. Although we will not go into the details of how F is calculated, it's important to understand how inbreeding affects genotypic frequencies.

When inbreeding occurs, the frequency of the genotypes will be:

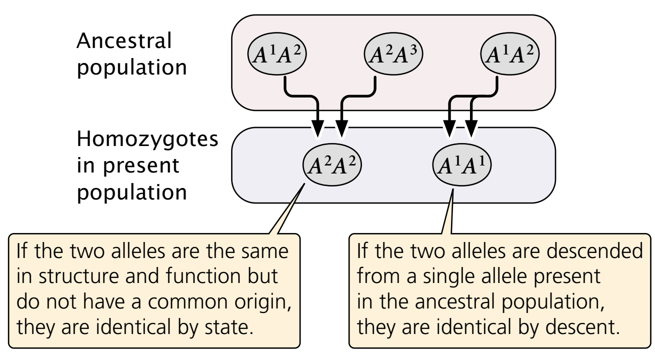
$$f(AA) = p^2 + Fpq \qquad (23.11)$$

$$f(Aa) = 2pq - 2Fpq$$

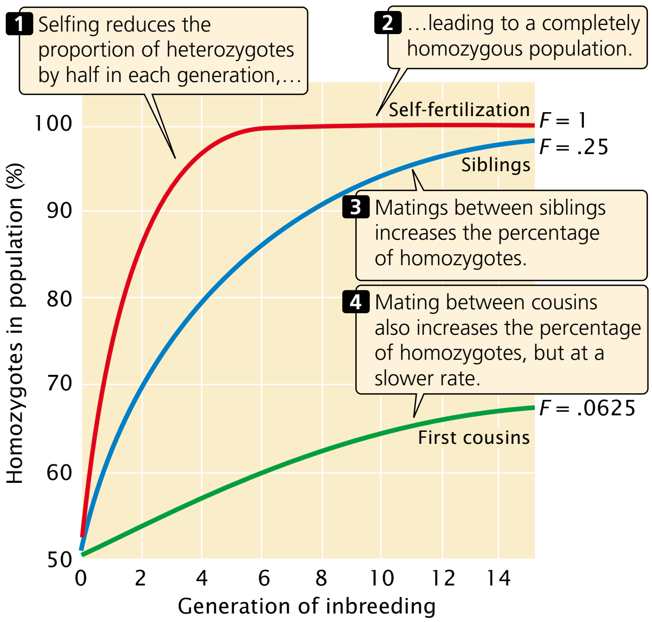
$$f(aa) = q^2 + Fpq$$

With inbreeding, the proportion of heterozygotes *decreases* by $2Fpq$, and half of this value (Fpq) is *added* to the proportion of each homozygote.

Consider a population that reproduces by self-fertilization (so $F = 1$). We will assume that this population begins with genotypic frequencies in Hardy-Weinberg proportions (p^2, $2pq$, and q^2). With selfing, each homozygote produces



23.4 **Individuals may be homozygous by state or by descent.** Inbreeding is a measure of the probability that two alleles are identical by descent.



23.5 Inbreeding increases the percentage of homozygous individuals in a population.

progeny only of the same homozygous genotype (*AA* × *AA* produces all *AA*; and *aa* × *aa* produces all *aa*), whereas only half the progeny of a heterozygote will be like the parent (*Aa* × *Aa* produces $^1/_4$ *AA*, $^1/_2$ *Aa*, and $^1/_4$ *aa*). Selfing therefore reduces the proportion of heterozygotes in the population by half with each generation, until all genotypes in the population are homozygous (Table 23.1 and FIGURE 23.5).

For most outcrossing species, close inbreeding is harmful because it increases the proportion of homozygotes and thereby boosts the probability that deleterious and lethal recessive alleles will combine to produce homozygotes with a harmful trait. Assume that a recessive allele (*a*) that causes a genetic disease has a frequency (q) of .01. If the population mates randomly ($F = 0$), the frequency of individuals affected with the disease (*aa*) will be $q^2 = .01^2 = .0001$; so only 1 in 10,000 individuals will have the disease. However, if $F = .25$ (the equivalent of brother–sister mating), then the expected frequency of the homozygote genotype is $q^2 + 2pqF = (.01)^2 + 2(.99)(.01)(.25) = .0026$; thus, the genetic disease is 26 times as frequent at this level of inbreeding. This increased appearance of lethal and deleterious traits with inbreeding is termed **inbreeding depression;** the more intense the inbreeding, the more severe the inbreeding depression.

The harmful effects of inbreeding have been recognized by humans for thousands of years and are the basis of cultural taboos against mating between close relatives. William Schull and James Neel found that, for each 10% increase in *F*, the mean IQ of Japanese children dropped six points. Child mortality also increases with close inbreeding (Table 23.2); children of first cousins have a 40% increase in mortality over that seen among the children of randomly mated people. Inbreeding also has deleterious effects on crops (FIGURE 23.6) and domestic animals.

Inbreeding depression is most often studied in humans, as well as in plants and animals reared in captivity, but the negative effects of inbreeding may be more severe in natural populations. Julie Jimenez and her colleagues collected wild mice from a natural population in Illinois and bred them in the laboratory for three to four generations. Laboratory matings were chosen so that some mice had no inbreeding, whereas others had an inbreeding coefficient of .25. When both types of mice were released back into the wild, the weekly survival of the inbred mice was only 56% of that of the noninbred mice. Inbred male mice also continously lost weight after release into the wild, whereas noninbred male mice regained their body weight within a few days after release.

In spite of the fact that inbreeding is generally harmful for outcrossing species, a number of plants and animals regularly inbreed and are successful (FIGURE 23.7). Inbreeding is commonly used to produce domesticated plants and animals having desirable traits. As stated earlier, inbreeding increases homozygosity, and eventually all individuals in the population become homozygous for the same allele. If a species undergoes inbreeding for a number of generations, many deleterious recessive alleles are weeded out by natural or artificial selection so that the population becomes homozygous for beneficial alleles. In this way, the harmful effects of inbreeding may eventually be eliminated, leaving a population that is homozygous for beneficial traits.

Table 23.1 Generational increase in frequency of homozygotes in a self-fertilizing population starting with $p = q = .5$

	Genotypic Frequencies		
Generation	***AA***	***Aa***	***aa***
1	$^1/_4$	$^1/_2$	$^1/_4$
2	$^1/_4 + {}^1/_8 = {}^3/_8$	$^1/_4$	$^1/_4 + {}^1/_8 = {}^3/_8$
3	$^3/_8 + {}^1/_{16} = {}^7/_{16}$	$^1/_8$	$^3/_8 + {}^1/_{16} = {}^7/_{16}$
4	$^7/_{16} + {}^1/_{32} = {}^{15}/_{32}$	$^1/_{16}$	$^7/_{16} + {}^1/_{32} = {}^{15}/_{32}$
n	$\frac{1 - (^1/_2)^n}{2}$	$(^1/_2)^n$	$\frac{1 - (^1/_2)^n}{2}$
∞	$^1/_2$	0	$^1/_2$

Concepts

Nonrandom mating alters the frequencies of the genotypes but not the frequencies of the alleles. Inbreeding is preferential mating between related individuals. With inbreeding, the frequency of homozygotes increases while the frequency of heterozygotes decreases.

Table 23.2 Effects of inbreeding on Japanese children

Genetic Relationship of Parents	F	Mortality of Children (Through 12 Years of Age)
Unrelated	0	.082
Second cousins	0.016 ($^1/_{64}$)	.108
First cousins	.0625 ($^1/_{16}$)	.114

Source: After D. L. Hartl, and A. G. Clark, *Principles of Population Genetics.* 2d ed. (Sunderland, MA: Sinauer, 1989), Table 2. Original data from W. J. Schull, and J. V. Neel, *The Effects of Inbreeding on Japanese Children.* (New York: Harper & Row, 1965).

Changes in Allelic Frequencies

The Hardy-Weinberg law indicates that allelic frequencies do not change as a result of reproduction; thus, other processes must cause alleles to increase or decrease in frequency. Processes that bring about change in allelic frequency include mutation, migration, genetic drift (random effects due to small population size), and natural selection.

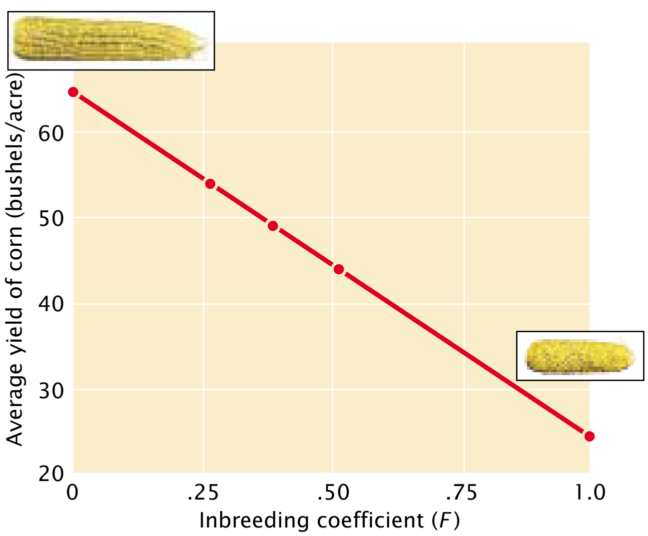
Mutation

Before evolution can occur, genetic variation must exist within a population; consequently, all evolution depends on processes that generate genetic variation. Although new *combinations* of existing genes may arise through recombination in meiosis, all genetic variants ultimately arise through mutation.

The effect of mutation on allelic frequencies Mutation can influence the rate at which one genetic variant increases at the expense of another. Consider a single locus in a population of 25 diploid individuals. Each individual possesses two alleles at the locus under consideration; so the gene pool of the population consists of 50 allelic copies. Let us assume that there are two different alleles, designated G^1 and G^2 with frequencies p and q, respectively. If there are 45 copies of G^1 and 5 copies of G^2 in the population, $p = .90$ and $q = .10$. Now suppose that a mutation changes a G^1 allele into a G^2 allele. After this mutation, there are 44 copies of G^1 and 6 copies of G^2, and the frequency of G^2 has increased from .10 to .12. Mutation has changed the allelic frequency.

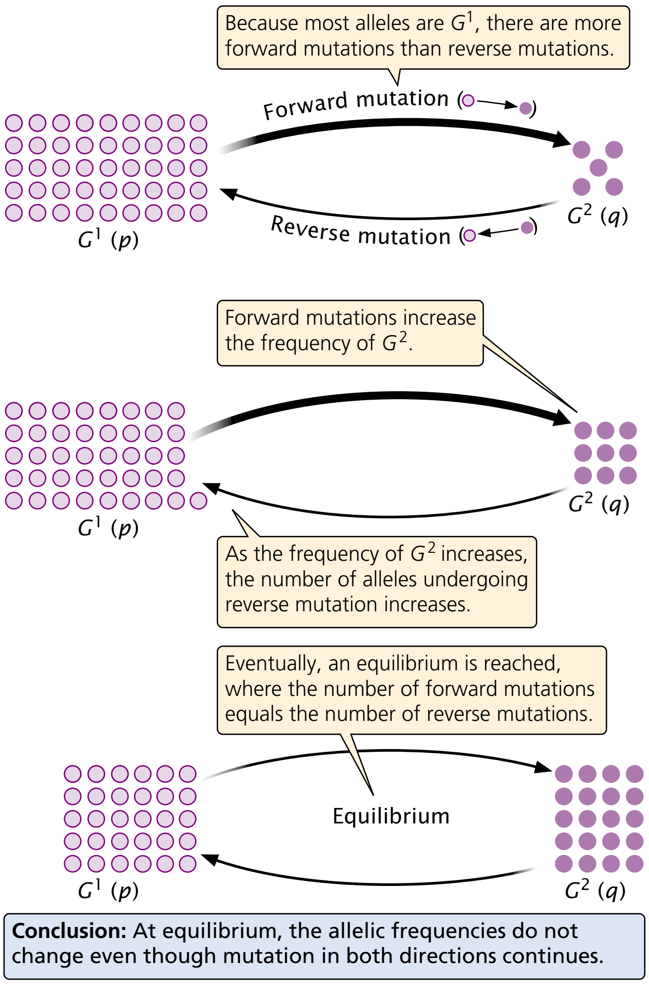
If copies of G^1 continue to mutate to G^2, the frequency of G^2 will increase and the frequency of G^1 will decrease (FIGURE 23.8). The amount that G^2 will change (Δq) as a result of mutation depends on: (1) the rate of G^1-to-G^2 mutation (μ); and (2) p, the frequency of G^1 in the population When p is large, there are many copies of G^1 available to mutate to G^2, and the amount of change will be relatively large. As more mutations occur and p decreases, there will be fewer copies of G^1 available to mutate to G^2. The change in G^2 as a result of mutation equals the mutation rate times the allelic frequency:

$$\Delta q = \mu p \qquad (23.12)$$



23.6 **Inbreeding often has deleterious effects on crops.** As inbreeding increases, the average yield of corn, for example, decreases.

23.7 **Although inbreeding is generally harmful, a number inbreeding organisms are successful.**



23.8 Recurrent mutation changes allelic frequencies. Forward and reserve mutations eventually lead to a stable equilibrium.

As the frequency of p decreases as a result of mutation, the change in frequency due to mutation will be less and less

So far we have considered only the effects of $G^1 \rightarrow G^2$ forward mutations. Reverse $G^2 \rightarrow G^1$ mutations also occur at rate ν, which will probably be different from the forward mutation rate, μ. Whenever a reverse mutation occurs, the frequency of G^2 decreases and the frequency of G^1 increases (see Figure 23.8). The rate of change due to reverse mutations equals the reverse mutation rate times the allelic frequency of G^2 ($\Delta q = \nu q$). The overall change in allelic frequency is a balance between the opposing forces of forward mutation and reverse mutation:

$$\Delta q = \mu p - \nu q \qquad (23.13)$$

Reaching equilibrium of allelic frequencies Consider an allele that begins with a high frequency of G^1 and a low frequency of G^2. In this population, many copies of G^1 are initially available to mutate to G^2, and the increase in G^2 due to forward mutation will be relatively large. However, as the frequency of G^2 increases as a result of forward mutations, fewer copies of G^1 are available to mutate; so the number of forward mutations decreases. On the other hand, few copies of G^2 are initially available to undergo a reverse mutation to G^1 but, as the frequency of G^2 increases, the number of copies of G^2 available to undergo reverse mutation to G^1 increases; so the number of genes undergoing reverse mutation will increase. Eventually, the number of genes undergoing forward mutation will be counterbalanced by the number of genes undergoing reverse mutation. At this point, the increase in q due to forward mutation will be equal to the decrease in q due to reverse mutation, and there will be no net change in allelic frequency ($q = 0$), in spite of the fact that forward and reserve mutations continue to occur. The point at which there is no change in the allelic frequency of a population is referred to as **equilibrium** (see Figure 23.8).

Factors determining allelic frequencies at equilibrium We can determine the allelic frequencies at equilibrium by manipulating Equation 23.13. Recall that $p = 1 - q$. Substituting $1 - q$ for p in Equation 23.13, we get:

$$\begin{aligned}\Delta q &= \mu(1 - q) - \nu q \qquad (23.14)\\ &= \mu - \mu q - \nu q\\ &= \mu - q(\mu + \nu)\end{aligned}$$

At equilibrium, Δq will be 0; so:

$$0 = \mu - q(\mu + \nu) \qquad (23.15)$$

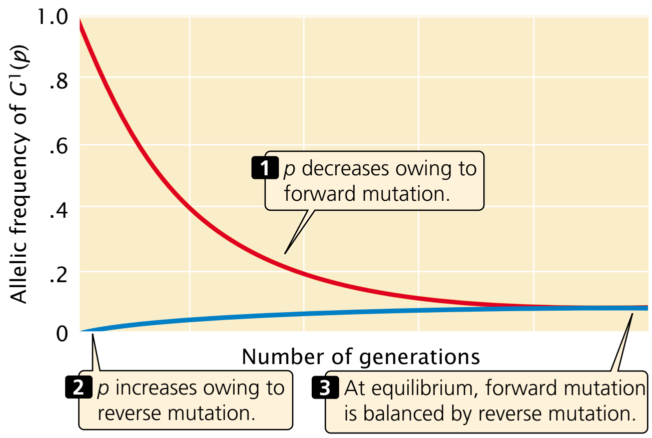
$$q(\mu + \nu) = \mu$$

$$\hat{q} = \frac{\mu}{\mu + \upsilon}$$

where $\hat{q}$ equals the frequency of G^2 at equilibrium. This final equation tells us that the allelic frequency at equilibrium is determined solely by the forward and reverse mutation rates.

Summary of effects When the only evolutionary force acting on a population is mutation, allelic frequencies change with the passage of time because some alleles mutate into others. Eventually, these allelic frequencies reach equilibrium and are determined only by the forward and reverse mutation rates. When the allelic frequencies reach equilibrium, the Hardy-Weinberg law tells us that genotypic frequencies also will remain the same.

The mutation rates for most genes are low; so change in allelic frequency due to mutation in one generation is very small, and long periods of time are required for a population to reach mutational equilibrium. For example, if the forward



23.9 Change due to recurrent mutation slows as the frequency of *p* drops. Allelic frequencies are approaching mutational equilibrium at typical low mutation rates. The allelic frequency of G^1 decreases as a result of forward ($G^1 \rightarrow G^2$) mutation at rate μ (.0001) and increases as a result of reverse ($G^2 \rightarrow G^1$) mutation at rate ν (.00001). Owing to the low rate of mutations, eventual equilibrium takes many generations to be reached.

and reverse mutation rates for alleles at a locus are 1×10^{-5} and 0.3×10^{-5} per generation, respectively (rates that have actually been measured at several loci in mice), and the allelic frequencies are $p = .9$ and $q = .1$, then the net change in allelic frequency per generation due to mutation is:

$$\begin{aligned}\Delta q &= \mu p - \nu q \\ &= (1 \times 10^{-5})(.9) - (.3 \times 10^{-5})(.1) \\ &= 8.7 \times 10^{-6} = .0000087\end{aligned}$$

Therefore, change due to mutation in a single generation is extremely small and, as the frequency of p drops as a result of mutation, the amount of change will become even smaller (Figure 23.9). The effect of typical mutation rates on Hardy-Weinberg equilibrium is negligible, and many generations are required for a population to reach mutational equilibrium. Nevertheless, if mutation is the only force acting on a population for long periods of time, mutation rates will determine allelic frequencies.

Concepts

Recurrent mutation causes changes in the frequencies of alleles. At equilibrium, the allelic frequencies are determined by the forward and reverse mutation rates. Because mutation rates are low, the effect of mutation per generation is very small.

Migration

Another process that may bring about change in the allelic frequencies is the influx of genes from other populations, commonly called **migration** or **gene flow.** One of the assumptions of the Hardy-Weinberg law is that migration does not take place, but many natural populations do experience migration from other populations. The overall effect of migration is twofold: (1) it prevents genetic divergence *between* populations and (2) it increases genetic variation *within* populations.

The effect of migration on allelic frequencies Let us consider the effects of migration by looking at a simple, unidirectional model of migration between two populations that differ in the frequency of an allele *a*. Say the frequency of this allele in population I is q_{I} and in population II is q_{II} (Figure 23.10a and b). In each generation, a representative sample of the individuals in population I migrates to population II (Figure 23.10c) and reproduces, adding its genes to population II's gene pool. Migration is only from population I to population II (is unidirectional), and all the conditions of the Hardy-Weinberg law apply, except the absence of migration.

After migration, population II consists of two types of individuals (Figure 23.10d). Some are migrants; they make up proportion m of population II, and they carry genes from population I; so the frequency of allele *a* in the migrants is q_{I}. The other individuals in population II are the original residents. If the migrants make up proportion m of population II, then the residents make up $1 - m$; because the residents originated in population II, the frequency of allele *a* in this group is q_{II}. After migration, the frequency of allele *a* in the merged population II (q'_{II}) is:

$$q'_{\mathrm{II}} = q_{\mathrm{I}}(m) + q_{\mathrm{II}}(1 - m) \qquad (23.16)$$

where $q_{\mathrm{I}}(m)$ is the contribution to q made by the copies of allele *a* in the migrants and $q_{\mathrm{II}}(1 - m)$ is the contribution to q made by copies of allele *a* in the residents. The change in the allelic frequency due to migration (Δq) will be equal to the new frequency of allele *a* (q'_{II}) minus the original frequency of the allele (q_{II}):

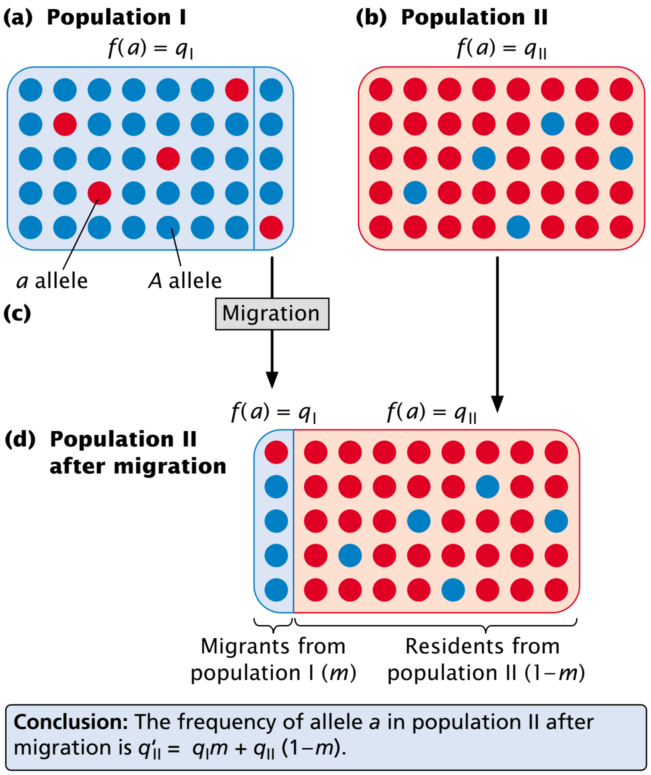
$$q_{\mathrm{II}} = q'_{\mathrm{II}} - q_{\mathrm{II}}$$

In Equation 23.16, we determined that q'_{II} equals $q_{\mathrm{I}}(m) + q_{\mathrm{II}}(1 - m)$. Substituting this value for q'_{II} into the preceding equation, we get:

$$q = q_{\mathrm{I}}(m) + q_{\mathrm{II}}(1 - m) - q_{\mathrm{II}}$$

Expanding the term $q_{\mathrm{II}}(1 - m)$, we get:

$$q = q_{\mathrm{I}}m + q_{\mathrm{II}} - q_{\mathrm{II}}m - q_{\mathrm{II}}$$



23.10 The amount of change in allelic frequency due to migration between populations depends on the difference in allelic frequency and the extent of migration. Shown here is a model of the effect of unidirectional migration on allelic frequencies. (a) The frequency of allele *a* in the source population (population I) is q_I. (b) The frequency of this allele in the recipient population (population II) is q_{II}. (c) Each generation, a random sample of individuals migrate from population I to population II. (d) After migration, population II consists of migrants and residents. The migrants constitute proportion m and have a frequency of *a* equal to q_I'; the residents constitute proportion $1 - m$ and have a frequency of *a* equal to q_{II}.

In this last equation, we are subtracting q_{II} from q_{II}, which gives us zero; so the equation simplifies to:

$$\begin{aligned} q &= q_I m - q_{II} m \\ &= m(q_I - q_{II}) \end{aligned} \tag{23.17}$$

Equation 23.17 summarizes the factors that determine the amount of change in allelic frequency due to migration. The amount of change in q is directly proportional to the migration (m); as the amount of migration increases, the change in allelic frequency increases. The magnitude of change is also affected by the differences in allelic frequencies of the two populations ($q_I - q_{II}$); when the difference is large, the change in allelic frequency will be large.

With each generation of migration, the frequencies of the two populations become more and more similar until, eventually, the allelic frequency of population II equals that of population I. When $q_I - q_{II} = 0$, there will be no further change in the allelic frequency of population II, in spite of the fact that migration continues. If migration between two populations takes place for a number of generations with no other evolutionary forces present, an equilibrium is reached at which the allelic frequency of the recipient population equals that of the source population.

The simple model of unidirectional migration between two populations just outlined can be expanded to accommodate multidirectional migration between several populations (FIGURE 23.11).

The overall effect of migration Migration has two major effects. First, it causes the gene pools of populations to become more similar. Later, we will see how genetic drift and natural selection lead to genetic differences between populations; migration counteracts this tendency and tends to keep populations homogeneous in their allelic frequencies. Second, migration adds genetic variation to populations. Different alleles may arise in different populations owing to rare mutational events, and these alleles can be spread to new populations by migration, increasing the genetic variation within the recipient population.

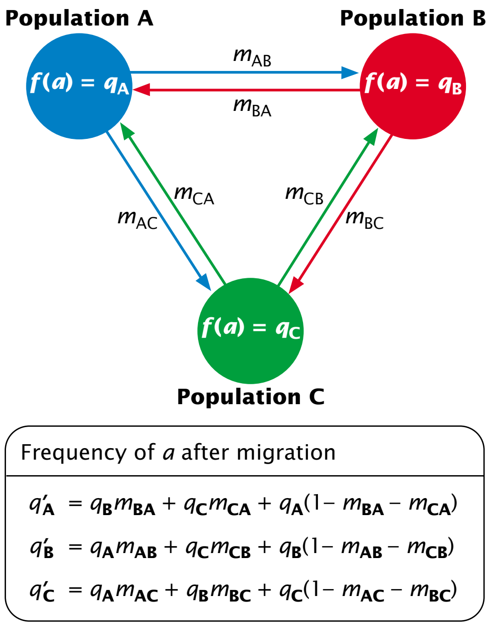
Concepts

Migration causes changes in the allelic frequency of a population by introducing alleles from other populations. The magnitude of change due to migration depends on both the extent of migration and the difference in allelic frequencies between the source and the recipient populations. Migration decreases genetic differences between populations and increases genetic variation within populations.

Genetic Drift

The Hardy-Weinberg law assumes random mating in an infinitely large population; only when population size is infinite will the gametes carry genes that perfectly represent the parental gene pool. But no real population is infinitely large, and when population size is limited, the gametes that unite to form individuals of the next generation carry a sample of alleles present in the parental gene pool. Just by chance, the composition of this sample may deviate from that of the parental gene pool, and this deviation may cause allelic frequencies to change. The smaller the gametic sample, the greater the chance that its composition will deviate from that of the entire gene pool.

The role of chance in altering allelic frequencies is analogous to flipping a coin. Each time we flip a coin, we have a 50% chance of getting a head and a 50% chance of getting a



23.11 Model of multidirectional migration among three populations, A, B, and C, with initial frequency of allele *a* equal to q_A, q_B, and q_C, respectively. The proportion of a population made up of migrants from other populations is designated by m, where the subscripts represent the source and recipient populations. For example, m_{AC} represents the proportion of population C that consists of individuals that moved from A to C. The allelic frequencies in populations A, B, and C after migration are represented by q'_A, q'_B, and q'_C.

tail. If we flip a coin 1000 times, the observed ratio of heads to tails will be very close to the expected 50:50 ratio. If, however, we flip a coin only 10 times, there is a good chance that we will obtain not exactly 5 heads and 5 tails, but rather maybe 7 heads and 3 tails or 8 tails and 2 heads. This kind of deviation from an expected ratio due to limited sample size is referred to as **sampling error.**

Sampling error occurs when gametes unite to produce progeny. Many organisms produce a large number of gametes but, when population size is small, a limited number of gametes unite to produce the individuals of the next generation. Chance influences which alleles are present in this limited sample and, in this way, sampling error may lead to changes in allelic frequency, which is called **genetic drift.** Because the deviations from the expected ratios are random, the direction of change is unpredictable. We can nevertheless predict the magnitude of the changes.

The magnitude of genetic drift The amount of sampling error resulting from genetic drift can be estimated from the variance in allelic frequency. Variance is a statistical measure that describes the degree of variability in a trait (see p. 000 in Chapter 22). Suppose that we observe a large number of separate populations, each with N individuals and allelic frequencies of p and q. After one generation of random mating, genetic drift expressed in terms of the variance in allelic frequency among the populations (s_p^2) will be:

$$s_p^{\,2} = \frac{pq}{2N} \tag{23.18}$$

The amount of change resulting from genetic drift (the variance in allelic frequency) is determined by two parameters: the allelic frequencies (p and q) and the population size (N). Genetic drift will be maximal when p and q are equal (each .5) and when the population size is small.

The effect of population size on genetic drift is illustrated by a study conducted by Luca Cavalli-Sforza and his colleagues. They studied variation in blood types among villagers in the Parm Valley of Italy, where the amount of migration between villages was limited. They found that variation in allelic frequency was greatest between small isolated villages in the upper valley but decreased between larger villages and towns farther down the valley. This result is exactly what we expect with genetic drift: there should be more genetic drift and thus more variation among villages when population size is small.

For ecological and demographic studies, population size is usually defined as the number of individuals in a group. The evolution of a gene pool depends, however, only on those individuals who contribute genes to the next generation. Population geneticists usually define population size as the equivalent number of breeding adults, the **effective population size** (N_e). Several factors determine the equivalent number of breeding adults. One factor is the sex ratio. When the numbers of males and females in the population are equal, the effective population size is simply the sum of reproducing males and females. When they are unequal, then the effective population size is:

$$N_e = \frac{4 \times n_{\text{males}} \times n_{\text{females}}}{n_{\text{males}} + n_{\text{females}}} \tag{23.19}$$

Table 23.3 gives the effective population size for a theoretical population of 100 individuals with different proportions of males and females. Notice that, when the number of males and females is unequal, the effective population size is smaller than it is when the number of males and females is the same. For example, when a population consists of 90 males and 10 females, the effective population size is only 36, and genetic drift will occur as though the actual population consisted of only 36 individuals, equally divided between males and females. A population with 90 males and 10 females has the same effective population size as a population with 10 males and 90 females—it makes no difference which sex is in excess.

Table 23.3 Effective population size (N_e) in theoretical populations of 100 individuals, each with a different sex ratio

Sex Ratio*	Number of Males	Number of Females	N_e
1.00	50	50	100
3.00	75	25	75
0.33	25	75	75
9.00	90	10	36
0.10	10	90	36
99.00	99	1	3.96
0.01	1	99	3.96

*The sex ratio is the ratio of the number of males to the number of females.

The reason that the sex ratio influences genetic drift is that half the genes in the gene pool come from males and half come from females. When one sex is present in low numbers, genetic drift increases because half of the genes are coming from a small number of individuals. In a population consisting of 10 males and 90 females, the overall population size is relatively large (100), but only 10 males contribute half the genes to the next generation. Sampling error therefore affects the range of genes present in the male gametes, and chance will have a major effect on the allelic frequencies of the next generation.

Other factors that influence effective population size include variation between individuals in reproductive success, fluctuations in population size, the age structure of the population, and whether mating is random.

Concepts

Genetic drift is change in allelic frequency due to chance factors. The amount of change in allelic frequency due to genetic drift is inversely related to the effective population size (the equivalent number of breeding adults in a population). Effective population size decreases when there are unequal numbers of breeding males and females.

Causes of genetic drift All genetic drift arises from sampling error, but there are several different ways in which sampling error can arise. First, a population may be reduced in size for a number of generations because of limitations in space, food, or some other critical resource. Genetic drift in a small population for multiple generations can significantly affect the composition of a population's gene pool.

A second way that sampling error can arise is through the **founder effect,** which is due to the establishment of a population by a small number of individuals; the population of Tristan da Cuna, discussed in the introduction to this chapter, underwent a founder effect. Although a population may increase and become quite large, the genes carried by all its members are derived from the few genes originally present in the founders (assuming no migration or mutation). Chance events affecting which genes were present in the founders will have an important influence on the makeup of the entire population. The small number of founders of Tristan da Cuna included two sisters and a daughter who suffered from asthma; the high incidence of asthma on the island today can be traced to alleles carried by these founders.

A third way that genetic drift arises is through a **genetic bottleneck,** which develops when a population undergoes a drastic reduction in population size. A genetic bottleneck developed in northern elephant seals (FIGURE 23.12). Before 1800, thousands of elephant seals were found along the California coast, but the population was devastated by hunting between 1820 and 1880. By 1884, as few as 20 seals survived on a remote beach of Isla de Guadelupe west of Baja, California. Restrictions on hunting enacted by the United States and Mexico allowed the seals to recover, and there are now more than 30,000 seals in the population. All seals in the population today are genetically similar, because they have genes that were carried by the few survivors of the population bottleneck.

The effects of genetic drift Genetic drift has several important effects on the genetic composition of a population. First, it produces change in allelic frequencies within a population. Because drift is random, allelic frequency is just as likely to increase as it is to decrease and will wander with the passage of time (hence the name genetic drift). FIGURE 23.13 illustrates a computer simulation of genetic drift in five populations over 30 generations, starting with $q = .5$ and maintaining a constant population size of 10 males and 10 females. These allelic frequencies change randomly from generation to generation.

A second effect of genetic drift is to reduce genetic variation within populations. Through random change, an allele may eventually reach a frequency of either 1 or 0, at which point all individuals in the population are homozygous for one allele. When an allele has reached a frequency of 1, we say that it has reached **fixation.** Other alleles are lost (reach a frequency of 0) and can be restored only by migration from another population or by mutation. Fixation, then, leads to a loss of genetic variation within a population. This loss can be seen in northern elephant seals. Today, these seals have low levels of genetic variation; a study of 24 protein-encoding genes found no individual or population differences in these genes.

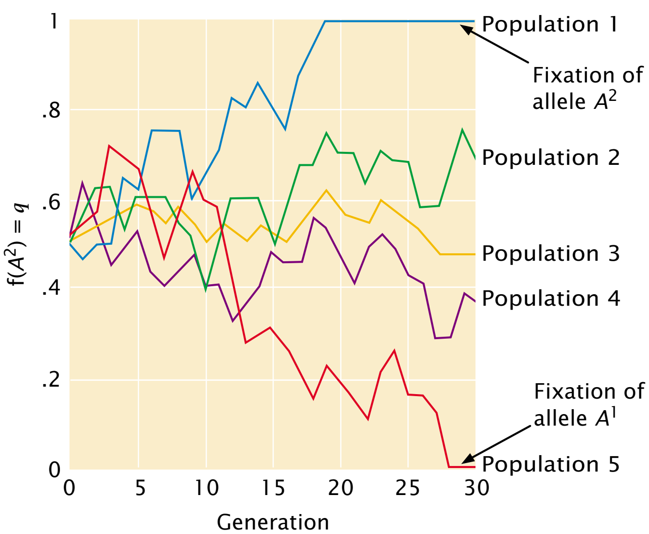
Given enough time, all small populations will become fixed for one allele. Which allele becomes fixed is random and is determined by the initial frequency of the allele. If

23.12 **Northern elephant seals underwent a severe genetic bottleneck between 1820 and 1880.** Today, these seals have low levels of genetic variation. (Lisa Husar/DRK Photo.)

the population begins with two alleles, each with a frequency of .5, both alleles have an equal probability of fixation. However, if one allele is initially common, it is more likely to become fixed.

A third effect of genetic drift is that different populations diverge genetically with time. In Figure 23.13, all five populations begin with the same allelic frequency (q = .5) but, because drift occurs randomly, the frequencies in different populations do not change in the same way, and so populations gradually acquire genetic differences. Notice that, although the variance in allelic frequency among the populations increases, the average allelic frequency remains basically the same. Eventually, all the populations reach fixation; some will become fixed for one allele and others will become fixed for the alternative allele. This divergence of populations through genetic drift is strikingly illustrated in the results of an experiment carried out by Peter Buri on fruit flies (FIGURE 23.14).

The three results of genetic drift (allelic frequency change, loss of variation within populations, and genetic divergence between populations) occur simultaneously, and all result from sampling error. The first two results occur *within* populations, whereas the third occurs *between* populations.



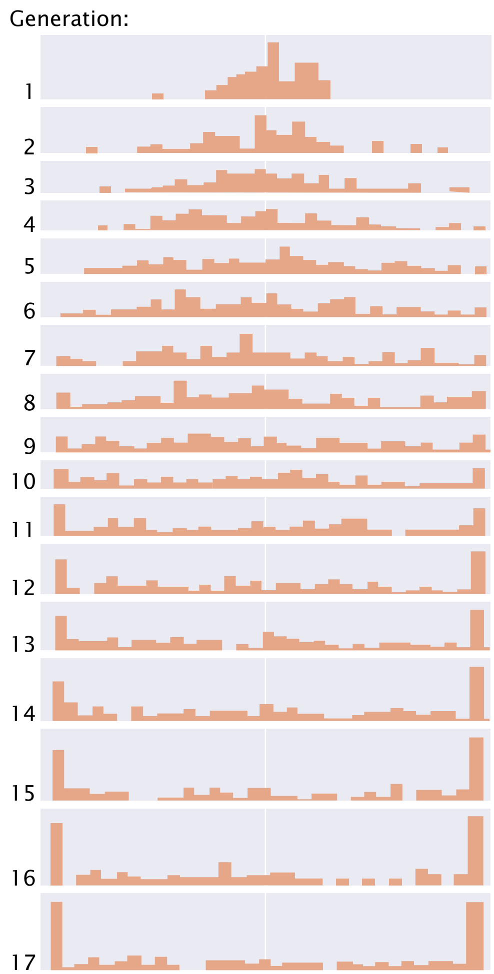
23.13 **Genetic drift changes allelic frequencies within populations, leading to a reduction in genetic variation through fixation and genetic divergence among populations.** Shown here is a computer simulation of changes in the frequency of allele A^2 (q) in five different populations due to random genetic drift. Each population consists of 10 males and 10 females and begins with q = .5.

Concepts

Genetic drift results from continuous small population size, founder effect (establishment of a population by a few founders), and bottleneck effect (population reduction). Genetic drift causes change in allelic frequencies within a population, loss of genetic variation through fixation of alleles, and genetic divergence between populations.

Natural Selection

A final process that brings about changes in allelic frequencies is natural selection, the differential reproduction of genotypes (see p. 000 in Chapter 22). Natural selection takes place when individuals with adaptive traits produce more offspring. If the adaptive traits have a genetic basis, they are inherited by the offspring and appear with greater frequency



23.14 Populations diverge in allelic frequency and become fixed for one allele as a result of genetic drift. In this experiment, Buri examined the frequency of two alleles (bw^{75} and bw) that affect *Drosophila* eye color in 107 replicate populations. Each population consisted of 8 males and 8 females; each population began with the frequency of bw^{75} equal to .5.

in the next generation. A trait that provides a reproductive advantage thereby increases over time, enabling populations to become better suited to their environments—to become better adapted. Natural selection is unique among evolutionary forces in that it promotes adaptation (FIGURE 23.15).

Fitness and selection coefficient The effect of natural selection on the gene pool of a population depends on the fitness values of the genotypes in the population. **Fitness** is defined as the relative reproductive success of a genotype. Here the term *relative* is critically important: fitness is the reproductive success of one genotype compared with the reproductive successes of other genotypes in the population.

Fitness (W) ranges from 0 to 1. Suppose the number of viable offspring produced by three genotypes is:

Genotypes:	A^1A^1	A^1A^2	A^2A^2
Mean number of offspring produced:	10	5	2

To calculate fitness for each genotype, we take the average number of offspring produced by a genotype and divide it by the mean number of offspring produced by the most prolific genotype:

$$\text{Fitness } (W): \quad \overset{A^1A^1}{W_{11} = \frac{10}{10} = 1.0} \qquad \overset{A^1A^2}{W_{12} = \frac{5}{10} = .5}$$

$$\overset{A^2A^2}{W_{22} = \frac{2}{10} = .2} \tag{23.20}$$

The fitness of the genotype A^1A^1 is designated W_{11}, that of A^1A^2 is W_{12}, and that of A^2A^2 is W_{22}. A related variable is the **selection coefficient** (s), which is the relative intensity of selection against a genotype. The selection coefficient is equal to $1 - W$; so the selection coefficients for the preceding three genotypes are:

23.15 Natural selection produces adaptations, such as those seen in polar bears that inhabit the extreme Arctic environment. These bears blend into the snowy background, which helps them in hunting seals. The hairs of their fur stay erect even when wet, and thick layers of blubber provide insulation, which protects against subzero temperatures. Their digestive tracts are adapted to a seal-based carnivorous diet. (Tom and Pat Leeson/DRK Photo.)

	A^1A^1	A^1A^2	A^2A^2
Selection coefficient $(1 - W)$:	$s_{11} = 0$	$s_{12} = .5$	$s_{22} = .8$

We usually speak of selection for a particular genotype, but keep in mind that, when selection is *for* one genotype, selection is automatically *against* at least one other genotype.

Concepts

Natural selection is the differential reproduction of genotypes. It is measured as fitness, which is the reproductive success of a genotype compared with other genotypes in a population.

The general selection model Differential fitness among genotypes over time leads to changes in the frequencies of the genotypes, which, in turn, lead to changes in the frequencies of the alleles that make up the genotypes. We can predict the effect of natural selection on allelic frequencies by using a general selection model, which is outlined in Table 23.4. Use of this model requires knowledge of both the initial allelic frequencies and the fitness values of the genotypes. It assumes that mating is random and the only force acting on a population is natural selection.

We have defined fitness in terms of relative reproduction, but it will be easier to understand the logic behind the general selection model if we think of the fitness of the genotypes as differences in survival. It applies equally to fitnesses representing differential reproduction.

Let's apply the general selection model outlined in Table 23.4. Imagine a flock of sparrows overwintering in Rochester, New York. Assume that we can determine the genotypes for a locus that affects the ability of the birds to survive the winter; perhaps the genes at this locus determine the amount of fat that a bird accumulates before the onset of winter. For genotypes A^1A^1, A^1A^2, and A^2A^2, p represents the frequency of A^1 and q represents the frequency of A^2. On the first line of the table, we record the initial genotypic frequencies before selection has acted, before the onset of winter. If mating has been random (an assumption of the model), the genotypes will have the Hardy-Weinberg equilibrium frequencies of p^2, $2pq$, and q^2. On the second row of the table, we put the fitness values of the corresponding genotypes. Some of the birds die in the winter; so here the fitness values represent the relative survival of the three genotypes. The proportion of the population represented by each genotype after selection is obtained by multiplying the initial genotypic frequency times its fitness (third row of Table 23.4). Now the genotypes are no longer in Hardy-Weinberg equilibrium.

The mean fitness $(\overline{W})$ of the population is the sum of the proportionate contributions of the three genotypes:

$$\overline{W} = p^2W_{11} + 2pqW_{12} + q^2W_{22} \tag{23.21}$$

The mean fitness $\overline{W}$ is the average fitness of all individuals in the population and allows the frequencies of the genotypes after selection to be obtained. In our flock of birds, these frequencies will be those of the three genotypes after the winter mortality. The frequency of a genotype after selection will be equal to its proportionate contribution divided by the mean fitness of the population ($p^2W_{11}/\overline{W}$ for genotype A^1A^1, $2pqW_{12}/\overline{W}$ for genotype A^1A^2, and $q^2W_{22}/\overline{W}$ for genotype A^2A^2), as shown in the fourth line of Table 23.4. When the new genotypic frequencies have been calculated, the new allelic frequency of A^1 (p') can be determined by using the now-familiar formula (Equation 23.4):

$$p' = f(A^1) = f(A^1A^1) + {}^1\!/_2\, f(A^1A^2)$$

and that of q' can be obtained by subtraction:

$$q' = 1 - p'$$

Table 23.4 Method for determining changes in allelic frequency due to selection

	A^1A^1	A^1A^2	A^2A^2
Initial genotypic frequencies	p^2	$2pq$	q^2
Fitnesses	W_{11}	W_{12}	W_{22}
Proportionate contribution of genotypes to population	p^2W_{11}	$2pqW_{12}$	q^2W_{22}
Relative genotypic frequency after selection	$\dfrac{p^2W_{11}}{\overline{W}}$	$\dfrac{2pqW_{22}}{\overline{W}}$	$\dfrac{q^2W_{22}}{\overline{W}}$

Note: $\overline{W} = p^2W_{11} + 2pqW_{12} + q^2W_{22}$
Allelic frequencies after selection: $p' = f(A^1) = f(A^1A^1) + {}^1\!/_2\, f(A^1A^2)$
$q' = 1 - p$

The last step is to determine the genotypic frequencies in the *next* generation. In regard to our birds, these genotypic frequencies are those of the offspring of the birds that survived the winter. If the survivors mate randomly, the genotypic frequencies in the next generation will be p'^2, $2p'q'$, and q'^2.

The general selection model can be used to calculate the allelic frequencies after any type of selection. It is also possible to work out formulas for determining the change in allelic frequency when selection is against recessive, dominant, and codominant traits, as well as traits in which the heterozygote has highest fitness (Table 23.5).

Concepts

The change in allelic frequency due to selection can be determined for any type of genetic trait by using the general selection model.

The results of selection The results of selection depend on the relative fitnesses of the genotypes. If we have three genotypes (A^1A^1, A^1A^2, and A^2A^2) with fitnesses W_{11}, W_{12}, and W_{22}, we can identify six different types of natural selection (Table 23.6). In type 1 selection, a dominant allele A^1 confers a fitness advantage; in this case, the fitnesses of genotypes A^1A^1 and A^1A^2 are equal and higher than the fitness of A^2A^2 ($W_{11} = W_{12} > W_{22}$). Because the heterozygote and the A^1A^1 homozygote both have copies of the A^1 allele and produce more offspring than the A^2A^2 homozygote does, the frequency of the A^1 allele will increase over time, whereas the frequency of the A^2 allele will decrease. This form of selection, in which one allele or trait is favored over another, is termed **directional selection.**

Type 2 selection (Table 23.6) is directional selection against a dominant allele A^1 ($W_{11} = W_{12} < W_{22}$). In this case, the A^2 allele increases and the A^1 allele decreases. Type 3 and type 4 selection also are directional selection, but in these cases there is incomplete dominance and the heterozygote has a fitness that is intermediate between the two homozygotes ($W_{11} < W_{12} < W_{22}$ for type 3; $W_{11} > W_{12} > W_{22}$ for type 4). When A^1A^1 has the highest fitness (type 3), over time the A^1 allele increases and the A^2 allele decreases. When A^2A^2 has the highest fitness (type 4), over time the A^2 allele increases and the A^1 allele decreases. Eventually, directional selection leads to fixation of the favored allele and elimination of the other allele, as long as no other evolutionary forces act on the population.

Two types of selection (types 5 and 6) are special situations that lead to equilibrium, where there is no further change in allelic frequency. Type 5 selection is referred to as **overdominance** or heterozygote advantage. Here, the heterozygote has higher fitness than the fitnesses of the two homozygotes ($W_{11} < W_{12} > W_{22}$). With overdominance, both alleles are favored in the heterozygote, and neither allele is eliminated from the population. Initially, the allelic frequencies may change because one homozygote has higher fitness than the other; the direction of change will depend on the relative fitness values of the two homozygotes. The allelic frequencies change with overdominant selection until a stable equilibrium is reached, at which point there is no further change. The allelic frequency at equilibrium ($\hat{q}$) depends on the relative fitnesses (usually expressed as selection coefficients) of the two homozygotes:

$$\hat{q} = f(A^2) = \frac{s_{11}}{s_{11} + s_{22}} \tag{23.22}$$

where s_{11} represents the selection coefficient of the A^1A^1 homozygote and s_{22} represents the selection coefficient of the A^2A^2 homozygote.

The last type of selection (type 6) is **underdominance,** in which the heterozygote has lower fitness than both

Table 23.5 Formulas for calculating change in allelic frequencies with different types of selection

	Fitness Values			
Type of Selection	A^1A^1	A^1A^2	A^2A^2	**Change in q**
Selection against a recessive trait	1	1	$1 - s$	$\frac{-spq^2}{1 - sp^2}$
Selection against a dominant trait	1	$1 - s$	$1 - s$	$\frac{-spq^2}{1 - s + sq^2}$
Selection against a trait with no dominance	1	$1 - \frac{1}{2}s$	$1 - s$	$\frac{-\frac{1}{2}spq}{1 - sq}$
Selection against both homozygotes (overdominance)	$1 - s_{11}$	1	$1 - s_{22}$	$\frac{pq(s_{11}p - s_{22}q)}{1 - s_{11}p^2 - s_{22}q^2}$

Table 23.6 Types of natural selection

Type	Fitness Relation	Form of Selection	Result
1	$W_{11} = W_{12} > W_{22}$	Directional selection against recessive allele A^2	A^1 increases, A^2 decreases
2	$W_{11} = W_{12} < W_{22}$	Directional selection against dominant allele A^1	A^2 increases, A^1 decreases
3	$W_{11} > W_{12} > W_{22}$	Directional selection against incompletely dominant allele A^2	A^1 increases, A^2 decreases
4	$W_{11} < W_{12} < W_{22}$	Directional selection against incompletely dominant allele A^1	A^2 increases, A^1 decreases
5	$W_{11} < W_{12} > W_{22}$	Overdominance	Stable equilibrium, both alleles maintained
6	$W_{11} > W_{12} < W_{22}$	Underdominance	Unstable equilibrium

Note: W_{11}, W_{12}, and W_{22} represent the fitnesses of genotypes A^1A^1, A^1A^2, and A^2A^2, respectively.

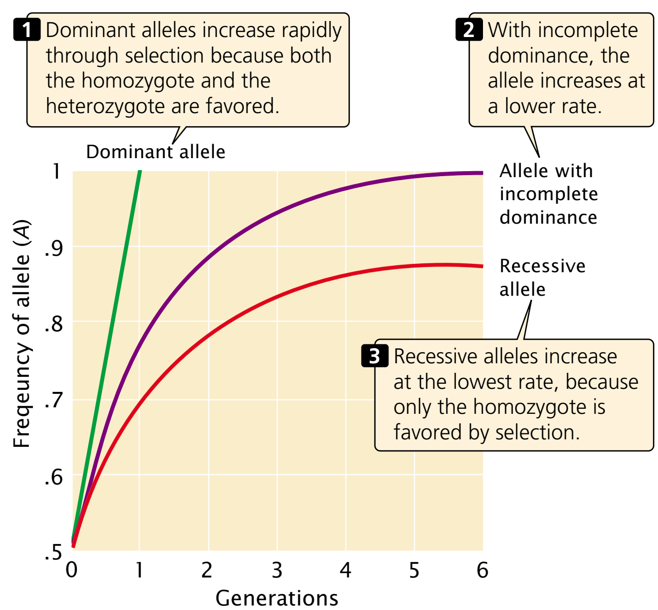
homozygotes ($W_{11} > W_{12} < W_{22}$). Underdominance leads to an *unstable* equilibrium; here allelic frequencies will not change as long as they are at equilibrium but, if they are disturbed from the equilibrium point by some other evolutionary force, they will move away from equilibrium until one allele eventually becomes fixed.

Concepts

Natural selection changes allelic frequencies; the direction and magnitude of change depends on the intensity of selection, the dominance relations of the alleles, and the allelic frequencies. Directional selection favors one allele over another and eventually leads to fixation of the favored allele. Overdominance leads to a stable equilibrium with maintenance of both alleles in the population. Underdominance produces an unstable equilibrium because the heterozygote has lower fitness than those of the two homozygotes.

The rate of change in allelic frequency due to natural selection The rate at which an allele changes in frequency owing to selection depends on the intensity of selection and the dominance relations among the genotypes (FIGURE 23.16). Under directional selection, dominant alleles will increase much more rapidly than recessive alleles, because homozygotes and heterozygotes are favored. With incomplete dominance, the heterozygote has a selective advantage, but not as much as the homozygote; so incompletely dominant alleles increase in frequency at a lower rate than that of dominant alleles. Recessive alleles increase at the lowest rate, because only the homozygote is favored by selection.

The rate at which selection changes allelic frequencies also depends on the allelic frequency itself. If an allele (A^2) is lethal and recessive, $W_{11} = W_{12} = 1$, whereas $W_{22} = 0$. The



23.16 **The rate of change in allelic frequency due to selection depends on the dominance relations among the genotypes.** Here, change in the frequency of an allele is shown for different types of dominance with a constant selection coefficient.

frequency of the A^2 allele will decrease over time (because the A^2A^2 homozygote produces no offspring), and the rate of decrease will be proportional to the frequency of the recessive allele. When the frequency of the allele is high, the change in each generation is relatively large but, as the frequency of the allele drops, a higher proportion of the alleles are in the heterozygous genotypes, where they are immune to the action of natural selection (the heterozygotes have the same phenotype as the favored homozygote). Thus, selection against a rare recessive allele is very inefficient and its removal from the population is slow.

The relation between the frequency of a recessive allele and its rate of change under natural selection has an important implication. Some people believe that the medical treatment of patients with rare recessive diseases will cause the disease gene to increase, eventually leading to degeneration of the human gene pool. This mistaken belief was the basis of eugenic laws that were passed in the early part of the twentieth century prohibiting the marriage of persons with certain genetic conditions and allowing the involuntary sterilization of others. However, most copies of rare recessive alleles are present in heterozygotes, and selection against the homozygotes will have little effect on the frequency of a recessive allele. Thus whether the homozygotes reproduce or not has little effect on the frequency of the disorder.

Mutation and natural selection Recurrent mutation and natural selection act as opposing forces on detrimental alleles; mutation increases their frequency and natural selection decreases their frequency. Eventually, these two forces reach an equilibrium, in which the number of alleles added by mutation is balanced by the number of alleles removed by selection.

Table 23.5 shows that the change in allelic frequency due to selection against a recessive allele is $-spq^2/(1-sq^2)$. When q is very low, q^2 is near zero; so $1-sq^2$ will be approximately $1-0=1$. Thus, when q is very low, the decrease in frequency due to selection is approximately $-spq^2$. The increase in frequency of an allele due to forward mutations is μp (Equation 23.12). At equilibrium, the effects of mutation and selection are balanced; so

$$spq^2 = \mu p$$

This equation can be rearranged:

$$q^2 = \frac{\mu p}{sp} = \frac{\mu}{s}$$

Taking the square root of each side, we get

$$\hat{q} = \frac{\mu}{s} \tag{23.23}$$

The frequency of the allele at equilibrium ($\hat{q}$) is therefore equal to the square root of the mutation rate divided by the selection coefficient. With the use of the equation for selection acting on a dominant allele (see Table 23.5) and similar reasoning, the frequency of a dominant allele at equilibrium can be shown to be

$$\hat{q} = \frac{\mu}{s} \tag{23.24}$$

Achondroplasia (discussed in Chapter 17) is a common type of human dwarfism that results from a dominant gene. People with this condition are fertile, although they produce only about 74% as many children as are produced by people without achondroplasia. The fitness of people with achondroplasia therefore averages .74, and the selection coefficient (s) is $1-W$, or .26. If we assume that the mutation rate for achondroplasia is about 3×10^{-5} (a typical mutation rate in humans), then we can predict that the equilibrium frequency for the achondroplasia allele will be $\hat{q} = (.00003/.26) =$ .0001153. This frequency is close to the actual frequency of the disease.

Concepts

Mutation and natural selection act as opposing forces on detrimental alleles: mutation tends to increase their frequency and natural selection tends to decrease their frequency, eventually producing an equilibrium.

Connecting Concepts

The General Effects of Evolutionary Forces

You now know that four processes bring about change in the allelic frequencies of a population: mutation, migration, genetic drift, and natural selection. Their short- and long-term effects on allelic frequencies are summarized in Table 23.7. In some cases, these changes continue until one allele is eliminated and the other becomes fixed in the population. Genetic drift and directional selection will eventually result in fixation, provided these forces are the only ones acting on a population. With the other evolutionary forces, allelic frequencies change until an equilibrium point is reached, and then there is no additional change in allelic frequency. Mutation, migration, and some forms of natural selection can lead to stable equilibria (see Table 23.7).

Table 23.7 Effects of different evolutionary forces on allelic frequencies within populations

Force	Short-Term Effect	Long-Term Effect
Mutation	Change in allelic frequency	Equilibrium reached between forward and reverse mutations
Migration	Change in allelic frequency	Equilibrium reached when allelic frequencies of source and recipient population are equal
Genetic drift	Change in allelic frequency	Fixation of one allele
Natural selection	Change in allelic frequency	Directional selection: fixation of one allele
		Overdominant selection: equilibrium reached

The different evolutionary forces affect both genetic variation within populations and genetic divergence between populations. Evolutionary forces that maintain or increase genetic variation within populations are listed in the upper-left quadrant of FIGURE 23.17. These forces include some types of natural selection, such as overdominance in which both alleles are favored. Mutation and migration also increase genetic variation within populations because they introduce new alleles to the population. Evolutionary forces that decrease genetic variation within populations are listed in the lower-left quadrant of Figure 23.17. These forces include genetic drift, which decreases variation through fixation of alleles, and some forms of natural selection such as directional selection.

The various evolutionary forces also affect the amount of genetic divergence between populations. Natural selection increases divergence among populations if different alleles are favored in the different populations, but it can also *decrease* divergence between populations by favoring the same allele in the different populations. Mutation almost always increases divergence between populations because different mutations arise in each population. Genetic drift also increases divergence between populations because changes in allelic frequencies due to drift are random and are likely to change in different directions in separate populations. Migration, on the other hand, decreases divergence between populations because it makes populations similar in their genetic composition.

Migration and genetic drift act in opposite directions: migration increases genetic variation within populations and decreases divergence between populations, whereas genetic drift decreases genetic variation within populations and increases divergence among populations. Mutation increases both variation within populations and divergence between populations. Natural selection can either increase or decrease variation within populations, and it can increase or decrease divergence between populations.

It is important to keep in mind that real populations are simultaneously affected by many evolutionary forces. This discussion has examined the effects of mutation, migration, genetic drift, and natural selection in isolation so that the influence of each process would be clear. However, in the real world, populations are commonly affected by several evolutionary forces at the same time, and evolution results from the complex interplay of numerous processes.

	Within populations	Between populations
Increase genetic variation	**Mutation Migration Some types of natural selection**	**Mutation Genetic drift Some types of natural selection**
Decrease genetic variation	**Genetic drift Some types of natural selection**	**Migration Some types of natural selection**

23.17 Mutation, migration, genetic drift, and natural selection have different effects on genetic variation within populations and on genetic divergence between populations.

www.whfreeman.com/pierce Web addresses for a number of resources on evolution

Molecular Evolution

For many years, it was not possible to examine genes directly, and evolutionary biology was confined largely to the study of how phenotypes change with the passage of time. The tremendous advances in molecular genetics in recent years have made it possible to investigate evolutionary change directly by analyzing protein and nucleic acid sequences. These molecular data offer a number of advantages for studying the process and pattern of evolution:

1. **Molecular data are genetic.** Evolution results from genetic change over time. Anatomical, behavioral, and physiological traits often have a genetic basis, but the relation between the underlying genes and the trait may be complex. Protein and nucleic acid sequence variation has a clear genetic basis that is easy to interpret.
2. **Molecular methods can be used with all organisms.** Early studies of population genetics relied on simple genetic traits such as human blood types or banding patterns in snails, which are restricted to a small group of organisms. However, all living organisms have proteins and nucleic acids; so molecular data can be collected from any organism.
3. **Molecular methods can be applied to a huge amount of genetic variation.** An enormous amount of data can be accessed by molecular methods. The human genome, for example, contains more than 3 billion base pairs of DNA, which constitutes a large pool of information about our evolution.
4. **All organisms can be compared with the use of some molecular data.** Trying to assess the evolutionary history of distantly related organisms is often difficult because they have few characteristics in common. The evolutionary relationships between angiosperms were traditionally assessed by comparing floral anatomy, whereas the evolutionary relationships of bacteria were determined by their nutritional and staining properties. Because plants and bacteria have so few structural characteristics in common, evaluating how they are related to one another was difficult in the past. All organisms have certain molecular traits in common, such as ribosomal RNA sequences and some fundamental proteins. These molecules offer a valid basis for comparisons among all organisms.
5. **Molecular data are quantifiable.** Protein and nucleic acid sequence data are precise, accurate, and easy to quantify, which facilitates the objective assessment of evolutionary relationships.
6. **Molecular data often provide information about the process of evolution.** Molecular data can reveal important clues about the process of evolution. For example, the results of a study of DNA sequences have revealed that one type of insecticide resistance in mosquitoes probably arose from a single mutation that subsequently spread throughout the world.
7. **The database of molecular information is large and growing.** Today, this database of DNA and protein sequences can be used for making evolutionary comparisons and inferring mechanisms of evolution.

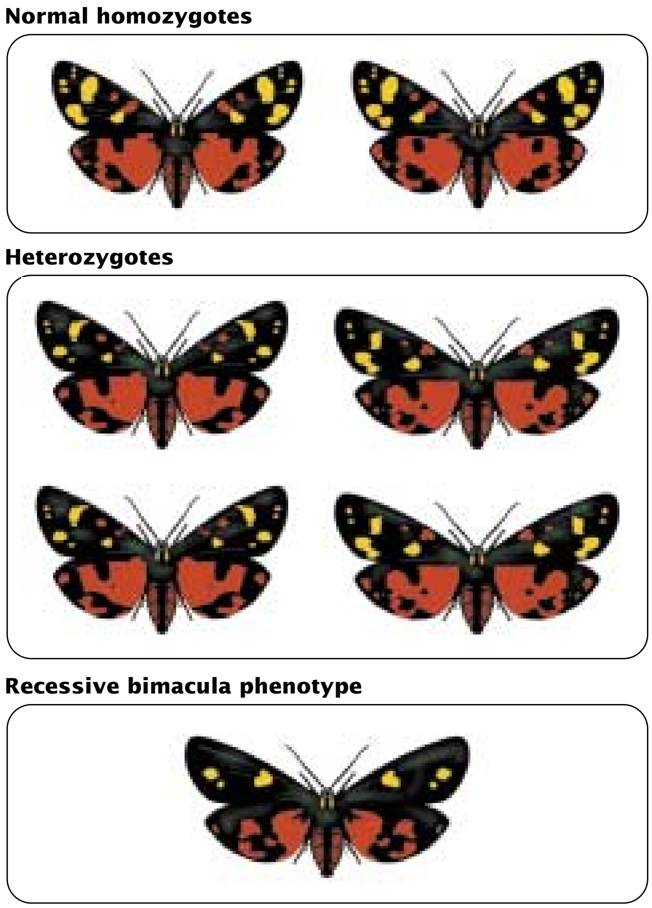
Studies of molecular evolution fall into three primary areas. First, much past research has focused on determining the extent and causes of genetic variation in natural populations. Molecular techniques allow these matters to be addressed directly by examining sequence variation in proteins and DNA. A second area of research examines molecular processes that influence evolutionary events, and the results of these studies have elucidated new mechanisms and processes of evolution that were not suspected before the application of molecular techniques to evolutionary biology. A third area of research in molecular evolution applies molecular techniques to constructing **phylogenies** (evolutionary trees) of various groups of organisms. A detailed evolutionary history is found in the DNA sequences of every organism, and molecular techniques allow this history to be read.

Concepts

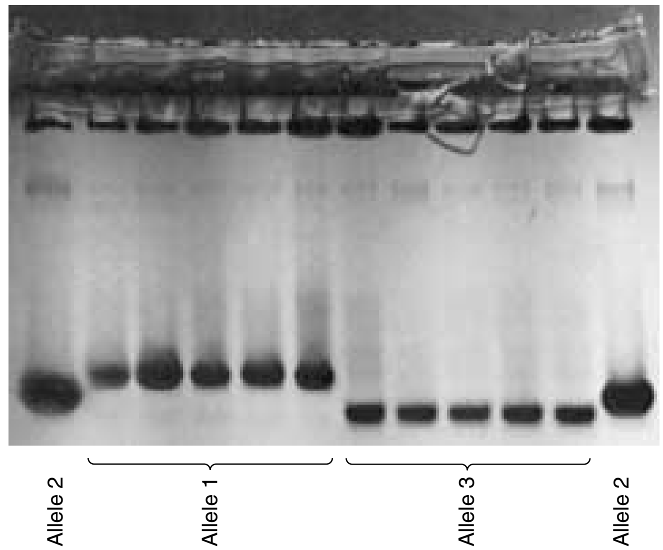
Molecular techniques and data offer a number of advantages for evolutionary studies. Molecular data (1) are genetic in nature and can be investigated in all organisms; (2) provide potentially large data sets; (3) allow all organisms to be compared, by using the same characteristics; (4) are easily quantifiable; and (5) provide information about the process of evolution.

Protein Variation

The study of the amounts and kinds of genetic variation in natural populations is central to the study of evolution. For many traits, a complex interaction of many genes and environmental factors determines the phenotype, and assessing the amount of genetic variation by examining phenotypic variation was difficult. Early population geneticists were forced to rely on the phenotypic traits that had a simple genetic basis, such as human blood types or spotting patterns in butterflies (FIGURE 23.18). The initial breakthrough that first allowed the direct examination of molecular evolution was the application of electrophoresis (see Figure 18.4) to population studies. This technique separates macromolecules, such as proteins or nucleic acids, on the basis of their size and charge. In 1966, Richard Lewontin and John Hubby extracted proteins from wild fruit flies, separated the proteins by electrophoresis, and stained for specific enzymes. Examining the pattern of bands on gels enabled them to assign genotypes to individual flies and to quantify the amount of genetic variation in natural populations. In the same year, Harry Harris quantified genetic variation in human populations by using the same technique. Protein variation has now been examined in hundreds of different species by using protein electrophoresis (FIGURE 23.19).



23.18 Early population geneticists were forced to rely on the phenotypic traits that had a simple genetic basis. Variation in the spotting patterns of the butterfly *Panaxia dominula* is an example.



23.19 Molecular variation in proteins is revealed by electrophoresis. Tissue samples from three fruit flies have been subjected to electrophoresis and stained for malate dehydrogenase. Homozygotes are represented as single bands; heterozygotes as triple bands. The genotype of each fly is given below each sample.

Measures of genetic variation The amount of genetic variation in populations is commonly measured by two parameters. The **proportion of polymorphic loci** is the proportion of examined loci in which more than one allele is present in a population. If we examined 30 different loci and found two or more alleles present at 15 of these loci, the percentage of polymorphic loci would be 15/30 = 0.5. The **expected heterozygosity** is the proportion of individuals that are expected to be heterozygous at a locus under the Hardy-Weinberg conditions, which is $2pq$ when there are two alleles present in the population. The expected heterozygosity is often preferred to the observed heterozygosity because expected heterozygosity is independent of the breeding system of an organism. For example, if a species self-fertilizes, it may have little or no heterozygosity but still have considerable genetic variation, which will be detected by the expected heterozygosity. Expected heterozygosity is typically calculated for a number of loci and is then averaged over all the loci examined.

The percentage of polymorphic loci and the expected heterozygosity have been determined by protein electrophoresis for a number of species (Table 23.8). About one-third of all protein loci are polymorphic, and expected heterozygosity averages about 10%, although there is considerable diversity among species. These measures actually underestimate the true amount of genetic variation, though, because protein electrophoresis does not detect some amino acid substitutions; nor does it detect genetic variation in DNA that does not alter the amino acids of a protein (synonymous codons and variation in noncoding regions of the DNA).

Explanations for protein variation By the late 1970s, geneticists recognized that most populations possess large amounts of genetic variation, although the evolutionary significance of this fact was not at all clear. Two opposing hypotheses arose to account for the presence of the extensive molecular variation in proteins. The **neutral-mutation hypothesis** proposed that the molecular variation revealed by protein electrophoresis is adaptively neutral; that is, individuals with different molecular variants have equal fitness. This hypothesis does not propose that the proteins are functionless; rather, it suggests that most variants revealed by protein electrophoresis are functionally equivalent. Because these variants are functionally equivalent, natural selection does not differentiate between them, and their evolution is shaped largely by the random processes of genetic drift and mutation. The neutral-mutation hypothesis accepts that

Table 23.8 Proportion of polymorphic loci and heterozygosity for different organisms, as determined by protein electrophoresis

		Proportion of Polymorphic Loci		Heterozygosity	
Group	**Number of Species**	**Mean**	**SD***	**Mean**	**SD***
Plants	15	0.26	0.17	0.07	0.07
Invertebrates (excluding insects)	28	0.40	0.28	0.10	0.07
Insects (excluding *Drosophila*)	23	0.33	0.20	0.07	0.08
Drosophila	32	0.43	0.13	0.14	0.05
Fish	61	0.15	0.01	0.05	0.04
Amphibians	12	0.27	0.13	0.08	0.04
Reptiles	15	0.22	0.13	0.05	0.02
Birds	10	0.15	0.11	0.05	0.04
Mammals	46	0.15	0.10	0.04	0.02

* SD, standard deviation from the mean.
Source: After L. E. Mettler, T. G. Gregg, and H. E. Schaffer, *Population Genetics and Evolution,* 2d ed. (Englewood Cliffs, NJ: Prentice Hall, 1988), Table 9.2. Original data from E. Nevo, Genetic variation in natural populations: patterns and theory, *Theoretical Population Biology* 13(1978):121–177.

natural selection is an important force in evolution, but views selection as a process that favors the "best" allele while eliminating others. It proposes that, when selection is important, there will be *little* genetic variation.

The **balance hypothesis** proposes, on the other hand, that the genetic variation in natural populations is maintained by selection that favors variation (balancing selection). Overdominance, in which the heterozygote has higher fitness than that of either homozygote, is one type of balancing selection. Under this hypothesis, the molecular variants are not physiologically equivalent and do not have the same fitness. Instead, genetic variation within natural populations is shaped largely by selection, and, when selection is important, there will be *much* variation.

Many attempts to prove one hypothesis or the other failed, because precisely how much variation was actually present was not clear (remember that protein electrophoresis detects only *some* genetic variation) and because both hypotheses are capable of explaining many different patterns of genetic variation. The controversy over the forces that control variation revealed by protein electrophoresis continues today, but the results of more-recent studies that provide direct information about DNA sequence variation demonstrate that much variation at the level of DNA has little obvious effect on the phenotype and therefore is likely to be neutral.

Concepts

The application of electrophoresis to the study of protein variation in natural populations revealed that most organisms possess large amounts of genetic variation. The neutral-mutation hypothesis proposes that most molecular variation is neutral with regard to natural selection and is shaped largely by mutation and genetic drift. The balance hypothesis proposes that genetic variation is maintained by balancing selection.

www.whfreeman.com/pierce Information on genetic variation in natural populations

DNA Sequence Variation

The development of techniques for isolating, restricting, and sequencing DNA in the 1970s and 1980s provided powerful tools for detecting, quantifying, and investigating genetic variation. The application of these techniques has provided a detailed view of molecular variation.

Restriction enzymes are one tool that can be used to detect genetic variation in DNA and examine patterns of genetic variation in nature. Each restriction enzyme recognizes and cuts a particular sequence of DNA nucleotides, known as that enzyme's restriction site (see Chapter 18). Variation in the presence of a restriction site is called a restriction fragment length polymorphism (RFLP; see Figure 18.26). Each restriction enzyme recognizes a limited number of nucleotide sites in a particular piece of DNA but, if a number of different restriction enzymes are used and the sites recognized by the enzymes are assumed to be random sequences, RFLPs can be used to estimate the amount of variation in the DNA and the proportion of nucleotides that differ between organisms.

Methods for determining the complete nucleotide sequences of DNA fragments (see p. 000 in Chapter 19) provide the most detailed evolutionary information, although they are both time consuming and expensive. DNA sequencing in evolutionary studies is therefore usually limited to a few individuals or to short sequences. Nevertheless, the high resolution of information provided by sequencing is often invaluable for understanding molecular processes that influence evolution and for determining phylogenies of closely related organisms. For example, DNA sequencing has been used to study the evolution of human immunodeficiency virus (HIV), the virus that causes AIDS. Like many other RNA viruses, HIV evolves rapidly, often changing its sequences within a single host over a period of several years. Evolutionary comparisons of HIV sequences in a dentist and seven of his patients who had AIDS demonstrated that five of the patients contracted AIDS from the dentist, whereas the other two patients probably acquired their HIV infection elsewhere.

Concepts

Restriction fragment length polymorphisms and DNA sequencing can be used to directly examine genetic variation.

Molecular Evolution of HIV in a Florida Dental Practice

In July 1990, the U.S. Center for Disease Control (CDC) reported that a young woman in Florida (later identified as Kimberly Bergalis) had become HIV positive after undergoing an invasive dental procedure performed by a dentist who had AIDS. Bergalis had no known risk factors for HIV infection and no known contact with other HIV-positive persons. The CDC acknowledged that Bergalis might have acquired the infection from her dentist. Subsequently, the dentist wrote to all of his patients, suggesting that they be tested for HIV infection. By 1992, 7 of the dentist's patients had tested positive for HIV, and this number eventually increased to 10.

Originally diagnosed with HIV infection in 1986, the dentist began to develop symptoms of AIDS in 1987 but continued to practice dentistry for another 2 years. All of his HIV-positive patients had received invasive dental procedures, such as root canals and tooth extractions, in the period when the dentist was infected. Among the seven patients originally studied by the CDC (patients A–G, Table 23.9), two had known risk factors for HIV infection (intravenous drug use, homosexual behavior, or sexual relations with HIV-infected persons), and a third had possible but unconfirmed risk factors.

To determine whether the dentist had infected his patients, the CDC conducted a study of the molecular evolution of HIV isolates from the dentist and the patients. HIV undergoes rapid evolution, making it possible to trace the path of its transmission. This rapid evolution also allows HIV to develop drug resistance quickly, making the development of a treatment for AIDS difficult.

Blood specimens were collected from the dentist, the patients, and a group of 35 local controls (other HIV-infected

Table 23.9 HIV-positive persons included in study of HIV isolates from a Florida dental practice

			Average Differences in DNA Sequences (%)	
Person	**Sex**	**Known Risk Factors**	**From HIV from Dentist**	**From HIV from Controls**
Dentist	M	Yes		11.0
Patient A	F	No	3.4	10.9
Patient B	F	No	4.4	11.2
Patient C	M	No	3.4	11.1
Patient E	F	No	3.4	10.8
Patient G	M	No	4.9	11.8
Patient D	M	Yes	13.6	13.1
Patient F	M	Yes	10.7	11.9

Source: After C. Ou, et al., *Science* 256(1992):1165–1171, Table 1.

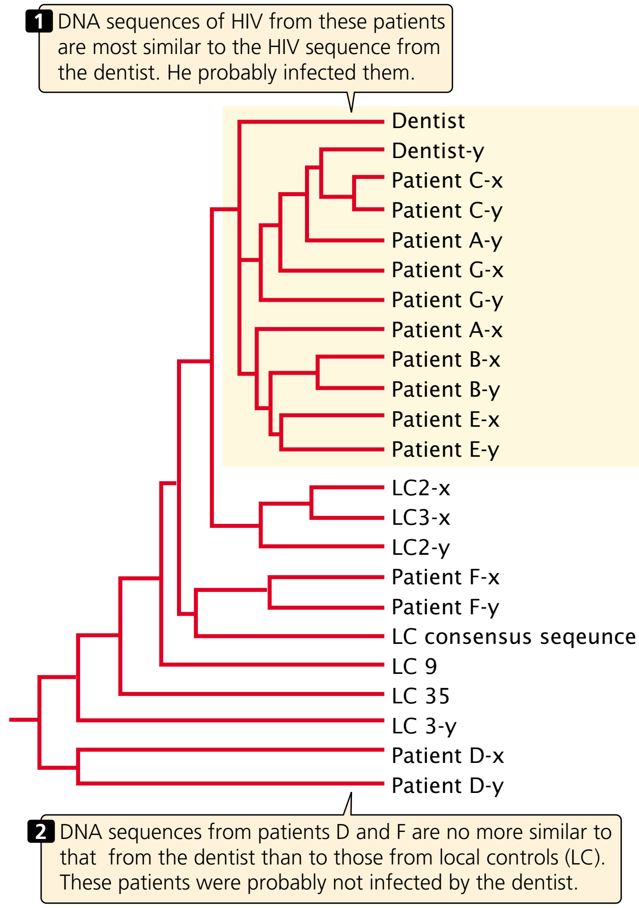
people who lived within 90 miles of the dental practice but who had no known contact with the dentist). DNA was extracted from white blood cells, and a 680-bp fragment of the *envelope* gene of the virus was amplified by PCR (see p. 000 in Chapter 16). The fragments from the dentist, the patients, and the local controls were then sequenced and compared.

The divergence between the viral sequences taken from the dentist, the seven patients, and the controls is shown Table 23.9. Viral DNA taken from patients with no confirmed risk factors (patients A, B, C, E, and G) differed from the dentist's viral DNA by 3.4% to 4.9%, whereas the viral DNA from the controls differed from the dentist's by an average of 11%. The viral sequences collected from five patients (A, B, C, E, and G) were more closely related to the viral sequences collected from the dentist than to viral sequences from the general population, strongly suggesting that these patients acquired their HIV infection from the dentist. The viral isolates from patients D and F (patients with confirmed risk factors), however, differed from that of the dentist by 10.7% and 13.6%, suggesting that these two patients did not acquire their infection from the dentist.

A phylogenetic tree depicting the evolutionary relationships of the viral sequences (FIGURE 23.20) confirmed that the virus taken from the dentist had a close evolutionary relationship to viruses taken from patients A, B, C, E, and G. The viruses from patients D and F, with known risk factors, were no more similar to the virus from the dentist than to viruses from local controls, indicating that the dentist most likely infected five of his patients, whereas the other two patients probably acquired their infections elsewhere. Of three additional HIV-positive patients that have been identified since 1992, only one has viral sequences that are closely related to those from the dentist.

The study of HIV isolates from the dentist and his patients provides an excellent example of the relevance of molecular evolutionary studies to real-world problems. How the dentist infected his patients during their visits to his office remains a mystery, but this case is clearly unusual. A study of almost 16,000 patients treated by HIV-positive health-care workers failed to find a single case of confirmed transmission of HIV from the health-care worker to the patient.

www.whfreeman.com/pierce Web site of the U.S. Center for Disease Control



23.20 Evolutionary tree showing the relationships of HIV isolates from a dentist, seven of his patients (A through G), and other HIV-positive persons from the same region (local controls, LC). The letters x and y represent different isolates from the same patient. The phylogeny is based on DNA sequences taken from the *envelope* gene of the virus. Viral sequences from patients A, B, C, E, and G cluster with those of the dentist, indicating a close evolutionary relationship. Sequences from patients D and F, along with those of local controls, are more distantly related. [C. Ou et al. Molecular epidemiology of HIV transmission in a dental practice, *Science* 256(1992): 1167.]

Patterns of Molecular Variation

The results of molecular studies of numerous genes have demonstrated that different genes and different parts of the same gene often evolve at different rates. Rates of evolutionary change in nucleotide sequences are usually measured as the rate of nucleotide substitution, which is the number of substitutions taking place per nucleotide site per year. To calculate the rate of nucleotide substitution, we begin by looking at homologous sequences from different organisms. We compare the homologous sequences and determine the number of nucleotides that differ between the two sequences. We might compare the growth-hormone sequences for mice and rats, which diverged from a common ancestor some 15 million years ago. From the number of different nucleotides in their growth-hormone genes, we compute the number of nucleotide substitutions that must have taken place since they diverged. Because the same site may have mutated more than once, the number of nucleotide substitutions is larger than the number of nucleotide differences in two sequences; so special mathematical methods have been developed for inferring the actual number of substitutions likely to have taken place.

When we have the number of nucleotide substitutions per nucleotide site, we divide by the amount of evolutionary time that separates the two organisms (usually obtained from the fossil record) to obtain an overall rate of nucleotide substitution. For the mouse and rat growth-hormone gene, the overall rate of nucleotide substitution is approximately 8×10^{-9} substitutions per site per year.

Nucleotide changes in a gene that alter the amino acid sequence of a protein are referred to as nonsynonymous substitutions. Nucleotide changes, particularly those at the third position of the codon, that do not alter the amino acid sequence are called synonymous substitutions. The rate of nonsynonymous substitution varies widely among mammalian genes. The rate for the α-actin protein is only 0.01×10^{-9} substitutions per site per year, whereas the rate for interferon γ is 2.79×10^{-9}, about 1000 times as high. The rate of synonymous substitution also varies among genes, but not to the extent of variation in the nonsynonymous rate. For most protein-encoding genes, the synonymous rate of change is considerably higher than the nonsynonymous rate because synonymous mutations are tolerated by natural selection (Table 23.10). Nonsynonymous mutations, on the other hand, alter the amino acid sequence of the protein and in many cases are detrimental to the fitness of the organism, so most of these mutations are eliminated by natural selection.

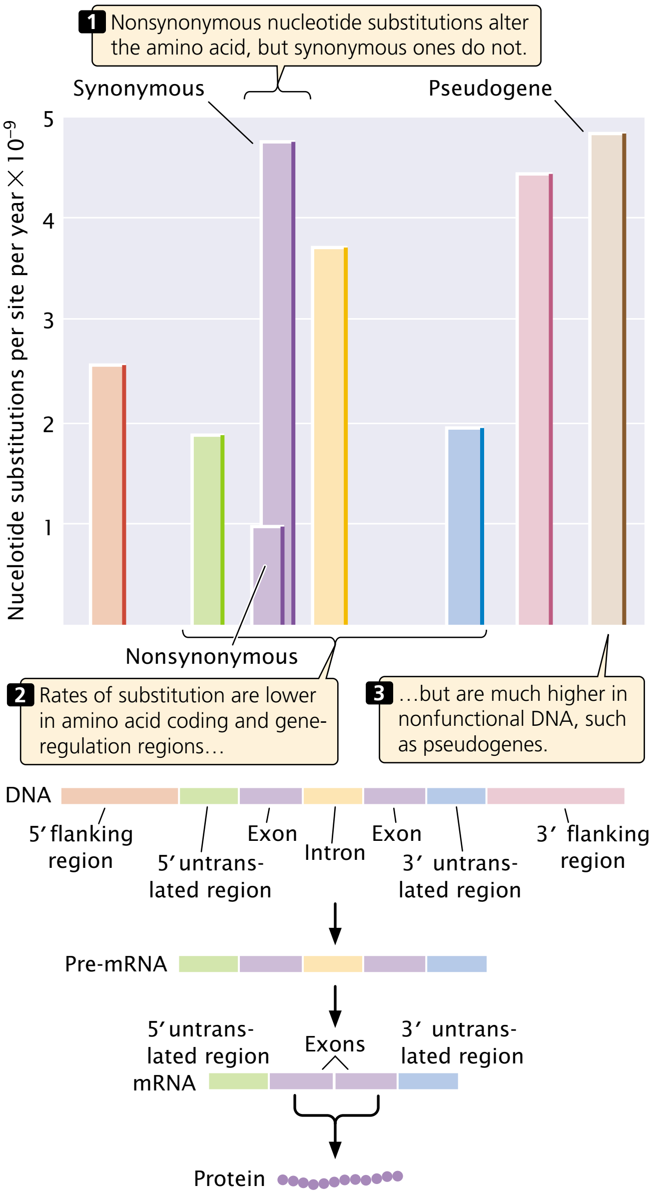
Different parts of a gene also evolve at different rates, with the highest rates of substitutions in regions of the gene that have the least effect on function, such as the third position of a codon, flanking regions, and introns (FIGURE 23.21). The 5′ and 3′ flanking regions of genes are not transcribed into RNA, and therefore substitutions in these regions do not alter the amino acid sequence of the protein, although they may affect gene expression (see Chapter 16). Rates of substitution in introns are nearly as high. Although these nucleotides do not encode amino acids, introns must be spliced out of the pre-mRNA for a functional protein to be produced, and particular sequences are required at the 5′ splice site, 3′ splice site, and branch point for correct splicing (see Chapter 14).

Substitution rates are somewhat lower in the 5′ and 3′ untranslated regions of a gene. These regions are transcribed into RNA but do not encode amino acids. The 5′ untranslated region contains the ribosome-binding site, which is essential for translation, and the 3′ untranslated region contains sequences that may function in regulating mRNA stability and translation; so substitutions in these regions may have deleterious effects on organismal fitness and will not be tolerated.

Table 23.10 Rates of nonsynonymous and synonymous substitutions in mammalian genes based on human–rodent comparisons

Gene	Nonsynonymous Rate (per Site per 10^9 Years)	Synonymous Rate (per Site per 10^9 Years)
α-Actin	0.01	3.68
β-Actin	0.03	3.13
Albumin	0.91	6.63
Aldolase A	0.07	3.59
Apoprotein E	0.98	4.04
Creatine kinase	0.15	3.08
Erythropoietin	0.72	4.34
α-Globin	0.55	5.14
β-Globin	0.80	3.05
Growth hormone	1.23	4.95
Histone 3	0.00	6.38
Immunoglobulin heavy chain (variable region)	1.07	5.66
Insulin	0.13	4.02
Interferon α1	1.41	3.53
Interferon γ	2.79	8.59
Luteinizing hormone	1.02	3.29
Somatostatin-28	0.00	3.97

Source: After W. Li and D. Graur, *Fundamentals of Molecular Evolution* (Sunderland, MA: Sinauer, 1991), p. 69, Table 1.



23.21 Different parts of genes evolve at different rates. The highest rates of nucleotide substitution are in sequences that have the least effect on protein function.

The lowest rates of substitution are seen in nonsynonymous changes in the coding region, because these substitutions always alter the amino acid sequence of the protein and are often deleterious. The highest rates of substitution are in pseudogenes, which are duplicated nonfunctional copies of genes that have acquired mutations. Such a gene no longer produces a functional product; so mutations in pseudogenes have no effect on the fitness of the organism.

In summary, there is a relation between the function of a sequence and its rate of evolution; higher rates are found where they have the least effect on function. This observation fits with the neutral-mutation hypothesis, which predicts that molecular variation is not affected by natural selection.

The Molecular Clock

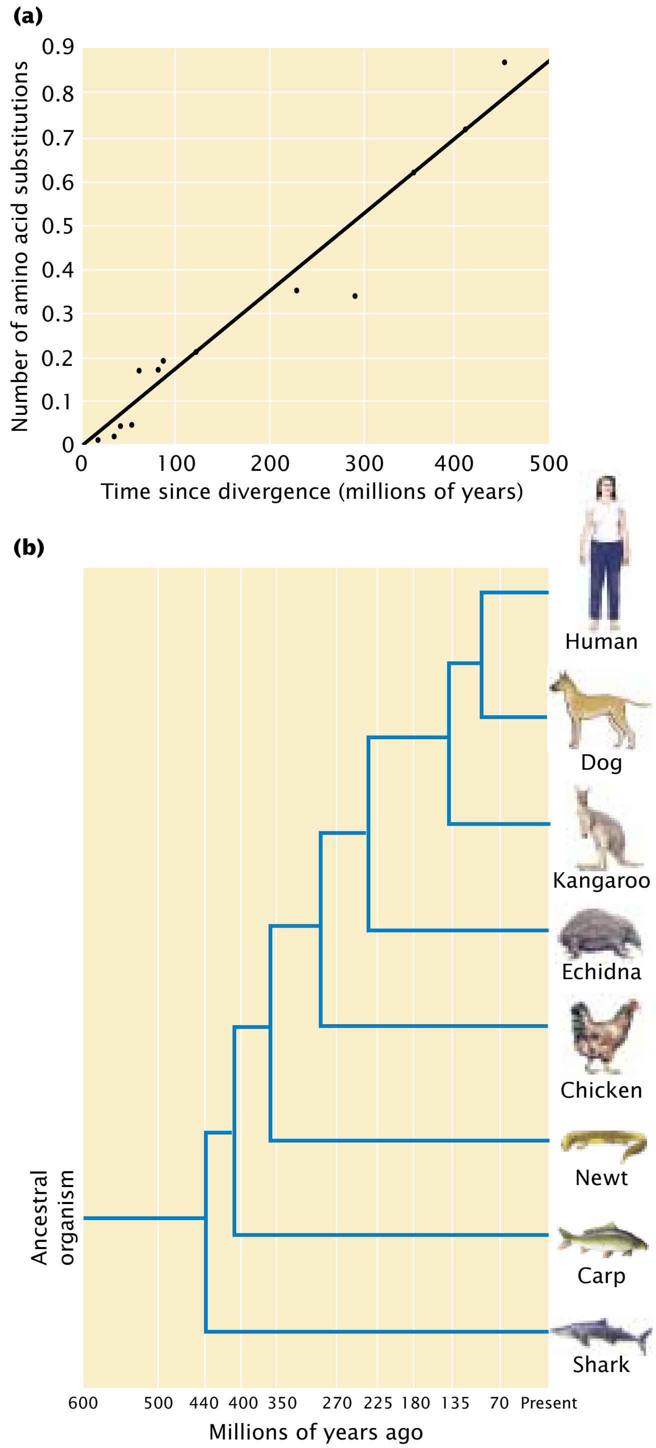
The neutral-mutation theory proposes that evolutionary change at the molecular level occurs primarily through the fixation of neutral mutations by genetic drift. The rate at which one neutral mutation replaces another depends only on the mutation rate, which should be fairly constant for any particular gene. If the rate at which a protein evolves is roughly constant over time, the amount of molecular change that a protein has undergone can be used as a **molecular clock** to date evolutionary events.

For example, the enzyme cytochrome *c* could be examined in two organisms known from fossil evidence to have had a common ancestor 400 million years ago. By determining the number of differences in the cytochrome *c* amino acid sequences in each organism, we could calculate the number of substitutions that have occurred per amino acid site. The occurrence of 20 amino acid substitutions since the two organisms diverged indicates an average rate of 5 substitutions per 100 million years. Knowing how fast the molecular clock ticks allows us to use molecular changes in cytochrome *c* to date other evolutionary events: if we found that cytochrome *c* in two organisms differed by 15 amino acid substitutions, our molecular clock would suggest that they diverged some 300 million years ago. If we assumed some error in our estimate of the rate of amino acid substitution, statistical analysis would show that the true divergence time might range from 160 million to 440 million years. The molecular clock is analogous to geological dating based on the radioactive decay of elements.

The molecular clock was proposed by Emile Zuckerandl and Linus Pauling in 1965 as a possible means of dating evolutionary events on the basis of molecules in present-day organisms. A number of studies have examined the rate of evolutionary change in proteins (FIGURE 23.22), and the molecular clock has been widely used to date evolutionary events when the fossil record is absent or ambiguous. However, the results of several studies have shown that the molecular clock does not always tick at a constant rate, particularly over shorter time periods, and this method remains controversial.

Concepts

Different genes and different parts of the same gene evolve at different rates. Those parts of genes that have the least effect on function tend to evolve at the highest rates. The idea of the molecular clock is that individual proteins and genes evolve at a constant rate and that the differences in the sequences of present-day organisms can be used to date past evolutionary events.

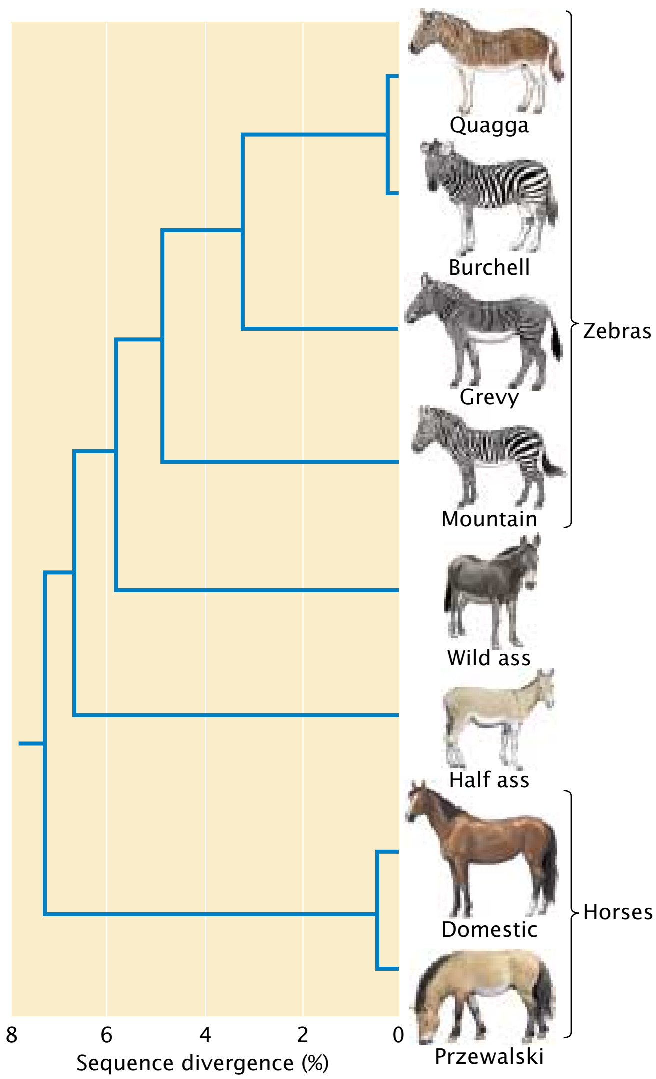


23.22 The molecular clock is based on the assumption of a constant rate of change in protein or DNA sequence. (a) Relation between the rate of amino acid substitution and time since divergence, based on amino acid sequences of α hemoglobin from the eight species shown in part *b*. The constant rate of evolution in protein and DNA sequences has been used as a molecular clock to date past evolutionary events. (b) Phylogeny of eight species and their approximate times of divergence, based on the fossil record.

Molecular Phylogenies

As already mentioned, a phylogeny is an evolutionary history of a group of organisms, usually represented as a tree (FIGURE 23.23). The branches of the phylogenetic tree represent the ancestral relationships between the organisms, and the length of each branch is proportional to the amount of evolutionary change that separates the members of the phylogeny.

Before the rise of molecular biology, phylogenies were based largely on anatomical, morphological, or behavioral traits. Evolutionary biologists attempted to gauge the relationships among organisms by assessing the overall degree of similarity or by tracing the appearance of key characteristics of these traits. The first phylogenies constructed from molecular



23.23 A phylogeny is the evolutionary history—the ancestral relationships—of a group of organisms. This branching diagram shows the phylogeny of horses based on mitochondrial DNA sequences. DNA of the extinct quagga was extracted from skins from preserved museum specimens.

(a)

The number 1 indicates an invariant position in the cytochrome c molecule (i.e., all the organisms have the same amino acid in this position). The position is probably functionally very significant.

Side chains marked by red arrows interact with the heme group.

Position in sequence	1				5					10					15					20					25					30				
Number of amino acids in different organisms at the position shown	1	3	4	3	2	1	2	3	3	1	4	3	2	1	3	3	1	1	1	3	1	3	2	2	3	2	1	3	1	1	2	1	3	1
Human	G	D	V	E	K	G	K	K	I	F	I	M	K	C	S	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Monkey	G	D	V	E	K	G	K	K	I	F	I	M	K	C	S	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Horse	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Donkey	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Pig	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Dog	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Rabbit	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Kangaroo	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	N	G
Chicken	G	D	I	E	K	G	K	K	I	F	V	Q	K	C	S	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Pigeon	G	D	I	E	K	G	K	K	I	F	V	Q	K	C	S	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Duck	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	S	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Turtle	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	N	G
Rattlesnake	G	D	V	E	K	G	K	K	I	F	T	M	K	C	S	Q	C	H	T	V	E	K	G	G	K	H	K	T	G	P	N	L	H	G
Tuna	G	D	V	A	K	G	K	K	T	F	V	Q	K	C	A	Q	C	H	T	V	E	N	G	G	K	H	K	V	G	P	N	L	W	G
Samia cynthia (moth)	G	N	A	E	N	G	K	K	I	F	V	Q	R	C	A	Q	C	H	T	V	E	A	G	G	K	H	K	V	G	P	N	L	H	G
Screwworm fly	G	D	V	E	K	G	K	K	L	F	V	Q	R	C	A	Q	C	H	T	V	E	A	G	G	K	H	K	V	G	P	N	L	H	G
Saccharomyces (baker's yeast)	G	S	A	K	K	G	A	T	L	F	K	T	R	C	E	L	C	H	T	V	E	K	G	G	P	H	K	V	G	P	N	L	H	G
Candida krusei (yeast)	G	S	A	K	K	G	A	T	L	F	K	T	R	C	A	E	C	H	T	I	E	A	G	G	P	H	K	V	G	P	N	L	H	G
Neurospora crassa (mold)	G	D	S	K	K	G	A	N	L	F	K	T	R	C	A	E	C	H		E			N	L	T	Q	K	I	G	P	A	L	H	G

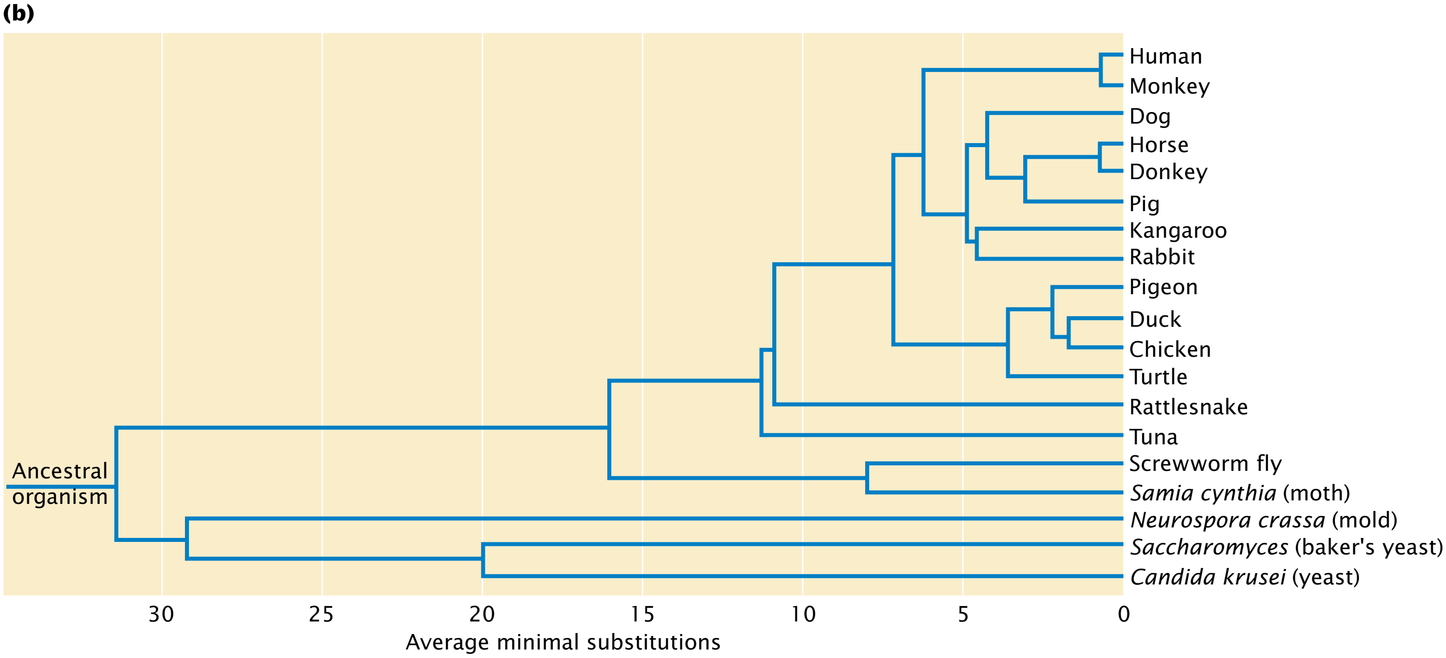
Acidic side chains:
- D Aspartic acid
- E Glutamic acid

Basic side chains:
- H Histidine
- K Lysine
- R Arginine

Hydrophobic side chains:
- F Phenylalanine
- I Isoleucine
- L Leucine
- M Methionine
- V Valine
- Y Tyrosine
- W Tryptophan
- A Alanine

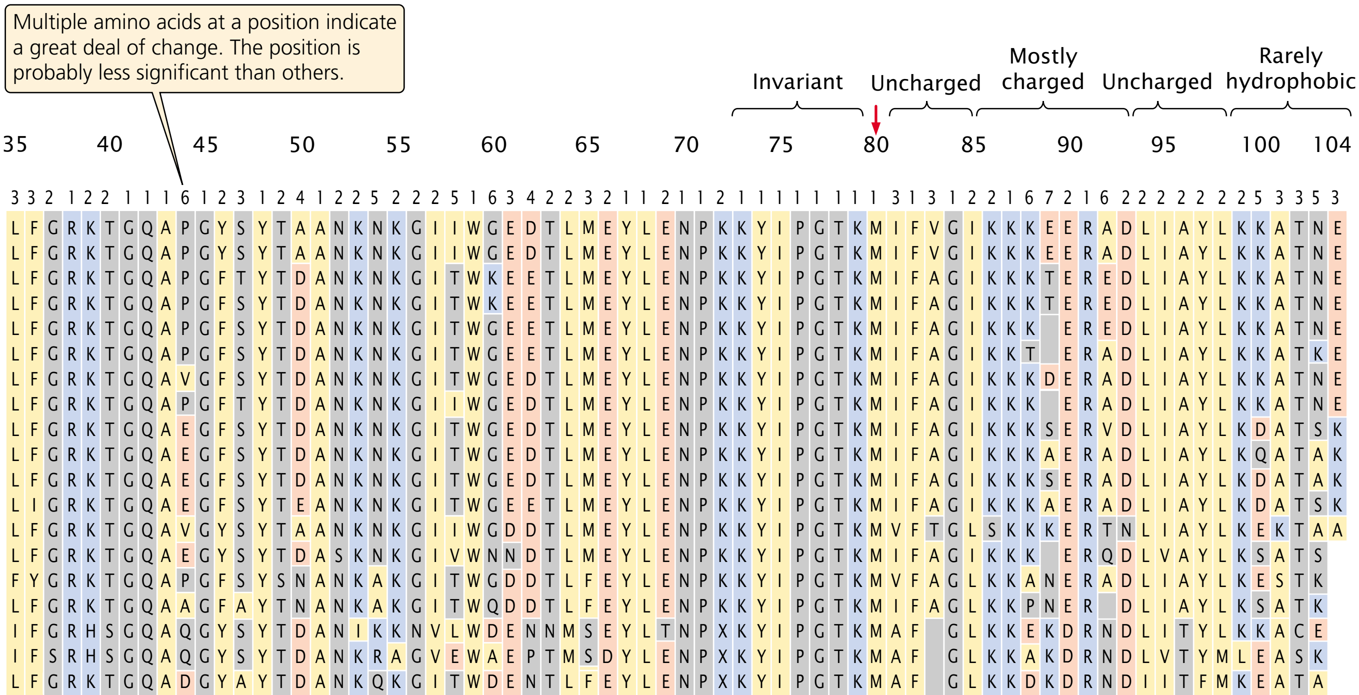
Other:
- C Cysteine
- P Proline
- Q Glutamine
- N Asparagine
- S Serine
- T Threonine
- G Glycine

(b)



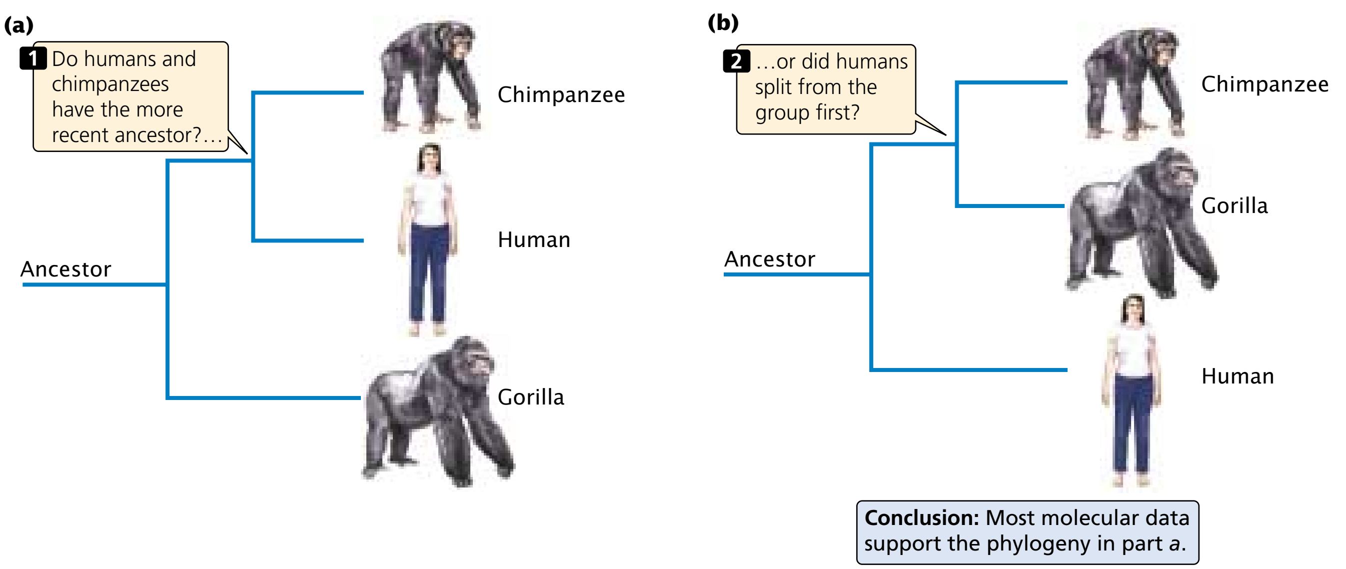
23.24 A phylogeny based on amino acid sequences of the cytochrome *c* molecule.

data were based on amino acid sequences of proteins such as cytochrome *c* (FIGURE 23.24), but, more recently, phylogenies have been based on DNA sequences. One example is the use of DNA sequences to study the relationship of humans to the other apes. Charles Darwin originally proposed that chimpanzees and gorillas were closely related to humans. However, subsequent study has placed humans in the family Hominidae and the great apes (chimpanzees, gorilla, orangutan, and gibbon) in the family Pongidae. Some researchers suggested that gibbons belong to a third family; others proposed that humans are most closely related to orangutans. Molecular data support the hypothesis that humans, chimpanzees, and

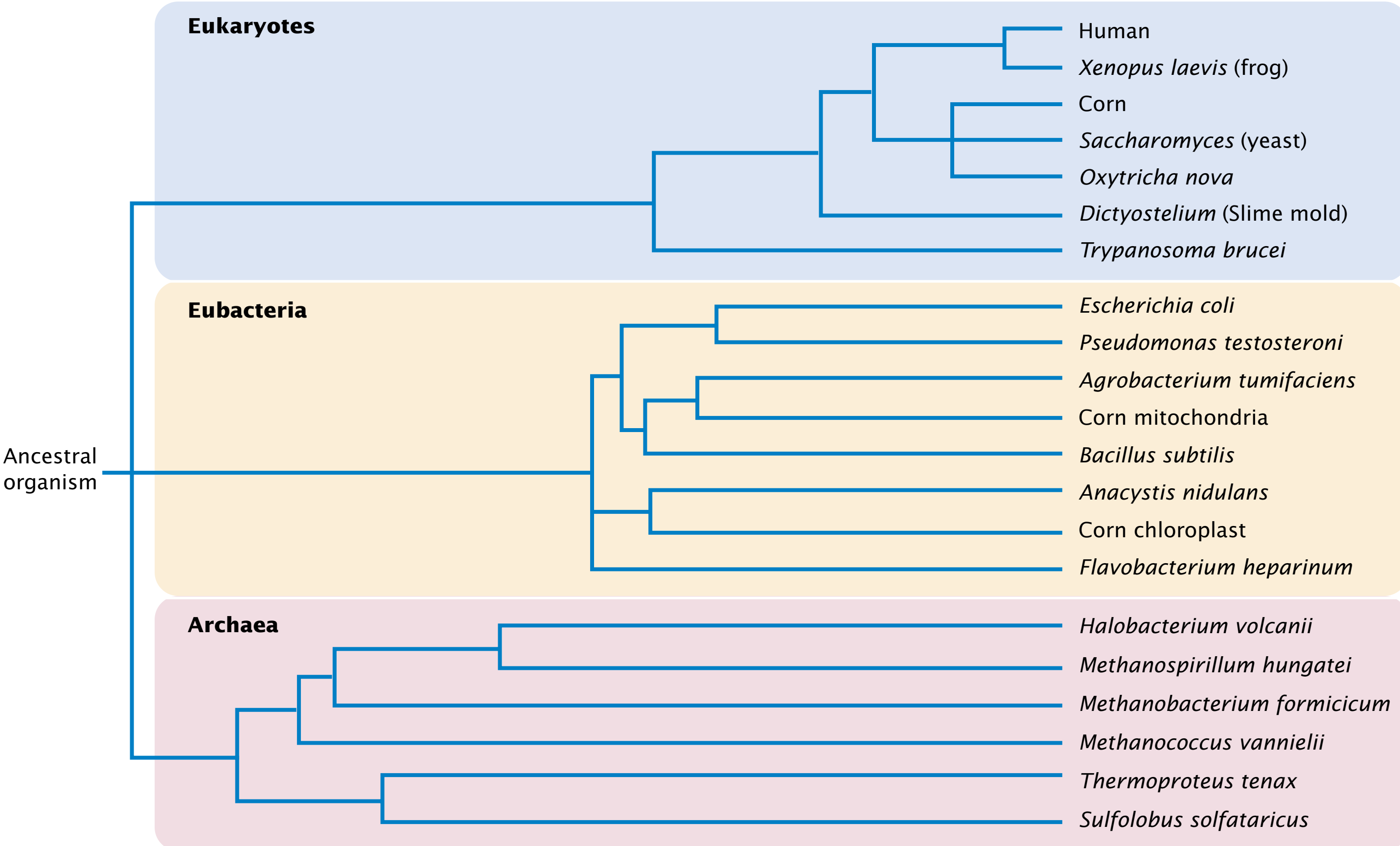


gorillas are most closely related and that orangutans and gibbons diverged from the other apes at a much earlier date. Growing evidence favors a close relationship between humans and chimpanzees (FIGURE 23.25).

Because molecular data can be collected from virtually any organism, comparisons can be made between evolutionary distant organisms. For example, DNA sequences have been used to examine the primary divisions of life and to construct universal phylogenies. On the basis of 16S rRNA, Norman Pace and his colleagues constructed a universal tree of life that included all the major groups of organisms (FIGURE 23.26). The results of their studies



23.25 **Two possible phylogenies of the human, chimpanzee, and gorilla relationships.** One phylogeny suggests (a) that humans and chimpanzees have the more recent ancestor. The other phylogeny suggests (b) that humans split from the group first and that chimpanzees and gorillas have the more recent ancestor.



23.26 A universal tree of life can be constructed from 16S rRNA sequences. Note that sequences from corn mitochondria and chloroplasts are most similar to sequences from eubacteria, confirming the endosymbiotic hypothesis that these eukaryotic organelles evolved from bacteria (see Chapter 20).

revealed that there are three divisions of life: the eubacteria (the common bacteria), the archaea (a distinct group of lesser-known prokaryotes), and the eukaryotes.

Concepts

Molecular data can be used to infer phylogenies (evolutionary histories) of groups of living organisms.

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Connecting Concepts Across Chapters

The central theme of this chapter has been genetic evolution—how the genetic composition of a population changes with time. Unlike transmission and molecular genetics, which focus on individuals and particular genes, this chapter has focused on the genetic makeup of *groups* of individuals. To describe the genes in these groups, we must rely on mathematics and statistical tools; population genetics is therefore fundamentally quantitative in nature. Mathematical models are commonly used in population genetics to describe processes that bring about change in genotypic and allelic frequencies. These models are, by necessity, simplified representations of the real world, but they nevertheless can be sources of insight into how various factors influence the processes of genetic change.

Our study of population genetics depends on and synthesizes much of the information that we have covered in other parts of this book. Describing the genetic composition of a population requires an understanding of the principles of heredity (Chapters 3 through 5) and how genes are changed by mutation (Chapter 17). Our examination of molecular evolution in the second half of the chapter presupposes an understanding of how genes are encoded in DNA, replicated, and expressed (Chapters 10 through 15). It includes the use of molecular tools, such as restriction enzymes, DNA sequencing, and PCR, which are covered in Chapter 18.

CONCEPTS SUMMARY

- Population genetics examines the genetic composition of groups of individuals and how this composition changes with time.
- A Mendelian population is a group of interbreeding, sexually reproducing individuals, whose set of genes constitutes the population's gene pool. Evolution occurs through changes in this gene pool.
- Genetic variation and the forces that shape it are important in population genetics. A population's genetic composition can be described by its genotypic and allelic frequencies.
- The Hardy-Weinberg law describes the effect of reproduction and Mendel's laws on the allelic and genotypic frequencies of a population. It assumes that a population is large, randomly mating, and free from the effects of mutation, migration, and natural selection. When these conditions are met, the allelic frequencies do not change and the genotypic frequencies stabilize after one generation in the Hardy-Weinberg equilibrium proportions p^2, $2pq$, and q^3, where p and q equal the frequencies of the alleles.
- Nonrandom mating affects the frequencies of genotypes but not alleles. Positive assortative mating is preferential mating between like individuals; negative assortative mating is preferential mating between unlike individuals.
- Inbreeding, a type of positive assortative mating, increases the frequency of homozygotes while decreasing the frequency of heterozygotes. Inbreeding is frequently detrimental because it increases the appearance of lethal and deleterious recessive traits.
- Mutation, migration, genetic drift, and natural selection can change allelic frequencies.
- Recurrent mutation eventually leads to an equilibrium, with the allelic frequencies being determined by the relative rates of forward and reverse mutation. Change due to mutation in a single generation is usually very small because mutation rates are low.
- Migration, the movement of genes between populations, increases the amount of genetic variation within populations and decreases differences between populations. The magnitude of change depends both on the differences in allelic frequencies between the populations and on the magnitude of migration.
- Genetic drift, the change in allelic frequencies due to chance factors, is important when the effective population size is small. Genetic drift occurs when a population consists of a small number of individuals, is established by a small number of founders, or undergoes a major reduction in size. Genetic drift changes allelic frequencies, reduces genetic variation within populations, and causes genetic divergence among populations.
- Natural selection is the differential reproduction of genotypes; it is measured by the relative reproductive successes of genotypes (fitnesses). The effects of natural selection on allelic frequency can be determined by applying the general selection model. Directional selection leads to the fixation of one allele. The rate of change in allelic frequency due to selection depends on the intensity of selection, the dominance relations, and the initial frequencies of the alleles.
- Mutation and natural selection can produce an equilibrium, in which the number of new alleles introduced by mutation is balanced by the elimination of alleles through natural selection.
- Molecular methods offer a number of advantages for the study of evolution. The use of protein electrophoresis to study genetic variation in natural populations showed that most natural populations have large amounts of genetic variation in their proteins. Two hypotheses arose to explain this variation. The neutral-mutation hypothesis proposed that molecular variation is selectively neutral and is shaped largely by mutation and genetic drift. The balance model proposed that molecular variation is maintained largely by balancing selection.
- Different parts of the genome show different amounts of genetic variation. In general, those that have the least effect on function evolve at the highest rates.
- The molecular-clock hypothesis proposes a constant rate of nucleotide substitution, providing a means of dating evolutionary events by looking at nucleotide differences between organisms.
- Molecular data are often used for constructing phylogenies.

IMPORTANT TERMS

Mendelian population (p. 000)
gene pool (p. 000)
genotypic frequency (p. 000)
allelic frequency (p. 000)
Hardy-Weinberg law (p. 000)
Hardy-Weinberg equilibrium (p. 000)
positive assortative mating (p. 000)
negative assortative mating (p. 000)
inbreeding (p. 000)
outcrossing (p. 000)
inbreeding coefficient (p. 000)
inbreeding depression (p. 000)
equilibrium (p. 000)
migration (gene flow) (p. 000)
sampling error (p. 000)
genetic drift (p. 000)
effective population size (p. 000)
founder effect (p. 000)
genetic bottleneck (p. 000)
fixation (p. 000)
fitness (p. 000)
selection coefficient (p. 000)
directional selection (p. 000)
overdominance (p. 000)
underdominance (p. 000)
phylogeny (p. 000)
proportion of polymorphic loci (p. 000)
expected heterozygosity (p. 000)
neutral-mutation hypothesis (p. 000)
balance hypothesis (p. 000)
molecular clock (p. 000)

Worked Problems

1. The following genotypes were observed in a population:

Genotype	Number
HH	40
Hh	45
hh	50

(a) Calculate the observed genotypic and allelic frequencies for this population.

(b) Calculate the numbers of genotypes expected if this population were in Hardy-Weinberg equilibrium.

(c) Using a chi-square test, determine whether the population is in Hardy-Weinberg equilibrium.

• Solution

(a) The observed genotypic and allelic frequencies are calculated by using Equations 23.1 and 23.3:

$$f(HH) = \frac{\text{number of } HH \text{ individuals}}{N} = \frac{40}{135} = .30$$

$$f(Hh) = \frac{\text{number of } Hh \text{ individuals}}{N} = \frac{45}{135} = .33$$

$$f(hh) = \frac{\text{number of } hh \text{ individuals}}{N} = \frac{50}{135} = .37$$

$$p = f(H) = \frac{2n_{HH} + n_{Hh}}{2N} = \frac{2(40) + (45)}{2(135)} = .46$$

$$q = f(h) = (1 - p) = (1 - .46) = .54$$

(b) If the population is in Hardy-Weinberg equilibrium, the expected numbers of genotypes are:

$$HH = p^2 \times N = (.46)^2 \times 135 = 28.57$$

$$Hh = 2pq \times N = 2(.46)(.54) \times 135 = 67.07$$

$$hh = q^2 \times N = (.54)^2 \times 135 = 39.37$$

(c) The observed and expected numbers of the genotypes are:

Genotype	Number observed	Number expected
HH	40	28.57
Hh	45	67.07
hh	50	39.37

These numbers can be compared by using a chi-square test:

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

$$= \sum \frac{(40 - 28.57)^2}{28.57} + \frac{(45 - 67.07)^2}{67.07} + \frac{(50 - 39.37)^2}{39.37}$$

$$= 4.57 + 7.26 + 2.87 = 14.70$$

The degrees of freedom associated with this chi-square value are $n - 2$, where n equals the number of expected genotypes, or $3 - 2 = 1$. By examining Table 3.4, we see that the probability associated with this chi-square and the degrees of freedom is $P < .001$, which means that the difference between the observed and expected values is unlikely to be due to chance. Thus, there is a significant difference between the observed numbers of genotypes and the numbers that we would expect if the population were in Hardy-Weinberg equilibrium. We conclude that the population is not in equilibrium.

2. A recessive allele for red hair (r) has a frequency of .2 in population I and a frequency of .01 in population II. A famine in population I causes a number of people in population I to migrate to population II, where they reproduce randomly with the members of population II. Geneticists estimate that, after migration, 15% of the people in population II consist of people who migrated from population I. What will be the frequency of red hair in population II after the migration?

• Solution

From Equation 23.16, the allelic frequency in a population after migration (q'_{II}) is

$$q'_{\text{II}} = q_{\text{I}}(m) + q_{\text{II}}(1 - m)$$

where q_{I} and q_{II} are the allelic frequencies in population I (migrants) and population II (residents), respectively, and m is the proportion of population II that consist of migrants. In this problem, the frequency of red hair is .2 in population I and .01 in population II. Because 15% of population II consists of migrants, $m = .15$. Substituting these values into Equation 23.16, we obtain:

$$q'_{\text{II}} = .2(.15) + (.01)(1 - .15) = .03 + .0085 = .0385$$

This is the expected frequency of the allele for red hair in population II after migration. Red hair is a recessive trait; if mating is random for hair color, the frequency of red hair in population II after migration will be:

$$f(rr) = q^2 = (.0385)^2 = .0015$$

3. Two populations have the following numbers of breeding adults:

Population A: 60 males, 40 females

Population B: 5 males, 95 females

(**a**) Calculate the effective population sizes for populations A and B.

(**b**) What predications can you make about the effects of the different sex ratios of these populations on their gene pools?

• Solution

(**a**) The effective population size can be calculated by using Equation 23.19:

$$N_e = \frac{4 \times n_{\text{males}} \times n_{\text{females}}}{n_{\text{males}} + n_{\text{females}}}$$

For population A:

$$N_e = \frac{4 \times 60 \times 40}{60 + 40} = 96$$

For population B:

$$N_e = \frac{4 \times 5 \times 95}{5 + 95} = 19$$

Although each population has a total of 100 breeding adults, the effective population size of population B is much smaller because it has a greater disparity between the numbers of males and females.

(**b**) The effective population size determines the amount of genetic drift that will occur. Because the effective population size of B is much smaller than that of population A, we can predict that population B will undergo more genetic drift, leading to greater changes in allelic frequency, greater loss of genetic variation, and greater genetic divergence from other populations.

4. Alcohol is a common substance in rotting fruit, where fruit fly larvae grow and develop; larvae use the enzyme alcohol dehydrogenase (ADH) to detoxify the effects of this alcohol. In some fruit-fly populations, two alleles are present at the locus than encodes ADH: ADH^F, which encodes a form of the enzyme that migrates rapidly (fast) on an electrophoretic gel; and ADH^S, which encodes a form of the enzyme that migrates slowly on an electrophoretic gel. Female fruit flies with different *ADH* genotypes produce the following numbers of offspring when alcohol is present:

Genotype	Mean number of offspring
$ADH^F ADH^F$	120
$ADH^F ADH^S$	60
$ADH^S ADH^S$	30

(**a**) Calculate the relative fitnesses of females having these genotypes.

(**b**) If a population of fruit flies has an initial frequency of ADH^F equal to .2, what will be the frequency in the next generation when alcohol is present?

• Solution

(**a**) Fitness is the relative reproductive output of a genotype and is calculated by dividing the average number of offspring produced by that genotype by the mean number of offspring produced by the most prolific genotype. The fitnesses of the three *ADH* genotypes therefore are:

Genotype	Mean number of offspring	Fitness
$ADH^F ADH^F$	120	$W_{FF} = \frac{120}{120} = 1$
$ADH^F ADH^S$	60	$W_{FS} = \frac{60}{120} = .5$
$ADH^S ADH^S$	30	$W_{SS} = \frac{30}{120} = .25$

(**b**) To calculate the frequency of the ADH^F allele after selection, we can use the table method. The frequencies of the three genotypes before selection are the Hardy-Weinberg equilibrium frequencies of p^2, $2pq$, and q^2. We multiply each of these frequencies by the fitness of each genotype to obtain the frequencies after selection. These products are summed to obtain the mean fitness of the population ($\overline{W}$), and the products are then divided by the mean fitness to obtain the relative genotypic frequencies after selection as shown here:

	$ADH^F ADH^F$	$ADH^F ADH^S$	$ADH^S ADH^S$
Initial genotypic frequencies:	$p^2 = (.2)^2 = .04$	$2pq = 2(.2)(.8) = .32$	$q^2 = (.8)^2 = 0.64$
Fitnesses:	$W_{FF} = 1$	$W_{FS} = .5$	$W_{22} = .25$
Proportionate contribution of genotypes to population:	$p^2 W_{FF} = .04(1) = .04$	$2pqW_{FS} = (.32)(.5) = .16$	$q^2 W_{SS} = (.64)(.25) = .16$
Relative genotypic frequency after selection:	$\frac{p^2 W_{FF}}{\overline{W}} = \frac{.04}{.36} = .11$	$\frac{2pqW_{FS}}{\overline{W}} = \frac{.16}{.36} = .44$	$\frac{q^2 W_{SS}}{\overline{W}} = \frac{.16}{.36} = .44$

$\overline{W} = .04 + .16 + .16$
$= .36$

To calculate the allelic frequency after selection, we use Equation 23.4:

$$p = f(ADH^F) = f(ADH^F ADH^F) + \tfrac{1}{2} f(ADH^F ADH^S)$$
$$= .11 + \tfrac{1}{2}(.44) = .33$$

We predict that the frequency of ADH^F will increase from .2 to .33.

The New Genetics

MINING GENOMES

Population Genetics: Analyses and Simulations

In this exercise, you will analyze real molecular data, primarily generated by high-school and college students, to learn how allele frequencies and genotype distributions can be used to study human populations. To do so, you will use the databases and statistical tools at the Dolan DNA Learning Center of Cold Spring Harbor Laboratory. In addition, you will use simulations to explore how factors such as population size, selection pressure, and genetic drift interact to cause allele frequencies to change.

COMPREHENSION QUESTIONS

1. What is a Mendelian population? How is the gene pool of a Mendelian population usually described? What are the predictions given by the Hardy-Weinberg law?
*2. What assumptions must be met for a population to be in Hardy-Weinberg equilibrium?
3. What is random mating?
*4. Give the Hardy-Weinberg expected genotypic frequencies for (**a**) an autosomal locus with three alleles, and (**b**) an X-linked locus with two alleles.
5. Define inbreeding and briefly describe its effects on a population.
6. What determines the allelic frequencies at mutational equilibrium?
*7. What factors affect the magnitude of change in allelic frequencies due to migration?
8. Define genetic drift and give three ways that it can arise. What effect does genetic drift have on a population?
*9. What is effective population size? How does it affect the amount of genetic drift?
10. Define natural selection and fitness.
11. Briefly discuss the differences between directional selection, overdominance, and underdominance. Describe the effect of each type of selection on the allelic frequencies of a population.
12. What factors affect the rate of change in allelic frequency due to natural selection?
*13. Compare and contrast the effects of mutation, migration, genetic drift, and natural selection on genetic variation within populations and on genetic divergence between populations.
14. Give some of the advantages of using molecular data in evolutionary studies.
*15. What is the key difference between the neutral-mutation hypothesis and the balance hypothesis?
16. Outline the different rates of evolution that are typically seen in different parts of a protein-encoding gene. What might account for these differences?
*17. What is the molecular clock?

APPLICATION QUESTIONS AND PROBLEMS

18. How would you respond to someone who said that models are useless in studying population genetics because they represent oversimplifications of the real world?
*19. Voles *(Microtus ochrogaster)* were trapped in old fields in southern Indiana and were genotyped for a transferrin locus. The following numbers of genotypes were recorded.

T^ET^E	T^ET^F	T^FT^F
407	170	17

Calculate the genotypic and allelic frequencies of the transferrin locus for this population.

20. Orange coat color in cats is due to an X-linked allele (X^O) that is codominant to the allele for black (X^+). Genotypes of the orange locus of cats in Minneapolis and St. Paul, Minnesota, were determined and the following data were obtained.

X^OX^O females	11
X^OX^+ females	70
X^+X^+ females	94
X^OY males	36
X^+Y males	112

Calculate the frequencies of the X^O and X^+ alleles for this population.

21. A total of 6129 North American Caucasians were blood typed for the MN locus, which is determined by two codominant alleles, L^M and L^N. The following data were obtained:

Blood type	Number
M	1787
MN	3039
N	1303

Carry out a chi-square test to determine whether this population is in Hardy-Weinberg equilibrium at the MN locus.

22. Genotypes of leopard frogs from a population in central Kansas were determined for a locus that encodes the enzyme malate dehydrogenase. The following numbers of genotypes were observed:

Genotype	Number
M^1M^1	20
M^1M^2	45
M^2M^2	42
M^1M^3	4
M^2M^3	8
M^3M^3	6
Total	125

(a) Calculate the genotypic and allelic frequencies for this population.

(b) What would be the expected numbers of genotypes if the population were in Hardy-Weinberg equilibrium?

23. Full color (D) in domestic cats is dominant over dilute color (d). Of 325 cats observed, 194 have full color and 131 have dilute color.

(a) If these cats are in Hardy-Weinberg equilibrium for the dilution locus, what is the frequency of the dilute allele?

(b) How many of the 194 cats with full color are likely to be heterozygous?

24. Tay-Sachs disease is an autosomal recessive disorder. Among Ashkenazi Jews, the frequency of Tay-Sachs disease is 1 in 3600. If the Ashkenazi population is mating randomly for the Tay-Sachs gene, what proportion of the population consists of heterozygous carriers of the Tay-Sachs allele?

25. In the plant *Lotus corniculatus,* cyanogenic glycoside protects the plants against insect pests and even grazing by cattle. This glycoside is due to a simple dominant allele. A population of *L. corniculatus* consists of 77 plants that possess cyanogenic glycoside and 56 that lack the compound. What is the frequency of the dominant allele that results in the presence of cyanogenic glycoside in this population?

*26. Color blindness in humans is an X-linked recessive trait. Approximately 10% of the men in a particular population are color blind.

(a) If mating is random for the color-blind locus, what is the frequency of the color-blind allele in this population?

(b) What proportion of the women in this population are expected to be color-blind?

(c) What proportion of the women in the population are expected to be heterozygous carriers of the color-blind allele?

*27. The human MN blood type is determined by two codominant alleles, L^M and L^N. The frequency of L^M in Eskimos on a small Arctic island is .80. If the inbreeding coefficient for this population is .05, what are the expected frequencies of the M, MN, and N blood types on the island?

28. Demonstrate mathematically that full sib mating ($F = {}^1\!/_4$) reduces the heterozygosity by ${}^1\!/_4$ with each generation.

29. The forward mutation rate for piebald spotting in guinea pigs is 8×10^{-5}; the reverse mutation rate is 2×10^{-6}. Assuming that no other evolutionary forces are present, what is the expected frequency of the allele for piebald spotting in a population that is in mutational equilibrium?

*30. In German cockroaches, curved wing (cv) is recessive to normal wing (cv^+). Bill, who is raising cockroaches in his dorm room, finds that the frequency of the gene for curved wings in his cockroach population is .6. In the apartment of his friend Joe, the frequency of the gene for curved wings is .2. One day Joe visits Bill in his dorm room, and several cockroaches jump out of Joe's hair and join the population in Bill's room. Bill estimates that 10% of the cockroaches in his dorm room now consists of individual roaches that jumped out of Joe's hair. What will be the new frequency of curved wings among cockroaches in Bill's room?

31. A population of water snakes is found on an island in Lake Erie. Some of the snakes are banded and some are unbanded; banding is caused by an autosomal allele that is recessive to an allele for no bands. The frequency of banded snakes on the island is .4, whereas the frequency of banded snakes on the mainland is .81. One summer, a large number of snakes migrate from the mainland to the island. After this migration, 20% of the island population consists of snakes that came from the mainland.

(a) Assuming that both the mainland population and the island population are in Hardy-Weinberg equilibrium for the alleles that affect banding, what is the frequency of the allele for bands on the island and on the mainland before migration?

(b) After migration has taken place, what will be the frequency of the banded allele on the island?

*32. Calculate the effective size of a population with the following numbers of reproductive adults:

(a) 20 males and 20 females

(b) 30 males and 10 females

(c) 10 males and 30 females

(d) 2 males and 38 females

33. Pikas are small mammals that live at high elevation in the talus slopes of mountains. Populations located on mountain tops in Colorado and Montana in North America are relatively isolated from one another, because the pikas don't occupy the low-elevation habitats that separate the mountain tops and don't venture far from the talus slopes. Thus, there is little gene flow between populations. Furthermore, each population is small in size and was founded by a small number of pikas.

A group of population geneticists propose to study the amount of genetic variation in a series of pika populations and to compare the allelic frequencies in different populations. On the basis of biology and the distribution of pikas, what do you predict the population geneticists will find concerning the within- and between-population genetic variation?

34. In a large, randomly mating population, the frequency of the allele (*s*) for sickle-cell hemoglobin is .028. The results of studies have shown that people with the following genotypes at the beta-chain locus produce the average numbers of offspring given:

Genotype	Average number of offspring produced
SS	5
Ss	6
ss	0

(a) What will be the frequency of the sickle-cell allele (*s*) in the next generation?

(b) What will be the frequency of the sickle cell allele at equilibrium?

35. Two chromosomal inversions are commonly found in populations of *Drosophila pseudoobscura:* Standard (*ST*) and Arrowhead (*AR*). When treated with the insecticide DDT, the genotypes for these inversions exhibit overdominance, with the following fitnesses:

Genotype	Fitness
ST/ST	.47
ST/AR	1
AR/AR	.62

What will be the frequency of *ST* and *AR* after equilibrium has been reached?

*36. In a large, randomly mating population, the frequency of an autosomal recessive lethal allele is .20. What will be the frequency of this allele in the next generation?

37. A certain form of congenital glaucoma results from an autosomal recessive allele. Assume that the mutation rate is 10^{-5} and that persons having this condition produce, on the average, only about 80% of the offspring produced by persons who do not have glaucoma.

(a) At equilibrium between mutation and selection, what will be the frequency of the gene for congenital glaucoma?

(b) What will be the frequency of the disease in a randomly mating population that is at equilibrium?

CHALLENGE QUESTIONS

38. The Barton Springs salamander is an endangered species found only in a single spring in the city of Austin, Texas. There is growing concern that a chemical spill on a nearby freeway could pollute the spring and wipe out the species. To provide a source of salamanders to repopulate the spring in the event of such a catastrophe, a proposal has been made to establish a captive breeding population of the salamander in a local zoo. You are asked to provide a plan for the establishment of this captive breeding population, with the goal of maintaining as much of the genetic variation of the species as possible in the captive population. What factors might cause loss of genetic variation in the establishment of the captive population? How could loss of such variation be prevented? Assuming that it is feasible to maintain only a limited number of salamanders in captivity, what procedures should be instituted to ensure the long-term maintenance of as much of the variation as possible?

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